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Smart Education and e-Learning 2021

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Editors

Smart Education and e-Learning 2021

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Preface

Smart education, smart e-learning, and smart universities are emerging and rapidly growing areas. They have a potential to transform existing teaching strategies, learning environments, educational activities, and technology in a classroom. Smart education and e-learning are focused at enabling instructors to develop new ways of achieving excellence in teaching in highly technological smart classrooms and smart universities and providing students with new opportunities to maximize their success and select the best options for their education, location, learning style, and mode of learning content delivery.

The ongoing COVID-19-related crisis is forcing various institutions of higher education to investigate and find new forms of modern education process, innovative pedagogy, active use of the state-of-the-art technology and systems, and propose new business models of operation. “The pandemic is speeding up changes in a tremendous way,” says Bert van der Zwaan, former rector of Utrecht University in the Netherlands, and author of *Higher Education in 2040: A Global Approach*. The author argues that “the phoenix of an entirely new type of university will rise from the ashes of the classical system: less tied to buildings and set locations, the new university will embed itself more deeply in society by offering innovative forms of digital knowledge and making customized teaching available on demand.”

We—the members of the international Smart Education and Smart e-Learning (SEEL) professional research and academic communities—believe that the traditional universities should be transformed to smart universities (SmU). The smart universities, smart education, and smart e-learning are characterized by various distinctive smart features, including (1) sensing and data collecting, (2) inferring or data processing and generating information, (3) self-analysis and information processing, (4) adapting to new conditions/restrictions/limitations, (5) anticipating and getting knowledge, and (6) self-optimization or self-organization and active use of obtained knowledge. Particularly:

1. “Sensing” smartness feature deals with SmU’s ability to automatically use various sensors and monitoring/control devices (robots) to identify, recognize, understand, and/or become aware of various events, processes, objects,

- phenomenon, etc., that may have impact (positive or negative) on SmU's operation, infrastructure, or well-being of its components—students, faculty, staff, software and hardware systems, computer network, etc.
2. “Inferring” (or, logical reasoning) smartness feature deals with SmU's ability to automatically make logical conclusion(s) on the basis of raw data, processed information, observations, evidence, assumptions, and/or established/implemented rules.
 3. “Self-learning (self-exploration, self-assessment, self-analysis, self-discovery, self-description)” smartness feature deals with SmU's ability to automatically obtain, acquire or formulate new or modify existing knowledge, experience, or behavior to improve its operation, business functions, performance, effectiveness, etc.
 4. “Adaptation” smartness feature deals with SmU's ability to automatically modify its teaching/learning strategies, administrative, safety, technological, and other characteristics, infrastructure, network, systems, etc., to better operate and perform its main business functions such as teaching, training, e-learning, safety, management, maintenance, and control.
 5. “Anticipation (awareness)” smartness feature deals with SmU's intelligence and predictive analytics software systems' ability to automatically collect raw data, process it in real time and predict what is going to happen, and how to address a specific event.
 6. “Self-organization (self-optimization, reconfiguration, restructuring, and self-recovery)” smartness feature deals with SmU's ability to automatically change its internal structure (components), self-regenerate and self-sustain in a purposeful (non-random) manner under appropriate conditions but without an external agent/entity. (A note: Self-protection, self-matchmaking, and self-healing are a part of self-organization).

From June of 2014 the enthusiastic and visionary scholars, faculty, Ph.D. students, administrators, and practitioners from all over the world have an excellent opportunity for a highly efficient and productive professional meeting—the annual international conference on smart education and smart e-learning. The members of SEEL international professional research and academic communities actively perform research and share their ideas and research outcomes in various related areas such as smart education, smart e-learning, smart universities, smart classrooms, smart pedagogy, and smart campus.

The KES international professional association initiated SEEL conference as a major international forum for the presentation of innovative ideas, approaches, technologies, systems, findings and outcomes of research; and design and development projects in the emerging areas of smart education, smart e-learning, smart pedagogy, smart analytics, applications of smart technology, and smart systems in education and e-learning, smart classrooms, smart universities, and knowledge-based smart society.

The inaugural international KES conference on Smart Technology-based Education and Training (STET) has been held at Chania, Crete, Greece, June 18–20, 2014.

The 2nd international KES conference on Smart Education and Smart E-Learning took place in Sorrento, Italy, June 17–19, 2015, the 3rd KES SEEL conference—in Puerto de la Cruz, Tenerife, Spain, June 15–17, 2016, the 4th KES SEEL conference—in Vilamoura, Portugal, June 21–23, 2017, the 5th KES SEEL conference—in Gold Coast, Australia, June 20–22, 2018, the 6th KES SEEL conference—in St. Julians, Malta, June 17–19, 2019, and the 7th KES SEEL conference was held in the online mode by the KES Virtual Conference Center, June 17–19, 2020.

The 8th international KES conference on smart education and e-Learning (SEEL-2021) will be held as a virtual conference by the KES Virtual Conference Center, June 14–16, 2021.

The main topics of the SEEL international conference are grouped into several clusters and include but are not limited to

- **Smart Education (SmE cluster):** conceptual frameworks for smart education; smart university; smart campus; smart classroom; smart learning environments; stakeholders of smart university; mathematical modeling of smart university; academic or institutional analytics; university-wide smart systems for teaching, learning, research, management, safety, security; research projects, best practices, and case studies on smart education; partnerships, national and international initiatives, and projects on smart education; economics of smart education.
- **Smart Pedagogy (SmP cluster):** innovative smart teaching and learning technologies; learning-by-doing; active learning; experiential learning, games-based learning and gamification of learning; collaborative learning; analytics-based learning; flipped classroom; crowdsourcing-based learning; project-based learning; adaptive learning; badging-based learning; productive failure-based learning; smart learning analytics; research projects, best practices, and case studies on smart pedagogy; smart curriculum and courseware design and development; smart assessment and testing; smart university's student/learner modeling; faculty modeling, faculty development, and instructor's skills for smart education; university-wide smart systems for teaching and learning; learning management systems; smart blended, distance, online and open education; partnerships, national and international initiatives and projects on smart pedagogy.
- **Smart e-Learning (SmL cluster):** smart e-learning: concepts, strategies, and approaches; massive open online courses (MOOC); small personal online courses (SPOC); assessment and testing in smart e-learning; serious games-based smart e-learning; smart collaborative e-learning; adaptive e-learning; smart e-learning environments; courseware and open education repositories for smart e-learning; smart e-learning pedagogy, teaching and learning; smart e-learner modelling; smart e-learning management, academic analytics, and quality assurance; faculty development and instructor's skills for smart e-learning; research, design and development projects, best practices, and case studies on smart e-learning; standards and policies in smart e-learning; social, cultural, and ethical dimensions of smart e-learning; economics of smart e-learning.
- **Smart Technology, Software, and Hardware Systems for Smart Education and e-Learning (SmT cluster):** smart technology-enhanced teaching and

learning; adaptation, sensing, inferring, self-learning, anticipation, and self-organization of smart learning environments; Internet of Things (IoT), cloud computing, RFID, ambient intelligence, and mobile wireless sensor networks applications in smart classrooms and smart universities; smart phones and smart devices in education; educational applications of smart technology and smart systems; mobility, security, access, and control in smart learning environments; smart gamification; smart multimedia; smart mobility.

- **“From Smart Education to Smart Society” Continuum (SmS cluster):** smart school; applications of smart toys and games in education; smart university; smart campus; economics of smart universities; smart university’s management and administration; smart office; smart company; smart house; smart living; smart health care; smart wealth; smart lifelong learning; smart city; national and international initiatives and projects; smart society.
- **“Smart University as a Hub for Students’ Engagement into Virtual Business and Entrepreneurship (SmB cluster):** entrepreneurship and innovation at university: student role and engagement; student engagement with virtual businesses and virtual companies; virtual teams and virtual team working (technology, models, ethics); university curricula for entrepreneurship and innovation (core and supplemental courses); new student goal—start his/her own business (instead of getting a job in a company); students and start-up companies (approaches, models, best practices, and case studies).

This year several very dynamic subgroups in our international research and academic communities proposed in-depth discussion on a number of specific topics in smart education and smart e-learning. We strongly support those pioneering initiatives and are very thankful to the organizers and chairs of the following special sessions at SEEL-2021 international conference:

- IS01: Smart Education and Smart Universities and their Impact on Students with Disabilities (chair—Prof. Jeffrey P. Bakken);
- IS02: Digital Education and Economics in Smart University (chair—Prof. Natalia A. Serdyukova);
- IS03: Recent Advances in Online Delivery of Courses (chair—Prof. Chee Peng Lim, co-chairs—Prof. Tamara Shikhnabieva and Prof. Lakhmi Jain);
- IS04: Smart University Development: Organizational, Managerial and Social Issues (chair—Prof. Svetlana A. Gudkova, co-chair—Prof. Lyudmila V. Glukhova);
- IS05: Online Education Empowered with Artificial Intelligence Techniques (Chair—Dr. Aleksandra Klasnja-Milicevic, co-chair—Dr. Mirjana Ivanovic).

One of the advantages of the SEEL conference is that it is organized in conjunction with several other smart digital futures (SDF) high-quality conferences, including agents and multi-agent systems technologies and applications (AMSTA), human-centred intelligent systems (HCIS), intelligent decision technologies (IDT), innovation in medicine and health care (InMed), and smart transportation systems (STS). This provides SEEL conference participants with unique opportunities to attend also

AMSTA, HCIS, IDT, InMed, and STS conferences' presentations, meet, and collaborate with subject matter experts in those “smart” areas—fields that are conceptually close to smart education and smart e-learning.

This book contains the contributions presented at the 8th international KES conference on smart education and e-learning. It comprises forty-one high-quality peer-reviewed papers that are grouped into several interconnected parts:

- Part I—Smart Education,
- Part II—Smart e-Learning,
- Part III—Smart Education: Systems and Technology,
- Part IV—Smart Education: Case Studies and Research,
- Part V—Digital Education and Economics in Smart University,
- Part VI—Smart University Development: Organizational, Managerial and Social Issues, and
- Part VII—Smart Universities and Their Impact on Students with Disabilities.

We would like to thank many scholars—members of the SEEL-2021 international program committee—who dedicated many efforts and time to make SEEL international conference a great success, namely:

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It is our sincere hope that this book will serve as a useful source of valuable collection of knowledge from various research, design and development projects, useful information about current best practices and case studies, and provide a baseline of further progress and inspiration for research projects and advanced developments in smart education and smart e-learning areas.

Peoria, IL, USA
 Shoreham-by-Sea, UK
 Sydney, Australia
 June 2021

Vladimir L. Uskov
 Robert J. Howlett
 Lakhmi C. Jain

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Smart Education

Smart Education: Predictive Analytics of Student Academic Performance Using Machine Learning Models in Weka and Dataiku Systems



Vladimir L. Uskov, Jeffrey P. Bakken, Prasanthi Putta,
Deepali Krishnakumar, and Keerthi Sree Ganapathi

Abstract Smart education and smart universities are based on active use of descriptive, diagnostic, predictive and prescriptive analytics as prescribed by the Gartner's Data Analytics Ascendancy Model. This paper presents the up-to-date findings and outcomes of the research, design and development project at the InterLabs Research Institute at Bradley University (U.S.A.) aimed at application of a quantitative approach to student academic performance data analytics in general, and innovative Machine Learning (ML) models-based approaches and systems to predictive academic and learning analytics in particular. The goal of this research is to identify the best ML models in the Weka and Dataiku data processing systems based on various forms of student data representation and multiple evaluation criteria for quality of predictive analytics. The analyzed ML models included Support Vector Machine, Naïve Bayes, Random Forest, Random Tree, Linear Regression, Logistic Regression, k-Nearest Neighbors, Multilayer Perceptron, J48, and Decision Stump models. The evaluation criteria for predictive analytics included Correlation Coefficient, Mean Absolute Error, Mean Absolute Percentage Error, Root Mean Squared Error, Root Mean Squared Logarithmic Error, R2 Score for regression ML models and Correctly Classified Instances and Incorrectly Classified Instances for classification ML models. The obtained research outcomes provide a well-validated recommendation about what ML models should be used in student academic performance predictive analytics in smart education and smart universities.

Keywords Predictive analytics · Machine learning models · Weka system · Dataiku system · Evaluation criteria for data prediction · Quality of data prediction · Student data representation forms

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1 Introduction and Literature Review

Smart education and smart universities are based on active use of main “smartness” features such as (a) adaptation, (b) sensing (i.e. getting data from sensors), (c) inferring (i.e. data processing and getting logical conclusions), (d) self-description and self-learning, (e) monitoring of incoming data and anticipation, and (g) self-organization and self-optimization. Smart academic and learning data analytics are based on active use of descriptive, diagnostic, predictive and prescriptive types of analytics as prescribed by the Gartner’s Analytics Ascendancy Model. As a result, the institutional data, student data, academic and learning data analytics and corresponding advanced software systems and technologies are crucial components of smart education and smart universities.

Importance of student data and data analytics. In accordance with the recent EDUCAUSE report “Top 10 IT Issues, 2020: The Drive to Digital Transformation Begins”, “... Institutions [of higher education] that haven’t implemented data governance and data architecture will need to do so in order to begin using AI [artificial intelligence] and analytics to deliver personalized, timely student services” [1].

“Each year, colleges lose up to a third of the students they accept to “summer melt,” and the global pandemic has only made the problem worse in 2020. However, using data analytics and business intelligence programs to make smarter student enrollment choices helped colleges reduce their summer melt by about 1 percent this fall, a new analysis suggests. The report, from AI and business analytics firm Othot, also argues that colleges using these technologies performed three times better than the national average in terms of their fall 2020 enrollment figures” [2].

“In the wake of COVID-19, demand for student success analytics has risen significantly” [3].

Literature review. The outcomes of literature review completed by our research team clearly shows that the researchers in Student Academic Performance (SAP) data analytics area primarily concentrate on analysis of student performance based on a few specific Machine Learning (ML) models and a very limited number of quality evaluation criteria used for quality of SAP data prediction; very often the authors even do not mention what evaluation criteria have been used.

Hamsa et al. in [4] described the developed “... student’s academic performance prediction model, for the Bachelor and Master degree students in Computer Science and Electronics and Communication streams using two selected classification methods: Decision Tree and Fuzzy Genetic Algorithm”.

Hasan et al. in [5] described how “... WEKA data mining tool is used to evaluate the decision tree algorithm for discovery of student’s performance along with Moodle access time. Simulation results demonstrate that Random Forest Tree algorithm showed better accuracy than comparative decision tree algorithms”.

Imran et al. in [6] wrote “... Three supervised learning algorithms (J48, NNge and MLP) are employed in this study for experimental purposes. The results showed that J48 achieved highest accuracy 95.78% among others”.

Adejo and Connolly in [7] wrote: “This research study advances the understanding of the application of ensemble techniques to predicting student performance using learner data. It used three state-of-art data mining classifiers, namely, decision tree, artificial neural network and support vector machine for the modeling”.

The completed analysis of the designated and multiple additional publications relevant to SAP data, Educational Data Mining (EDM) and Learning Analytics (LA) areas (for example, [8–10]) clearly shows that, unfortunately, those and many other additional publications do not provide a systematic approach to (1) use of various forms of SAP data representation, for example, scores for course assignments in absolute or relative (%) numbers, in detailed letter grade or regular letter grade forms, (2) use and comparison of obtained outcomes for multiple available regression and classification ML models, and (3) use and comparison of obtained research outcomes for multiple evaluation criteria [11] of quality of SAP predicted data. As it is shown below, these topics make significant contributions to the high quality accuracy of SAP data predictive analytics.

Our relevant past work. In our past publications, for example, in [12–14], we (1) identified at least 7 main sources of student data in academic environment [12]; (2) identified specifics of original SAP data and analyzed the main types of SAP data inconsistencies [12]; (3) identified the main types of SAP data representation to be used in smart LA systems [12]; (4) designed and developed the architectural model—i.e. components and links between them—of smart LAS systems for smart university [13, 14], and (5) identified at least 5 hierarchical levels (with their organization similar to “nested” loops in programming) of SAP data processing needed for a high quality of SAP predictive analytics in smart LA systems [13, 14].

2 Research Project Goal, Objectives and Environment

Project goal. The goal of the current phase of a multi-aspect research, design and development project at the InterLabs Research Institute at Bradley University (U.S.A.) is to identify the best ML models in the Weka and Dataiku systems for SAP data analytics, based on (a) various forms of SAP data representation, (b) multiple regression and classification ML models in designated systems, and (c) multiple evaluation criteria for quality of SAP data predictive analytics.

Project objectives. To achieve this goal, the project team concentrated on the following project objectives: (1) analyze the main characteristics of Weka and Dataiku systems—those that are relevant to SAP data analytics; (2) perform benchmarking of ML models in terms of quality of SAP data predictive analytics based on various SAP input data sets, and (3) arrive with well-thought recommendations for researchers and practitioners in academia regarding the obtained quality of ML models for SAP predictive analytics in designated systems.

Project research environment—SAP input datasets. SAP input data sets for ML testing are based on student performance data collected by one of the co-authors while teaching the advanced computer science course in 2015–2020. A total of 188 students took this course in the past (“past students”); as a result, the corresponding SAP data was cleaned and used to train ML models using “90–10” training–testing ratio to arrive with the SAP master dataset.

A total of 21 students were in the “special” group during the current semester—“current students”.

18 learning assignments are used in the designated course such as assignments for homework, in-classroom exercises, tests, labs, comprehensive multi-aspect and multi-task course project, midterm and final exams.

For the purpose of this research, we used SAP input data set of variant A—Var A. It contains SAP data obtained by current students for the first 9 learning assignments, including midterm exam. As a result, various ML models were trained and used to predict current students’ final scores (final grades) in the course, based on (1) SAP data for all 18 course learning assignments obtained by 188 past students in this course, and (2) SAP data of 21 current students in the course—those who completed just the first 9 course learning assignments in the course.

Variant B—Var B—contains SAP data obtained by the current students for the first 14 course learning assignments.

Variant C—Var C—contains SAP data, obtained by the current students for all 18 course learning assignments. This dataset was available after the completion of course by all 21 current students. It was not used in training/testing of ML models at all; it was used solely by research team to evaluate quality (accuracy and adequacy) of SAP predictive analytics by ML models in Weka and Dataiku systems.

One of the main goals of this research was to (1) compare the quality of predictions of students’ final scores (or, final grades) in the course by various ML models based on available Var A and Var B SAP input datasets, and (2) evaluate quality of SAP data predictions based on calculation of a difference between those predictions, i.e. (Var B–Var A) values for each ML model.

Project research environment—SAP input data forms. SAP data in predictive analytics may be stored and used in various forms in LA systems. This is because (1) different universities may have different policies on the evaluation of learning assignments, submitted by students, and (2) different forms of SAP data may have an impact on the accuracy of SAP data processing and prediction in data analytics systems [12]. The main SAP data forms include (1) “SAP data in Absolute Numbers” (ABS) form, where maximal score for each learning assignment is represented by the absolute number of points, for example, 15, 50, 100, etc. points; (2) “SAP data in Relative Numbers” (REL) form, where each course learning assignment is represented by a percentage amount of its portion of (contribution to) the course final grade, for example, for example, Test 1 may contribute 2.42% to the final grade, midterm exam—10.00%, etc.; (3) “SAP data in Detailed Letter Grades” (DLG) form, where assignment is represented by the detailed letter grade such as A+, A, A–, B+, B,

B—, etc., and (4) “SAP data in Regular Letter Grades” (RLG) form, where each assignment is represented by the letter grade such as A, B, C, D, etc.

Project research environment—systems with ML models used. We actively use the WEKA and Dataiku systems for a comprehensive benchmarking of ML algorithms’ effectiveness and accuracy in terms of described SAP data predictive analytics.

The WEKA system is an “excellent open-source collection of plug-in algorithms for a machine learning workbench including artificial, neural network algorithms, and artificial, immune system algorithms” [15].

The Dataiku system is “the platform democratizing access to data and enabling enterprises to build their own path to AI in a human-centric way” [16].

Table 1 contains a summary of main characteristics of both systems.

ML models used. We actively used the following 10 regression and classification ML models in this research: (1) multilayer perceptron (MLP), (4) k-Nearest Neighbors (kNN), (5) decision stump (DS), (6) random forest (RF), (7) random tree (RT), (8) logistic regression (LoR), (9) naïve Bayes (NB), and (10) J48 ML models.

Prediction quality evaluation criteria used. We have actively used 10 data prediction evaluation criteria for quality of SAP data predictive analytics (as compared to the actual data outcomes in Var C datasets), including: (1) correlation coefficients (CC) with numeric values in the range from 0 to 1, where “1” corresponds to the best result; (2) mean absolute error (MAE) meaning the average of the magnitude of the individual errors without taking account of their sign, where lesser numeric value means better quality; (5) root mean squared logarithmic error (RMSLE), where lesser numeric value means better quality; (6) relative absolute error (RAE), where lesser numeric value means better quality; (7) root relative squared error (RRSE), where lesser numeric value means better quality; (8) R2 score (R2), where greater numeric value means better quality; (9) correctly classified instances—CCI (in %), and (10) incorrectly classified instances—ICI (in %).

Table 2 contains a summary of data prediction evaluation criteria used in Weka and Dataiku systems in our project.

3 Project Research Outcomes

In accordance with the project goal, our research team conducted hundreds of experiments focused at identification of the best ML models in the Weka and Dataiku systems for SAP data predictive analytics. (A note: Due to the limited space available in this paper, we present below just a few examples of obtained research outcomes; other relevant obtained outcomes are available upon written request).

Table 1 Weka and Dataiku systems: a summary of main characteristics

Characteristics	Weka system	Dataiku system
Most important features and functions	• Data pre-processing • Classification • Regression • Clustering • Association • Visualization	• Interactive Data Pipeline • Interactive Python, R, and SQL, Notebooks • Code and Share Your own Recipes • Interactive Data Visualization • Model and Predict • Code or Click • Deploy and Run • Integrated documentation and knowledge sharing • Change management • Team activity monitoring
Strengths and Opportunities	• Powerful tool for ML analysis • Installing the software is simple; it needs Java 8 to be installed as a pre-requisite • Free online courses and course materials that teach machine learning and data mining using Weka are available • Easy to learn • Easy to use • Meets almost all user requirements	• The tool offers a dashboard that makes it simple for users to generate visualizations and interactive charts from their datasets • With Dataiku DSS, users can explore, generate, and do preparations without dealing with storage, access, and format issues • Availability of free online tutorials and User-friendly documentation
Possible weaknesses and threats	• Memory upper bound is of 2.5 GBs. As a result, “too slow” and “not enough memory” problems often occur	• License cost is on the higher end • Less support on platform installation and maintenance
Technical Platform	• Windows • Mac OS • Linux	• Linux • Mac OS
Price (if any)	• Free	• Free Community Edition • Commercial with various option such as Discover (For Small Teams), Business (For Mid-Sized teams), and Enterprise (Scalable Automation and Governance)

Table 2 WEKA and Dataiku systems: evaluation criteria available and used

	Prediction evaluation criteria used	Weka	Dataiku
For regression ML models	Correlation coefficient (between 0 to 1, where 1 is the best)	+	+
	Mean absolute error (lower the better)	+	+
	Root mean squared error (lower the better)	+	+
	Relative absolute error (lower the better)	+	
	Root relative squared error (lower the better)	+	
	Mean Absolute Percentage Error (lower the better)		+
	Root Mean Squared Logarithmic Error (lower the better)		+
	R2 Score (higher the better)		+
For classification ML models	Correctly Classified Instances	+	+
	Incorrectly Classified Instances	+	+

3.1 Predictive Analytics of Student Academic Performance Using Weka System

The examples of obtained research numeric outcomes regarding quality of SAP predictive analytics by ML models in the Weka systems are presented in Tables 3 and 4. The outcomes of two particular cases are presented in graphical form on Fig. 1.

3.2 Predictive Analytics of Student Academic Performance Using Dataiku System

The examples of obtained research outcomes regarding quality of SAP predictive analytics by ML models in the Weka systems are presented in Tables 5 and 6, and Fig. 2.

4 Discussion and Recommendations

A summary of the obtained SAP predictive data analytics is presented in Table 7. It contains information about (1) ML models in both Weka and Dataiku systems—those that provided the best SAP data predictive analytics outcomes (as compared by outcomes of Var A and Var B datasets versus Var C datasets), (2) our ranking of

Table 3 Regression ML models in the Weka system: A comparison of quality of SAP data prediction and identified best (in green color) and worst (in red color) predicted outcomes, i.e. students' finals scores/grades in the course

SAP data in REL form	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C)							
		MLP	LiR	SVM	KNN(=5)	DS	RF	RT
	CC (Fig.1a)	0.754	0.809	0.828	0.758	0.074	0.799	0.184
	MAE	8.038	4.296	3.968	5.368	8.613	5.716	9.695
	RMSE	12.128	5.388	5.141	6.591	12.888	6.697	13.060
	RAE	118.50%	63.33%	58.50%	79.13%	126.97%	84.27%	142.93%
	RRSE	136.37%	60.58%	57.80%	74.11%	144.91%	75.30%	146.85%
	"Var B"-based prediction of SAP final scores/grades by ML models (as compared to Var C)							
		MLP	LiR	SVM	KNN(=5)	DS	RF	RT
	CC (Fig. 1b)	0.576	0.869	0.858	0.783	0.000	0.885	0.424
	MAE	8.694	3.852	4.001	5.089	6.830	4.284	8.092
	RMSE	13.181	4.425	4.581	6.177	8.919	4.923	10.533
	RAE	128.18%	56.79%	58.98%	75.03%	100.69%	63.16%	119.30%
	RRSE	148.20%	49.76%	51.50%	69.45%	100.28%	55.35%	118.43%
	(Var B - Var A) increase of quality of SAP data prediction by ML algorithms							
		MLP	LiR	SVM	KNN(=5)	DS	RF	RT
	CC	0.177	0.061	0.030	0.025	0.074	0.086	0.240
	MAE	0.657	0.444	0.033	0.278	1.783	1.432	1.603
	RMSE	1.053	0.962	0.560	0.414	3.969	1.774	2.527
	RAE	0.097	0.065	0.005	0.041	0.263	0.211	0.236
	RRSE	0.118	0.108	0.063	0.047	0.446	0.199	0.284
SAP data in ABS form	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C)							
		MLP	LiR	SVM	KNN(=5)	DS	RF	RT
	CC	0.758	0.841	0.827	0.758	0.074	0.784	-0.083
	MAE	40.634	20.276	19.922	26.838	43.063	30.056	41.627
	RMSE	60.308	25.412	25.808	32.957	64.439	35.651	52.550
	RAE	119.81%	59.78%	58.74%	79.13%	126.97%	88.62%	122.74%
	RRSE	135.62%	57.15%	58.03%	74.11%	144.91%	80.17%	118.17%
	"Var B"-based prediction of final scores/grades by ML models (as compared to Var C)							
		MLP	LiR	SVM	KNN(=5)	DS	RF	RT
	CC	0.602	0.869	0.858	0.783	0.106	0.898	0.534
	MAE	44.435	19.258	20.017	25.448	34.148	19.944	33.575
	RMSE	66.626	22.128	22.901	30.885	44.563	23.090	37.749
	RAE	131.02%	56.78%	59.02%	75.03%	100.69%	58.80%	99.00%
	RRSE	149.83%	49.76%	51.50%	69.45%	100.21%	51.92%	84.89%
	(Var B - Var A) increase of SAP data prediction's quality by ML algorithms							
		MLP	LiR	SVM	KNN(=5)	DS	RF	RT
	CC	0.156	0.029	0.031	0.025	0.032	0.114	0.617
	MAE	3.801	1.018	0.094	1.390	8.915	10.112	8.051
	RMSE	6.317	3.285	2.907	2.072	19.876	12.562	14.801
	RAE	0.112	0.030	0.003	0.041	0.263	0.298	0.237
	RRSE	0.142	0.074	0.065	0.047	0.447	0.282	0.333

identified best regression and classification ML models in both used systems, and (3) our recommendations regarding particular ML models to be used for each specific combination of SAP data form and type of ML model.

Table 4 Classification ML models in the Weka system: A comparison (by tenfold cross validation) of quality of SAP data prediction and identified best (in green color) and worst (in red color) predicted outcomes, i.e. students' finals scores/grades in the course

SAP data in DLG form	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C) in %								
		NB	LoR	MLP	SVM	kNN(k=5)	J48	RF	RT
	CCI	34.78	52.17	39.13	39.13	47.83	39.13	52.17	21.74
	ICI	65.22	47.83	60.87	60.87	52.17	60.87	47.83	78.26
	"Var B"-based prediction of final scores/grades by ML models (as compared to Var C) in %								
		NB	LoR	MLP	SVM	kNN(k=5)	J48	RF	RT
	CCI	34.78	21.74	39.13	30.43	30.43	43.48	43.48	21.74
	ICI	65.22	78.26	60.87	69.57	69.57	56.52	56.52	78.26
	(Var B - Var A) increase of SAP data prediction quality by ML algorithms in %								
		NB	LoR	MLP	SVM	kNN(k=5)	J48	RF	RT
	CCI	17.39	0.00	17.39	4.35	26.09	47.83	8.70	13.04
	ICI	17.39	0.00	17.39	4.35	26.09	47.83	8.70	13.04
SAP data in RLG form	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C) in %								
		NB	LoR	MLP	SVM	kNN(k=5)	J48	RF	RT
	CCI	52.17	47.83	43.48	47.83	34.78	60.87	30.43	43.48
	ICI	47.83	52.17	56.52	52.17	65.22	39.13	69.57	56.52
	"Var B"-based prediction of final scores/grades by ML models (as compared to Var C) in %								
		NB	LoR	MLP	SVM	kNN(k=5)	J48	RF	RT
	CCI	69.57	47.83	60.87	43.48	60.87	13.04	39.13	56.52
	ICI	30.43	52.17	39.13	56.52	39.13	86.96	60.87	43.48
	(Var B - Var A) increase of SAP data prediction quality by ML algorithms in %								
		NB	LoR	MLP	SVM	kNN(k=5)	J48	RF	RT
	CCI	17.39	0.00	17.39	4.35	26.09	47.83	8.70	13.04
	ICI	17.39	0.00	17.39	4.35	26.09	47.83	8.70	13.04

5 Conclusions and Next Steps

Conclusions. The obtained research findings and outcomes enabled us to make the following main conclusions:

1. Predictive analytics of SAP data is a complex multi-level and multi-task hierarchical process; its components and organization (sequencing) has a significant impact on the effectiveness and quality of SAP data predictive analytics.
2. The quality of SAP data predictive analytics significantly depends of (a) a system or library of ML models, (b) forms of SAP data representation, (c) sets of available regression and classification ML models, (d) evaluation criteria for quality of SAP data predictive analytics, and (e) size and quality of SAP input datasets.
3. For the SAP data in "relative numbers" and "absolute numbers" forms, (a) linear regression, (b) random forest and (c) support vector machine ML models in both Weka (Table 3) and Dataiku (Table 5) systems demonstrated the best outcomes in terms of quality of SAP data predictive analytics.

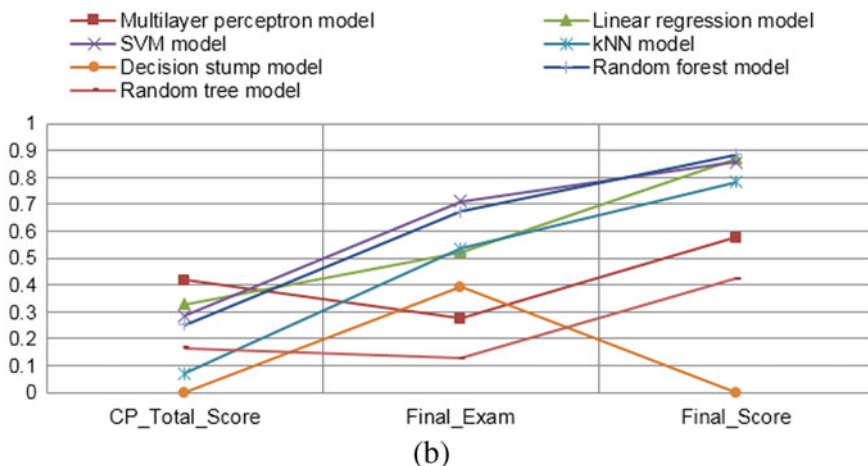
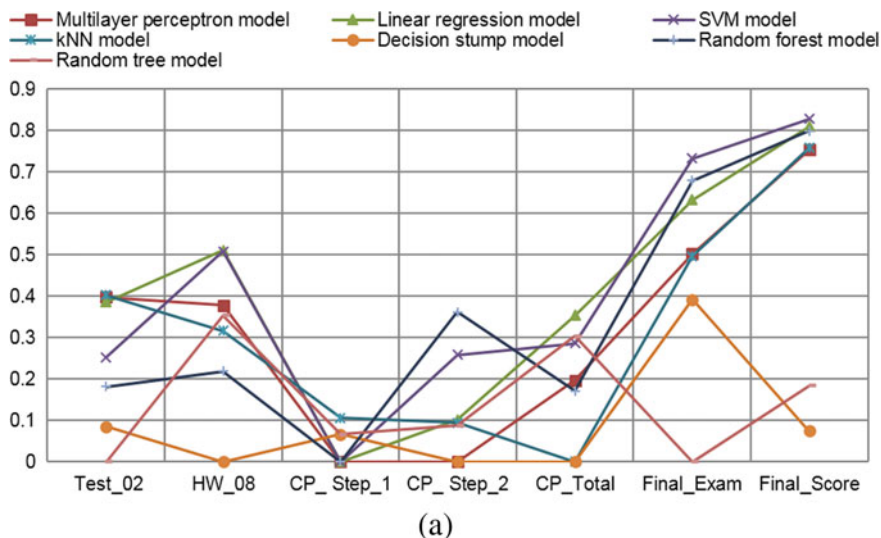


Fig. 1 Regression ML models in Weka system: the obtained outcomes for quality of SAP data processing in REL form and prediction using calculated values of “correlation coefficient” evaluation criteria (Y-axis) for 2 cases: **a** for the “Var A versus Var C” case based on available SAP data for 9 out of 18 course learning assignments; **b** for the “Var A versus Var C” case based on available SAP data for 14 out of 18 course learning assignments (X-axis)

4. There is no a clear leader among best ML models in analyzed systems for the SAP data in “detailed letter grades” and “regular letter grades” SAP data forms. Partially the best outcomes in Weka system (Table 4) were demonstrated by (a) logistic regression, (b) random forest, and (c) J48 ML models; in Dataiku

Table 5 Regression ML models in the Dataiku system: A comparison of quality of SAP data prediction and identified best (in green color) and worst (in red color) predicted outcomes, i.e. students' finals scores/grades in the course

SAP data in REL form	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LiR	kNN(k=5)	DT
	CC	0.865	0.824	0.849	0.610	0.536
	MAE	4.496	4.238	3.850	6.089	9.765
	MAPE	0.052	0.049	0.044	0.071	0.113
	RMSE	5.614	5.290	4.907	7.086	13.690
	RMSLE	0.028	0.026	0.026	0.036	0.073
	R2S	0.749	0.679	0.721	0.372	0.287
	"Var B"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LiR	kNN(k=5)	DT
	CC	0.855	0.891	0.873	0.824	0.618
	MAE	4.749	3.811	3.745	4.492	8.184
	MAPE	0.056	0.043	0.042	0.053	0.099
	RMSE	6.180	4.301	4.344	5.318	10.561
	RMSLE	0.032	0.021	0.021	0.027	0.041
	R2S	0.731	0.794	0.762	0.678	0.382
	(Var B - Var A) increase of SAP data prediction quality by ML algorithms					
		SVM	RF	LiR	kNN(k=5)	DT
	CC	0.010	0.067	0.024	0.214	0.082
	MAE	0.252	0.426	0.105	1.598	1.582
	MAPE	0.004	0.006	0.002	0.018	0.015
	RMSE	0.566	0.988	0.563	1.768	3.128
	RMSLE	0.003	0.006	0.005	0.008	0.032
	R2S	0.018	0.115	0.042	0.306	0.094
SAP data in ABS form	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LiR	kNN(k=5)	DT
	CC	0.874	0.813	0.846	0.715	0.536
	MAE	30.947	25.203	19.200	25.480	42.439
	MAPE	0.073	0.057	0.045	0.059	0.099
	RMSE	40.350	29.948	24.322	32.374	58.496
	RMSLE	0.041	0.030	0.026	0.033	0.074
	R2S	0.764	0.661	0.716	0.512	0.288
	"Var B"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LiR	kNN(k=5)	DT
	CC	0.873	0.887	0.886	0.758	0.521
	MAE	31.361	32.574	20.005	27.371	40.541
	MAPE	0.074	0.071	0.044	0.062	0.091
	RMSE	41.051	39.405	23.851	34.453	48.360
	RMSLE	0.042	0.038	0.023	0.034	0.050
	R2S	0.762	0.786	0.785	0.575	0.271
	(Var B - Var A) increase of SAP data prediction quality by ML algorithms					
		SVM	RF	LiR	kNN(k=5)	DT
	CC	0.001	0.073	0.040	0.043	0.016
	MAE	0.414	7.371	0.804	1.891	1.898
	MAPE	0.001	0.014	0.000	0.003	0.008
	RMSE	0.700	9.457	0.472	2.079	10.136
	RMSLE	0.001	0.008	0.003	0.001	0.024
	R2S	0.002	0.125	0.069	0.063	0.016

system (Table 6)—(a) k-Nearest Neighbors ($k = 5$), (b) support vector machine, (c) logistic regression and (d) random forest ML models.

- The research team arrived with a final set of recommendations regarding utilization of most effective particular ML model for each specific combination of SAP data form and type of ML model—regression or classification (Table 7).

Table 6 Classification ML models in the Dataiku system: A comparison (by tenfold cross validation) of quality of SAP data prediction and identified best (in green color) and worst (in red color) predicted outcomes, i.e. students' finals scores/grades in the course

SAP data in DLG form	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LoR	kNN(k=5)	DT
	CCI	34.78%	21.74%	26.09%	39.13%	4.35%
	ICI	65.22%	78.26%	73.91%	60.87%	95.65%
	"Var B"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LoR	kNN(k=5)	DT
	CCI	34.78%	13.04%	34.78%	34.78%	8.70%
	ICI	65.22%	86.96%	65.22%	65.22%	91.30%
	(Var B - Var A) increase of SAP prediction quality by ML algorithms					
		SVM	RF	LoR	kNN(k=5)	DT
SAP data in RLG form	CCI	0.00%	8.70%	8.70%	4.35%	4.35%
	ICI	0.00%	8.70%	8.70%	4.35%	4.35%
	"Var A"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LiR	kNN(k=5)	DT
	CCI (Fig. 2a)	43.48%	0.478	43.48%	43.48%	0.261
	ICI	56.52%	0.522	56.52%	56.52%	0.739
	"Var B"-based prediction of final scores/grades by ML models (as compared to Var C)					
		SVM	RF	LoR	kNN(k=5)	DT
	CCI (Fig. 2b)	0.478	0.522	0.609	0.304	43.48%
	ICI	0.522	0.478	0.391	0.696	56.52%
	(Var B - Var A) increase of SAP data prediction quality by ML algorithms					
		SVM	RF	LoR	kNN(k=5)	DT
	CCI	4.35%	4.35%	17.39%	13.04%	17.39%
	ICI	4.35%	4.35%	17.39%	13.04%	17.39%

Next Steps. Based on the obtained research/design/development findings and outcomes, the next steps in the "SAP Data Predictive Analytics" research project are aimed at (1) integration of the developed InterLabs Smart Learning Analytics system [14] with ML models in Weka and Dataiku systems, and (2) continuation of analysis of SAP data predictive analytics using various ML models in the Scikit-Learn library, Azure ML, Goggle autoML and SAS platforms.

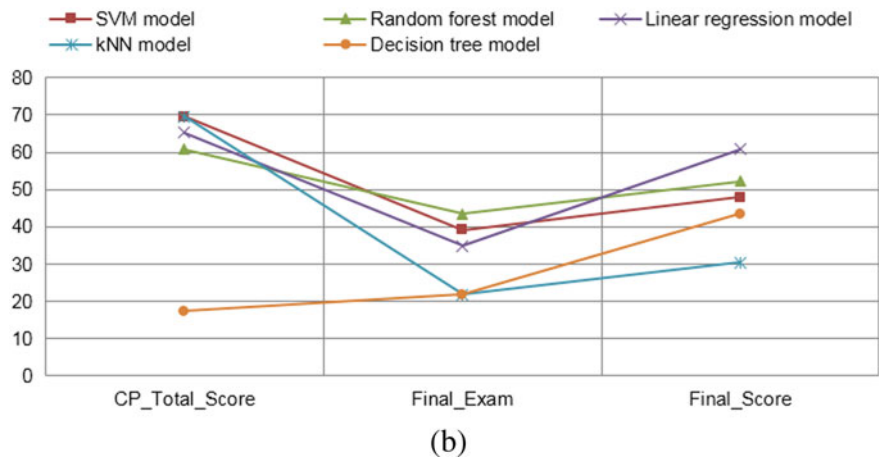
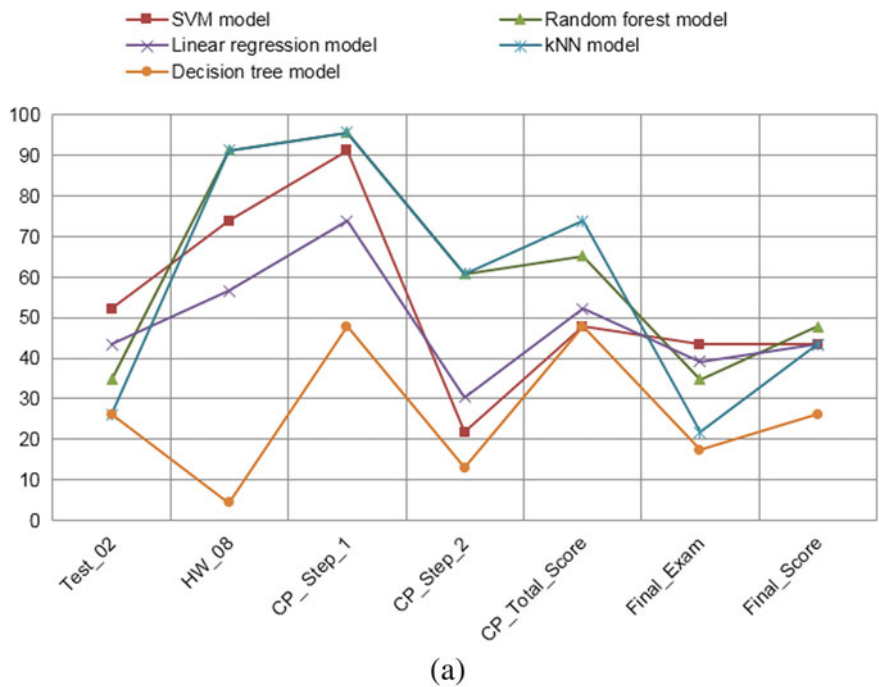


Fig. 2 Classification ML models in the Dataiku system: the obtained outcomes for quality of SAP data processing in RLG form and prediction using calculated values of “correctly classified instances” evaluation criteria (Y-axis) for 2 cases: **a** for the “Var A versus Var C” case based on available SAP data for 9 out of 18 course learning assignments, **b** for the “Var B versus Var C” case based on available SAP data for 14 out of 18 course learning assignments (X-axis)

Table 7 SAP data processing and prediction with ML models in the WEKA and Dataiku systems: our recommendations

Type of SAP data	ML models	Best outcomes were calculated by the following ML models for Var A-based prediction		Best outcomes were calculated by the following ML models for Var B-based prediction		Our ranking of ML models	Our recommendations
Data form		Weka	Dataiku	Weka	Dataiku		
SAP data in REL form	Regression ML models	SVM	SVM	LiR	RF	Best	SVM, LiR
		LiR	LiR	SVM	LiR	2nd best	RF
		RF	RF	RF	kNN (k = 5)	3rd best	–
SAP data in ABS form	Regression ML models	LiR	LiR	LiR	LiR	Best	LiR
		SVM	RF	RF	RF	2nd best	RF
		RF	kNN (k = 5)	SVM	kNN (k = 5)	3rd best	kNN (k = 5)
SAP data in DLG form	Classification ML models	LoR, RF	kNN (k = 5)	J48, RF	kNN (k = 5)	Best	kNN (k = 5), RF
		–	SVM	–	SVM	2nd best	J48, SVM
		kNN (k = 5)	LoR	MLP	LoR	3rd best	LoR
SAP data in RLG form	Classification ML models	J48	RF	NB	LoR	Best	NB, LoR
		NB	LoR, SVM	MLP	RF	2nd best	RF, J48
		LoR, SVM	–	kNN (k = 5)	SVM	3rd best	kNN (K = 5), SVM

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16. Dataiku system: <https://www.dataiku.com/>

Educational Trajectories Modeling for Practice-Oriented Higher Education



Elena A. Boldyreva and Lubov S. Lisitsyna

Abstract This article is devoted to modeling and managing educational trajectories for competitive graduates learning in knowledge-intensive professional areas of practice-oriented higher education. The authors propose the educational space and educational trajectories models of the practical-oriented programs of study (PoS). These models consider numerous factors of influence—changes in the regulatory framework of profiles and trends in the professional field, planned and achieved student learning outcomes. The educational space’s main feature is its redundancy—it contains the maximum possible number of planned learning outcomes. The model involves detailing elementary ones’ primary competence and then comparing them with the educational space Q ’s expected learning outcomes. The model in the form of a hyper-graph (plan-graph) establishes causal relationships between learning outcomes, including planned ones. The authors propose a method of managing educational trajectories based on a multi-layer plan-graph of the workshop. This research result is the educational space and the plan-graph of PoS “Computer Systems and Technologies” of ITMO University. With these models’ help, two versions of the “Network Protocols” course workshop were developed and implemented in the educational process (in 2019 and 2020 years). The workshop was tested by students of the 3rd semester of the master’s program. The expert community in network technologies evaluated the achieved results of training in this discipline in the form of a course project. To assess the effectiveness of the workshops’ updated versions, the authors used the degree of closeness of students’ tasks in the course project to real professional tasks. For the last academic year, it was 73%, and after the implementation of the modified version of the workshop in the current academic year—90%.

Keywords Workshop design · Management of educational trajectories · Educational space model · Hypergraph · Modeling of educational trajectories

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1 Introduction

The trend of modern higher education is the construction of individual educational trajectories. Research in the field of pedagogy [1, 2] allows us to define an individual educational trajectory as a personal trajectory of developing students' personal, intellectual, creative, and other abilities. When moving along this educational trajectory, the development and realization of the student's potential occur.

The educational trajectory is an ordered sequence of elements of the student's educational activity to achieve personal, educational goals that correspond to their abilities, capabilities, motivation, interests, carried out with the teacher's coordinating, organizing, and consulting activities [3].

The formation of an educational program in a university as an individual educational trajectory is a specific model for achieving higher professional education. The individual educational trajectory at the university can be implemented in three main directions: content (theoretical and practical content of the Program of Study (PoS), the PoS variability and learning directions), activity (expressed in the individualization of pedagogical technologies), and organizational aspects.

In the conditions of constantly changing job (labor) market needs in knowledge-intensive professional areas and student-oriented higher education, the individual educational trajectory is implemented in the form of a purposeful PoS, which allows the student to choose and implement the graduate's competence model with the help of pedagogical support. It is necessary to allow the student to study according to the labor market's latest trends to prepare him as much as possible for his future professional activity.

It is about considering the significant number of continually changing labor market requirements, which must be considered when designing the PoS of multidisciplinary education and many professional disciplines that require systematic revision for updating. To solve this problem, it is necessary to offer models for managing educational trajectories within students' practice-oriented education.

This article is devoted to modeling educational trajectories in the redundant educational space of the subject (professional) field. The authors propose the models for managing educational trajectories based on a hypergraph, which allows us to reflect the redundancy and variability of educational content.

1.1 *The Task of the Educational Trajectories' Modeling*

The basis of the methodology for modeling and managing educational trajectories is the diagnosis and monitoring of all stages of its passage for compliance with the PoS regulatory documentation requirements. Since educational trajectories exist within a competency-based approach to education, it is undoubtedly essential to have an orderly gradual formation of mastering competencies' necessary results. The

employers expect these decision models as learning results (outcomes) (step-by-step passing of the training trajectory in the format “from simple to complex”).

Thus, the success of mastering the PoS workshop in the multidisciplinary area depends not only on the selection of relevant tasks for specific profiles but also on their optimal location on the educational process schedule, in compliance with the principle of the sequence of development and formal restrictions.

The complexity of modeling and managing the educational trajectories in the multidisciplinary PoS is that the number of learning tasks of the PoS workshop can reach several hundred tasks. Without proper modeling and visualization of its results, it is quite challenging to consider all the features of the disciplines’ location in the accounting plan, all the limitations of regulatory documentation and at the same time minimize the redundancy of the educational trajectory. That appears from repeated duplication of the same theoretical and practical materials.

This task leads to the fact that these theoretical and practical aspects are studied repeatedly in several disciplines. The data of the results of a survey of 97 students for existing PoS “Computer systems and technologies” of the University ITMO showed that the basics of computer arithmetic that are part of the discipline “Informatics” in the first year, then re-read (again) also in the disciplines of “Computer Architecture”, “Low-level programming”, “Functional circuitry” and partially in “Principles of professional activity”. It happens because teachers are sometimes unable to keep track of changes in the content of related disciplines. Sometimes it is considered that students did not fully master the material earlier. Sometimes the same topics and learning tasks are included in the work programs—this means that the teacher is forced to spend contact training time on repeating what he has already completed, instead of ensuring that students complete the training course optimally.

Thus, the task of research is to provide a tool for modeling and managing educational trajectories of the practice-oriented PoS from the point of view of optimal theoretical and practical content for each profile of multidisciplinary training, depending on the formalized requirements for educational trajectories of a particular profile and the restrictions imposed by the curriculum.

2 Research Background

The latest trend in the implementation of the educational process at the university is the creation of an internal and external practice-oriented educational environment in order to acquaint the student with the professional field, with the actual requirements of potential employers and allow him to form his vision of professional activity, to realize his potential in it. Several papers in this area do not imply the direct influence of the professional field’s requirements on forming the educational trajectory and tracking all stages of its passage. An ontological approach describes the competence model in the IT field [4, 5]. The educational process and educational programs models are based on the graph model [6, 9, 10] and expert analysis [7, 8]. In research [11] the author introduces information needs. He characterizes the personal information

need by the absence of a model interpretation of the processes taking place in the surrounding world. Any human activity (including professional) is based primarily on decisions made by a person—on decision models. Furthermore, it manifests itself due to the lack of a person's model of a decision within a specific situation. Simultaneously, the decision model will reflect what a person will do, what tools and technologies to use.

In terms of the labor market, a sought-after specialist differs from an unclaimed one in that he has many solution models for a variety of professional problems. He does not need time to form them. Since practice-oriented higher education aims to place the student in conditions in which he can form these decision models, the cost of error will not be so high. The learning tasks that a student solves during education form an idea of professional activity processes and formulate the necessary decision models in advance if the educational task is as close as possible to the labor one. The generated decision models (algorithms for specific actions and skills in using specific tools) are the potential graduate's employers' planned learning outcomes (LO).

We should also note that specialists of different majors sometimes have different decision models for the same professional tasks. So educational trajectories must meet the information needs of each student—they should be flexible. Thus, the problem arises of educational trajectories managing based on the planned LO, considering the professional field's requirements and students' information needs.

3 Model of Redundant Educational Space for the Practice-Oriented PoS

The totality of all LOs identified by specific learning systems, tools, and technologies form an educational space model Q —L.S. Lisitsyna first proposed the model [10].

This discrete space defines the set of states of the student's competence $X = \{x_0, x_1, x_2, x_3 \dots\}$ (shown in Fig. 1b), each of which is $x_i \in X$ is "responsible" for obtaining a specific LO in terms of knowledge and skills (shown in Fig. 1a). The basic properties of the educational space Q and the restrictions imposed on this space are described in [10]. The model assumes detailing the core (frame) competence to elementary ones and then comparing them with the educational space's expected learning outcomes. However, this study proposes to move from specially established expected results and, step by step, link them to professional competence and, accordingly, to the curriculum disciplines PoS. We define the educational process in the educational space Q context as a transition from one state of the set X to another.

The state x_0 determines the starting point of the process of preparing students for PoS. The educational space Q 's learning outcomes come from the excess content of the professional area's labor market requirements using the LO identification method and following specific educational tasks. For this comparison, the teacher of the discipline acts as an expert and implements it independently. Solving educational problems (the obtained model of solutions) and the LO required by the employer are

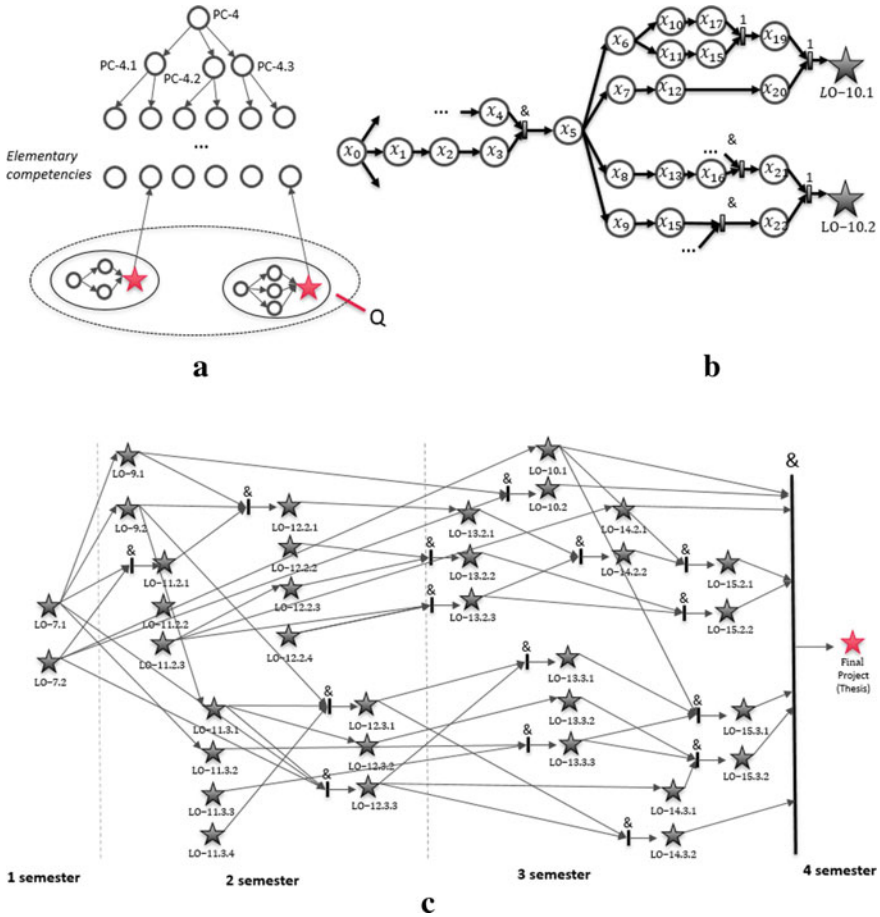


Fig. 1 **a** Hierarchical model of the relationship between the learning outcome and the competence displayed on the educational space. **b** Plan-graph (educational trajectories) of discipline workshop on the task level. **c** Plan-graph of the PoS “Computer systems and technologies” (masters’ level) at the level of planned learning outcomes.

considered a single, indivisible whole. So, the corresponding knowledge and skills formation is carried out throughout the entire PoS workshop.

It is also necessary to note the following feature of the discrete educational space. The set of states of the educational process $X = \{x_1 \cup x_2 \cup x_3 \cup \dots\}$ contains subsets $x_i \neq \emptyset$, describing the states of the educational process of the PoS workshop, which are responsible for mastering the professional competency (PC).

Let W_i be the set of paths that connect the expected LO of the PoS with the specific results of mastering the competence PC_i . Let us establish the mapping ω when $\omega: W_i \rightarrow X$ determines the correspondence between mastering the competence and the expected LO through the set of variable states X in the educational space Q .

The route $w_{ij} \in W_i$ simulates the planned result of mastering the competence PC_i , expected by LO, and the trajectory of movement in the educational space Q . Moving along the trajectory of ordered elements leads to the final state of the subset X_i , which we marked with a star sign.

If we talk about the educational space from the point of view of designing an PoS workshop, then the final state is a solved educational problem (a successfully created decision model). The previous states of the subset X_i correspond to auxiliary educational tasks that are solved using specific tools, technologies and methods, and their solution is the result of mastering detailed (components) competencies. The planned LO, modeled by the specific trajectory, is mapped into many interchangeable elements of the subset X_i , where $X_i \in X$. Interchangeable states meet the needs of potential employers and PCs (professional competencies) for specific skills and level of proficiency in tools but differ in the specifics of the formation of these skills, depending on the profile of multidisciplinary training. A hierarchical model of the relationship between the learning outcome and professional competence is shown in Fig. 1a.

This section's result is as follows: the educational space Q is structured based on planned LO and represents a set of variable states $X_i \in X$, each responsible for a set of tools and technologies for achieving LO.

4 Development of an Educational Trajectories Graph Model

This section presents a model in the form of a hyper-graph (plan-graph), which establishes causal relationships between learning outcomes, including those planned (partially showed in Fig. 1b). This model is scalable, i.e. multilayer. The upper layer of the model is the causal relationship between the planned learning outcomes (the totality of these results forms the core of the educational program workshop) (shown in Fig. 1c). At this level of representation, the model is well understood by the employer. The lower layers of the model are associated with splitting the hyper-graph vertices and replacing them with sub-hyper-graphs until each vertex—the result of training—is associated with the results of solving the training problems of the workshop. This level is clear to teachers and allows to manage educational trajectories at the level of the PoS workshop content in general and the workshops of disciplines (shown in Fig. 1b).

Modeling the educational process in the form of a plan graph can be performed based on the statement “the formation of the planned LOs of one of the variable states of the educational space X_i is based on the formed results of mastering the basic and constituent competencies of a non-empty subset of states $X_{(i-1)}, X_{(i-2)}, X_{(i-3)}, X_{(i-4)} \dots$ ”. It defines the n -fold relationship in the state space X .

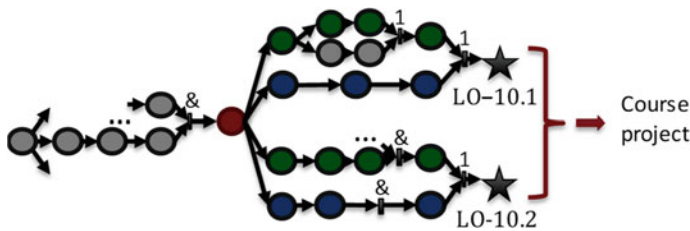


Fig. 2 The plan-graph fragment with different educational trajectories for the discipline “Network Protocols”

Thus, we build a plan-graph of the educational space Q , which determines the order of forming the development of competencies, planned LOs, and cause-and-effect relationships between them. Accordingly, the states that are the “predecessors” of the current one are the so-called prerequisites—that is, what the learner should know and do before starting to master new information.

The plan-graph of the educational space Q is a weighted directed hypergraph $H(X, P)$, where the set X is the set of vertices of the hypergraph representing the states of the educational space Q . P is the set of oriented hyper-edges, where $P = \emptyset$. In this case, the hyper-edges’ directions allow an ordered transition between states and lead to the planned LO. Using the weight w_i of transitions (hyper-edges), one can consider the normative parameters of labor intensity (academic hours). A fragment of the plan-graph with different educational trajectories for different years of the PoS “Computer systems and technologies” is shown in Fig. 2.

According to such a plan-graph, the educational trajectories management is reduced to the formal task of constructing a subhypergraph to the planned LOs. The achieved LOs meet the needs of potential employers of PoS graduates choosing alternative vertices at the hyperedge source. Moreover, reaching at least one such vertex will not require reaching other vertices of this subset. The variability of connections of the drain vertex x_i with the source vertices of the i -th hyper-arc is determined by the Boolean function in DNF (disjunctive normal form), in which each variable $y_1, y_2, \dots$, takes two values: “1”—the corresponding LO immediately precedes the i -th LO, “0”—otherwise. So, for example, in Fig. 1b the transition to the state x_5 is possible only as a result of the composition of two sequences passing through x_3 and x_4 (corresponds to the “logical AND” function). And the transition to planned LO 10.1 can be carried out in three alternative ways. The variation vertices in this example are x_6 and x_7 , x_{10} and x_{11} . In this case, the top of the drain x_{19} and the planned LO 10.1 corresponds to the “logical OR” function.

It is shown that the characteristic features of the plan-graph are its scalability and variability. Some layers simulate the causal relationships between the planned LOs, between the LO of the entire PoS and its disciplines. The variability of the model is associated with the revision of the planned LOs.

These models formed the basis of the approach to the management of educational trajectories, namely the computer-aided design model of the workshop profile [12] based on the analysis of the labor market.

5 Practical Results

For experimental research, it was decided to take a workshop of the discipline “Network protocols”, which is implemented in the third semester of the master’s program, forms professional competence number 4, and ends with preparing and defense of the course project.

Figure 2 describes this discipline’s plan-graph with variable educational trajectories for 2019 (blue track) and 2020 (green track) years. Table 1 describes the variable part of the educational trajectory of the discipline.

Table 1 clearly shows how, based on the professional field analysis and the LO identification method’s application, six unique tools were selected for the expected learning outcomes. Furthermore, it was decided to supplement the existing practice of the discipline—to the existing Ptolemy modeling environment add laboratory work on network modeling in the PacketTracer program, which allows gaining experience in configuring networks and real network equipment, and on analyzing network traffic using the Wireshark program. All trainees completed the course, the average score for the course was 85.1%.

Table 1 Educational space of the discipline “Network Protocols” with two alternative trajectories

Learning Outcomes	Educational space of the discipline “Network Protocols”	
	Original Version (2019—blue track)	Updated Version (2020—green track)
LO-10.1 Analyzes network traffic in a multi-segment network	1. Simulates the data flow of the TCP/IP protocol stack 2. Builds models to study various mechanisms for managing the flow of data in the network 3. Analyzes the data transfer rate in the network with different protocol parameters (tool: Ptolemy)	1. Configures network traffic capture, configures traffic filters 2. Analyzes network traffic using various data flow management mechanisms 3. Analyzes the performance of network connections 4. Models and identifies threats in network traffic (tool: Wireshark, PingPlotter)
LO-10.2 Administers and analyzes equipment in multi-segment networks	1. Models and configures network elements 2. Simulates the interaction process for multiple TCP connections 3. Develops a simulation model of a multi-segment network of a high level of abstraction (tool: Ptolemy)	1. Configures the Cisco network equipment 2. Simulates the interaction process for wired and wireless connection of network elements 3. Configures static and dynamic routing parameters 4. Develops a simulation model of a multi-segment network with Cisco equipment (tool: Packet Tracer)

Of particular interest is how to assess as objectively as possible the effectiveness of the implementation of this workshop. The main difficulty in assessing the effectiveness of graduates' competitiveness is that the labor market is "filled" with graduates gradually. It will be possible to assess the effect of the implemented workshops and educational trajectories through feedback from graduates' employers. However, such statistics on dynamics will become available only in the long term and covered in my doctoral dissertation.

Until then, an assessment of the effectiveness of the application of updated versions of workshops can be the degree of closeness of students' tasks in the course project to real professional tasks.

To control the achievement of the learning outcomes, we used the course project of the discipline, which combined all the educational tasks of the discipline. To assess the two achieved LO of discipline "Network protocols" we invited 15 experts in network technologies. They assess students' course projects for 2019 and 2020 for consideration. The final score of proximity was calculated as a weighted average, considering the degrees of confidence in each expert's opinion [12]. Over the past academic year degree of closeness was 73%. After implementing the modified version of the workshop in the current academic year degree of closeness became 90%. The slight discrepancy in the estimates shows that the workshop was quite relevant in the past but still benefited from its modification.

6 Conclusion

Based on the outcomes of the conducted research, the authors describe the educational trajectories management models—the educational space Q model for a practice-oriented training program and the plan-graph for the educational trajectories of the workshop.

The relationship between learning outcomes and professional competencies is provided by the learning outcomes hierarchy model, first proposed by prof. Lisitsyna L.S. in her Thesis. The model involves detailing the main competence to elementary ones and then comparing them with the educational space Q's expected learning outcomes. However, in this study, it is proposed to move away from the specially established planned learning outcomes in the educational space and step by step to link them to professional competence, and, accordingly, to the curriculum disciplines. The peculiarity of the educational space is its redundancy—so we take into account the variability of the professional field and give space for varying learning paths.

The model in the form of a Hyper-graph (the plan-graph) establishes the causal relationship between learning outcomes, including expected. The top layer of the model is the causal relationship between the expected learning outcomes. At this level of representation, the model is well understood by the employer. The lower layers of the model are associated with splitting the hyper graph vertices and replacing them with sub-hyper graphs until each vertex—the result of training—is associated with the results of solving the workshop's training problems. This level is clear to teachers

and allows them to manage educational trajectories at the level of the workshop's content in general and the workshops of the disciplines in particular. A method for managing educational trajectories according to this plan graph is proposed.

This research result is the educational space and the plan-graph of PoS “Computer Systems and Technologies” of ITMO University. With these models' help, two versions of the “Network Protocols” course workshop were developed and implemented in the educational process (in 2019 and 2020). The workshop was tested by students of the 3rd semester of the master's program. The expert community in network technologies evaluated the achieved results of training in this discipline in the form of a course project. To assess the workshops' versions' effectiveness, the authors used the degree of closeness of students' tasks in the course project to real professional tasks. For the last academic year, it was 73%, and after the implementation of the modified version of the workshop in the current academic year—90%. This positive trend indicates a reduction in the gap between education and the professional areas, an increase in this PoS graduates' competitiveness.

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A Hybrid Online Laboratory for Basic STEM Education



Karsten Henke, Johannes Nau, Robert Niklas Bock,
and Heinz-Dietrich Wuttke

Abstract The following article describes a hybrid online lab that was developed at the TU Ilmenau and is used in basic education for technical computer science as well as in further education. Based on the tasks that can be solved with its help, an overview is given of learning scenarios in which the lab is used, and it is shown what benefits students can derive from it. The hardware and software architecture underlying the lab as a platform is briefly described. Special features here are the communication structure based on web services and the interactive tools belonging to the lab, which can also be integrated into learning management systems as Interactive Content Objects (ICO). Cloud-based networking guarantees the easy maintainability of the instances of the GOLDi-Lab (GOLDi—Grid of Online Devices Ilmenau), which are currently distributed among ten international universities. The exchange in this international network leads to new ideas for the further development of the laboratory, which are presented in a concluding section.

Keywords Control education engineering · Distance learning · Web-based education · Virtual and remote labs

1 Introduction

Students must be prepared for a working environment characterized by teamwork, interdisciplinary work, global action, and the agile handling of projects. New technologies and concepts demand a wide range of skills from future engineers, creativity

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in the safe handling of permanently available multimodal sensor technology, and the responsible use of artificial intelligence in the analysis of the resulting data volumes.

The practical use of virtual interactive resources accessible via the Internet is intended to promote such competencies. In the education of students, online laboratories make an important contribution to this [1]. They can be divided into virtual labs, which are based on simulations and virtual objects, and remote labs, which allow experimentation with real objects at remote locations via the Internet. A combination of both variants is called a hybrid online lab [2].

The following paper discusses the use of a hybrid online laboratory in basic computer science education for students of technical disciplines. It describes in which learning scenarios it can be used advantageously and how its architecture supports these versatile applications. The advantages of cloud-based work and the lab's instantiation and remote maintenance capabilities are also discussed.

2 Tasks in Basic Computer Science Education

The laboratory described in the following chapters is intended to support computer science fields of studies as offered in the basic studies of the TU Ilmenau for all engineering courses where the foundations for the understanding, systematic design, and formal verification of digital systems are laid.

The learning objective of the course is to enable students to independently design and build digital controls and demonstrate their correctness. Students should acquire the skills to record sensor values using digital circuits and process them to make actuators react in a manner specified in the task, e.g. the control of an elevator or a simple 3-axis portal.

To achieve this goal, the mathematical fundamentals of Boolean algebra are taught first. Following this is the systematic, formally verifiable functional description of digital controls, such as the formulation of dependencies of sensor values in the form of Boolean equations. Another concept that is taught, which is particularly important for the construction of sequential circuits, are finite state machines (FSM). Building on this knowledge, it is shown how these descriptions can be implemented in digital circuits. To prove the correctness of their behaviour, a practical construction by interconnecting elementary basic circuits, which correspond to the primitive operators of Boolean algebra, is necessary. This is conventionally done in a laboratory experiment. In the following, it is described how this teaching concept is accompanied and supported with the help of the hybrid online laboratory "GOLDi" [3].

Several online tools are integrated into the online lab, giving students the opportunity for Internet-based experimentation leveraging knowledge taught in lectures. For example, normal forms of Boolean algebra can be explored interactively with the help of the tool "SANE" (in German: Schaltsysteme Arbeitsblätter im Netz) [4]. Figure 1 shows a screenshot of the SANE tool, illustrating the application of Boolean algebra to the minimization of digital circuits.

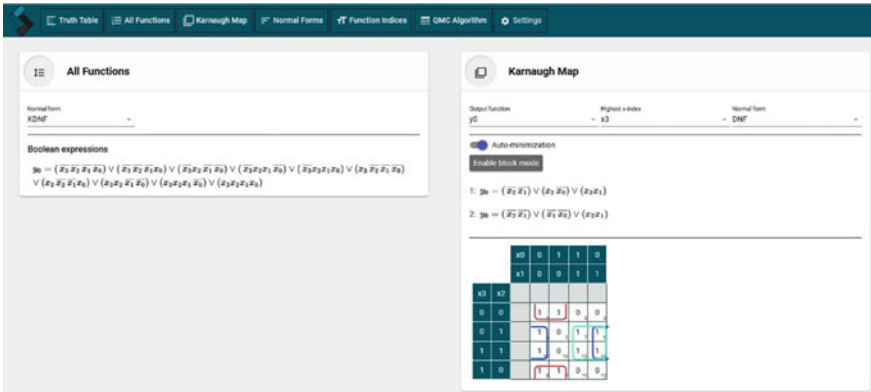


Fig. 1 Minimization of Boolean expressions

Students can experiment with given tasks or freely with any expressions of algebra and thus thoroughly understand connections of algebraic normal forms in the interaction. For this purpose, the tools are designed to perform calculations in realtime and present the results in different freely selectable views and contexts—e.g. Boolean expressions in different normalized forms or as a value table.

For sequential circuits and control algorithms that are systematically designed on the theoretical basis of finite state machines, the tool “**GIFT**” (Graphical Interactive FSM-Tool) is available [5]. With the help of a graphic editor, control algorithms can be designed and analyzed here as automaton graphs. The analysis is performed using selectable animated waveforms of the input and output signals of the design or by experimenting in the hybrid online laboratory GOLDi.

The link between Boolean algebra, switching algebraic expressions, and their circuit implementation is provided by the online tool “**BEAST**” (Block Diagram Editing and Simulation Tool) [6]. In this tool, students can virtually create digital circuits and observe their behavior in realtime using waveforms and color-coded connections that symbolize live and inactive virtual lines. Figure 2 shows a screenshot of this tool.

This can be used to develop both simple combinatorial circuits and more complex sequential circuits as control algorithms.

The presented tools SANE, GIFT, and BEAST are used for practical experimentation based on the skills acquired in individual sections of the training concept. This allows the students to solve problem-oriented subtasks interactively at the respective level of knowledge. Since the solutions of these subtasks are also essential components of control algorithms, the tools can also be used in the preparation of laboratory experiments in the GOLDi laboratory.

The degree of difficulty of the tasks can be adapted to the respective level of knowledge in the semester. At the beginning of the semester, for example, only simple tasks such as working with Boolean constants and variables are possible, but these can be tested immediately in the online lab by setting variables (actuators) to constant values

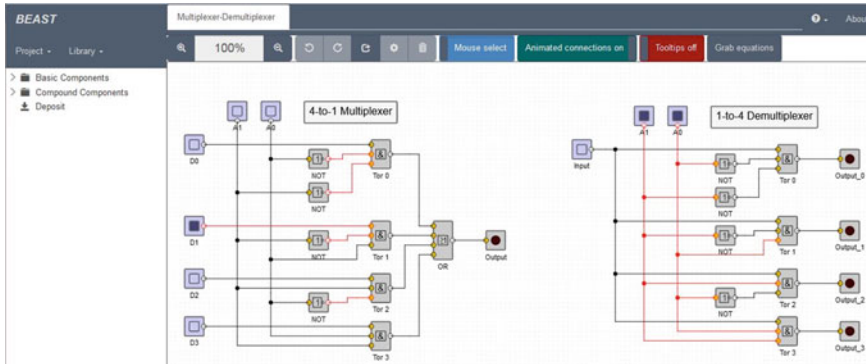


Fig. 2 Multiplexer/demultiplexer circuit with the BEAST tool

and triggering an immediate reaction in the lab. Appropriate protective devices in the laboratory and feedback messages prevent the laboratory devices from performing damaging operations (e.g. controlling motors beyond certain limit switches). In the next training step, tasks on Boolean expressions follow, with the help of which more complex dependencies of sensor signals can be taken into account for the control of actuators. Finally, to control sequential processes, competencies for the design of sequential circuits or finite state machines (FSM) are required, which can also be processed with the online tools described above and exported to the GOLDi lab.

To prove the acquired competencies, the functionality and correctness of a designed control algorithm have to be demonstrated in a laboratory test. In the GOLDi lab, the students' task is to design a control algorithm for an electromechanical model, the so-called physical system.

In the basic training, virtual interpreters are initially used as control devices, which are easy to operate with the skills acquired in theory. The variables of the Boolean expressions are coupled to the corresponding sensors or actuators in the laboratory and calculated clock-based. The behavior of the physical system and the assignment of the input and output variables, i.e. the current digital sensor and actuator values, can be observed via a web camera.

In higher semesters, microcontrollers, programmable circuits (FPGAs), or industrial programmable logic controllers (PLCs) are then used as control devices. The necessary skills for programming in higher programming languages (e.g. C++) or hardware description languages (e.g. VHDL), as required for programming FPGAs, are acquired in separate courses. Challenging tasks that can be solved with this knowledge are, for example, parallel control algorithms for an elevator control system including the control of the control panels inside and outside the elevator model. The comparison between a software solution and a hardware solution either via an FPGA or a PLC is also interesting for the education in higher semesters. For the editing of source codes in the browser and their compilation, professional design tools are available as a cloud service, which can be accessed via the web interface WIDE

(Web-Integrated Design Environment), which provides a uniform user interface for all programming languages [7].

The further development of the laboratory is always carried out with the involvement of students from higher semesters within the framework of software projects and annual student projects as well as in bachelor’s and master’s theses. In particular, competencies for the challenges posed to future graduates by the concept of Industry 4.0 are developed. To outline the diversity of such tasks, the architecture of the GOLDi lab will be described in the following section.

3 Architecture

The GOLDi lab flexibly implements all variants of remote labs—remote, virtual, and hybrid labs. This makes it possible to conduct all experiments either completely virtually or on real devices or in a combination of both.

3.1 Hardware

The hardware core of the hybrid online laboratory “GOLDi” consists of a laboratory server, physical systems that can be operated in parallel (electromechanical models such as an elevator, a high-bay warehouse, a manufacturing cell with conveyor belts), and control units (e.g. microcontroller, FPGA board). The conversion of the Ethernet protocol data to the input/output ports (GPIO) of the control units is carried out via the on-board RasPi Compute Module. Figure 3 provides an overview of the hardware components and structure.

Physical systems and control units are connected via the internet and can be coupled as desired for experiments, each consisting of any number of physical systems control units. This flexible architecture allows easy expansion of the available physical systems and units without having to make hardware changes to the overall architecture.

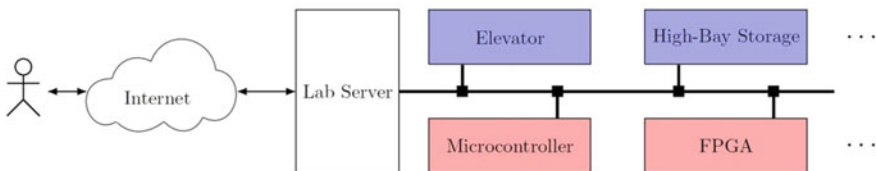


Fig. 3 Hardware grid structure of the GOLDi lab

3.2 Software

The software of the GOLDi lab is managed via a database. The tools and the management software for users and experiments as well as documents are stored in it. The user interfaces of the services are managed via a cloud. Figure 4 shows the software structure of the GOLDi-Lab. Currently, a total of 10 GOLDi infrastructures are running in Armenia, Australia, Georgia and Ukraine.

The cloud-based software architecture allows efficient remote maintenance and updating of the lab software. All changes in the cloud are immediately available to all partners without the need for on-site installations or updates. Control units and physical systems differ at the individual sites and are adapted to the local requirements of teaching. They are available to all users of the GOLDi Cloud according to the available capacities.

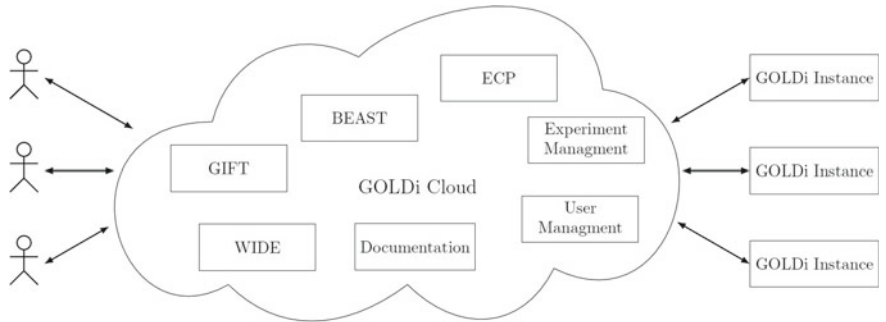


Fig. 4 Software structure of the GOLDi Lab

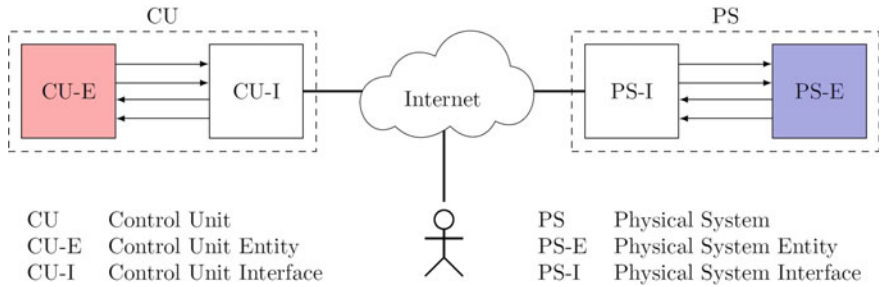


Fig. 5 GOLDi lab infrastructure

3.3 *Experimenting with the Hybrid Online Lab GOLDi*

The architecture of the GOLDi lab (see Fig. 5) allows any combination of real and virtual devices. The control unit (CU, e.g. microcontroller, FPGA, PLC) receives sensor signals from the physical system (PS, elevator, 3-axis portal, high storage warehouse) and converts them according to the control algorithm into actuator signals that are sent to the physical system. While virtual control units and physical systems are running in the user's browser, the real devices are present as hardware in the lab.

The virtual CU interprets and simulates an abstract description of the control algorithm based on a finite state machine. This abstract form of the control algorithm is independent of a concrete implementation. How this flexibility of the hybrid lab is used in different learning scenarios is the subject of Sect. 4. Here, a general description of experimentation in the GOLDi lab is given first.

Virtual devices are realized as JAVA scripts and run in a browser. Virtual physical systems consist of an animation part, which is responsible for the visualization of the movements of the physical systems and a simulation part, which simulates the movement behavior of the real physical system and its sensor/actuator reactions. The virtual control unit is an interpreter for FSM equations that cyclically computes the values of the control signals for the actuators from the equations and current sensor values.

The real devices are located in the laboratory and are connected to the laboratory server via a local Ethernet (LAN). This realizes the communication with the browsers on the client devices (PCs, laptops or mobile devices) via web sockets.

The lab server is connected to the real CU and PS via Ethernet. The students' control algorithm is loaded onto the CU-I with the help of the programmer and processed during the experiment.

By configuring the experiment, students decide between four types of experiments:

- (A) Virtual experiments (PS virtual, CU virtual),
- (B) Abstract experiments (PS real, CU virtual),
- (C) Real remote experiments (PS real, CU real) and
- (D) Implementation test experiments (PS virtual, CU real).

Details of the realization of these experiment configurations are described in [8]. At this point, the next section will only discuss the use of the different configurations in learning scenarios.

4 Current and Future Teaching Methods

The task in an experiment is to control a physical system with a self-designed algorithm so that it performs a given motion sequence. The use of the digital-twin concept enables the exploration of side effects that may occur in incorrect designs already

during the design phase. In this process, partial concepts of the control algorithm can be tried out on a virtual emulation of a physical system independently of the later implementation. As soon as the algorithm works on the digital twin, it will also work in the real environment.

The following examples discuss which of the experiment configurations (A) to (D) described in Sect. 3.3 can be usefully applied to which learning scenario.

4.1 Lecture Hall

Configuration (B) is suitable for demonstrating the design process in the lecture hall because abstract experiments can be used to show the step-by-step elaboration of FSM-based control algorithms. Starting with concepts of Boolean constants, variables, and expressions, an understanding of the concept of finite state machines is gradually established. It is shown how to program a virtual control unit using basic elements of control algorithms. For example, setting actuators y to constant Boolean values sets them in motion or stops the motion again depending on sensor values x and their logical connection. In this way, the effects on real physical systems can be demonstrated directly in the experiment during the lecture. For first-year students, it is very motivating to see in this way an application of what they are taught in theory.

4.2 Reflexion/“Flipped Classroom”

Virtual experiments (configuration A) are beneficial for both self-study and flipped classroom scenarios. These types of experiments run offline in the browser once they have been configured and started. This allows students to run their experiments independently of the Internet connection and prepare questions for discussion with the instructor in the seminars. They can try different variants of control algorithms or parts of them and explore the differences or repeat the experiments shown in the lecture hall. Since the virtual experiments run entirely on client computers, many students can perform experiments simultaneously. In conjunction with a learning management system (LMS), the course content can be directly coupled with the tools described in Sect. 2 (SANE, BEAST, GIFT) and virtual experiments, making it possible to immediately try out what has been read interactively. The interactive content objects in the LMS are also used for exercises and tests—but so far only with a few examples.

4.3 Guided Designs

In workshops for vocational training with 10–20 people, teachers first use virtual experiments (configuration (A)) and let the learners follow each of the demonstrated steps on their computers. This is initially done independently of any subsequent implementation at the abstract level of the FSM, which was also taught in theory. In this way, learners become familiar with the design process and the operation of the GOLDi-Lab. Afterward, they are given a similar task to solve on their own. Different solutions proposed by the learners are used as a basis for discussion on a presentation board and for selecting the best solution. It is used after changing the configuration (B) to control the real physical system with it. Experience shows that this motivates the learners and encourages them to do their best in competition with the others.

In the further course of the workshop, the control algorithms are implemented in hardware and software. To be sure that the implementation was successful, the learners can first work with virtual physical systems in test configuration (D). Finally, the real experiment is used in configuration (C).

4.4 Laboratory Exercises

To replace real experiments by using online labs, configuration (C) is the adequate choice. The GOLDi booking system allows the reservation of a dedicated configuration for a specific time window. The architecture of the laboratory allows the connection of several identical physical systems and control units without having to consider this when configuring the experiment. When reserving experiments, available devices are accessed dynamically. For example, if three instances of the same physical system are installed, this is transparent to users. They do not see which specific device is connected to the experiment. Only if all devices are occupied, they have to search for another time slot. There are different priority classes for the reservation so that it is guaranteed that teachers can make the necessary reservations for their course.

5 Further Development

Future focus areas for the further development of the GOLDi Lab aim in three directions:

1. Further integration of ICOs and experiments into an LMS,
2. Collecting anonymized data during experiment processing for feedback and analysis purposes to improve the lab; and
3. Using the data to develop concepts for formative assessment of students and for greater individualization of training.

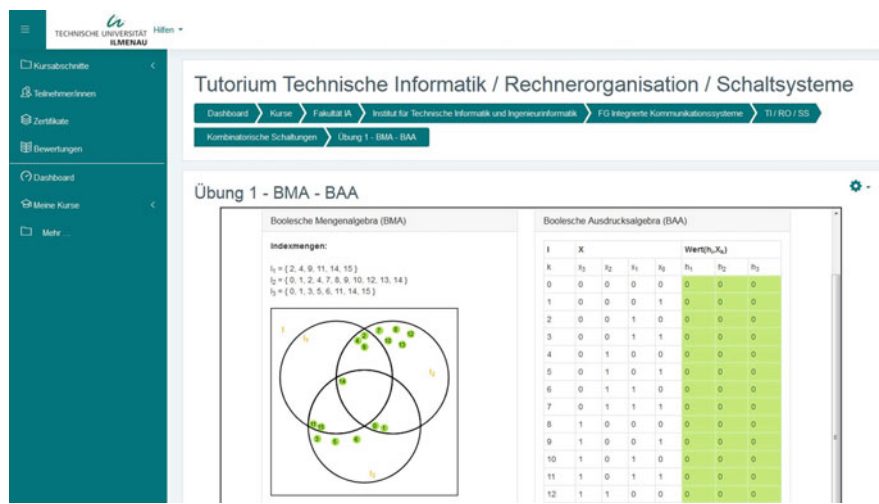


Fig. 6 Integration of ICOs into the LMS Moodle

Learning management systems have entered higher education in recent years. They offer the possibility to track arbitrary actions of students on the computer and to check the level of knowledge they have reached. In currently common LMSs, multiple-choice questions, cloze tests, and the like are used for this purpose. However, these testing methods can only verify lower levels of the mentioned taxonomy, so-called LOTS (lower-order thinking skills). These include remembering, understanding, and applying what has been learned. Higher levels of knowledge such as analysis, synthesis, and evaluation of facts are called higher-order thinking skills (HOTS). The goal of further developments of the GOLDi-Lab is to achieve approaches to verify these higher levels of knowledge in conjunction with interactive teaching software and online labs coupled to an LMS. As a first example, Fig. 6 shows the integration of tasks from the SANE tool into moodle.

Another direction of further development of the GOLDi-Lab is the design of new task types, which do not only refer to the design of control algorithms but also to more complex tasks such as resource optimization and tasks that deal with the integration of virtual and real physical systems. Here, for example, tasks are also suitable that do not have a complete algorithm as a goal but focus on error or behavior analysis of given algorithms using the black-box method [9].

6 Conclusion

With the GOLDi lab described in this paper, a hybrid online laboratory is available, which has been realized with modern web technologies and can be used for the education of students of technical disciplines. Experiments that can be performed in the

lab relate to the systematic design of digital combinatorial and sequential circuits and control algorithms for controlling virtual and/or real physical systems using various programmable hardware control units such as microcontrollers, circuits and PLCs. The design of the online laboratory allows for a variety of use scenarios in education and training, where both real and virtual devices can be combined in the experiments. In the future, an international network of several instances will enable diverse cooperation possibilities in cross-disciplinary and cross-cultural collaboration.

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Approach to Relevant Data Providing for the Pedagogical Design in Knowledge-Intensive Areas



Vadim D. Kholoshnia and Elena A. Boldyreva

Abstract This article is devoted to the issue of creating an information system for obtaining data necessary for the pedagogical design. The proposed system is implemented in the form of an application programming interface that provides data in a standardized form, which allows it to be used not only as an independent data source but also, for example, to be embedded in the information systems of universities. The proposed system provides the following standardized data: information on educational and professional standards, employers' requirements, and key skills of applicants in accordance with the standards, events related to the professional sphere. Educational and professional standards are downloaded from open government sources. Data on employer requirements and key skills of the applicant is extracted from vacancies obtained using the hh.ru application programming interface in accordance with the standards. The data is processed using natural language processing technologies and cluster analysis. The text data is normalized using Porter's stemmer, a TF-IDF weight matrix is created and cluster analysis is performed using the DBSCAN algorithm in order to determine the most necessary and relevant skills in a particular professional field. An additional function is a selection of events related to the professional field. The data obtained make it possible to form the relevant content of the educational tasks of the workshops and adjust the implementation of the educational process to the needs of the labor market, which increases the relevance of the skills acquired during the program of study.

Keywords Labor market needs · Professional standards · Educational standards · Machine learning · Natural language processing · Clustering

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1 Introduction

Currently, in a period of high competition in the labor market, the main goal of educational institutions is to prepare graduates competitive in the labor market. The constant program of study (PoS—Program of Study) revision in order to increase the relevance of the skills acquired by students is becoming not just a requirement of regulations, but a necessary condition for ensuring high-quality training of graduates. At the level of federal-state standards, measures are being taken to increase the influence of employers on the learning process in universities: to involve them in the development of educational standards, the creation of basic departments at enterprises, and the provision of the practical orientation of training [1–4].

Such measures are especially relevant when it comes to training specialists in high-tech knowledge-intensive fields. A distinctive feature of such areas is the constant updating of the technological and instrumental base and, as a result, the constant change in the requirements of employers.

This article is devoted to the issue of creating an information system for obtaining data necessary for the pedagogical design and PoS development. The data obtained make it possible to form the relevant content of the educational tasks of the workshops and adjust the implementation of the educational process to the needs of the labor market, which increases the relevance of the skills acquired during the PoS.

2 Review of Existing Solutions

At the moment, the direction associated with information support for the implementation of the educational process is actively developing. Information support includes numerous tools for modeling educational trajectories, data-mining, and web-mining for finding relevant data. These tools help to more fully model the subject area and the needs of participants in the educational process. Such support allows making the process of PoS designing dynamic, adapting to labor market trends.

The popularity of these trends in the PoS design is confirmed by numerous researches. In the review, the authors cite some of them. Curriculum modeling based on the ontological approach is described in the article [5]. Graph models and structural approaches to the formation of curricula are presented in the study [6]. Examples of works carrying out modeling and expert PoS analysis are described in studies [7, 8]. However, these works do not imply a direct influence of the requirements of the professional field on the formation of the educational trajectory.

Software implementation of information support for the educational process leads to the emergence of various information systems. These systems help PoS developers design more detailed and accurate PoS structure and content and help educators keep in touch with the labor market. The following works describe an automated system for monitoring and analyzing the personnel needs of universities [9], a qualification-oriented expert system for managing the educational process of a university [10],

and an intelligent system for supporting the PoS formation based on neural network language models, taking into account the requirements of the professional field [11]. These expert systems interact with the needs of the labor market and offer recommendations for modifying the educational process. Work [11] also describes interaction with the labor market through labor functions (and job descriptions), but the authors do not describe in detail how exactly this interaction is carried out. The emphasis is on the analysis of professional standards. This is an important aspect of the professional field. But professional standards are inert and cannot keep up with the constantly changing labor market, especially if we are talking about high-tech knowledge-intensive professional areas. In the article [12], the authors propose a search engine that filters the vacancies of the LinkedIn service by specific skills and requirements for the applicant. The work is quite solid, but there are several peculiarities—the system is not intended for use in the educational process, and LinkedIn analysis is difficult in the Russian labor market.

From here, the authors formulate the main drawback of documentation-oriented systems—recommendations are drawn up for PoS, but no specific recommendations for their implementation are given. At the same time, the content of PoS must be constantly updated and include new tools and technologies of the professional field. This is the only way we can prepare highly qualified specialists in the profile who will be able to meet the real needs of the labor market after graduation.

Currently, no one has proposed a centralized information system that provides the kind of up-to-date data required for PoS development. In this study, the authors developed and partially implemented an information system for analyzing professional and educational areas into the real educational process. The authors understand that labor market trends are updated faster than a student completes their studies. Therefore, the system being developed is an attempt not only to “catch up” with the market but also to predict its development in order to be proactive.

3 Proposed Information System

The proposed information system offers an approach to solving the task of obtaining relevant data for the development of educational standards. The theoretical basis for the system was presented in the work [13]. The system is divided into modules related to the external view, the database, and also to each other. A simplified structure of the system is shown in Fig. 1.

The software implementation of the system is described in the Python programming language. The software interface of the information system is developed using the Django library. Text data is processed using the natural language toolkit library, clustering is done using the scikit-learn library, pandas library is used to store intermediate data in the form of matrices and arrays, as well as standard libraries for data processing and retrieval, such as re and request. The system of storing systematized data is implemented using the PostgreSQL relational database.

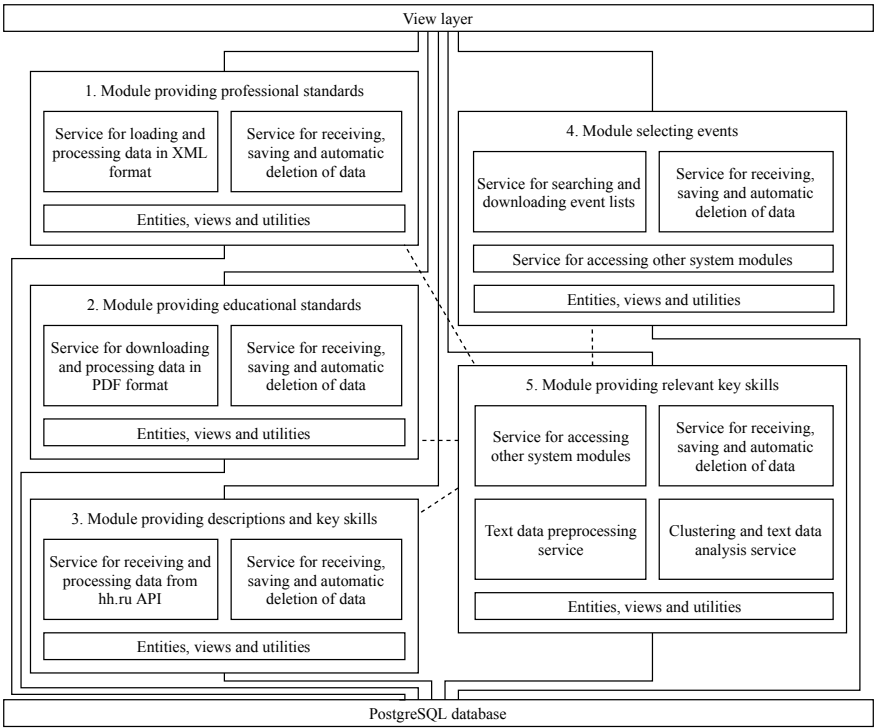


Fig. 1 Simplified structure of the proposed system

The data is provided to the view layer of the software interface in JSON format, which makes it possible to use it not only for sending to user interfaces but also for any other applications, for example, information systems of universities.

4 Modules Providing Data on Educational and Professional Standards

Currently, the sources providing data on educational and professional standards are user interfaces and do not have API alternatives. The data provided is available for download in formats such as XML, CSV, XLSX, PDF, while the most common format for transferring data between applications is JSON.

The module accepts educational and professional standards code or name as input. When a request is sent with a code, only one standard is returned, if it exists. If the request contains a name, the system searches for standards similar by name and returns a list.

The developed system accesses government websites, searches for and loads data using selenium webdriver, processes, and saves them to a relational database, which allows subsequently, with repeated requests, to access the data directly, bypassing loading. In order to maintain the relevance of the loaded information about the standards, a service that periodically removes irrelevant records from the database has been created.

4.1 Module Providing Descriptions and Key Skills

The module for obtaining descriptions and key skills is implemented as an application that requests data from the application programming interface of the hh.ru recruiting service.

This module accepts keywords as input to search for vacancies using the service. Also, the module can accept the internal ID of the loaded data to get a specific vacancy.

The hh.ru programming interface provides vacancies in the form of a list, for which descriptions and key skills are required corresponding subqueries. To reduce the data loading time when the user repeatedly accesses the system module, it was decided to save the loaded information to the database, access it, and also periodically delete it to maintain relevance.

4.2 Module Selecting Events

A prototype of a module for the selection of activities related to the professional field is being developed.

The request for the selection of events must contain the code or name of the educational standard. In the case of sending a standard code, the system selects measures for only one standard. If the title is sent, the system returns a list of events selected for all program of studies with a title similar to the one specified.

After that, the module for obtaining key skills is called up to obtain the relevant data. Then, with the help of the obtained data on the current key skills, a search for activities is carried out on the corresponding open sources. Such sources can be both specialized sites that provide such information and search engines.

After receiving and processing the data, the system returns a list of activities containing basic information and a link to the data source.

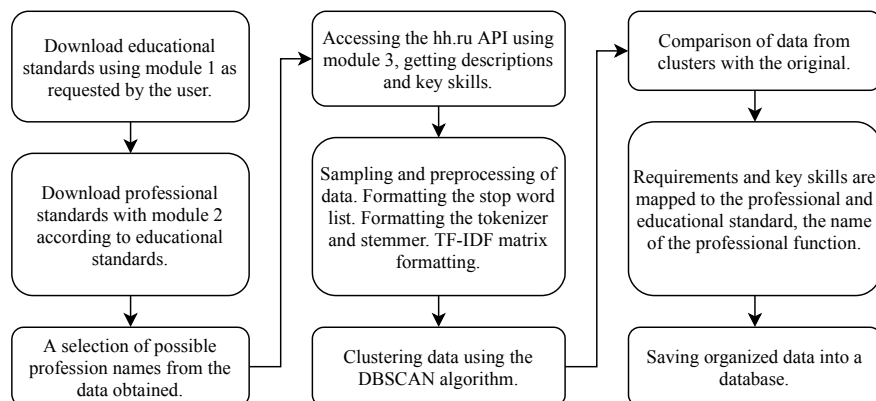


Fig. 2 The cycle of the module providing key skills

4.3 Module Providing Relevant Key Skills

The module providing relevant key skills allows one to receive relevant data on the key skills of applicants for professions that meet the educational standard. The cycle of the module for obtaining relevant key skills is presented in Fig. 2.

The module accepts the code or name of a training profile or PoS as input. This is followed by a call to module 1 in order to obtain a list of occupational standards codes using educational standards. Then, with the help of module 2, a list of possible vocational names is formed, derived from professional standards corresponding to the training profile. Further, using this list, a set of keywords is formed, necessary to obtain descriptions and key skills using module 3. The next stage is data processing using natural language processing technologies and cluster analysis.

Data processing consists of several stages:

1. Fetching and processing data using regular expressions.
2. Preparing data using natural language processing technologies.
3. Cluster analysis of prepared data.

Fetching and processing data using regular expressions. After loading, the selection takes place, as well as the processing of the received data using regular expressions. Key skills are stripped of unnecessary characters, such as punctuation marks, that might be posted by job applicants.

A regular expression for selecting requirements from job descriptions is formed from a replenished list of keywords containing expressions such as: “responsibilities, necessary, requirements, skills, looking for, waiting”. Before the formation of a regular expression, the list of keywords is normalized and processed using a stemmer. The resulting regular expression is used to select requirements from vacancy descriptions.

Preparing data using natural language processing technologies. Before conducting a cluster analysis, the received employers' requirements must be prepared using natural language processing technologies. Data preprocessing begins with the formation of an array of stop words. An array of stop words is formed not only from the standard sets of the `nltk.corpus.stopwords` library, but is also supplemented from an updated dictionary stored in the database for a more accurate selection of only meaningful expressions during preprocessing in accordance with the specifics of the professional field. After that, the tokenizer and stemmer are formed. Snowball-Stemmer is used as a stemmer—an addition to Porter's stemmer [14], which provides the ability to process, in addition to English, others, including Russian.

Next, a TF-IDF weight matrix is created. Each highlighted entry containing a request is considered a separate copy. The matrix is compiled using `TfidfVectorizer` (scikit-learn package), using a pre-prepared tokenizer and stemmer. The TF-IDF measure is used to represent collection documents in the form of numerical vectors reflecting the importance of using each word from a certain set of words (the number of words in the set determines the dimension of the vector) in each instance. Such a model makes it possible to compare texts by comparing the vectors that represent them in any metric, that is, by performing cluster analysis. The result of the algorithm's work is saved as a matrix of the pandas library for further use in clustering.

Cluster analysis of prepared data. The clustering algorithm is applied to the resulting matrix [15, 16]. In a comparative analysis of the results of such clustering algorithms as `MiniBatchKMeans`, `KMeans`, `AffinityPropagation`, `MeanShift`, `SpectralClustering`, `AgglomerativeClustering`, `DBSCAN`, `OPTICS`, `Birch`, `Spectral-Biclustering`, `SpectralCoclustering`, `FeatureAgglomeration`, the results of this clustering were chosen as the main clustering algorithm after clustering algorithm with a large amount of data turned out to be the most relevant, the separation of topics close in importance was the most correct. Also, the choice of the `DBSCAN` algorithm is due to the fact that, within the framework of this spatial clustering algorithm, it is not necessary to indicate the number of clusters in advance—within the diversity of employers' requirements, it is difficult to predict the number of thematic clusters in advance. In addition, in the lists of responsibilities and requirements, after preliminary processing, records of the form “Understand well what this is about” may remain. These records can be regarded as noise and are not included in any of the clusters. Clusters of the following types are distinguished: those corresponding to the subject area and requirements-noise (the latter are removed from the list of requirements). Table 1 provides examples of such clusters.

When choosing clustering algorithms such as, for example, `MiniBatchKMeans` and `AgglomerativeClustering`, one must specify the number of clusters. It is based on the number of basic labor tasks from professional standards for a given profile, doubled (since about 50% of the requirements for a vacancy are insignificant for this profile).

This increase makes it possible to take into account “noise” and allocate additional clusters for storing such records.

Table 1 Cluster content example (DBSCAN)

Cluster type	Cluster content
Noise	['business correspondence', 'business communication', 'business analysis', 'sociability', 'initiative', 'result orientation', 'business communication', ...]
Cluster 1	['1c: enterprise 8', '1c: production enterprise management', '1c: enterprise', '1c: enterprise management', '1c: enterprise 7', '1c: enterprise: retail', '1c: enterprise 8', ...]

After analysis and processing, the data is sorted and systematized. Then the systematized data is written to the database and presented to the user.

5 Practical Application

The proposed approach and the implemented tools were used to form a list of requirements and key skills for the “Embedded and Cyber-Physical Systems” profile. As a baseline dataset for the embedded software snippet, the authors took and analyzed the requirements and key skills for software-related professional standards for 6 different specializations—1210 vacancies and 3540 requirements in total.

The total execution time of the algorithm with data loading and processing for one professional standard “System Programmer” was 50.5 s. 180 vacancies were found and 802 key skills were identified, as well as 120 requirements. It took 3.35 min to load and process all the required professional standards. For this DBSCAN algorithm, the following parameters are used: the maximum distance between two samples is 0.01, the number of samples (or total weight) in the vicinity is 2, the sheet size is 100. Multithreading is also enabled to reduce the running time of the algorithm.

With the specified parameters, about 200 clusters are allocated, which contain data that is significant for this profile. Using AgglomerativeClustering clustering, 95 significant clusters and 90 clusters of the MiniBatchKMeans algorithm were identified. A visualization of the operation of the clustering algorithms DBSCAN and MiniBatchKMeans are shown in Fig. 3a, b.

Further, the clusters were correlated with the labor tasks of professional standards, the closest to the formulations of labor tasks were selected [7]. Since there is a difficulty in assessing the semantic proximity of texts in Russian, a “manual” assessment of the semantic proximity of the selected clusters and the main work tasks was carried out. Based on the results of this assessment, the DBSCAN clustering algorithm showed the best results - the selected clusters turned out to be more “thematic” and contained fewer insignificant requirements.

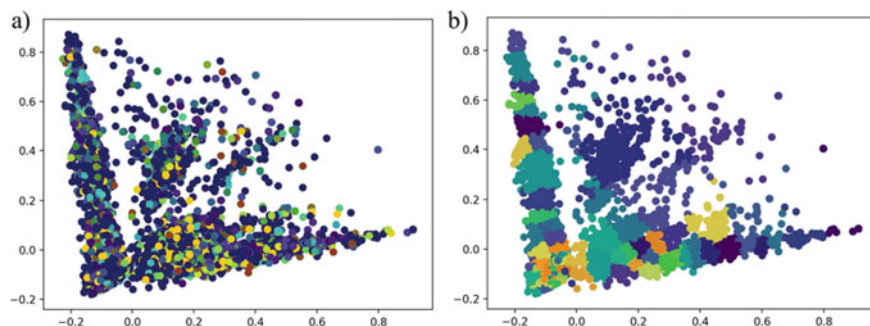


Fig. 3 Visualization of the DBSCAN and MiniBatchKMeans clustering algorithms

6 Conclusions

This paper discusses the issue of creating an information system to obtain relevant data necessary for the pedagogical design and PoS development.

The data provided by the system makes it possible to increase the relevance of the skills obtained after completing program of studies, as well as to speed up the process of their development.

The retrieved data show the relationship of requirements and key skills with professional standards, job functions, and specializations. The results of the analysis of the data obtained make it possible to form the relevant content of the educational tasks of the workshop and adjust the implementation of the educational process to the needs of the labor market.

As a direction for further research, it was decided to consider the creation of a system for automatic search and formation of educational tasks.

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Personalizing Older People Training in Modern Technologies for Successful Life in Smart Society



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Abstract Governmental social centers for the elderly occasionally organize courses related to modern digital technologies. However, practice shows they are not highly effective for improving senior citizens' life quality in a smart society. The paper presents the results of a study focused on characteristics of the “silver” clients of governmental social centers, their requests and expectations from this kind of training. The research was carried out to further determine the ways and means meant to personalize this process following the basic principles of smart education, i.e., to provide flexibility, variability, adaptability and feasibility of such training.

Keywords Teaching the elderly · Smart education · Personalization of training · Silver generation · Andragogy and gerontology

1 Introduction

Within a short time, humanity has come from mechanization to the automation of production. Due to the development and implementation of information technologies, it has entered a new stage—the stage of informatization, which is also rapidly supplemented by widespread digitalization. In turn, the transition from analogue

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systems to digital cyber-physical ones could not but entail the need to change the public worldview, which is a natural consequence of the fourth industrial revolution of the twenty-first century. New conditions create new challenges for the theory and practice of teaching older people. The question of how this can be done, following the trends of the evolution of smart society and education, is the basis for the presented research.

1.1 New Conditions for the Elderly

According to Voronkova and Kyvliuk [1] the next stage in the development of human civilization will be a super-intelligent smart society, where the possession of modern digital technologies and the ability to interact with various automated cyber systems will directly affect the people's life quality. Such theories and reasoning sound quite optimistic unless one takes into account that humanity is aging at a rapid pace. In the work [2] Boulton-Lewis calls this phenomenon "the Silver Tsunami" to emphasize that the aging of the population affects all aspects of human life, including the social, economic, cultural and political spheres. According to the report of the United Nations Population Fund (UNFPA), already in 2022, the number of people over 60 years old will exceed 15% of the total population of the Earth, and by 2050 there will be a lot more elderly people than children under 15 years old [3]. The fact that the population around the world is rapidly ageing and the predictions of the UNFPA experts on the growth of population over the age of 80 from 4.272 to 7.514 million go against the bright prospects of creating a super-intellectual society since for this particular age group it is very difficult to master the technological innovations constantly replacing each other.

Achievements of gerontology and andragogy determined the general principles of teaching people of the "silver wave" at the end of the twentieth century: dividing the material into small "portions" (microlearning); frequent repetition of information; reinforcement of oral teaching with audiovisual materials, written materials and practice; using analogies and examples to illustrate abstract notions; creating the most comfortable psychological environment, implying multiple situations of success [4]. At the same time, the training itself was not directly related to older people's needs to continue a full life in society.

At the end of the twentieth century, few courses taught older people to solve everyday problems, e.g., sending telegrams, setting up a TV and an answering machine, handling fax, a microwave oven, a washing machine and a vacuum cleaner. Today, due to the significant penetration of digital technologies into everyday life, both house-hold appliances and telecommunications are becoming more technological. Maintaining social activity requires the ability to interact in virtual environments and social networks. Even communicating with relatives in the modern digital world involves e-mail, instant messengers (IM), video calls. Interaction with government,

utilities, banking, housing and postal organizations requires work in automated information systems. The existing experience and knowledge can allow enhancing the quality of seniors' training by the implication of smart education principles [5].

1.2 Peculiarities of Teaching the Elderly: Literature Review

In 2015, while teaching the computer technology in the context of a 20-h course in basic skills, Gonzalez et al. [6] conducted a study devoted to defining the behavior model of older people taught to use digital tools. Researchers in [6] argue that people of the third age do not resist learning computer skills and do not believe that the learning process is difficult or inaccessible. In contrast, senior citizens suppose they can learn if they stay healthy and have an adequate level of cognitive function. Accordingly, educators who work with older people should remember that teaching computer literacy can stimulate silver citizens' self-confidence and increase their self-esteem. Computer literacy is, therefore, a tool that can be used to maintain their physical and mental health, to provide active ageing.

Russian researchers argue that, in contrast to the younger trainees, people of pre-retirement and retirement age in Russia come to study at state social centers with specific needs. Thus, Shalashova and Smirnova [7] note that "... for this category of the population it is important: to study something that has always been interesting for them, but not previously mastered because of full-time employment; and/or to acquire skills and abilities that contribute to improving the quality of life (skills of using mobile applications, the basics of legal and financial literacy, etc.)".

However, there arises a discrepancy between older people's needs and the existing practice of implementing various courses in state organizations for people of the third age. In current conditions, the transformation of older people's demands for the certain content of education should be considered as a natural consequence of changes associated with the development of mobile technologies, the expansion of e-government services, the emergence of new technologies for the provision of multi-media content, etc., as well as the emergence of new information threats [4, 5] ... The training system can be built in collaboration of state social centers with universities, and based on a cluster-distributed model [8].

Considering the above, it is significant to find and justify ways to diversify older people's education in modern technologies in order to improve their quality of life in a smart society via the organization of courses and events on the basis of state social centers in Russia. At the same time, it can be assumed that the efficiency of such training can be ensured through personalization [9] and active using technologies of m-learning [10], provided the needs of the elderly population are previously identified and defined as typical patterns of expectations of older people.

1.3 Research Goal

The paper explores the case of teaching older people information and communication technologies. It states the preliminary objective of the on-going study—to identify the needs, attitudes and experience of third-age people in the field of using information and communication technologies in the current stage of development of a smart society.

2 Materials and Methods

The research was carried out in 2016–2019 in the Municipal State Institution “Active City” (Novosibirsk, Russia) among 214 elderly people who were invited to study under the “Academy of Computer Literacy” program, dedicated to the basics of working with computers and the Internet. The program was organized with the support of the Foundation of Social Programs named after L.I. Sidorenko. The following tasks and stages of research work were specified to identify the needs, attitudes and experience of people of the third age in the field of information and communication technologies.

2.1 The Main Tasks of the Research

First, it was necessary to find out what kind of digital devices older people have at home. Second, it was important to state the leading motivational factors that determine the need for computer literacy training. The third task was to specify the possible structure and content of training courses in accordance with the requests of the studied group of elderly people. Finally, the researchers were to analyze how requests vary in the following four age groups: (1) 55–60 years old; (2) 61–70 years old; (3) 71–80 years old; (4) 80 years and older. To solve the above problems, a complex questionnaire was developed.

2.2 Characteristics of the Subjects Studied

At the pre-research stage of the study, due to individual interviews, it was found that people aged 71–80 (53.3%) take the greatest interest in training in modern information and communication technologies and digital tools for solving various kinds of problems. The interviewees accounted for their interest in technology by a desire to master new things and a large amount of free time. The majority of the respondents mention the decrease in family responsibilities since their children have

grown up and do not need constant help. Also, most of the interviewees (52.6%) have grandchildren, mostly late adolescents.

The second most motivated group (36.4%) are the participants aged 61–70. These are mainly recently retired people, who have resigned or have given up a part-time job. The overwhelming majority of respondents in this age group used to work in industries, mainly involved in manual labor, which did not require any knowledge or skills in the use of modern information technology. During the interview, they admitted that they wished to make up for it through additional training. Only 7% of 55–60-year-old interviewees admitted they needed additional training in modern technologies. During the interview, the majority stated that they were confident users of computer devices, constantly used a smartphone, had a smart TV at home and claimed that they owned other smart devices and gadgets (smartwatches, smart vacuum cleaners, programmable washing machines and dishwashers etc.). However, the interviewed did not want to miss the opportunity to take up free training at the state Centre to deepen their knowledge and develop their skills in modern technologies.

The least motivated group (3.3%) were people aged 80 and older. During the interview, it was found that the low level of interest is caused by the difficulty of moving around to visit the Centre, high fatigue and health problems, but not by the reluctance to learn new things. The interview shows there is no relationship between the marital status of older people and the level of interest in learning modern information and communication technologies, and digital tools for solving various kinds of problems.

3 Results

Availability of digital devices for the elderly. The analysis of the data shows that 16 out of 214 respondents (7.5% of the total number of respondents) do not have a personal computer or laptop at home. The remaining 92.5% (198 people) have either a personal computer or a computer for joint use of the family, or other devices and computer equipment. 7.9% of respondents use a mobile phone (without Internet access) as a means of communication. 10.7% of respondents prefer to work with a tablet. 92.1% daily interact using a smartphone with Internet access. 92.1% (197 people) have access to the Internet at home.

14 respondents do not have a computer or laptop at their disposal, but they get access to the Internet via a smartphone (13 people) and a tablet (1 person). Only two respondents out of all the surveyed do not have any access to a personal computer or wired Internet. However, they can use smartphones and tablets with mobile 3G/4G Internet. In general, 85.5% (183 people) have access to two devices for receiving and processing information at home, all of them had a smartphone or tablet (see Fig. 1).

Unexpectedly, 100% of respondents have digital devices at home. Most of them have access to the Internet and use computer or mobile technologies for their needs. At the same time, 86.4% experience difficulties when working with a computer or other devices, 4.7% do not work with a PC at all, while everyone has a computer

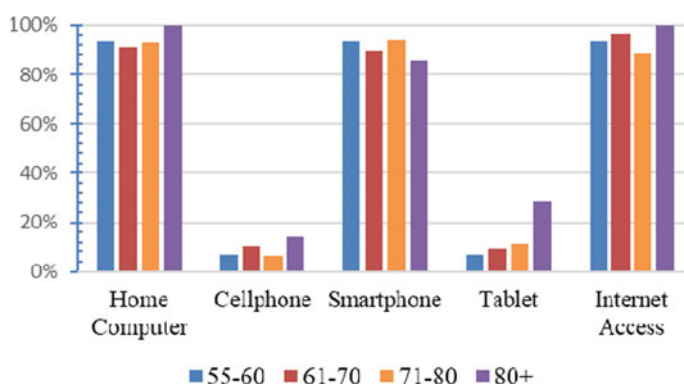


Fig. 1 Availability of digital devices at home, percentage in each age group

with Internet access. Such data indicate that potentially modern pensioners have all the conditions for using the information space and digital services necessary to solve everyday problems and satisfy information and communication needs. At the same time, 5.1% note that they face minor problems when working with digital devices, 3.3% have practically no difficulties, 0.5% of respondents do not experience any difficulty. Some respondents find using mobile and smart devices more difficult than working with computers and laptops (7 people aged 71–80 and 1 aged 61–70).

Reasons for learning and requests of the elderly. The survey reveals that 47 of 214 respondents (about 22%) have previously taken computer and information literacy courses, but did not get the desired result. 71% of the respondents admit that they turn to younger relatives and friends for help when using digital devices; 6.1% rarely resort to outside help, trying to overcome difficulties on their own, and only 0.9% do not feel the need for anyone's help. The survey also identified the motives for the retired to undergo training in state social centers. The participants of the study were asked to rank the relevance of possible causes from 1 to 8. (See the results of the averaged rank values in Table 1).

Table 1 Results of ranking the reasons for learning

Rating	Response (cause)	Average rank
1	Learn to solve everyday problems in a new way	2.21
2	Why not, it's free of charge	3.29
3	To expand the possibilities of digital communication	3.36
4	Assert myself in the eyes of my relatives and friends	4.07
5	Ability to find a new occupation (profession)	5.00
6	The opportunity to spend time as a leisure option	5.50
7	Satisfy interest in the new and unknown	5.79
8	The opportunity to resume the career	6.79

The peak position in the rating of learning motives is occupied by a possibility to solve everyday problems in a new way (average 2.21). The second reason for learning is a possibility to take advantage of a free course offered by the state (average 3.29). The third most popular reason is a wish to expand social communication (communication with relatives, friends, acquaintances) in the digital environment (average rating 3.36).

The respondents placed the need for self-affirmation in the eyes of the loved ones and the desire to keep up with the times at the fourth position on the list, (average 4.07). The fifth position in the ranked list is a search for new employment opportunities (jobs) and earnings in the digital environment (average rating 5.00). The least significant reasons for training in the state social Centre are an opportunity to spend free time, a course as an option for leisure (average 5.5); an opportunity to learn something new and unknown (average 5.79); an opportunity to resume the career (average rating 6.79).

The needs of older people in mastering new technologies for a smart society were also surveyed via open question questionnaires. The participants of the study were asked to state 4–6 most troubling difficulties associated with modern digital devices. The results were interpreted in terms of two age groups: up to 65 and over 65 years old. This criterion was chosen because the average retirement age in Russia is 65.

About 66% of the respondents (141 people out of 214) were over 65 years of age. The responses received were clustered and ranked according to the relative frequency of occurrence. As a result, we identified the following needs (Table 2).

About 44% of respondents (73 people out of 214) fell into the category of 55–65-year-old people. 45 people out of 73 (62%) keep working in retirement. The responses received were also clustered and ranked according to the relative frequency of occurrence. As a result, it was possible to identify the following content domains (Table 3).

The final interview based on the questionnaire data clarified that respondents under 65 are actively using Internet services to solve their problems: they can search for the necessary information, watch videos online, have accounts on social networks, use state electronic services, can order and pay for goods in online stores, manage housing and communal services. And all 100% of the participants in this group noted that they do not want to lag behind the younger generation in the use of modern technologies, including more advanced ones (individual virtual assistants, gadgets, smart home tools, digital ecosystems). The final block of questions was aimed at identifying attitudes towards training prospects. For this, the study participants answered 2 questions. The first was “Do you think that without digital education, an elderly person may feel socially excluded, living on the margins of modern society?”. The second question was “Do you agree that older people cannot do without digital technologies in today’s life?”.

The processing of the results showed that 95.8% believe that without digital technology training, an elderly person may be excluded from modern life, 1.4% of those surveyed do not agree with this opinion and 2.8% found it difficult to answer. The answers to the second question showed that 83.6% of respondents believe that older

Table 2 Ranking of identified requests for the course content in group 65 +

Rating	Domains of training content	Relative frequency (%)	Count
1	Communication on social networks, messengers, finding friends and relatives, sharing information	65.25	92
2	Virtual greetings to relatives and friends	59.57	84
3	Ordering medicines, searching for contacts of pharmacies, delivery and payment of medicines by bank cards	57.45	81
4	Troubleshooting problems with a personal computer, mobile device, Internet connection	57.45	81
5	Operation of household appliances (washing machines, TVs, refrigerators, etc.)	55.32	78
6	Security of cashless payments, use of ATMs	51.77	73
7	Video communication and mobile devices, cellular communication management (tariff plans, balance control, etc.)	51.06	72
8	Search for objects in the city and contact information of organizations	48.94	69
9	Appointment to a doctor via computer and mobile devices	45.39	64
10	Unwanted calls and messages, advertising	41.84	59
11	Search for special offers and discounts in stores, ordering food delivery (payment by bank cards), other discount opportunities	36.17	51
12	Search for news and information on topics of interest	36.17	51
13	Getting transport services (timetables and routes of public urban and regional transport, metro, taxi)	33.33	47
14	E-payments, interaction with public utilities	28.37	40
15	Getting reference information (hospitals, government agencies, postal services, housing and communal services)	25.53	36
16	Getting information on cultural events and social life, search/booking tickets to theaters and exhibitions	25.53	36
17	Getting and registering certificates, documents, applications through electronic government services	24.82	35
18	Search for travel services (vouchers, transport tickets)	22.70	32
19	Opportunities for games, entertainment, online cinemas	19.15	27

(continued)

Table 2 (continued)

Rating	Domains of training content	Relative frequency (%)	Count
20	Using Office applications (word processing, spreadsheets, presentations)	9.93	14

Table 3 Ranking of identified requests for the course content in group under 65

Rating	Domains of training content	Relative frequency (%)	Count
1	Using mobile devices when traveling abroad (Wi-Fi, roaming, settings, instant messengers, IP-telephony)	57.53	42
2	E-signature, certificates, exchange of digital documents with confirmation of their authenticity	50.68	37
3	Specialized software (BIM, CRM, 3D, electronic document management systems)	42.47	31
4	SMM, social media promotion, audience engagement, blogging and vlogging (Instagram, YouTube, VK, Facebook)	39.73	29
5	Online commerce technologies, domains and hosting (how to open your own online store)	36.99	27
6	E-payments, interaction with public utilities	35.62	26
7	Effective ways to format text documents, PDF editing	32.88	24
8	Advanced features of spreadsheet editors	28.77	21
9	Information and data visualization tools, multimedia presentations	26.03	19
10	Search for travel services (vouchers, booking transport tickets, hotels)	26.03	19
11	Operation of household appliances (washing machines, TVs, refrigerators, etc.)	24.66	18
12	Troubleshooting problems with a personal computer, mobile device, Internet connection	20.55	15
13	Appointment to a doctor via computer and mobile devices	17.81	13
14	Security of cashless payments, use of ATMs	15.07	11
15	Opportunities for games, entertainment, online cinemas	12.33	9

people cannot do without digital technologies in modern life, 6.1% disagree with this statement, and 10.3% found it difficult to answer.

4 Discussion

The practical work of Municipal State Institution “Active City” in Novosibirsk (Central Siberia, Russia) shows that people aged 61–80 are increasingly seeking help for additional training in the field of computer and information literacy. Presumably, this is because people of this age group are already retired and have more free time for self-development and self-realization. Besides, in contrast to the 55–60-year-olds, they have less experience using information and communication technologies in employment. Also, the study shows that in 2016–2019 clients of the state social Centre were mainly manual workers with secondary vocational education. And this explains the fact that their requests for the training content were mostly of social and domestic nature.

Unexpectedly, 100% of the respondents had the necessary tools (hardware and software) to solve problems using information and communication technologies. However, many said that they experienced difficulties or could not at all effectively use the available resources. More than half of the participants admitted that they had received those devices as gifts from relatives. Also, many relatives did not provide consulting support on the use of the devices they had given. The interaction with the group under study showed that, on average, about 70% of older people come to the second meeting at the social Centre with their tablets, laptops, smartphones and ask specific questions related to solving the problems they need, learn about the *functionalities* of their devices. For those people, additional training is a means of self-improvement and self-realization, as well as some opportunity to keep up with the young and feel more confident in the digital world. Analysis of the contingent structure of different age groups in 2016–2019 showed that most people who seek help in state social centers in Russia (for example, Novosibirsk) are over 65.

This can be explained by the fact that many Russian citizens under 65 are still employed. People aged 55–60 are no longer surprised by smartphones, tablets, wearable electronics, “smart” household appliances as they already use them. It seems that over time, the digital divide between generations may disappear. Since the data concerning people of 55–65 years old indicate a fairly high level of technology skills for solving social and everyday problems, they can be considered “digital citizens” equipped with modern means. However, due to their age characteristics, there is a danger for many competencies developed in employment to fade away over time.

The older generation may need additional preparation for using innovations. The revealed interests in the content of training indicate that the requests of different age groups differ. It actualizes the need to develop approaches to personalizing the training of the retired. However, state social centers in Russia implement universal educational programs based on the principle “everything for everyone”.

5 Conclusions

Today it is necessary to reconsider the means and ways of implementing older people's additional training in modern technologies for a successful life in a smart society. We suggest this can be achieved through the following changes.

1. The content of such educational programs should be developed according to a modular principle and microlearning methods. In this case, it will be possible to work out individual education programs.
2. It is effective to teach people in heterogeneous age groups, where representatives of a younger age can provide consulting support to older ones, and demonstrate examples of solving typical problems.
3. Older people need to learn in a comfortable psychological environment, using a BYOD approach (Bring Your Own Device). The obtained data showed that this is possible to implement.
4. Learners should have a chance to attend meetups that help people experience success and realize advancement in modern technologies.
5. It is necessary to design and implement additional education programs for the older generation, taking into account the data obtained on their causes and requests for training in modern technologies.

These will ensure the key principles of smart education: flexibility, variability, adaptability and feasibility. As the next step the authors plan to apply the results of the presented research in order to change the educational programs for the elderly.

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Method of Planned Learning Outcomes Identification in Higher Education Based on Intellectual Analysis of Labor Market Needs



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Abstract This article describes the formation and identification of planned learning outcomes in higher education to prepare competitive graduates of various profiles in high-tech technical fields. The authors propose a method for identifying planned learning outcomes, which allows selecting relevant tasks in the workshop's core based on intellectual analysis of the labor market and ranking them based on expert employers' differentiated opinions. Identification of the planned learning outcomes base on the implementation of several stages: the formation of an excessive list of suitable PoS professional tasks from online recruitment systems, an expert community, and differentiation of their opinions based on the confidence coefficient in the opinion of each expert, ranking of selected tasks based on expert assessments and selection of the top list of the most relevant tasks that fit into the PoS complexity, and setting by each task of professional competence (PC) and complexity in the curriculum. This method's practical implementation helped to analyze more than 11 thousand vacancies in the online recruitment system hh.ru. Of these, 94 names of appropriate methods, technologies, tools, etc., which should be owned by its graduates were selected, including 6 names for the discipline "Network Protocols". The workshop of this discipline was tested by students of the 2nd year of the master's program. The expert community evaluates the achieved learning outcomes on the degree of closeness of the tasks solved by graduates to real professional ones, which confirmed the positive dynamics and the effectiveness of the proposed method.

Keywords Educational design · Learning outcomes identification · Employer's expert opinion · Intellectual job market analysis · Web data extraction

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1 Introduction

Indicators of compliance of the educational process's real results with the expected (planned) ones are an essential criterion for higher professional education effectiveness. With their help, the management bodies of universities' educational activities can monitor the quality of implementation of educational programs (PoS—Program of Study), future employers of their graduates—to understand what specialists they can “get” at the exit from the educational institution. Students know what requirements the employers impose on them, i.e., what knowledge, skills, and abilities they need to be competitive in the labor market.

Planning the expected learning outcomes is a complex process in which, along with university teachers, representatives of potential employers of university graduates should also participate. Universities should repeat this process annually to update the educational process's content aimed at reducing the gap between the requirements of teachers and employers of university graduates. It is crucial for knowledge-intensive education areas, where there is an annual update of technologies and tools that students should be familiar with. The identifiers of expected learning outcomes for this direction and profile of training are the competencies and labor intensity allocated in the educational program for their development.

In this paper, within the framework of the previously proposed approach to the design of workshops [1], developing a method for identifying the expected learning outcome using elementary competencies that determine a set of the most important educational tasks—the core of the PoS workshop.

1.1 The Task of Identifying the Planned Learning Outcomes Based on the Intellectual Analysis of the Labor Market

Far from solving “real problems”, the graduate becomes unable to compete with more experienced specialists for a higher-level position and salary. Thus, preparing graduates for real professional conditions in the learning process to make them competitive in the labor market is particularly acute. When developing educational trajectories and training materials, it is necessary to focus on a full-fledged participant in the educational process—future employers of students.

Thus, the tasks of systematic analysis of the professional field (labor market) and identification of the planned learning outcomes consist of selecting specific professional tasks that can be adapted for implementation in the educational process of practice-oriented training of students of knowledge-intensive profiles. The research aims to reduce the gap between the professional sphere requirements and the requirements of teachers in higher education.

2 Research Background

In the context of a large amount of data that can influence the formation of educational content and the implementation of the educational process, the direction associated with its information support based on modeling technologies, analysis of regulatory documentation, and the labor market's needs is actively developing. Also, universities tend to attract existing labor market specialists as consultants and conduct labor market analysis to determine which competencies are currently relevant and in-demand [2–5].

The development of a system for monitoring the needs of the labor market, proposed in the studies [6, 7], is designed to bridge the gap between the labor market's specific requirements and the general formulations of regulatory documents of educational standards. An attempt is made to adapt the traditional competence model of a graduate to the continually changing professional needs. In their research, the authors use the semantic neural network model of the word2vec language to determine how educational programs' content meets the current needs of the labor market. As a result of the study, the authors also proposed a forecast of professional field changes. However, the professional field in IT is knowledge-intensive. There are difficulties in predicting the market entry of new technologies and programming languages. Such a forecast can negatively affect the inert regulatory documentation of educational programs and directions if a large reserve for the future is made based on this forecast results. Also, these articles do not provide specific algorithms and methods for analyzing texts, and it isn't easy to assess the effectiveness of the technique used. There is not enough data on the testing of the proposed algorithms.

The following development is also an example of a tool for analyzing specific skills and requirements [8]. The study's authors propose a search engine that filters vacancies in the LinkedIn system through the applicant's particular skills and requirements. They form the list of skills based on the LinkedIn system's data, then processed by clearing out insignificant general vocabulary. The system allows you to determine whether specific vacancies correspond to the skills selected from the list of the general requirements. However, these studies contain insufficient data on the testing of the proposed system, and the use of the LinkedIn service in the Russian market is problematic.

In [9], the authors search for the most popular vacancies within specific locations (regions) based on data extracted from online recruitment systems for vacancies and particular requirements for them. The method is based on lexical-semantic analysis (LSA) of job descriptions and allows you to create a rating of vacancies with the appropriate skills. A similar approach is presented in the study [10]. In this case, lexical and semantic analysis allows you to create a list of vacancies with specific professional requirements (averaged over this area). The study results are recommendations for finding a "reference" specialist for particular professions, corresponding to the job description's maximum number of requirements.

However, these studies and systems either do not directly relate to the educational process or offer general recommendations in the regulatory documentation. The

proposed method of identifying the intended learning outcomes uses the results of mining jobs in online recruitment. It allows taking in the workshop's core to rank and professional tasks of the PoS graduates based on differentiated experts-employers. In contrast to the known approaches, it will enable to provide recommendations for the educational process not at the level of "general" for regulatory documentation but at the level of specific academic tasks and particular tools for solving them.

2.1 Method of Planned Learning Outcomes Identification Based on Intellectual Analysis of Labor Market Needs

The following stages characterize the method of identification of planned LO proposed by the authors:

1. Selection of labor tasks (requirements of employers) corresponding to the education major from job descriptions in online recruitment systems (algorithm for specifying the planned training results).
2. Collecting and processing expert opinions according to the degree of task significance (considering the confidence coefficient in the i -th expert k_i for differentiating opinions).
3. Ranking tasks by significance with establishing a minimum threshold of significance, the formation of the core of the workshop.
4. Calculation of the weight (weight coefficient) of the j -th task W_j in the frames of the total labor intensity of the PoS as a whole.
5. Establishing the "labor task—competence" relation for competencies from the PoS's general characteristics for further distribution of tasks by discipline and recalculating the weight coefficient for a specific discipline. It allows to calculate the recommended complexity of the training task and the score for its implementation.

A unique feature of the proposed method is the possibility of systematically identifying the expected learning outcomes for the annual redesign of the PoS's educational content.

Let's consider the stages of the proposed method in more detail.

2.2 Selection of Labor Tasks (Employers' Requirements) from Job Descriptions in Online Recruitment Systems

The primary source of relevant requests from employers is online recruitment systems-job aggregators. The main advantage of online recruitment systems is the availability and openness of information about vacancies, from which you can extract the employer's requirements. According to the TalentTech report [10], only 18% of

applicants do not use online recruitment systems to find a job, 26% of applicants prefer online channels for job search, and the remaining 56% combine offline use and online media. Thus, online recruitment systems are currently the leading aggregator of information on specialists' employment in various fields, primarily technical.

Also, the online recruitment market is growing all time, including the Russian market. The largest segment in the Russian online recruitment market was occupied by the company "HeadHunter", wherein 2018, according to TalentTech [11], 45% of vacancies of all open vacancies of the labor market were placed. The authors selected headHunter Service (hh.ru) for further research.

The selection of employers' requirements from the texts of job descriptions is carried out by performing a software implementation of the algorithm for specifying the planned learning outcomes [12]. The algorithm's base is the lexical and semantic analysis (LSA) of employers' current requirements for specific projects (tasks) in knowledge-intensive areas. Based on these analysis results, the head labor tasks that a graduate of the profile should solve to be competitive in the labor market are identified and specified. Data collection is carried out using syntactic analysis of vacancies corresponding to the specialization from online recruitment systems. From the job description, the algorithm highlights the required skills and responsibilities of applicants. And it processed the obtained data with the help of lexical and semantic analysis technologies, clustered by topic, and the most popular groups of professional tasks and technologies (tools) are identified, with the help of which they are preferably solved.

The scheme for extracting and processing information from the job description is shown in Fig. 1.

The texts of job descriptions in various online recruitment systems have a similar structure—each vacancy contains a block describing the applicant's duties and requirements. Depending on the employer, the difference may be that different words may indicate the data we are interested in (responsibilities and requirements), for example, "Requirements" "Required" "It is necessary that...".

The program module functions according to the following scheme: a lexical and semantic analysis of normative documentation of educational and professional areas, search for the necessary vacancies in the online recruitment system by specialization/by possible names of professions and their processing in several stages: a selection of requirements and key skills using regular expressions, preliminary data preparation using natural language processing technologies and further cluster analysis. In terms of the F-measure, the authors selected the DBSCAN spatial clustering algorithm for additional work.

The extracted formulations are included in the redundant list of professional tasks. As a result, we get a complete list of workshop tasks with advanced tools, models, methods, and technologies that the employer expects to use. This list is pervasive, and the complexity of PoS has significant limitations. To assess the significance and ranking of tasks in the approach, the authors propose an expert community of these PoS graduates' potential employers. After the evaluation, we get the core of the workshop—the top list of the most significant expected learning outcomes of graduate education.

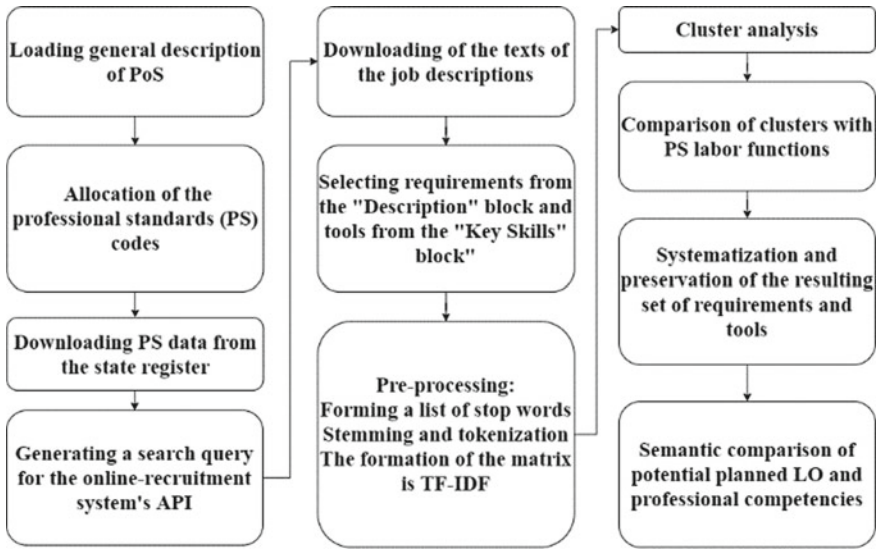


Fig. 1 The algorithm for automated analysis of labor market requirements in online recruitment systems

2.3 Collecting and Processing Expert Opinions on the Task Significance Degree (Considering the Experts Confidence Coefficient)

For PoS “Computer systems and technologies”, the authors invited 39 specialists as experts—representatives of leading companies in the Russian (75%) and foreign (25%) IT labor markets. Also, we invited two heads of educational programs of the higher educational institution Olin College Engineering, Massachusetts, USA, and recent graduates to the experts’ network community. Obviously, the opinions of these experts should have different weight when making decisions. Therefore, a proposed method for differentiating expert opinions [1, 12] includes determining the expert’s professionalism coefficient, the coefficient of similarity of the expert’s assessments with those of the reference expert, and the coefficient of confidence in the opinion calculated on their basis. This coefficient is a kind of correction for the estimates that the expert will make in the future.

The role of the expert in the method of identification of planned LO is as follows: the expert is asked to select for each expected result from the primary and redundant list of those that, in his opinion, are relevant at the moment, if desired, to supplement this list, and evaluate their significance on a 10-point scale. The value of the assessment is adjusted for the coefficient of confidence in the opinion of each expert.

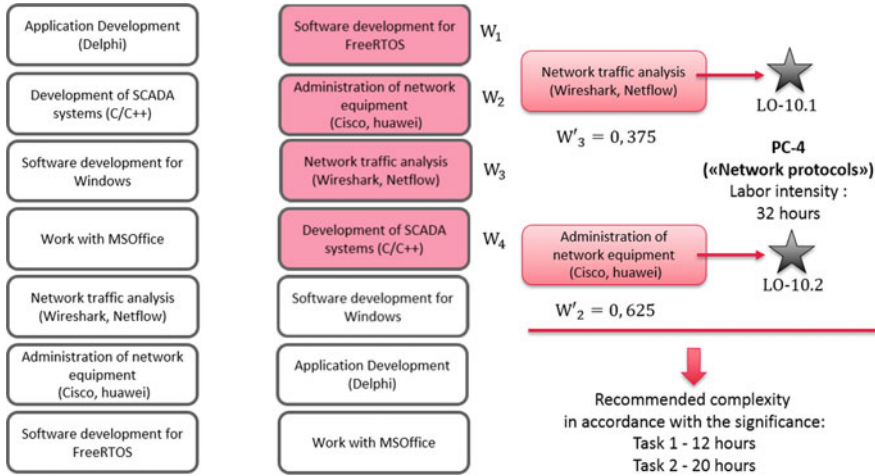


Fig. 2 The process of identifying planned learning outcomes on the example of the discipline “Network Protocols”

2.4 Ranking Tasks by Significance, the Formation of the Workshop’s Core

The selection of expected learning outcomes in the core of the PoS workshop is carried out by ranking according to the assessment of significance, establishing a threshold of entry, for example, not less than 60% of the maximum possible assessment of significance. Figure 2 shows the process of identifying the planned learning outcomes and shows the formation of a rating of professional tasks to select the top list of the most significant tasks that will need to be considered.

2.5 Weight Coefficient Calculation for the j -th Task W_j in the Frames of the Total Labor Intensity of the PoS as a Whole

Further, within the formed workshop’s core, the algorithm (software) calculates the weights [1]. This weighting factor affects the share of the current learning outcome (tasks for its formation) in the total labor intensity of the practical training program and the task’s location in the learning path. The weight coefficient is characterized using the formula (1)

$$W_i = \frac{\sum_{j=1}^m C_{i,j} * k_j}{\sum_{i=1}^n \sum_{j=1}^m C_{i,j} * k_j} \quad (1)$$

where

- W_i weight coefficient of i -th learning task,
- C_i score of j -th experts for i -th learning task,
- k_j confidence coefficient for j -th experts,
- n a number of the workshop's learning tasks,
- m a number of experts.

The weighting factor also determines the complexity of the recommended tasks in a particular discipline workshop, forming appropriate professional competence. A score for the achievement of the learning outcomes for this weighting factor W'_i is recalculated for each discipline separately after establishing the relationship between the professional objectives of the core workshops and professional competencies of the discipline. The process of forming weight coefficients is also shown in Fig. 2.

2.6 Establishing the “Labor Task-Competence” Relation for Professional Competencies (PC) from the PoS General Characteristic

Experiments on semantic comparison of employers' requirements (actually, planned learning outcomes) and competencies from the general characteristics of the PoS “Computer systems and technologies” were conducted with the formed top-list. As a result of the study, each competence formulation was aligned with 5 to 10 planned LOs with specific tools, making it possible to adjust the content of the workshops of the disciplines that form these competencies.

3 Practical Results

The authors used the method of identifying planned learning outcomes for the PoS “Computer systems and technologies”. For this PoS we analyzed:

1. One educational standard for the academic major “Informatics and computer technic”;
2. Two general descriptions of the PoS “Computer systems and technologies” (bachelors' level and masters' level);
3. Eleven relevant professional standards for the education area.
4. More than 11 thousand vacancies and 120 thousand requirements.

The software implementation allowed to extract and process more than 100 thousand requirements of employers-potential planned learning outcomes. It also identifies intended learning outcomes of graduates' multidisciplinary learning by field of study “Informatics and computer engineering” using the 94 unique tools and

technologies professional field, including six tools and technologies for the course “Network protocols”. We used allocated requirements for skilled and basic labor tasks to identify expected learning outcomes in the kernel development project of the workshop of specialization “computer science” and training of the discipline “Network protocols”. For the discipline “Network Protocols”, examples of identified planned learning outcomes compared with competencies are presented in Table 1.

The authors created the workshop’s project (core) and updated the educational content (content) of the PoS disciplines, to review the location and complexity of the disciplines, following the approach to project management of the workshops [1]. The modified program has been implemented in the educational process of ITMO University since the 2020/2021 academic year.

The compliance of the planned and achieved training results for the discipline “Network Protocols” was evaluated by an expert community consisting of 15 experts

Table 1 Comparison of the planned learning outcomes and competencies for the discipline “Network Protocols”

Competence code	Competence achievement indicator(s)	Planned learning outcomes (skills) of the graduate
PC-4. The graduate is able to develop simulation models of interaction of computer systems, simulate network protocols, determine and evaluate interaction indicators	PC-4.1 Defines and evaluates the performance of the protocols, building protocols by combining basic mechanisms	LO-10.2 Calculates the optimal TCP window size Defines the metric of the RIP routing protocol Evaluates the cost of the OSPF routing protocol interface Performs testing of Ethernet channels Implements and maintains network equipment of cisco, huawei, mikrotik in multi-segment networks Configures and administers network equipment (switches, routers) Cisco, Juniper, Huawei, HP, IBM Configures DNS servers within a multi-segment network
	PC-4.3 Performs debugging and verification of software models of interaction between computer systems and their components	LO-10.1 Analyzes network traffic using Wireshark, Tcpdump, and Netflow tools Performs network diagnostics based on the ICMP protocol Processes IP traffic as part of software and hardware complexes (routing, NAT, firewall, VPN, ...)

in network technologies. Experts confirmed the positive dynamics (17%) of the degree of proximity of the educational tasks solved by graduates in the course project of the discipline to the professional sphere's tasks. It indicates that the research aims has been achieved.

4 Conclusions

This paper describes the method of identifying the planned LO for automation of collection, systematization and processing of expert opinions from potential employers of graduates, and developing a workshop's core for the PoS "Computer systems and technologies" (direction "Informatics and computer technique"). This method allows selecting relevant tasks in the core of the workshop and ranking them based on expert employers' differentiated opinions. The authors propose an algorithm based on the text's lexical-semantic analysis (LSA) to implement the method. This algorithm and its realizing software allow for cluster analysis based on job ads in online recruitment systems and concretize planned LO using methods, technologies, tools, etc. required for a graduate of this PoS in the labor market. The authors used this software tool in the 2019/2020 academic year to identify the planned training programs for graduates of the university profile "Computer Systems and Technologies". It analyzed more than 11 thousand vacancies in the online recruitment system hh.ru. Among them, the software selected 94 names of appropriate methods, technologies, tools, etc., which should be owned by our graduates. To choose the planned learning outcomes, the faculty created a network community of potential employers of graduates in the PoS "Computer systems and Technologies". As part of this community for the 2019/2020 academic year, 32 experts participated, including 24 representatives of the Russian and 8 representatives of IT specialists' foreign labor markets.

The final assessment of the proximity of the achieved training results to real professional tasks was calculated as a weighted average, taking into account the degrees of confidence in each expert's opinion. For the last academic year, it was 73%, and after the implementation of the modified version of the workshop in the current academic year, it increased by 17%. A small discrepancy in the estimates shows that the workshop was quite relevant earlier but still benefited from its modification.

Prospects for further research are related to the promotion of the proposed approach in developing new and modernizing existing workshops for practice-oriented PoS. We will also improve methods and technologies for supporting expert communities, monitoring the competitiveness of graduates in the labor market, and managing individual educational trajectories of students following their professional interests.

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Smart e-Learning

A Smart e-Learning System for Data-Driven Grammar Learning



Hengbin Yan and Yinghui Li

Abstract In foreign language teaching, grammar is an important subject that has benefited relatively little from recent advances in computer-assisted language learning. To address this bottleneck, we propose in this paper a data-driven approach to grammar teaching/learning in a smart learning environment. We describe the design and implementation of a smart e-learning system and propose a new algorithm for unsupervised extraction of grammar patterns. The system enables autonomous learning by allowing users to plug in customized corpora for personalized analysis. A preliminary evaluation on system-generated grammar patterns reveals that the quality and quantity achieved are comparable to previous supervised systems that rely on precompiled pattern lists and hand-coded rules. We discuss the potential application of the system in a smart learning environment and its integration into grammar teaching curriculums to assist teachers and students both in and outside the classroom.

Keywords Smart e-learning · Data-driven learning · Grammar pedagogy

1 Introduction

As a relatively recent concept, and despite multiple attempts by researchers, the precise definition of smart learning appears to have remained elusive [1]. Recent years have seen some convergence on consensus about the conceptual dimensions of smart learning [2]. The first dimension is “smart technologies” through which students and teachers employ intelligent and customized technologies to facilitate effective, personalized learning. The second dimension refers to the cultivation of

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“smart learners” whereby learners master learning by developing such skills as self-monitoring and knowledge organization [3]. The third dimension, “smart pedagogy”, proposes the creation of a learning environment in which the teaching strategies actively adapt to the integrated smart technology and the changing needs of the students [1].

Of the many areas which have seen success in the application of the smart learning paradigm, foreign language learning is one that can reap further benefits. While technology-enhanced instruction empowered by computer technologies and intelligent tutoring systems has been found to be generally effective [4, 5], evidence for the efficacy of Computer-Assisted Language Learning (CALL) appears relatively limited, despite years of development and abundant publications in the field [6]. Moreover, the scope of investigations in CALL has traditionally been confined to several popular topics. For example, a recent survey on mobile-device-assisted language learning found that most of the studies have focused on the instruction of vocabulary-related knowledge (e.g. vocabulary, pronunciation), while none has investigated the issue of grammar knowledge [7].

1.1 Data-Driven Grammar Learning

As a “fundamental linguistic resource to communication” [8], grammar is an essential component of foreign language ability. However, grammar is also a difficult subject to master, especially for foreign language learners [9]. To address existing bottlenecks in grammar pedagogy, Data-Driven Learning (DDL) [10] has been proposed as a viable approach to improving the effectiveness of grammar learning. The DDL approach makes use of vast collections of authentic corpus data to allow learners to actively explore the intricacies of language [11, 12]. Previous experimental studies suggest that DDL can lead to raised awareness, better teaching/learning of lexico-grammatical items, and improved learner centeredness and self-learning skills [13]. DDL has been found to be particularly effective in helping students master grammatical structures and formulaic sequences, where the pairings between forms and meanings are discovered by learners through active learning [11, 14]. This complements existing technology-enhanced language learning, where the grammatical level has traditionally been neglected [7].

Despite the development of DDL both in research and practice, however, a number of issues have been raised as impediments to its widespread adoption as a pedagogical model [12]. These include: (1) the traditional interface featuring chopped-off concordance lines have proven inadequate and potentially counterproductive in helping learners grasp lexico-grammatical patterns; (2) teachers often lack sufficient technical knowhow for the use of sophisticated DDL corpus tools; (3) DDL systems often present one-size-fits-all materials regardless of varying proficiency levels, often resulting in hampered comprehension; (4) existing systems still require overt teacher intervention, which has led to students’ over-reliance on teachers to the detriment of

autonomous study. An improved system is needed to alleviate some of these existing problems in data-driven learning.

1.2 Grammar Pattern Extraction

Recent developments in cognitive and psycholinguistic theories have established grammatical constructions, or learned pairings of form and meaning, as the central components of language learning. It is argued that language learning is, in essence, the acquisition of grammatical constructions [15]. While there has been differing accounts of the exact form and nature of constructions, recent studies have shown that constructions can be operationalized as grammatical patterns, such as those laid out in the Collins COBUILD English Dictionary, as potentially important manifestation of constructions [16].

To build a grammar learning system, we adopt the descriptive representation of Pattern Grammar, which represents grammatical patterns using sequences of abbreviated notations of lexical, phrasal and clausal elements [17]. For example, the pattern *V n to-inf* denotes a verb followed by a noun and a to-infinitive, and can represent examples such as *ask somebody to do something*. This representation scheme presents a straightforward, flexible and transparent description of grammar patterns, rendering them particularly suited to language teaching [17]. Drawing attention to grammar patterns can facilitate form-focused instruction, where conscious and focused attention to target structures is found to result in substantial and durable target-oriented gains compared with implicit instruction, especially for L2 learners [15].

There have been several attempts to apply pattern grammar to data-driven learning [18–20], but they suffer from similar shortcomings: (1) they are all supervised, i.e. rely on the pre-existence of a manually compiled pattern list, which can be time-consuming and expensive to build (2) even armed with a compiled list, individual hard-coded rules are still needed to extract each pattern programmatically.

1.3 Project Goal

Consistent with the cognitive proposal that construction learning is usage-based, we present in this study a data-driven application for the learning of grammatical patterns following a smart learning paradigm. We propose a prototype of an intelligent data-driven system for discovering grammar patterns from large-scale corpora that can be used in a self-regulated smart e-learning environment. We describe the process of developing the system, evaluate the quality and quantity of the automatically retrieved grammar patterns, and discuss its potential integration into a grammar learning curriculum.

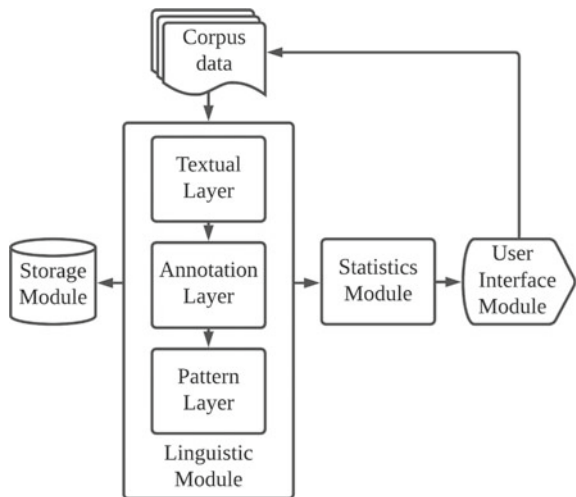
2 Method

2.1 System Architecture

The system features a simple and efficient architecture consisting of four main modules, each responsible for an encapsulated function in the pattern extraction pipeline (Fig. 1).

- The storage module is responsible for the persistence of all system-related data, including textual data, multilayered annotations and user customizations etc. For textual annotations, a generic multilayered format is adopted to ensure maximal compatibility and exchangeability with mainstream schemes in accordance with the Text Encoding Initiative (TEI).
- The linguistic module provides the main pattern extraction functionalities. The module is comprised of three separate layers in a linguistic processing pipeline. The textual layer deals with raw input texts, segmenting them into tokens and sentences. The annotation layer applies syntactic taggers and parsers to the stored tokens and adds linguistic annotations (lemmas, parts-of-speech, constituent and dependency parses etc.) on top of the textual layer. The pattern layer performs searches for grammar patterns, scoring, ranking and evaluating the patterns before outputting them to the statistics module.
- The statistics module collects and aggregates statistics related to the extracted patterns. It performs descriptive analysis and generates visual summaries in the form of tables and graphs.
- The user interface module provides a web-based frontend that handles interactions with users. It takes in user queries, forwards them to the system backend before presenting the statistical data generated from the backend on the web interface.

Fig. 1 Overall architecture of the proposed system



2.2 *Corpus Data*

With a generic design in algorithm and architecture, the proposed system accepts as input textual data of arbitrary sizes and taken from different sources. The simplicity and unsupervised nature of the algorithm makes it ideal for big-data processing without requiring manual data annotation. Although English is currently the primary focus of investigation, the algorithm is language-agnostic and thus supports a wide range of major languages (as supported by the syntactic parser in use).

As a data-driven learning tool, the system features a flexible plug-and-play architecture that allows users to provide their own data sources for pattern mining, enabling for teachers and students to tailor the system to their individual purposes. For example, ESL (English-for-Specific-Purposes) teachers may plug in course-related reading materials and retrieve pattern lists central to the targeted genres. On the other hand, students can plug in their own essays written as part of the course requirements, utilizing professional jargons and grammatical expressions specifically taught in the course. The list of patterns detected in students' essays can be compared with that of course texts, obtaining statistics that can be used to quantify learner progress.

To test the feasibility of our proposed algorithm for pattern extraction, we perform evaluations on a general-purpose corpus, the British National Corpus (BNC) [21], a freely available corpus covering a wide range of sources, both written and spoken, in British English. The BNC corpus samples 100 million words from a balanced list of genres representative of authentic language use, which makes it well suited for language teaching (one of the stated purposes for its compilation) in British English, the dialect typically taught around the world to English as a Foreign/Second language (ESL/EFL) learners. For a manageable evaluation of the pattern results, we used in this study a sub-corpus of the BNC, the XML version of the BNC Baby, which contains four one-million-word, balanced subsets of the BNC covering four genres (news, academic, fiction and conversation). The Python programming language was used to preprocess the data. Tokens were first extracted from the XML files of the BNC corpus, removing the original layers of annotations in the corpus. Token and sentence boundaries defined in the corpus were kept (such segmentation could also be performed by a syntactic parser). For transcriptions of spoken conversations, filler words such as (*um*, *uh*, etc.) were removed to improve accuracy of subsequent parsing and analysis.

2.3 *Pattern Extraction Algorithm*

The pattern extraction algorithm extends the application of existing Association Measures (AMs) commonly used in computational and corpus linguistics to pairwise associations between units beyond the word level. The past decades have seen the extensive applications of statistical measures to quantify association strengths between words that co-occur with probabilities beyond chance (an indication of

potential syntactic/semantic relationships between them). Despite decades of development, existing studies have mostly focused on measuring associations between the co-occurrence/collocation of words, while the co-occurrences of multiword sequences have been largely neglected. However, in Pattern Grammar and other approaches to constructionist investigations, the object of study involves sequences whose lengths are not predefined. For example, in the pattern *V from n to n*, *n* is a noun phrase that can be realized lexically as a sequence consisting of one or more words, from single word nouns to long phrases modified by complex noun modifiers. Even in a simple clause, the number of pairwise relationships between adjacent words and multiword units can be potentially huge.

The problem of pattern extraction is concerned with the identification of statistically significant relationships between elements that may be part of a grammar pattern. To investigate all possibilities of pairwise combination, a search space is created by aligning different layers of structural information (lexical, syntactic etc.) at the word level. For each unit (both single words and multiword spans) in the search space, a decision will be made about whether the unit can be part of a grammar pattern formed with neighboring units.

To perform the search for patterns, the algorithm starts with plain texts segmented into words and sentences using an automatic syntactic parser. The parser adopted in this study is the Stanford Parser [22], a neural multilingual parser with state-of-the-art performance and available as part of the CoreNLP toolkit [23]. The parser performs constituent parsing on the tokenized text, yielding the lemmas, parts-of-speech and a constituent parse tree for each sentence. A constituent parse for an example sentence containing the pattern *n range from n to n* is given in Fig. 2.

The tokens, lemmas, parts-of-speech as well as phrasal labels (NP, VP etc.) from the parse tree are then aligned at the word level to represent the multi-layered structure of the search space for grammar patterns. The algorithm iterates through each unit at each position in each layer and store the positions of each possible combination between the unit and its neighboring unit. Bigrams are formed by successively pairing neighboring units within the same and across different layers.

After all unit combinations have been exhausted, the AM score for each combination is computed using an association measure. The chosen measure is log-likelihood ratio, a measure of collocation known for its good performance in a wide variety of linguistic scenarios [24]. We rank the combinations based on AM scores from high to low. The top combination is highly associated relative to other combinations, and may serve as a component of a new candidate pattern. The combination is merged into one single unit, which will combine with other units for AM computation and ranking. The process of computing, ranking and merging will iterate until a predefined number of rounds have been reached, or no remaining combinations in the current round have an AM score above a predefined threshold.

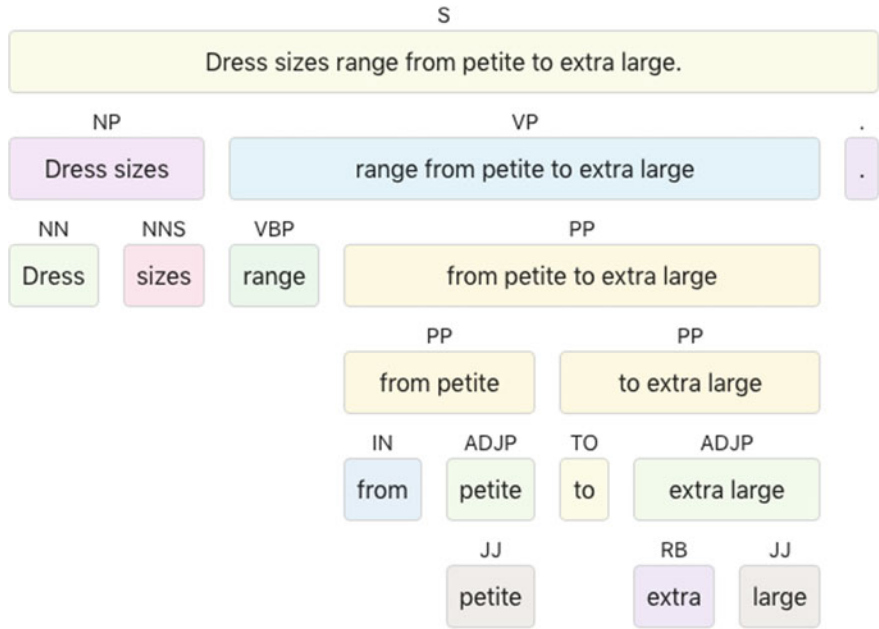


Fig. 2 Constituent parse tree for an example sentence

3 Results and Discussion

3.1 Pattern Extraction Results

The evaluation of the results of pattern extraction will be performed over two stages: the first stage where selected samples from the system output will be evaluated qualitatively; the second stage where a more comprehensive quantitative assessment on the BNC Baby is performed to obtain statistics on the overall accuracy of the system. The two stages are an ongoing process, requiring collaboration between several linguistic experts. Here we report our preliminary evaluation on several verbal patterns extracted using the system. The verbs surveyed are commonly used verbs with a variety of potential patterns. We compare the results of the unsupervised extraction with the patterns obtained from a manually compiled pattern list as reported in [19]. The list of verbs and associated patterns are presented in Table 1.

In general, for the verbs surveyed, the number of automatically extracted pattern types exceed that of manually identified patterns. For example, for the verb *vary*, in addition to correctly identifying the patterns in the manual list, the system produced unique patterns such as *vary according to*, *vary in n*, *vary from n to n* etc., which are valid and sufficiently attested by authentic samples in the corpus. Such patterns are valuable because usually they are not described in a dictionary entry on collocational usage. However, the results also reveal a number of erroneous patterns, such as *n*

Table 1 Comparison of automatic extracted patterns with manually compiled patterns

Target verb	Extracted patterns (count)	Patterns in [19]
apply	n apply n (91), n apply to n (85), n apply (58), n be apply to n (33), n apply for n (32), n be apply (32), to apply (18), * n apply n of (16), be apply (14), by applying n (13), apply n to (9), apply in (8), n apply for (3)	Apply n, apply n to n, apply to n, apply for n
assume	Assume that clause (75), n assume (58), n assume that clause (50), assume n (47), n be assumed that clause (24), to assume that clause (24), to assume (17), n be assumed to be (14), assume that (13), n be assumed (13), n assume clause (13), n be assumed to (12), be assumed (12), n be assumed to v (11), assume to be (8), *n n assume clause (7)	Assume n Assume that be assumed to
emerge	n emerge (46), n emerge from n (20), emerge from n (13), emerge as n (12), n emerge in n (12), to emerge (9), n emerge as (7), *n n emerge as n (7), n emerge that clause (7), *clause n emerge (6), n emerge that (5)	Emerge, emerge from n, emerge as, emerge that
require	n require n (141), require n (107), n be required (93), n require (85), n require n of n (50), n required to v (38), n be required to v n (24), n required for n (24), require n to (20), that n be required (19), n be required to v (15), n require that (19), be required to v (15)	Require n, require n to, require that, be required of
vary	n vary (70), n vary in n (16), vary n (14), n vary with n (10), n vary according to n (10), n vary from n to n (10), *that n vary (8), vary in n (8), n vary from n (7), *n vary with n of n (7), vary from n (7), n vary from (3)	Vary n, vary, vary from n

* Indicates incorrect pattern

vary with n of n. In this example, while *n vary with n* is itself a valid pattern, it is erroneously merged with an additional prepositional phrase (*of n*), presumably due to the high frequency of the nominal pattern *n of n*. While such errors can be trivially corrected with manual checking, they reveal potential shortcomings with the completely unsupervised approach and merits further fine-tuning in the algorithm.

The results of this preliminary evaluation show that the pattern extraction system can extract valid grammar patterns in addition to expert-compiled patterns. The overall quality and quantity of the patterns are comparable to, and in some cases, even exceed those of manually-compiled and hard-coded patterns.

3.2 *Integration into a Grammar Teaching Curriculum*

The proposed system can be integrated into a grammar teaching curriculum, serving several important functions as a smart learning tool. One obvious function is assisting the design of teaching materials at various stages of learning, providing a thesaurus-like resource for learners and teachers. Learners at different stages of learning may benefit from personalized learning materials suited to their proficiency levels. With example usage and summary statistics for identified patterns, teachers can easily design targeted teaching materials and prioritize the teaching of frequent grammar patterns. Teachers can engage students in form-focused instruction conducive to raising the awareness of the target grammar structures. With its interactive features, the system enables a personalized environment for self-regulated learning. Users can freely explore the intricacies and complexities of language and learn according to one's proficiency level, learning style and interest. Deployed outside of the classroom, the system allows students to learn at their own pace without temporal or geographical restrictions. Another potential area where the system may prove invaluable is language for specific purposes, which calls for the pedagogy of different materials from different genres. Traditionally, language teaching in this field has relied on pre-compiled materials which only target very specific areas at a particular proficiency level. With the proposed data-driven model, students are now empowered by the system to engage in open-ended exploratory self-learning of grammatical constructions.

4 Conclusion

In this paper, we have described the design and development of a smart grammar learning system that utilizes unsupervised algorithms for the automatic extraction of grammar patterns from corpora. The system features a modularized architecture that allows teachers and students to plug in arbitrary pieces of textual materials and receive immediate feedback through a web-based interface. Our preliminary evaluation on pattern extraction for common verbs reveals that the quality and quantity of the retrieved results are comparable to, and in some cases, even exceed those of manually constructed systems. The flexibility and ease-of-use of the system smoothens its integration into a traditional language learning classroom, where students conventionally play a largely passive role in the learning process. The system enables students to engage in autonomous, data-driven learning which has been found to increase learning efficiency and motivation. Despite the initial promising experimental results, the system, as a work in progress, still needs further development and additional empirical validation before it can be effectively deployed to a real-world smart learning environment.

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Experience in Smart e-Learning System Application When Switching to Distance Education to the Fullest Extent: The Case of the Moodle LMS



Leonid L. Khoroshko, Maxim A. Vikulin, and Alexey L. Khoroshko

Abstract Due to the existing epidemiological situation, the question of smart e-learning organisation has become ever more crucial. In the twinkling of an eye all educational organisations faced many problems, of which nobody has ever thought before. How to organise full-time education on a completely distance basis? What technologies will implement such transfer? And above all: how to preserve quality of learning under such difficult conditions? In this paper authors tell how they themselves answered these questions, and what solutions were implemented at the Moscow Aviation Institute (National Research University).

Keywords e-Smart learning · Interactivity · Moodle LMS · e-Learning

1 Introduction

Despite the popularity of smart e-learning systems, most people did not consider them as one of the primary instruments in the learning process organisation. In full-time academic education such systems were an auxiliary element for a long time accompanying basic classroom training. In 2020, due to the necessity of switching to distance education to the fullest extent, the situation has changed fundamentally.

However, smart e-learning systems application in a new way gives rise to new questions, which have not been of topical interest before. Probably, the major question is: How a smart e-learning system may help in switching to distance education with minimal losses in quality of learning?

In this paper authors tell how they themselves answered this question, as well as other questions, and what solutions were implemented at the Moscow Aviation Institute (National Research University), the case of LMS MOODLE smart e-learning system.

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The fundamental objective of this paper was the organization of classroom learning online. For this purpose, it was decided to use the BigBlueButton videoconferencing system, which has certain competitive advantages. The main advantages for the authors were open-source code and the possibility of hosting on their own servers, which allowed integrating video conferencing into the education system, as well as ensuring that all information remains within the corporation.

The solutions known to the authors imply the use of separate videoconferencing platforms that are not related to the e-learning system in any way. From there, the novelty in this paper is the result of the flexible integration of the education system with videoconferences, whereby all the necessary learning materials are available in one place, which makes it easier for students to access them.

2 BigBlueButton Video Conferencing System

Clearly, the most important moment in organisation of full-time education on a distance basis is delivery of classes. Video conferencing technologies have been used by no means the first year, and there will be no trouble to use them for organisation of in-class online learning, subject to availability of required equipment.

There are different types of in-class learning: lectures, practical training, and laboratory classes. Everything is quite simple with delivery of online lectures. A prearranged presentation is required, by which a lecturer gives a class; voice and video communications may be implemented through logging on to the conference even from a mobile device. Organisation of such online lectures may not even require any special equipment; however, its availability and use make the process much more ergonomic [1].

The situation is rather more difficult as to practical training and laboratory classes. Such classes may be completely different in form, which will depend on major in which training is taken. E.g., organisation of practical training in the form of a seminar for humanitarian subjects will not differ much technically from delivery of lectures. For specialised information technology subjects it is possible to install dedicated software on computers of students or to use online equivalents available for common use. It allows to approximate practical training and laboratory classes to full-time education, and provides each participant with the possibility to apply hand-on experience.

The situation is completely different with technical disciplines that involve delivery of laboratory classes on practice pieces. E.g., such disciplines as physics or chemistry. Organisation of laboratory classes in this case will differ from lectures, and it is usually impossible to repeat experiments at home. To solve this problem shooting equipment will be necessary for organisation of video conferencing with demonstration of laboratory classes. Another solution is the use of online laboratory classes in LMS MOODLE smart e-learning systems. Nevertheless, development of such laboratory facilities is a labour-consuming task, and if an educational institution

has no online laboratory classes already developed, then this option will not do as an operative solution of the problem.

Having considered different approaches to organisation of classes in online environments and requirements for their implementation, it follows that different classes have unique features for them to be organised in online environments. Nevertheless, it is important for an educational institution to organise as many types of classes in online environments as possible, according to a fixed schedule. It brings in distance education the required element of a full-time education discipline, which is so much missed upon a sudden change of environment. Moreover, the more classes are actually delivered, the stronger is the contact of a teacher with students, which is also a sore spot of online education [2].

There are many different platforms, which allow organising video conferencing. All of them have specific features, benefits and drawbacks. A video conferencing system should be user-friendly, but installation of any auxiliary applications complicates everything. Moreover, online classes should be combined with a smart e-learning system, where supplementary materials are placed. For the Moscow Aviation Institute (National Research University) an ideal solution was to integrate one of the video conferencing platforms directly into LMS MOODLE smart e-learning system. On the one hand, it allowed making the smart e-learning system central in the educational process, as far as any learning activity is delivered through it. On the other hand, it allowed delivering video conferences directly in the browser, which solves the problem of the necessity to install any additional software.

BigBlueButton was selected as a video conferencing platform. Below you will find a detailed review of the selected platform and its capabilities that allow organising online classes.

BigBlueButton is a web-conferencing system with an open-source code for online learning; however, it may be used to deliver briefings, presentations and webinars as well. BigBlueButton supports in the online mode shared use of audio and video, slides, a chat room, a screen, a multiuser board, online surveys, breakout rooms, session records and their reproduction for subsequent viewing.

The system allows operation with the following user types:

- *Observer* is a user (a student, as a rule), who may take part in a chart session, send/receive audio and video, respond to surveys and to show emoticons (e.g., to raise a hand).
- *Tutor* has all functionalities of an observer, moreover, he/she may appoint presenters, download presentations, activate a multiuser screen mode, share the screen, and manage parameters of the presentation and user area.

BigBlueButton platform was integrated as a module into LMS MOODLE smart e-learning system, which allows to add and to use it in any smart educational course. It also means that a new module has its own settings, just as the others. When using BigBlueButton Video Conference module, you may specify name, description, a calendar event (a date range, when the participation is possible), groups and parameters of online session recording, necessity to send a notice to users registered for the video conference (Fig. 1).

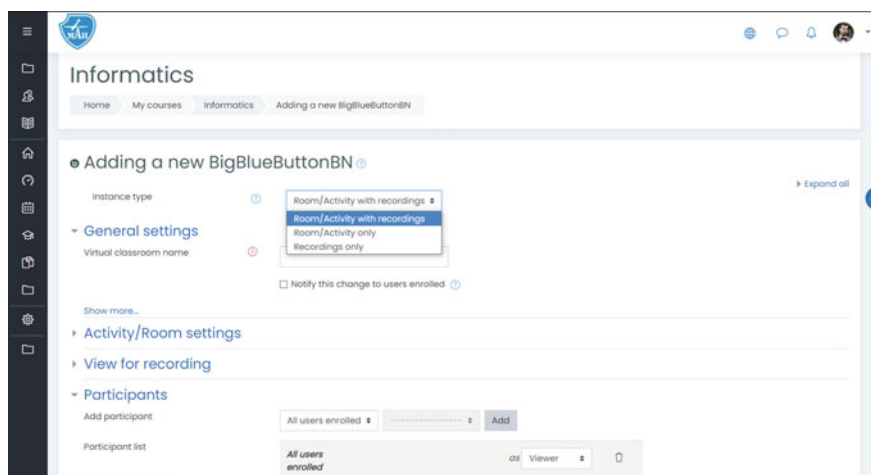


Fig. 1 Adjustment of BigBlueButton video conference module

In *Registration of Participants* setup section the users are registered, where necessary. By default, all users registered for the course have access to the conference, which differs in accordance with access rights. To notify users registered for the course of the forthcoming conference, mark *Send a Notice* in *Basic Settings* setup section. A notification specifying the video conference time will be sent to all users registered for the course. A notification for participants will be sent in the form of a personal message. The course expert will be indicated as a message sender [3].

When setting up the conference, particular attention should be paid to the *Group Mode* parameter included into the *General Module Settings* group. This setting allows creating one conference for several groups with an option of their differentiation. With the setting of the *Visible Groups* group mode activated, when linking up you will be offered to select a group, for which to deliver the conference. However, if you fail to select a group, the conference participants will not see the *Tutor*, as well as the *Tutor* will not see the conference participants. The list of participant groups is determined by group settings in the course. When selecting settings of the *No Groups* group mode, the conference will be available to all course participants.

For participation in a video conference an output device is required, such as headphones or speakers. For participation in voice communication a microphone is required. Connection to the conference with the use of desk top speakers and microphone is not recommended as far as it results in echo and degradation of communication quality. It is recommended to use a separate microphone and headphones, or a special computer head set. Audio settings are made each time when logging on the conference. When logging on the conference, the browser will request permission to use the microphone. Upon confirmation the system will offer to say a few words to check the microphone operation.

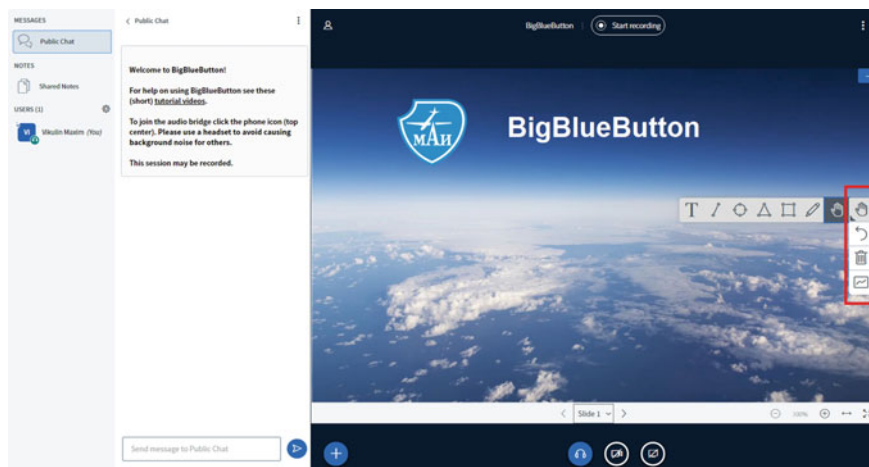


Fig. 2 Interactive whiteboard functionality

In order to be seen by other video conference participants, you should press the webcam button. Then you should adjust webcam settings according to the offered instruction and to press the *Allow* button. If all settings are made in a proper way, your webcam will be reflected in the *Webcam* window.

To approximate an online class to full-time education, use of an interactive whiteboard may be required. On the right part of the *Presentation* window buttons of the interactive whiteboard instruments are located (Fig. 2).

The following instruments of the interactive whiteboard are available:

- Pencil
- Square figure
- Circle figure
- Triangle figure
- Draw a line
- Text
- Clear all inscriptions on the whiteboard, delete an inscription on the whiteboard
- Multiuser mode.

One more instrument of video conference participants' involvement in joint activities is a possibility to conduct an online survey. During the video session the *Tutor* may switch over to a prearranged slide with a survey on the presentation page and ask the participants to complete the survey. For this purpose he should create a survey slide on the presentation page in advance and download this file.

Last but not the least reviewed functionality is video lecture recording. The record will be stored in the smart e-learning system, which will allow students to listen to a lecture once again at any time. In order to start video session recording, you should press the *Record* button. After confirmation, online class recording will start. To

stop recording you should press the record button once again, following which the conference may be completed.

3 Educational Course Organisation

Despite the fact that online classes constitute a foundational part of an educational process in any form of education, organisation of students' individual work is of importance as well.

As mentioned above, integration of the video conferencing platform into LMS MOODLE smart e-learning system allowed to connect to online classes directly from the system and to store video conference records in LMS MOODLE smart e-learning system. Therefore, it becomes possible to implement online classes and their records into the material structure previously organised in educational courses and intended to support blended learning.

After nearly two semesters of distance education, we identified and successfully put into practice the following rules for using functionality of LMS MOODLE smart e-learning system to organise distance education.

First, an educational course should contain learning materials, which facilitate consolidation of the lectures taken and offer an opportunity to review the material. These may be full courses of lectures or lecture notes, presentations, study guides and lists of supplementary references. It is critically important in distance education to provide students with an opportunity to independently handle any questions, which may arise, as far as interaction with a teacher became more complicated than in full-time education. Learning materials may be placed in different ways: in files, by references to external resources, or web pages. We recommend using the *Lecture* module, which is more complicated for implementation, but it has significant advantages [4].

The second important point in educational process organisation is delivery of practical training and laboratory classes. Posted guidelines for implementation thereof will help to improve distance work. It is recommended to use the *Task* module for collection of answers, which will allow not only to evaluate students' papers directly in the smart e-learning system, but also to store all answers in the system in an orderly fashion [5].

The third recommended rule is availability of the *Forum* module [6] in each educational course. On the one part, it will allow students to post questions, and teachers to answer them. On the other part, where necessary, you may maintain questions and answers base available to all course participants for review.

The final recommendation is the use of the *Test* module to organise testing [7]. Testing may serve not only as an instrument of final control in the educational course, but also as lecture material digest check, as well as an intermediate step for access to other learning materials. Moreover, training testing format is possible as a means of teaching, rather than control.

4 Conclusions

As mentioned above, at the time of writing the Moscow Aviation Institute (National Research University) delivered two semesters of distance education already. At these hard times smart e-learning systems have become of interest for all, and for educational institutions in particular, as never before. Organisation of the educational process became a problem, which was resolved by the Moscow Aviation Institute (National Research University) through integration of BigBlueButton video conferencing platform and the smart e-learning system. Interaction of these two systems allowed organising both planned online classes according to schedule and individual work of students at any time convenient for them. Drive of the Moscow Aviation Institute (National Research University) for the use, support and development of the smart e-learning system allowed being one step closer to readiness to switching to distance education to the fullest extent.

Further development of e-learning is inevitable; this is shown by the educational activity experience over the past year. According to the authors, the prospects for the e-learning development in terms of distance working boil down to improving the quality of the developed educational and methodological materials, i.e. training courses.

At the moment, we already have all the necessary technologies and means for the implementation of high-quality online learning. But it is important to understand that the available technologies are disjointed and the arrangement of their interaction requires individual solutions for implementation in the educational process. The authors consider the main directions of the development of e-learning in educational institutions to be the competent use and integration of existing technologies and the development of interactive courses that will not only contain the necessary information and tasks for students, but also be interesting to study.

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Gamification Model for Developing E-Learning in Libyan Higher Education



Entisar Alhadi Al Ghawail, Sadok Ben Yahia, and Joma Rajab Alrzini

Abstract This paper presents the utilization of the gamification model in the Libyan higher education context and investigates its contribution in creating new learning procedures to complete a task, and to develop the learners' inspiration, accomplishment and motivation for inclusion of gamification. It also aims to incorporation of gamification into E-Learning within the institutions of the Libyan higher education, for using this model with students' motivation and perception. Additionally, it presents this learning model in terms of ADDIE, these five key elements of the ADDIE model include: Analysis, Design, Development, Implementation, and Evaluation. The model includes important elements in e-learning. The results of this study show that motivation platform of gamification enables each student to recognize another student for a certain task related to learning process sharing, and to use different gamification elements (leader boards, points, scoring, levels, challenges to solve, incentives, student picture, team rewards, and badges).

Keywords Gamification · E-learning environment · Libyan higher education · Motivation · Game-based learning.

1 Introduction

Globalisation has, effectively, led us to perceive the world as a sort of unitary community, and this has been caused by increased technology use over the course of the last

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hundred years or so in the educational context; it is also because of this that learning, teaching, and coaching, as individual concepts, have all altered our perception of students [1]. Rhema and Sztendur [2] showed “Many of today’s students have spent their learning through video games, online environment, and social networks rather than traditional classrooms, techniques of gamification can include setting up precise rules, point, rewards, and competitions”. Also the benefits of looking to games to discover effective techniques to increase engagement and enjoyment have been recognised. According to [2], email, audio conferencing, teleconferencing, radio broadcasting, teaching via TV, communicative voice response systems, communicative radio-based therapy, CD ROMs, and audio cassettes are just a handful of examples of the variations of ICT-based products accessible to the public in the educative arena. Indeed, what with the fairly new phenomenon of laptops and mobile phones sweeping across the public, it should perhaps come as no surprise that gamification is now classed as one of the most renowned phenomena within the E-Learning arena. According to [3], gamification is actually being implemented in certain E-Learning contexts. In turn, it is being reported that content is more interesting, motivational, and eye-catching post-implementation. According to [1], Libya ranks first in terms of training strategies within Africa; this is due to the fact that it possesses a sufficient, comprehensive evaluation strategy in terms of its education system. It is with this in mind that the following research questions have been formulated:

- What is the model of gamification which may be designed in e-learning environment in Libya?
- Are the rewards and motivation for students positively influence within a gamification using?

This study designed a research model that explains the user’s engagement with gamification in higher education; rewards and motivation in this context refer to the extent to which students have a pleasurable experience when engaging with gamification. Researchers propose the following hypothesis: the rewards and motivation for students do not influence gamification. This study is organised as follows: firstly, review the literature adoption on ICT, gamification in the education process and motivation in the gamified learning environment are discussed. Secondly, present the methodology and proposed research methods. Thirdly, address discussion on the model of gamification. Finally, conclusions and future research are presented.

2 Literature Review

2.1 *The Gamification*

Gamification is the utilisation of game components and game deduction in non-game situations to expand target conduct and commitment [4]. Gamification can be defined as the incorporation of elements of gaming to an E-Learning environment, and we can say that this establishes atmospheres whereby gamification is incor-

porated within such learning processes. Indeed, it is involved in the establishment of those approaches in which students learn most effectively [5, 6]. Gamification enables us to help the understudy's inspiration and possibly impact on practices as a result [7, 8]. These persuasive insights and results can permit individuals to incorporate with and bring into their regular day to day existence that which they realised and experienced through the gamification disclose how to bring the basic mental drivers of the commitment into the gamified experience: **Autonomy**: the feeling of will and nearness when playing out an errand. It is encouraged by giving chances to pick, utilising positive input and not controlling directions. **Competence**: the need of individuals to partake in (ideal) difficulties and feel equipped and proficient, for gaining new information or aptitudes, accepting positive criticism and accordingly natural inspiration. **Relatedness**: the need of all people to feel associated with others; a need which current coordination among games and informal organisations is enormously misused.

2.2 *E-Learning Gamification*

One of the biggest things occupying university professors at the moment is the need to establish a method for enhancing student engagement, as this has been a known problem area for lecturers internationally and, as per this paper, gamification within such courses can act as such an approach to combat this issue. According to [9], audio, text, images, animation and learning-based video games are amongst the most commonly used educational gamification sources being used although, notably, it is important to bear in mind that tech-based games are not the only things used in terms of gamification. As much as such video gaming is based on largely yielding positive effects, insufficient gadget guidance, government initiatives, materials, and groundwork are some of the key issues of E-Learning in the context of its use within Libyan higher education [2, 10]. In a similar vein, it is worth noting the Libyan Education load is especially leading to further education and functional research as priorities taking the back seat for lecturers [11]. Some of this includes its bite-sized activities, the importance placed on positive reinforcement, challenges formulated to urge academic advancement, and instantaneous feedback; gamification possesses a few of the characteristic elements of behaviourist learning theory [12]. The growth and reformation of Libyan higher education were kick-started in the 1980s as a result of the suddenly increased population. The vast majority of students within Libyan higher education demonstrate a partiality toward the utilisation of technology.

2.3 *The Classroom's Game Elements Layout Design*

The selection of the educational approaches to be taken (i.e. the methodology that is going to be utilised whilst completing tasks and exercises, and done with the aim

of advancing in-game skills), the description of the student (i.e. the establishment of the students' aptitude, degree of organisation, and traits in order to construct the layout of the task accordingly), the educational circumstances (i.e. bearing in mind the opposing methods, substances, strengths and weaknesses), and the interpretation utilised (i.e. determined via lists in terms of the degree of drenching, commitment, and cleverness of the player), are the four dimensions that game design has been founded upon [13]. Such a setup is the result of a variety of elements (including programming apps and computer-based reality).

- The Mechanics are “the guidelines and ideas that officially determine the game-as-framework”.
- The Dynamics depicts the clients' communications with the mechanics. From a specialised perspective, it depicts the run-time conduct of the game-as-framework.

2.4 Motivation and Enthusiasm for Learning

According to [14], motivation in this context can be defined as a student's inclination towards obtaining an objective, which may or may not include those within an academic context. In turn, serious game implementation (i.e. games formulated specifically in mind of furthering education rather than for entertainment), off-sale game implementation (i.e. games formulated to expand on those on-sale to enhance their effectiveness in an educational context), and encouraging students to formulate their own games (i.e. for the purpose of furthering design and heuristic skills) are the three key approaches of video games within the educational context in order to encourage this motivation [11, 15]. The study sample consisted of (15) of sixth grade students at a school in Southern Ontario in Canada. The researcher adopted a quasi-experimental one group approach as a methodology of the study where the educational computer games were used to inspire students of the experimental sample to learn, and the study instruments on the note card and the trend metre were reflected. The results suggested that the efficacy of the games and educational computer games improved the incentive of students to learn.

3 Methodology

The researchers used experimental design based on two groups: an experimental group that was studied using Gamification as video games and educational computer games, with a control group that was studied in the normal way in the class. The researchers performed the following testing of two independent samples with the SPSS program. A t-test was conducted to find differences in effect learning between the control group which uses traditional methods and the experimental group which uses gamification. The following research questions have been formulated:

- Q1. What is the model of gamification which may be designed in e-learning environment in Libya?
- Q2. Are the rewards and motivation for students positively influenced within gamification?

In this study, the researchers used the experimental approach and design to develop e-learning by using the gamification model in Libyan higher education [16]. This study designs a research model that explains a student's engagement with game-based learning. A literature review also reveals that the application game mechanics to the classroom may increase students' intrinsic motivation.

3.1 The Study Sample

The study sample consisted of 29 students from the Economic and commerce college at Alasmarya Islamic University in Libya. The researchers divided the random sample into two groups, an experimental group of 15 students who studied using electronic educational games, and a control group of 14 students studied in the normal way (traditional) in the class. The study is limited by the nature of the participants of only female students, as it is a girls' college. In the structure of the study, an educational content analysis process is defined as "all actions taken by the teachers and their educational mission is to divide the educational tasks into the elements that make up the academic content" such as concepts, encouraging e-learning, and dealing with computer-related technological concepts, such as processor unit, memory unit, input and output units, etc. The researchers analysed whether the relationships between the proposed variables were statistically significant through content analysis using the ADDIE model.

3.2 The Study Tools

In order to achieve the aim of this study, the researchers designed the following educational tools: content analysis and design, technological concept testing, learning motivation, and educational content distribution to educational games. Also, two separate games were conceived with 5 levels. The first game is "Fishing concepts" with 3 level and the second game is "Quick concepts" with 2 levels. In addition, the researchers distributed educational content at different rates of play, and at different levels of the game after setting out prototypes for electronic games and deciding the number of levels in each game. The educational content was written by using Microsoft word 2013 (Table 1).

Table 1 The distribution of technological concepts at the game levels

The name of game	The level	Technological concepts
Fishing concepts	1	Computer definition—computer components
	2	Input units—output units—additional storage units—central processor
	3	Software and hardware—networks
Quick concepts	1	Networks
	2	E-Commerce

3.3 *The Gamification Model's Design into the E-Learning Environment*

The researcher followed a series of practical measures to develop an educational environment for teaching using gamification. When formulating our model for E-Learning in the context of higher education, sufficient management should be at the core of all of the factors of gamification, in order to achieve this sufficiency. In game-based learning, rewards given to students are received as a pay-off for completing pre-assigned tasks. Rewards stimulate students to learn, in order to earn more points, increase their levels, and achieve more badges or trophies for their performance (see Fig. 1).

3.4 *Game-Based Education Development*

Analysis It is at this point that the information required for the learning model is pinpointed by the investigators; it is important to note that it is essential that we arrange the content that students will learn in the aspect of exam results and the educational method being implemented, due to the fact that the model at hand has been based on the grounds of putting forward some of the key elements of gaming [17]. The suitable programs, curricula, and sources will be formulated which will be done adhering to a list of recommendations (see Table 2).

Design The design stage of this study puts forward the parts of the game that are potentially included in the E-Learning setup; we have formulated such a setup in order to incorporate a range of approaches to gamification [17, 18] trophies, badges, and a leader board all of which are discussed in more detail below: **Trophies:** The setup we have designed will provide users scoring the best with trophies; such reward schemes are proven to motivate students in this educational context. **Badges:** There are several situations in which users will be given the reward of badges: to students who use the software most frequently, to those who finish a topic, and to students when they finish a task/level. **Leader Board:** According to Zaric et al. [17] the implementation of leader boards results in students feeling driven and inspired toward participating

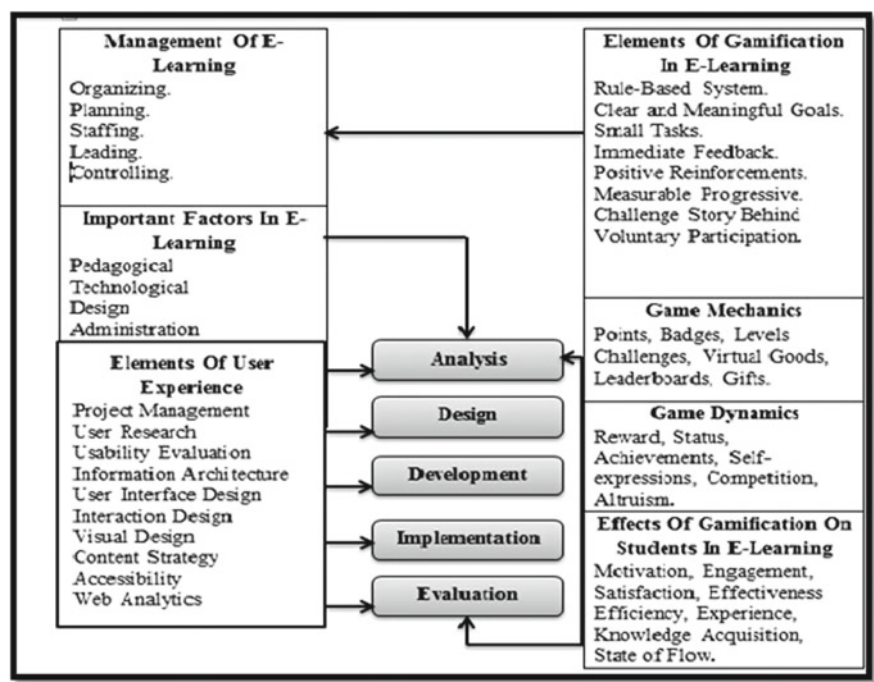


Fig. 1 Introduction of Gamification into the E-Learning environment model

Table 2 The task approaches based on the educative environment

Approach	Task
Active	Solid for-instances, such as practical set tasks, case studies, and experiments
Reflective	Open-ended demonstrations, allowing for pupils' expansion
Verbal	Text-based papers; lessons and literature; essay questions
Sequential	The sources at hand need to be consumed in a certain sequence when it involves two or more steps
Global	The used resolutions and educational blueprints are formulated by the pupils
Sensor	Practical activities, firm communication, and solid for-instances
Intuitive	Conceptual problems and hypothetical bases

more actively in their own educations. To clarify, a leader board would be employed as a method of specifying where each student stands in terms of their educational achievements (signified by the ‘levels’ they are at and the badges they have received). **Development** The gamification environments in need of development incorporating gamification into approaches include educative media, educative sources, and the educative administrative system; it was over the course of such development that we consulted a variety of professionals in order to effectively weigh up our learning environment.

Table 3 Elements leading to success of gamification in E-Learning

Criteria	Description
Failure	Allows for failures that will take the user by surprise
Reasoning	Urges the use of logical reasoning
Emotionality	Creates feelings of emotion within the user
Exploration	Encourages inquisitiveness, questioning, and examination
Doing	Urges practical work
Observation	Urges users to observe things independently
Motivation	Encourages the birth and/or expansion of motivation within users

Implementation It was at this stage that E-Learning, as a concept, was presented to students; this was done in order to establish who our users would actually be. Indeed, the most optimal way to realistically determine the effectiveness of E-Learning is to provide it to the individuals who would be using it and to take their direct feedback, altering it if necessary [17].

Evaluation The capability and efficiency of game-based learning can be determined via the weighing up of users' levels of drive and fulfilment upon use; according to [19], E-Learning is a highly esteemed source. Within such an evaluation, it is our primary objective to determine whether motivation has been enhanced within the students by analysing the system's stored information concerning such users. Table 3 shows the elements influencing success rates in students.

3.5 The Researchers Designed the Interface for Gamification, Namely the Learning Platform's Gamification Model Layout

The gamification model in question has been formulated in mind of the overall aim of enhancing students' motivation and interaction within their education and has been created with a variety of different gaming mechanics. This, in turn, has led to this platform having several 'levels', the software providing mini motives for the completion of the next task whether that be to get a high number of points or to unlock another achievement, thus encouraging more effective study time. Gamification represents an innovative and engaging methodology to motivate students and enhance their learning process. Through this exploratory study, we shed light on students' attitude towards gamification and the actual use of gamification in higher education (see Fig. 2).

To enhance participation and motivation for students within higher education, the gamification model has been notably created in adherence to the academic perspective currently governing rules for student motivation. This model now incorporates a layered structure which, as mentioned above, provides mini-goals for (see Fig. 3).

Fig. 2 Educational environment “user interface”

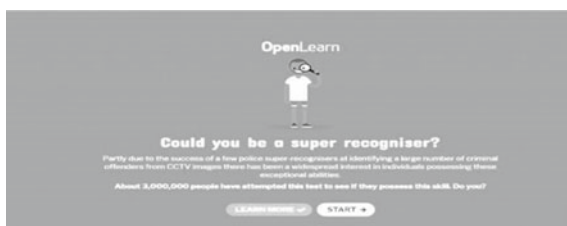


Fig. 3 Question selection

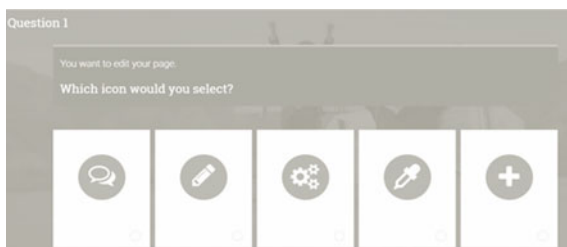


Fig. 4 The winning interface in the game levels

There is a minimum of one task to be done within each level of the system; ‘points’ can be gained by users in all of these levels (as a result of its positive reinforcement/reward scheme) (Fig. 4).

Elements of this process of game-based learning include things such as the establishment and handling of the system and its materials, to the designing stage, selecting the people who will aid in the system design, and development. On the other hand, the name of the game used was Fishing Concepts; this includes: Computer Definition, Computer Components, Input Units, Output Units, Additional Storage Units, Central Processor, Software, Hardware and Networks. Also, the Quick concepts game was used which includes: Definition of Networks and E-Commerce. Figure 5 Shows the layout that users will encounter when receiving ‘points’ for successfully doing an activity.

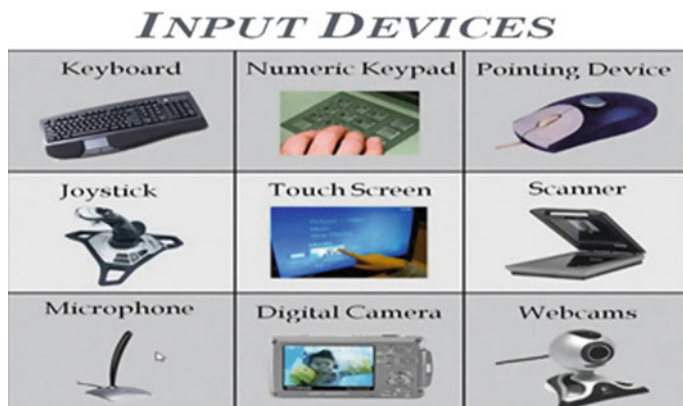


Fig. 5 Information of concepts

4 Results and Discussion

The first question of the study states: “What is the model of gamification which may be designed in an e-learning environment”? To answer this question, the researcher analysed and designed ADDIE model for the educational games that can be used in a higher education environment. The data analysis is a two-stage analytical procedure. The first stage design of the ADDIE model to the use of gamification for higher education, and the second stage examined the structural model to test the research hypothesis which said: rewards and motivation do not influence the use of gamification. The researchers performed the following testing of two independent samples by SPSS programs. The study set the test scores for two groups in the computer principles subject by checking that there were statistically significant differences in the averages of the test scores between the students of the control and experimental groups. In examining Table 4, of “Normality of Tests”, in all cases the value of Sig. = 0.200 was greater than 0.05. Thus, the null hypothesis, that is, the data in two classes had a normal distribution, was accepted. In this study, rewards signify the provision of choice, because students can trade the earned points for other virtual items or gifts (Table 5).

Table 4 Tests of normality

The group		Kolmogorov-Smirnov			Shapiro-Wilk		
		Statistic	Df	Sig.	Statistic	Df	Sig.
The grade	Experimental group	0.126	15	0.200*	0.944	15	0.432
	Control group	0.114	14	0.200*	0.955	14	0.640

* This is a lower bound of the true significance. a. Lilliefors Significance Correction at a significance level of $\alpha = 0.05$

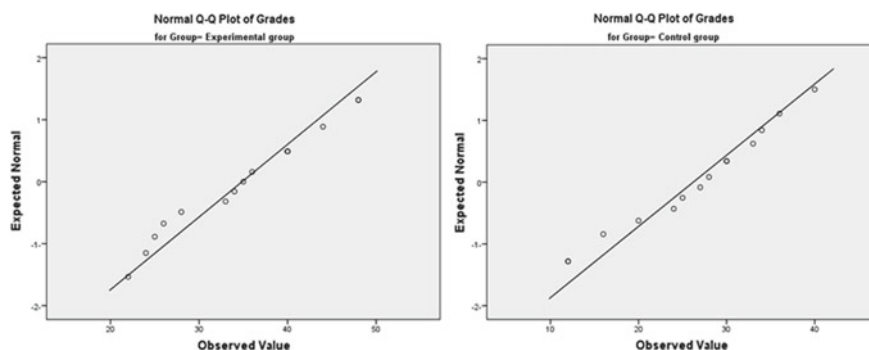


Fig. 6 The test of normality

Table 5 Mean of test grades for two groups by use of educational games

Mean of scores	Control group	27.64
	Experimental group	34.29
Std. deviation	Control group	10.566
	Experimental group	10.759

Table 5 shows the results of the t-test of the mean scores for the experimental group (mean = 34.29) and the control group (mean = 27.64); the results of the study confirmed the hypothesis which said that motivation, engagement and rewards positively influence gamification in an E-learning environment.

The homogeneity test was carried out and the Levene test was applied in order to check the accuracy of the results obtained from the two study groups. Since the value of Sig. (0.980) was greater than 0.05, the researchers accepted the imposition of nothingness and the homogeneity of the two groups, which indicates the similarity in the level between the two groups (Table 6). A t-test was conducted in order to find differences scores for the test between the experimental group and control group. A t-test was conducted (2.707), and Sig. = 0.012 was less than 0.05. Thus, the mean scores of the two groups was not equal; that is, there was a disparity between the scores of the students from the experimental and control groups in the two classes. Table 6 shows differences in student scores by educational games. Thus, researchers accept the following alternative hypothesis: Motivation positively influences engagement with gamification using.

Gamification is presented in the computer principles curriculum, which includes pictures and graphics, in a fun way, as it aroused the students' interest, and increased their motivation towards learning. The researcher believes that the variations between the study group and the control group are due to the successful use of gamification to provide computer concepts to students and to promote E-learning. After each student's responses, students received immediate feedback, encouraging them to continue playing to win, which enhanced their interaction and motivation to carry out game-level educational missions.

Table 6 The results of the t-test for the average of test scores in computer principles subject for the two study groups

		Levens's test for equality of variances		t-test for equality of means						
		F	Sig.	T	Df	Sig. (2-tailed)	Mean differ- ence	Std. dif- ference	95% confidence interval of the difference	
									Lower	Upper
The grade	Equal vari- ances assumed	0.001	0.980	2.707	27	0.012	8.65238	3.19680	2.0931	15.2116
	Equal vari- ances not assumed			2.705	26.793	0.012	8.65238	3.19867	2.0868	15.2178

Independent samples test

5 Conclusion and Recommendations

This study has explained the advantages of gamification incorporation within an E-Learning environment of Libyan higher education as a way to enhance student motivation and provide teachers with further materials. As we can see, both from our literature review and findings, gamification directly leads to the education process being improved, for access to further content and tutorials for students, as this enhances student participation. Gamification mainly serves to make user activities more motivational and rewards are a pivotal factor with which to determine student engagement with gamification. The present study contributes to the gamification literature by explaining the mediation role of motivation between the game dynamics and engagement. Competitions enable students to challenge each other to achieve the highest result for an activity. The leader board is central to displaying results and celebrating winners.

6 Limitations and Future Research

This study is limited by the fact that we ran it in a single college. It would be interesting to involve more colleges to see what other factors (e.g., satisfaction and culture) could impact the overall results. Furthermore, there are some participants who had never used the gamification system in place, which could have some influence on the results. Although our study is longitudinal in design, a three-month period may not be an ideal timeframe for measuring gamification effects. We suggest further research that will explore how satisfaction is influenced by different motivation drivers and ultimately investigate the relationship between the satisfaction of educational process and homework engagement. Future researchers should conduct their studies with a larger sample so that results can be more generalisable and indicative of the further improvements that should be made in order to create the best possible model.

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Digital Divide and Social Media Related to Smart e-Learning in Obstetrics During the Health Emergency by COVID-19 in Peru



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Abstract The objective of the research was to identify the types of digital divide and the use of social media related to the results of the application of smart e-learning in obstetrics during the health emergency due to COVID-19 in a public management university located in a remote place of Peru, such as the Santiago Antúnez de Mayolo National University (UNASAM). A cross-sectional research was developed by applying an online questionnaire to obstetric students between September and October 2020. It was found that the moderately satisfactory results achieved by the majority of students with the use of smart e-learning are related to digital divide of access, use and quality of information, as well as with the use of WhatsApp, Facebook and YouTube, findings that mean a contribution to the study of the application of digital smart e-learning tools in contexts of technological inequality and health crisis caused by COVID-19.

Keywords Digital divide · Social media · Smart e-learning · Obstetrics

1 Introduction and Literature Review

Virtual teaching–learning environments have been used for more than two decades in various countries, with successful teaching and learning experiences worldwide through the use of technological resources and smart education [1, 2], a reality of which Peru no has been part of, among other reasons, the existing digital divide, mainly in the most remote and poor populations, where only 52.5% of families have access to the Internet and the technological means necessary for their employment [3].

Although the benefits of the use of technological resources for smart education in the medical area have been highlighted in various research [4–7], there is still

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not enough information on the use of digital tools that facilitate smart e-learning in obstetric students from public universities located in remote places in the interior of Peru, especially during the health emergency caused by COVID-19, which has evidenced the existing digital divide in various countries [8–10].

In this sense, regrettably during the last decades, Peruvian authorities have not paid enough attention to the problem of the digital divide in state-run university institutions such as UNASAM, which is located in a Andean geographical region of the Peruvian highlands (3052 m.a.s.l.), whose obstetric students are mostly in a situation of social and economic vulnerability, due to this situation and the lack of experience in distance education at UNASAM before the COVID-19 pandemic, teachers have had to resort to the use of social media on smartphones as a smart e-learning strategy, and no research has been carried out on the results of this strategy in the learning of obstetric students in the circumstances described above.

2 Project Goal and Research Methodology

As a result of the aforementioned considerations, this research was carried out with the objective of identifying the types of digital divide and the use of social media related to the results of the application of smart e-learning in obstetrics during the health emergency by COVID-19 at UNASAM.

According to what was established in the objective of the research, it was considered to address as study variables the types of digital divide, the use of social media and the results of the application of smart e-learning by obstetric students, also considering their sociodemographic characteristics, for which a questionnaire created by the study authors was used as a data collection instrument.

The questionnaire consisted of 11 questions, the first part of which comprised 4 questions about the characteristics of the students: age (18–20 years, 21–23 years, ≥ 24 years), origin (rural area, urban area), family's monthly income (<S/. 930.00, $\geq$ S/. 930.00) and academic cycle (I–III cycle, IV–VI cycle, VII–X cycle).

The following 3 questions addressed the digital divide through access to electronic devices with permanent Internet connection (yes, no), digital divide of use on the lack of digital skills to manage devices and technological applications (yes, no) and digital divide of quality information (yes, no).

Regarding the social media used by obstetric students as digital smart e-learning tools during the teaching–learning process in the different subjects that were enrolled in the academic semester 2020-I, 3 questions were considered about the use of WhatsApp (yes, no), Facebook (yes, no) and YouTube (yes, no); ending with 1 question about the results of the application of smart e-learning according to the only parameter established in UNASAM according to the final average achieved during the academic semester 2020-I: unsatisfactory (≤ 10), minimally satisfactory (11–13), moderately satisfactory (14–16) and satisfactory (17–20). It should be clarified that in order to ensure the veracity of the answers and avoid any bias and/or inconsistency in the data collected, the response of the students to this last question was

corroborated by the authors of this research in the academic management system of UNASAM.

The content validity of the questionnaire was carried out by means of expert judgment followed by the evaluation of the results obtained with the Kendall's concordance test, demonstrating said validity with a significance level of 0.001. Likewise, after conducting a pilot test on 20 students, the reliability of the instrument was determined using Cronbach's alpha index, with a result of 0.862; applying the questionnaire between September and October 2020 through an online form, whose link was sent to the institutional emails of the students.

With regard to statistical analysis, it was used the statistical package SPSS version 22.0 to process the information collected, performing a descriptive statistical analysis according to the type of variable, using absolute frequencies and percentages. The Chi square test was applied with a significance level of $p < 0.05$, to identify the relationship of the types of digital divide and the use of social media with the results of the application of smart e-learning in obstetrics during the emergency health by COVID-19.

On the other hand, in compliance with the ethical considerations of the World Medical Association and the Declaration of Helsinki, voluntary participation and the virtual filling of the informed consent form were requested via email, guaranteeing the right to privacy, anonymity and confidentiality of the data provided by students. Likewise, all the procedures established by UNASAM were followed for the approval of the respective research protocol, which was presented to the Ethics Committee of said university, authorizing its execution.

2.1 Population Under Study

It was decided to carry out a cross-sectional research, with a population of 189 obstetrics students from UNASAM, who met the inclusion criteria of being enrolled during the 2020-I academic semester and having a minimum attendance of 70%; while student dropout was the only exclusion criterion considered, deciding to work with the entire population due to the number of students, as well as the possibility that not all students would agree to participate in the research and/or do not complete the informed consent form.

Of the total population (189), 20 students participated in the pilot test; 10 did not accept the invitation, 7 did not respond to the email and 152 accepted to participate voluntarily, completing the information requested in the online questionnaire, whose characteristics of the participants are presented in Table 1, where it is evidenced that the majority of students were between 18 and 20 years old (66.5%), coming from urban areas (52.6%), with a family's monthly income lower than the minimum living wage, which in Peru is equivalent to approximately 254.52 USD (71.1%), corresponding mostly to students between the I–III cycle of studies (56.6%) of the professional career of obstetrics.

Table 1 Characteristics of students

Characteristic	n	%
<i>Age</i>		
– 18–20 years	101	66.5
– 21–23 years	30	19.7
– ≥ 24 years	21	13.8
<i>Origin</i>		
– Rural area	72	47.4
– Urban area	80	52.6
<i>Family's monthly income:</i>		
– < S/. 930.00	108	71.1
– $\geq$ S/. 930.00	44	28.9
<i>Academic cycle</i>		
– I–III cycle	86	56.6
– IV–VI cycle	54	35.5
– VII–X cycle	12	7.9

3 Research Outcomes

The majority of obstetric students achieved moderately satisfactory results with the application of smart e-learning during the academic semester 2020-I (45.4%) (Table 2). Students with unsatisfactory results presented the highest percentages of digital divide regarding access to electronic devices with permanent Internet connection (17.8%), lack of digital skills to use technological tools (16.5%) and absence quality information (18.4%); evidencing that in all the types of digital divide mentioned above, there was a statistically significant relationship with the results of the application of smart e-learning in obstetrics during the health emergency due to COVID-19 ($p < 0.05$).

Regarding the use of social media (Table 3) as digital smart e-learning tools in obstetrics, the highest proportion of students with moderately satisfactory results used WhatsApp (36.8%), Facebook (34.2%) and YouTube (31.6%) in the virtual teaching–learning sessions of the courses in which they were enrolled during the 2020-I academic semester, finding that the use of these social media presented a statistically significant relationship with the results of the application of smart e-learning in obstetrics ($p < 0.05$).

4 Discussion

The results found in the present research show similarity with the findings of previous studies [8, 11, 12], in which the effects and consequences of the digital divide on the

Table 2 Digital divide according to results of application of smart e-learning

Digital divide	Results of smart e-learning								Total		<i>p</i> -value ^a
	Unsatisfactory		Minimally satisfactory		Moderately satisfactory		Satisfactory				
	n	%	n	%	n	%	n	%	n	%	
<i>Of access</i>											
– Yes	27	17.8	24	15.8	18	11.8	5	3.3	74	48.7	< 0.001
– No	1	0.6	11	7.2	51	33.6	15	9.9	78	51.3	
<i>Of use</i>											
– Yes	25	16.5	21	13.8	15	9.9	2	1.4	63	41.6	< 0.001
– No	3	1.9	14	9.2	54	35.5	18	11.8	89	58.4	
<i>Of quality information</i>											
– Yes	28	18.4	26	17.1	20	13.2	8	5.3	82	54.0	< 0.001
– No	0	0	9	5.9	49	32.2	12	7.9	70	46.0	
Total	28	18.4	35	23.0	69	45.4	20	13.2	152	100	

^aChi square test

Table 3 Use of social media according to results of application of smart e-learning

Social media	Results of smart e-learning								Total		<i>p</i> -value ^a
	Unsatisfactory		Minimally satisfactory		Moderately satisfactory		Satisfactory				
	n	%	n	%	n	%	n	%	n	%	
<i>WhatsApp</i>											
– Yes	4	2.6	15	9.9	56	36.8	17	11.2	92	60.5	
– No	24	15.8	20	13.1	13	8.6	3	2.0	60	39.5	< 0.001
<i>Facebook</i>											
– Yes	2	1.3	10	6.6	52	34.2	14	9.2	78	51.3	
– No	26	17.1	25	16.4	17	11.2	6	4.0	74	48.7	< 0.001
<i>YouTube</i>											
– Yes	1	0.6	8	5.2	48	31.6	11	7.2	68	44.6	
– No	27	17.8	27	17.8	21	13.8	9	6.0	84	55.4	< 0.001
Total	28	18.4	35	23.0	69	45.4	20	13.2	152	100	

^aChi square test

learning outcomes of the students were determined; considering that the lack of access to electronic devices and innovative educational software typical of smart education, is a complex problem due to the lack of knowledge of the real meaning of virtual education [13] by teachers themselves, who see their training possibilities limited permanently, due to multiple reasons, including the lack of support and interest from the authorities.

In this sense, due to the aforementioned situation, alternatives should be implemented [9, 10] according to the students' digital skills [14, 15], which unfortunately was lacking in more than a third of the participants in the present research, a finding similar to that reported in other investigations [11, 13] but in lower percentages; while the importance of the digital divide of quality information was also highlighted by other studies [8, 12], which revealed that it is not only important to have adequate technological resources, but also to know how to use them to access truly useful and necessary information for the training of students, which is why the support and guidance of teachers is needed during the application of smart e-learning as references of professionalism in the academic world, in order to guide what resources or bases data are adequate for the search for reliable and quality information.

Regarding the deficiencies evidenced and their relationship with the results of the application of smart e-learning in obstetric students of UNASAM during the health emergency due to COVID-19, in addition to the aforementioned factors, this situation could be explained in the economic, social and cultural barriers that afflict the student population of said university, which, because it is state-run, does not have sufficient financial resources to reduce the enormous digital divide that has become evident due to the COVID-19 pandemic, not having taken the technological support measures for the students in time that allow them to overcome the existing problems, owing to deficiencies in university management.

Regarding social media, their use was related to the results of the application of smart e-learning in obstetric students during the academic semester 2020-I, a finding similar to that found by other researchers [2, 15], who also determined high levels of student satisfaction regarding the improvement of their learning through the use of networks in smart education, which can be complemented with the implementation of collaborative social networks at the international level, reasons that would explain the high percentages of use of said social media by UNASAM obstetrics teachers, due to the knowledge of their use before the start of the COVID-19 pandemic, the lack of training regard to other online educational methodologies and the digital divide of students, aspects highlighted in students from other countries [2].

Due to the aforementioned, it is necessary to train teachers with respect to other technological tools that could be used for the development of smart education in university students, whose scope and importance have been demonstrated worldwide [4–7], especially in circumstances as complicated as those caused by COVID-19, becoming tools that will surely continue to be applied with greater emphasis in the coming years.

In this sense, the learning results of the students related to the use of social media as digital tools of smart e-learning in obstetrics, would be due to the greater interaction, affinity and collaborative learning of the participants, aspects highly

valued among young people, reducing the problems of isolation, loneliness and lack of motivation, associated with distance education [12], and that according to some studies could be reduced through the use of WhatsApp, Facebook and YouTube [11, 12]; always under the support and guidance of the teacher [2], in order to avoid the dissemination of unreliable information on these platforms, as well as to prevent the possible occurrence of cyberbullying between students and people outside the process of teaching–learning.

On the other hand, it is important to highlight the existence of several limitations in the present research, such as the non-description of the characteristics of use of each of social media as part of the development of smart e-learning in obstetrics, due to the quantity and variability of subjects in which the students were enrolled, with an average of 5 per student, as well as the cross-sectional design developed, which limited the long-term evaluation of the variables under study, not having deepened the results of the learning achieved by students, due to the failure to record data on formative assessment at UNASAM; aspects that could be considered in future research and that do not affect the importance of the information provided on the situation of smart education in obstetrics in a Peruvian public university in difficult health circumstances; information that could be taken into account to carry out new studies with university students from other medical specialties, where topics related to, for example, smart technology applications in medical education are discussed, as well as their relationship with other aspects outside the academic field such as smart society, smart city, among others.

5 Conclusions and Future Steps

Conclusions The types of digital divide identified and the use of social media are related to the results of the application of smart e-learning in obstetric students of UNASAM during the health emergency due to COVID-19, findings that highlight the use of digital smart e-learning tools in contexts of technological inequality and of health crisis caused by COVID-19.

Future Steps Future research on the subject under study should be aimed at identifying and analyzing the conditions in which smart e-learning is carried out with university students from various scientific disciplines, including areas related to medical education in general, in which it is important to take into account the particularities of the subjects and the limitations of the students, as well as the participation of teachers, especially in circumstances of social, economic and health vulnerability.

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Smart Education: Systems and Technology

Learning Smart Behaviors Through Digital Simulations: Combining Individual-, Firm- and System-Level Complexity



Andrea Montefusco, Federica Angeli, and Nunzio Casalino

Abstract Modern days are characterized by increasingly complex challenges, faced by highly interconnected actors, organisations and systems. Now more than ever, understanding and learning complex reasoning is a crucial skill to navigate complex scenarios. At the same time, complex reasoning also includes the capability to update competencies and skills if, when and how the situation requires, through double loop feedback and learning. We argue that advanced flight simulation technology can provide the ideal space where the capabilities and cognitive frames underpinning complex reasoning can be stimulated, consolidated and nurtured. We describe the case of FLYINGAN/EXPERIENCE lab technology, adopted in business educational curricula. We show how, by providing a psychologically safe environment, this technology allows to educate complex reasoning, by creating an inductive, constructivist learning process founded on goal-based scenarios. As learners need to execute tasks to fly the plane without any prescription about their behaviors and roles, complexity is appreciated while it emerges at multiple levels—individuals, team, and macro-system. We contribute to the literature on gamification and simulations, by highlighting how virtual reality can not only be widely employed in human resource management and training, but can also be crucial in stimulating complex reasoning and in gathering behavioural data—both quantitative and qualitative—with central scientific implications for both policy and practice.

Keywords Business simulations • Learning complexity • Training processes • Gamification • Virtual reality • Flight simulation • Managerial skills • Leadership • Decision making

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1 Introduction

Thanks to advanced digital technologies, virtual environments allow managers and workers to fully experience learning and knowledge, from a subjective perspective, where multisensory interaction is provided. The innovative and multimedia delivery of training and experiential contents can also improve managers' impact on social business context. Virtual environments are suitable to create online social spaces that, in organizational disciplines, can be interpreted with strong connections to social sciences. Interactive business learning environments can deliver contents by simulating unique social contexts for learning. Virtual reality has the power to help leaders experience and manage emotional leadership situations in a way that no other tool or learning technique can. Virtual reality tools enable students to come into contact with a more dynamic representation of reality than a traditional case study. Dynamic role play, or situational learning, could enhance teaching by requiring students to solve problems in an immersive environment. Immersion is becoming a fantastic learning opportunity, but another advantage of these tools is that they can be widely personalized. This manuscript highlights the multifaceted benefits of digital simulations for management education and research, through the case of flight simulation technologies. The manuscript is composed of eight sections. The first summarizes the learning complexity challenges. The second the roles and the impacts of simulations as pillars of modern gamification theories. The third section analyzes the complexity challenges. The fourth illustrates the FLYINGANEXPERIENCE solution, tools, and technologies to enhance experiences and create effective learning. The last sections list perspectives and limits.

2 Learning Complexity Presents a Twofold Challenge

Learning challenges both teachers and instructors. Teaching is an active process, based upon task execution through technical and human skills, performed by either a person or a small team: teachers and instructors. On the contrary, learning is a transformation process that involves some experience and ends up in a sort of permanent change in the state of the learner. We should consider learning as both an individual and organizational ability to evolve while exploiting and exploring simultaneously, having ambidexterity capacity [1]. Ambidexterity poses a double challenge: learn to innovate and innovate how to learn. This second property, the double loop learning ability [2], is most salient in dynamic and complex contexts, where it is hard to detect and make sense of variations through the frames and the basic assumptions of the current culture [3]. FLYINGANEXPERIENCE laboratory allows the recording of quantitative performance while it facilitates observations of many behaviors relevant to teamwork, which can be analyzed through rigorous research methods. Some simulation ingredients are: complex systems; processes and procedures; time; social pressures. It provides many variables that the team has to manage simultaneously

in order to achieve the final goal. To recreate a learning environment that simulates complexity and engages people in an immersive experience, in 2006 one of the authors introduced the flight simulation training for non-aviation people in Executive Education and Universities courses. The FLYINGANEXPERIENCE simulation lab offers a broad set of exercises designed according to goals-based scenario learning [4]. Its learning tools are flexible and can be easily adapted to meet the learning requirements through a rigorous design process. By January 2021, 172 teams have attended FLYINGANEXPERIENCE, either in the static or dynamic simulator.

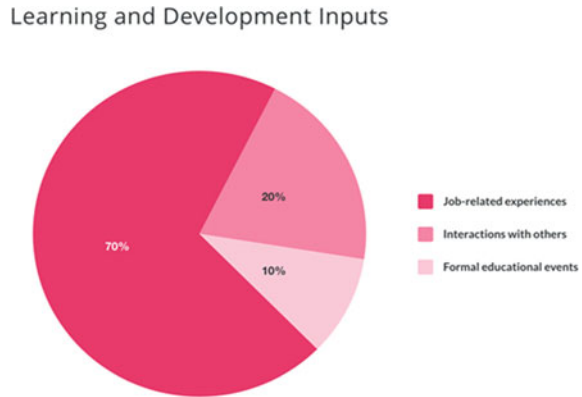
3 Gamification and Simulations

Simulations are one of the pillars on which modern gamification theories are based, as they share elements with multimedia games that are essential and foundational for every gamification solution. Two essential concepts underlying the usefulness of simulations are those of “safe circle” and “fail to learn”. Simulations are first and foremost a safe context [5], where the user can make mistakes without worrying about the cost of the error (which reality, on the other hand, often entails). Therefore, they represent a safe circle, a space where one can test different approaches and behaviors to verify the consequences or a place where one can train to improve one’s skills. In games, people learn from mistakes. Moreover, they have the opportunity to continue making mistakes until they understand the winning tactic or until they have gained a specific skill necessary to win. Gamification allows people to take up these concepts, and it allows creating experiences through which people can iteratively learn and even fail to learn. For this very reason, a simulation experience does not require a learning phase to be used because it is based on an interactive dynamic that incentivizes the user (through competition and rewarding) to get involved [6]. Therefore, one of the most significant advantages given by simulations is to allow people to experiment and learn by doing. Experiments and trials are critical when we consider that most things we learn are from the field, through experience and through learning by doing and interaction (Fig. 1).

The “70:20:10” model developed by some leadership researchers [7, 8] is used within the learning and development sector to describe the sources of successful managers’ educational experiences. This framework comes from a series of studies aimed at understanding how people learn in the workplace. The results showed that learning is divided as follows [8]:

- 70% experiential learning: people learn through experience, practice, and challenges, therefore, through doing;
- 20% social learning: people learn through interaction and collaboration with others, e.g., through mentoring and training sessions;
- 10% formal learning: represents what people learn through study (courses or studies of materials, formal training interventions, and structured courses).

Fig. 1 The weight of experiences, interactions and formal educational events



In a company, simulations are used to recreate situations that workers may encounter during their work, giving them the ability to try and retry the management and resolution of situations [9], even complex, to increase their productivity and reduce the risks and costs related to the error. The use of the simulations brings benefits to the employee, the company, and the end customer:

- the employees have the chance to develop their skills, being able to test different strategies without being concerned about the consequences of their actions;
- the company reduces the risks of costs (or lost revenues) deriving from the employees' failures;
- the customer receives higher quality services and improved skills.

Simulations are capable of quickly enriching individual experiences and skills. Thanks to simulations, people will be able to train and increase their skills and experience in managing emergencies [10]. A key feature is that simulations can create whatever scenario is necessary, thus allowing to adapt the context to different needs. The use of simulations allows people to practice in a safe context where the error is no longer a criticality but the basis of learning. These particular platforms allow people to practice with standard protocols in the most varied and complicated contexts to help them face even the most critical scenarios. Participants will be able to increase their skills by experimenting every day with different scenarios and situations [11], relieving and trying the management of an emergency again, trying to do better and better. Simulations also require less effort on teachers, as they offer student ready-made practice to reproduce complex emergencies. An additional advantage is found in the cost–benefit ratio. The use of virtual reality compared to live training [12] has reduced costs as it needs fewer teachers, and there is no need to create specific settings for the exercise. Besides, not everything can be reproduced live. In a virtual environment, it is possible to train participants in scenarios like a fire in a building with many people inside who need to be rescued or a car accident involving multiple cars, or even a plane crash. What is more important, therefore, is always the amount of experience that participants can gather: they will be able to live complex and

difficult-to-manage experiences that rarely happen. They will have the opportunity to develop the skills to deal with the situation at their best, avoiding putting people at risk and saving lives [13]. Simulations allow for increasing people's skills through the recreation of various environments and scenarios that ultimately allow teaching how to manage situations that are both ordinary and extraordinary.

4 Complexity in FLYINGANEXPERIENCE™

Complexity has various definitions. Organizations, firms, and markets have been described as complex systems from diverse research streams [14–16]. Integrating and simplifying them, we focused on 5 factors to generate organizational complexity in a learning experience:

Complex Reasoning We focused primarily on two sides of managing complexity: making sense of contexts and making decisions. Both of them require complex reasoning skills [16, 17]. The literature considers the Complex Adaptive System a robust perspective to model the world complexity and foster complex reasoning [16–18]. The learning experience challenges some pillars of complex system theory: non-linear behavior, loop effects, delays, dependencies, interactions, and emerging properties [16].

Complex Technology Many research streams consider technology a normal part of human (working) life for years to come, especially in managing complex systems [19]. Though technology is its primary tool to generate complex scenarios, FLYINGANEXPERIENCE integrates technology challenges into the learning experience. People must understand and operate some airplane's systems quickly, and they have to use technology for sharing information and strengthening the sense- and decision-making processes. Exercises allow facilitators to debrief human-machine integration behaviors of the teams, especially their ability to exploit *augmentation* systems when required by learning aims.

5 FLYINGANEXPERIENCE™ Learning Goals and Processes

Lab Learning Goals The lab structure provides a flexible complex scenario fruitful for various learning or research goals. The most requested scenarios are used to train collaborative decision-making, learning group dynamics, experiencing loop effects, exploring complex adaptive systems, experiencing leadership in an ambiguous and uncertain context, understanding routines evolution, experiencing ambidexterity in a complex system, understanding the human-machine integration, managing

augmented and intelligent systems. The lab is also beneficial for conducting experiments on the above subjects and its technology offers the opportunity to record behaviors and collect performance data at the same time. That is especially fruitful for the quasi-research approach.

A Non-factual Analogy The learning experience is non-factual, as attendees are non-pilots. Unlike business games and business simulations, learners cannot directly apply their technical knowledge and competencies that they have developed at work. We argue that non-factual simulations have three strengths:

1. Performance anxiety decreases. As learners are requested to carry out tasks that are from their professional capabilities, they do not feel they are being evaluated about their technical knowledge and competencies.
2. Participants have to deal with an unknown scenario, sense new situations and contexts they are not familiar with, and learn fast by adapting behaviors and competencies through reflections about their frames. Moreover, the simulation forces them to face uncertainty and ambiguity, which are common challenges nowadays [20].
3. This learning environment is likely to be more psychologically safe. As routines and roles are broken, competencies are reset, learners feel more confident to be themselves and explore learning objectives through a trial-and-error approach. Psychological safety and learning from errors strategy are two important antecedents of team learning [21].

The features listed above are pre-requisites for inductive learning and also are essential components for any constructivist learning process founded on a so-called *goal based scenario* [4, 17, 22]. In most role-playing, especially in learning activities, the players receive behavioral prescriptions [23]. In the FLYINGANEXPERIENCE lab, the learners need to execute tasks to fly the plane without any prescription about their behaviors. Roles are task-based. E.g., the navigation team has mainly to set the autopilot for following the correct route. Consequently, people receive only role goals e.g., navigate the plane and the necessary information set. Nobody is either asked to follow specific behaviors or receive suggestions about that.

A “Team Pilot” FLYINGANEXPERIENCE is a team learning laboratory. This directly follows from its primary aim: immersing attendees in a complex system experience. The lab is both adopted for running experiments in team behavior and teaching. The team organization and roles vary to fit the actual laboratory goals. Most of the time, the whole team is split into four sub-teams, which have to fulfill four roles:

1. Navigation, making sense of instruction and position information, this team presides over the route, called *lateral navigation*;
2. Aircraft Management ensure the right speed and manage altitude, also called *vertical navigation*;
3. Radio communicate with Air Traffic Controller and their airline *dispatcher*;

4. Command is the “captain in command” of the flight.

Surprise is the most common reaction participants show after they have realized that they have to fly a plane in a team. As the experience proceeds, learners understand this organization’s rationale: set the scenario to stress the learning objectives.

6 Learning Through Reflection upon Evidence

The experience is grounded in the constructivist theory of learning [24], follows an inductive learning agenda [4, 25], and it develops through three stages:

1. Briefing. set the learning goals and rules, deliver task instruction, assign roles, explain how to manage the flight simulator autopilot and how to read the actual flight parameters on the cockpit instruments.
2. Experience, where people fly the plane and facilitators collect pieces of evidence about their behaviors.

De-briefing, which is the pillar of this approach, activates learning and cognitive modifications through systematic reflections [4, 25–27]. We emphasize that simulation debriefing, starting from performance gaps, needs to integrate two processes: the reflections about behaviors and the exploration of people frames [22, 27, 28]. That constitutes the difference between a simple game and a gamification learning experience. Behaviors are rooted in frames, so transforming behaviors is of the utmost importance to explore people’s cognitive frames. FLYINGANEXPERIENCE adopts the approach sometimes named as *good judgment*, based on the double loop learning approach [27, 29] and proposed by Rudolph et al. [22]. It is also grounded in Evidence-based Training methodology [30].

7 FLYINGANEXPERIENCE: The Lab Technology and Stages

The Experience Goals and Stages Just after 2 min of flight time a radio message breaks the cockpit noise: “Flight 1, this is Rome Departures. You are cleared to 10.000 feet, heading to MEDAL waypoint, speed 250 knots. You will call us 1 min before MEDAL. Please, read the instructions back”. The “ATC sub-team”, which manages communication between the “virtual plane” and the virtual ATC (Air Traffic Control), has to share the information with the other sub-teams. All people must strive to integrate exploration and execution of their tasks until landing. The time in “Flying an Experience”, as in business, does not wait.

The attendees’ mission is to fly the 737-400 from Rome Fiumicino to Milan Linate airport, following a point-to-point route and managing the autopilot to keep the flight

Table 1 Tasks, goals and limits of the system

Complexity and learning goals in the FLYINGANEXPERIENCE™
High variance tasks
High interdependent tasks
Non-linear behaviors
Simultaneous exploration/exploitation
Internal (system behavior) and external (compliance with regulatory rules) boundaries and limits

path and control variables within specific ranges. The participants have to monitor speed, vertical speed, flight path, and other variables of Table 1, and eventually set the final approach to land using the auto-landing system. According to specific learning goals, attendees have to manage unexpected events that challenge their cognitive and cooperation skills. The descriptions and the pictures try to give the reader the sense of engagement, immersion, and urgency that FLYINGANEXPERIENCE attendees feel during the laboratory time.

- That is attained through four stages:
- Stage 1 (1 h) in four steps:
 - Pre-flight briefing: the facilitators instruct the participants on how to manage the flight. The facilitator follows the “pre-flight checklist”;
 - Sub-team split;
 - Flight plan and instruction explanation (general);
 - Flight plan and instruction explanation (specific for sub-teams):

Aircraft Management: Speed, Vertical Speed, Flaps, (Spoiler);
Navigation (lateral): Heading, Course, Radio VOR frequency;
Air Traffic Control: Radio Com Frequency, ATC communication;
Command: Speed, (Vertical Speed), Flaps, Heading, Course, Radio VOR, Radio Com, (navigation mode, communication content).
 - Stage 2 (30') in three steps:
 - Pre-flight sub-teams briefing: sub-teams have 15 min to talk about task, to discuss roles, to organize cooperation, to exchange information, to define five common questions to ask the facilitators;
 - Test (5' of flight);
 - Q&A (max 5).
 - Stage 3 (around 1 h) Flight Simulation;
 - Stage 4 (around 3 h) Debriefing (Fig. 2 synthesizes the lab stages).

Evidence-Based Reflections The simulation allows the collection of very diverse types of information. The detailed data allows for observing, understanding, and

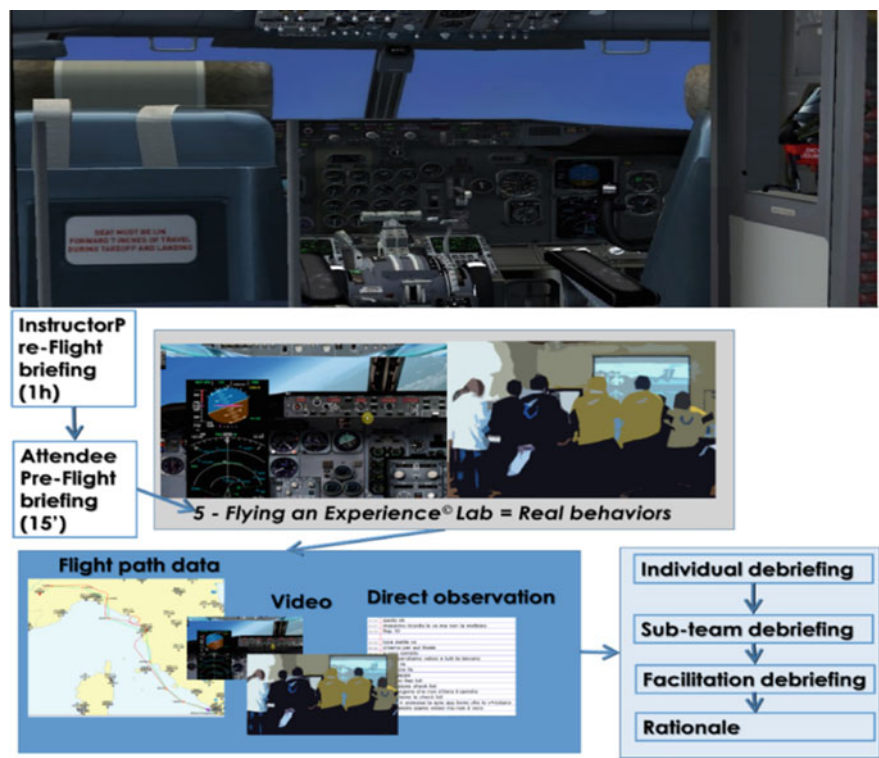


Fig. 2 The experience's stages

explaining decision-making in connection with specific events; the videos show the group dynamics and the individual behaviors, also expressed through non-verbal language (Fig. 3).

Table 2 lists the variables the participants must set to manage the flight and the available measurements. The human mind prefers *direct* measures, like the car indicator that directly shows the car speed.

An Extensive Set of Performance and Behavioral Data Past data interpretation benefits teachers, researchers, and learners. Because the lab has been adopted for diverse aims and hosted attendees with a wide range of knowledge, professional experience, disciplinary background and cultures, the behavioral and performance database is rich and allows for exciting comparison. Table 3 shows the distribution of sessions for diverse curricula.

Every session has its performing and behavioral data set with teachers' notes. Including the mean time spent for debriefings, teachers' lab time is more than 600 h.

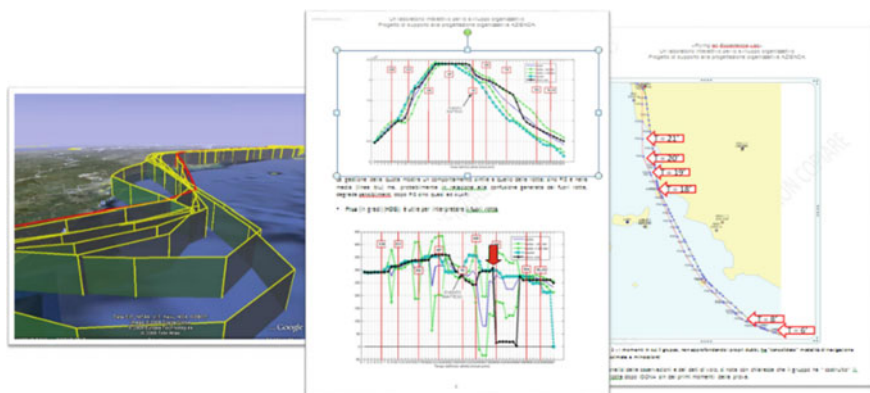


Fig. 3 Diverse data allow diverse sights of performance

8 The Diverse Simulation Environments and the Technology

Combining various simulators' types and characteristics makes it possible to create diverse simulation environments in line with learning aims, organizational cultures, and organizational constraints like the budget. Four variables should be kept into consideration and can set in various ways:

1. **Static versus dynamic flight simulator.** The first ranges from a powerful PC with simulation software, two screens, and simple autopilot interfaces to mockups reproducing a realistic cockpit with all controls, interfaces, and instruments. The static simulators have realistic flight models but do not simulate airplane movements. Instead, the dynamic flight simulator perfectly reproduces the real plane, including motions, so the virtual crew feels body acceleration as if they were flying the actual plane (see Fig. 4).
2. **Online versus in person.** On-line has been proven effective by performance data, is appreciated by learners, and the vast majority of exercises are available. The only unrelaxable limit is that a *full dynamic flight simulator* (see below) is not available.
3. **On-site versus simulation center.** Static flight simulators differ from each other mainly for interface fidelity. Most simulation centers have a rich offering of high-fidelity static simulators, with a realistic cockpit, very close to the real aircraft.
4. **Exercise complexity** ranges from simply following a flight path to managing complex situations, dealing with flight augmentation, and managing the ultimate autopilot technology enriched with trend indicators. The diverse types of simulators support a set of possible research questions and learning objectives.

The lab technology and organization provide clear performance evidence, quantitative, like flight path and parameters, and qualitative, like behaviors' recordings.

Table 2 Control variables and measurements available to attendees

Variable	Variable code	Variable control mode		Variable measurement type		Value available in flight instruction	
		Direct setting	Indirect	Direct	Indirect	As value to set in order to arrive at the next waypoint	As target value at the waypoint
Speed (Knots)	S	X		X		X	
Altitude (Ft.)	A	X		X			X
Vertical speed (Ft./min)	VS	X		X	X (by EnergyCircle EC)	$V_s = f(S, A, OCD, TWE)$	
Heading (Degree)	H	X		X		X	
Off course distance (Nautical miles)	OCD		X		X		
Track waypoint error (Nautical miles)	TWE		X		X		
Fuel (Gallon)	F		X		X		
Fuel consumption (Gallon/h)	FH		X		X		
Communication radio frequency	FRC	X (System state)		X (Actual setting)		X	
Navigation radio frequency	FRV	X (System state)		X (Actual setting)		X	
Course (Degree)	COU	X (System state)		X (Actual setting)		X	

Table 3 Lab sessions classified by curricula

	Lab sessions	Flight time
Degree curricula	13	11:46:00
Business school master	21	21:24:00
Business school open	24	25:43:00
Business school custom	84	82:28:00
Research/Consulting	32	32:12:00
Total	174	173:33:00

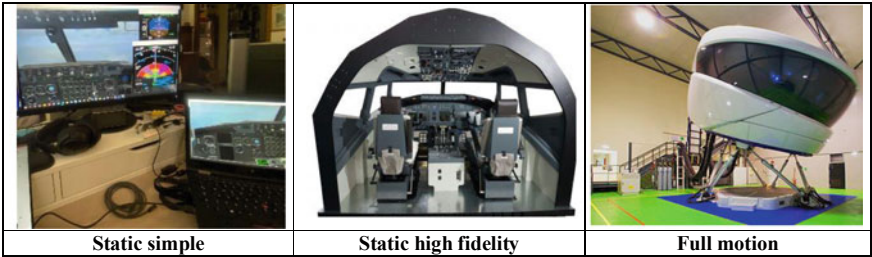


Fig. 4 Static and full-motion flight simulators

So, either the teacher can close the learning loop through rigorous feedback, or the researchers can collect accurate data. The lab integrates various technical tools to create a virtual environment, record performance data, and collect information about behaviors. Table 4 lists the mandatory ones and highlights their purposes. A third column suggests how to use them in the three primary simulation environments: in-person static, in-person full-flight, and online. We included only the mandatory tools, as a detailed description of the diverse environment’s peculiarities would require an extensive discussion, which is out of scope. We do not indicate any brands, as it would call for a comparative performance discussion, out of our scope. Static flight simulators all require powerful personal computers that must benefit from powerful GPUs.

The online experience is remarkable. Since the beginning of the Covid-19 pandemic, we have taught twenty courses with FLYINGANEXPERIENCE lab, sometimes along with other simulations. Teams showed performances close to the in-person experience. To allow team cooperation on line, we split every team into two classroom meetings. The simulator transmits the virtual cockpit for the command team and the co-pilot team. Figure 5 shows the teacher’s workstation. The 1 and 2 are the screens sent through two online channels to attendees (Command and Co-pilot view). The screen 3 shows the actual flight path and performance parameters. The desktop 4 is the simulator control panel: hidden to attendees, it allows to manage scenario events, see all flight data and record flight performance. Panels 5 and 7 serve autopilot, virtual radio, and system switches. Desktop 7 is the whiteboard (touch-screen) and hosts the conference (reduced in the picture for privacy), allowing the

Table 4 The lab equipment

Main purpose	Tools description and notes	Suggested environment
Virtual environment	Flight simulator—static	In-person, static online
	Flight simulator’s high-fidelity add-on—airplanes	In-person, static online
	Flight simulator’s high fidelity add-on—landscape scenarios	In-person, static online
	Flight simulator’s high fidelity add-on—air traffic control system and ai traffic generator	In-person, static online
	Flight simulator’s control panels	In-person, static
	Flight simulator’s flight controls	In-person, static
	Cockpit mock-up	Simulation center
	Flight simulator—dynamic	Simulation center
	HD monitors	In-person, static
	Virtual ATC transceivers	In-person, static
	Multichannel video conferences systems	Online
Performance data recording	Virtual black-box	In-Person, static online
	Flight path recorder and analyzer	In-person, static online
	Integrated performance analyzer	In-person, static online
Behavior’s recording	HD cameras—wide angle and 360°	In-person, static full flight
	Digital audio and notes SW with timestamp	ALL
	Virtual voice recorder	In-person, static



Fig. 5 The teacher’s workstation

Fig. 6 Online command classroom screen



Fig. 7 Online co-pilot classroom screen



teacher to observe the team's cooperation. It also has the timestamp notes software to synchronize observation with video.

Figures 6 and 7 show the participants' screens.

It is important to note that the two teams have different flight data, as half of them have the Command sight, while the others have the co-pilot sight, so they must cooperate to build a unitary sense of what is going on and make effective decisions. A third communication channel allows the command and co-pilot teams to cooperate with each other. FLYINGANEXPERIENCE is grounded in deep knowledge of three fields. Andrea Montefusco, who designed the FLYINGANEXPERIENCE lab, is a pilot, a flight safety officer, and a teacher and researcher in organizational behavior. He has operated simulators for twenty-two years and knows flight simulator's technology in depth. These three competencies are paramount for a teaching or research team to set engaging scenarios and manage the simulation fruitfully. This is especially important within the design of experimental scenarios that enable the study of team behaviors and human-machine integration in complex environments. Designing and operating such a lab is a far more advanced experience than buying a static or renting a dynamic flight simulator. If these competencies are unbalanced, scenarios are likely to be trivial, mainly based on common sense and poor outcomes. To design scenarios either for non-pilot achieving learning goals or for researchers collecting relevant data goes far beyond implementing aviators' storytelling. The purpose has always been pursuing authentic learning through scenarios grounded in scientific knowledge and rigorous feedback.

9 Conclusions

Work environments change, as do employees, through innovation. In particular, virtual reality and the savvy use of simulations often make the difference within a business context. The technology described above provides a simulated reality, a three-dimensional environment built on the computer that can be accessed through computer devices that project those who use them into a realistic scenario. Virtual reality aims to make those who benefit from it participate more, bringing them into an abstract dimension that is far more advanced than one in which the content is merely and passively enjoyed. Using virtual reality technologies in human resource management, and therefore in decision-making, leadership supporting systems, and recruiting phases, interviews and hiring processes, means to stimulate greater involvement and deeper understanding of all the actors involved. Simulations can be applied during conferences, meetings, presentations, performance reviews, brainstorming activities and team building/social events to promote communication and an orientation towards innovation. The most relevant area of application of virtual reality and simulations is learning, in which the results are tangible and prolonged over time. Companies that believe in innovation by exploiting virtual reality technology in human resources can create their equipment or use external companies. The technological means can ensure a revolutionary approach in both the recruitment and training of human resources.

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Implementing Virtual Reality in K-12 Classrooms: Lessons Learned from Early Adopters



Espen Stranger-Johannessen and Siw Olsen Fjørtoft

Abstract There is very little research on K-12 education outside designated research studies on how teachers use VR for learning. This study addresses gap in research by asking: How did teachers start with VR and how do they use it? We interviewed nine teachers from different primary, lower secondary, and upper secondary schools—who had one to five years of experience with VR—among the first to include VR in their teaching, making them early adopters. All schools used HMDs, but the number, brand, and model varied. VR was used to teach various subjects and topics, but mostly for a limited part of the lesson. Due to the limited number of HMDs, the teaching was often organized as stations, where the students took turns using the HMDs and working on other tasks related to the theme. This requires good planning from the teacher and a certain degree of self-regulation among the students, as the teacher’s eyes and ears are on the student(s) wearing the HMDs. The early adoption of VR sometimes encountered challenges related to technical (e.g., firewalls) or ethical issues (e.g., privacy), but some argued that for innovation to happen, it was necessary to “act first and ask later”.

Keywords Virtual reality · K-12 education · Technology-enhanced learning · Early adopters

1 Introduction

Virtual reality (VR) is frequently heralded as a new technology with great potential in education—much like TV, desktop computers, and cell phones have been before [16]. Like other technological innovations with a potential for student learning, VR is both praised and questioned, and is receiving much attention both in the media and among researchers. Although most research and application (e.g., industry and

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medical training) appears to focus on post-secondary education and adults [19], there is a growing body of research on the use of VR in K-12 education, and a burgeoning interest among educators to introduce VR in schools [9, 11, 14]. There are also indications of better learning effects in K-12 than post-secondary education [19]. However, there is very little research on K-12 education outside designated research studies on topics such as students' experiences or learning outcomes or how teachers use VR for learning at their own initiative. This paper addresses the latter, specifically by posing the research question: How did teachers start with VR and how do they use it? This study provides valuable insight for anyone interested in the background of ongoing practices among early adopters—some of the first to adopt a new technology, innovation, or product, and tend to serve as opinion leaders [12].

2 Research on VR in Schools

Commercially available head-mounted displays (HMDs) are only a few years old. Google Cardboard was launched in 2014 and has been used in several research studies (e.g., [17]). More than two million school children were estimated to have tried Google Expeditions, which points both to the interest and accessibility of this kind of VR [3]. In 2016 HMDs with more advanced graphics, such as Oculus Rift and HTC Vive were introduced, but they rely on cable connection to a powerful computer. Two years later, stand-alone HMDs became available, and by 2019 some of these also offer six degrees of freedom, allowing for three-dimensional movement of both the body and the head [14]. Most current research rely on more advanced forms of VR, including all schools in this study.

There is no single agreed-upon definition of VR [5], but a key distinction is between immersive (such as HMDs) and non-immersive technology (such as desktop virtual reality). Non-immersive VR in K-12 education has been investigated for a long time [9], but these findings cannot readily be generalized to also apply for immersive VR, and Wu et al. [19] found a higher learning effect for immersive VR. Kwon [7] concluded that authentic VR—such as using handheld controllers rather than joysticks, gave students a better sense of experientiality and higher score on analysing, evaluating, and creating questions. Taken together, these studies point to the benefits of authenticity for learning.

Immersion is widely seen as a central affordance of VR that contributes to a sense of presence, engagement, motivation, and improved learning perceptions [2, 5]. The effect on learning is less clear, as the research reports different findings. Makransky et al. [8] claimed that HMDs had psychological effects, such motivation, but not on learning performance. In a review of research on HMDs, Jensen and Konradsen [6] found a higher effect on skills training than on knowledge learning. Wu et al.'s [19] review of quasi-experimental and RTCs concluded that only one third of the studies had a positive effect on learning.

VR in education raises a number of ethical issues, including age. As children develop, so do they cognitive and psychological abilities as well as their physiology,

all of which are central to how VR is perceived. For instance, primary school children are more susceptible to audio and visual stimuli [1], and children are not always able to separate VR experiences from real life [13]. Cybersickness can be a challenge in VR. Reports vary on the extent of cybersickness [6], but some approaches to reduce cybersickness have been identified [18].

Privacy is also a concern, since some VR systems require login and collect biometric data, such as hand movements. This is particularly relevant in the context of the rapid development of AI, which might combine biometric and other data for personalized learning [14]. Southgate et al. [15] raised the question of the learning outcomes of students who did not use VR, including students who functioned as “spotters” to guide students who moved around wearing HMD, and noted that some girls were reluctant to try VR, and urged researchers and practitioners to consider possible effects of gender.

Most of the recent studies on the use of VR in K-12 education are intervention studies where researchers introduce VR to schools to measure learning or other effects. Fransson et al. [4] interviewed teachers about their thoughts on using VR after a workshop, and identified organisational, institutional, contextual and practical challenges and opportunities. Few studies are based on researcher–teacher collaboration over time to provide an account of both pedagogical and technical/practical challenges and solutions (e.g., [11, 14]). While these studies as well as those with research-driven experimental approaches point to the potential for the use of VR in schools, they also clearly demonstrate a range of challenges that need to be overcome, and beg the question of how teachers will make use of VR without the technical and theory-based pedagogical support from researchers. This is the case of almost all the teachers in this study—the early adopters of a new technology on the frontier of research.

3 Methods

For the purpose of data collection, we attempted to identify all schools in Norway where VR was used by searching online, as early adopters tend to influence others by sharing experiences and reviews in blogs and social media. We also used professional networks, as well as snowball sampling (asking interviewees about other schools that used VR). This search yielded 18 different schools, all of whom were contacted. Some replied that they did not have enough experience to meaningfully contribute to the research, but nine agreed to be interviewed. All nine interviewees had personal experience of using VR as a teacher (including one as a substitute teacher), but some had since changed jobs and worked with leadership or support at the time of the interview.

The interviews were done via video conferencing due to the Covid-19 pandemic, and lasted between 17 and 53 min, with an average of 37 min (the shortest interview experienced technical difficulties). The audio recordings were transcribed

verbatim and the teachers’ responses were analysed using thematic analysis to develop categories of salient issues [10].

4 Findings

4.1 The Schools, the Teachers and the Equipment

The schools where VR was used, were geographically spread and represent the whole school system: two primary schools, four lower secondary schools, three upper secondary schools (see Table 1). In three cases the equipment was shared between the schools, which means that the interviewees represent 20 schools in total. In the primary schools, VR was used in the upper grades, which roughly correspond with the lowest recommended age for using VR (usually 12 or 13 years).

All schools used HMDs, but the number, brand and model varied, from five at one school up to a full set for the whole class (approx. 30) at one school and a municipality (where the HMDs circulated between schools), with the majority somewhere in between. In most cases, the HMDs were used in groups, so some students used VR while the others did other activities. Five of the schools had more than one brand/make, such as both Oculus Quest and HTC Vive, which gave them an opportunity to use different (and usually more advanced) software, but also required more effort to learn and maintain the equipment. Because the technology improves so

Table 1 Overview of teachers and schools

ID	Gender	School	VR equipment
1	Male	Lower secondary school	10 Oculus Quest, 10 Oculus Quest 2
2	Female	Upper secondary school	5 HTC Vive, 5 Oculus Quest, 2 360° cameras
3	Male	Upper secondary school	17 Oculus Go, 1 Oculus Rift, 1 360° camera
4	Male	5 primary, 2 lower secondary schools	Approx. 30 Samsung Gear VR
5	Female	Primary school	10 Oculus Quest
6	Male	Lower secondary school	Approx. 30 Oculus Quest, 5 HTC Vive (can upgrade to Oculus Rift through 5 computers)
7	Male	Lower secondary school	5 Oculus Quest, 7 HTC Vive, 1 HMD for PlayStation
8	Male	Upper secondary school	5 Oculus Quest connected to gaming computers
9	Male	4 Lower secondary schools	10 HMDs for the four secondary schools in the municipality

Table 2 Overview of teachers and schools

ID	Subjects/topics with VR	Initiative	VR use
1	English, mathematics, religion and ethics, social studies	A teacher	About 1 year
2	Anatomy, Norwegian, nursing	A teacher	4–5 years
3	Anatomy, architecture, chemistry, electrical installation, English, history, Norwegian, religion and ethics	A teacher	2 years
4	Mathematics	A VR company	2–3 years
5	Interdisciplinary, religion	A teacher	2–3 years
6	English, Norwegian, physical education, religion and ethics, social studies	Two teachers	3–4 years
7	Mathematics, music, physical education, religion and ethics, science, Social science	A teacher	3–4 years
8	Anatomy, history	The principal	2–3 years
9	Physical education, Science	A regional research project	About 1 year

quickly and prices go down, some schools purchased new, more advanced hardware over time (Table 2).

VR was used in various subjects and topics (as well as interdisciplinary), including foreign language (English), language arts (Norwegian), humanities, social studies, pure and applied science, nursing, first aid, music, and physical education. In other words, most subjects were covered, except art and crafts, but one school had plans for that subject. All “core subjects” were taught using VR by at least three schools, while the subjects/topics mentioned by just one teacher, are electives in upper secondary schools which not all students study (architecture, chemistry, electrical installation, and nursing) or not usually taught in upper secondary school (music).

Out of the nine teachers, seven were men and only two women. While this difference may not be significantly different statistically speaking, it seems to reflect assumptions that men are more likely to be early adopters of new technology.

4.2 Initiative and Implementation

In six of the nine schools, the initiative to start using VR in the school came from the teacher. Three of these teachers had been to an annual fair—two technology fairs in London and one local VR fair, which the principal had asked this teacher to attend. At the other three schools, the initiative came from the principal, a VR company that entered an agreement with the municipality, and lastly, a teacher started experimenting with VR because the school was involved in a research project on VR. Personal initiative, mostly from teachers, but also from school leadership, is

clearly the main entry point for starting with VR. The majority of these teachers expressed interest in and often experience with technology, and a few have or have had responsibility for IT as part of their work alongside teaching.

Seven out of nine schools did not have one HMD for each student in a class, which meant that not all students in a class could use VR simultaneously. Therefore it was most common to organize the learning sessions as stations, where one group of students used VR while the others worked with something else related to the same topic. Several schools had also organized technology/VR labs, which meant that they had to move to a specific room to use the equipment. This required planning in advance.

The following are examples from use of VR in different subjects/topics and with different approaches.

Anatomy. Two of the teachers used VR in teaching anatomy. This application is available in the game portal, for example in HTC Vive. Due to limited access to equipment, the class is divided into two or three, depending on the size. One teacher said that a typical anatomy session starts with a theoretical review before groups of 5–7 students enter the VR lab, while the rest work in the classroom with different tasks. Those who are in the VR lab get about 15–20 min, depending on the specific topic. Finally, there is a whole class summary, sometimes with assignments. The advantage of VR in anatomy is that students get to see internal organs, the heart pumping blood, size and body functions in a completely different way than what is possible by watching a movie or using a dummy.

Social studies or interdisciplinary. Several teachers have used the game *Ghost Giant*, where you are a friendly giant who helps a village. For example, students can write their own stories of the people they meet in the game, such as their background and more. The students can also analyse the literary tools in the game, including the use of music. One of the teachers who used this application said that the students could play the game for several hours straight.

Religion and ethics. Two teachers related how they let the students try a VR application that was not about content, but they linked the experience itself to a theme in the subject religion and ethics. The plan was to let one student at a time try a virtual roller coaster while the other students watched (for example, in 2D on a large screen in the classroom). For those who had not yet tried the application, it was fun to see how the students with the HMDs reacted with movements and sounds and how they tried to tell the other students how incredible this experience was—but without being able to imagine it. When all the students had tried a minute or two of the roller coaster, the teachers used this experience as a starting point in discussing the inside/outside perspective of a religion. It is difficult for those who are not part of a denomination or religion to understand the genuine enthusiasm of the believers who view it from the inside.

Other use. Other VR games that they mentioned were geometry app in mathematics, *Beat Saber* for rhythm and movement in physical education and music, and various

social awareness games in social studies and religion and ethics. In addition, one used a 360° first aid application on tablets.

4.3 Challenges

Much of VR is novel technology and relies on an Internet connection. One teacher described HTC Vive as technically complicated, and several pointed out that teachers need to spend time learning both hardware and software before introducing it to the students. The schools' firewalls and networks caused technical challenges, and in one case the use of VR had violated the municipality's security regulations. In spite of this, the novelty of the technology makes it hard to navigate technical, practical and regulatory bottlenecks. Some teachers argued that for innovation to happen, it is sometimes necessary to "act first and ask later", i.e., to buy the VR equipment first and manage issues with Internet connection, firewalls and security measures later. Cybersickness, however, was not raised as a concern by any of the teachers.

5 Discussion

5.1 Early Adopters and Development of Practice

Getting started with VR requires commitment and investment from the school, and arguably more importantly, someone to take initiative to acquire hardware, software, negotiate technical, legal, infrastructural challenges, and develop a VR pedagogy for one's own subjects. Often the teacher who takes the lead, the early adopter, ends up assisting colleagues with several of these issues as well. No teachers mentioned resistance from the leadership. Instead, they report support ranging from a "hands-off" approach, to active, financial support, including the school principal effectively being the initiator for using VR at the school. In only two out of nine schools did the initiative come from outside (a company and a research project).

Funding is central to implementing VR. Although the schools had somewhat different ways of securing funding, it seems that funding was ultimately about priorities and willingness on behalf of the school and/or school owner (municipality or county). The VR hardware collection expanded over time in several schools, pointing to a development in the use of VR technology. Some teachers pointed to the need to explore and learn before going to scale, while others mentioned the benefits of more advanced technology, such as 360° cameras, that allowed for production of their own content. When asked about recommendations for other teachers or schools who are thinking about starting with VR, the most common response was the importance of sense of commitment on part of the teacher and the need to explore.

5.2 Curriculum, Content, or Motivation as Drivers?

VR was used in a range of subjects, except in the two cases with an external collaborator. To the extent there is a limiting factor to the subjects where VR is used, it is the number of teachers who use it. If only one teacher uses VR, the number of subjects with VR is naturally limited. Most of the use was based on specific content or topics that may be related to the curriculum, e.g., anatomy, or the solar system. VR applications are rarely developed for or adapted to a specific curriculum [6, 7]. Hence, it requires both experience and confidence as a teacher to see the benefit of implementing VR experiences as part of the subject. As one stated, “The fact that we use VR is merely a feature to spur students’ engagement in the subject.”

With one exception, all the teachers used VR for a limited part of the lesson. Some emphasized that they never played for entertainment. As one of them said, “VR is a tool, and the tool is only as good as the lesson plan you make.” Limitations due to equipment and the number of students who can receive guidance from the teacher, require careful planning and organization of the session. The teacher’s focus is on the student(s) wearing HMDs. The other students must be given tasks that keep them busy, which requires a certain degree of self-regulation and maturity.

Some teachers mentioned overcrowded curriculum as a potential inhibitor of use (cf. Southgate et al. [15]). This was possibly a reason why some of their colleagues were sceptical about using applications that cannot be directly linked to the curriculum. The use of HMDs and other technical equipment requires some time for preparation. In many cases, they had to move to another location at the school. Therefore, it is crucial to ensure that VR provides added value, which requires both will, ability, and curiosity from the teacher—typical features of an early adopter [12].

5.3 Technical and Ethical Challenges

Technical challenges can give rise to ethical challenges, such as secure networks and issues of privacy, which teachers are not always aware of (cp. Southgate et al. [15]). Depending on the equipment, some devices require a personal account to buy and download games, e.g., the second-generation Oculus Quest requires login with Facebook. It would be a violation of GDPR to use learning tools where teachers or students have to use their private profiles.

Other ethical challenges can be related to age and reactions that occur when using HMDs. In the research literature there are few studies on primary schools, and VR is not recommended for young children due to their limited ability to cognitively regulate and process the VR experiences [15].

6 Conclusion

This study has shown how different Norwegian schools in K-12 education have implemented VR. The schools represented were well equipped with digital devices, and all had access to quality HMDs, but only one had a complete set for the whole class. We interviewed nine early adopters who provided insight into how the acquisition came about and how they use VR within various subjects. We found that VR is used as a tool to engage students in a specific theme or content, mostly as a limited part of the lesson. The students are usually divided into groups in which the ones that are not wearing the HMDs are working on other tasks related to the theme. This requires good planning from the teacher and a certain degree of self-regulation among the students, as the teacher's eyes and ears are on the student(s) wearing the HMDs. The teacher should be conscious about whether and how VR can provide added value within a subject. The early adoption of VR sometimes encountered challenges related to technical (e.g., firewalls) or ethical issues (e.g., privacy), but some argued that for innovation to happen, it was necessary to "act first and ask later". There is need for more research on didactical approaches and the pedagogical value of using VR in K-12 classrooms.

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Software Testing Education Experiences Using Collaborative Platforms



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Abstract Software testing is an important phase in the software development process. Testing skills acquirement in higher education is hindered by factors related to academic curricula, lack of adequate tools or knowledge. The paper presents the software testing course design, where the performed activities are meant to help participants gaining hard and soft skills in testing. The participants' perception of the importance of different learning activities was investigated. The results indicate that the software industry collaboration should aim to add value to the teachers' endeavors. The consequences of communication and collaborative tools integration that was studied on several groups of students suggest grades improvement and valuable soft skills gaining.

Keywords Software testing education · Technology in the classroom · Collaborative learning · Testing tools

1 Introduction

Testing is an important step in the software development process that allows obtaining relevant information about the quality of the tested work product [1]. In Computer Science (CS) undergraduate programs, actual software development activities are emphasized over testing activities. Therefore, designing and teaching verification and validation activities are challenges for educators.

The Software Systems Verification and Validation (VVSS) course [2] at Babeș-Bolyai University of Cluj-Napoca is currently a compulsory course held during the second semester of the last year in CS bachelor's degree. It is a general verification

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and validation course, advocating for a wide range of practical and formal activities. In this way, participants start their journey approaching various degrees of inspection activities and dive into the complex process of software testing.

In a world in continuous change, in which the face-to-face contact is often replaced by virtual meetings, especially in the context of the new COVID-19 pandemics, the teaching process needs to adapt and embrace students' learning needs in the virtual space. This involves the integration of various communication and collaboration tools in the teaching and learning process. The main contributions of the paper refer to the following topics: students' perception on learning activities during course length, efficiency improvement while working on subsequent lab assignments, and consequences of adopting online tools in teaching.

This paper is composed of five sections. In Sect. 2, relevant related research on teaching software testing is considered. Section 3 presents the VVSS course activities and their alignment to the learning objectives. The research questions and inquiry methodology are detailed in Sect. 4. Section 5 includes the results analysis. The paper ends with conclusions and future work.

2 Related Work

The literature on software testing teaching identifies the following research aspects: (1) when and how to include software testing in CS curriculum, (2) tools used in software testing education and (3) teacher preparation in software testing education. The first topic in the list is summarized in the following.

Software testing is an important phase in the software development process. Still, there is a struggle in gaining testing related skills because many academic curricula focus on the initial development phases. Astigarraga et al. [3] argue that many CS undergraduate programs highlight development and teach testing as a formal engineering discipline. The skills which are retained in the testing area are mainly learned because of ad-hoc activities, while working on solutions for an implementation-oriented course. The lack of formal test education among graduates requires an additional effort from the industry demanding additional resources to provide a proper knowledge in software testing.

Clarke et al. [4] highlight the aspect of addressing a large number of topics in software engineering (SE) programs with the immediate drawback of giving little attention to software testing. They propose a minimally disruptive approach of integrating software testing into SE courses. By that, students access a collaborative learning environment that contains learning materials on testing techniques and testing tools. In this way, the testing skills are acquired along with the development activities. This perspective is recommended by Jones et al. in [5] and [6]. The authors introduce an approach that gives the students testing experiences throughout the curriculum, rather than the usual approach of offering a separate course in software testing.

Software testing can be introduced in CS curriculum following various approaches, which indicate that testing activities are applicable to a wide range

of software engineering courses. Therefore, testing can be performed by students on their own assignments [7, 8] or other students' work [5]. More, enhanced testing can be performed by using black-box testing on software that is seeded with errors [9]. Test-driven approach in software development [10] is another way to consider testing concepts. Next section discusses VVSS course in great detail.

3 VVSS Course Design

This section presents the educational objectives and the assessment rules stated for VVSS course, together with the activities attended by students.

3.1 Educational Objectives

The *Software Systems Verification and Validation* course [2] at Babes-Bolyai University of Cluj-Napoca is currently a compulsory course held in the 6th semester of the last year in CS bachelor's degree. The course takes 12 weeks and provides 5 credit points (ECTS) that requires a total of 125 h (= 5 * 25 h) of study.

The course follows the guidelines of ACM (Association for Computing Machinery) Curricula Recommendations [11] for CS. It covers a wide range of topics, from concepts extensively used in practical software testing to the more abstract ones, like formal methods used for program specification and verification.

The course is designed as two hours lecture each week, supported by two hours seminar and lab activities every two weeks. Students can attend a non-mandatory project activity that allows them to deepen their knowledge in software verification topics at their choice.

Assessment. The final grade is obtained by using the following formula: $FinalGrade = 0.50 * LabGrade + 0.10 * SeminarGrade + 0.40 * WrittenExam + ProjectGrade$. The *ProjectGrade* helps the students to increase their final grade up to 2 points.

The following subsection details the course activities.

3.2 Course Learning Activities

VVSS course consists of activities organized as seminars and laboratories.

Seminar Activities The course covers various topics in the six allocated seminar activities. The main goal is to detail, experiment and reflect on the concepts approached during lectures. Students can further investigate the testing-related

concepts during lab activities, while other notions concerning correctness and program refinement are approached during seminars only.

Lab Activities There are five lab activities designed for this course. Each of them focuses on different verification tasks employed during software development, having different weights in the *LabGrade*. The tasks associated with each assignment are applied to an already existing software product. The topics covered by the lab assignments are:

- **Lab01** (10%): document inspection, human and tool-based code review;
- **Lab02** (15%): create test cases based on black-box testing (BBT) techniques at method/class level such as: equivalence class partitioning (ECP), boundary value analysis (BVA), best representative (BR), domain testing (DT), etc.; use of testing frameworks (JUnit) and test management tools (Testlink);
- **Lab03** (15%): create test cases based on white-box testing (WBT) coverage criteria, as control flow graph (CFG) and various coverage criteria, e.g., statements, decisions, conditions, multiple conditions, loops;
- **Lab04** (30%): create test cases to test items that emphasize dependencies, apply integration strategies; use of mocking frameworks (mockito); use of continuous integration/ continuous delivery (CI/CD) tools (Jenkins) and report the test results to the test management tools (Testlink);
- **Lab05** (30%): create test cases to test web applications using different black-box test design techniques; use of powerful dedicated frameworks, e.g., Serenity BDD framework with Selenium WebDriver.

Specific task requirements, tutorials and guidelines prepared by teachers are available for each assignment. The following section details our research questions.

4 Research Methodology

In this paper, we investigate the following research questions:

- **RQ1:** What activities do students find helpful while learning software testing?
- **RQ2:** Can efficiency be improved throughout specific lab assignments?
- **RQ3:** What are the consequences of integrating online tools in VVSS classes?

To answer our research questions, we have considered an exit survey filled in by students and the teachers' feedback on the learning activities. The questionnaire had both open and closed questions, asking students' opinion about topics discussed over the course, learning experiences, and tools used to communicate and collaborate in and outside classes. We analyzed 32 complete answers out of 202 students enrolled in VVSS during the academic year 2019–2020.

Closed questions required students to assess specific statements, the answers being formulated following a five points Likert-scale [12], ranging from strongly disagree to strongly agree. The answers to the open questions were collected and manually

analyzed. We have used an adapted qualitative analysis procedure [13, 14], as follows: (i) each open question was uniquely coded, indicating a category; (ii) each answer to a specific open question was subsequently coded, based on the main idea provided by the respondent; (iii) complex answers that addressed more than one aspect were broken in many answers, assessed and coded subsequently; (iv) answers that do not match to existing codes, required a new code. To assess the content of each answer, the goals and the expectations for the designed activities were used.

The tasks for lab assignment Lab02–Lab04 required Testlink as a test management tool and Jenkins as an automation server. The students designed the tests and ran build jobs using this continuous integration (CI) system. The data regarding the test cases and build jobs was exported at the end of the semester and parsed using a python script to extract relevant statistical data.

The results of our inquiry are presented in Sect. 5.

5 Research Findings

This section analyzes the resulting data for the addressed research questions.

5.1 *RQ1. What Activities Do Students Find Helpful While Learning Software Testing?*

Throughout the course students are engaged in various types of activities, especially designed to gain hard and soft skills. Table 1 shows how students perceived the various activities intended to help them acquire software testing knowledge. The values are expressed as percentages on the five-point Likert-scale. The activities are presented from the most to the least relevant, the Top column being computed by

Table 1 Importance of learning software testing activities (Scale: S/D = strongly disagree, D = disagree, N = neutral, A = agree, S/A = strongly agree)

Activity	S/D	D	N	A	S/A	Top
Discussion and interactions	0	0	13	34	53	87
Support from teachers	0	0	13	20	67	87
Lectures	0	0	16	47	37	84
Lab work	0	0	16	31	53	84
Live demos	0	3	12	22	63	82
Prepare for the exam	0	0	19	31	50	81
Guest lectures	6	5	30	26	33	48
Bibliography	6	6	47	16	25	29

adding the positive answers, subtracting the negative ones, and ignoring the neutral answers.

Students consider most relevant for their learning experience those activities that involve interaction and support for practical work. Live demos were appreciated by 63% of students as being particularly important in learning software testing. To a higher extent, the support from teachers for various activities is perceived as especially important by 67% of students. The results suggest a group of students focused on problem solving, rather than theoretical aspects. This is confirmed by the 37 and 25% of students that appreciated important lectures and bibliography, e.g., papers, reference books on software testing.

The survey reported interesting data for guest lectures as learning activities. We expected to have a clear relevance of the practitioners' involvement in VVSS course. Still, 6% of the participants that provided answers appreciated them least important. We have further addressed this issue by investigating the question regarding the collaboration of software companies in teaching software testing. This was an open question, where the students could bring personal thoughts on their opinions. We have discovered that students being reluctant to guest lectures have provided additional details. The disagreement on the software companies' involvement was narrowed to the following: to not include marketing details in the course, to prepare more in advance to similar presentations, to discuss fewer aspects that concern a specific software company. Otherwise, 48% of the students appreciated the collaboration with the software industry in the VVSS course and 25% of them provided further feedback on why they like it. Some of the students pointed out the goal of the invited lectures: "... a really nice experience, it gave us an overview about how companies apply testing on applications..." or "... I think it is a good way to see how it works in the real world. It makes us understand why we learn all these things..." or "... good, it closes the gap between the university course and real career options...".

5.2 RQ2. Can Efficiency Be Improved Throughout Specific Lab Assignments?

A total of 202 students participated in the 2019–2020 VVSS course and they created 206 test plans with 1019 test cases in Testlink. Each student was required to implement continuous integration on their projects, either solo or as part of a team. Most of the students chose to work as part of a team, with a number of 79 unique teams being formed. A team could contain a maximum of 3 people, the minimum size of an Agile team [15].

Jenkins was used as an automation server, creating a bridge between the code repositories of the students (GitHub, GitLab and Bitbucket) and the test cases designed in Testlink. In each job the server would execute the designated tests and transfer the test results in Testlink. At the end of the course there were 1024 Build Jobs created by the students in Jenkins.

In this section we are focused on activities for Lab02, Lab03 and Lab04, where actual data was added in Testlink so we can easily compare them.

Lab02 covered black-box testing (BBT) techniques, followed by white-box testing coverage criteria (WBT) in Lab03 and integration strategies (INT) in Lab04.

In Fig. 1 we can observe the distribution of the Jenkins builds per each lab activity. There was a total of 7880 builds, with a large portion on the first lab (BBT) and fewer builds on the next two.

The BBT lab had 4150 builds for 397 test cases, with an average of 10.45 builds per test case. We can notice an improvement in efficiency for the WBT lab, with 1103 builds for 197 test cases and an average of 5.59 builds per test case. The INT lab in this series resulted in 2627 builds for 425 test cases, with an average of 6.18 builds per test case, like the WBT lab.

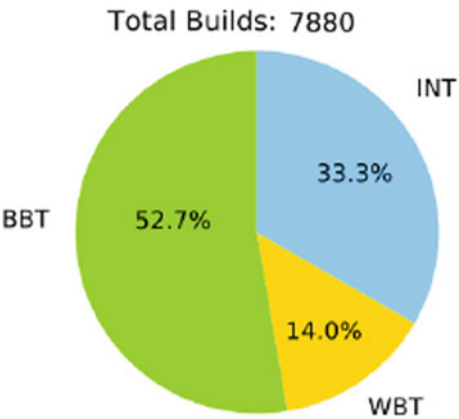


Fig. 1 Builds distribution exported from Testlink

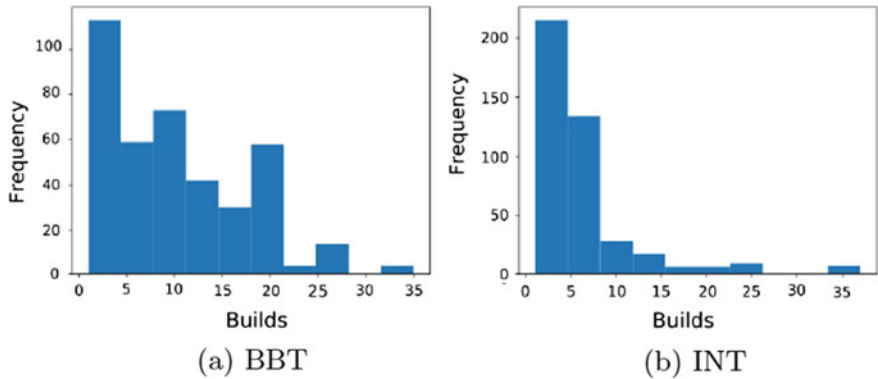


Fig. 2 Testlink build frequency

This pattern can also be observed in Fig. 2 where there is a large difference in build frequency for the first lab and the last lab in this series. Comparing the histograms, we notice that in (a) BBT there were multiple test cases with a large number of builds ($10 \rightarrow 20$) and in (b) INT most of the builds were less than 10 per test case.

In the first lab of this series (BBT), more builds mean the students put more effort to complete a successful test case scenario, perhaps spending more time on bug fixes and environment setup.

The overview of the build status for all the test cases, distributed per labs: BBT had 69.26% passed tests, WBT 76.64% and INT 80.23%, suggesting small improvements throughout lab assignments.

5.3 RQ3. What are the consequences of integrating online tools in VVSS classes?

To increase quality in teaching software testing, several tools that might ease communication and collaboration were employed for different groups of participants. We have studied the lab grades average and final grades of students enrolled in VVSS course during years 2017–2020. The results of the formative assessment (Lab Mean) and summative assessment (Final Grade Mean) are shown in Table 2.

Students enrolled in the 2016–2017 academic year attended traditional teaching classes, where the resources were available over the main instructor's web page. The following three groups made use of Slack as a communication platform, with actual face to face classes for years 2018 and 2019. During the academic year 2019–2020, COVID-19 pandemic circumstances called up VVSS teachers to move all classes and final assessment to online learning environments. Skype platform was mainly used to replace the face-to-face interaction.

The results in Table 2 indicate a general increase of grades as the Slack tool was employed. During the first three years that were investigated, students worked individually on their assignments. The improved support for students throughout the course activities is reflected by higher grades from year 2018–2020.

During the year 2020 team tasks were assigned for the lab activities in order to improve participants' soft skills. For the lab grades average, we have noticed an

Table 2 Average results from different academic years

Year	#Students	Slack	Skype	Lab work in groups	Lab grade mean	Lab grade mean
2017	180	No	No	No	5.86	6.88
2018	202	Yes	No	No	7.76	7.47
2019	201	Yes	No	No	7.71	7.46
2020	202	Yes	Yes	Yes	8.53	8.98
2020	Self-assess					7.13

increase from 7.46 to 8.98. So far, our analysis on this grade boost is associated with team working activities, since the projects designated to work on during lab activities had similar levels of complexity over all investigated years. Further investigation is required to check whether other factors contributed to this raise.

At this end of the course, students were asked to self-assess their knowledge in software testing with a grade ranging from 1 to 10. The average grade on their evaluation is included in the last row of Table 2. The value 7.13 is below the actual final grade average by almost 2 points. This gap may suggest that their self-assessment considered real world complex software projects, compared to the ones used for the learning purposes. More, the estimation was made before sitting the final exam.

The survey included questions regarding the tools used over the course duration. Figures 3 and 4 present the feedback provided by students, indicating their agreement to several statements on Slack and Skype.

The answers show that participants adapted quickly to the communication platform proposed to them. Over 70% of the students *strongly disagreed* on the difficulty to use Slack platform. More, participants *strongly agreed* on the benefits of using Slack for communication (75%) and getting help when needed (72%).

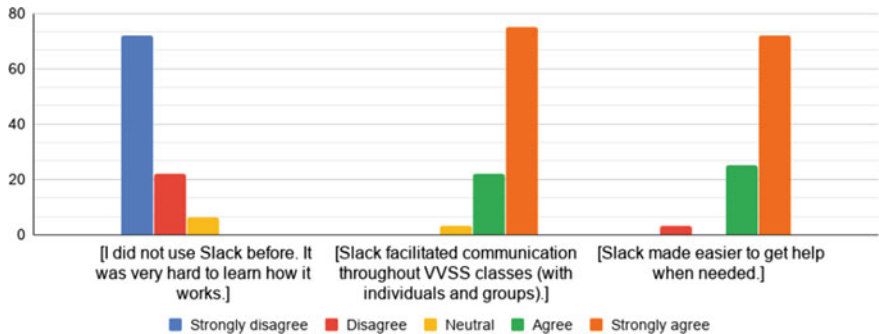


Fig. 3 Student feedback on using slack throughout VVSS course

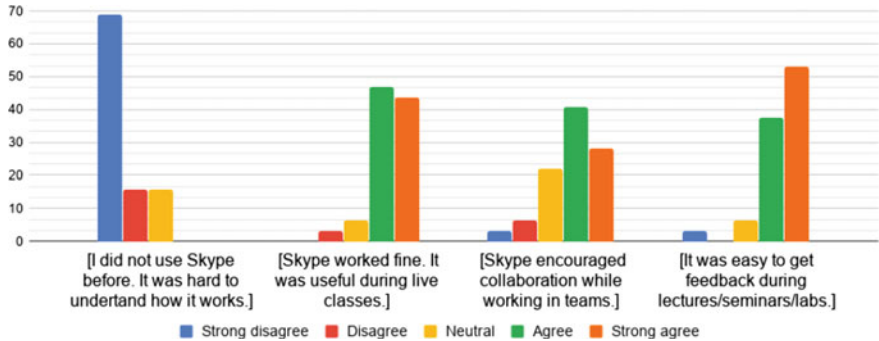


Fig. 4 Student feedback on using Skype for online activities

The aim of using Skype was to create a virtual classroom for the online activities. Participants indicated Skype was an easy use to work with tool by 69% of survey participants. Team working during online classes, in terms of communication and collaboration over Skype was assessed positively as well.

Slack and Skype seem to overlap on some features, e.g., text/chat-based ones. Still, students were notified from the beginning that the main communication will be held on Slack. This aspect is important to be stated when several tools that expose any similar functionality are employed over course duration and each of them is chosen for different purposes.

6 Conclusions

This paper addresses important aspects related to teaching and learning software testing in current days. Research on this topic indicates the need for a comprehensive approach that manages the learning objectives, the associated activities, tools employed in software testing and the proper training of educators. We have discussed the design of the VVSS course, considering the activities and the employed assessment rules.

A survey analysis showed that students favor practical learning experience on software testing, where software industry contribution is clear, aiming to add value to the teachers' endeavors. The efficiency improvement from one lab assignment to another is possible and pushes for skill acquirement. Integration of communication and collaborative tools such as Slack and Skype during the course activities improved student grades on formative and summative assessment and helped students gain essential soft skills.

Future work on the investigated issues aims to deepen the research on increasing testing tool diversity and learning tools during collaborative activities.

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Interactive Theorem Prover Based on Calculational Logic to Assist Finite Difference and Summation Learning



Federico Flaviani

Abstract Universities show difficulties and resistance to carry out virtual mathematics instruction, especially due to the high probability that cheating will occur among students, at the time of assessments. In this work an interactive theorem prover based on calculational logic is presented, adjusted to the topic of finite differences and summation, which aims to reduce the probability of cheating in virtual education, forcing the student to understand each step of a given exercise during the interaction with the aforementioned theorem prover. In this work, data and positive educational experiences of the use of the tool are shared, including virtual assessments experiences, where it was shown that there was no cheating among students.

1 Introduction

CalcLogic <https://calclogic.herokuapp.com> is a web interactive theorem prover, based on the calculational logic popularized by Gries Schneider's book [1]. An interactive theorem prover is a software application that is capable of verifying a mathematical reasoning that is entered, and making a judgment about its correctness. CalcLogic works by verifying one-step inferences, according to the Dijkstra-Scholten notation, which interactively with the user, notifies an error, or performs the inference by printing the formula resulting from it.

The calculational logic works based on the replacement of an expression by another equal, within a formula. The proof process is the same as the way algebra exercises are solved in the early years of high school. The student of a calculational logic course advances rapidly, because he associates his already existing skills in algebraic manipulation, with the proof mechanism of this formal system.

For the purposes of assisting calculations of finite differences and summation, it is sufficient to work only with equalities $Exp_1 = Exp_2$, so they are the only formulas

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that the application supports, which are all indexed by an identification (id). Although calculational logic has different proof methods, due to the previous restriction, only one method was implemented, that consists of starting from one side of the equation to reach the other side, all this by means of intermediate equalities organized vertically. For example a proof of the equality $Exp_1 = Exp_n$, has the following form:

$$\begin{aligned}
 &Exp_1 \\
 &= \langle \text{justification}_1 \rangle \\
 &Exp_2 \\
 &= \langle \text{justification}_2 \rangle \\
 &\vdots \\
 &= \langle \text{justification}_{n-1} \rangle \\
 &Exp_n
 \end{aligned}$$

Each justification refers to the id of the statement used, which instance of it was used, and optionally, the details of an inference rule called Leibniz's rule. The instance of the statement that is used in the inference is determined by specifying the parallel substitution operator $[\bar{x} := \overline{Exp}]$ (with $\bar{x}$ list of variables and $\overline{Exp}$ list of expressions) that will be applied to instantiate the statement.

For example, if statement $a + b = b + a$ is an axiom indexed with 1, and you want to select instance $(a + b) + 0 = 0 + (a + b)$ of this statement, in the justification it must be written $\langle \text{st 1 with } a, b := a + b, 0 \rangle$, to indicate that statement 1 will be used, but substituting the variables a and b with the expressions $a + b$ and 0 respectively. By specifying the above, two possible inferences are determined as follows (by the symmetry of $=$):

$$\begin{array}{ccc}
 (a + b) + 0 & & 0 + (a + b) \\
 = \langle \text{st 1 with } a, b := a + b, 0 \rangle & \text{or} & = \langle \text{st 1 with } a, b := a + b, 0 \rangle \\
 0 + (a + b) & & (a + b) + 0
 \end{array}$$

Only one of the above inferences can be used in one single step, therefore the most convenient is selected. CalcLogic can identify this convenience, depending on the state of the user proof.

Leibniz's rule consists of replacing a subexpression of an expression Exp with another equal subexpression. You must specify the sub-expressions Exp_1 and Exp_2 that will be replaced by each other, and the place in Exp where the replacement will occur. For this, using the previous mechanism, an instance of some statement such that $Exp_1 = Exp_2$ must be chosen, and for the place in Exp , an expression E is used, equal to Exp , except that a z occurs in the place where the replacement will occur.

For example, if in $a.(0 + 0) + (-(a.0))$ you want to replace $0 + 0$ by 0 and you have that statement $a + 0 = a$ is indexed with 2, then the inference would look like this:

$$\begin{aligned}
 &a.(0 + 0) + (-(a.0)) \\
 &= \langle \text{st 2 with } a := 0 \text{ and } E : a.z + (-(a.0)) \rangle \\
 &a.0 + (-(a.0))
 \end{aligned}$$

2 Contribution

Using calculational logic is natural for undergraduate students, because it is much like the process of solving a math exercise in high school. This naturalness is inherited by the application CalcLogic, which together with an inference input mechanism, through clicking and underlining, has made the user experience simple and fast.

The typical resolution exercises of finite difference and summation have been simulated, as proofs of equalities with the method of starting from one side and reaching the other. Therefore, a student who is not familiar with mathematical proofs, but is familiar with solving exercises in calculus, adapts quickly and does not have the feeling of proving something.

The application can be used to carry out formation activities and non-classroom assessments, in which case it was shown (Sect. 5), that textual cheating between students is very unlikely or evident if it occurs.

3 Related Jobs

Calculational logic was described for the first time in the book by Dijkstra and Scholten [2] and later in the book of Gries and Schneider [1], who popularized it as the textbook of courses of logic, in the computer science program of many universities. However, from the rigorous point of view, the description of this logic was open to misunderstandings, in fact Vladimir Lifschitz in [3] writes the following about it: “One of the reasons why the contribution of Dijkstra and Scholten has not been appreciated by logicians is that what the authors call formal proofs are not really formal proofs in the usual sense; they are informal proofs written in a relatively strict format.”

Works that studied the foundations of the system, were carried out in the years following the publication of the work of Dijkstra and Scholten. Among these are [4–16].

The application presented in this work is an interactive theorem prover. There are many applications within this category. The most famous interactive theorem prover are Isabelle/HOL [17] (with its interactive window *PG Tips* [18]) and Coq [19].

There are two Interactive Theorem Provers for Calculational Logic; CalcCheck stand alone [20], and CalcCheck web [21]. The last two applications are basically the same, the only exception is that the first one requires installation and the second one runs with a browser and internet connection. The difference of these applications with the one presented in this work, is that the previous ones require that the proof be written in $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ in order to be processed and verified, while a CalcLogic user does not need to know any language in particular, since almost everything is done by means of the mouse.

4 Application Description

The application manages a session for each student and teacher, therefore it is necessary to create an account in it.

4.1 List of My Theorems

The student has the possibility to visualize his progress by seeing the complete list of theorems that he must prove. In the list are all the statements: the axioms, the proved theorems, and the ones he has not yet proved. The theorems that has already proven and the axioms are accompanied by an open lock symbol, indicating that said statement can be used to make another proof. The theorems that have not yet been proven are accompanied by a closed lock symbol, indicating that said statement cannot be used in another proof, until the proof of itself was made (which is when the lock will appear open).

The student's goal is to open as many locks as possible (prove as many theorems as possible), adding a gamification element to the learning process.

4.2 Proofs

The list of statements to be proved is displayed and the user is free to select the one they want to start a proof. The theorems in the application are all equations $LeftSide = RightSide$. When one is selected to be proved, the user must then select from which of the two sides of the equation the proof will begin. If *LeftSide* is selected, then from that expression, the user must make several one-step inferences until reaching *RightSide*, at which point the application automatically recognizes that the theorem has been proven. The same happens at the moment to reach *LeftSide* if *RightSide* is the start point selected.

4.2.1 One Step Inference

Once a proof has started, a list appears on the right hand side with the already proven axioms and theorems. In order to carry out a one-step inference, one of the axioms or theorems already proved must be selected with the mouse, so that the substitution fields and Leibniz's rule appear (see Fig. 1). Once the details of the substitution and Leibniz rule to be performed, have been entered correctly, the inference is made by pressing the Infer button.

The application allows the user to delete the last one-step inference made, by pressing the Go Back button.

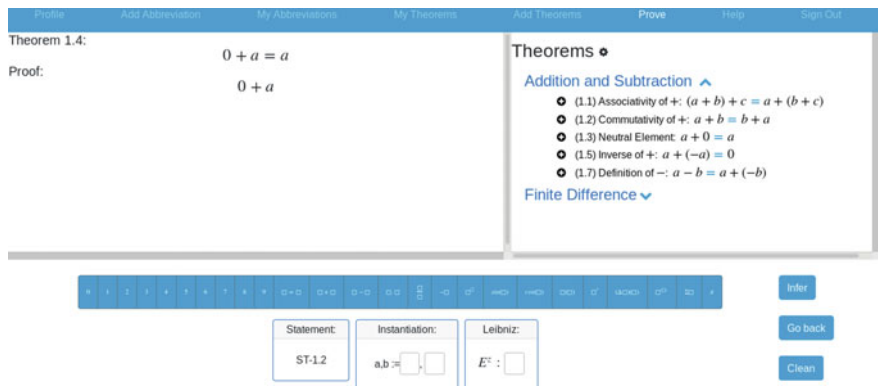


Fig. 1 Forms to specify statement, substitution and Leibniz rule

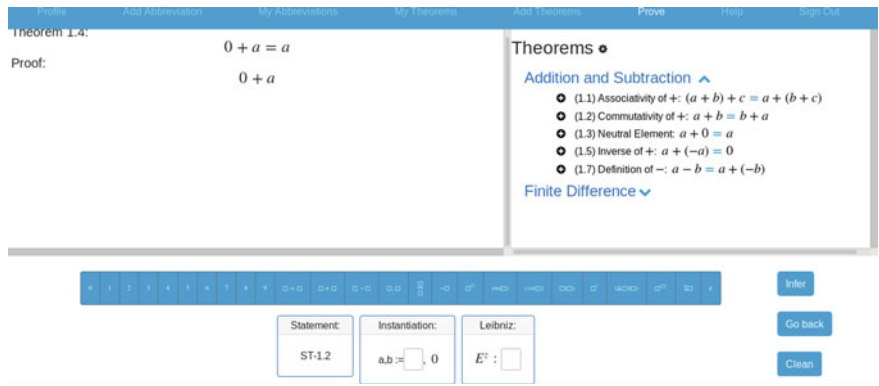


Fig. 2 Filled with 0 only the substitution form corresponding to b

4.2.2 Instantiation of a Theorem or Axiom

Once the theorem or axiom has been selected, the substitution field appears with the lists of variables that occur in the theorem and a form for each of these variables. By entering the expression to be instantiated in each of these forms, the theorem or axiom is instantiated in the way that the user wishes. If any of these forms is left blank, the application understands that the variable corresponding to the blank form is not instantiated in the inference. Figure 3 shows the result of leaving blank the instantiation form of Fig. 2, corresponding to the variable a .

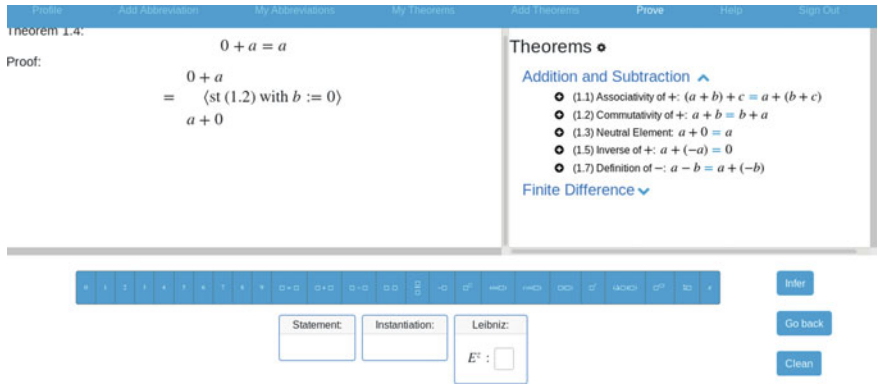


Fig. 3 Result of filling only the form of b in the substitution of Fig. 2

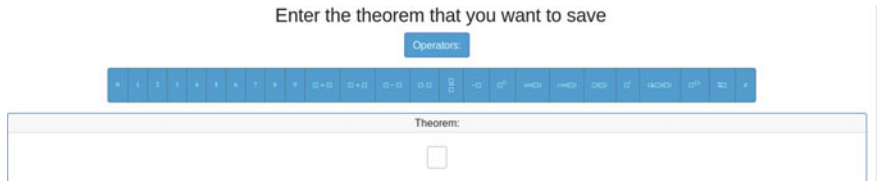


Fig. 4 Table of buttons associated with each symbol to be able to enter a theorem

4.2.3 Assisted Entry of Expressions

In every field of a form where it is necessary to enter a formula or mathematical expression, the application has a table of buttons for assisted entry. In Fig. 4 you will see the table for each button corresponds to a symbol. Included are 1) the integer constants 0, 1, . . . , 9 and the constant π , 2) the operations addition, subtraction, multiplication, unary minus, division, exponentiation, functions sin and cos, 3) the function application, derivation, finite difference (Δ), factorial polynomial exponentiation, and summation (Σ).

To enter a formula in a form, you must place the cursor on that form and use a button from the table. If the button corresponds to an operation of arity n , when pressing it, the operator is printed on the form and n forms are generated, located in the position where the operator's arguments should be (for example, in Fig. 5 the effect of pressing the $+$ operator button is shown in the theorem input form). To continue entering the desired formula or expression, you must place the cursor on the form of one of the arguments of the previously selected operator, and repeat the process recursively (For example, Fig. 6 shows the effect of pressing the button “exponentiation” when the cursor was on the second argument of the operator $+$).

To reverse the effect of pressing one of the buttons in the table, the user must place the cursor on one of the forms generated by the button and press Back Space

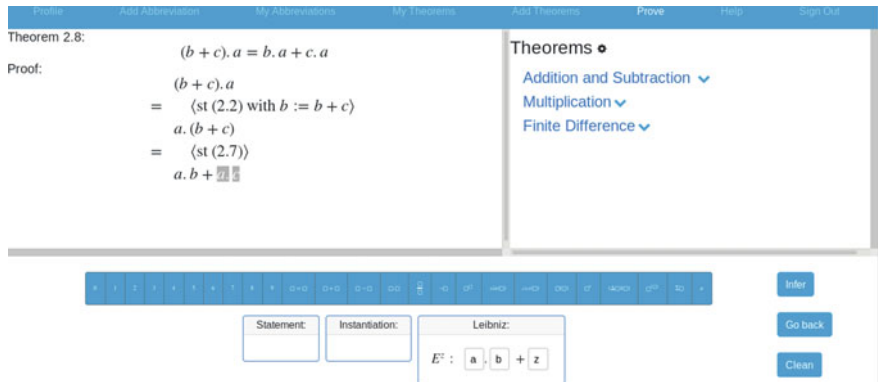


Fig. 7 Auto-completion of Leibniz form after underlining sub-expression $a.c$

5 Experience with the Assessments

During the COVID-19 pandemic which forced teaching from classroom to virtual, this application was used to complete a Discrete Math course at Simón Bolívar University in Venezuela. This course included topics related to finite difference and summation.

The class consisted of students with little or no knowledge of calculational logic. Therefore, an introduction phase on this logic was required prior to instructing students in it's use. This phase lasted one week, exhibiting the following aspects:

1. Exercises during the introduction phase focused on topics on high school algebra and trigonometry, aimed at achieving the maximum student concentration in the new logical aspects that the application incorporates, working in an area familiar and comfortable for them.
 - (a) The introduction phase began with the field axioms of real numbers, without mentioning this algebraic structure, and the first proofs were planned to be one-step length, where it was enough to use theorems without instantiation.
 - (b) Later, the complexity was increased by incorporating proofs with the use of instantiation, and then continued with proofs with the use of instantiation combined with Leibniz's rule.
2. After performing the first exercises on field axioms, when the student understood how to make all kinds of inferences, they proceeded to practice what they had learned, solving typical high school problems such as the addition, multiplication and simplification of fractions, multiplication and division of expressions with exponents of equal bases, exponentiation of exponentiations, etc.
3. Finally, the introduction phase was completed with proofs of complicated trigonometric identities.

Once the teacher verified the completion of the training exercises for all his students, he began using the application to dictate content for the Discrete Math course. The course content was delivered by sharing the teacher's screen with an online meeting platform, and the interactive model of the application allowed it to be used as a virtual blackboard.

The assessment consisted of a series of exercises of considerable length, which were loaded into the application from a certain point in time, and only for the duration of the assess. The results of the students were stored in the database and afterwards, the teacher was able to review them.

A relevant aspect that was observed in the experiment, is that despite what was expected, verbatim cheating between students is easier to identify when using this tool rather than with a written assess. One can argue that the student is forced by the application to perform atomic steps, which are not usually done in a written exam, then the universe of combinations in which they can be done is greater than that of a written exam. Therefore, it is almost impossible for two students to do exactly the same. For example, the following is the first 6 steps of the solution registered in the database, of theorem 13.4 (on finite differences), of two students. Note that since the exercise is long, many associative, commutative, distributive properties, sign rule, simplification lemmas, etc. were used. They both used these properties in different order, to reach the same final expression ($\sin(\frac{\pi}{2} \cdot x)$). Because of the number of combinations, it is highly unlikely that two proofs will match, unless it is a copy.

Theorem 13.4:

$$(\Delta(-(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \sin(\frac{\pi}{4})}))) (x) = \sin(\frac{\pi}{2} \cdot x)$$

Proof of student A (first 6 steps):

$$\begin{aligned} & (\Delta(-(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \sin(\frac{\pi}{4})}))) (x) \\ &= \langle \text{st (10.3) with } f, x := \frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \sin(\frac{\pi}{4})}, x \rangle \\ & \quad - (\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \sin(\frac{\pi}{4})}))(x) \\ &= \langle \text{st (7.00) and } E : -(\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot z}))(x) \rangle \\ & \quad - (\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \frac{2^{\frac{1}{2}}}}))(x) \\ &= \langle \text{st (4.4) with } a, b, c := 2, 2^{\frac{1}{2}}, 2 \text{ and } E : -(\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{z}))(x) \rangle \end{aligned}$$

$$\begin{aligned}
& -(\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{\frac{2 \cdot 2^{\frac{1}{2}}}{2}}))(x) \\
& = \langle \text{st (4.7) with } a, b := 2, 2^{\frac{1}{2}} \text{ and } E : -(\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{z}))(x) \rangle \\
& \quad -(\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2^{\frac{1}{2}}}})(x) \\
& = \langle \text{st (7.105) with } x := x \text{ and } E : -(\Delta(\frac{z}{2^{\frac{1}{2}}}}))(x) \rangle \\
& \quad -(\Delta(\frac{\frac{2^{\frac{1}{2}}}{2} \cdot (\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x))}{2^{\frac{1}{2}}}}))(x) \\
& = \langle \text{st (4.3) with } a, b := 2^{\frac{1}{2}}, 2 \text{ and } E : -(\Delta(\frac{z \cdot (\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x))}{2^{\frac{1}{2}}}}))(x) \rangle \\
& \quad -(\Delta(\frac{(\frac{1}{2} \cdot 2^{\frac{1}{2}}) \cdot (\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x))}{2^{\frac{1}{2}}}}))(x)
\end{aligned}$$

Below are the first 6 steps of the proof of the same theorem, made by another student:

Proof of student B (first 6 steps):

$$\begin{aligned}
& (\Delta(-(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \sin(\frac{\pi}{4})}))) (x) \\
& = \langle \text{st (10.3) with } f := \frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \sin(\frac{\pi}{4})} \rangle \\
& \quad -(\Delta(\frac{\cos(\frac{\pi}{2} \cdot (x - \frac{1}{2}))}{2 \cdot \sin(\frac{\pi}{4})}))(x) \\
& = \langle \text{st (7.18) and } E : -(\Delta(\frac{z}{2 \cdot \sin(\frac{\pi}{4})}))(x) \rangle \\
& \quad -(\Delta(\frac{(\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x)) \cdot \frac{2^{\frac{1}{2}}}{2}}{2 \cdot \sin(\frac{\pi}{4})}))(x) \\
& = \langle \text{st (7.16) and } E : -(\Delta(\frac{(\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x)) \cdot \frac{2^{\frac{1}{2}}}{2}}{2 \cdot z}))(x) \rangle \\
& \quad -(\Delta(\frac{(\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x)) \cdot \frac{2^{\frac{1}{2}}}{2}}{2 \cdot \frac{2^{\frac{1}{2}}}{2}}))(x) \\
& = \langle \text{st (4.22) with } a, b, c := \cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x), \frac{2^{\frac{1}{2}}}{2}, 2 \text{ and } E : -(\Delta z)(x) \rangle \\
& \quad -(\Delta(\frac{\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x)}{2}))(x)
\end{aligned}$$

$$\begin{aligned}
&= \langle \text{st (4.3) with } a, b := \cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x), 2 \text{ and } E : -(\Delta z)(x) \rangle \\
&\quad - (\Delta(\frac{1}{2} \cdot (\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x))))(x) \\
&= \langle \text{st (20) with } a, f := \frac{1}{2}, \cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x) \text{ and } E : -z \rangle \\
&\quad - (\frac{1}{2} \cdot (\Delta(\cos(\frac{\pi}{2} \cdot x) + \sin(\frac{\pi}{2} \cdot x))))(x)
\end{aligned}$$

Another aspect that was observed is that although the application does not allow an atomic step to be done incorrectly, it can be observed as in written exams, when a student did not study the materia. In fact, the teacher can discriminate excuses such as: “the service and the connection were slow”, “with more time I would have finished it”, “the application threw an error so I could not finish”. This is due to the fact that the application does not prevent the student from taking many correct steps that lead him down a path where it is impossible to reach the result, which is evidence that he did not study the technique that would guide him on the right path. This phenomenon occurs in this form of assessment exactly as it occurs in written exams, as well as the elements to discriminate the veracity of the students’ excuses.

6 Conclusions

In the experiments carried out to make non-classroom assessments with 33 students, it is concluded that for exercises of medium length or more, there was no verbatim cheating, because no answer from one student is the same as another. On the other hand, it is easy to determine the reasons why a student cannot solve an exercise. This is done by inspecting your solution in the incomplete state he finished. In the experiment carried out, 100% of the students who made excuses, blaming the application for its poor performance, demonstrated not knowing how to solve the exercise, when a detailed inspection was made of what they had in their solution. Because of these arguments, I recommend the assessment mechanism used, in situations in which it is impossible assess in the classroom.

It is also concluded that learning using this technology is effective for the student. Based on experience, all the students commented that they learned in a fun way. They were forced to internalize each step, because if they did not understand an inference step, they could not proceed. I maintain that in order to advance to the next section in CalcLogic, almost all the previous exercises must be done, then the student is obligated to do them, achieving training that often is not achieved with paper and pencil.

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Smart Education: Case Studies and Research

The Effect of Emergency Remote Teaching from a Student's Perspective During COVID-19 Pandemic: Findings from a Psychological Intervention on Doping Use



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Abstract The coronavirus (COVID-19) pandemic affected teaching modality worldwide. Indeed, there was an immediate urge to transition from face-to-face to online teaching to avoid the COVID-19 spread. The speed of such process led to the employment of emergency remote teaching. While the effectiveness and the advantages of planned online teaching modalities were well-assessed, little is known about implementing emergency remote teaching on learning processes. Accordingly, the present study aimed to evaluate the differences between face-to-face teaching and remote modality in terms of engagement, perceived learning, and the goodness of the learning environment from a student's perspective. Results showed that students perceived no differences between face-to-face and remote modalities. Such findings suggest that a remote intervention can provide high engagement, perceived learning, and a positive learning environment in emergencies, such as the COVID-19 pandemic.

Keywords Media Literacy intervention · Emergency remote teaching · COVID-19

1 Introduction

The coronavirus (COVID-19) pandemic, started in December 2019, caused harmful effects on people's health and radically changed daily life activities (e.g., work, sport,

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education). Indeed, the restrictive measures adopted in many countries to contain the transmission of the COVID-19 (e.g., quarantine, lockdowns, social distancing) influenced many aspects of life, including teaching modalities [1, 2]. The schools of every degree have found themselves in the complicated role between fulfilling their original mission to teach future generations and the current and imperative need to contain COVID-19 infections.

The United Nations Educational, Scientific and Cultural Organization (UNESCO) recently estimated that over 144 countries had suspended in-person education, affecting approximately 1.2 billion students [3]. In such conditions, the transition from face-to-face teaching to online learning seems to be the only viable solution to ensure an adequate education level and avoid COVID-19 spread [2, 4, 5]. This transition has created new challenges for teachers and students [6, 7] who have had to familiarize themselves with technological tools they usually do not employ.

The speed with which the transition to online teaching modalities has taken place is unprecedented and has generated the immediate need to transfer face-to-face teaching to online modalities. Given the circumstances, it seems appropriate to differentiate online learning modalities, which could be employed as planned online education, and emergency remote teaching [4]. While the first modality has been characterized by a relatively long period devoted to planning the teaching and learning environment, the second has been defined as “a temporary shift of instructional delivery to an alternate delivery mode due to crisis circumstances” [4]. Accordingly, teachers had to respond to an emergency context that may have influenced online courses’ quality, given the lack of time to build a planned online education. While in scientific literature many studies attest that online modalities are comparable to face-to-face teaching as regard learning efficacy [8–11], with also significant advantages in terms of saved time [12, 13], few studies assessed the students’ perception of the emergency remote teaching in terms of engagement, perceived learning and the goodness of the learning environment. Indeed, few case studies (e.g., [5]), qualitative (e.g., [14]), and quantitative surveys (e.g., [2]) evaluated emergency remote teaching, showing, overall, that the use of this online modality was not optimal as well as the traditional face-to-face one. Given the current situation and the uncertainty about the length and the future development of the pandemic, it seems imperative to comprehend its effects on emergency remote teaching.

Considering this background, the present study evaluated the students’ perception of media literacy interventions employed both in face-to-face modality and emergency remote one. Indeed, in the last few years, the present research group conducted face-to-face media literacy interventions regarding the use of Performance and Appearance Enhancing Substances (PAES) in adolescents and young people [14, 15]. Such interventions aimed to inform the students how media could influence their attitudes and beliefs about the use of doping substances. Due to the current COVID-19 emergency, the present research group adapted the media literacy interventions from face-to-face modality to the emergency remote one. This scenario gave us the opportunity to compare these two teaching modalities in terms of the student’s evaluation of the engagement, perceived learning, and goodness of the learning environment.

The present data derived from a larger study focusing on assessing an anti-doping media literacy intervention's effectiveness.

2 Materials and Methods

2.1 Participants

The study relied on two samples of Italian students participating to a PAES use media literacy intervention carried out respectively in a traditional face-to-face modality ($n = 99$; Mean age = 17.76, SD age = 0.59; 42.1% female) and in a emergency remote modality ($n = 157$; Mean age = 16.92, SD age = 0.88; 31.8% female). According to the Italian school system, participants represented the last three grades of high schools: third grade (16-year-olds; $n = 31$; 13.6%); fourth grade (17-year-olds; $n = 92$; 40.3%) and fifth grade (18-year-olds; $n = 105$; 46.1%).

The participating students were identified via a convenience sampling procedure. Students' parents were fully informed in advance about the aims of the study, and written consent was obtained by both the schools and the students' parents. All students filled out a questionnaire (see below for the details) evaluating different aspects of the intervention after its conclusion during school hours, except for 2 students of the remote modality and 4 students of the face-to-face one, who did not respond to questionnaires. The attendance rate for the sessions of the intervention is described in Table 1.

The study was approved by the Ethics Review Board of the Department of Social and Developmental Psychology, "Sapienza" University of Rome.

Table 1 Attendance rate across the two modalities for each session

Sessions	Intervention modality	Attendance rate		Total
		Present (n, %)	Absent (n, %)	
Sports science expert	Face-to-face Remote	81 (85.30%)	14 (14.70%)	95
		150 (96.80%)	5 (3.20%)	155
		231 (92.4%)	19 (7.6%)	250
Communication expert	Face-to-face Remote	84 (88.40%)	11 (11.60%)	95
		147 (94.80%)	8 (5.20%)	155
		231 (92.4%)	19 (7.6%)	250
Sport psychologist	Face-to-face Remote	91 (95.80%)	4 (4.20%)	95
		151 (97.40%)	4 (2.60%)	155
		242 (96.8%)	8 (3.2%)	250

2.2 Description of the PAES Use Media Literacy Intervention

The intervention was developed, coordinated, supervised, and monitored by researchers from the “Sapienza” University of Rome and the “Foro Italico” University of Rome. The intervention’s design, evaluated in the present study, followed the structure of Lucidi and colleagues [15] intervention with high school students and the Mallia and colleagues [16] intervention with sports science university students. The intervention was carried out in face-to-face modality from January to March 2020, before the Italian lockdown due to the COVID-19 pandemic. The emergency remote modality was carried out from October to December 2020, during which the Italian government imposed the closing of all schools, leading the students to attend online lessons.

Specifically, the present study focused exclusively on the first phase of the intervention, comprised of six 90 minutes sessions that relied on three experts (a sports science expert, a sport psychologist, and a communication expert). Noteworthy, the experts were the same for both teaching modalities (i.e., face-to-face and remote one). Each student has been requested to attend two sessions of each expert.

Specifically, the experts focused on media messages concerning PAES use in sport from their expertise viewpoint. Following, there is a description of the exact content for each expert’s sessions: (a) the sports science expert focused on the moral, ethical, and health consequences of PAES use and how media may offer arguments to minimize these issues; (b) the communication expert addressed the role that media messages may have in promoting biased beliefs about sport and PAES use, underlining and emphasizing, for instance, unrealistic sport and aesthetic goals; and finally, (c) the sport psychologist focused on the ways cognitive strategies may help students in managing and correctly evaluating media messages, as well as on the awareness of their personal sports goals and capacity to counteract temptations toward PAES use.

2.3 Measures

Students participating in the experts’ sessions in both modalities (i.e., face-to-face and emergency remote) were asked to evaluate each of the three experts’ sessions by eight items evaluating how: (a) the intervention resulted interesting (i.e., Interest); (b) students enjoyed the intervention (i.e., Enjoyment); (c) the intervention was informative for students (i.e., Instructive); (d) students learned new knowledge (i.e., Perceived Learning); (e) the sessions’ educational material was clear and easy to follow (i.e., Education Material Clarity); (f) the sessions were well organized and managed (i.e., Learning Environment); (g) students perceived the expert as competent (Perceived Expert Competence); (h) the expert was clear and easy to follow for students (i.e., Expert Clarity). Students were asked to rate their agreement to each of the eight questions using a five-point scale ranging from 1 (“Totally disagree”) to 5 (“Totally

agree"). High values in each dimension revealed a positive evaluation of the specific expert block of sessions on that dimension.

2.4 Statistical Analysis

A two-way mixed multivariate analysis of variance (MANOVA) was carried out using SPSS 26 version. Such MANOVA considered the dependent variable as a combination of all the eight dimensions that students evaluated ("Interest", "Enjoyment", "Instructive", "Perceived Learning", "Education Material Clarity", "Learning Environment", "Perceived Expert Competence", and "Expert Clarity") and as independent variables the "teaching modality" (two-levels between-subjects factor: "face-to-face" and "remote") and the "sessions" (three-levels within-subjects factor: "sports science", "communication", and "sport psychology"). The Pillai's Trace and the Wilks' Lambda (Wilks' Λ) were used to assess multivariate effects. The significance level was set at $\alpha = 0.05$. In the case of significant multivariate effects, a two-way mixed analysis of variance (ANOVA) with the above-mentioned between and within-subjects factors was conducted to test for univariate effects (i.e., for each of the eight dependent variables). The significance level for each univariate ANOVA was set at $\alpha = 0.006$ ($0.05/8 = 0.006$) to compensate for family-wise error rate inflation. In the case of significant univariate effects, post-hoc tests were employed using Bonferroni corrections to determine any statistically significant differences between the means. Finally, analyses were repeated, also considering outliers, to assess for potential findings' differences.

The advantage of employing MANOVA rely on the possibility to investigate both multivariate and univariate effects, without inflating Type I error rate, and at the same time ensuring robustness against violation of normality assumptions and homogeneity of variance-covariance matrices [16, 17].

3 Results

3.1 Assumptions Check for Mixed MANOVA and Mixed ANOVA

As preliminary steps, assumptions check for mixed MANOVA was performed [17]. Twenty-eight univariate outliers (face-to-face: $n = 11$; remote: $n = 17$) were detected and removed from subsequent analysis. MANOVA analysis appears appropriate since the dependent variables seemed to be related between them in each category of the within-subjects factor, without multicollinearity issues, since the correlation matrices for each level of the between-subjects factor reported Pearson-correlation coefficients between 0.3 and 0.9. According to Kolmogorov-Smirnov, and Shapiro-Wilk tests,

none of the dependent variables were normally distributed (all $p < 0.001$). Nevertheless, the MANOVA is commonly robust against the violation of the normality assumption [17]. The Box's M test result showed that the assumption of homogeneity of variance covariance matrices was violated ($p < 0.001$). However, Box's M test is sensitive to sample size, and MANOVA is robust also against violations of this assumption [16].

For what concern univariate mixed ANOVA assumptions check, Mauchly's W tests findings showed that the sphericity assumption was violated for enjoyment, instructive, education material clarity, perceived expert competence, and expert clarity dependent variables. Accordingly, Huynh-Feldt corrections were employed when assessing within-subjects effects of such variables [18]. Finally, Levene's tests findings showed no significant F values (all $p > 0.05$), assuming homogeneity of variances across "teaching modality" groups for all the eight dependent variables in each "sessions" level.

3.2 Mixed MANOVA and Mixed ANOVA Results

Results showed that the multivariate effect of the "teaching modality" on the combination of all the eight dependent variables, regardless of the factor "sessions", was not significant ($F(8, 178) = 1.099$, $p = 0.366$; Pillai's Trace = 0.047; Wilks' $\Lambda = 0.953$; partial eta squared = 0.047). The multivariate effect of the "sessions", regardless of the factor "teaching modality", instead, was significant ($F(16, 170) = 3.548$, $p < 0.001$; Pillai's Trace = 0.250; Wilks' $\Lambda = 0.750$; partial eta squared = 0.250). Moreover, the multivariate interaction between the teaching modality and the sessions was significant ($F(16, 170) = 2.564$, $p = 0.001$; Pillai's Trace = 0.194; Wilks' $\Lambda = 0.806$; partial eta squared = 0.194). To deeply explore the multivariate effects, a two-way mixed ANOVA was employed for each dependent variable.

Results showed that the "teaching modality" effect was not significant for each of the eight dimensions (all $p > 0.006$; see Table 2).

Conversely, the "sessions" effect was significant ($p < 0.006$) for all the eight dimensions, also considering sphericity correction of Huynh-Feldt, except for instructive ($p = 0.059$) and education material clarity ($p = 0.008$) variables (see Table 3). There were no statistical effects for each of the eight dimensions of students' feedback regarding the interaction effects since all the p values associated with each F were above 0.006. Following the univariate ANOVA results, only the main within-subject effect was further analyzed through post-hoc tests. Looking at specific dimensions' differences (see Table 3), regardless of teaching modality, the sports science and sport psychologist experts' sessions were evaluated by students as equally interesting but more interesting than the communication expert's sessions. Regarding the perceived expert competence, students attributed higher values both to the sports science expert and sport psychologist, without differences between them, than the communication expert. The sport psychologist's sessions were also evaluated as the

Table 2 Average students scores for each dimension feedback across the two teaching modalities

Dimensions of student feedback	Intervention modality		F	P value	Partial eta squared
	Face-to-face	Remote			
	(Mean $\pm$ SE)	(Mean $\pm$ SE)			
Interest	4.44 $\pm$ 0.07	4.37 $\pm$ 0.05	0.588	0.444	0.003
Enjoyment	4.30 $\pm$ 0.07	4.23 $\pm$ 0.05	0.623	0.431	0.003
Instructive	4.59 $\pm$ 0.07	4.55 $\pm$ 0.05	0.204	0.652	0.001
Perceived learning	4.45 $\pm$ 0.07	4.48 $\pm$ 0.05	0.202	0.653	0.001
Education material clarity	4.41 $\pm$ 0.07	4.34 $\pm$ 0.05	0.731	0.394	0.004
Learning environment	4.33 $\pm$ 0.07	4.38 $\pm$ 0.05	0.271	0.604	0.001
Perceived expert competence	4.64 $\pm$ 0.06	4.60 $\pm$ 0.05	0.189	0.664	0.001
Expert clarity	4.46 $\pm$ 0.07	4.38 $\pm$ 0.05	0.890	0.347	0.005

Note The degrees of freedom for the reported F values are (1.185)

Table 3 Average students scores for each dimension feedback across the sessions

Dimensions of student feedback	Sessions			F	P value	Partial eta squared
	1	2	3			
	(Mean $\pm$ SE)	(Mean $\pm$ SE)	(Mean $\pm$ SE)			
Interest	4.44 $\pm$ 0.05	4.28 $\pm$ 0.06	4.50 $\pm$ 0.05	10.40	0.000	0.053
Enjoyment	4.22 $\pm$ 0.05	4.16 $\pm$ 0.06	4.42 $\pm$ 0.05	12.92	0.000 ^a	0.065
Instructive	4.60 $\pm$ 0.04	4.51 $\pm$ 0.05	4.60 $\pm$ 0.05	2.88	0.059 ^a	0.015
Perceived learning	4.42 $\pm$ 0.05	4.41 $\pm$ 0.05	4.56 $\pm$ 0.05	5.31	0.005	0.028
Education material clarity	4.35 $\pm$ 0.05	4.31 $\pm$ 0.06	4.47 $\pm$ 0.05	4.95	0.008 ^a	0.026
Learning environment	4.35 $\pm$ 0.05	4.26 $\pm$ 0.06	4.46 $\pm$ 0.05	7.05	0.001	0.037
Perceived expert competence	4.65 $\pm$ 0.04	4.55 $\pm$ 0.05	4.66 $\pm$ 0.04	5.46	0.005 ^a	0.029
Expert clarity	4.43 $\pm$ 0.05	4.29 $\pm$ 0.06	4.53 $\pm$ 0.05	11.88	0.000 ^a	0.060

Note 1 = Sports science expert; 2 = Communication expert; 3 = Sport psychologist; Several "sessions" average scores, for each dependent variable, differ from the other average scores according to the post-hoc tests (using Bonferroni correction). In details "Interest": (1 $\simeq$ 3 > 2); "Enjoyment": (3 > 1 $\simeq$ 2); "Instructive": (1 $\simeq$ 2 $\simeq$ 3); "Perceived Learning" (3 > 1 $\simeq$ 2); "Education material clarity" (1 $\simeq$ 2 $\simeq$ 3); "Learning environment" (3 > 1 $\simeq$ 2); "Perceived expert competence" (1 $\simeq$ 3 > 2); "Expert clarity" (3 > 1 > 2)

^aHuynh-Feldt correction employed

most enjoyable, with the highest perceived learning and the best learning environment. Finally, according to students' feedback, the sport psychology was clearer than the sports science expert, which was clearer than the communication expert. No significant differences emerged between sessions for what concerns the education material clarity and values of instructiveness.

Note that the repetition of the analysis with the outliers showed slightly different findings only in terms of the main within-subject effects of univariate ANOVA. Indeed, the inclusion of outliers showed no differences across sessions in terms of perceived learning, learning environment, and speaker's perceived competence. The other results, instead, were not different from the findings without outliers.

Summarizing, the present study results showed that the attendance rate was higher for remote teaching modality. At the same time, there are no differences between face-to-face and remote modalities in terms of students' feedback. Lastly, there were differences across sessions regarding interest, enjoyment, perceived learning, learning environment, perceived expert competence, and expert clarity scores. Overall, the sport psychologist and the sports science expert were evaluated through higher scores in these dimensions than the communication expert.

4 Discussion

During the COVID-19 emergency, education has shifted from face-to-face to online modality to avoid large gatherings and crowds for blocking the transmission of the virus.

The present study aimed to evaluate the differences between face-to-face and emergency remote teaching using a media literacy intervention [14, 15] in terms of engagement, perceived learning, and the goodness of the learning environment perceived by students. The importance of media literacy interventions relies on the fact that the widespread use of multimedia devices by young people has created the need to develop a critical attitude to grasp the methods of persuasion present in the messages conveyed by the media [19]. Besides, the restrictive measures adopted by many countries to prevent the spread of COVID-19 (e.g., quarantine, lockdown, social distancing) led many students to spend most of their time online, a situation that may conduct to problematic internet use [20–22], with a potentially stronger influence of the media, and overall risk of psychological distress [23].

The study focused on the sessions delivered by experts within the “Media Literacy intervention” project (funded by the Italian Ministry of Health, Anti-Doping Vigilance Committee). The three experts were respectively a former elite athlete expert in sports science, a communication expert, and a sport psychologist. The study results showed that, overall, there were no significant differences between face-to-face and emergency remote intervention. It is essential to note that the current study assessed an intervention that was very time-bound and different in its purpose than an established school curriculum administered in an online version. Nevertheless, these findings are similar to the previous literature analyzing the students' satisfaction with

online teaching versus the face-to-face modality [24]. Accordingly, the present study supports the idea that an emergency remote intervention, given its limitations in planning, interaction, and social communications, can still provide the students a high engagement, high perceived learning, and a positive learning environment comparable to in-person teaching. Such aspects are relevant for the educational process's effectiveness (cf. [13]).

Regarding the differences in students' feedback scores considering the intervention sessions, the results showed that students overall had given higher scores to the sport psychology and the sports science expert than the score attributed to the communication expert. It could be speculated that the schools' curricula (i.e., sporting high schools) influenced the students' feedbacks, leading them to attribute higher scores to those experts (i.e., the sport psychologist and the sports science expert) who dealt with topics closer to the ones that student usually study. However, it is essential to note that none of the experts scored less than an average of 4 in each of the eight students' feedback dimensions, showing that the sessions described in the current study were highly evaluated. Given all our results, the prolongation of the pandemic, and the frequent school interruption necessary to prevent contagion, future studies should also be focused on the efficacy of online education versus the face-to-face one.

Our study's limitations are related to (a) the generalization of the results to the entire student population, considering our sample's specificity and the intervention. Indeed, the described intervention referred to a specific psychoeducational theme different from traditional school curricula; (b) the small number of the lessons, considering that other studies evaluated entire academic and school curricula in a longitudinal design; (c) our data could be influenced by a so-called "ceiling-effect", that could potentially hide slight differences between teaching modalities. It could probably depend on the tool employed for the present students' feedback; (d) the inequality of sample size between teaching modalities. Indeed, the remote teaching group had almost twice as many students as the face-to-face group. Although it was a limitation for the current study, this result should be taken in account when considering the difference between these two modalities in terms of attendance rate.

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Relationship Between Teacher's Teaching Expertise and Digital Literacy



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Abstract The purpose of this study is to find out how teachers' digital literacy affects teaching expertise. A teacher's teaching expertise is defined as a teacher's classroom teaching behavior observed as the tool used in International Comparative Analysis of Learning and Teaching (ICALT) research. In addition, digital literacy was measured using the TK survey tool developed by So Yeon-hee (2013) for Korean teachers according to the TPACK model. A learning model for Korean elementary school teachers' teaching expertise was constructed using the observation data of 297 elementary school teachers. As a result of applying this model to the class observation data of 18 elementary school teachers who participated in participatory observation, it was found that the digital literacy of elementary school teachers positively influenced the teaching expertise.

Keywords Teaching expertise · Digital literacy · ICALT · TPACK · Decision tree analysis · Machine learning

1 Introduction

Medley [1] and Schalock et al. [2] found that teaching expertise is a system of competence as a whole in which a teacher possesses a single ability such as knowledge and skills. It is said that it has a complex and professional meaning that includes instructional performance and instructional effectiveness. In addition, Yu [3] stated that teaching expertise can be defined as a teacher's ability to use expertise and skills to prescribe a means for achieving instructional objectives in terms of 'professionalism'.

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With the advent of the knowledge and information society in the twenty-first century, there has been a growing voice that new abilities should be added to the teaching expertise of teachers. Teacher competency required in the digital society includes creativity, problem solving, communication, social competence, flexibility, technology literacy, ethical awareness, and passion in the basic literacy area. Relationship formation, instruction design and development, learning affordance creation, evaluation and reflection, external cooperation, and work performance management should be included [4]. In particular, it is noteworthy that technology literacy was included in the basic literacy area. This is one of the competencies that should be equipped even for instructors who foster a new generation of learners who have easy access to the use of cutting-edge technology, and that the correct use and effective use of technology is one of the competencies that must be equipped.

Technology literacy here can be understood as digital literacy in the end, and accordingly, how the teacher's digital literacy relates to the teacher's teaching expertise becomes an interesting research topic.

2 Theoretical Background

2.1 *Teaching Expertise and Its Measurement*

Teaching expertise can be defined differently depending on whether the approach is focused on 'lesson' or 'professionalism'. Here the lesson is defined as the classroom teaching in the conventional elementary or secondary schools, and the professionalism as the designating teaching job as a profession like a medical doctor. In the case of focusing on professionalism, teaching expertise is defined as a teacher's ability to use the knowledges and skills necessary to achieve the purpose of the lesson. However, in measuring the expertise objectively, it depends on how you view the constituent factors of the lesson. Prior studies have generally classified areas such as the planning-execution-evaluation stage according to the time flow of the lesson, or attempted an environmental approach including the teacher's traits approach, interactive approach, and school situation that influences on the lesson. Among them, the study of the International Comparative Analysis of Learning and Teaching (ICALT) has developed and utilized an observation tool to scientifically analyze the classroom behavior of teachers that can be directly observed during the classroom teaching [5].

The ICALT approach began in the 1990s by a research team at the University of Groningen in the Netherlands. They derived 151 instructional behaviors of teachers that could be observed by meta-analyzing prior studies on teaching effectiveness, and classified the derived instructional behaviors into nine categories [6]. The nine categories includes (1) Provide opportunities to learn minimum goals, (2) Check student achievement, (3) Special measures for difficult students, (4) Learning climate, (5) Classroom management, (6) Clear and structured instruction, (7) Intensive and activated learning, (8) Teaching learning strategies, and (9) Differentiated instruction:

the latter six ones only are adopted as the observable domains of expertise since the former three cannot be easily and objectively observed in the classroom teaching.

2.2 *Digital Literacy for Teachers*

Teachers in the twenty-first century are required to have expanded and integrated competencies than existing roles. It continues to argue that the level of computer use in the classroom should be improved in order to devise an appropriate curriculum and evaluation method rather than the mechanical skills of teachers and provide better learning opportunities for students [7].

Mishra and Koehler [8] proposed a model called TPACK (Technological Pedagogical and Content Knowledge) to explain the interactive knowledge about technology, education, and content. This framework emerged as a necessary competency for teachers in situations where the use of ICT in the classroom is inevitable due to the technological development of the twenty-first century. CK (content knowledge), PK (pedagogical knowledge), TK (technological knowledge), but also PCK (pedagogical content k.), TCK (technological content k.), TPK (Technological pedagogical k.) and TPACK (technological pedagogical and content knowledges) organically interacts together.

Research on the development of tools to measure teachers' TPACK levels has also been conducted from various angles [9–11], and is used in various studies on TPACK. Based on the TPACK model, many studies have been conducted to measure the skills and subject integration knowledge of pre-service teachers [12–14].

It is clear that the ICT element in the digital literacy is an important element for the teacher's teaching expertise. However, TPACK is applied differently in teaching content knowledge for each subject, and the method of measurement is difficult due to different approaches such as teaching models. So [15] developed a measurement tool based on TPACK, consisting of five areas: perception of digital-based technology, general knowledge, utilization knowledge, pedagogical knowledge, and education content knowledge as a digital literacy required for teachers.

2.3 *Relationship Between Digital Literacy and Teaching Expertise of Teachers*

The research on how technology and digital literacy affect teaching expertise can be found in research related to classroom effectiveness in many studies. Jung [16] analyzed the background variables related to teachers that affect TPACK while investigating teacher factors that affect TPACK of elementary and secondary teachers. Through a review of prior research for this study, 9 teacher factors of gender, teaching experience, training experience, educational beliefs, self-efficacy, attitude

to technology, ambient pressure, motivational support, and technical support were extracted.

From the various studies, it can be concluded that technology is helping to improve teaching expertise. However, there are few studies on how digital literacy affects the expression of teaching expertise in the scene of class execution. Digital literacy factors that influence the expression of teaching expertise need to be studied in a scientific method to examine how the factors of digital literacy relates in the real classroom teaching environment.

3 Research Method

3.1 Research Samples

The study samples used in this study consisted of an experimental group of 18 people and a learning sample group of 297 people. To find out the relationship between teaching expertise and digital literacy, the experimental group selected for this study was grouped by 18 elementary school teachers in Daejeon. They agreed to film the class video, and their teaching expertise was measured with the lesson video by ICALT observation experts, and the same teacher's digital literacy was also measured. The learning sample group was used as a criterion for comparing the level of teaching expertise of the 18 experimental groups. In this study, 297 elementary school teachers extracted from the database of the ICALT Research Center in Korea were composed of teachers in Daejeon, the same as the experimental group.

3.2 Measurement Tools

As a tool for measuring teaching expertise, ICALT was used. ICALT was developed by the research team of Professor Van de Grift of the University of Groningen in the Netherlands for the purpose of measuring the quality of teaching behavior, which is both pedagogical and didactical of teachers. The ICALT measurement tool consists of six domains hierarchically structured to measure the expertise of teacher behavior according to the teacher development stage. Six domains of expertise are consisted of high inferential 32 questions which are associated with low inferential 114 examples of teaching behaviors.

The teacher's digital literacy measurement tool is based on the TPACK(Technological Pedagogical and Content Knowledge) model presented by Mishra and Koehler [8], which explains interactive knowledge about technology, education, and content. This study employed the customized tool for Korean teachers [15] that reconstructed into 5 areas, 12 technology knowledge, and 35 measurement questions.

3.3 Analytical Methods

A two-stage cluster analysis was conducted using an unsupervised learning algorithm to check whether there is a sub-specialty cluster in each teaching expertise domain according to the teacher development stage. In addition, through the analysis of the decision tree, the variables of teacher backgrounds that influence the classification of the expertise cluster in each lesson specialty area were extracted, and a classification model was derived. And this classification model was applied to 18 participants' observational data, and the difference in digital literacy between clusters of the participatory observation data classified through the scoring was analyzed. In order to find out whether the digital literacy competency can explain this group classification, the average value of each lesson of expertise type and the average value of each area of digital literacy were quantitatively compared. Digital literacy factors influencing were explored.

4 Results and Discussions

4.1 Overall Descriptions About Digital Literacy Level of the Sampled Students

A decision tree analysis was conducted to search for teacher background variables that influence the classification of teaching expertise clusters by domains according to the teacher development stages. The analysis results are as follows (Table 1).

First, a two-stage cluster analysis by ICALT domains of teaching expertise for the 297 learning sample group returned in two or three sub-clusters as shown in Table 1, domain 1, 2, and 7 by two clusters and domain 3, 4, 5, and 6 by three clusters.

Decision tree analysis was conducted to confirm the predictive factors for the background variables that affect the expertise cluster classification shown above. By the analysis method, the higher located node, shown in Fig. 1, is defined as the more important classifying variable.

Decision tree analysis from domain 2 to 6 were also conducted in the same way as in domain 1. Their result figures are omitted here and the node classification results are shown as in Table 2.

4.2 Teachers' Digital Literacy and Its Effects on Teaching Expertise

The digital literacy for 18 participating teachers is shown in Table 3.

Table 1 Teaching expertise clusters in learning sample

Expertise domains	Clusters	Mean	Std. dev.	min	max
1 (learning climate)	1	3.85	0.20	3.50	4.00
	2	2.92	0.27	2.25	3.25
2 (classroom management)	1	3.81	0.22	3.50	4.00
	2	2.93	0.31	2.00	3.25
3 (clear instruction)	1	3.83	0.17	3.57	4.00
	2	3.28	0.11	3.14	3.43
	3	2.73	0.27	2.00	3.00
4 (activated learning)	1	3.63	0.21	3.43	4.00
	2	3.12	0.11	3.00	3.29
	3	2.60	0.26	1.86	2.86
5 (learning strategies)	1	2.90	0.08	2.83	3.00
	2	3.49	0.28	3.17	4.00
	3	2.41	0.27	1.50	2.67
6 (differentiated instruction)	1	3.26	0.08	2.83	3.00
	2	3.06	0.28	3.17	4.00
	3	2.63	0.27	1.50	2.67

When analyzing the digital literacy of participating teachers, as the number of cases was small and the homogeneity of variance was not the same, a nonparametric verification analysis was conducted to verify whether there is a difference according to the teacher's background variable for each sub-areas of digital literacy. Among the various background variables, it was found that there was a significant difference in the general knowledge of digital literacy according to the teacher's career. However, there were no significant differences in digital literacy according to gender, academic background, observation grade, and subject. When examining the digital literacy of teachers by background variable, the result was found to be insignificant in most areas. This means that the background variable of teachers is not related to digital literacy, or that the difference in the capabilities of digital literacy among all members is not very large.

Main research question of this study is to find the relationship between teaching expertise and digital literacy of 18 experimental teachers, by comparing the scores of their teaching expertise with digital literacy. Results in Table 4, significant differences of ICALT teaching expertise in only Domain 4, activating teaching and learning, were found among groups of tech_perceptions, tech_pedagogical knowledge, and tech content knowledge of digital literacy.

In parallel with this, the result of an additional analysis using the decision tree technique to determine which variable of digital literacy affects teaching expertise is shown in Fig. 2.

In domain 4 of teaching expertise, activated teaching and learning, the type of teacher is divided into two types of node 1 and node 2 depending upon the differences

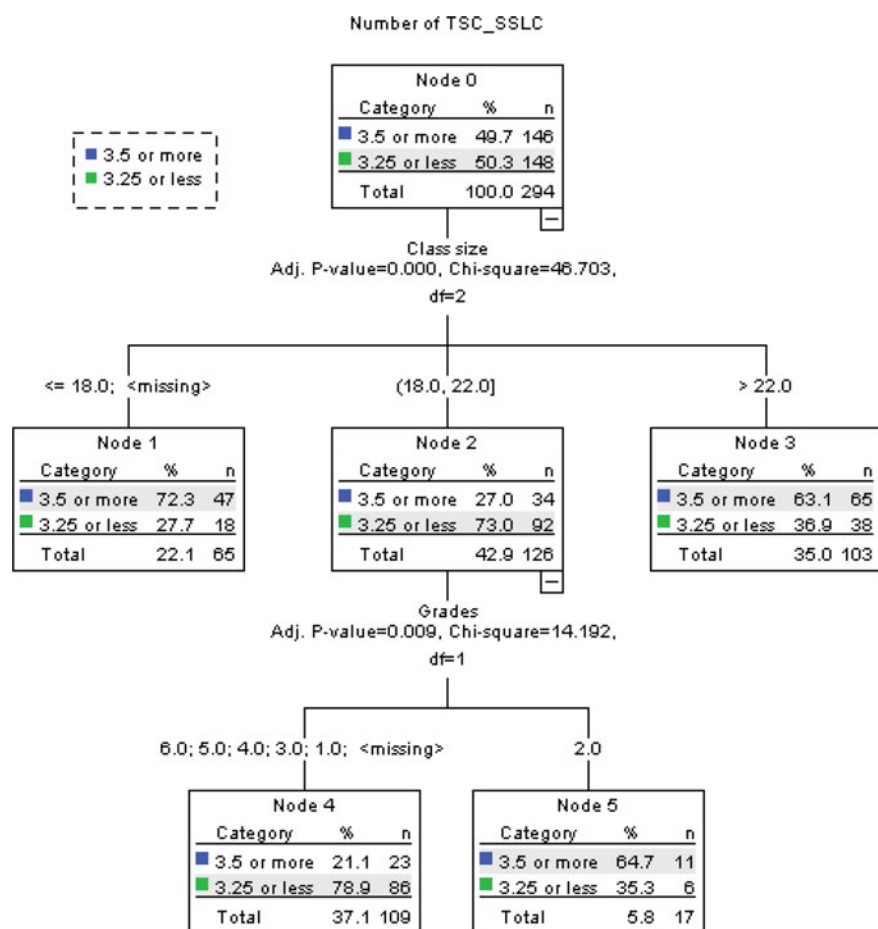


Fig. 1 Decision tree for ICALT domain 1

of digital literacy by its sub-area of technology of technology perception (chi square = 8.471, $df = 1$, $p = 0.032$). This means that the higher the awareness of technology, the higher the teaching expertise in domain 4 for activating learning in classroom. This domain is measuring teaching expertise based on students' immersion and intensive engagement in classroom, which can be concluded that good perception and pedagogical application of teacher's digital knowledge contribute to improve skills and expertise in related domains of teaching.

Table 2 Decision gree for ICALT domains

ICALT domain	Order	Nodes	Background	χ^2	d/f	p_value
1 (learning climate)	1	3	Class size	46.703	2	0.000
	2	2	Grades	14.192	1	0.009
2 (classroom management)	1	4	Class size	26.584	3	0.000
3 (clear instruction)	1	3	Class size	48.366	3	0.000
4 (activated learning)	1	3	Class size	29.251	2	0.000
	2	5	Teacher career	32.361	4	0.004
5 (learning strategies)	1	3	Class size	30.411	4	0.000
6 (differentiated instruction)	1	4	Class size	23.814	2	0.000
	2	4	Grades	19.600	6	0.011
		2	Subjects	15.287	1	0.008

Table 3 Digital literacy scores (N = 18 teachers)

Areas of Tech_knowledge	Mean	Std. dev.	min	max
Total	3.6058	0.43193	2.58	4.42
Tech_perceptions	3.9292	0.65367	2.24	4.57
Tech_general knowledge	3.1508	0.83941	1.57	4.43
Tech_utilization k	3.4444	0.98352	2.25	5.00
Tech_pedagogical k	2.8519	0.71031	1.67	4.17
Tech_education contents k	4.6528	0.37514	3.75	5.00

Table 4 Differences of digital literacy by ICALT exptercise clusters in domain 4

Areas of Tech_knowledge	Clusters	N	Mean	Std. dev.	t	p
Tech_perceptions	2	16	3.85	0.65	-4.394	0.001***
	3	2	4.57	0.01		
Tech_general k	2	16	3.18	0.89	0.811	0.463
	3	2	2.93	0.30		
Tech_utilization k	2	16	3.52	1.00	0.952	0.481
	3	2	2.88	0.88		
Tech_pedagogiac l k	2	16	2.94	0.70	3.801	0.007**
	3	2	2.13	0.18		
Tech_education contents k	2	16	4.61	0.38	-4.155	0.001***
	3	2	5.00	0.00		

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

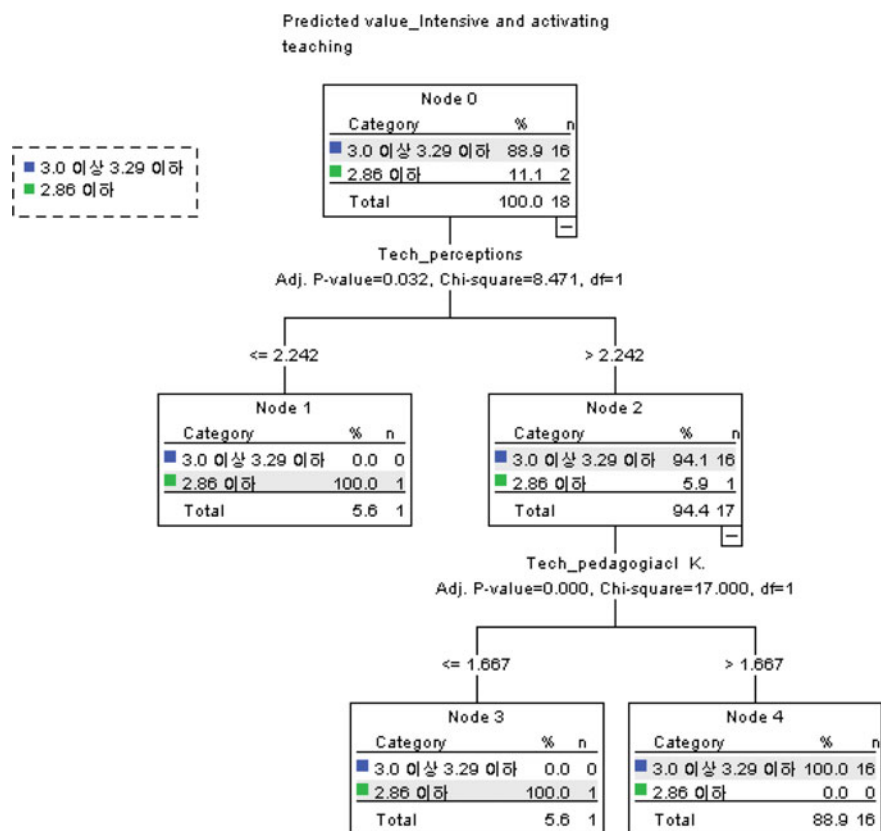


Fig. 2 Decision tree of ICALT domain 4 affected by digital literacy

5 Conclusions and Suggestions

In this study, in order to analyze the factors of digital literacy affecting the teaching expertise, 18 elementary school teachers' teaching expertise and digital literacy were measured. Main findings are as follows.

First, it was confirmed that 297 learning group teachers were formed into 2 to 3 clusters according to the scores for each domain of teaching expertise. The two clustered areas are the domain 1, safe and stimulating classroom climate and domain 2, efficient classroom organization. Domain 3, clear instruction, domain 4, activating teaching, domain 5, teaching learning strategies and domain 6, differentiated instruction are divided into 3 clusters.

Second, as a result of analyzing the characteristics of the teaching expertise of the learning group teachers using the decision tree technique, the teacher background variables that influence the teaching expertise in each area were extracted. The most

powerful background variables extracted were the classroom size, teaching career. Grades and subjects of teaching.

Third, digital literacy had a significant effect on only the domain 4 among 6 of teacher's teaching expertise measured by ICALT tool. The sub-factors of teacher's digital literacy that influence intensive and active instruction are technology_perception, technology_pedagogical knowledge, and technology_education content knowledge.

In terms of technology perception, technology pedagogy, and content knowledge, the digital literacy of teachers has an influence on teaching expertise. However, in the factors of general knowledge of technology and knowledge of technology use, the influence on teaching expertise was not revealed, which would mean that the digital literacy measures of teachers' digital literacy and technology use are already in line with other teacher backgrounds. This suggests that the daily use ability and technical problems of digital literacy are factors that do not have a separate effect on teacher teaching expertise, and that there is no significant difference by type of teacher development stage. However, technology education knowledge and technology education content knowledge, which are the sub-factors of digital literacy that are influencing the development of teaching expertise, require more attention and professional development depending on the type of teachers, and are empirical about the digital literacy measurement tools of teachers.

This supports the theories of existing scholars that the technical ability of the teacher or the ability to use ICT to improve students' ICT ability, and it can be seen that the background variables of teachers are applied to the teacher's teaching expertise through the diagnosis tool of recent digital literacy. However, when looking at the results of this study, the teacher's digital literacy should be measured in its specific manner differently from a general one so that it can contribute more effectively to the higher level of teaching expertise.

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Inspiring the Organizational Change and Accelerating the Digital Transition in Public Sector by Systems Thinking and System Dynamics Approaches



Nunzio Casalino, Stefano Armenia, and Primiano Di Nauta

Abstract Digitalization is enabling a higher productivity across the organizations, facilitating the creation of new and better products and services with fewer resources in the workplace. The progressive increase of digitalization of documents and processes can allow the digital transformation by the redesigning of all organizational processes to increase efficiency and improve structural performances. Digitalization is not a mere replacement of a paper-based document with its digital version. It means designing and managing all organizational processes in an integrated and collaborative way changing the business models, operational processes, and customer experiences. To understand how digital transformation deeply impacts to the organizational processes in Public Administrations, we utilize the Systems Thinking (ST) and System Dynamics (SD) approaches. They permit to analyze the benefits that can be pursued through the digitalization of processes, in terms of organizational change, productivity and economic savings. For those by the digital transformation it will be possible to offer to organizations several opportunities to effectively improve the working processes and outcomes.

Keywords Digital transformation · Organizational change · Digital transition · System dynamics · System thinking · e-learning · Public sector · Business innovation

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1 Introduction

Digital transformation is the integration of digital technology into all areas of a business, fundamentally changing how to operate and deliver value to customers. It is also a cultural change that requires organizations to continually challenge the status quo, experiment, and reach results. In the last few years, digital transformation has become a new relevant research stream [1–5] and the debate about dematerialization is still ongoing [6]. The dematerialization of documents, the progressive increase of digital documents and processes, are key drivers of organizational digital transformation.

Digital transformation cannot occur without eliminating paper documents and dematerialization without a new digital technologies' strategy, which inevitably requires a new organizational business model. Indeed, there is no digital transformation without dematerialization [7]. In Italy, for years now, public administrations have been at the center of an impressive series of digital changes aimed at creating structures oriented towards the culture of effectiveness and efficiency. These changes have led to e-Government, which is to say, strong computerization of the public administration, both CPA (central public administrations) and LPA (local public administrations) levels.

In Europe, the implementation of e-Government has required the dematerialization of all documents and the redesign all organizational processes to increase efficiency through cost reduction and rise effectiveness by improving the services offered. Besides, in the last three years, some new educational activities aim to meet the training needs of public administrations, companies, and other organization which plan to acquire or train the profiles of the Digital Transition Manager and the Manager of Documents Management System, providing innovative knowledge on organizational, technological, and legal skills [8].

To better understand how dematerialization implied a digital transformation and, therefore a more substantial effect on the Italian Public Administration's organizational processes, we use Systems Thinking (ST) and System Dynamics (SD) approaches. In particular, the purpose of our research is to highlight, in the context of a systemic approach to digital transformation through the right training activities, modeling and simulation, the advantages that Public Administrations can carry out. This by the digitalization of their processes, in terms of organizational changes and economic savings.

2 The Impact of Digital Transformation and Digitalization in the Organizational Processes

Although managers often use digitalization as an umbrella term for digital transformation, the terms are very different. Digital transformation requires a much broader adoption of digital technology and cultural change. Digital transformation is more about individuals than it is about digital technology, and it needs organizational

changes that are customer-centric, backed by leadership, driven by radical challenges to corporate culture, and the leveraging of technologies that empower and enable employees.

The terms Digital Transformation have not a unique meaning [9]. New digital technologies demand different mindsets and skillsets than previous waves of transformative technology [10] that in turn transform the organizations. At the same time, the term “transformation” expresses the comprehensiveness of the actions that need to be taken when organizations face these new technologies. Therefore, digital transformation goes beyond merely digitizing resources, it will take place when organizations embrace all the potential of social learning in the design and the process of delivering contents, and it involves a company-wide digital strategy. To ensure that an organization captures the business value of a digital transformation, it should carefully formulate a digital transformation strategy that coordinates the many independent threads of it, supporting the navigation into complexity and ambiguity of identifying its own digital “sweet spots” [9]. Every organization needs to manage documents to carry out and support its operational processes [11], regardless of what the organizational scope is or what is the contest in which it operates. Most of the valuable information in organizations is in the form of documents, such as business forms, reports, letters, memos, policy statements, contracts, agreements, etc. [12]. Their gathering, storage, management, and research represent for an organization a considerable cost that becomes even more significant if they are of paper.

In our research, the concept of dematerialization indicates the progressive increase in digital and computerized management of documents and processes within public and private bodies, with the consequent takeover of dedicated solutions at the expense of traditional (paper) supports. Digital documents allow money, time, and labor savings since, if they are sent, received and stored in electronic format, then they do not have to be transcribed, recorded, inserted in files, classified, moved, and searched between cabinets, drawers, folders, and boxes. Therefore, the processes become more efficient. The shift towards document dematerialization and the practical implementation of the digitalization process requires a step-by-step process and the need for a progressive internal adaptation. It is a very complex phase, which involves the evaluation of many critical components: redefining some stages of the document life cycle and its processing, as well as reorganizing the management, processing, storage, and research phases in the archive. As a general macro-process in document management, four distinct sub-phases can be identified (Fig. 1). Such phases encompass the classic activities typical of document management as presented on Fig. 1.

3 Interoperability and Effective e-Government

The EU’s 2019 eGovernment Benchmark Report underlines that the gap between countries remains significant and Europe, despite good results on digitalization, is penalized by the low use of online services. It shows a situation of general evolution of the digitalization of public services in European countries, but in a context of

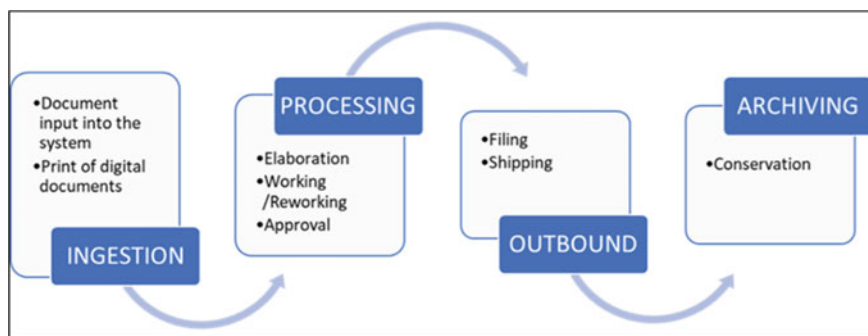


Fig. 1 The process of document management

insufficient security and poor ease of use by citizens, especially in consideration of a presence that is not yet adequate of digital skills. The new quality public digital services and really easy to use, not only avoid requesting them several times from citizens (respecting the “Once only” principle also stated in the Tallinn declaration), but also make certain flows completely automated, reducing for citizens and businesses the need to continuously interact.

The European Commission defined e-government as: “the use of information and communication technologies in public administrations combined with organizational change and new skills in order to improve public services and democratic processes and strengthen support to public policies”. In another communication, n.179/2016 of April 19th, 2016, the European Commission established the basic principles on which the action for e-government must be based for the years 2016–2020. The action plan was defined as follows: “By 2020, public administrations and public institutions in the European Union should be open, efficient and inclusive, providing borderless, personalized, user-friendly, end-to-end digital public services to all citizens and businesses in the EU. Innovative approaches are used to design and deliver better services in line with the needs and demands of citizens and businesses. Public administrations use the opportunities offered by the new digital environment to facilitate their interactions with stakeholders and with each other”.

The paper digitalization establishes a great opportunity for the Public Administration to spread several benefits and to resolve difficulties such as the extraordinary expenses of treatment, space, searching, packing, and timing [13]. It can be applied in the reengineering of organizational processes and can be a goal that could rapidly reached.

4 The Role of the Digital Transition Manager

The Digital Transition Manager (DTM or RTD in Italian) is a new managerial profile whose main role and function are that of operationally guaranteeing the digital transformation of the administration, coordinating it in the development of digital public services and in the adoption of new transparent and citizens' open relationships models. Today each public administration is required to entrust to a single general management office, allowing the transition to the digital operating modes and the consequent reorganization aimed at creating a digital and open public administration (Table 1).

Table 1 Responsibilities of the DTM-RTD office inside PAs

Area of responsibility	Activities for which the DTM-RTD Office is responsible
Planning	Planning and coordination of relevant initiatives for the purposes of a more effective provision of online services to citizens and businesses through the tools of application cooperation between public administrations, including the preparation and implementation of service agreements between administrations for the creation and sharing of cooperative information systems
	Planning and coordination of the process of diffusion, of identity systems and digital domicile, e-mail, digital protocol, digital signature, and of the rules on accessibility and usability as well as the process of integration and interoperability
	Planning and coordination of the purchase of IT, telematic and telecommunication solutions and systems, in order to ensure their compatibility with the implementation objectives of the digital agenda
	Guidance, planning, coordination and monitoring of IT security in relation to data, systems and infrastructures, also in relation to the public connectivity system
Coordination	Strategic coordination for development of telecommunication and voice systems
	Direction, coordination and monitoring of the planned planning for the development and management of telecommunication and voice information systems
	Cooperation in reviewing the reorganization of the administration
	Direction and coordination of services development, both internal and external, provided by the administration's telecommunications and information systems
Other	Continuing analysis of the consistency between the organization and the use of information technologies, in order to improve user satisfaction and the quality of services as well as to reduce the time and costs of administrative action
	Access of disabled people to IT tools and promotion of accessibility

5 The First Innovative Online Learning Initiative for the Digital Transition Manager

Innovative educational and training initiatives, based on the experiential learning techniques, have been launched from Luiss Business School in the last five years. These courses, updated to the most recent issued EU legislation, are oriented to executives of public organizations, professionals, entrepreneurs, and managers interested in acquiring, on the one hand, the necessary legal skills, and on the other, the organizational and technological competences for the digital transition or the document management system. A considerable knowledge and awareness of the most recent digital innovations applicable in existing processes and of document and content management tools, as well as their interdependencies and organizational aspects, how to choose them, adopt them and the best way to use them is therefore necessary. The new Digital Administration Code introduces a set of innovations that concretely affects the behavior, the organizational practices of the quality of services rendered. The Luiss Business School course (<https://businessschool.luiss.it/conservazione-e-dematerializzazione-documentale/>) focuses mainly on aspects such as: the methods of reorganizing processes and document flows, the organizational rationalization and simplification of procedures, the effective introduction of the IT protocol, the adoption of online filling forms, online public payments, data exchanging methods among companies, professionals and public administrations, the suitable adoption of the PEC (certified electronic mail), the ways of dematerializing documents; the evaluation of the effectiveness of document management systems, the enrichment of the contents of corporate and institutional portals in terms of transparency and traceability. The 21 modules of the learning path are the following (Table 2).

The course aims to meet the new training needs of Public Administrations, companies and professionals that plan to create the profiles of a Digital Transition Manager or a Manager of Documents' Management, transferring them innovative knowledge, experiences plus the needed organizational, technological, and legal skills (see Fig. 2).

Besides the course focuses also on: the reorganization of processes and documental flows, the rationalization and simplification of procedures, the effective use of analytics services, the IoT and cloud technologies, blockchain and privacy management, accessibility and usability in digital organizations, management of online services' security, etc. (see Figs. 3 and 4).

6 A Systems Approach to Digitalization

To better understand the organizational impacts of dematerialization in PAs, and its enabling effect on the overall digital transformation, we choose to use a systemic perspective which is provided by the System Dynamics modeling and simulation approach. Based on Systems Theory and System Thinking, System Dynamics (SD)

Table 2 The new Luiss Business School learning path for the digital transition managers

Module ID	Module title
1	The European Digital Agenda and the national digital strategy
2	Digital rights for citizens and businesses: the digital public administration
3	The current regulatory framework
4	The profile of the Digital Transition Manager (DTM or RTD in Italian)
5	The profile of the document manager
6	The economic and organizational benefits of the digital transformation
7	Management of digital archives and document management system
8	The digital document and digital files
9	Conservation systems and technologies
10	Communication tools: e-mail, certified e-mail and delivery systems
11	Electronic signatures
12	Online access to governmental documents
13	The General Data Protection Regulation (GDPR)
14	New digital technologies for the enterprises and the public sector
15	Public administration data: interoperability and new online services
16	Websites and online portals of public administrations
17	The cloud computing for enterprises and public online services
18	Digital tools for the online identification and integrated multi-service cards
19	From the document management system to effective digital working processes
20	CSCW and CMS for optimizing internal and external communication
21	The electronic invoicing to the public organizations and between professionals

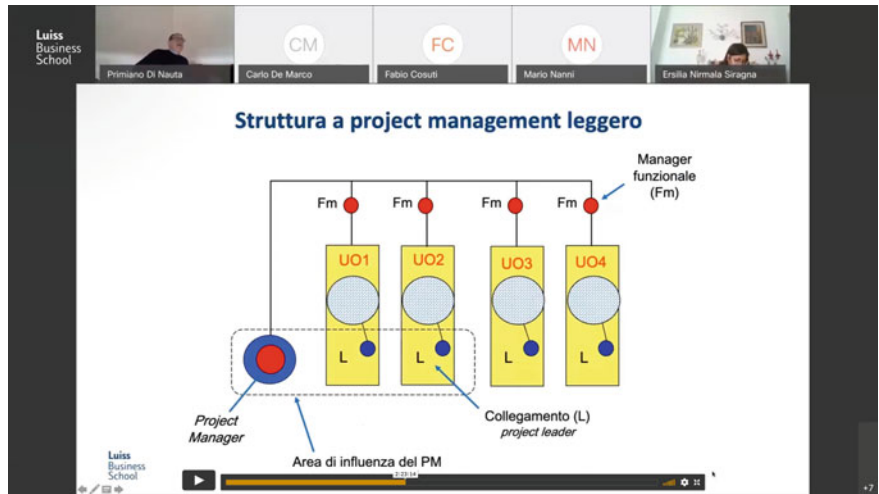


Fig. 2 A video module of a document management process design

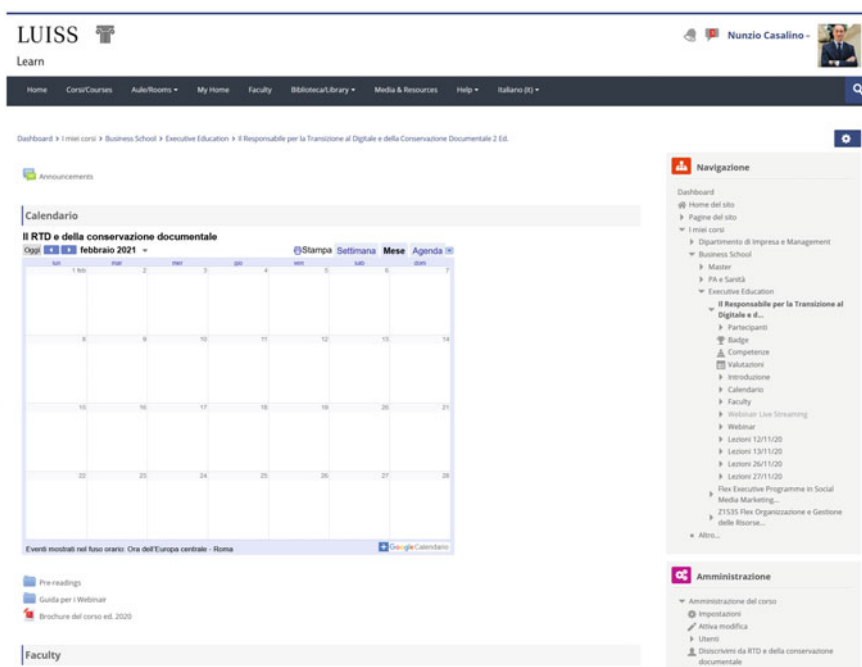


Fig. 3 The e-learning platform with its multimedia contents—<https://learn.luiss.it/>

is a computer-based modeling and simulation approach that allows defining the mathematical relations between different variables and instructs a computer to make the discrete-step computational effort of solving the differential set of equations [14]. The trends of all variables out of computer simulations are plotted over a specified period into the future. The validation of the model is based on historical data and sensitivity analyses. SD provides an understanding of the overall performance behavior of the system and the influence of the various factors to the problem to support policy design by making simulations of different scenarios.

Systems Thinking and its operational form, System Dynamics, is a way of looking at systems from a holistic point of view. Its purpose is to determine what is the system's structure and in what way the structure affects its behavior over time [14]. The use of System Dynamics allows a different approach to the analysis, over time, of the functioning of complex systems and their formalization, since it is able to manage intrinsic characteristics of real-world systems, such as non-linearity, presence of delays, self-organization, dependencies on past behavior, feedback processes and resistance to change [15]. Thanks to these features, the SD approach will allow for the definition and analysis of simulation results of a digital transformation process like document management, hence proving to be versatile and representative of operating methods, also related to non-homogeneous organizational structures. In other words,

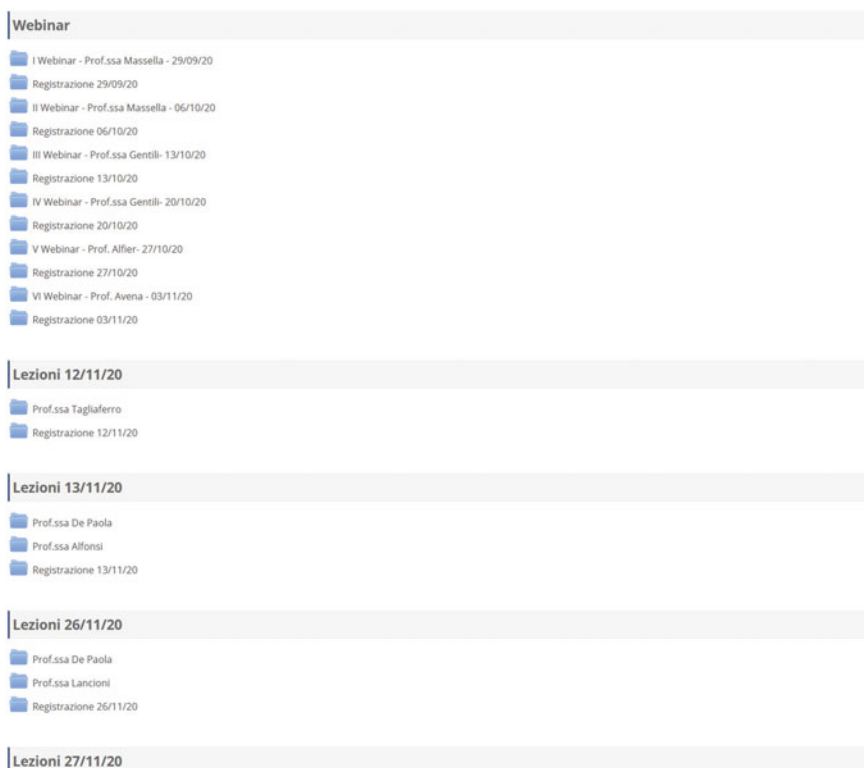


Fig. 4 The e-learning platform with its multimedia contents—<https://learn.luiss.it/>

through a systemic approach, it is possible to capture the complexity of the PAs and, therefore, the behavioral dynamics of these organizations.

By means of a System Dynamics simulation model, it is thus possible to evaluate these dynamics and make scenario analysis and eventually identify the points with high leverage towards change and improvement (policy levers). By considering the four phases in the document management process (see Fig. 5), we have examined the organizational impacts deriving by the introduction of digitalized document management, by comparing the simulation results of two different SD models. In fact, the first model describes the situation in which the organization works using only documents in paper format, from the admission to the archiving process; conversely, the second model represents an organization that creates and manages digital documents only.

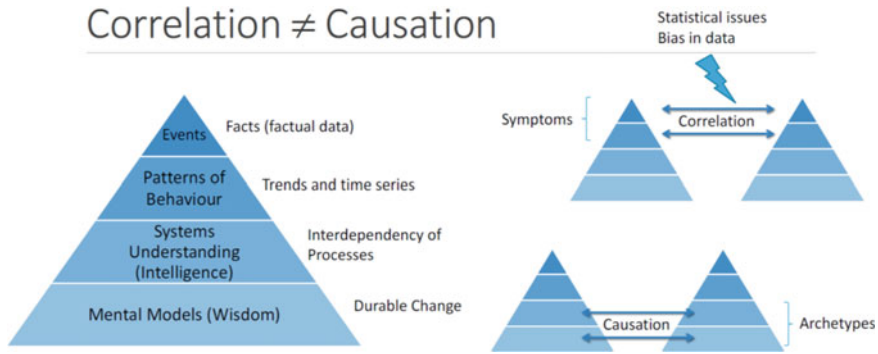


Fig. 5 Correlation differs from causation. A systems approach aimed at understanding how data is driven by the organizational structure can allow the application of effective decisions

The structural differences between the two models are due to the unavoidable organizational need to redesign processes that a digital solution needs, if compared to a traditional paper-format solution. In fact, in the digital context, it is possible to automate business processes and manage the work of resources thanks to workflow management software that allows to design, implement, and automatically manage document processes within the organization. As already mentioned, digitizing company activities means creating and managing the entire internal and external processes in an integrated and collaborative way: it is not a mere replacement of the paper document with the electronic but a moment of redesign of the entire process in order to obtain an increase in performance and an improvement in the operations carried out in their entirety [16–18]. In other words, it is an intrinsically systemic activity. The transition to digital management requires specific solutions for the organization under study with the primary objective of guaranteeing the validity, legitimacy, and compliance with the current national and international regulations of the new digital process [19, 20]. The considerable paper volumes linked to the analogic management of the documents generated daily, from the initial “ingestion” phase, through the “processing” phase, up to the “outbound” phase with the last “conservation” of the document (which generally takes place over a long period of time), has a considerable impact on all the cost items related to the logistic management of a document: the filling, the storage, the research, and the shipment. The number of produced documents also influences the storage times which, despite being already quite long due to the limits imposed by the laws, end up lasting even more due to the need of maintaining a direct control [21, 22] over the document life cycle, with a consequent increase in the costs of renting the storage rooms, spaces, security, etc.

Case Study. With the partnership of Infocert S.p.A., a leading company in the digitalization services market, we have defined and analyzed the document management process of the Public Administration. We designed two SD models representative of the management of the generic document process with the support of the

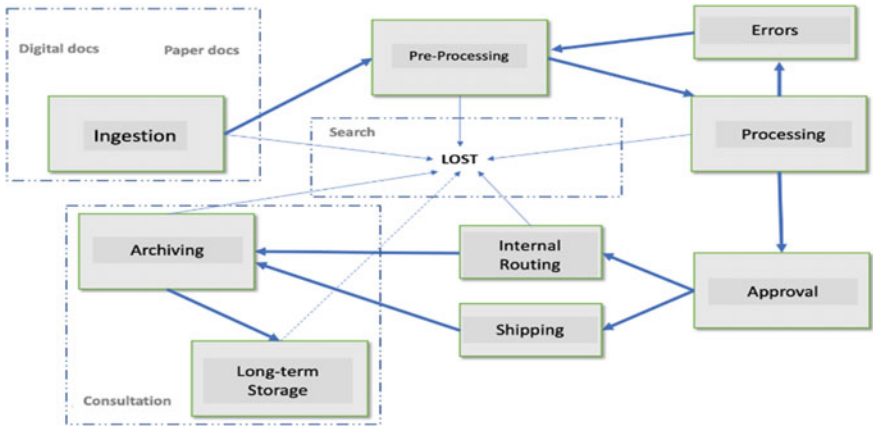


Fig. 6 An evolution of the process model aimed at introducing the various steps and states in which a document can be found inside a typical document workflow

Powersim Studio® software. The first model, called “As-Is”, represents the situation in which the reference organization operates, is a traditional model of “paper-intensive,” where there is no implementation of solutions for the dematerialization of document management (see Fig. 6).

The second model, called “To-Be”, describes the future situation. The “To-Be” model represents the situation in which an organization should tend to optimize its processes and sub-processes by increasing its performance, reducing paper consumption, and the costs associated with managing the entire system. All the sub-processes and activities have been treated as black boxes. The main objective is not to understand the specific functioning, but the behavior of each of the phases at a given input solicitation. Our conceptual model considers the aspects, issues and objects described in Table 3.

7 Simulation and Results

The work that has been carried out on the analysis of the transition to the digital management of a documented process and the dematerialization of paper documents has had as its primary objective to develop a simulation model to demonstrate the organizational advantages connected with digitalization. In qualitative terms the changes from the comparison between the two models (“As-Is” and “To-Be”) concern:

- The number of incoming digital documents to be managed. These, by hypothesis, are greater in the “To-Be” scenario and tend to increase based on the digital culture implemented by the organization, leading to an increase in dematerialization processes which in turn trigger a reinforcement mechanism on the request

Table 3 Conceptual model that considers aspects, issues, and objects

Aspects included in our model	Description
Documents	Can be paper or digital
Human resources	Employees are the core point for the development of effective dematerialization processes
Processing	Document management activity that includes both inbound documents, the incoming flow from both outside and inside the company, as well as outbound documents, that is shipping, archiving, waste
Management	Refers to long-term management and current management, which concerns the definitive and non-definitive documents, respectively
Archiving	Allows daily and non-current documents to be consulted
Preservation	Maintenance of papers over the long term, which refers to an extended time horizon, both for legal reasons and for the value of the documents
Costs	Linked to all activities related to materials (from the use of the card for reproduction to the cost of shipments to the cost of research, etc.). They represent the main advantage derived from the implementation of digitalization policies that bring numerous savings
Time	Linked to the timing of each activity, reducing the costs related to the loss of efficiency (a concrete example is a time required for the search, or the waiting time to receive an authorization signature);
Productivity	Refers to the number of documents in a time interval, which each employee can process and work. This indicator, which is a fundamental element for the continuity and speed of the management flow, with the new solutions has a value almost higher than 100%

for digital documents which inevitably changes the flow of information and the ways of working within the organization;

- In the ingestion phase, in the “To-Be” context, the printing processes of documents are smaller, making the flow of information more fluid;
- In the “To-Be” context, printing costs are almost totally zero;
- The digital filing of digital documents takes place wholly and directly automated through specific software and without the need for resources dedicated to this activity, with a saving of funds used for this activity and the time required;
- In the “To-Be” context, the transmission of documents outside takes place mainly through certified e-mails with a substantial reduction in total shipping costs.

The results of simulation showed that the reduction in total costs of documents management process is almost 60% (see Fig. 7). In detail:

- total costs from paper-form model (per year): 641.061 Eur;
- total costs from digital-form model (per year): 263.161 Eur.

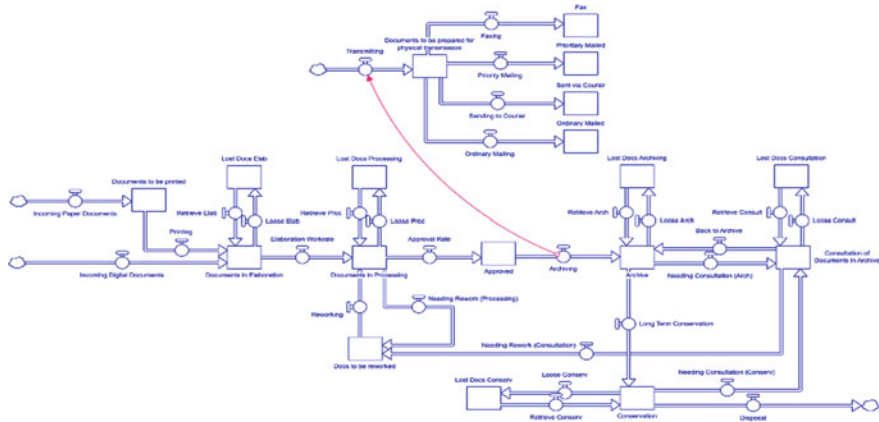


Fig. 7 As-Is workflow overall description (simplified stocks and flows process view)

It is worth noticing that the analysis took into account only the direct costs related to the activities envisaged in the four processes. It did not evaluate the savings achievable in terms of recovery of working time of human resources in the organization. It is worth mentioning that a system dynamics simulation approach allows for the development of effective decision support systems or even Interactive Learning Environments, which can be presented very effectively to the users (or decision makers) under the form of intuitive dashboards, like in the example shown in Figs. 8 and 9, where a simulator related to the decisions is presented (SUSTAIN project, funded under the EU Erasmus + Programme, 2017-1-EL01-KA203-036303, <http://sustainerasmus.eu/wp/>). In particular, it shows an example of a general dashboard that can be

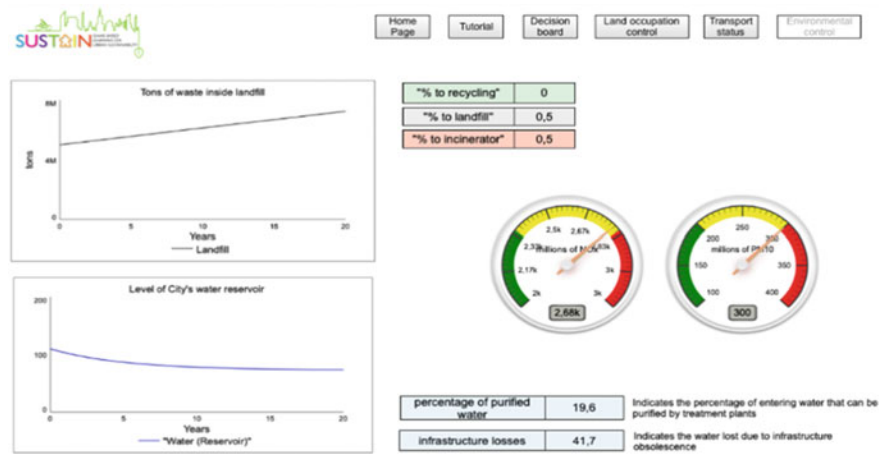


Fig. 8 Dashboards of the behaviors of variables in environmental sector—SUSTAIN project

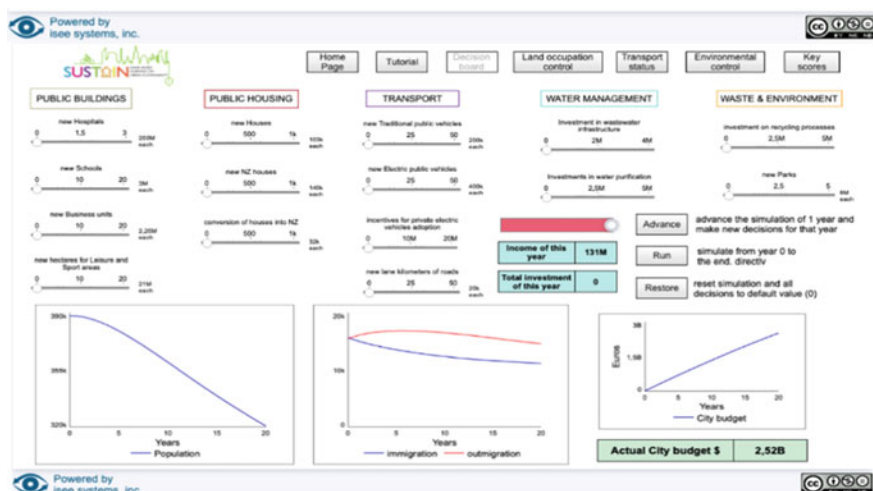


Fig. 9 Dashboards of the behaviors of variables in environmental sector—SUSTAIN project

developed by a System Dynamics model in order to keep under control, the behavior of all relevant variables (of which it is also possible to get additional insights) and eventually to steer them towards the desired goals by means of the various available decisional levers.

This extends the concept of Industry 4.0 and Digital Twin by introducing the wider concept of smart model-based governance, which implies the connection of organizational knowledge (mainly the internal databases) with the processes modelled through a system thinking approach, leading to the possibility to manage an organization by anticipating undesirable behaviors and thus economic and financial risks. It is worth highlighting some of the main smartness features of the proposed systemic simulation approach. Namely these characteristics are the following:

- **Sensing** (*collecting data and/or getting data from users and about users*): this is of course a fundamental step for building the simulation model and goes through a co-creative design approach typical of systems thinking, named Group Model building, where specific design sessions to build causal relationship diagrams are carried out with users from which useful information is extracted. Additionally, reconstructing historical data from the system will be very useful in the next quantitative modeling and simulation step;
- **Self-learning** (*self-investigation*): the group model building session mentioned before is the first, qualitative, step of the whole modeling process, where tacit knowledge is extracted and where mental models of different types of users get aligned, thus building an overall awareness on the extent of the real system being modeled, which implies an important learning process for users; the quantitative system dynamics modeling step is also very important as many of the

hypotheses derived in the previous qualitative step get put to the test, thus building additional information on the way the system should work;

- **Inferring** (*data processing and logical conclusions = obtained new knowledge*): a fundamental step is the one following the simulation of the system dynamics model: the information obtained from the simulations is reviewed and checked against the initial hypotheses for the expected behavior of the system: this is the typical step in which unexpected behavior emerges and where eventually new knowledge on the real system's behavior gets consolidated;
- **Adaptation** (*changes in behavior due to obtained new knowledge*): following the emergence of unexpected behavior, it is possible to adapt the decisions related to the available policy choices and eventually experiment, in a risk-free environment, so to develop further new knowledge on the management of the system;
- **Self-optimization** (*changes in structure due to obtained new knowledge*): if some behavior not in line with historical data emerges, it is also possible to iterate the modeling process in order to explore adapting the current model structure so to reproduce the real historical behavior (model validation);
- **Anticipation** (*use of obtained new knowledge to prepare for unexpected events and forces, to react on expected and unexpected events, and to protect entity*): this is the ultimate advantage of a system dynamics modeling and simulation approach: following the various iterations through which it is possible to go with reference to the previous features, once the model is validated it is ultimately possible to explore the outcomes of different policy choices and evaluate their potential impacts (as said, in a risk free environment), hence anticipating potential drawbacks from the implementation of wrong or counterintuitive policies.

8 Conclusions

The member countries of the European Union are committed to pursuing objectives of innovation and transformation of the decidedly ambitious e-Government processes. Their goal is to allow the relations between citizens, organizations, and institutions more transparent and effective on the basis of the principle of mobility within the single market. The digital technologies are playing a decisive role, also from a social and environmental sustainability point of view. In the first place there are the creation of “user-centric” services based on the principles of openness, flexibility, interoperability and application cooperation. These new services are helping to create a reality in which, for example, in an easier way than today a citizen can go to study in a country other than his own, create a professional activity in yet another and spend years of retirement into yet another. Alongside solutions developed for the benefit of institutions, citizens, and companies of individual countries, therefore, services to facilitate the opening of commercial activities from one country to another. Among the other possible solutions, suffice to mention the digital signature valid in all countries, the electronic identity cards recognized by all administrations, the authorization and payment services based on common standards. These

innovative infrastructures are making the single European market more competitive in the world economic scenario. Another common goal on which there has been much discussion is the use of ICT to achieve environmental sustainability purposes. The ICT can contribute to the achievement of these objectives changing the way of working and conducting business. The use of digital tools allows changes in the way public administrations deliver their activities, communicate, and provide services, but can also have a much more extensive impact such as changing the structure and culture of an organization, or engaging and integrating citizens and other partners into the co-design and co-delivery of public services. Under such a perspective, the shift towards document dematerialization and the practical implementation of the digitalization process requires a step-by-step process and the need for internal and external adaptation; it is a very complex phase that involves the evaluation of many critical components, it is essential for example:

- redefine some steps of the document life cycle;
- reorganize the management, processing, storage and research phases;
- evaluate the impact of these changes on human resources and organizational models in relation to the wide variety of existing documents;
- assess the organizational difficulties related to the use of new technological tools, the implementation of new procedures and the relationship difficulties outside the organization with reference to the end customers of the process;
- confer full legal validity to electronic substitutive documents in relation to the updated laws continuously in force.

Through a systemic approach, aimed at modelling and simulating the As-Is (paper) versus To-Be (digital) document workflow, we were able to assess the savings that are obtainable in a small context and that hence preludes to the wider savings that can be achieved from public administrations. Although it represents only a small improvement that can be implemented in the digital reengineering of administrative processes, dematerialization is a goal that could be achieved quickly. Digitalization, for its cross-cutting importance, is in fact nowadays present in a stable form into many aspects of the public system and that is why it is necessary to operate it with even more significant policy aimed at raising awareness of the use of the public digital transformation. In conclusion, our research confirms once again that the paper digitalization is a major challenge for the benefits that public administrations can realize but of course is just a single step (though pretty relevant) on their path to digital transformation. A continuous specific training activity will be needed to train Digital Transition Managers [8, 22, 23] considering that such profiles need to have a systemic perspective and systemic skills like Systems Thinking with also others focus on the tools, like decision support systems [8, 15, 24] that the public sector is slowly adopting to improve [25] its performances and offer better services to citizens.

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A Case Study on Teaching a Brain–Computer Interface Interdisciplinary Course to Undergraduates



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Abstract The construction of an environment appropriate for information technology education is still challenging, especially in countries such as North Africa and the Middle East. Interdisciplinary courses that keep undergraduate students updated about emergent technologies are thus crucial for information technology education in these regions. Brain–computer interface (BCI) is a promising method that combines contemporary science, emerging technologies, and neuroeducation to establish a scientific grounding for teaching and learning. However, teaching multidisciplinary courses to undergraduates demands a combined learning approach that is challenging. Students must engage in active learning, contribute skilled participation, and imbibe additional knowledge as well as skills from traditional-type lectures. Further, they must also comprehend brain functions and use new measurement methods, advanced signal processing algorithms, and classification/control methods. This paper presents a mixed approach to undergraduate instruction that is theoretically and practically tethered to BCI aspects and utilizes a suitable mix of a BCI expert and teaching resources, such as slides, videos, and the Unicorn Education Kit. Thirty female students were taught the theoretical aspects of BCI and were asked to apply their BCI knowledge via original projects taken from conception to implementation in a single semester. The principal outcomes of this interdisciplinary course encompassed the development, implementation, and assessment of electroencephalogram (EEG)-based BCI education projects. Undergraduate students applied the theories acquired in class to observe and evaluate electrical signals generated by brain activity and measured via the Unicorn Education Kit. The efficacy of this project-based learning (PjBL) experiment was evaluated through student responses to a questionnaire and the analysis of examination results. The participants acquired the requisite knowledge and evinced higher interest in the fields of study and were able to build their own BCI applications. They were thus motivated to engage in further BCI research.

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Keywords Brain–computer interface (BCI) · Learning theories · Interdisciplinary education · Unicorn Education Kit · Electroencephalogram (EEG) · Project-based learning (PjBL)

1 Introduction

Formal education is a process through which trained and qualified teachers employed by accredited or legal institutions (e.g., schools, colleges, universities, etc.) inculcate knowledge, values, skills, and attitudes based on certain established standards to students [1, 2]. This structured and systematic form of learning can ameliorate the economic, social, and intellectual development of any individual. Conversely, learning is the continual, lifelong, conscious, or unaware cognitive activity of adopting knowledge, values, and skills [3, 4]. The facilitation of learning is critical to the fostering of intellectuals, experts, and highly skilled persons [5]. Proffering an informed curriculum designed to provide an appropriate educational environment can aid students in attaining diverse perspectives. Teachers can propose holistic and comprehensive curricula integrating numerous knowledge domains into higher education programs to encourage multidisciplinary or interdisciplinary learning [6].

Scholars have suggested that multi- and inter-disciplinary courses aiming to amalgamate theories and its practice from different disciplines represent the future of education and research [7]. Therefore, education systems should treat skills from different subjects as overlapping rather than distinct. Moreover, it is increasingly important and highly recommended to encourage projects requiring cross-disciplinary curricular material, especially concerning cutting-edge technologies in higher education [8]. For instance, teaching interdisciplinary courses that ensure students are updated about emergent technologies is crucial to information technology education, especially in the countries such as the Middle East and North Africa (MENA), where inculcating the appropriate environment remains challenging [9].

This paper focuses on the brain–computer interface (BCI), an interdisciplinary course that incorporates the promising domains of modern science, emerging technologies, and neuroeducation to effectively combat neuromyths and establish a basis for teaching and learning [10].

BCI is an increasingly popular technology that is utilized for replacing, restoring, enhancing, supplementing, or improving natural central nervous system (CNS) outputs, thereby modifying ongoing interactions between the CNS and its external or internal environment by using measured brain activity [11–17]. This technology is primarily employed to assist and improve communication in incidences of motor paralysis due to stroke, spinal cord injury, cerebral palsy, and amyotrophic lateral sclerosis. However, healthy people can also benefit from BCI technology by enhancing their control over smart home appliances, robots, drones, and videogames. The development of BCI applications requires expertise in multiple disciplines, such as medicine, biology, neuroscience, information technology (IT), signal processing, and

artificial intelligence. Therefore, teaching this emergent technology to undergraduate students so they can develop their own assistive, adaptive, and rehabilitative BCI applications poses a challenge. The addition of some neuroscience-based integrated hardware/software system courses to IT curricula may help to contest neuromyths and oppose the industry of largely pseudoscientific brain-based products that target parents, teachers, schools, and even local governments. A combination of methods may remedy such problems, such as courses that specifically address field-related neuromyths, inculcate skills that transcend the curricular content, provide consistent training, and offer access to reliable sources of information. Neuroscientists and educators can collaborate to guide, prepare future generations, and contribute to the proper allocation of resources.

BCI courses require instructors to cover numerous multidisciplinary conceptual aspects for their students; therefore, comprehending the complex course content becomes difficult, especially for undergraduate students [18–20]. Therefore, the current study proposes a solution that utilizes a mixed methodology grounded in the theoretical and practical aspects of BCI—applying an accurate combination of BCI expertise and teaching materials (such as slides, videos, and the Unicorn Education Kit) to effectively teach undergraduates. The Unicorn Education Kit includes eight electroencephalogram (EEG) apparatuses (Unicorn Hybrid Black) and the complete Unicorn Suite Hybrid Black software environment to prepare students for neuroengineering and neuroscience courses.

Learning theories can be classified into five fundamental types [3], namely, behaviorism, cognitivism, constructivism, experimentalism, and the social/constructivist theory. Figure 1 illustrates the general framework for teaching undergraduate BCI courses using multiple learning theories. This framework demonstrates that a BCI expert (instructor) should teach students to think critically and analytically as well as implement and validate innovative ideas to develop a BCI application over a single semester of theoretical and practical education. The expert should also teach how to minimize effort waste and maximize productivity by learning from and with others, working just enough to move to the next step, and then iterate further using lean user experience (lean UX) strategy.

The development of BCI applications is centered on collaborative teamwork and continuous learning through timely feedback from the BCI expert. Both the specialist and the students must work together to rapidly develop multiple BCI applications (proof of concept) before generating the final version of a prototype. The remainder of this paper is structured as follows: Sect. 2 presents the methodology and explains the main components utilized for evaluating student learning during a single BCI semester. Section 3 outlines the results of the questionnaire analysis, student outcomes evaluation, and the project evaluation. Section 4 discusses the results of this BCI case study from the development, implementation, and assessment of EEG-based BCI education projects. The paper concludes with Sect. 5, which details the potential of the proposed BCI teaching framework and notes its challenges and limitations.

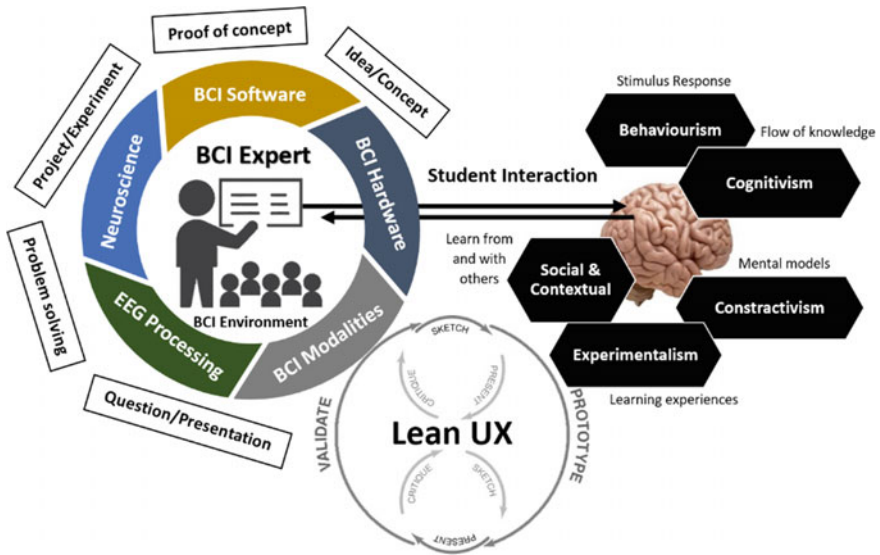


Fig. 1 General framework of teaching brain-computer interface course for undergraduate students using multiple learning theories for developing their own assistive, adaptative, rehabilitative BCI applications

2 Material and Methods

This paper presents a mixed approach using theoretical and practical aspects of BCI to teach neuro-engineering features to undergraduates to develop their own applications in a short span of time. However, teaching BCI to undergraduate students remains a challenging process because it mandates multi- or inter-disciplinary learning. In comparison to the learning associated with traditional lecture-based teaching (LT), this approach requires active and skilled student participation, and students must acquire numerous abilities, such as understanding brain functions, using new brain measurement methods, applying advanced signal processing algorithms, and deploying classification and control methods. The primary hypothesis of this study is that constructing a suitable environment for MENA students can offer a powerful means of apprising them of many current and pressing research themes. This environment entails the perfect combination of highly skilled BCI practitioners and appropriate teaching materials, such as slides, videos, and an EEG and BCI hardware/software kit for enhancing students' ability to develop multidisciplinary and innovative capstone projects designed to resolve open real-world problems. Such an assignment would enhance students' opportunities to compete against their international peers and work in science-rich environments in the future. This BCI course is thus aimed to help build a new awareness in students of the meaningful connections that exist among the disciplines of neuroscience and computer engineering.

Thirty female students participated in this study. They were taught theoretical and practical aspects of BCI over one semester of selected topics in a computer engineering course (CENG 580) at the United Arab Emirates University (UAEU). The curriculum of this course was structured in three equal parts, namely, content, skills and thinking processes, and assessments. The participants were instructed to build and test their own BCI applications using the Unicorn Education Kit (g.tec medical engineering, Graz, Austria). This kit comprises eight wearable and wireless instruments for measuring EEG signals. Each instrument includes a user and developer software that enables students to measure, record, and process EEG data in real time. The Unicorn Education Kit includes eight Unicorn Hybrid Black and the complete Unicorn Suite Hybrid Black software environment containing eight of Unicorn Hybrid Black, Unicorn gel and dry electrodes, Unicorn Python API Hybrid Black, Unicorn Simulink Interface, and Unicorn Speller Hybrid Black. The course content was provided by a BCI specialist with more than 10 years of experience in the field. The study material was uploaded on the Blackboard platform of the UAEU. The BCI expert was instructed to engage students, design practice activities, recommend appropriate textbooks, and use commensurate BCI technology and tools.

On average, students expended 3 h per week for one semester on the BCI course, watching video tutorials of EEG equipment and working on their project. The students progressed gradually from the fundamental concepts of BCI and signal processing algorithms to using specialized hardware and software for developing BCI applications to resolve problems of daily life. The course content was devised based on published research papers [18] but was slightly modified to adapt to the IT department curriculum. This study aimed to investigate the challenges and perspectives of teaching a BCI course to undergraduate students in the MENA region. To this end, a research questionnaire was formulated to obtain a comprehensive understanding of the effects of neuromyths and the different approaches used by students to imbibe the content of multidisciplinary courses. A total of 30 students responded to the online questionnaire, which was created using Blackboard Forms and comprised numerous items segmented into four sections, as presented in the Appendix section Table 1. The online questionnaire was estimated to take 15 min to complete (the questions displayed in Sect. 3 of Table 1 were presented in random order).

3 Results

The assessment of the proposed multidisciplinary course was based on three phases, namely, subjective questionnaire analysis, objective students' outcomes evaluation, and project evaluation (examining the path from conception to proof of concept). The efficacy of this mixed problem- and project-based learning methodology was assessed via student responses to the questionnaire and the analysis of the examination/project results. This approach promoted the evolution and application of the content imbibed by students over the semester. Using this interdisciplinary approach, the students were expected to synthesize their learning in responding to several questions by

drawing on evidence from multiple disciplines, such as medicine, neuroscience, engineering, computer science. The students were instructed to respond in various ways to augment developmentally appropriate benchmarks for student learning, such as critical thinking, imagination, analysis, and engineering skills.

3.1 Questionnaire

Figure 2 presents the results of the research questions asked in Table 1. Belief in neuromyths was found to be quite common in most respondents before they began the BCI course, and only 35% of the answers were correct. After completing the BCI course (one semester), the questionnaire was re-administered to ascertain whether the students could distinguish between facts and neuromyths. The correct answers increased by 64%, demonstrating that the students absorbed a lot of neuroscientific knowledge. Further, almost all the students changed their answers from “difficult” to “easy” for the rest of the questions after completing the BCI course. All students evinced high interest in the BCI course and were motivated and engaged in the same classroom during the entire semester, because according to them, BCI connected as a topic with their desired leisure activities (working on emerging technology, controlling drones and robots, coding, etc.). The students were also engaged in the classroom because of the instructor’s (BCI expert’s) preparedness in affording students a conducive environment and appropriate course materials combined with the right topics to attract their interest.

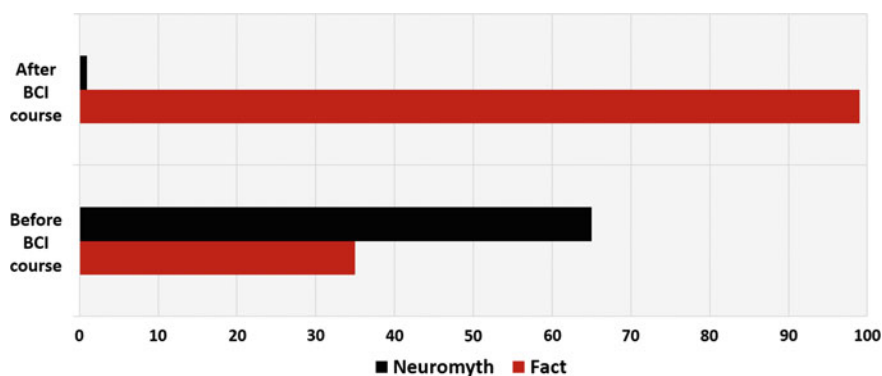


Fig. 2 The results of the research questions asked in Sect. 3. These results exhibit the analysis of questionnaire responses to common myths related to neuroscience. These items were queried before beginning the BCI course and after the semester ended. The correct answers are marked in red as “fact,” and the wrong answers in black are labeled “neuromyths”

3.2 Theoretical Assessment

The BCI expert was instructed to maximize the percentage of time all students were to be engaged with theoretical content. The evaluation of students was based on the BCI course learning outputs using the Blackboard platform. A camera was used with the Respondus LockDown Browser to discourage cheating in the online examinations. As most undergraduate students are concerned about their grades, the BCI expert tried to teach them how to earn as by focusing on learning rather than grades. The students had to take six quizzes in one semester. Three of them were given before the midterm exam, and the rest were administered before the final exam. The students secured an average of 70% on their first quiz. Thereafter, their quiz grades began increasing until they averaged 90% in the last quiz.

The BCI expert increased the complexity of the quiz questions with time; however, the students did not evince any negative responses to this difficulty. Figure 3 displays the results of the BCI quizzes, midterm examination, and final examination. Multiple-choice questions were given to the students, which provide respondents with multiple answer options. The statistics of each question and each answer could be inspected through Blackboard. For instance, the students were asked, “Which of the following techniques is considered an invasive brain measurement?” The students’ answers were EEG (76.66% correct and 23.33% incorrect answers), Intracranial Electroencephalography (iEEG) (40% correct and 60% incorrect answers), Electrocorticography (ECoG) (93.33% correct and 6.66% incorrect answers), and Magnetoencephalography (MEG) (83.33% correct and 16.66% incorrect answers).

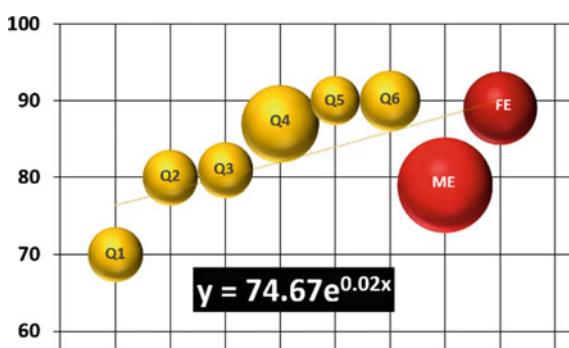


Fig. 3 The results of BCI assessments to evaluate learning outcomes using quizzes, midterm examination, and final examination. Quizzes (Q1–Q6) are represented in yellow; the midterm examination (ME) is shown in light red, and the final examination (FE) is displayed in dark red. The y-axis is the average value of the exam and the size of the circle is the standard deviation value

3.3 *Project-Based Learning Assessment*

Students were made to practice some lean UX ideas and apply them to their project design and development processes. The 30 students were divided into 7 groups. Each group was instructed to envision an original BCI concept and implement it over the semester. The BCI expert used numerous evaluation criteria to evaluate each student group, including knowledge combined with practical abilities, creative thinking, problem-solving, project presentation aptitude, value-based teamwork, leadership skills, and independent learning. Additionally, the BCI expert was instructed to accord grades for four items (idea, report, presentation, and implementation) to appraise the path from BCI idea to proof of concept and evaluate project development and management skills. The seven groups received an average grade of 14.28 out of 20 points for proposing relatively original ideas, 16.71 for structuring their reports, 16.14 for making comprehensive presentations, and 16.28 for executing the proof of concept. The BCI expert applied the flipped classroom philosophy to encourage self-directed learning practices in students. During the course of the semester, the students independently learned many new technical skills, such as programming languages and network communication. All the participants demonstrated high self-motivation and interest in their topics in the implementation phase of their BCI project.

Some even participated in international competitions to present their BCI project and enhance their experiential learning in collaboration with international peers. The BCI expert provided regular feedback to improve their performance and to help students pause and rethink or recalibrate their project to further advance the outcomes. Figure 4 shows an example of a group project undertaken by three UAEU undergraduate students of the College of Information Technology who controlled a drone using brainwaves. This group utilized the wireless Unicorn Hybrid Black EEG headset to stream and record EEG signals to extract the P300 characteristic and translate it to commands to control drone movements in real time. By the end of the project, the students acquired numerous technical skills, such as recording brainwaves in real time, using P300 Speller, sending commands from Unicorn to the drone using socket, and connecting the BCI with a drone (Parrot Bebop 2) to remotely pilot the drone and control its flight pattern. Additionally, five teams of the participating 30 students (each team comprised four or five undergraduates) participated in the virtual international BCI hackathon [The Brain–Computer Interface Designers Hackathon (BR41N.IO), <https://www.br41n.io/>] to build EEG-based BCI applications. In 24 h, they constructed original BCI applications to control drones, robots, or videogames using only brain activity. One of the teams won the third place in this competition (BR41N.IO summer school-2020) among 50 international teams across the world.



Fig. 4 An example of undergraduate students' BCI projects. The students could control a drone and video game using EEG signal and they have participated in an international BCI hackathon

4 Discussion

This paper presented several assessment results based on questionnaires, examinations, and projects to investigate the impacts and challenges confronted by instructors and undergraduates when teaching and learning multidisciplinary courses, such as the BCI module. The proposed module intended to inculcate in the participating students the practical skills and theoretical knowledge necessary for them to successfully develop their own BCI systems (see Fig. 5).

The primary challenge was designing a suitably complex and multidisciplinary BCI course whose content would engage undergraduates. To accomplish its purpose, the approach of problem- and project-based learning was used. The authors of this paper believe that developing students' creativity requires nothing more than the right environment that is conducive to undergraduate students by according them the means of apt study resources and equipment such as the Unicorn Education Kit along with the services of a fitting BCI instructor in any emerging technology. The main outcomes of this case study of teaching an interdisciplinary course pertain to the development, implementation, and assessment of EEG-based BCI education projects. The projects required undergraduate students to apply learned theories and improve their knowledge and technical skills in observing and evaluating electrical

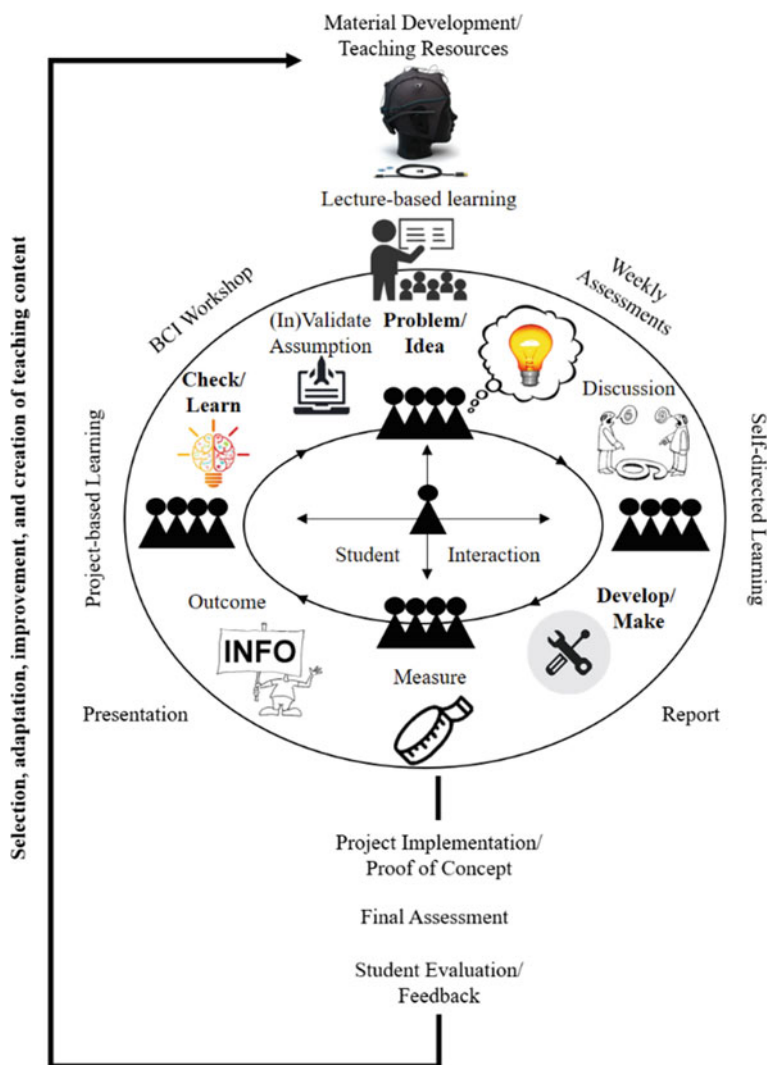


Fig. 5 The proposed student-centered learning framework for inculcating in the participating students the practical skills and theoretical knowledge necessary for them to develop their projects

signals generated through brain activity and measured by the Unicorn Education Kit. Students who participated in this study were able to build original and innovative BCI applications from conception to proof of concept. They acquired the necessary knowledge and evinced higher levels of interest in the field. In turn, they became more engaged in further BCI research. Two teams wrote conference papers and presented them at international forums [21, 22]. All groups stated that they had enjoyed their learning journey and expressed their satisfaction with the rewards they received. The

BCI expert applied some gamification techniques with the students to enhance their sense of accomplishment. Prospective researchers and instructors are encouraged to involve undergraduates in more multidisciplinary courses. However, what seems like an unusual choice now could well become the norm for the education systems of the future. Additionally, complementary studies should be executed to further develop and evaluate the proposed BCI course, undertake large-scale projects, and assess the influence of such modules on the achievement of learning goals [23].

5 Conclusion

This paper presented a case study of a mixed approach module that combined the theoretical and practical aspects of BCI via problem-oriented projects to teach undergraduates the entire path of development from idea to proof of concept. The predominant outcomes of this BCI course include content development, project implementation, and assessment of EEG-based BCI education projects. The participating undergraduate students applied the theories they had learned and improved their technical skills. It is posited that the addition of some neuroscience modules to education curricula could contribute to combat and remedy neuromyths that prevail in the MENA region. It is recommended that prospective researchers and instructors should involve undergraduates in more multidisciplinary projects.

Appendix

See Table 1.

Table 1 Questionnaire for undergraduate students

Section 1: <i>Demographic questions</i>
– Age
– Gender
– Education level
(continued)

Table 1 (continued)**Section 2:** *True/false questions on imbibing multidisciplinary knowledge (administered at the beginning and at the end of the BCI course)*

Q1. Learning multidisciplinary material that requires knowledge of medicine, neuroscience, and information technology is:

- (a) Easy
- (b) Challenging, but doable
- (c) Difficult

Q2. What are the challenges that you may face or did face in learning the material set for the BCI course?

- (a) Complex topic required multiple fields of knowledge
- (b) Too much scientific and technical information
- (c) No adequate BCI textbooks
- (d) Appropriate equipment and tools for implementation are difficult to find
- (e) Others: (please specify)

Section 3: *True/false questions to evaluate the effects of neuromyths on students (administered in the beginning and at the end of the BCI course)*

Q1. I am right-brained, which makes me more creative (True or False)

Q2. I am left-brained, which makes me more creative

Q3. Women's brains are completely different from men's brains

Q4. Men are smarter than women

Q5. Bigger brains are smarter (relative to body size)

Q6. The bigger the brain the smarter you are

Q7. The human brain is the largest

Q8. Brain size does not correlate with intelligence

Q9. We only use 10% of our brain

Q10. We all need eight hours of sleep to function

Q11. Some people have an evil spirit in their brain; we can drill a hole in the skull and let it out

Q12. An evil spirit can control my brain

Q13. Working out crossword puzzles can keep your brain young

Q14. Individual personalities display right- or left-brain dominance

Q15. Brain damage is always permanent

Q16. Drinking alcohol kills brain cells

Q17. Eating sugary snacks results in hyperactivity and reduced focus and attention

Q18. Hemispheric dominance (whether you are "left-brained" or "right-brained") determines how you learn

Q19. Teenagers lack the ability to control their impulses in the classroom

Q20. Being in a coma is like being asleep

Q21. Playing classical music to infants makes them smarter

Q22. Adults cannot grow new brain cells

Q23. Male brains are biologically better suited for math and science, and female brains are oriented toward empathy

Q24. Students learn best when teaching styles match their learning styles

Q25. We know what you are thinking: Extrasensory perception (ESP) is a scientific certainty

Q26. Some people are left-brained (logical) and some are right-brained (creative)

Q27. Men are logical while women are emotional

Q28. Men are good at mathematics but women are not

(continued)

Table 1 (continued)

Section 4: *Questions related to developing brain–computer interface applications (asked at the end of the BCI course)*

Q1. Developing a BCI application from scratch to proof of concept is:

- (a) Easy
- (b) Challenging, but doable
- (c) Difficult

Q2. Did the course materials help you implement your BCI idea?

- (a) Yes, but it was not enough
- (b) Yes, it made a big difference
- (c) No

Q3. How can the BCI course be improved?

- (a) Engaging students
- (b) Designing more practice activities
- (c) Giving students adequate BCI textbooks
- (d) Have more BCI equipment and tools for every student
- (e) Others: (please specify)

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Information Technology in Teaching Future Pop Vocalists to Promote Their Creativity at the University



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and Svetlana V. Ward

Abstract One of the urgent problems in the system of training students, whose profession is connected with creativity, is learning how to present their creative achievements to society. The lack of demand for creativity, the impossibility of demonstrating it leads even great musicians to lose motivation to create works, to translate them on the stage and even to depression. In this regard, at present, courses are being introduced into the system of training musicians and artists at universities to promote their creativity and the possibilities of its presentation to the public. This promotion is extremely important for the system of training pop vocalists, whose creativity, associated with current and innovative trends in song fashion, should soon reach the consumer and be properly appreciated by him. Today, university teachers where pop vocalists are trained to include topics related to the promotion of the future artist's creativity in certain disciplines. However, being carried away by the problems of management and the problems of quality of performance, teachers' pay insufficient attention to such modern means of promoting creativity as information technology. The purpose of this article is to show that minimum of knowledge about information technologies, including social networks, programs, music platforms, video editors, which will allow an aspiring pop vocalist to successfully promote their creativity. The presented information technologies were introduced into the process of preparing future pop performers during one semester and showed their effectiveness. The main result of the experiment was the emergence of more invitations of students to new and interesting projects.

Keywords Information technology · Social networks · Programs · Music platforms and video editors · University education · Pop vocals

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1 Introduction

In show business today there are many ways to promote the creativity of pop vocalists to attract listeners and viewers. A huge staff of managers is engaged in such work if it is a famous pop singer or only one manager if a pop vocalist is still little known in wide musical circles.

The promotion of creativity is an obligatory component of the real existence of show business, because no matter how talented the performer is, his work is always aimed at the listener and viewer, and it is they who, first of all, determine the degree of the performer's talent and the degree of his popularity and demand. Sometimes only chance can reveal interesting pop artists. The most famous example, as pointed out by Eric Nichols, Charles Duheadway, Rishikesh Aradhai and Richard F. Lyon, is Canadian vocalist Justin Bieber, whose songs have been revealed on YouTube. However, many talented performers will never be discovered if they simply post their song to YouTube. Consequently, the pop vocalist needs to learn to develop his creative abilities [1].

Unfortunately, during the period of study at the university, little attention is paid to training a pop vocalist for this type of activity. All the time is devoted to the development of his vocal abilities, musical abilities, stage skills, general cultural development. However, little time is devoted to the ways of promoting one's creativity, which undoubtedly primarily include information technologies. All this does not make it possible to present to future pop vocalists those information technologies that can be successfully used by them to promote their creativity in society. Studying at a pedagogical university on a pop profile, students accumulate not only performing but also pedagogical baggage. And this means that the knowledge gained about the ways of using information technologies in promoting creativity, they will pass on to their future students—schoolchildren and students engaged in pop vocals.

Today, researchers use a “communication strategy” (V. Yu. Korsak) to reveal the content of the promotion process of a pop vocalist [2]. The purpose of its development is to inform the listening audience about the performer, his repertoire, status, awards, achievements, etc. Several tasks for the implementation of a communication strategy include those that contribute to the identification and selection of certain information tools, without which the promotion of creativity today is impossible. These tools include microblogging, for example, Twitter and V. Kontakte, which are especially popular today in Russia [3]. Given this circumstance, managers, like advertisers, spend a lot of money to promote certain products, including concerts of rock and pop vocalists.

Students studying at the University for Variety Performers master such informational means on their own, sometimes without having an idea of their wide possibilities.

2 Theoretical and Methodological Basis

Currently, both in Russia and abroad, various studies are being carried out on ways to promote the creativity of various performers and on the characteristics of such performers. Researchers Kubacki and Croft find a definition for musicians who perceive themselves in the unity of their artistic and managerial market identity—“Musicpreneur” [4]. Analytical conclusions on the methods and content of advertising are disclosed by Roy et al. [5] However, this article deals with already popular performers, whose advertisements were studied by authors in three countries, and from the standpoint of the preferences of listeners from different social strata of society. The role of social networks in advertising is considered today from approximately the same positions, namely from the position of influence on the viewer. So Mendez-Vazquez Vega and López-Cuevas [6] disclose in the article how the social influence in online social networks occurs on the opinions and interactions of people, which, undoubtedly, can be taken into account when promoting the creativity of a particular pop artist. Kirchner [7] considered the level of success of a pop musician not only at the social but also at the local, regional, national and international level. From the point of view of advertising consumers, its content, the introduction of musical content, and the use of information technologies are considered [8]. But vocalists-students studying at universities are so far interested not so much in determining the social composition of consumers of their creativity, as in learning how to spread their creativity in general and, in particular, using such effective means as information technology.

Another aspect of considering the problem is the quality of creativity advertised through the network when network music does not attract pop vocalists to create a product for promotion from an aesthetic point of view (“sound freezes”, “sound lags”, etc.) Researchers found that collaborative network performance is still not high-quality, this requires special programs that improve such performance [8]. And since joint performance on the network can be a form of promoting the creativity of pop vocalists, then such a program should be well known to them.

And, finally, the issues that are being discussed today by both the pedagogical community and managers are about the need to prepare student pop vocalists to get acquainted with new information media both for the process of high-quality learning and for promoting their creativity [9]. Possessing such technologies today, a pop musician can compete with large record companies through self-promotion [10]. The importance of self-government in promoting their creativity among pop vocalists was blurred from the 1960s to the 1990s, today the opposite is true: the importance of intermediaries is being blurred by the focus on self-promotion [11]. All of the above made it possible to develop an experiment plan for the introduction of information technologies into the educational process of future pop vocalists.

3 An Experiment on the Implementation of Information Technology in the Educational Process of Future Pop Vocalists

3.1 Organizational Basis of the Experiment

The main purpose of the pedagogical experiment was to familiarize students with the possibilities of information technology to promote their creativity in the social environment. An analysis of the content of some disciplines included in the curricula of modern Russian universities that train pop vocalists has shown that such an active introduction can be carried out in such disciplines as “Methodology for teaching pop vocal”, “Directing a pop performance”, “Management in culture and art”, “Imageology in pop performance”, “Class of solo pop-jazz singing”. In all these disciplines there is a reserve of time for solving the problems of promoting one’s creativity.

To improve students’ knowledge about the possibilities of information technology in promoting their creativity at the Ural State Pedagogical University under programs 44.03.01. Teacher Education. Profiles: Music education and Additional education (pop-jazz vocals) an experiment was carried out to familiarize bachelor students with the possibilities of information technologies that contribute to the promotion of their creativity.

The experiment involved 86 bachelor students studying pop vocal according to the specified educational program. This group of participants was divided into control and experimental groups, respectively, 42 and 44 people, according to the number of students in academic groups. In the experimental group, the following disciplines were introduced into the content of the above disciplines: tasks to familiarize themselves with information technologies that allow for the active promotion of their creativity, tasks that form the ability to create personalized content and distribute it in various ways in society, tasks that form the ability to create a presentation of events for which, after acquaintance with the student’s content, he would be invited to projects, competitions, concerts, or he would receive an interesting offer to perform in certain events.

The purpose of the ascertaining stage of the experiment was to identify students’ knowledge about the possibilities of information technology to promote their creativity, the ability to create their advertising product using information technology, the ability to present their creative product to society in various ways. The criteria for the professional formation of a pop vocalist in terms of promoting their creativity were: cognitive (knowledge of modern media technologies, the ability to use Smart-education technologies and modern media technologies to promote oneself as an artist), activity-based (creating one’s professional creative content), quantitatively creative (number of invitations to projects after broadcasting their content). The results of this stage made it possible to identify the problem of the experiment and confirm its relevance. So, it turned out that in the experimental and control groups, when questioning, students demonstrated knowledge about the possibilities of such

social networks as Instagram, TikTok, youtube, when composing certain content or advertising their creativity, students used the same programs that improve their advertising product Vegas, Pinnacle studio, after the presentation of this advertising product within a month, the number of interesting creative proposals for each student was counted. Unfortunately, only about 4 and 6% (respectively, the experimental and control groups) had interesting proposals for participation in various creative projects and performances. The general results on the three designated criteria allowed us to conclude that, although half of the students participating in the experiment showed an average level, nevertheless, these results demonstrated a low percentage of advertising of their creativity in society. All this served as the basis for the inclusion in the above-mentioned disciplines of material related to familiarizing students with information technologies.

3.2 The Search Stage of the Experiment

The content of the experimental work (its formative stage) was structured by components, the appearance of which in lectures and practical classes was due to the technology of creating an advertising product by students and its distribution. These components included questions of knowledge formation (1) About creating a high-quality advertising product and music platforms; (2) About the technology of creating an advertising video and developing content for a pop vocalist (3) About solving creative problems of creating advertising. The topics, the content of which included these information technologies in the designated disciplines, were: “Characteristics of the image of a modern pop artist”, “Professional and creative formation of a pop artist”, “Professional formation of a pop artist and promotion in social networks.”

Let us present a summary of each component of the experimental work from the standpoint of the characteristics of certain information technologies.

3.2.1 Creation of a Quality Advertising Product

To create high-quality vocal compositions, the pop performer was presented with the capabilities of such computer programs as Cubase, Logic, FL studio. With their help, students could create their own author's compositions, make high-quality arrangements of their musical material, as well as create remix arrangements for already known pop compositions. Thanks to these programs, the student could change the genre basis of the well-known pop composition, the texture of the musical material, the rhythmic and tempo implementation, while the melodic line of the vocal work remains unchanged.

Presentation of music platforms. Creating their own author's music, each of the novice pop performers places it on music platforms, so the students were presented with the platforms: Spotify, Yandex music, V. K. music, TikTok, OOMUSIC.RU [12].

Students were given recommendations on the placement of their author's compositions, having previously emphasized that they first need to register on music platforms, then register copyright on it, and after that, only the composition appears in the public domain for listening to a wide range of people.

3.2.2 Technology for Creating an Advertising Video

The students were offered technologies for creating commercials, clips and mini-films. In the process of training, students were asked to create an advertising video on the topic: "I am a professional vocal performer", "I am an artist." They created a clip of their author's song or a mini-film about their work. Themes of the video films: "One Day for a Beginning Artist", "My Meeting with Famous People". To complete this task, students had to study programs—video editors, namely Vegas, Pinnacle studio, Adobe premiere pro.

Creation and development of pop vocalist content. For students, the possibilities of presenting their audio content were revealed, namely, recording of copyright songs, fragments of recordings from the radio, recording of solo concerts and professional photographs, for this, such programs as adobe premiere pro, Cubase, vegas were used, which included the personal verbal characteristics of their work, and also viewers' reviews of the performance or reviews of any media persons. Such content included: checklists of concerts, as well as components of visual content—photos, screenshots of reviews about creativity, etc. It was proposed to post the created content on Vkontakte, Facebook, Youtube, TikTok, Telegram or Instagram.

3.2.3 The System of Creative Tasks to Create an Advertising Product

In the process of teaching students, the following creative tasks were offered to them, which were carried out using information technologies:

- develop an individual route map "I am an artist". This map was supposed to include: author's material—media stage (presenting oneself as an artist through media technologies)—concert activities. The student could position himself to society in reliance on the comprehension of personal SWORT analysis;
- develop ways of distribution to promote your creativity. It was explained to the students how to release their author's songs for official releases from a major or small label.
- develop an advertisement for your work and present it first to your fellow students who will give it an assessment, then on Vkontakte, Instagram, Facebook, Youtube, TikTok.
- develop ads that include collaborations with other artists. For this, the Smule Sing program was used, where it is possible to record a song with a famous pop artist, short videos with media personalities are shot in TikTok, Instagram can broadcast live with famous people, as well as upload photos, videos and stories with them.

3.3 The Control Stage of the Experiment

The control stage of the experiment was carried out after a semester of the experimental work. Its purpose was to test the effectiveness of the implementation of information technologies, modern media technologies in the educational process of university training of a pop artist. Comparative analysis of the initial and final sections showed significant dynamics in the professional development of students in terms of their readiness to promote their creativity. The main positive result of the experiment was a significant increase in invitations of students, novice pop vocalists to various interesting and creative projects: “New Star”, “Songs on TNT”, “Voice”, “X-factor”, “Come on together.” The number of such invitations in the experimental group increased by almost 30%, while in the control group by only 19%.

The results of the ascertaining and control stages are shown in the table.

Levels	Experimental stages			
	Initial assessment (%)		Final assessment (%)	
	EG ^a	CG ^b	EG ^a	CG ^b
Low	43	42	14	23
Medium	53	52	65	62
High	4	6	21	15

^aEG experimental group, ^bCG control group

The quantitative analysis of the data of the experimental work was carried out using the capabilities of the statistical analysis apparatus. The hypothesis being tested was that the differences between the data in the initial and final assessments are significant. Here the Pearson criterion was used. The theoretical value of the Pearson criterion is $X^2_t = 9.21$. The experimental value of the criterion X^2_e EG = 37.2. The experimental value of the criterion X^2_e of the CG group = 10. The result is: $(X^2_e \text{ EG} = 37.2) > (X^2_e \text{ CG} = 10)$. The hypothesis is confirmed. Analysis of the theoretical and experimental value of the Pearson criterion for group A showed that $(X^2_e \text{ EG} = 37.2) > (X^2_t = 9.21)$, which means between the levels of becoming a professional pop vocalist and the introduction of Smart-education technologies of modern media technologies in this group there is a dependency. A similar analysis of the data for the control group showed that between the levels of formation of a professional pop vocalist and the stages of experimental work there is $(X^2_e \text{ CG} = 10) > (X^2_t = 9.21)$. The analysis of the obtained data for the EG and CG groups allowed us to assert the significance of the difference between these groups in terms of the studied characteristics.

4 Key Findings

The information technologies considered in the article, the development of which must be mastered by a pop vocalist while still studying at a university to promote his creativity, can be grouped according to the following indicators:

- Social networks—Vkontakte, Facebook, Youtube, tik-tok, Telegram, Instagram;
- Programs—Cubase, Logic, FL studio, Smule Sing;
- Music platforms—Spotify, Yandex music, V. K. music, TikTok, OOMUSIC.RU;
- Video editors—Vegas, Pinnacle studio, Adobe premiere pro.

At the end of the experiment, an analysis of advertising products sold by students using information technology was made. 14% of students who showed a low level posted only 4 author's compositions on music platforms. An examination of their content showed that the quality of the recordings and photos presented in it was low. The recordings contained extraneous noises and overtones, the photos were taken indistinctly. In the group of students whose work was rated as intermediate, content was presented, some of which was of poor quality. The results of the experiment to familiarize bachelor students with the possibilities of information technologies that contribute to the promotion of their creativity, making it possible to determine the prospects for continuing to work with pop vocalists to familiarize themselves with information technologies: the inclusion of technologies that allow you to "clean up" your recordings made at a concert, methods of eliminating glare in photos and videos, as well as recommendations for placing author's compositions and arrangements on the Internet services Spotify, Yandex music, etc.

In general, according to the experiment, it can be concluded that the information technologies offered to students for acquaintance, namely, social networks, programs, music platforms and video editors, create the basis for the promotion of each future pop vocalist of their creativity. The participation of such vocalists, in addition to performing and teaching activities, will make it possible to familiarize children and youth with these information technologies, i.e. all those who will study pop vocals under their guidance.

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Digital Education and Economics in Smart University

Validating Development Indicators for Smart University: Quality Function Deployment



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Abstract The article shows the application of a quality management tool called “Quality Function Deployment” at university. This tool allows managers to justify the choice of controlled parameters for their monitoring and further assessment as the most priority evaluated characteristics of the competitiveness for a smart university from the external consumers of educational services known as stakeholders. In contrast to already known studies in the area of education system development, this research reflects the possibility of predicting the effectiveness of further activities of smart organization performers. The results obtained allow managers developing a mechanism for assessing the level of a smart organization development in the context of existing classification characteristics. The practical importance is the possibility of system monitoring for the estimated characteristics of competitiveness.

Keywords Smart university · Development indicators selection · Quality function deployment

1 Introduction and Literature Review

Nowadays there are many studies describing the concept of smart education and the smart university [1], the peculiarities of its functioning in different conditions [2]. The analysis of innovative technological strategies of learning and teaching for smart education [3, 4] clearly shows that in the near future intelligent smart pedagogy will be actively implemented by the world’s leading academic institutions to teach local and remote students in the same class.

Quality management techniques are often used to assess the quality of an organization’s performance. The quality function deployment (QFD) [5, 6] is a managerial

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tool that is considered to be useful for monitoring the target achieving. It is well known that the use of innovative learning strategies including (1) learning by doing, (2) flipped classroom, (3) play-based learning, (4) adaptive learning, (5) contextual learning, (6) collaborative learning, (7) analytics of learning, (8) “bring your own device” (BYOD) strategy, (9) face-to-face learning, (10) cross-learning, (11) robotic learning and other advanced technological approaches to learning and teaching represented in many modern studies [8, 9]. It additionally confirms the necessity to predict the required indicators of the university’s smart development and to assess the quality of its activities in the near future [10].

The analysis of modern studies [10–12] has shown that in order to introduce innovative learning technologies, a teacher is to have a set of certain competencies and skills that are in high demand today [13–15]. In the article the quality deployment function is revealed and suggested to be used for selecting a set of special competencies that employees of a smart university should possess.

The idea of deploying the quality function was represented in Japan in the late 1960s. QFD as a method was developed in 1966 by Japanese engineer and scientist Akao [5] and first applied in 1972 at Mitsubishi and then in 1977 at Toyota. Since 1988 the method has been applied in Germany. QFD is a flexible decision making method used in product development [6]. According to the creators of the methods, QFD is used to identify the most important characteristics of new or existing products for specific customer groups, market segment, company or development technology. An important result of developing these methods, which include various schemes and matrices, is the possibility of reusing them to obtain other results. QFD transforms customer needs (customer feedbacks) into different product characteristics, prioritizes for each product/service and simultaneously defines product or service development objectives.

In 1977 a combination of tables-panels named “house of quality” was represented in Japan. This “house of quality” is a set of different matrices, tables, lists and it is today considered as being a tool to support individual transformation steps in the QFD process. While using QFD managerial tool four key phases of the project for creating a new product/service are to be defined and considered. These phases correspond to the QFD approach according to the American Supplier Institute’s standard, which is most formalized and schematized.

These 4 key stages include:

Phase 1—the product planning;

Phase 2—the product’s parts planning;

Phase 3—the system of the product manufacturing processes planning;

Phase 4—the production equipment planning for product manufacturing.

Thus Professor Akao identified four important areas for designing a new product or service:

1. quality development;
2. technology development;
3. development of accounting and cost reduction system;

4. development of product reliability.

According to professor Akao ideas each new product or service development is to be based on an individual quality function deployment. Nowadays this approach is also considered to be relevant and in demand for many areas of manufacturing and services [6, 14, 15].

2 The Research Problem

Modern society and its challenges make top managers designing a new emergent property for teachers and employees at smart university and organizations. This property consists of “the staff readiness to use smart technologies” in the educational environment. For this purpose, the required skills of teachers in the electronic education smart system were selected.

These are considered to be in demand for modern electronic educational systems for the period up to 2021. This Respondent Survey Content was proposed by V. Uskov and his team of smart pedagogy researchers. Later on, we will choose the results of the work [7] as a basis for the research where the authors developed and proposed to international experts the following list of possible requirements (11 ideas) to the skills of a modern teacher for distant education and e-learning.

The key notes for the “consumer voice and requirements” are represented in Table 1 as the following:

All these abilities are related to the stakeholders’ requirements and considered to be very important for effective management at smart university. Thus, the authors consider that the **research problem of the study** is to identify the possibility of applying QFD to justify the choice of the most important indicators that have to be monitored and assessed by managers for strategic development of smart university.

3 Methodology

3.1 The Key Idea of Choosing the QFD

According to the moder studies the key idea of choosing the QFD method and its application for education is the following:

Step 1. 11 important characteristics are selected as the consumer’s demands according to the tenets offered by V. Uskov [7] in his studies. Among these 11 characteristics six priorities which are considered to be important and relevant for higher education in Samara region of the RF are chosen by the authors. The are denoted as tenets and correspond to $T_1, T_2, T_3, T_4, T_5, T_6$.

Step 2. In the research [12] the authors defined 6 key indicators promoting smart university as a self-learning smart organization and the research hypothesis is turned

Table 1 Stakeholders' Requirements

Requirement	Requirement content
Requirement 1	The ability to teach E-learning courses by using advanced information, communication, multimedia, collaborative and networking technologies (R1)
Requirement 2	The ability to collaborate with participants in the educational process by using advanced technologies (e.g., collaborative whiteboard web applications, web tools for brainstorming, web tools for creating and annotating web documents and developing web applications, network file sharing tools, etc.) (R2)
Requirement 3	The ability to quickly involve students in the E-learning process (R3)
Requirement 4	The ability to quickly establish contacts in the electronic environment with all participants in the E-learning process (R4)
Requirement 5	The ability to adapt to individual needs and requirements of students (R5)
Requirement 6	The teacher as an innovator in the use of advanced e-learning technologies (R6)
Requirement 7	The teacher as an adapter of developed innovative methods and means of modern e learning(R7)
Requirement 8	The teacher as a creator of educational content for e-learning education by using different modern approaches and technologies (R8)
Requirement 9	The ability to manage the E-learning course (R9)
Requirement 10	The ability to communicate by using advanced technologies including forums, chat, videoconferencing, audio conferences, blogs, wikis, etc. (R10)
Requirement 11	The ability to motivate students for studying and self-development in the e-learning environment (R11)

out that the controlled factors T_i are to be correlated and have a cause-effect relation with the key characteristics of a smart university (K_i).

These indicators usually include:

Relationship convergence coefficient (K_1)

The employee intellectual activity index of, K_3

The employee innovation indicator, K_4

Knowledge Transfer Ratio (K_6)

Conformity of available knowledge to the requirements of the environment (K_7)

Coefficient of knowledge increment (K_{10}).

The authors' task is to estimate the correlation dependence between the indicators T_i and K_i . If the correlation link is strong, then a new emergent feature for both the smart university and university teachers ensuring their self-development and self-improvement is required. Similarly to Step 1 the authors introduce new symbols: $K_1 = Q_1$, $K_3 = Q_2$, $K_4 = Q_3$, $K_6 = Q_4$, $K_7 = Q_5$, $K_{10} = Q_6$.

Step 3. Designing the quality house allows managers in higher education taking into account all the causal relations.

Relying on the conclusion of the researchers represented in modern studies [5, 6, 10–12] the following author's vision for solution of the quality problem for higher education is offered.

3.2 Designing the Quality House

Figure 1 represents the Quality House simulated by the authors according to the algorithm and modern trends [5]. The authors’ algorithm for designing the Quality House is as follows:

- Step 1. Determining consumer expectations (room 1).
- Step 2. Comparison of products with competitors (room 2).
- Step 3. Identify and quantify improvement goals and objectives (room 3).
- Modeling: Level of improvement (LoI) represented formula (1)

$$LoI = \frac{TV}{PA},$$

(1)

where:

- LoI level (degree) of improvement;
- TV target value;
- PA product assesment weightiness (W) represented formula (2)

$$W = \frac{R}{LoI},$$

(2)

where:

- W weightiness;

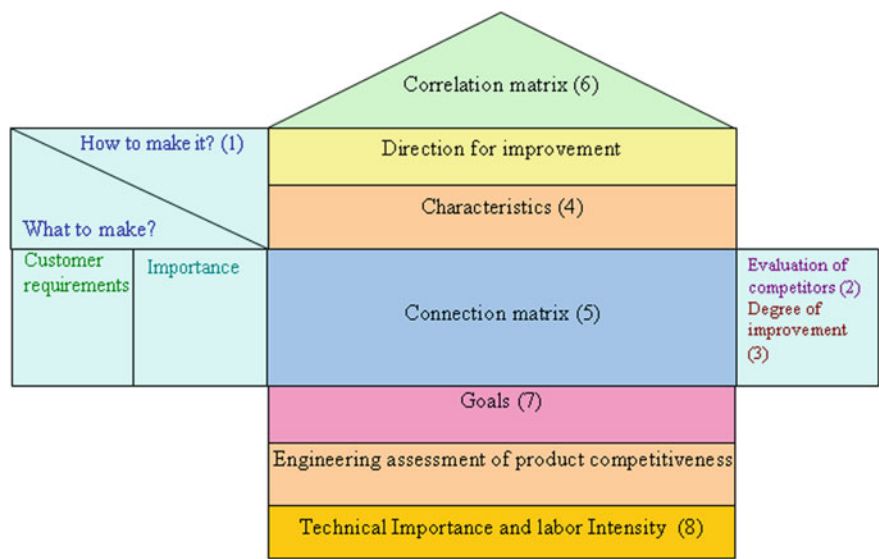


Fig. 1 The quality house (the author’s simulation)

R relevant (importance).

weightiness in % represented formula (3)

$$W(\%) = \frac{W * 100}{\sum_{i=1}^k W_i}, \quad (3)$$

where $i = 1, 2, \dots, k$ (number of characteristics to be evaluated).

Step 4. Determine technical specifications (room 4).

Step 5. Determine the relationship between consumer expectations and specifications (room 5).

Calculations are performed according to formula (4)

$$CR = C * W(\%), \quad (4)$$

where:

CR connection relevance;

C connection.

Step 6. Determine the interaction between the technical specifications using the links (room 6, roof).

Step 7. Calculate the total score and priority of specifications (room 7), formula (5)

$$P(\%) = \frac{TPA * 100}{\sum PA}, \quad (5)$$

where:

P priority

TPA total score of the product.

Step 8. Determine the target characteristics of improved products by testing (room 8).

3.3 Hypothesis of Research

According to some modern studies [10] the controlled factors T_i should have a causal relationship with the key characteristics of a smart university (Q_i). If the correlation relation is strong there is the necessity to design new emergent features and demand for smart university teachers and employees, providing their self-development and self-improvement. The higher education human resources possessing new hard and soft skills are considered as a necessary condition for the self-development of the university and its pressing forward to a smart organization.

Table 2 Customer requirements for the emergent quality of a teacher in a smart learning environment

Customer Requirements	Importance of expectation (expert evaluation)
Requirement 1 (T1)	4.3
Requirement 2 (T2)	4.6
Requirement 6 (T3)	4.9
Requirement 7 (T4)	4.9
Requirement 8 (T5)	4.5
Requirement 10 (T6)	4.8

4 Results

4.1 Benchmarking for Smart University: Experiment

The authors used benchmarking for using QFD tool at three universities of Samara region. Togliatti State University (TSU), The Volga Orthodox Institute named after St.Alexis, Metropolitan of Moscow (VOI), Volzhsky University named after V.N. Tatischev (VUIT) and reveal the following outcomes. Additionally the possibilities of the universities to work as self-study organizations were analyzed.

Table 2 represents the survey results for the above mentioned requirements represented in many scientific studies and which are known to be important for effective smart university [7].

5 Experiment: QFD Simulation

5.1 The Emergent Property Planning

The first stage for QFD includes “the emergent property planning”.

Figure 1 reveals quality indicators for the kernel of information-pedagogical system which are considered to be the indicators for smart university development. The indicators are identified and based on Regulations [10] representing the key notes the successful smart organization is to be based on (Tables 2 and Table 3).

Tables 4 represent benchmarking for 3 universities.

According to the represented results and client’s requirements, weighted scores and results of comparison are transferred to the “quality houses” represented (Table 5).

Table 3 The cause-effect diagram

Customers' or stockholders' requirements	The Kernel quality indicators for the information and pedagogical system (System Development Indicators + Characteristics)	Units of measurement the range from 0 to 1		
		TSU	VOI	VUIT
Requirement 1	The convergence indicator, K1	0.6	0.7	0.5
Requirement 2	The risk indicator of the loss convergence, K2	0.5	0.5	0.6
Requirement 3	Intellectual activity index of the employee, K3	0.8	0.6	0.5
Requirement 4	Employee readiness for innovation indicator, K4	0.8	0.7	0.5
Requirement 5	Knowledge transformation indicator, K5	0.6	0.7	0.5
Requirement 6	Knowledge transfer indicator, K6	0.6	0.6	0.5
Requirement 7	Indicator of conformity of available knowledge to the requirements of the environment, K7	0.8	0.6	0.6
Requirement 8	Effectiveness for knowledge core, K8	0.8	0.7	0.6
Requirement 9	Knowledge volume, K9	0.8	0.8	0.8
Requirement 10	Knowledge growth, K10	0.8	0.7	0.6
Requirement 11	Timing for knowledge core, K11	0.7	0.6	0.5

Table 4 Benchmarking for information-pedagogical kernel at smart university

Factors	Assessment			target value	improvement level	weightiness	weightiness, %
	VOI	TSU	VUIT				
T1	+	+	+	5	1	4	12.4
T2	—	+	—	5	1	4	12.4
T3	+	+	—	5	1.25	6.25	19.4
T4	—	+	—	5	1.25	6.25	19.4
T5	+	+	+	5	1.25	5.32	20.2
T6	+	+	—	4	1.33	6.65	16.2
						32.47	100

Table 5 The correlation links between rooms (the author's vision)

Consumer expectation	Importance of expectation	Q1	Q2	Q3	Q4	Q5	Q6
T1	4.3	 9					
		111.6			37.2		
T2	4.6					 9	
						111.6	
T3	4.9	 3		 9	Δ 1		Δ 1
		58.2		174.6	19.4		19.4
T4	4.9		 9				 9
			174.6				174.6
T5	4.5	Δ 1	 9				
		20.2	181.8				
T6	4	Δ 1			 9		
		16.2			145.8		

5.2 Designing the Quality House

Correlation links between the presented indicators should be defined and calculated.

The following symbols are represented:

the triangle symbol represents (1) a weak link;

the oval sign (3) represents moderate link;

the double oval sign (9) represents high correlation.

When filling out matrices, the QFD team can numerically estimate the priority of a quality characteristic in terms of its contribution to overall customer satisfaction. The main advantage of the proposed method is that the matrices allow comparing the organization with other institutions in a demonstrative form. It is also possible not only to calculate the most important for the project characteristics, but also to analyze the results taking into account the market requirements.

In our example, they are as follows:

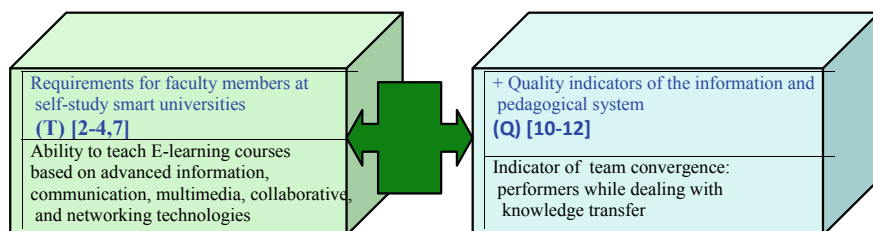


Fig. 2 Result: implementation of consumer requirements for a smart university specialist (fragment, author's vision)

- (1) The possibility to teach courses using advanced information, communication, multimedia, collaboration and networking technologies (corresponds to Q1).
- (2) A teacher as an innovator in the use of advanced EE technologies (corresponds to Q4).
- (3) Teacher as an adapter of developed innovative methods and means of modern EE (corresponds to Q5).
- (4) the ability to communicate using advanced technology (corresponds to Q6).

Thus, the resulting cortege of indicators IPS (Q1, Q4, Q5, Q6) can serve as an assessment of the formation of a new emergent property of the teacher, consisting in the willingness to function in a self-learning intelligent organization.

Due to the calculations the authors define substantial integration between the indicators. $T1 \text{ and } Q1 = > 111.6$; $T2 \text{ and } Q5 = > 111.6$; $T3 \text{ and } Q3 = > 174.6$; $T4 \text{ and } Q2 = > 174.6$; $T5 \text{ and } Q2 = > 181.8$; $T6 \text{ and } Q4 = > 145.8$

Thus, the following conclusion based on the results of the simulation and integration can be made: nowadays there is a necessity for a smart university to foster a specialist possessing skills due to stakeholders' requirements consisting of the following features (Fig. 2).

According to the above mentioned simulations and the results of integration the employees possessing skills and features corresponding to the stakeholders' requirements are considered to be a must at smart university. The skills of being able to apply smart technology to the knowledge transfer at the international educational are the key ones.

6 Conclusion and Future Trends

Conclusions. Modern management tools allow managers to identify various cause-and-effect relationships that characterize the activities of the analyzed structures in the terms of a systematic approach and modern trends.

1. The paper proposes such management tools as quality deployment function and benchmarking, which allow managers justifying the choice of those quality indicators for fostering and development of an intellectual university.
2. The practical importance of the research is that the timely identification of the needs for the required competencies will allow training and fostering the specialists demanded in the educational market in advance, "to order".

The novelty of the research is that QFD toolkit has been applied to education needs.

The future trends

- (1) Developing the methodology of QFD application for university managers who are engaged in its strategic development;
- (2) Testing the system of indicators for assessing the competence of QFD application for educational purposes.

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Comprehensive Unified Indicator of the Smart System's Quality: Application to e-Learning



Natalia A. Serdyukova and Vladimir I. Serdyukov

Abstract This paper discusses the issue of algebraic methods in the digitalization of smart systems that allow building complex characteristics of qualitative and quantitative measures of the system, and on this basis, a comprehensive unified indicator of the system's some properties is introduced. Within the framework of the Zermelo–Fraenkel set theory axiomatic, the existence of such an indicator is proved. In fact, thereby the solution of the generalization of multicriteria Operation Research problem has been obtained within the framework of the Zermelo–Fraenkel set theory axiomatic. Received results can be used in e-learning and in smart learning theory in assessment of control function of smart learning system's functioning.

Keywords Algebraic methods · Digitalization of smart systems' functioning outcomes · Qualitative and quantitative measures

1 Introduction

In modern conditions of the covid-19 pandemic, distance learning is becoming increasingly important, in which one of the essential places is given to the system of monitoring learning outcomes. For the successful implementation of the control function in the smart-learning system, constant monitoring of students' knowledge is required, which, in turn, leads to the need of processing a huge amount of statistical data. Another serious problem is the lack of a unified complex of qualitative and quantitative measures that characterize the control function (feedback) in the smart learning system, despite the huge number of mathematical methods used in this area.

Other areas closely related to the problem of finding a comprehensive unified indicator relate to technical areas of knowledge, for example, the problem of optimal design of IPsec-based virtual private networks (VPN). Optimal design of IPsec-based

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virtual private networks depends on multiple factors and parameters, such as the architectural model, hardware and software setups and the technical platform solutions, network topology models, modes of the tunnel's operation, levels of the Open Systems Interconnection model, encryption/decryption algorithms, modes of cipher operation, security protocols, security associations and key management techniques, connectivity modes, parameters of security algorithms, computer architectures, the number of tunnels in the virtual private networks, and other factors.

The present paper discusses the technique of using of algebraic methods in the digitalization of smart systems theory that allows to build a unified complex characteristic of qualitative and quantitative measures of the system, and on this basis, a comprehensive unified indicator of the system's properties. In fact, thereby the solution of the generalization of multicriteria Operation Research problem has been obtained within the framework of the Zermelo–Fraenkel set theory axiomatic. Within the framework of the Zermelo–Fraenkel set theory axiomatics, the existence of such an indicator is proved. The need to search for new mathematical methods when solving, for example, planning and forecasting problems in the field of innovative systems is explained by some shortcomings of the mathematical methods used at present, [1–11]. Here we shall use the designations as in [1, 2].

2 Problem Statement

In [1, Chap. 3], we have defined P —property of the system, where P is a unary predicate, given on the class of smart systems, and have shown how to formalize P —properties of the smart systems using the notion of a group of factors G_S , that describe the system S . Also we have considered such properties as sustainability, efficiency, innovativeness, compensational possibilities of a system [1], structural sustainability, [2] and so on. Each of these properties can be described by different qualitative and quantitative characteristics and measures.

For example, in [1], we proposed an algebraic formalization of innovation and effectiveness concepts of smart systems and constructed a tensor assessment of the efficiency of smart—systems functioning. Let's remind it for convenience. We propose to use homomorphisms of the group G_S of factors defining the system S into the group $GL(n, \Delta)$ of linear homogeneous transformations of the vector space R^n as tensor estimates of the efficiency of the functioning of the system. Also, we can consider homomorphisms of the group of factors G_S that define a system S in the group $GL(n, \Delta)$ of linear homogeneous transformations of the vector space Δ^n over an arbitrary field Δ .

So, the huge number of such indicators needed to describe smart system's functioning maximal accurate rise a problem:

If there exists one comprehensive unified indicator of the smart system characterizing P —property well-enough?

The stated problem is in tight connection with the solution of the multicriteria Operation Research (the last task is the stated problem's discrete case).

Fig. 1 Boolean ring of two elements

$\ominus$	$\emptyset$	A
$\emptyset$	$\emptyset$	A
A	A	$\emptyset$

$\cap$	$\emptyset$	A
$\emptyset$	$\emptyset$	$\emptyset$
A	$\emptyset$	A

We shall prove within the framework of the Zermelo–Fraenkel set theory axiomatics, that such an indicator can be constructed.

3 Main Results

Let's consider P —property of a smart system S modelled by a group of factors G_S defining the system S . Let $k_1, \dots, k_n$ be the criteria¹ for evaluating the P —property of the smart system, and let $z_1, \dots, z_n$ be the indicators characterizing the criteria for evaluating the property P of the smart system $k_1, \dots, k_n$ respectively. Let's consider the Boolean algebra $\mathbf{K}_P = K_P = \langle \{k, \dots, k_n\}, \vee, \wedge, \neg, 0, 1$ generated by the set of criteria $\{k, \dots, k_n\}$. That is the basic set K_P of the algebra $\mathbf{K}_P$ is closed under all principal operations $\vee, \wedge, \neg, 0, 1$ of algebra $\mathbf{K}_P$.

The P —property can be identified by the criterion of the first moment of time of its manifestation in the functioning of the smart-system (the first moment of its appearance) and by the nature of the functioning of the smart-system—the nature of the functioning of the smart-system changes with the appearance of the property P (the scale of functioning of the property P). The question arises:

How can one identify changes in the “level” of P —property of a smart system S modelled by a group of factors G_S defining the system S .

Let's consider some examples concerning e-learning.

Example 1 Let's consider P —property of a smart system S modelled by a group of factors G_S defining the system S . Let k_1, k_2 be the criteria² for evaluating the P —property of the smart system, and let z_1, z_2 be the indicators characterizing the criteria for evaluating the property P of the smart system k_1, k_2 respectively. Let's consider the Boolean algebra $\mathbf{K}_{P(2)} = K_{P(2)} = \langle \{k_1, k_2\}, \vee, \wedge, \neg, 0, 1$ generated by the set of criteria $\{k_1, k_2\}$. That is the basic set $K_{P(2)}$ of the algebra $\mathbf{K}_{P(2)}$ is closed under all principal operations $\vee, \wedge, \neg, 0, 1$ of algebra $\mathbf{K}_{P(2)}$.

Let's consider Boolean ring $\mathbf{B} = \langle B, \ominus, \cap, \setminus, 0, 1 \rangle$ of two elements given by its tables of operations (Fig. 1):

Here $\ominus$ —symmetric difference of sets: $A \ominus B = (A \setminus B) \cup (B \setminus A)$.

We shall construct completion $\widehat{\mathbf{K}_{P(2)}}$ of $\mathbf{K}_{P(2)}$.

Let's consider ring $\mathbf{Z}_2 = Z_2, \oplus, \ominus, 0, 1$ of residue classes of integers modulo 2 (Fig. 2).

¹A sign on the basis of which an assessment, definition or classification of something is made <https://znachenie-slova.ru/criterion>.

²See Footnote 1.

Fig. 2 Ring of residue classes of integers modulo 2

$\oplus$	0	1
0	0	1
1	1	0

$\odot$	0	1
0	0	0
1	0	1

So, from here one has that $\widehat{K_{P(2)}} \cong \widehat{B} \cong \widehat{Z_{2^\infty}} \cong Q_2$ —field of 2-adic numbers, [3].

To monitor indicators z_1, z_2 one may need a huge of measures, so the following question arises: how can we simplify the system of indicators without losing the quality of system's S management?

Example 2 Let P —property determines the quality of the system of education S . In [1] we decomposed system S into the following three components:

1. Knowledge system S_1 (information subsystem)
2. Methodological and methodical complex S_2 of the knowledge system S (an adaptive subsystem)
3. The system of students S_3 (target subsystem, target audience).

Among the most important questions concerning the functioning of these subsystems there are questions related to:

- knowledge system S_1 (information subsystem): how can one provide the necessary and sufficient level of knowledge system required for successful work in modern conditions?
- methodological and methodical complex S_2 of the knowledge system S (an adaptive subsystem): how can one provide the necessary and sufficient level of methodological and methodical training required for assimilation of the knowledge system and successful work in modern conditions?
- system of students S_3 (target subsystem, target audience): how can one to track and evaluate the level of knowledge of the target audience?

There exist a huge set of indicators to provide these requirements, [4, 5], but the sharp problem is how not to overlook and survey all this array of indicators to manage rightly the system of education. The following question arises: how can we simplify the system of indicators without losing the quality of system's S management.

Example 3 The P —property determines the property determines whether it belongs to one of the e-learning areas.

As in [1] we can change a little construction for evaluating the efficiency of smart system gen in [1, Chap. 6] and to build by analogy with it a tensor assessment of P —property of smart-systems functioning. Again, the question arises: how can we simplify the system of indicators without losing the quality of system's S management.

Let's remind for convenience that in book [1], we proposed a tensor assessment of the efficiency of smart-systems functioning. Let's remind how it was done

for convenience of reading. We propose to use homomorphisms of the group G_S of factors defining the system S into the group $GL(n, \Delta)$ of linear homogeneous transformations of the vector space R^n as tensor estimates of the efficiency of the functioning of the system. Also, we can consider homomorphisms of the group of factors G_S that define a system S in the group $GL(n, \Delta)$ of linear homogeneous transformations of the vector space Δ^n over an arbitrary field Δ .

Examples 2 and 3 lead to the question of the existence of one complex indicator on a scale with values in R^+ , assessing the property P .

To construct such measure let's consider a finite set of criteria for evaluating P —property of a smart system. Let P —be a property of a smart system depends (characterized by factors³ that determine system S) on factors $a_1, \dots, a_n$ that determine system S , and let $x_1, \dots, x_n$ be numeric indicators that characterize factors $a_1, \dots, a_n$ respectively. Let $k_1, \dots, k_{n_m}$ be the criteria⁴ for evaluating the P —property of the smart system, and let $z_1, \dots, z_{n_m}$ be the indicators characterizing the criteria for evaluating the property P of the smart system $k_1, \dots, k_{n_m}$ respectively. Let's consider the Boolean algebra $\mathbf{K}_P = K_P = \{\{k_1, \dots, k_{n_m}\}, \vee, \wedge, \neg, 0, 1\}$ generated by the set of criteria $\{k_1, \dots, k_{n_m}\}$. That is the basic set K_P of the algebra $\mathbf{K}_P$ is closed under all principal operations $\vee, \wedge, \neg, 0, 1$ of algebra $\mathbf{K}_P$. Let's note that the elementary theory $Th(\mathbf{K}_P)$ describes the P —property of the system S .

Now we shall try to find a complex indicator on a scale with values in R^+ , assessing the property P . To do this let's introduce a metric on the algebra $\mathbf{K}_P$.

We need the following corollary from [6–8],

Stone's Theorem

Any two finite Boolean algebras of the same power are isomorphic.

A finite Boolean algebra by Stone's theorem is an algebra of sets; therefore, there are 2^r in it for some natural. Any finite Boolean algebra is atomic. If it has 2^r elements, then r is the number of its atoms.

Also, we shall need the definition of metric space. It runs as follows.

Definition 1 A metric space is an ordered pair $\langle M, \mu \rangle$, where M is a set and μ is a function $\mu : M \times M \rightarrow R$ such that for any $x, y, z \in M$ the following conditions holds:

1. $\mu(x, y) = 0 \Leftrightarrow x = y$;
2. $\mu(x, y) = \mu(y, x)$;
3. $\mu(x, z) \leq \mu(x, y) + \mu(y, z)$.

It is well-known that each Boolean algebra is a distributive lattice with complements, and, conversely, each distributive lattice with complements, is a Boolean algebra. [7, 8].

In [6] the following theorem is proved:

³A factor is a cause, a driving force of a process, which determines its nature or its individual features. <https://ru.wikipedia.org/wiki/Factor>.

⁴a sign on the basis of which an assessment, definition or classification of something is made <https://znachenie-slova.ru/criterion>.

Theorem 1 [6] *The Stone space of a Boolean algebra A is metrizable if and only if A is no more than countable. So, we can view completion $\widehat{K_P}$ of the metric space.*

$$K_P = K_P = \{\{k, \dots, k_{n_m}\}, \vee, \wedge, \neg, 0, 1\mu >,$$

with respect to the metric μ .

Recall that a metric compact is a metric space that is compact as a topological space with a topology induced by the metric. A point $a \in X$ is called a fixed point of the map: $X \rightarrow X$, if $f(a) = a$.

Recall also that a map $T : X \rightarrow X$ is called a weakly contraction map if the following inequality $\mu(f(x), f(y)) < \mu(x, y)$ holds for any different points $x, y \in X$.

It is known (Browder's fixed-point theorem) that any weakly contraction map of a compact to itself has a single fixed point, [9].

We apply now Browder's fixed point theorem to the completion $\widehat{K_P}$ of the metric space $K_P = K_P = \{\{k, \dots, k_{n_m}\}, \vee, \wedge, \neg, 0, 1\mu >$, with respect to the metric μ , and we obtain that the fixed point of the contraction map f of this completion determines an indicator which is a complex indicator on a scale with values in R^+ , assessing the property P , where P is the property for a system to be innovative one, and P is identifying by two criteria, k_1, k_2 , where k_1 is scientific and technical novelty and k_2 is production applicability, for example. Now, let's consider the diagram.

$f : \widehat{K_P} \rightarrow \widehat{K_P} \rightarrow^\mu R^+$. For a fixed-point a of the contracting map f we have:
 $f(a) = a, \quad \mu f(a) = \mu(a)$.

So, we have the following definition:

Definition 2 We call a the universal indicator of the property P , and.

$$\mu f(a) = \mu(a) \text{ is its estimate.}$$

Remark Using embedding $K_P \hookrightarrow \widehat{K_P}$ one can clarify the quantitative estimation of property P .

Let's now return now to Example 2, where property P defines determines the quality of the system of education S .

Then P —property can be characterized using the following criteria (or features, or conditions), for example:

- k_1 —the necessary and sufficient level of knowledge system required for successful work in modern conditions k_1 relates to the system of knowledge S_1 ,
- k_2 —the necessary and sufficient level of methodological and methodical training required for assimilation of the knowledge system and successful work in modern conditions? k_2 relates to the system of methodological and methodical complex S_2 of the knowledge system S (an adaptive subsystem),
- k_3 relates to the system of students S_3 (target subsystem, target audience): the measure which show how can one to track and evaluate the level of knowledge of the target audience. Herewith, k_3 can be represented by a set of several indicators.

Remark As to concern measure k_3 one can use the technique of quasi-fractal algebraic systems worked out in [2] to construct a unified complex of indicators.

Now let's construct a metric space as in previous case

$$K_3 = K_3 = \langle \{k_1, k_2, k_3\}, \vee, \wedge, \neg, 0, 1; \mu \rangle.$$

Then, as in previous case, we apply theorem about the Stone space of a Boolean algebra, [9], and Browder's fixed-point theorem to the completion.

$$\widehat{K_3} = \widehat{K_3} = \langle \{k_1, k_2, k_3\}, \vee, \wedge, \neg, 0, 1; \mu \rangle.$$

with respect to the metric μ . We obtain, that the fixed point a of the contraction map f of this completion determines an indicator which is a complex indicator on a scale with values in R^+ , assessing the property P , (maximization of a function or system quality indicator).

Again, as in previous case, we call a the universal indicator of the property P , and

$$\mu f(a) = \mu(a) \text{ is its estimate.}$$

Remark One can repeat this algorithm for any finite number $k_1, \dots, k_m$ of criteria of the property P . So, we get the main result of the paper.

Theorem 2 *Let P —be a property of a smart system depends (characterized by factors⁵ that determine system S) on factors $a_1, \dots, a_n$ that determine system S , and let $x_1, \dots, x_n$ be numeric indicators that characterize factors $a_1, \dots, a_n$ respectively. Let $k_1, \dots, k_{n_m}$ be the criteria⁶ for evaluating the P —property of the smart system, and let $z_1, \dots, z_{n_m}$ be the indicators characterizing the criteria for evaluating the property P of the smart system $k_1, \dots, k_{n_m}$ respectively. In the frame of the Zermelo–Fraenkel set theory axiomatics, there exists a universal indicator of the property P , and it can be clarified by quantitative methods.*

Example 4 Let's consider the abbreviation SMART. It is well known that the management uses the abbreviation SMART, which has been proposed by G. T. Doran in 1981. The abbreviation SMART means a smart target and combines capital letters from English words indicating what the real goal should be: Specific, Measurable, Attainable, Relevant, Time-bounded. Thus, SMART is a well-known and effective technology for setting and formulating goals. Let's construct Boolean algebra $K_P = K_P = \langle \{k_1, \dots, k_m\}, \vee, \wedge, \neg, 0, 1 \rangle$ generated by the set of criteria $\{k, \dots, k_m\}$, which characterize the property "SMART", and apply the supposed in Theorem 2 assessment with the help of one indicator.

⁵A factor is a cause, a driving force of a process, which determines its nature or its individual features. <https://ru.wikipedia.org/wiki/Factor>.

⁶a sign on the basis of which an assessment, definition or classification of something is made <https://znachenie-slova.ru/criterion>.

The following question arises: how can one refine and regulate the level of the smart system property using the estimate defined in the Theorem 2. To do this one can consider the countable set of criteria $\{k, \dots, k_m, \dots\}$, which characterize the property “SMART”, after that the countable Boolean algebra.

$$K_{\aleph_0} = \langle K_{\aleph_0} = \langle \{k, \dots, k_m, \dots \mid m = 1, 2, \dots\}, \vee_{\aleph_0}, \wedge_{\aleph_0}, \neg, 0, 1 \rangle$$

with countable infinite operations $\vee_{\aleph_0}, \wedge_{\aleph_0}$ considered, for example in [10] should be constructed. After that one should apply a construction similar to that used in the proof of the Theorem 2. The usage of countable infinite operations $\vee_{\aleph_0}, \wedge_{\aleph_0}$ can be used to improve the accuracy of assessment of the level of the property “SMART” of a system and to link qualitative and quantitative measures of the system.

4 Application in E-Learning

One of the most urgent problems in the current conditions of the Covid-19 pandemic is the problem of SMART (in the sense proposed by G. T. Doran) organization of distance learning.

There are many works devoted to this problem. In [4] the following vision on smartness property has been declared: “Our vision for engineering of smart learning analytics—the next generation of systems and tools for learning analytics—is based on the concept that this technology should strongly support “smartness” levels of smart academic institutions such as adaptivity, sensing, inferring, anticipation, self-learning, and self-organization”. Again, the same problem is how to measure the level of these properties and how to use these measurements for clarifying these levels? Here we can use Theorem 2 to simplify the measuring process at the final stage and to use one universal indicator instead of a set of characteristics.

Another urgent problem in the current conditions of the Covid - 19 pandemic is the problem of controlling the knowledge of the training contingent when using distance learning. In [5] the following “The availability of tools that measure, collect, clean, organize, analyze, process, store, visualize and report data about student academic performance in an academic course and/or student overall academic progress in the selected program of study has given rise to the field of learning analytics for student academic success. Student data representation, processing and prediction, as a central part of learning analytics system. Other studies on CSVA (abbreviation as in [5], 38 articles, 9.50%) focused on methods to visually explore data (using interactive graphs) to highlight useful information and produce accurate and data-informed decisions. The completed analysis of designated and multiple additional publications relevant to SAP (student academic performance) data, EDM and LA (learning analytics) areas clearly shows that, unfortunately, those publications do not provide a systematic approach to (1) possible forms of SAP data representation

and (2) control flow for SAP data processing in LA systems.” is stated. Here we can use Theorem 2 to simplify controlling process of the knowledge of the training contingent, at the final stage and to use one universal indicator instead of a set of characteristics, and on this basis to build a hierarchical system of indicators that simplifies monitoring of learning outcomes.

5 Conclusions. Future Steps

Conclusions. Within the framework of the Zermelo–Fraenkel set theory axiomatic, the existence of a comprehensive unified indicator of the smart system’s properties is proved. In fact, thereby the solution of the generalization of multicriteria Operation Research problem has been obtained within the framework of the Zermelo–Fraenkel set theory axiomatic (the last task is the stated in this paper problem’s discrete case).

Future Steps. In [1] we have defined the notion of P —pseudo-innovative systems using the notion of duality. The definition runs as follows, [1], that is:

Definition 6.6 Algebra $\bar{G} = G[\{f_\alpha^{n_\alpha} | \alpha \in \Gamma\}]$ is called P —pure projective (P —pseudo-innovative system) if every diagram with exact P —pure string.

$$\begin{array}{c} \bar{G} = \langle G[\{f_\alpha^{n_\alpha} | \alpha \in \Gamma\}] \\ \swarrow \varphi \quad \downarrow \psi \\ 0 \rightarrow \bar{B} = \langle B[\{f_\alpha^{n_\alpha} | \alpha \in \Gamma\}] \xrightarrow{\alpha} \bar{A} = \langle A[\{f_\alpha^{n_\alpha} | \alpha \in \Gamma\}] \xrightarrow{\pi} \bar{C} = \langle C[\{f_\alpha^{n_\alpha} | \alpha \in \Gamma\}] \rightarrow 0 \end{array}$$

that is $\text{Im } \alpha$ is P —pure in $\bar{A}$ can be extended to commutative one that is $\psi\pi = \varphi$. Also, we have received the following result, [1]:

Theorem 6.8 A system that is both P —innovative and P —pseudo-innovative, is P —degenerate, that is, it loses property P and cannot function.

In the same way as we have done it for P —innovative systems, one can get the universal indicator of the P —pseudo-innovative system and its estimate.

So, the following question arises: how the assessments constructed in Theorem 2 for P —innovative and P —pseudo-innovative systems are connected?

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Classroom of the Future Realization in the Industrialization Era: Towards 4.0 Learning



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Abstract The momentum at which fourth industrial revolution (Industry 4.0) as a growing and expanding component of Information Technology (IT) with a focus on integrating the physical-to-digital and digital-to-physical world cannot be overemphasized. Industry 4.0 has attracted great attention in science and engineering, technology, industry and the society at large so much that change is inevitable even within the education sector, hence making education 4.0 fashionable. Education 4.0 is a response to industry 4.0 needs where humans and technologies are aligned to enable new capabilities. The education sector can be dynamic role players in fourth industrial revolution by leveraging enabling technologies such as IoT or cyber-physical systems, analytics, additive manufacturing, robotics, HPC, artificial intelligence and cognitive technologies, advanced materials, and augmented reality to realize the classroom of the future. This paper takes a visionary approach—to rethink education for providing education with a purpose. We envision an integrated Education 4.0 Architecture as a guideline for the implementation of Education 4.0 in a South African context. This architecture will be informed by African National and Regional Industrialization Strategic Objectives.

Keywords Education 4.0 · Conceptual architecture · Road map

1 Introduction

The future scenarios of the political, social, cultural and economic sectors will depend on the contributions of the Students of our schools today. More than ever before, education must be visionary and future-oriented, in the face of stunning scientific and

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technological innovations and changes, unprecedented socio-economic challenges and opportunities, Surprising socio-political reforms, and amazing cultural reawakening. Educational innovations are imperative, and would no doubt be effective if they are research-based and instilled with technology of education (i.e. Systematic Approach to the teaching–learning process); and technology in Education (e.g. Use of hardware and software's) [1]. Education has been shaping our future since very early times in the history of humanity. In the twenty-first century, the fast pace imposed by the digital era impacts not only the way we work or run our day to day life, but also the way we learn. Education is being transformed by technology to create new teaching models and inspire new ways of thinking, to adapt to our constantly changing society. Hence constantly challenged to keep up with these changes to better serve individuals and behaviour [2].

Our constant search for knowledge is one of the reasons that education is squeezed to change with times, through the multitude of communication channels and the enormous amount of information that surrounds us.

The recent study done by IBM, suggest that the changes that the education sector experiences will be improved when data can accompany the individual throughout their lifelong learning journey [3]. Amongst the commotion that comes with continuous technology change—Industrial Revolution has given a new motivation to educational transformation [4], one that emphasize integration analysis and use of data as key capabilities for the Industrial Internet [5]. In recent years, education experts recognize the profound impact that numerous technological innovations in ICT is having on education. They agree that Education 4.0 will be shaped by innovations and will indeed have to train teachers to produce students that are innovation inclined. The major influence of Education 4.0 is the industrial revolution concept which has become the buzzword that will affect almost all spheres of the human race. The world has to date experienced three industrial revolutions [6]:

1. The First Industrial Revolution was based on the introduction of mechanical production equipment driven by water and steam power.
2. The Second Industrial Revolution was based on mass production achieved by division of labour concept and the use of electrical energy.
3. The Third Industrial Revolution was based on the use of electronics and IT to further automate production.

The main concept driving Industry 4.0 is the use of cyber-physical systems which extends the IoT relationship of the act of physical-to-digital and digital-to-physical leaps that are somewhat unique to industrial processes [7]. There is still no consensus regarding what Education 4.0 is. In the context of this paper it is defined as a school of thought that encourages non-traditional thinking when it comes to imparting education by essentially using technology-based tools and resources to drive education in non-traditional ways [8].

Predictions say that most of the routine activities including monitoring will be entirely or partly taken over by machines [9]. One example is the IBM Watson machine that has developed AI-based expert system that can replace junior lawyers. AI system has also been developed, having potential to replace basic-level medical

practitioners. This may mean fewer jobs for entry-level professionals in these areas, although specialist jobs may [5]. In this context, it is vitally important to impart appropriate education to the future workforce. Based on the trends so far, researchers predict that Industry 4.0 will necessitate profound changes in major aspects of education, i.e.-content, delivery or pedagogy, and structural management of education. Industry 4.0 demands changes in the contents of not only technical education, but also education in general. New emphasis will have to be given on certain skills and new contents have to be added, therefore new educational programmes will have to be developed to meet changing demands across disciplines [10].

2 Background

In the era of Industry 4.0, jobs that require creativity are likely to stay, irrespective of discipline, hence Education 4.0 must be able to produce highly creative graduates with the ability to think critically. Graduates must be innovative and entrepreneurial and have cognitive flexibility to deal with complexity as many of them will be co-working with not only a man but also robots. The need for better communication and collaborative skills will be far more important than ever, demanding graduates to acquire self-learning skills to remain relevant in the era of rapid changes [11].

2.1 *The Evolution of Digital Education*

In this section we are noting interesting advancement of education, based on the Industrialization approach [12].

Education 1.0. The foundation of essentialist curriculum is based on traditional disciplines such as mathematics, natural science, history, foreign language, and literature where classrooms are teacher oriented.

Education 2.0. Kim [13] suggests that Web 2.0 has really been the flowering of new relationships between individuals and businesses and reflects new ways of thinking that the technology has facilitated or created. It is about engaged conversations that take place directly, and do not rely on top-down management, but peer feedback and mentoring. It is also an incredibly effective restructuring of how learning takes place, and somehow, we somehow have to figure out how to bring this experience into our learning institutions before they become obsolete.

Similar to Web 2.0, Education 2.0 includes more interaction between the teacher and student; student to student; and student to content or expert. Some school administrators and educators seem to have taken some steps and moved into a more connected, creative Education 2.0 through using cooperative learning, global learning projects, Skype in the classroom, and shared wikis, blogs and other social networking in the

classroom. Industrial and education experts say that in 2013, Education 2.0 should have been the norm not the exception.

Education 3.0. Education 3.0 is based on the belief that content is freely and readily available. It is self-directed, interest-based learning where problem-solving, innovation and creativity drive education. Education 3.0 is characterized by rich, cross-institutional, cross-cultural educational opportunities within which the learners themselves play a key role as creators of knowledge artefacts that are shared, and where social networking and social benefits outside the immediate scope of activity play a strong role.

Education 3.0 is a constructivist, heutagogical approach to teaching and learning. The teachers, learners, networks, connections, media, resources, tools create a unique entity that has the potential to meet individual learners, educators, and even societal needs.

2.2 *Education 4.0*

The explosive hype around Industry 4.0 is a warning that the education sphere is tapping into Education 4.0 which is suggested to affect all the domains (Cognitive, Affective and Psychomotor) in the Bloom's model. In the cognitive domain, Application, Analysis, Evaluating and Creating will become way more important relative to the lower level cognitive skills [10].

Industry 4.0 will require the human resources in the Industry 4.0 to be adequately equipped with digital and data literacy. Students across disciplines will, therefore, need to gain digital and data literacy during their studies [14]. The convergence of Man and machine in Industry 4.0 will mean that the disciplinary distance between science and technology, and humanities and social sciences will be reduced.

An important segment of Industry 4.0 will feasibly be situated at the intersection of disciplines such as electrical engineering, mechanical engineering, business administration and computer science. This requires universities in collaboration with industry to come up with new interdisciplinary programmes [15].

3 Motivation and Problem Statement

In a South African education system, most government schools are still operating on education 1.0. Although some are denying it. The introduction of learning devices, like tablet does not equate to industrialized education, there are additional components needed to arrive to education that is fully industrialized. Due to the dramatic change brought by Industry 4.0, it is undeniable that nearly 50% of the core curriculum content subject knowledge acquired during the first year of a technical degree will be outdated by the time a student graduate. In all, 65% of children,

entering primary school today, will work in completely new job types that do not yet exist.

A framework for twenty-first century learning has been developed and 10 public primary schools (sandbox schools) have been selected in the Waterberg district of Limpopo, where a pilot study was initiated in 2018. The Funda UJabule School on the UJ's Soweto Campus will also participate. The project is meant to inform stakeholders about how twenty-first century competencies could be infused into schools. The pilot project may demonstrate possibilities and challenges that come with implementing this initiative in everyday practice, as well as how to proceed in everyday practice with a feasible curriculum for the future that our children face.

Our work is an attempt to design a generic integrated system framework that take into considerations the holistic vision of the South African government in ensuring that SA Education system compete with other parts of the world.

Our proposed architecture is a result that emanates from concepts borrowed in the 5-level architecture and the 4 Stage IoT Solutions Architecture, which when integrated activates the cyber-physical world (a critical component towards industrializing Education).

4 Methodology for Developing the Conceptual Architecture

Our study adapts the U.S. Department of Defence Architecture Framework (DoDAF) [16] harmonizing approach to design our proposed systems architectures.

Systems Architecture is a generic discipline to handle objects which either exist or to be created called “systems”, in a manner that supports reasoning about the structural properties of these objects, most often using architecture framework tool. It responds to the conceptual and practical difficulties of the description and the design of complex systems. An architecture of a system is a model consisting of properties, relationships between various elements, behaviours and dynamics and multiple views of a system, which are complementary and consistent.

Depending on the context, System's Architecture can in fact refer to [17]:

- an architecture of a system that describe or analyse a system architecture;
- a method to build the architecture of a system while meeting business needs;
- a body of knowledge for “architecture” systems while meeting business needs, methods, heuristics, and practices.

According to [18], System's Architecture helps to describe consistently and design efficiently complex systems such as:

- an industrial system (the original meaning of Systems Architecture);
- an IT infrastructure (Enterprise Architecture);
- an organization (Organizational Architecture);
- a business (Business Architecture), or
- a project (Project Architecture).

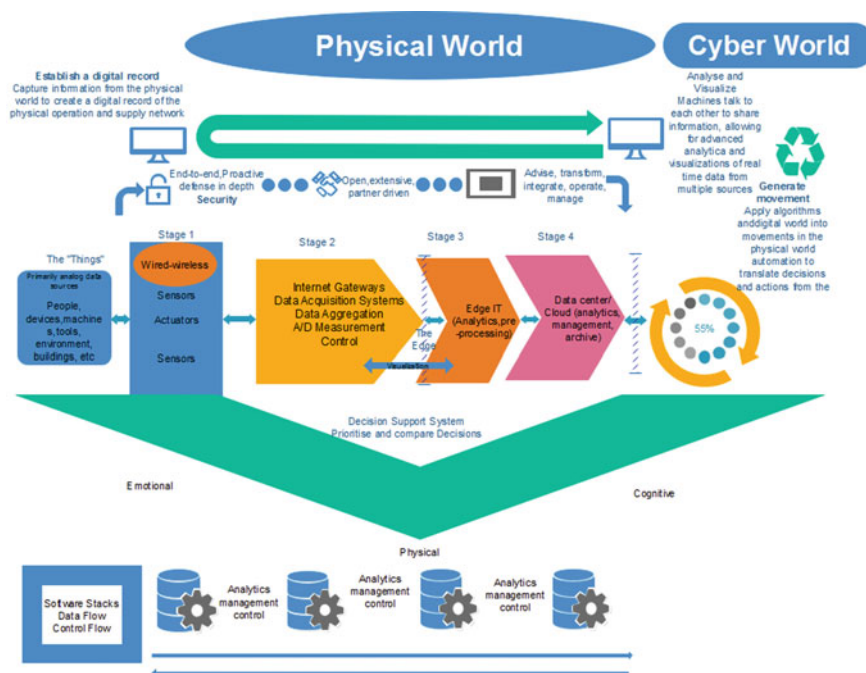


Fig. 1 Proposed Education 4.0 System Architecture

System's architecture will consider any system with a socio-technical approach, even when dealing with a "purely technical" system. Socio-technical approach is particularly considered during the design or transformation of a system, the systems in the scope of this design or transformation can be divided in two separated systems in interaction: the product which consists of (the system being designed or transformed the project, the socio-technical system in charge of the design or transformation of the product. teams, tools, other resources) and their organization following strategies [18].

4.1 Education 4.0 System Architecture

To realize our proposed Education 4.0 System Architecture, we take a systems architecture approach, originally known as Industrial system approach. The complementary tool we are to use in organizing the various elements of our integrated Education 4.0 system architecture is the U.S.A Department of Defense Architecture Framework (DoDAF) which has its primary focus on the description of the architecture.

Figure 1 presents the proposed System Architecture that provides a step-by-step guideline for developing and deploying an Education 4.0 application. The System architecture consists of three main functional components:

- (1) The physical world that ensures the connectivity of objects to machines; this is where IoT related mechanism take place;
- (2) The Cyber-world where the real value of the IoT for manufacturers is analyzed from cyber-physical models of machines and systems; and
- (3) The dimensions of labor to enable innovative ideas necessary in industry 4.0 era [19].

Physical world. In the physical world the IoT mechanism is at play, the data coming from objects (people, devices, machines and other tools) is relayed by sensors and actuators. The connections are made possible by secure implementation of computer networks, internet and communication protocols to the cyber-world. This communication is based on normal internet protocols or dedicated protocols such as MTConnect.

Two factors to be considered at this level. The various types of data and various methods to aggregate, measure and control data to enable data streaming at the edge of the network where the data is being generated. At the edge, the efficient data processing is possible since large amounts of data can be processed in real-time near the source, with no latency and reduced Internet bandwidth. The edge allows smart applications and devices to respond to data almost instantaneously, as its being created, eliminating lag time, therefore invoking the cyber world. This both eliminates costs and ensures that applications can be used effectively in remote locations.

Cyber World. In the cyber world, data analytics, management and archive takes place through cyberphysical mechanism where the virtual twin of the physical machine exists in the cyber world. That is, the physical assets from IoT and its digital twin (software model) that imitates the behavior of the physical assets establish a digital record working in parallel with the physical component but with a huge difference. They are not bounded by time or location. Information from the physical world is captured to create a digital record of the physical operation and supply network to run complex analytics.

Dimensions of Labor. The three dimensions of labor (cognitive, emotional, and physical labor) plays a critical role in the industrialization era. Due to industrialization, most, physical work that only humans can do was gradually performed by machines. It helps in identifying the important components that will be at play in the fourth industrial revolution. We are at an era where some jobs are going to disappear or rather take a whole new direction, at the same time many jobs are to be created because of Industry 4.0. The cognitive component plays a critical role in implementing knowledge of the monitored system to enable it to prioritize and compare decisions.

5 The Conceptual Education 4.0 Road Map: South Africa's Case

The education 4.0 research intends to prepare interested stakeholders for disruptive changes at the connection of learning and skills change landscape. To customize the proposed Education 4.0 System Architecture to the case of South Africa, a conceptual road map was designed that will assist different education stakeholders identify action steps to prepare organizations for the future (public and government sector).

We took a learner centered approach in designing our road map, which is known to be effective in realizing Education 4.0. The learner-centered approach encourages practical engagement, innovation, collaboration, ubiquitous transparency amongst other things.

The major components of this approach are as follows (Fig. 2).

1. **Visible World:** the use of ubiquitous approach to achieve detailed, real-time reporting that will influence all aspect of organization decision making. Meaning the organization viability can be tracked and documented.
2. **Business and Data Intelligence:** Simplifying business and data intelligence for human comprehension requires an approach that acknowledges the role of both machines and people.
3. **Science at Work:** The data ecosystem is by far a critical aspect in designing everything, including systems, hiring algorithms that will influence skills of the future. Hiring practices, management and training will have to rely on deeper understanding of neuroscience and complex behavioral algorithms.
4. **Trans disciplinary Rules:** A world cultivated by diversity is innovative and will likely survive the disruptive fourth industrial revolution. Diversity makes groups

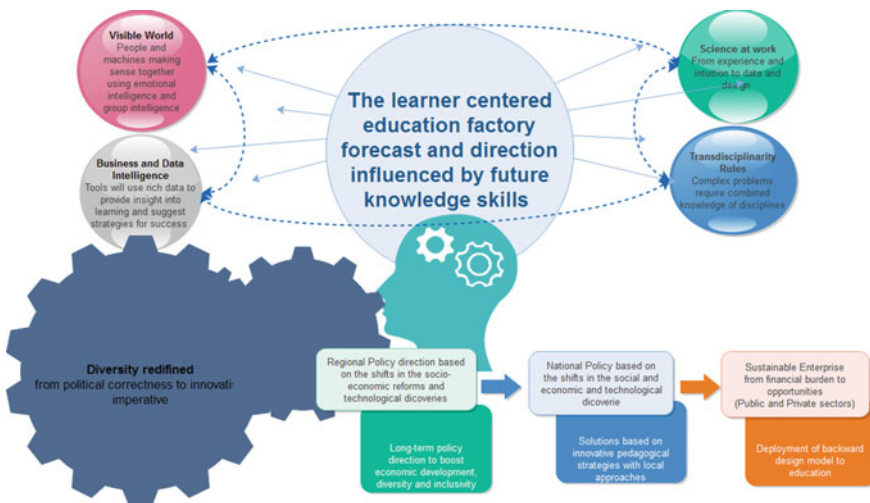


Fig. 2 Education 4.0 Conceptual Road Map: Case of South Africa

smarter and innovative to solve complex problems, may require universities to take an anti-disciplinary approach.

Governments, businesses and individual learners must work in harmony for real, comprehensive change to close the gap of between education outcomes and expected industry skills as the world enters the Fourth Industrial Revolution through efficient Fourth Industrial Revolution-inspired strategies. Implementation of curricula and technologies for learning programs for the benefit of people with no access to quality education in Africa at large is necessary to enable inclusion creativity, cognitive skills, and design and systems thinking. This is to prepare them for the fourth industrial revolution and empower them to unlock economic prosperity and social justice in their communities and beyond. A system design thinking approach that make use of backward design model in deploying education may lead to achieving more sustainable designs.

A combination of focused investment strategies and practical government policies in line with the fourth industrial revolution is the key to breaking free from high fixed cost models and ensuring accessible quality education.

6 Conclusions and Next Steps

Conclusions. South Africa is taking a decisive approach to ensure that the country become major role players in industry 4.0 commotion.

The Department of Science and Technology has currently approved a new white paper replacing the one adopted in 1996, which identifies the fourth industrial revolution as a key focus. The sandbox curriculum project initiative together with our piece of contribution is a good starting point that will attract interest from researchers, whose focus is on education, hence showing some level of readiness for Education 4.0 in South Africa.

The limitations of this study are that it is a concept that has not been tested. It stands an opportunity to trigger more research questions and ideas on how Education 4.0 can change the Educational Landscape in South Africa.

Next Step. For future research direction, we plan to deploy and evaluate the backward design model in education, particularly, particularly by distinguishing a case study for testing the model.

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The Concept of Transition from Smart University to Smart Business in Digital Economic Environment



Anna A. Sherstobitova, Lyudmila V. Glukhova, Svetlana A. Gudkova,
Elena N. Korneeva, Olga A. Filippova, and Tatiana G. Lyubivaya

Abstract The concepts of “Digital economy” and “Internet of Things” have now become common directions of modern development where the “Industry 4.0” concept is considered as of prior importance. Higher education drives this aspect by means of practice-oriented approach in teaching where the required competences are studied and applied in practice for certain professional targets. The authors proposed three basic models reflecting the conceptual vision of the digital transformation of the smart environment management competencies required for effective activities of state unitary enterprise: Model of data collection and processing during the digital transformation; Model of audit by the IoT using, Model of three-stage implementation of the IoT System in the state unitary enterprise “Kupinskoye”.

Keywords Digital economy · Digital transformation · Practice-oriented approach · Organizational management · Indicators · Innovation

1 Introduction

Nowadays the digital economy is considered as a way of life where all forms of business and industrial relations are converted into digital information. The importance of digitalization and the development of the digital economy is defined and described in Russian regulatory documents.

The society’s demand for a new generation of employees possessing the skills of applying smart technologies and their implementation in business and production is fully represented in many modern studies [1–3]. Smart universities and smart businesses follow the practice-oriented approach that drives the employer-sponsored targeted training and constant development of the required hard&soft skills for special purposes [4, 5]. In other scientific studies the importance of feedback parameters and

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indicators for monitoring the achievement of sustainable targets for smart university and business development are emphasized [6–8].

The designed and implemented “work flow charts” reflect the necessary actions of the Russian Government to support digital transformation for all the facilities in the RF. In particular clause 4.1 states that in the fourth quarter of 2020 all digital communication opportunities in rural and county areas are to be provided and it is noted that “...all settlements with a population from 500 to 10,000 people are to have broadband access to the Internet” (as part of the investment program of PJSC Rostelecom “Rural Communication”) [9], and consistent widespread introduction of LPWAN communication networks using domestic equipment in small towns and urban-type settlements in Russia [9] are to ensure the narrow-band communication on LPWAN technology for the collection and processing of telematic information (p.4.7, the implementation date—2024). All these requirements are of great importance for the development of state unitary enterprises which activities are aimed at extracting profits in favor of the state and cover their own costs [7, 8]. The practice of smart technologies implementation in the activities of state unitary enterprises in a county of the RF is shown. The empiric base for the research is the state unitary enterprise “Kupinskoye” located in Bezenchuksky district of Samara region in the RF.

The novelty of the proposed solutions is to design the system for training the personnel of state unitary enterprises located in counties or outskirts of the RF to use modern management methods based on the means of the digital economy for their professional goals and activities.

The methodological basis of the study is a set of scientific approaches, allowing to determine the ways for the identified problems solving.

2 The Concept of Digital Transformation: Literature Review

The authors of the article agree with the conclusions about the current trends and the requirements to transform the digital skills acquired in a smart university according to the Internet of Things concept for the new society (Industry 4.0) described in papers [10–12]. Currently the concepts of IoT (Internet of Things) and IIoT (Industrial Internet of Things) are coming deeper to all the spheres of modern environment including farming and agriculture and not associated.

The IoT is known as a connected set of technical devices and computing facilities capable of transmitting information between each other without human involvement. Such control can already be provided by a smartphone or smart speakers or other smart devices appearing on the market recently.

IIoT implies a system of interconnected computer networks and connected production facilities with built-in sensors and specialized software. It allows continuous system monitoring for remote monitoring and control in an automated mode, without human involvement. Also the authors of the article follow current trends in the

application of The Internet of Things for industry and business Concept described in a number of modern studies on the society and industry digitalization issues. [13, 14].

The analysis of modern studies has shown that there are a lot of effective IoT applications for agriculture. They allow the intelligent data collection on temperature, rainfall, soil moisture, wind speed, soil pest infestation; soil quality, and more. The use of IoT devices in “smart farms” makes it possible to make predictions of yield growth through the use of quality seeds, agrochemicals, fertilizers, water, consumed electricity “as needed” [14]. The monitoring and control are performed by various smart sensors, sensors, and robotic complexes.

The results obtained can be used to automate farming methods, make informed management decisions to minimize risks and waste, increase labor productivity. In order to solve the tasks set by the Government of the Russian Federation to develop digitalization in the outskirts of the RF regions a number of mandatory technical and technological solutions are to be implemented and provided. In 2019, the Government of the Samara region and the Russian telecommunications operator and provider of digital services PJSC “MTS” announced a strategic partnership for the implementation of projects aimed at the digitalization of the Samara region economy, urban economy and social sphere of the region.

3 Research: The Problem, Goals and Objectives

Research problem. The problem of the study can be described by the current situation in the RF that many industrial enterprises in the agricultural industry are not ready for the real implementation of digital transformation tools. One of the reasons is the lack of specialists skilled in the implementation and promotion of Smart tools.

The object of the study is the unitary state enterprise in the Samara region. It is an advanced and developing enterprise and its key activity is milk production.

The other related activities include the provision of fodder base, plant growing and livestock breeding for dairy production.

Research Goal of the study is to find the reasons for the slow adaptation of the enterprise to the digital transformation and suggest tools for performance increasing for the case study.

Research Objectives. The main objectives of the study are the following:

- (1) analysis of economic activity and identification of key factors to define promising development directions of digital transformation for milk production (profiling activity of the test enterprise) and provision of quality fodder (related activity of this enterprise);
- (2) conducting a survey of employees, statistical processing of the results, simulation for new processes, taking into account the needs of digital transformation;

- (3) development of recommendations for discussion and strategic planning of enterprise development in the the digital transformation age, aimed at reducing the risks of production activities.

The research, testing and assessment were carried out on the basis of Togliatti State University due to the initiative assignment of the unitary enterprise management “Kupinskoye” in the period May–July 2020. For a comprehensive comparative analysis the documents of accounting and financial statements were studied and analyzed for the audited period 2016–2020.

Research Hypothesis. The hypothesis of the study is the following: the conceptual approach to simulation and designing the digital transformation models for the internal processes of such enterprises will identify the necessary management tools to reduce the financial and social risks for the enterprise.

A survey of employees of the enterprise, the main suppliers and consumers was conducted. The results of processing were classified and grouped. After statistical processing of the results (Fig. 1) the following conclusions were obtained:

The main financial risk is the risk of “financial under-profit” due to the slow digital transformation of internal processes within the enterprise.

The main social risk is the risk of untimely and slow adaptation of employees and administrative and managerial staff of the enterprise to the transformation processes of the digital economy. One of the main reasons that initiate the emergence and development of risks is the insufficient level of knowledge, skills and abilities of the enterprise’s employees, as well as the insufficient level of understanding the importance and inevitability of the ongoing innovative processes of organizational change.

The joint assessment activity of Togliatti State University and the state unitary enterprise “Kupinskoye” is represented in the Fig. 2.

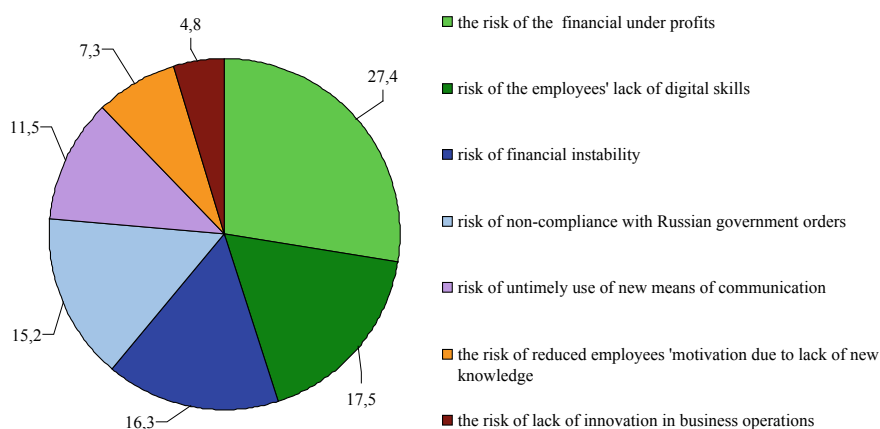


Fig. 1 The results of processing (author's vision)

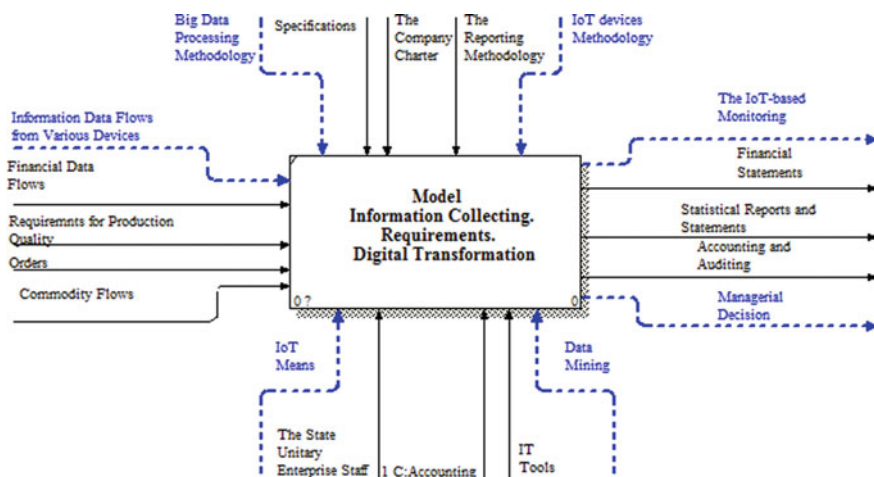


Fig. 2 The simulated model for data collection and processing in the digital age (the authors' vision)

4 Results

4.1 Information Flow Model: Current

As a result of the auditing and analysis of the information flow at this enterprise the model of existing information flows and information management for the enterprise functioning is designed. The author's vision for the processes of collecting, processing and storing information in two aspects is shown: Fig. 1 represent current information flows and Fig. 2 the projected ones for the near future (Figs. 1 and 2). The dotted line show the projected flows.

System and process approaches are used to for simulation. Instrumental base is structural analysis methodology SADT which is widely used in process approach for the process activity visualization.

4.2 Information Flow Model: Simulation

The analysis of the projected and simulated model revealed that its implementation makes it possible to process big data through the use of those modern digitalization tools that are aimed at the development of end-to-end digital technologies, monitoring systems based on IoT and artificial intelligence systems. This solution makes it possible to improve the quality of classification and systematization for all the data coming from various sources of the production cycle, its collection, storage and processing. The use of modern digital information processing tools reduces the risk

of data loss or the risk of untimely processing. In contrast to the existing standard approach used today it is possible to create the added value for all the participants involved in the production process. This is the novelty and practical significance of the authors' proposed model.

4.3 The Internal Audit Methodology

Taking into account that nowadays there is a great need for rapid implementation of digitalization tools in the agriculture production environment the authors propose a methodology for internal audit of the enterprise by using IoT reflected in the form of a table model (Table 1). The model is designed by taking into account the requirements of GOST R ISO 9001-2015. The model reflects a consistent assessment of compliance to the organization's own requirements and to its quality management system; to the requirements of modern conditions of economic development and compliance with the requirements of the standard.

For analyzing the activities of a large agro-food unitary enterprise "Kupinskoye", a survey among the employees of milk production on the demand to introduce IIoT in the activities of the enterprise for the near future (period 2022–2025) was conducted. The survey included analyzed groups of questions united by a single name, such as: "labor organization", "management organization", "milk production", "crop production", "feed base".

Statistical processing of the survey results revealed poor trends of the enterprise's performance although at present there is a positive dynamics for its activity as a

Table 1 The model for internal audit (the author's vision)

Level	IoT means	IT	Activity
Preparatory	IoT sensors (pressure, lighting, humidity, fire, water, electricity, etc.)	Skype, Zoom, e-mail, specialized software, network technology LPWAN / NB-IoT, in some cases - 2G and satellite communications and others	Collection and analysis of information, its classification and systematization
Main	Robotic complex, smart farm, smart warehouse etc	Office information technology, Data Mining, mathematical and simulation modeling packages, statistics, analysis, etc	Verification of compliance with the requirements of GOST R ISO 9001–2015
Final	Monitoring and control system	Technologies of information transfer and team discussion	Control activities

whole, taking into account the results obtained in June 2020 and the forecast for 2021 (Fig. 3, the author's vision).

The figure shows that the analyzed enterprise works steadily, makes a profit although it is considered to be insignificant, and the number of costs and revenues are approximately comparable. The authors' model obtained on the basis of the submitted statistical reports (report "About Profit and Losses") at the audited enterprise "Kupinskoye" for the analyzed period and for the forecasted perspective.

Figure 4 shows the trend line of logarithmic type. It reflects the possibility of predicting the results of the available raw data. In this case it shows a positive trend of income growth.

The results of the company's employees and partners survey revealed the following situation (Fig. 5) reflecting the needs of implementation of the industrial internet of things.

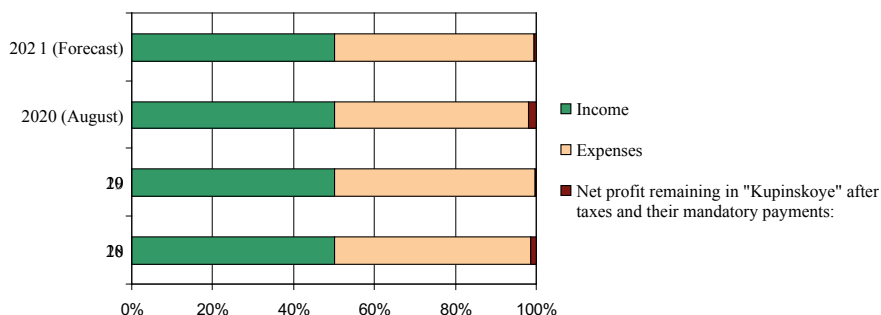


Fig. 3 Dynamics of the consolidated indicators of financial and economic activity for the state unitary enterprise "Kupinskoye"

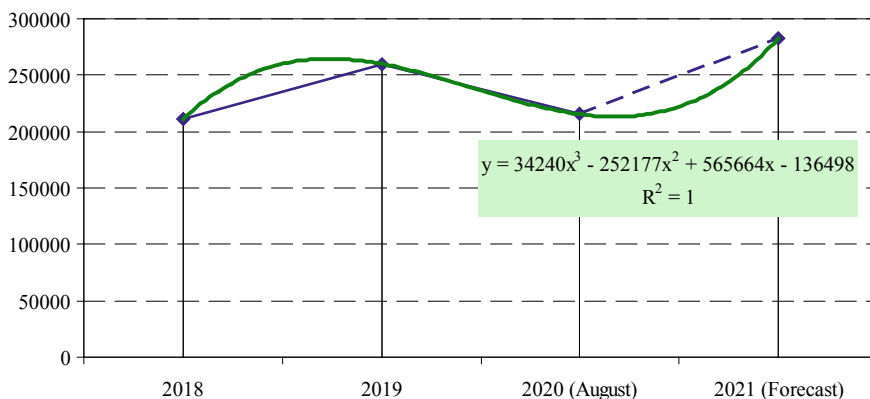


Fig. 4 Dynamics of profit in the analyzed period (the author's vision)

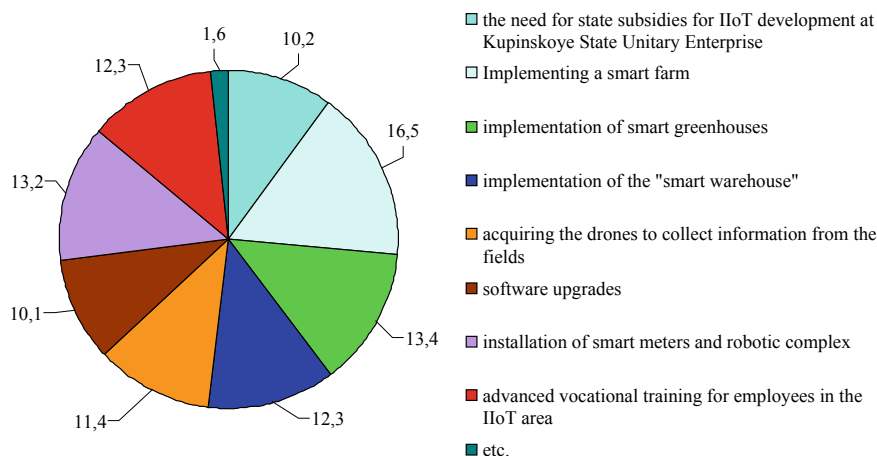


Fig. 5 The employees' assessment for digitalization transformation of the state unitary enterprise "Kupinskoye" (the author's vision)

4.4 The Model for Assessing the Needs of the Enterprise's Digitalization and Smartness

According to the results of statistical processing of the survey it was found that the largest number of responses associated with a group of questions in the section "organization of labor" and "forage".

The section "labor organization" included questions about the need to improve working conditions by introducing "smart farm", "smart greenhouses", "robotic complexes", "smart meters". In total this section "work organization" caused the greatest interest and took the 1st place in the needs for the further development of the enterprise, gaining in total 45,1%. The second place was taken by the nomination "fodder supply" (33.6%). A lot of answers related to the section "management organization" are not shown on the diagram. The diagram reveals the answers that took the highest position, reflecting the need to improve the employees skills in obtaining new knowledge about IIoT (7.3%).

In the "dairy production" section, the greatest acceptance was given to the installation of a robotic complex (13.2%).

5 The Model of Digital Transformation for the State Unitary Enterprise

At the end of 2020 and while making strategic planning for the future, the enterprise's managers decided to conduct the employees questionnaire and partners survey

about the priority implementation of IIoT. The questionnaire with clear indicators, criteria for their evaluation, short workshops for the employees in the digitalization of agriculture subject area were designed and conducted.

5.1 Digital Transformation by Means of IIoT

For this purpose the authors of the study proposed the Model of digital transformation by means of IIoT for the unitary enterprise “Kupinskoye” (Fig. 6).

The model conventionally reflects the authors’ first vision for the possible implementation of IIoT. The model does not specify specific tools. However here is the authors’ vision of what can be included in each of the implementation stages.

Since dairy production is the key area of “Kupinskoye” enterprise activity, the first priority tasks (1st stage) are the introduction of IoT sensors for temperature and humidity, IoT sensors for “smart feeding”, IoT sensors for lighting and so on. The priority task can also include the introduction of a robotic complex for the system monitoring of the health of cattle, robotic milking, quality control of incoming feed, etc.

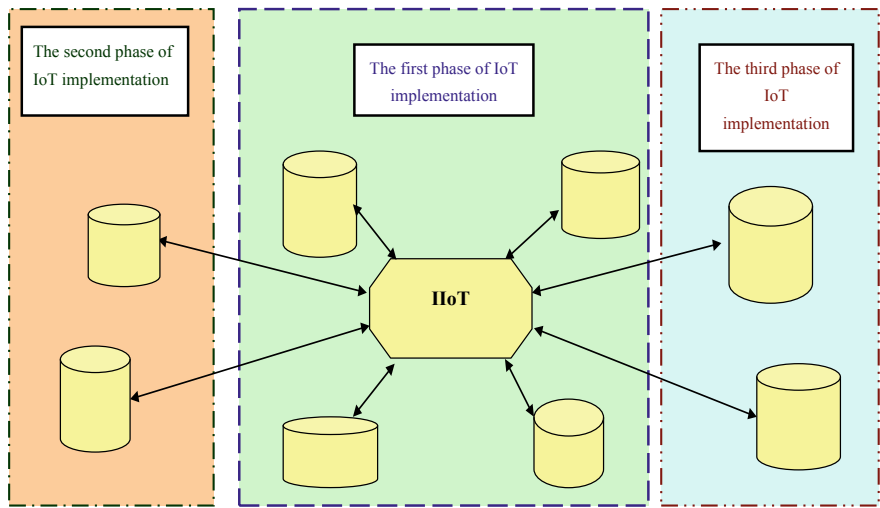


Fig. 6 Schematic design model of the stage-by-stage IIoT implementation for the unitary enterprise “Kupinskoye” (the author’s vision)

5.2 *Step-by-Step Technology for Implementing the Model*

The sequential and staged planning for the implementation of IIoT tools is justified by the following stages:

The first stage includes the implementation of four basic components: “smart farm”, “smart warehouse”, “energy”, “transport”. The “smart farm” includes a robotic complex for automated milking operations, various sensors controlling the required parameters of humidity and temperature regime in the room; “smart warehouse” solves the issues of quality control of fodder: forage, silage, haylage. A smart warehouse contains sensors for temperature conditions, fire safety, humidity, etc.; “energy” controls the efficiency of energy distribution in buildings and structures;

The second stage is the introduction of IoT sensors in the area of “Veterinary”, “Logistics (warehouse)”. This will allow remote management (diagnosis, control, regulation) of the processes of prevention for livestock.

The third stage is represented by the digital transformation of the enterprise’s activity for “Farming” and “Logistics “. These management solutions will provide remote monitoring and automated proctoring of land and crops. The solutions are aimed at improving crop yields, due to the timely processing of information coming from drones (monitoring of cultivated areas) IoT sensors “watering on demand,” IoT sensors of fuel consumption by agricultural machinery and etc.

Therefore in the strategic development plan for the state unitary enterprise “Kupinskoye” the mass staff training of in the digital transformation area and the auxiliary processes for enterprise’s smartness and its competitive advantages are to be introduced and implemented.

6 Conclusions and Next Steps

Conclusions. The conducted research has shown that there possibilities for digital transformation in the enterprises’ performance of Samara region. The case study is the state unitary enterprise “Kupinskoye” located in the rural area of the Samara region.

1. The study reveals that the obtained results can be applied to many enterprises located in rural areas to ensure their competitiveness in local markets.
2. Digital transformation is possible through the mass introduction and spread of cloud applications, broadband Internet, IoT technology, big data management ser-vices (Data Mining), interconnected IT solutions based on Internet of Things plat-forms (robotics, drones, sensors, etc.).
3. There is a need for a developed IoT ecosystem, including the partnership of a wide range of participants and data exchange between them; the availability of

- domestic developers with experience in creating digital integrated solutions and business analysts are considered to be necessary for rural areas' development.
4. The proposed original management models can be applied to different types of enterprises for strategic and organizational planning decisions.

Next steps. The next planned steps in this project include an implementation of new training programs for the system of professional education aimed to fostering the skills for digital transformation of society.

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The Potential of Smart Pedagogies for Sustainable Education in Foreign Language Teaching



Štěpánka Rubešová

Abstract Education for sustainability has its inherent place within the teaching methods in the Centre of Foreign Languages (CFL), University of Hradec Kralove (UHK). The main purpose of this paper is to analyze the employment of smart pedagogies and emerging didactics as a means of education for sustainability, focusing on quality education and fostering skills priorities. The methodology of this study discloses significant sustainability competences and skills, inevitable for students' future employability relying on the integration of four steps of SAMR model in second language classroom teaching in (CFL UHK). The classification of sustainability competences went under scrutiny in this study.

Keywords Smart pedagogies · Sustainability · Second language

1 Introduction

The smart pedagogical methods employed for the education for sustainability in this study highlights the priority of acquiring lifelong skills and competences from the second language classroom teaching [1]. The technology-enhanced learning, such as the usage of SAMR model [2–4], engage learners and make a real difference in students' active thinking, critical reflection, analysis, team collaboration, creativity and understanding. There have been twenty years of research into the life-long skills and competences for education for sustainability [5], which focuses on the improvement of the quality of life through acquiring the knowledge and shift towards active learning methods.

The SAMR model with its four steps of substitution, augmentation, modification and redefinition [3, 4, 6] was designed for the practical gradual usage of digital devices in the classroom. The first step, substitution, is the stage where functional change is affected. The second enhancement stage, augmentation, increases the productivity of learning with the use of digital technology. The last two stages, modification and

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redefinition, constitute transformation steps, while the modification step employs technology for redesigning the traditional tasks and redefinition develops educational techniques which have not been feasible without digital technology.

The lifelong universal literacy, creative literacy and literacy across disciplines advocate the critical position between more than 50% of adult students who consider the lifelong learning indispensable for their employability [7], concepts which have become the primordial aim in teaching languages in (CFL UHK).

2 Objectives

The aim of this study is to capture and investigate the impact of the learning process of second language acquisition on education for sustainability among university students from the Centre of Foreign Languages, University of Hradec Kralove (CFL UHK). The current situation of Covid-19 period led educators to rely on smart pedagogical approaches, employing the integration of digital devices [8–10] and pedagogical concepts, performed in the educational process by the SAMR model application. The paper analyses, gathers and classifies sustainability competences through the cognitive level of the SAMR model [3, 4] in a second language classroom. Zooming in on the future students' employability, the current Covid-19 learning approaches employed in the second language acquisition seminars in the (CFL) encourage to develop and expand students' personal ideas, reflections and enhance skills and capacities with the main goal to broaden competences for their future employability.

3 Methodology

Seminars of foreign language acquisition in (CFL UHK) focuses on competence-based education to develop professional and social skills, attitudes, knowledge and social abilities through SAMR enhancement and transformation model.

Sustainable Development Goals [11] rely on education as the main driver, through developing the adequate skills, knowledge and competences of students in the learning process, which contributes to their future decent employment and the increase of young people integration into the labor market. Teaching didactics employed in language seminars fastens the attitudes to the world cultural, language and ethical diversity and focuses on the absorption of knowledge and skills considering them as essential for students' future employment. The methodology draws on previous findings of studies on sustainability competences and pedagogical approaches for sustainable education [1, 12–16].

The correlation between the SAMR integration [17–19] into the second language learning methods and emerging outcomes for sustainable development, on the other hand, went under scrutiny in this study, grounding in the huge diversity of competence-based classifications [1, 13, 14] heading for employability.

The second language university seminars, such as Academic English language, Spanish ELE language, Spanish for Latin American Studies and Czech for foreigners have been studied for eight years and finally analyzed with the focus on the linkage between digital devices integration and educational outcomes; embracing skills, competences, attitudes and knowledge for students’ future employability [1, 12–16, 20, 21].

The proper classification of competences for sustainable education in second language seminars was based on previously existing classifications [1, 13, 14] with the aim to provide a snapshot of the (CFL) learning process.

4 Main Results

The employment of the SAMR model in foreign language learning in (CFL UHK) proved the development of the competences for sustainable development through the teaching process, encapsulating learning materials and drawn competences into four charts.

The first step of the SAMR model, *substitution*, is being integrated in students’ learning through the usage of digital language course-books published in *EdiEle*, *Oxford University Press*, *Edika*, etc., visual presentations and oral in-class assignments, employing *PowerPoint*, *Prezi* and others. Students are equally taught the strategies on how to prepare adequate and quality presentations, including digital techniques in order to make and keep contact with the audience and attract their attention, integrating poll and random picker mobile applications into their academic and professional presentations. These skills and competences such as digital literacy, personal involvement, future orientation and others gained in these second language seminars are undoubtedly beneficial and essential for students’ future employability and execution of their different job level positions (Table 1).

The pedagogical methods for sustainable education in second language teaching classrooms make a close linkage between digital SAMR *augmentation* and its

Table 1 The employment of SAMR—*substitution* in foreign language learning for sustainable education (SE)

	Learning material and sources	SE competences
SAMR—SUBSTITUTION	• Projectors, laptops, tablets, mobiles • Presentation programs <i>PowerPoint</i>, <i>Prezi</i> • Online course books <i>EdiEle</i>, <i>Edika</i> - Oxford University Press • Random Pickers	• Digital literacy • Communication and use of media • Future orientation • Personal involvement • Anticipatory thinking

output competences. In the classroom, students are motivated to practice grammatical and lexical units through interactive exercises where results are generated online. Supposing the majority of current students possess mobile phones supplemented by personal earphones, the active participation in the M-learning (mobile learning) through mobile applications (M-apps) such as *English Grammar in Use Tests*, *English daily Listening*, *English Conversation App*, etc. are highly recommended and supported. The M-learning substitutes the traditional way of paper self-testing for a mobile application and enhances the students' experience, develop objective evaluation and self-reflection of their knowledge and skills. The self-evaluation and self-correction are also encouraged by Word Review and Writing Assistance Software, such as *Grammarly*, *Scribendi*, etc. Similarly listening to English conversational extracts played through M-apps has a vital role in acquiring second language oral skills. Consequently, creativity, critical and systematic thinking is much easily developed by the employment of podcasts or various types of videos with or without concurrent questions (Table 2).

M-apps such as *Poll Everywhere* demonstrates that the model of SAMR *modification* stage represents a synthesis of academic and professional forum discussions, while sharing ideas, cooperation and voting in order to solve emerged issues among class peers is promoted. This web-based audience response system lets students interactively embed personal opinions directly into the presented and discussed subjects developing interdisciplinary reflection and critical thinking. The omnipresent video conferences performed through communication platforms such as *Microsoft Teams*,

Table 2 The employment of SAMR—*augmentation* in foreign language learning for sustainable education (SE)

	Learning material and sources	SE competences
SAMR—AUGMENTATION	- Interactive exercises through Mobile Apps <i>English Grammar in Use Tests</i>, <i>English daily Listening</i>, <i>English Conversation App</i> <i>Ver-taal.com</i> - Audio applications - Online dictionaries, thesaurus, Apps - Word review and writing assistance software <i>Grammarly</i> - Podcast with concurrent questions <i>Profedeele</i> <i>British Council podcasts</i> - Videos <i>TED</i>, <i>youtube</i> - Silent movies - Online crossword puzzles	- Complexity - Anticipatory thinking - Systemic thinking - Interdisciplinary work - Evaluation and self-evaluation - Self-correction - Self-reflection - Problem-solving - Critical thinking - Reflection or discussion - Cooperation - Structuring, ordering - Imagination, emotional thinking - Creativity - Case studies - Cross-cultural work - Responsibility - Future orientation

Zooms, *Skype* and others, have become the symbol of Covid-19 online education and contribute to the development of interpersonal, verbal and non-verbal communication (Table 3).

Integration of learning tool activities, which redefine a traditional task and which would not be possible without technology, is called SAMR *redefinition* stage. Online chatting with a virtual help assistant such as Spanish state-owned train company *RENFE*, the international travel agency *Invia*, discount website *Skrz*, and other software of chats for customer support are employed in the seminar didactics in order to encourage students of second language seminars to phrase their own ideas and real requests in a studied second language. By making direct queries students improve their structuring, problem-solving skills and finding solutions. In

Table 3 The employment of SAMR—*modification* in foreign language learning for sustainable education (SE)

	Learning material and sources	SE competences
SAMR—MODIFICATION	• C-learning—sharing documents <i>Microsoft Teams</i> <i>Moodle</i> <i>Google-Doc, CVC</i> • <i>Paddlet</i> • <i>Kahoot</i> • <i>Poll everywhere</i> • Visible concepts <i>Google Earth</i> • Virtual conferences <i>Skype, Microsoft Teams, Zoom</i>, etc • Graphic design tool website <i>Canva, Pixlr</i> • Gaming <i>Minecraft for Education</i>	• Digital literacy • Planning and realising innovative projects • Anticipatory thinking • Systemic thinking • Problem-solving • Critical thinking • Reflection or discussion • Structuring, ordering • Reflection or discussion • Online discussion forums • Complexity • Interpersonal communication • Cooperation • Team-work • Cross-cultural understanding • Non-verbal communication • Responsibility • Creativity • Imagination, emotional thinking • Game playing skills • Game programming • Agility • Strategizing • Risk-taking • Artistic skills • Self-evaluation • Self- reflection • Voting • Interdisciplinary work • Case studies • Future orientation

the redefinition stage, technologies allow for the creation of new tasks, recording and uploading videos through video makers, such as in *Renderforest*, *youtube* or others, or composing audio journals *Journify* and program stories in *Twine*, while cultivating rhetorical skills, performing, game programming and other sustainability competences. Students' virtual visits to the most famous museums or galleries in the world can be acquainted through the virtual guided tours using digital technological devices. The communicative software and applications such as *wonder.me* enables students to launch international discussions, take part in virtual projects while transcultural understanding and problem-solving are developed through the interdisciplinary and cross-cultural work environment (Table 4).

5 Conclusion

This study demonstrates that students of the Centre of Foreign Languages, University of Hradec Kralove (CFL UHK) are instructed and encouraged to develop competences for sustainable development in analyzed language seminars; Academic English language, Spanish ELE language, Spanish for Latin American Studies and Czech for foreigners. The mastery and literacy across disciplines gained through second language learning of the (CFL) are considered inevitable and essential in order to thrive in a lifelong career. Building on the previous studies, which demonstrated that more than half of adults (54%) [7], in the labor force believe that training and development of new skills are essential throughout their career, students at the (CFL) are encouraged to conceptualize, analyze, and evaluate diverse subjects.

The integration of the SAMR model into the learning materials of second language seminars in (CFL) contributes to the development of sustainability competences; such as critical thinking, creativity, transcultural understanding, self-reflection, self-evaluation, social-emotional perception, planning and applying solutions, processing and sharing information and a wide range of other skills and competences significant for students' future orientation and employability.

Table 4 The employment of SAMR—*redefinition* in foreign language learning for sustainable education (SE)

	Learning material and sources	SE competences
SAMR—REDEFINITION	• Video creation and uploading videos <i>Renderforest</i> <i>Youtube</i> <i>Vimeo</i> <i>Animaker</i> • Video contests <i>European video contests</i> • Recording audio • Audio journaling <i>Journify</i> • International forums, project creation, videoconferencing through continents <i>Microsoft teams, Skype, Zoom</i> <i>Flipgrids</i> <i>Wonder.me</i> <i>Google earth</i> <i>Wizer.me</i> <i>Genial.ly</i> • Chats for customer support <i>Renfe, Skrz, Invia, Ikea</i> • Augmented reality <i>Aumentaty</i> • QR codes • Virtual galleries, museum, field trips • Interactive nonlinear stories with international sharing <i>Twine</i>	• Advanced digital literacy • Responsibility • Creativity • Gamer programming competences • Map planning • Research content skills • Artistic feelings • Rhetorical skills • Performing • Empathy • Solidarity • Compassion • The social-emotional competences • Cooperation • Team-work • Reflection or discussion • Online discussion forums • Complexity • Anticipatory thinking • Systemic thinking • Structuring, ordering • Reflection or discussion • Critical thinking • Planning and realising innovative projects • Problem-solving in heterogeneous groups • Leadership • Adaptability • Interdisciplinary work • Interpersonal communication • Transcultural understanding • Non-verbal communication • Self-evaluation • Self- reflection • Voting • Case studies • Future orientation

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University Innovative Networking in Digital Age: Theory and Simulation



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Abstract The relevance of the chosen research topic is justified by the need to find the tools for remote collaboration of various educational teams in the process of the university development as an open university. The methods of research are methods of system analysis. The issues for the optimal self-adjustment of evolutionary process at smart university by using Hurwitz criteria and indicators are considered. The procedure of the smart university capacity evaluation is proposed. The study is based on the main methodological hypothesis: the implementation of smart technologies and environment will improve the quality of economic collaboration and cooperation between universities in the conditions of digital environment and challenges for innovation.

Keywords Digital innovations · System approach · Interaction between universities · Collaboration

1 Introduction and Literature Review

Nowadays the issues of “smart university” are increasingly discussed in the studies of both the foreign and domestic authors including a lot of definitions connected with the smart university such as “smart education”, “smart e-learning”, “smart technology”, “smart class”, “smart pedagogy” were introduced only a few years ago [1–3]. Leading academic institutions around the world are actively exploring ways to transform a traditional university into a “smart” one in order to adapt to changing society and economy. It should be noted that a “smart university” may have many components of a traditional university, but it is to possess several additional “smart” components to implement and actively use “smart” functions such as adaptation, self-learning, forecasting and self-optimization [4–8].

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Some studies reveal [9, 10] that any model of the universities' networking is the result of their organizational and legal changes, management efforts and development.

The University, as a complex system, requires integration of various ordinary and flexible models, restrictive hypotheses and tools for process structuring, use of cloud technologies, Big Data, Data Mining, smart technologies and modeling with the help of both classical (differential equations, algebraic structures, etc.) and non-classical representations of processes—neuro, fuzzy, fractal [11].

Universities which for one reason or another do not have enough investments for digital transformation of the educational process and do not create an infrastructure relevant to the digital economy can face a new danger of being pushed out of the educational market or taken over by the stronger competitor.

2 Theoretical Base

2.1 Modern Trends for Smartness

While considering a modern university that can be classified as a smart university we should study the peculiarities of its functioning from the standpoint of a complex system. The authors propose to consider the organizational structure of a smart university as a set of goals, capacity, environment, competencies of employees, processes and technologies (Fig. 1).

A lot of modern studies adhere the “working” interpretation of smart university as a unified and integrated system based on:

- clusters of educational institutions and partner structures;
- developed digital and media infrastructure of the system and its components (universities);
- highly motivated faculty and university employees;
- highly intellectual approach to education that allows to achieve higher and intellectually valuable results with the help of intellectual decisions

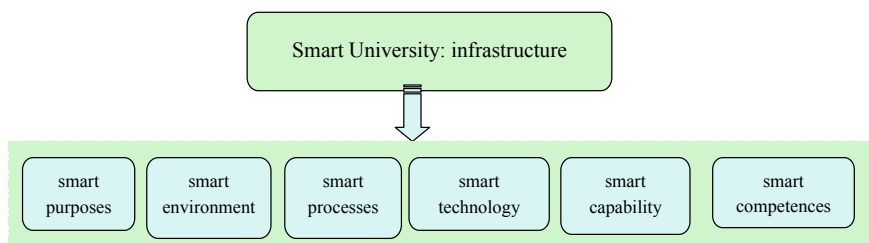


Fig. 1 The smart university system

- personal development and motivation.

This is an educational collaboration of intellectual agents, educational actors, supported by intellectual smart infrastructure and supported by the state, business, public organizations, partners and employers. The pedagogical functionality and structure of educational relationships are changing—from paradigms and choice of educational platforms to “digital footprint” and e-tracking of a student, from gamification to accounting for the design of a learning scenario, from distance learning systems to open education and social network consulting and training.

A lot of modern studies promote Smart Maturity Model (SMM) as a methodology used to design, develop, assess and continuously improve a smart university’s main business functions including teaching, learning, innovations and research, services, enrollment, management, administration, control, security, safety, etc. Nowadays SMM and its suggested maturity levels are known to be key note levels evaluating if the university ready for innovative networking. SMM defines 5 levels of university “smartness” maturity in terms of the university’s commitment to innovative collaboration and integration.

2.2 Simulation of Smart System Capacity: Its Complexity and Sustainability

The complexity of the tree structure and representation of the system can be set by the number of hierarchy levels, and the measure, according to the factors taken into account, can be determined integrally. For example the complexity of the tree structure for the system can be defined by the number of hierarchy levels and the measures according to the above mentioned factors. It can be determined integrally according to the formula (1):

$$s = \sum_{i=1}^n s_i k_i, \quad (1)$$

where n —the number for the system’s levels; s_i —complexity for i —hierarchy with k_i elements.

2.3 Simulation: Assessment at SMM Level

The decision maker can apply the principle of choosing an optimal solution by using different optimistic and pessimistic indicators including the criteria of Savage, Wald, Hodge-Lehman, Bayes-Laplace, Hurwitz, etc. [11]. For example, the Hurwitz

test reflects a balanced position between extreme optimism and extreme pessimism regarding to the assessment of SMART capacity.

Here the Hurwitz condition can be specified (2):

$$G_i = \gamma \min_{1 \leq j \leq n} \{a_{ij}\} + (1 - \gamma) \max_{1 \leq j \leq n} \{a_{ij}\}, \quad (2)$$

where $A = \{a_{ij}\}, i = 1, 2, \dots, m; j = 1, 2, \dots, n$ —winning matrix, $0 \leq \gamma \leq 1$ —subjectively selectable and identifiable parameter which is often assumed to be 0.5.

According to Hurwitz the “net strategy” whose performance indicator will take the highest value is considered as the following (3):

$$g = \max_{1 \leq i \leq m} \{G_i\} \quad (3)$$

or (4):

$$g = \max_{1 \leq i \leq m} \{\gamma \min_{1 \leq j \leq n} \{a_{ij}\} + (1 - \gamma) \max_{1 \leq j \leq n} \{a_{ij}\}\}. \quad (4)$$

Consider an example of choosing the best solution. Here the situation with the square matrix of wins with various SMART University solutions take the form (5):

$$A = \begin{vmatrix} 38 & 36 & 32 \\ 32 & 42 & 45 \\ 27 & 35 & 45 \end{vmatrix} \quad (5)$$

then with $\gamma = 0.5$ (often assumed, neutral, expectation) we get the vector (6)

$$G = \begin{vmatrix} 30 \\ 38.5 \\ 36 \end{vmatrix} \quad (6)$$

and an efficiency index equal to the maximum element of that vector or $g = 37.5$.

If you change the strategy to completely optimistic ($\gamma = 0$) or completely pessimistic ($\gamma = 1$), then, accordingly, we have vectors (7)

$$G = \begin{vmatrix} 38 \\ 45 \\ 45 \end{vmatrix}, \quad G = \begin{vmatrix} 32 \\ 32 \\ 27 \end{vmatrix} \quad (7)$$

and values $g = 45, g = 32$.

Therefore, the optimistic strategy in this situation turned out to be the most effective of the three considered. The education system, self-organizing, strives for synergistic processes and effects: self-development, self-discipline, self-education. Smart-educational (SmE) structure can become the base, infrastructure to ensure the growth of competitiveness and intellectual capital of society.

3 Our Results: Simulation

3.1 Smart Educational Environment of the University

Each university has its own corporate educational environment closely related to the socio-economic situation and demands of the region, smart models of interaction and their key factors. There are no standards, uniform methodology and architecture of the university information environment, therefore, according to information entropy, uncertainty and chaos are growing. The analysis of modern studies and the survey of the university staff at Togliatti State University (TSU) reveals the authors' vision for necessity of achieving the following development indicators (Fig. 2).

The main task of the teacher is to design the complex of basic competencies and culture of self-study, self-education. Figure 3 shows an analytical review of smart educational space revealed on the basis of Uskov's maturity model [2].

The authors examined the activities of three universities to assess their compliance with the maturity model. The universities involved in the experiment were:

- Kabardino-Balkarian State University (KBSU), city of Nalchik;
- Togliatti State University (TSU), city of Togliatti;
- V. N. Tatishchev Volga University (VUIT), city of Togliatti.

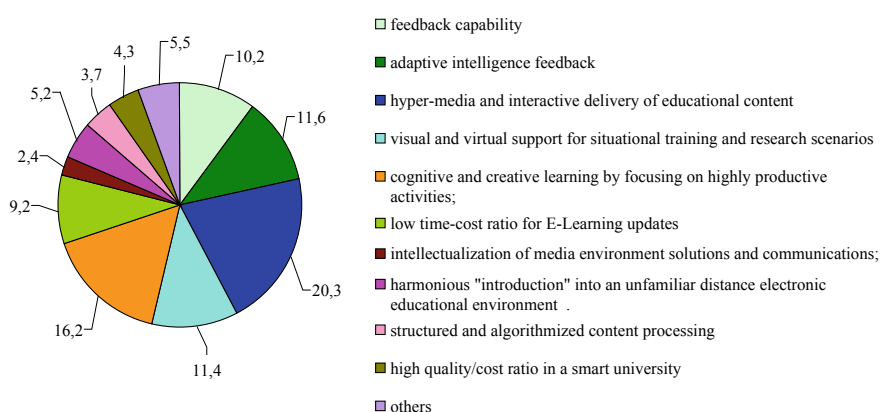


Fig. 2 The Development indicators (the author's simulation model)

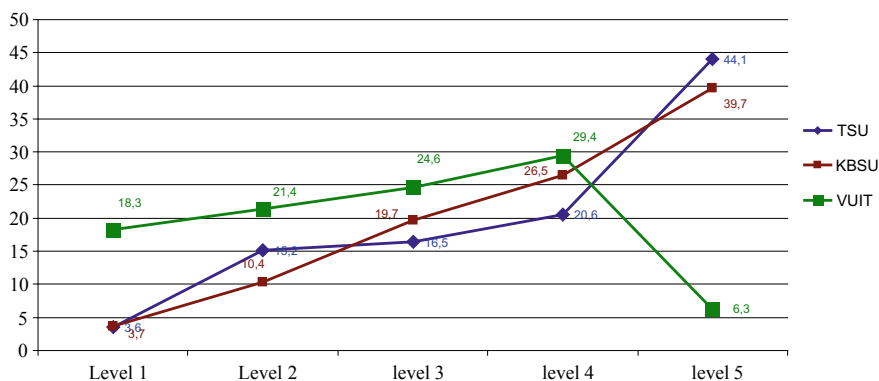


Fig. 3 The results of statistical processing of analytical data in 2020

The evaluation showed that KBSU and TSU are on the way to full transition to the sixth level of maturity, shown in the work of Uskov [2]

From here the authors conclude that in the designing and development of collaborative interaction in the conditions of digitalization universities should be at the same level of development for maturity models.

3.2 *Smart Models Simulation for Universities Cooperation: System Approach*

The authors consider the basic principles for collaborative interaction and cooperation of universities and the contradictions that need to be eliminated (Table 1).

Thus the digital economy needs systemic educational mechanisms and processes, a transition from teaching approach “what to do according to a formulated task?” to teaching approach “how to formulate a task and what, when, how and why to do it, to solve and use it?”.

3.3 *System Capabilities for Smart Collaboration*

There are problems in promoting smart cooperation and educational infrastructure, including psychological. Psychological factors affecting the transformation of a university into a smart university (Fig. 4).

Table 1 System principles and contradictions of smart cooperation (a fragment: the author’s vision)

Traditional education	System approach	Contradictions
Trainee–tutor	Learning activity is based on cognitive activity	The rate of growth in the availability of information and the rate of its depreciation
Knowledge is the sum of knowledge obtained	The professional level of education quality should be developed and maintained throughout the active life of an individual	Volume of information and possibilities of its actualization (including psychophysiological, technological)
The value of an employee is determined by the knowledge and skills acquired by him/her	Knowledge is not only knowledge but also the ability to adequately target resource-oriented activities in real and virtual situations	Public order for information resources and their use
The curricula and plans are focused on a student with an intermediate level of training and are quite static	The value of an employee is determined not only by his knowledge, but also by his attitude to corporate goals and decisions	Didactic and methodical possibilities of technologies and methodical, system-synergetic and analytical possibilities of the environment
The main requirement to a trainee is the availability of the required normative and criterion level of training	Programs and plans should be focused on the highest level of competence (Industry Level 4.0, Internet Level 2.0), be adaptable, flexible, dynamically adaptable	High educational and self-educational opportunities of the environment for teachers and students and a low degree of their involvement in the process of effective use of these opportunities

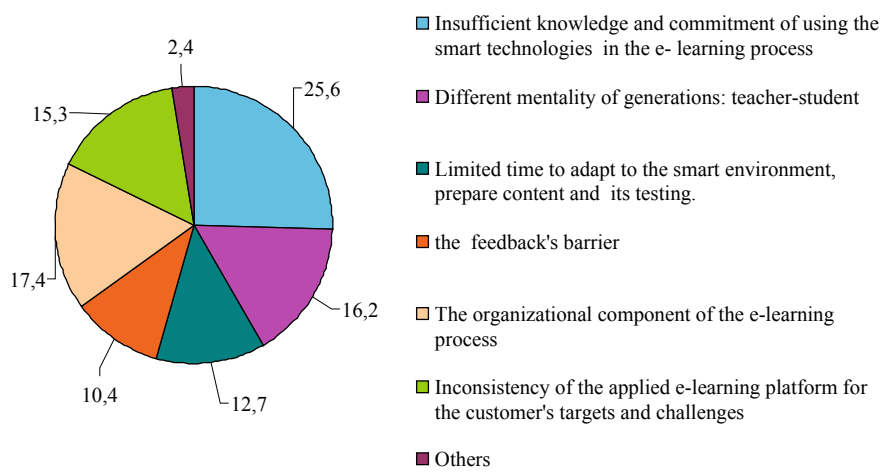


Fig. 4 Psychological factors (the author’s vision)

3.4 The Assessment Algorithm for Smart Capacity of the University

The authors propose the following procedure for smart-capacity assessment at the university:

Step 1: Identify a set of key controlling factors that form the university capacity;

Step 2: Collect data by using the developed monitoring procedure or survey and expert methods;

Step 3: Identify indicators by factors and coefficients representing intensity, networking, updating, etc.);

Step 4: Carry out a standard procedure of preliminary statistical processing that includes sampling, hypothesis testing, etc. It can also be done by means of socially oriented Data Mining packages or standard statistical and mathematical packages;

Step 5: Assess the significance, rank the factors, select the most significant factors and their deviations;

Step 6: Implement the capacity assessment procedure based on calculation of parameters, indicators, levels, scores, levels.

3.5 Universities Need to Create Advanced Integrated Educational Environments

Generalized groups of key factors and resources for developing and strengthening the information and cultural space of universities may include the following groups:

- (1) global networks, media resources and global open spaces;
- (2) national, regional, state, institutional, informational, educational, cultural, social spaces;
- (3) local, corporate, community, informal, family communication groups.

The negative aspects of networking include:

- (1) the possibility of individual isolation, which results in the withdrawal of a person into a “virtual” reality;
- (2) the emergence of educational “speculation”;
- (3) growth of added value of educational products, narrowing in unification of educational programs, insufficient attention to national educational systems, etc.

Innovative and attractive in terms of educational and scientific and methodological potential, the approach considers universities as independent agents. Such educational structures can independently or as part of a cluster enter the educational market with their directions, specialties and courses.

4 Target Interaction Model of Higher Education Institutions

Figure 5 reflects the main medium-term development goal and its private goals. The authors simulate Togliatti State University (TSU) becoming a smart university and get the status of a National University through its innovative networking with other educational agents in the both local and international educational markets.

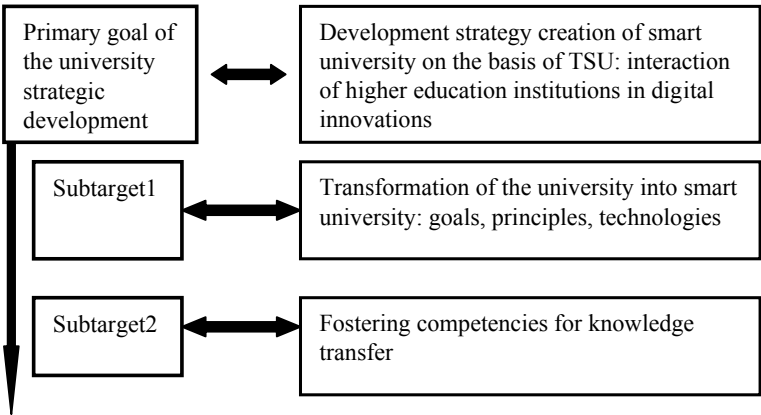


Fig. 5 The goal setting model for the university development (author’s model, fragment)

Table 2 Aggregation of Tasks: content component (a fragment)

Goals (G)	Tasks (T)	Decisions (D)
G1	T1. Smart conformity assessment of interaction between universities in different sections	D1.1 Financial and economic stability analysis of the university for effective collaboration
		D1.2 Formation of the base of criteria for assessment of collaborative interaction
		D1.3 Audit of SMART infrastructure of the university: identifying problems and developing measures to minimize their impact
	T2. Use of smart tools of the university for realization of international educational activity	D.2.1 Marketing analysis of stakeholders’ requests
		D.2.2 Development of economic and management models of collaborative interaction
		D.2.3 Promotion of the university’s collaboration services to the international level of cooperation

In Table 2 a fragment of description for the economic essence of the considered tasks to achieve the goal (author's vision) is represented.

The described issues are relevant, because it is the innovative networking of the university that makes the chain “smart university - smart business - smart society” strong and effective for dealing with modern challenges.

5 Conclusions and Future Steps

Conclusions. This project reveals the evaluation of university performance in terms of compliance with the maturity model of transition from traditional university to smart university. The study was based on the statement:

1. The cooperation of a network of digital universities is considered to be effective if the universities have the same or similar level of maturity.
2. The results of the evaluation reveal that two universities can effectively cooperate in the organization of the network cooperation while the third university requires management consulting for further digital transformation.

Next Steps. The authors plan the further research in developing a monitoring system to assess the maturity level of university development and professional training for human resources.

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Innovative “Algebraic Methods in Digitalization of Smart Systems” Course



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Abstract The emergence of new intellectual (or smart) technologies changes not only the mentality of human society, but also all spheres of its life. This change is associated with the emergence and development of a powerful and expressive language—the language of numbers and digital technology. At present, intelligent technologies and intelligent systems and smart systems are becoming an integral phenomenon in almost all spheres of human life. This, in turn, requires new courses and new mathematical methods in the field of smart systems theory. The paper contains a description of such a course developed by the authors in the field of algebraic methods in the digitalization of economic smart systems and in the field of education. The course “Algebraic methods in digitalization of smart systems” was developed by the authors on the basis of works on algebraic formalization and identification of smart systems carried out in the period from 1998 to the present time. Almost simultaneously with the development, fragments of the course were tested at a seminar on financial mathematics at the Academy of Budget and Treasury of the Ministry of Finance of the Russian Federation.

Keywords Algebraic model · Algebraic formalization · Algebraic identification · Digitalization · Smart systems

1 Introduction

The paper offers a brief overview of the course, developed by the authors in order to clarify the results of forecasting the development of economic smart—systems obtained by the authors in the framework of two monographs [1, 2] and a number of articles on the topic “Algebraic methods of modeling smart systems”. In 1992–1993,

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based on the study of the works of Yu. L. Ershov, we proposed a method for isolating and studying servant or pure embeddings (connections with special properties that match the internal properties of a subsystem and a system) in a special class of algebraic systems—groups, which made it possible to transfer and generalize the well-known results of the theory of purities of Abelian groups to the case of arbitrary non-Abelian groups, [3]. It turned out that this technique can be used in general systems theory to model the links of the system. In the 2000s, applications of this technique to economic systems were also found [4].

2 Problem Statement: The Need to Develop New Methods to Improve the Digitalization of Smart Systems

The main problem in the area of digitalization of almost all spheres of smart systems is the need to develop and to study new methodology and methods to improve the accuracy of forecasting, quality of planning and training specialist which can use such new methodology and methods. So, the main problem of the present paper is to present a new course “Algebraic Methods in Digitalization of Smart Systems” aimed at this goal in the sphere of economic and finance and some technic domains.

Let’s explain this position. The economic activity of a smart university affects almost all, if not all areas of the financial and economic system. In this regard, the issues of development and implementation of new mathematical methods in the field of digitalization of economic and educational smart systems are relevant. One of the most important components of the successful functioning of a smart system is the availability of methods for predicting the processes occurring in smart systems methods for predicting the functioning of smart systems that are adequate to the changing external conditions of the existence of a smart system. In [5], for example, there was written:

Financial crises pose unique challenges for forecast accuracy. Using the IMF’s Monitoring of Fund Arrangements (MONA) database, we find that IMF forecasts add substantial informational value, as they consistently outperform naive forecast approaches. However, we also document that there is room for improvement: two-thirds of the key macroeconomic variables and six variables (growth of nominal GDP, public investment, private investment, the current account, net transfers, and government expenditures) exhibit significant forecast biases.

Let’s briefly outline the disadvantages of some classical methods currently used for forecasting and planning the functioning of economic smart systems.

2.1 Disadvantages of Some Classical Methods Currently Used for Forecasting and Planning the Functioning of Smart Systems

2.1.1 Neural Networks

In [6] we read: “Living systems—such as a cell, a person, an economy, or even the climate (which is produced by life)—resist the tidiness of mathematical laws. perhaps that awkwardness is just an expression of the fact that the system under analysis is not easily reconciled with simple equations. Methods such as neural networks are based on a set of equations, but if you write them out, they seem haphazard and strange.”

In [7] the following disadvantages of Neuron nets are marked: “Neural networks require significantly larger datasets for their training, compared to other machine learning algorithms. They also need significant computing power for training”. The foundational theorem for neural networks states that a sufficiently large neural network with one hidden layer can approximate any continuously differentiable functions. If we know that a problem can be modeled using a continuous function, it may then make sense to use a neural network to tackle it. A function f is said to be continuously differentiable if the derivative $f'(x)$ exists and is itself a continuous function. In real conditions, there are practically no such functions.

2.1.2 Expert Forecasting

In a crisis, market instability, variability of the characteristics of the external environment and a limited amount of information, the possibility of using statistical methods for medium- and long-term forecasting decreases.

Let's consider the scripting method. The scenario method is an effective tool for organizing forecasting, combining qualitative and quantitative approaches. The scenario identifies the main factors that need to be taken into account and indicates how these factors might affect the anticipated events. Scenarios are usually descriptions of events and assessments of indicators and characteristics during the time. In economics and finance, three scenarios are usually used: baseline, conservative, baseline + , or optimistic, and then target scenario should be chosen. In this regard, the question arises about the real number of different scenarios that provide consideration of all possible options for the development of the system. In book [1, Chapter 10], the concept of the final state of the system was introduced and it was shown that the possible number of different variants of the system depends on the number of elements of the group of factors G_S that determine the model of the system S . So, for example, if G_S is simple of the order less than 10^6 then the scenario for the development of the S system will be only 1. It is also possible that the number of different scenarios for the development of the system, far exceeding 3.

3 Main Results

3.1 *The Base of the Course*

The course “Algebraic methods in digitalization of smart systems” is based on the developments of the authors [1, 2]. Let us briefly touch upon the main results of the course. In [1] there are revealed some general patterns of the theory of smart systems using a theory of algebraic systems—as synthesis of algebra and logic, discovered by Malt’shev [8]. Book [1] presents a translation of the theory of intelligent systems from a verbal language into a language of algebraic formalization which is a synthesis of a theory of algebraic systems and basic postulates and results in theory of systems. In [2] we continue [1] and try to investigate smart systems with the help of the theory of fractals, developed by Mandelbrot [18] for a detailed refinement of the smart-system model obtained using algebraic formalization. The notion of a quasi-fractal algebraic system introduced in [2] is a generalization of the concept of an algebraic system to the level of fractals, with the difference that the conception of self-similarity is a little bit broader than in the case of fractals. The key to self-similarity in the case of quasi-fractals is the concept of signature and type of algebraic system [1, 2] contains our results received in works [1–17] and reported at all-Russian and International conferences for the period from 1998 till nowadays.

3.2 *Aim, Objectives and Structure of the Course*

The aim of the course. The course is designed to train specialists in the field of mathematical methods in the field of finance and economics, the theory of smart systems. The aim of the course is training of highly qualified specialists in the field of theory of development and application of economic and financial systems, and the practice of their use.

The objectives of the course are:

- study of the foundations of the theory of smart systems in the language of algebraic formalization of systems theory,
- studying the basics of constructing mathematical models of the factors that determine the model in the sphere of finance and economics, in the form of algebraic systems,
- mastering the skills of using the knowledge gained in solving practical problems of digitalization in the field of finance and economics.

The structure of the course is as follows.

1. Introductory part. Basic information, definitions and theorems of model theory, underlying the discipline “Algebraic methods in digitalization of smart systems”.
2. Algebraic Formalization of Smart Systems.
3. Algebraic Identification of Smart Systems.

3.3 *Discipline Content*

Let's begin with discipline content.

Let's give some explanations. The development of system approach and a lot of works devoted to the results in general system's theory brought up the question of what language these results should be expressed and how these results should be justified. System approach specifies General scientific methodology, so the justification of the results in this area should not be only empirical. More and more works of different complexity and different expressive means that offer various formal languages and approaches to describe the general system theory appear.

In 1937, Ludwig von Bertalanffy proposed the concept of a system approach and a General Theory of Systems and also the development of a mathematical apparatus for describing typologically dissimilar systems. His main idea is to recognize isomorphism, that is identity, sameness of laws governing the functioning of system objects.

The main methodology of Introductory part. Basic information, definitions and theorems of model theory, underlying the discipline “Algebraic methods in digitalization of smart systems” is Model theory and The Theory of algebraic Systems. Model theory is a branch of mathematical logic that studies the relationship between formal languages and their interpretations, or models. Model theory is devoted to the study of the fundamental relationship between syntax and semantics. The first one corresponds to a formal language, and the second one is a model—a mathematical structure that allows some description in this language. The main development of model theory was obtained in the works Tarsky, Maltsev and Robinson.

The main methodology of Algebraic Formalization of Smart Systems is explained by the following. In the 1970s, A. I. Mal'tsev developed a theory of algebraic systems that connects algebra and logic and which is a universal mathematical apparatus for studying both algebraic and logical objects. In 1990s the concept of purities by predicates was introduced by one of the authors, and later on there were found out some applications of the theory of purities by predicates to practice. This conception makes possible to get a new methodology for the study of systems theory based on the idea of formalizing a notion of a system using algebraic systems and methods of general algebra. It allows to clarify connections between quantitative and qualitative analysis of a system in order to specify the previously known concepts in the deepening of the study of qualitative properties.

The main methodology of Algebraic Identification of Smart Systems is the methodology of the theory of algebraic systems, as a synthesis of algebra and logic, discovered by A. I. Maltsev, and a generalization of the concept of an algebraic system to the level of fractals—quasi-fractal algebraic systems introduced in this book, based on the theory of fractals developed by Benoit Mandelbrot [18].

This concept allows us to obtain the following main results: a method for studying intelligent smart systems, a new concept of a structurally stable quasi-fractal intellectual smart system, the concept of parametric algebraic potential of a smart system, the concept of a quasi-fractal scale and level of measurability of a smart system, an explanation the occurrence of mutations.

The main kinds of algebraic systems we used for it are groups and Boolean algebras. Also, methods of probability theory have been used. Randomness of the system structure, system infrastructure, stability and integrity of the smart system are investigated from this point of view.

The concept of the Erdős–Renyi algorithm on finite graphs and the concept of the Cayley graph of a group are used. The concept of probability-isomorphic groups is introduced. The main tool we used here, quasi-fractal algebraic systems, helps us to see a smart system in more details by adding new factors to the model of factors determining the system to describe previously indivisible elements of the initial model that is as to use scaling and zoom.

The part “Examples” contains examples: application of the results obtained to issues of economics and finance and to the knowledge system which can be used in smart universities [16, 17, 19].

4 Approbation of the Course

Certain sections of the course offered to your attention were tested at the Academy of Budget and Treasury of the Ministry of Finance of the Russian Federation: at a seminar on financial mathematics, organized by the Department of Higher Mathematics of the Academy of Budget and Treasury, as well as within the course “Mathematical justification of financial decisions” in 1998–2013 [20]. The aim of the seminar on financial mathematics was to train specialists in the field of finance, well versed in the correct use of mathematical methods in the field of economics and finance. The meetings of the seminar on financial mathematics were held once every two weeks. The average number of listeners of the seminar in financial mathematics has been about 10–15 students and postgraduates of the Academy of Budget and Treasury plus 3–4 teachers of the Academy of Budget and Treasury and other Moscow universities.

Nowadays some examples concerning application of the obtained results to issues concerning the development trends of financial and economic systems are explaining in the courses of Finance, Public and municipal finance, Mathematical justification of financial decisions, in Plekhanov Russian University of Economics [21]. The course “Algebraic methods in digitalization of smart systems” was approved by the Department of Applied Mathematics of Bauman Moscow State Technical University.

5 Conclusion. Future Steps

In the future, it is planned to develop detailed courses “Algebraic Methods in Digitalization of Smart Systems” for students of economic areas with an emphasis on modeling in the field of the fiscal system, and for students of technical specialties with an emphasis on solving specific applied problems in the field of technology. For each of the sections of the course indicated in Table 1, it is planned to prepare

detailed short tutorials with the main provisions of the course, illustrated with specific examples, as well as the development of assignments for students to independently complete. It is also planned to prepare assessment materials.

Table 1 Discipline content

	Algebraic methods in digitalization of smart systems	
The name of the discipline section (topic)	Content	Development results (know, be able, own) Educational technologies
Introductory part. Basic information, definitions and theorems of model theory, underlying the discipline “Algebraic methods in digitalization of smart systems”	General concepts of model theory and algebraic systems Relationships and displays Models and algebras Classical algebras Groupoids and groups Rings and bodies Lattices (structures) First and second degree languages	Lectures; seminars; written homework; consultations of teachers; independent work of students
Algebraic Formalization of Smart Systems	The problem of General Systems Theory’s Formalization The performance of a system by using an algebraic system of factors determining the system. P -properties of a system The simulation of the system with the help of finite group of factors determining the system. P -properties of the system Cayley tables and their role in modeling associative closed system with feedback External and internal properties of a system Integrity and P -integrity of a System by predicate P Formalization smart systems’ axiomatic Formalization of system links: different approaches Duality in Smart Systems Theory P -innovative and P -pseudo-innovative systems on the predicate P and their properties Algebraic approach to the risk description Linear programming models with risk The transition from an infinite model of factors that determine the system to a finite model The model of algebraic formalization of risks of changing the scenarios of the long term development of a smart system of six factors on the example of a smart university Pro- P -groups and algebraically closed groups: application to smart systems P -Sustainability of a system Algebraic formalization of sustainability concept Sustainability of ranking systems in education	Lectures; seminars; written homework; consultations of teachers; independent work of students

(continued)

Table 1 (continued)

	Algebraic methods in digitalization of smart systems	
Algebraic Identification of Smart Systems	Problem statement The connection between the problem of smart systems identification and classical logic Quasi-fractal algebraic systems as a possible approach to solving the problem of smart systems identification General system functions Target subsystems of smart systems System predictability levels Smart system measurement scales Quasi-fractals and synergistic effects Quasi-fractal scale measurability Testing as smart system encoding (on the example of a knowledge system) Quasi-fractal measures and quasi-fractal homomorphisms The concept of a parametric algebraic potential of a system The potential of a quasi-fractal system Classification of smart systems by potential Probabilistic measures defining scales System structure Randomness and integrity Technique Erdős–Rényi The target subsystem of the system Cayley graphs and probability-isomorphic groups Verbal tensor estimates in the Smart Systems Theory Sustainability of a system modeled by a quasi- fractal algebraic system, substitution of system's functions; system's compensational possibilities The concept of structural sustainability of a closed associative system with a feedback, structurally sustainable and unsustainable smart systems, structurally sustainable systems with a minimum regulation range	Lectures; seminars; written homework; consultations of teachers; independent work of students
Examples	Examples in economics and finance	Seminars; consultations of teachers; independent work of students

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Using of the Taxonomic Structures in the Process of Studying the Foreign Languages



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Abstract The article discusses the didactic potential of using taxonomic structures in the process of learning foreign languages, in particular English. There are a number of topics in the English language that are difficult to learn. These include: an extensive temporal structure of verb forms, the construction of interrogative sentences, etc. The use of taxonomic structures in the study of such topics allows you to structure, formalize and visualize the educational material to be studied and thereby solve the indicated problems. Particular attention in the article is paid to the issue of visualizing the taxonomic structures of educational material for learning English using modern software, the use of which in the process of teaching foreign languages contributes to the improvement and improvement of the quality of acquired new knowledge.

Keywords Learning process · Complex topics when learning english · Taxonomy · Information technologies · Visualization · Software · Grammar

1 Introduction

Software tools allow to visualize the composition of the structure of complex tense forms of verb. Also allow visually represent the educational material for learning. Thus, using the capabilities of modern visualization tools would solve this problem. Visualization tools can be used not only when the teacher explains new educational

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material but also when students work independently, as well as when they control knowledge.

The quality of foreign language teacher training is an urgent task for modern teacher education. Numerous studies are devoted to various aspects of this training [1–9]. In the logic of the problems raised in the article, we pay attention to the following fundamental points of modern works of domestic and foreign scientists [5, 6, 10–12]. The importance of introducing professionally-oriented technologies in the process of teaching foreign languages is emphasized in [13, 14]. The Social cultural space of dialogue for the formation of foreign language competence suggests using Smol'yannikova [15]. N. A. Trubitsyna, in turn, draws attention to the use of pedagogical technologies to assess the quality of training of future foreign Language Teachers [16]. Along with pedagogical technologies, E. G. Kashina suggests using theatrical technologies to reveal the creative potential of a foreign language teacher and develop his/her creative personality.

Realizing the powerful potential of didactic technologies in higher education (D. V. Chernilevsky), a special place should be given to the application of various methods and models of semantic representation of information and knowledge (T. Sh. Shikhnabieva, V. L. Shamshurin). As well as the use of various visualization tools and methods that can be widely used in learning a foreign language (N. V. Izotova, E. Yu. Buglaeva, Z. N. Kodzova, E. V. Kravchenko, O. K. Titova).

The possibilities of using information technologies for the development and use of proprietary applications are disclosed in [17, 18]. The use of information and communication technologies for organizing independent work of students is shown in the study by E. V. Zakharova. The relevance of media education technologies in the professional training of future foreign language teachers is determined in [19].

However, there are a number of topics in English that cause difficulties in learning them. These include: the type-time structure of verb forms, the construction of interrogative sentences, and a number of other topics. Despite a lot of research, it should be noted that not enough attention is paid to solving this problem based on the use of modern visualization tools, which greatly facilitate the perception of educational material. These problems can be solved by using the taxonomic structures of the educational material with their subsequent visualization based on modern software tools, which are the subject of our article.

The aim of the study is to improve the training of future English teachers through the use of visualization tools for taxonomic structures of grammatical material. Expected results:

1. To develop methodological approaches to the use of visualization tools for taxonomic structures of grammatical material in the preparation of future foreign language teachers.
2. On the basis of the proposed methodological approaches, create a teaching aid for future English teachers based on the use of visualization tools for taxonomic structures.
3. Create a workshop on the use of visualization tools for taxonomic structures of the grammatical material of the English language and test it in practice.

2 Materials and Methods

Despite the considerable number of works dealing with various problems and develop the potential of ICT in improving the efficiency of professional training of future teachers of a foreign language, using the possibilities of visualization tools, namely, taxonomic structures of grammar of the English language has not received its proper lighting, and these problems remain out of sight.

Thus, there is a contradiction between the increasing possibilities of visualization tools for taxonomic structures of grammatical material and the insufficient use of their potential in the study of complex topics (verb forms, construction of interrogative sentences, word order in a sentence, etc.) of the English language, which causes difficulties for foreign language teachers.

Study and analysis of the literature on this topic [1, 2, 5, 9, 11, 12]. We have shown that there is a problem of studying English grammar, in particular, complex verb tenses, construction of interrogative sentences, etc. The difficulty in learning English verb forms is due to the existence of many verb tenses that differ in their formation. In English, there are the following verb tenses (Fig. 1):

- Present Simple
- Present Continuous
- Present Perfect
- Present Perfect Continuous
- Past Simple
- Past Continuous
- Past Perfect
- Past Perfect Continuous
- Future Simple

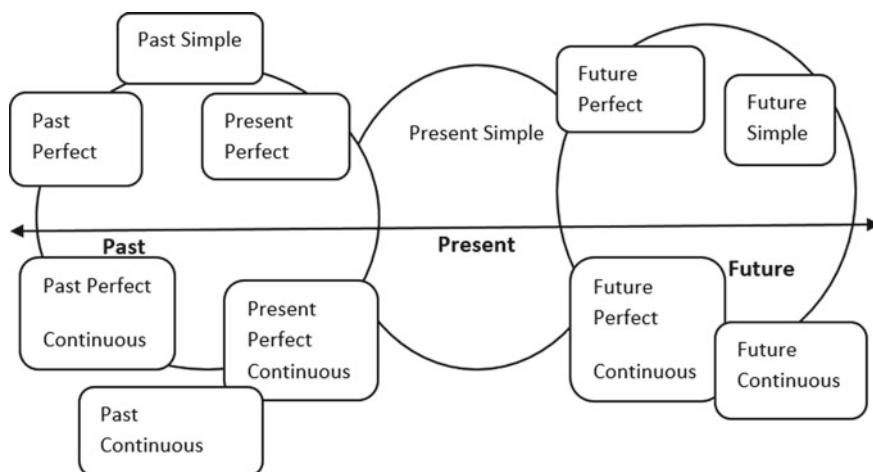
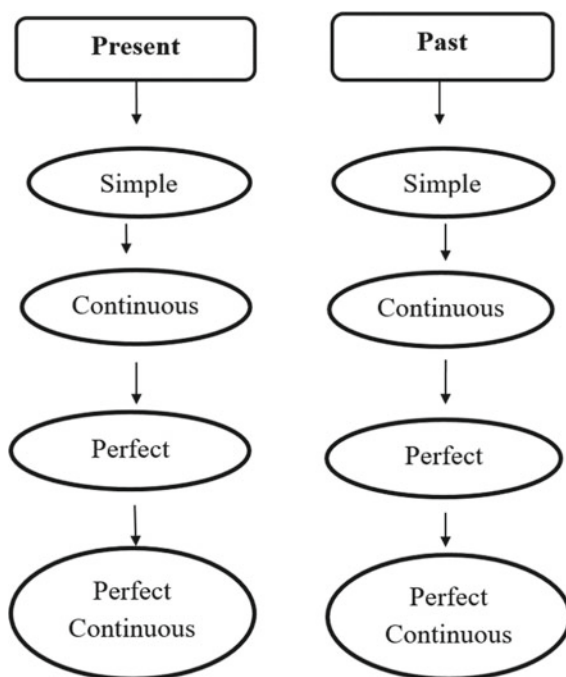


Fig. 1 English verb tenses

Fig. 2 English verb tenses of the present and past groups



- Future Continuous
- Future Perfect
- Future Perfect Continuous.

Figure 2 shows the generalized taxonomic structure of the Present and Past types of modern verbs.

As practice has shown, teaching English at different stages, presented in a declarative form of educational material about verb tenses, the construction of interrogative sentences and other complex topics causes difficulties in learning by students. One of the most effective methods of solving this problem is the construction of taxonomic structures of English grammatical material and their visualization using modern software tools.

3 Results

The use of taxonomic structures in the study of abstract and Humanities disciplines allows you to visualize and visualize the educational material, which in turn makes it easier to learn [14].

It should be noted that visualization tools can be used not only when explaining new educational material by the teacher, but also when students work independently, as well as when monitoring their knowledge [17].

Figure 3 shows an example of visualization of all types of modern forms of the English verb group Present in the active voice.

As we can see at Fig. 3, there are a number of forms of Present in English: Present Simple–present regular action, Present Continuous–present continued, Present Perfect–present perfect, Present Perfect Continuous–present perfect for a long time. It is quite difficult for a Russian-speaking native speaker to understand the difference in English verb tenses if you do not use the marker words shown in Fig. 4.

As Fig. 4 has shown, the simple present tense is when you use a verb to tell about things that happen continually in the present, like every day, every week, or every month. We use the simple present tense for anything that happens often or is factual.

Figure 5 shows that the present continuous verb tense indicates that an action or condition is happening now, frequently, and may continue into the future.

The present perfect tense refers to an action or state that either occurred at an indefinite time in the past (e.g., we have talked before) or began in the past and continued to the present time (Fig. 6).



Fig. 3 Verb tenses of the present group

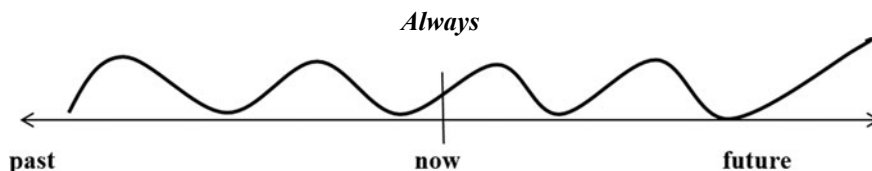


Fig. 4 Present simple tense



Fig. 5 Present continuous tense

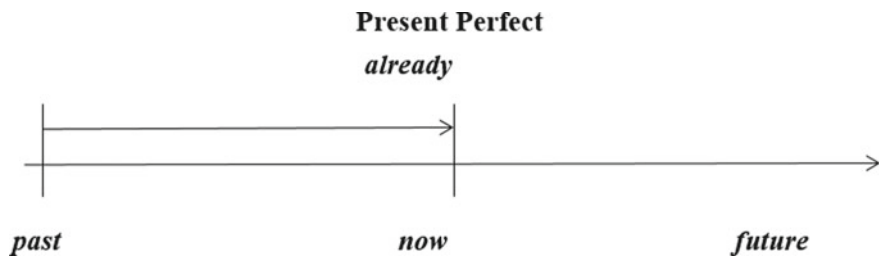


Fig. 6 Present perfect tense

The present perfect continuous tense (also known as the present perfect progressive tense) shows that something started in the past and is continuing at the present time. (Fig. 7).

In this case, the use of a taxonomic structure is an effective way for initial familiarization and further learning of the language. With proper use of visualization tools for taxonomic structures, students are expected to better assimilate a particular type of modern verb form, or any other similar aspect of grammar that may cause difficulties in learning a foreign language.

We present the results of a brief review of software tools that can be used for visualization of educational materials in General and taxonomic structures in particular (Table 1). Along with the list of software tools for visualizing educational material, the table provides a brief description of software tools and a description of their functionality in the process of teaching foreign languages.

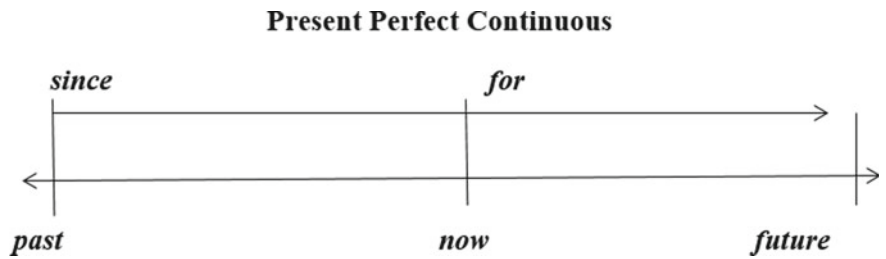


Fig. 7 Present perfect continuous tense

Table 1 Software tools for visualization of educational materials

Software tool	Brief characteristic
CourseLab http://www.courselab.ru/	A simple but powerful platform for creating interactive learning materials that can be used online or transferred to disk. This tool allows you to create and edit materials without knowledge of programming languages
MOS Solo http://www.mindonsite.com/en/product/mos-solo/	A tool that provides many opportunities for visualizing educational materials through multimedia. For example, you can create interactive graphic quizzes, tests, surveys, and so on
Zenler https://www.zenler.com/	One of the most productive platforms for creating educational resources. E-courses created on this platform will work on any device, including iOS and Android. Existing PowerPoint presentations can serve as the basis for the created material. The service also allows you to record video from the screen, add audio, animation, and so on
Lesson Writer http://www.lessonwriter.com/	A special service for creating a variety of English lessons, with which any information (article, excerpt from a book, table, etc.) can be easily turned into a handout containing all the necessary questions and exercises
Quizlet https://quizlet.com/ru	The web-based test creation platform recently launched two templates to create interactive tests in the Gravity and Scatter formats, based on maximum visualization of the training material. Software developers offer very easy-to-use game designers or game engines that allow language teachers to develop their own game scenarios and game task formats
Kotobee https://www.kotobee.com	A specialized electronic program for creating interactive textbooks containing video, audio, 3D, book widgets, questions, and more
Prezi https://prezi.com/	Cloud service that is used for creating interactive presentations that include interconnected structures—graphs; provides the possibility of almost infinite scaling
Mindmeister https://www.mindmeister.com/ru	Web service for creating connection diagrams, working together on them, with support for mobile devices, and presentations. Allows you to create “mental maps” using built—in tools-geometric fields for text and/or images connected to each other using connecting lines; provides the ability to use a variety of color schemes, add icons, links, and comments

(continued)

Table 1 (continued)

Software tool	Brief characteristic
Mind Manager https://www.mindjet.com/ru/	A program for creating intelligence maps and solving a variety of tasks. Allows you to structure data using tools similar to the previous application; provides the ability to present information in the form of a flowchart without unnecessary decorative elements
iMindMap https://app.imindmap.com/	Editor for creating intelligence maps. Allows you to colorfully visualize the material due to the fact that each branch that departs from the main concept is colored in its own personal color, and when detailing the structure, it becomes thinner. It provides an opportunity to most clearly reflect a type of graph—a tree
Bubbl https://bubbl.us/	Service-constructor for creating intelligence maps (mental maps, knowledge maps, mind maps), allows you to create visual effects. It is a way of expressing thoughts using graphic diagrams. These schemes are also very convenient for brainstorming sessions, where each of the participants can not only offer their own version, but also comment on someone else's, pointing out weaknesses or even making necessary changes
Wordle http://www.wordle.net/create	A tag cloud is a visual representation of a list of categories. Allows you to turn text into a “cloud” of the most frequently encountered words; allows you to choose the appropriate configuration of the “cloud”

As we can see from Table 1, there are many software tools that have a fairly high didactic potential. However, as the analysis of the study of modern practice of teaching foreign languages has shown, software tools, as well as other visualization tools, are not used sufficiently to visualize educational information. The methodology of teaching foreign languages offers various approaches to the study of topics that cause difficulties. The study of modern teaching experience shows that there are General didactically, specific methodological principles of teaching a foreign language, along with traditional and non-traditional teaching methods.

However, as the practice of using taxonomic structures of grammatical material and their visualization based on the use of modern software tools when learning English by native Russian speakers has shown, the quality of learning the educational material increases with this approach. This, in turn, is also associated with a clear, visual form of presentation of educational information, which increases its perception by students.

4 Conclusions and Next Steps

Conclusions. In the modern world, the importance of the quality of education as an important factor in the economic and social progress of society and the development of human creativity is noticeably increasing. Therefore, the education system also faces the task of achieving a new quality. One of the ways to implement this task is to introduce an effective system of training and control of acquired knowledge into the practice of schools and universities. One of the ways to improve the quality of teaching foreign languages, in particular, English, is to use the taxonomy of educational material with its subsequent visualization based on modern software tools. These approaches can also be used when organizing independent work and monitoring students' knowledge.

Next steps. We plan to continue developing a training and methodological complex based on the proposed approaches to teaching foreign languages.

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Smart University Development: Organizational, Managerial, and Social Issues

Modern Approach for Strategic Development of Smart Universities: Digitalization and Knowledge Export



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Abstract Nowadays the world economy follows the tendency of implementing new forms of educational activity and transition of higher education institutions to the form of open education facilities. The processes of globalization and digitalization make transfer of intercultural and cross border communications and knowledge export at the global educational market being very relevant. Challenges of modern society are also considered to be as drivers for the transition of higher education institutions to the open smart university system. It forces higher education employees to implement a synergy of all digital communication skills and acquire skills in new intellectual technologies of modern e-learning. The purpose of the study is to simulate and design the model for the training and assessment of professors' skills to work in the international digital educational environment.

Keywords Management · Open university education · CLIL technology · Global educational market · Professorial staff training model for open university

1 Introduction

Content Language Integrated Learning (CLIL) methodology has been widely used in the European education system since 1990. In Russia, CLIL can be very effective for the development of the international educational market and implementation of Russian universities for global clients. Today the fundamental conditions for higher education institutions are reflected in the theory of neoliberalism, holism and digitalization, which consider smart universities as autonomous organizations, able

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to promote their educational services to the competitive international educational market, striving to improve their efficiency and their competitive image and position to maximize their share in profits. The implementation of ideas and requirements of knowledge export in educational activities and the development of key skills and abilities becomes a necessary and priority area for a smart university. Many smart universities in the Russian Federation use various integration tools to promote and export knowledge through the design and development of open university status. Therefore, the search for mechanisms to manage the internal resources of the university for the implementation of opportunities for export educational activities remains relevant.

The aim of the research is to model and design a model of university export activities to work in the international digital educational environment. Particular objectives of the study are to design a model for the strategic development of the university.

2 Literature Review and Problem Statement

Currently there are many interesting approaches considering the essence and conceptual foundations for the development of smart universities. Some of them consider a smart university as an open system [1, 2]. The others reveal peculiarities of open educational systems based on knowledge transfer and teachers' readiness for self-development [3], including the export of their knowledge when teaching in a foreign language. There are a lot of studies promoting CLIL methodology [4] as an example of the best technology for the educational process and fostering both the soft skills & hard skills. Conditions for implementation of professional training programs on the basis of CLIL methodology and knowledge export are considered and modern features of globalization for higher education system requiring modern approaches for foreign students teaching [5] and processes of remote knowledge transfer are described [6]. Some studies identify the need to design and export knowledge for different target groups, including employer-sponsored training and on-line courses for international students and employees [7, 8]. Many studies represent new trends of designing effective educational activities, increasing the financial support of higher education institutions in terms of cross-border educational services, including language courses, graduate schools, internships and additional professional education [9]. Having analysed the above mentioned papers the authors reveal the plan of strategic development of the university for the near future, models of educational processes transfer for international interaction and implementation of CLIL technology in the system of open university in the global educational market.

Research problem: the search for effective ways for the university and higher institutions development in the conditions of digitalization and demand for knowledge export in higher education.

Main objectives:

1. To consider the reasons and opportunities for the development of open universities;
2. To consider the directions of possible knowledge export in universities.

The authors reveal the plan of strategic development of the university for the near future, models of educational processes transfer for international interaction and implementation of CLIL technology in the system of open university in the global educational market.

3 Methodology

3.1 Causes and Drivers for Open University Development

In accordance with the postulates of the theory of neoliberalism [7], universities are regarded as participants in the educational market, having a certain autonomy and the ability to independently strengthen and expand their position in the international market of higher education. It is assumed that the development of the respective fields of activity of a university, changes in the characteristics of offered educational services and the conditions of their implementation may contribute to the growth of indicators of export performance in the short term. Quantitative indicators of its profitability and scale have been used to assess the university performance. The level of the syllabus diversification, competitive advantages and the cost of educational services are considered as internal factors of export performance.

The expansion of the range for educational services offered by smart university increases the probability of increasing the number of foreign students and extending their stay in the status of consumers for the university's services. Russian higher education institutions offer foreign citizens to pass preliminary training for entry and further education in the tracks of higher education—bachelor's, master's, postgraduate, i.e. to pass the “continuous educational track” [8].

3.2 Causes and Drivers for Knowledge Export

Nowadays the task of developing the export of educational services is known as one of the priorities for education. The knowledge export at higher education services means their provision to foreign citizens; implemented both in the country where the university is located and in the other countries due to digitalization and implementation of distance learning. Innovations in education include collaboration as a process of joint efforts of several participants at project activities to achieve the target objectives of the university's strategic development. Export Knowledge syllabus is understood as a complex set of educational characteristics that allows

to transfer or export the required knowledge in different discourses of communication. Modern educational paradigm requires and digitalization allows the implementation of different interaction trajectories including “teacher- foreign student”, “student—educational Internet environment”, “mentor at the enterprise—graduate of the university”, “professors—employees at enterprise” and others [4, 5, 9].

3.3 Going to Open University at International Educational Market

Since 2014, Togliatti State University (TSU) has been introducing different forms of export educational syllabus and curriculum based on modern requirements from society and due to collaboration of the university and major regional enterprises including automobile, chemical, processing industries. Figure 1 represents the results of the analysis for export of educational services at the university for the period 2015–2020. The trend line reflects a 99% probability of further growth in exports of educational services. The author’s vision is represented.

Figure 2 shows the dynamics of growth in the involvement of foreign students in export and educational activities over the analyzed period. The increase in the foreign students involvement and participation over the last two years is represented.

The data and figure reveal that since 2016 the number of foreign students at the university has increased by 4.6 times thus requiring syllabus and tracks for further knowledge export at the international educational markets. The causes and drivers for further development of the University’s export activity are shown in the diagram: author’s vision (Fig. 3).

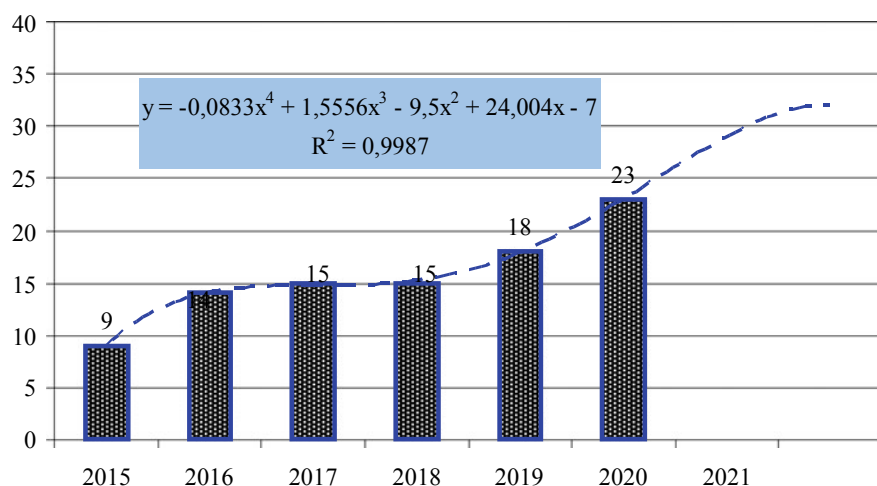
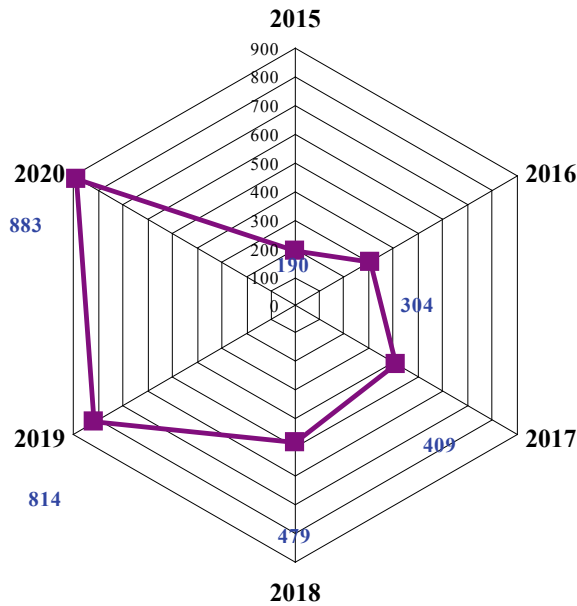


Fig. 1 Dynamics of educational services export

Fig. 2 Number of foreign students at the university (according to statistics from TSU)



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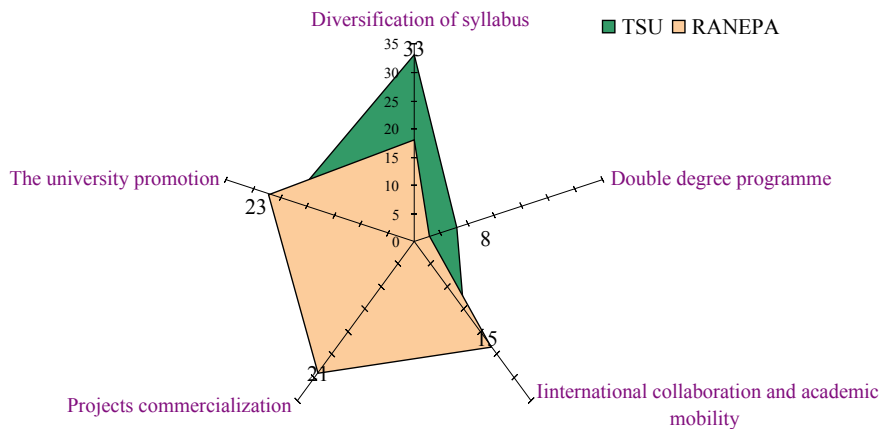


Fig. 3 Drivers for knowledge export activity at the university (TSU&RANEP)

Currently the knowledge export activity at the university (TSU) is provided for students from 23 countries including Germany, Israel, Congo, Korea, Serbia, USA, Croatia, the Netherlands, Tajikistan, Kazakhstan and Uzbekistan, Korea and China.

Due to digitalization the knowledge export is going to be increased by 12% in 2023 by expanding the targeted international educational markets and be attracting new stakeholders from Iran, Turkey. But modern challenges require the professors of the university dealing with the knowledge export at the international educational markets have a lot of soft skills which make them competitive at the educational market. Lifelong learning for digital literacy and effective communication in international professional teams (English speaking) are in high demand and key professors skills making both the professors and the university competitive at the international educational market.

3.4 CLIL for Knowledge Export at the University

CLIL is known as a subject and language integrated training and is recommended in Europe for the organization of subject teachers' training in English in their subjects. The Decree of the President of Russia (May 7, 2018, No 204 "The National Goals and Strategic Development Tasks of the Russian Federation until 2024") states the need for the Government to ensure the development of a national project in the of education requiring the number of foreign citizens studying in higher education and scientific organizations to be doubled, as well as to implement a set of measures to employ the best of them in the Russian Federation. Thus, in order to attract foreign students, it is necessary to prepare a number of curriculum and syllabus in a lingua franca. Nowadays English is known as lingua franca for scientific and business discussion. In this respect, the CLIL is the toolkit that allows the identified needs of the external environment to be implemented. The introduction of CLIL technologies in higher education is justified by the need for advanced training of teaching staff, whose personal characteristic is their readiness to teach and export knowledge of a subject in a foreign language which is a lingua franca for international students.

4 Methodological Approach for Strategic Planning of Smart Open University Development

The authors represent their vision for the Strategic Planning of the University development by introducing the key elements of the CLIL methodology. The overall goal setting for the development of a smart open university is shown in Fig. 4.

It can be seen from the model that in order to achieve the main objective (Goal) two private targets (T1 and T2) are to be achieved. For example, in order to form the faculty's readiness to export educational syllabus and curriculum, the following objectives targets are to be achieved (Fig. 5).

Table 1 represents the task tree describing the process in more details. A fragment of the goal setting and the process of achieving the goals is presented here. The

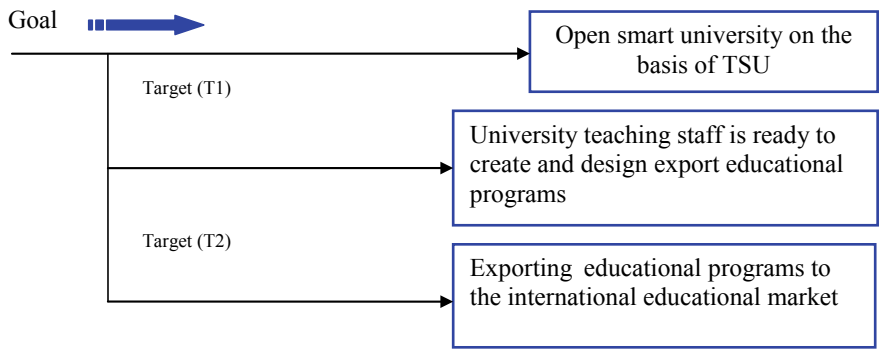


Fig. 4 University development goal setting model (the author’s model))

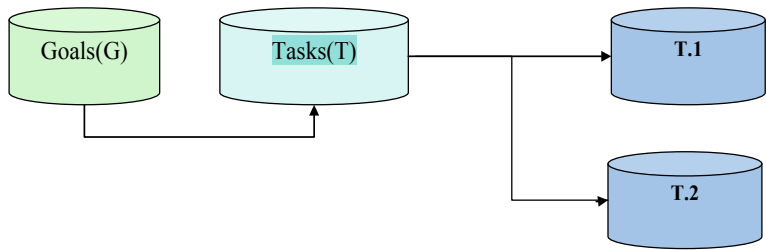


Fig. 5 Model for achieving strategic planning objectives and targets (the author’s vision, fragment)

Table 1 Goal and targets for knowledge export

Main purpose/goal	Tasks and targets	Subtasks/solutions
Goal	T.1. Implementation of CLIL methodology	T.1.1 CLIL for teachers and professors in the RF: theory, concepts, discussion
		T.1.2 CLIL implementation skills for subject teachers and professors
		T.1.3 CLIL content development for current challenges and requirements
		T.1.4 Pilot testing of CLIL
	T.2. Utilizing CLIL methodology for educational services: knowledge export for domestic and international educational markets	T.2.1 Market analysis of training requests from the International Education market
		T.2.2 Modeling, designing of syllabus and contents for exporting educational services based on the CLIL methodology
		T.2.3 Promotion of the developed knowledge export work curriculum and syllabus to the international educational market

solution of two targets (T1, T2) for achieving the main objective (Goal) is shown. The targets have subtasks, the solution of which is reflected in the corresponding column. The necessary financial, human and organizational or managerial resources must be identified in order to ensure that the solution is achieved.

For example, in order to target (T1) CLIL methodology study (T.1.1), it is necessary to have a syllabus describing the process of learning the methodology, as well as the availability of methodological aids to facilitate better understanding of the CLIL and to develop skills in its practical application. It is also necessary to develop a special test base for assessment the skills of the university professors for the designing export knowledge in the international professional environment.

According to the table it should be mentioned that all the described targets and steps for their implementation require collaboration between English language teachers with teachers of special subjects, specialists in their process area (chemistry, mathematics, mechanical engineering, electrical engineering, etc.). This mentoring enables them to organize joint work and acquire primary skills in presenting developed content in English to a wide range of future stakeholders and consumers.

5 Planning Knowledge Export at the Strategic Level

The planning of export educational activities is represented on the base of Togliatti State University (TSU) by target parameters (Table 2) determined by the authors based on the reference data received from the International Affairs Department of the university, taking into account the statistics on the distribution of indicators reflected in Figs. 1–3 of this study.

At the end of 2019, the export of education activities was planned by 11%, but taking into account the pandemic, the authors forecast the overall increase in smaller volumes, not exceeding 10% of the available figures. It should be noted that it is very difficult to make strategic planning for all types of activities; there is a necessity to solve the tasks represented in Table 1. Without achieving the stated targets, it

Table 2 Target values indicators for knowledge export at the university (TSU)

Indicator	Year (*-forecast, planning)			Notes
	2019	2020	2021*	
The share of foreign students studying the bachelor's, and master's degree in the total number of students at the university	883	957	1034	An expected minimum growth of 8.5% due to the expansion of export educational services
Knowledge export Syllabus/content/number	23	29	42	An expected minimum growth of 1.5% due to the expansion of export educational services

is difficult to develop export educational services further because the university's teaching staff are not ready for knowledge export in foreign language.

6 CLIL Methodology for Developing Staff's Soft Skills at the University

6.1 *Simulation*

The syllabus for lifelong teachers and professors training for their career development and knowledge export at the international educational market has been developed and tested by the authors. The training included the following steps:

- (1) English Test (TOEIC Framework): for knowledge export the teacher is required to have B1–B2 competence at least according to CEFR;
- (2) CLIL: theory, discussion, workshops;
- (3) Language training and accumulation of linguistic skills into the teacher's/professor's subject area;
- (4) Creating Mind maps and Testing.

This is followed by a discussion of the smart cards that have been developed, revision of the remarks and correction of stylistic or other mistakes made. This is followed by a cyclical process aimed at improving practical activities step by step.

6.2 *Testing*

During the teachers' training a website containing CLIL smart mind maps is used. The link to it opens and the teacher creates CLIL lessons by using the proposed template. Figure 6 represents a fragment of the CLIL mind map designed by a team of professors for knowledge export. Subject area: CATIA for engineering and car designing.

The designed smart mind maps have already been tested for knowledge export at on-line studying during lockdowns due to pandemic in 2020. The tested students group included Russian and foreign students. All the participants of the tested group demonstrated mastery of the subject, got a credit for the course and appraised the subject content highly. As a result of the suggested methodology, the rank of the considered universities has risen both in the international scientific community and in the international educational rankings.

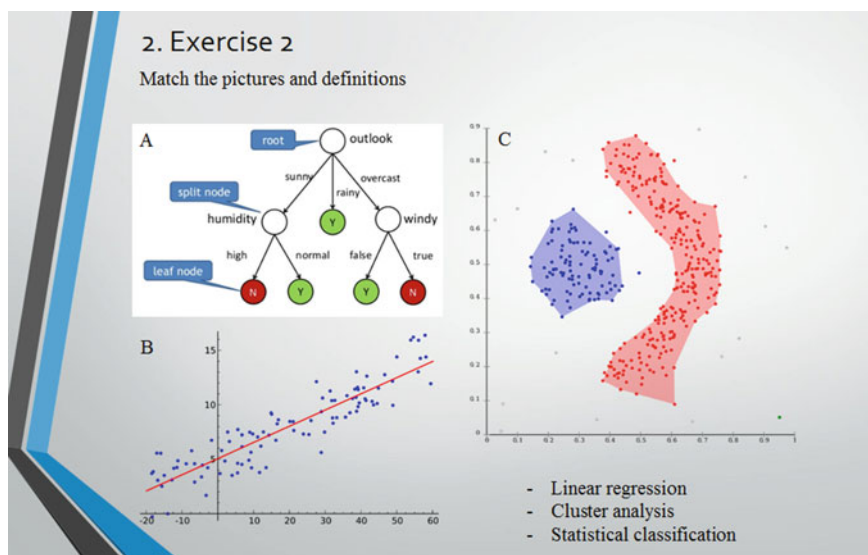


Fig. 6 Example of performing a practical task in CLIL lesson

7 Conclusion and Trends

Conclusions. Digitalization makes knowledge export easier and faster. Russian universities, both in the capital and in the regions, are entering the international education market.

1. Knowledge export corresponds to modern trends and challenges in science, pedagogy and management.
2. Knowledge export syllabus and curriculum promote the university into the international market and make it more competitive.
3. Knowledge Export is successful only if faculty members possess both hard and soft skills that are necessary for being successful at the international educational market.
4. CLIL is considered to be an effective tool for the lifelong learning and career development of the university teaching staff and graduates.
5. Knowledge export develops innovative potential of all university employees involved in the collaboration process and forms new knowledge.

Future Trends. In the future the suggested methodology is going to be implemented in the training of the universities faculty for knowledge export and creating new on-line courses for international students. Thus the rating of Russian universities in the international educational community will increase.

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Project Management of Smart University Development: Models and Tools



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Abstract Our research is based on the problem of insufficient knowledge of the hybrid approach to smart university project management and the possibility of joint classical (cascade) and flexible methodology use. In contrast to the existing publications, here we regard the possibility of managing the transformation processes of a classical university into a smart university which is considered on the basis of project management methods and means. At the same time, the essence of the smart university structure and its individual components are not considered. The university is evaluated as a business structure and a management object of many internal innovative projects presented in a portfolio form. The main attention is focused on project management tools that allow managing the timing, cost and quality of the project portfolio. The novelty of the solutions and the practical significance of the publication consist in the author's models application for managing a smart university development based on the implemented project office and the evaluation model of integration. Expert methods and the apparatus of linear and dynamic programming are used for modeling. The proposed tools have been tested in the activities of three universities. The optimal ratio of practical application of project management integration methods is obtained. A significant reduction in the appearance of unidentified processes in project activities that initiate the appearance of risks and uncertainties has been achieved.

Keywords Smart university · Project management · Digital transformation of the university · Mathematical model · Hybrid approach · Agile · COVID-19

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1 Introduction and Literature Review

1.1 *Smart University as a Research Object*

The huge speed of systemic change that is currently happening everywhere, as a result of the global COVID 19 pandemic, has accelerated the digitalization of all activities. Now there is the understanding of the fact that everything that can be digitized will be digitized. The restrictions imposed by the pandemic indicated the necessary directions for development and changes in the educational sphere. In conditions of the pandemic, there were smart universities which were able to adapt to the new learning environment in a short time, with a flexible set of educational smart technologies and tools [1]. Other universities which are in digital transformation on the way to the transition to a smart university, had to urgently reduce the terms of development projects. That case made them seek additional funding for the new projects and tasks implementation. For this reason, in this context, the most important task of university management is to select and master new approaches fast to manage digital transformation projects on the way to a smart university. The urgency of changing management models appeared even earlier in accordance with the technologies requirements and ideas of Industry 4.0, but in connection with the pandemic, this task has become more urgent. In 2018 Klaus Schwab and Nicholas Davis spoke about the necessity to use flexible management methods, in particular, the Agile methodology to obtain more effective results for Industry 4.0 projects [2]. Similar statements can also be seen in other studies [3–5].

1.2 *Analysis of Project Management Tools Capabilities*

The survey of the Project Management Institute (PMI) “Pulse of the Profession 2020” was conducted in 2020. It presents the feedback and points of view of 3060 project specialists, 358 senior managers and 554 directors of project management offices from various industries all around the world. The fact is that the speed of technological change is among the eight main threats, as stated by 29% of respondents [3]. It shows the necessity to accelerate digital transformation as fast as possible. Significant investments are also planned in digitalization over the next three to five years (44%), and 46% of organizations prioritize the management culture development that values project management. It should be noted that, in accordance with the majority of respondents, the project economy is already developed today. In many ways, most organizations, including educational ones, are associated with its projects that are implemented using different approaches and tools of project management. It should also be paid attention to such fact that the management of digital transformation projects faces many problems and questions. According to some surveys, only 10% of digital transformation projects meet the planned deadlines and budgets [4]. It is also impossible to say that the Agile methodology and its tools are 100%

universal and can be applied in all university projects, taking into consideration the peculiarities and differences between either the educational or business organization environment. Therefore, the identification of criteria on the basis of which a particular project management methodology can be used (cascade, flexible) or their optimal combination and applicability in project management is an urgent task in the framework of smart university creation and development.

It is also essentially important to understand that digital transformation within the framework of creating and developing a smart university is not a single project, but a permanent process. It is a set of projects (a portfolio of projects) that allows the main business processes to be digitized, creating additional value, while changing the entire the university educational ecosystem.

Nowadays, as part of the digital transformation and transition to the SmU model, the university top management is forced to combine structured and standardized approaches to project management. They are based on three project categories (cost, time, content), with flexible methodologies (Agile) based on such categories as speed, efficiency and value. For example, the Agile methodology is aimed at a rapid response to changes and satisfaction the project stakeholders interests. It is of great importance for universities when implementing projects to develop “smart” SmU components [5]. In this case, it is recommended to explore the possibility of using a hybrid approach. Its use will eliminate the disadvantages of the classical and flexible approaches and combine their advantages. That is why we further consider the possibility of an optimal combination of project management tools of classical, flexible and hybrid methodologies for a smart university.

1.3 Research Problem

Today, a lot of research is devoted to the use of project management tools. Therefore, the research problem can be described as follows: how we can determine the optimal combination of project management tools in order to manage unidentified processes of a university transformation into a smart university and requirements for the digital transformation project result in conditions of uncertainty.

Let us consider a smart university as a single object of research with a lot of (portfolio) digital transformation projects (projects for creating smart components) that require compliance with the conditions.

2 Choosing the Methodology for Project Management of Smart University Activities

2.1 The Methodology for Hybrid Approach Applying

From the point of view of implementing a hybrid approach the most valuable publication is the one of the Project Management Institute (PMI) and the Agile Alliance (Agile Alliance)—“Agile. It is a practical guide”, which contains recommendations and options for the hybrid approach applying and adapting various methods. This guide is structured to correlate with the leading edition of PMI, specifically, the “Guide to the Body of Knowledge on Project Management (PMBOK® Guide) (sixth edition)” [6].

It should be remembered that a smart university can also include other components. And project management tools (not necessarily “flexible” or “hybrid”) are also used for their development. Thus, the question of the optimal combination of different methodologies and tools in project management is actual.

As a part of the hybrid approach study, we were based on the works of various authors who consider the aspects of Agile application and its frameworks. These are D. Sutherland “Scrum. Revolutionary method of project management” [7], Yu. D. Ageev, Yu. A. Kavin, I. S. Pavlovsky “Project management methodologies. Agile and Scrum”, M. Kohn “Agile: project evaluation and planning” [8], Yu. Appelo “Agile management: Leadership and team management”, etc.

2.2 The Model of Project Portfolio Management in Smart University Under Conditions of Uncertainty

Our research is also based on the problem of insufficient knowledge of the hybrid approach to smart university project management and the possibility of joint classical (cascade) and flexible methodology use. In our opinion, the hybrid approach allows us to prevent the project content spread, manage unidentified processes and requirements for the digital transformation project result in the context of uncertainty projects development in smart university. An approximate model of university project management in the creation and development of smart and other components is shown in Fig. 1.

The ability to manage unidentified processes and requirements of the external environment appears due to the core integration of the classical, flexible and hybrid approaches. So, next, the evaluation model of their optimal integration will be considered.

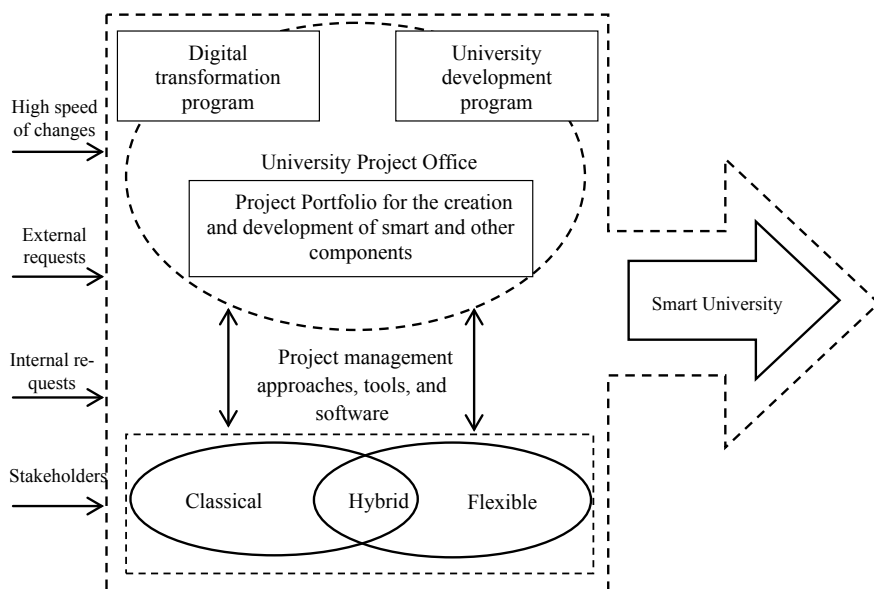


Fig. 1 A model of project management in university for the creation and development of smart and other components

2.3 The Evaluation Model of Tools Optimal Integration in Project Management

So, we can distinguish three main approaches to smart university project management: classic, flexible and hybrid but at the same time there is a practical question of choosing either one of them or their combination. The study of the above approaches allowed us to identify the main characteristics of the project and build a graphical evaluation model for determining the optimal combination of project management approaches for a particular project (Fig. 2).

The criteria for agile project are presented in the right part of the model (the example can be the project of software development services as one of the University smart component). The criteria of less flexible projects are shown in the left part. As a bright example here we can mention the construction of University smart campus. We regard it without any software campus smart elements, but the construction project only. In the middle of the figure there is a scale from 1 to 10 points for expert project evaluation. The higher the score on the scale, the greater the flexibility of the project for this criterion. A flexible project is characterized by maximum ratings, where the use of flexible methodology tools as Agile (Scrum, Kanban, etc.) will be appropriate. All intermediate ratings on a scale from 2 to 9 show the necessity to use hybrid project management models, combining the tools of flexible and classical methodology. Minimal points show the necessity to apply the classic project management tools. Based on this model, it is recommended to prepare questionnaires. They will help

Content rigidity											
There are strict regulations, the content of the project is not flexible	1	2	3	4	5	6	7	8	9	10	The project content may change along the way, the result is unclear at the start of the project
The cost and complexity of the changes											
It is difficult to implement technologically because the cost is high	1	2	3	4	5	6	7	8	9	10	Changes along the way are possible, the cost of changes is not high
Minimum viable product (MVP)											
The project product can be delivered entirely only	1	2	3	4	5	6	7	8	9	10	The project product can be divided into blocks and handed over in parts
External factors influence											
It is implemented in a fairly stable external and internal environment	1	2	3	4	5	6	7	8	9	10	External factors can change the project content, the environment is extremely dynamic
Stakeholder engagement											
The stakeholders' interests are clear, the impact is minimal	1	2	3	4	5	6	7	8	9	10	Interests are unclear, new ones may arise, and the level of influence is high

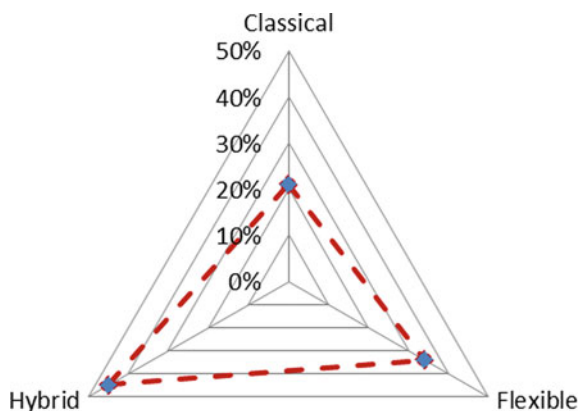
Fig. 2 A model for selecting smart university project management approaches based on the evaluation scale

project managers in their work on the creation and development of smart university components to receive materials for making the right decisions to use a particular project management tool.

2.4 The Evaluation Model of Project Management Tools
Optimal Integration: Data Analysis Results

A survey of respondents—the experts in the field of project management and managers of digital transformation projects (40 respondents) based on the developed evaluation materials allowed us to obtain the results on the optimal combination of project management tools of the studied methodologies (classical, flexible and hybrid). The recommended optimal combination of tools in the project office of smart university as a percentage is: Classical (21%), Flexible (34%), Hybrid (45%) (Fig. 3). This project management methodologies integration makes it possible to identify unrecorded processes and manage smart university projects more effectively in accordance with the external environment requirements.

Fig. 3 Recommended optimal tools combination in the project office of smart university



3 Project Activities of Smart University: Models of Optimal Resource Allocation

3.1 Classical Approach to the Allocation of Project Resources

Limited financial resources and the high speed of external changes pose another important task for the university in the transition to SmU. It is the distribution of resources between projects. There are two approaches to solving this problem [9]:

1. Static, based on the use of linear (LP) or non-linear programming (NLP) methods, where a single (one-stage) distribution of resources between smart university projects is considered;
2. Dynamic, where the optimal resources allocation is made taking time in consideration. The main methods in this case are dynamic programming (DP), based on the application of the Bellman optimality principle. Based on Vasiliev's conclusions [9], we propose to use the mathematical apparatus to solve the selected problem.

Generally, the above problem is mathematically formulated as follows:

- minimize (or maximize) the objective function

$$T = \sum_{j=1}^n C_j \times X_j \rightarrow \min(\max) \quad (1)$$

under probabilistic constraints

$$P \left\{ \sum_{j=1}^n a_{ij} X_j \leq b_j \right\} \geq \alpha_i \quad (2)$$

where $j = 1 \dots n$, $i = 1 \dots m$, $\forall X_j \geq 0$, α_i is a specified probability ratio, $0 \leq \alpha_i < 1$, «n» is the number of resource types, “ a_{ij} ” is the expenditure of «i» resource on «j» project, “ C_j ” is an income (expense) from «j» project.

In the future, the task of allocating resources between projects can be solved in various ways:

- (1) C_j , a_{ij} , and b_j are random variables,
- (2) a_{ij} and b_j are random variables, C_i are determinate variables.

To solve the problem of allocating resources between projects, the second option, which establishes the probabilistic nature of resources, will be the most acceptable.

3.2 Setting the Task of Allocating Project Resources Using Linear Programming Methods

Let us formulate the problem of resource allocation in a more concrete form.

As an objective function, we take the expression [9]

$$R_0 = \frac{\sum_{j=1}^n R_j}{R_{0_{\max}}} = \sum_{j=1}^n \overline{R_j}, \quad (3)$$

where $\overline{R_j} = \frac{R_j}{R_{0_{\max}}}$ is the relative rating of «j» project, $R_{0_{\max}}$ is a total rating of projects (project portfolio).

Then the value of $\overline{R_j}$ characterizes the project's contribution to the overall project rating. Omitting the line in the future, we will keep in mind R_j as the relative rating of the project. The value R_j is calculated as follows [9]:

$$R_j = U_{1j}R_{unj} + U_{2j}R_{mtj} + U_{3j}K_{nj}R_{ypj}, \quad (4)$$

where U_{1j} , U_{2j} , U_{3j} coefficients that take into account all the importance of a particular project rating at the moment; R_{unj} , R_{mtj} , R_{ypj} are the potentials of intellectual (innovation), smart infrastructure and educational «j» project.

You need to find the resource allocation where the target function R_0 takes the maximum value and the constraints are met [9]:

$$P_s \left\{ \sum_{j=1}^m C_{sj} \left(\frac{R_a}{R_n} \right)_{sj} \times R_{sj} \leq b_s \right\} \geq \alpha_s, \quad s = 1, 2, 3; \quad 0 \leq R_{sj} \leq 1, \quad (5)$$

where $s = 1, 2, 3$ is regarded as the index of activities (intellectual (innovation), infrastructure, education); P_s means the probability that resources allocated for improving ratings potentials do not exceed the total at the University; C_{sj} is the

unit cost of increasing capacity in «j» project; $(R_a/R_n)_{sj}$ is considered as the activity coefficients of utilization potential for this type of activity in the «j» project; b_s is the total value of assets (resource) supplied by the University for building improvement; α_s means the probability of receipt of funds to support the activities of the «s».

In this form, the task is a stochastic programming one. However, by some transformations, it is reduced to the standard LP task. Omitting some indexes, we can show this in general form and get a simpler formulation of the task.

3.3 The University Resource Management Model for the Projects Implementation by Type of Activity

Let b_s be a normally distributed random variable with mathematical expectation m_{bs} and mean square deviation σ_{bs} . Then any of the constraints can be written as [9]:

$$P_s \left\{ \frac{b_s - m_{bs}}{\sigma_{bs}} \geq \frac{\sum_{j=1}^m C_{sj} \left(\frac{R_a}{R_n} \right)_{sj} R_{sj} - m_{bs}}{\sigma_{bs}} \right\} \geq \alpha_s. \quad (6)$$

If we assume that the value on the left side of the inequality which is under the sign of “probability” operation, has a normal distribution law, which is partially justified by the repeated use of averaging and summation operations in obtaining these values, then the inequality can be fulfilled only if:

$$\frac{\sum_{j=1}^m C_{sj} \left(\frac{R_a}{R_n} \right)_{sj} R_{sj} - m_{bs}}{\sigma_{bs}} \leq \gamma(\alpha_s), \quad (7)$$

where $\gamma(\alpha_s)$ is the value of a normed (standard) random distributed quantity (quantile), for which: $\Phi_0(\gamma(\alpha_s)) = \alpha_s$ where $\Phi_0(\dots)$ is the integral function of the normal distribution.

Then the inequality reduces to the usual linear constraint [9]:

$$\sum_{j=1}^m C_{sj} \left(\frac{R_a}{R_n} \right)_{sj} R_{sj} \leq m_{bs} + \gamma(\alpha_s) \sigma_{bs}. \quad (8)$$

In this case, the task is reduced to maximization R_0 under condition (8). The above version of the problem statement assumes the possibility of managing the university's resources for the projects implementation by type of activity.

3.4 Modeling Predictive Situations of Project Resource Management

To implement the control, it is necessary to take into consideration the dynamics of changes in the coefficients of the objective function U_{sj} . Such task, in particular, is solved in the work, but in a very approximate formulation [9].

In this case, the dependency introduction $U_{sj}(t)$ leads to parametric linear programming (PLP). At the same time, there is a possibility that for some t , called critical (t_{kp}), the solution will be invalid, suboptimal. Thus, in PLP, it is necessary to determine not only the optimal solutions vector $R_x(t)$, but also the point t_{kp} , after which the problem is not solved.

The cases in PLP are investigated when time functions can be $U(t)$, $b(t)$ or $U(t)$ and $b(t)$ simultaneously. Let's limit to the cases when $U(t)$ or $b(t)$ are changed:

$$(1) \quad U(t) - V_{c2} \geq 0.$$

If $B_{ikp}(t)$ is the optimal basis matrix applicable to the critical value t_{ikp} , then the solution $R_{ikp} = B_{ikp}(t) \times b(t)$ will be optimal for all values $t > t_{ikp}$ where the conditions are held $R_i(t) - U_i(t) \geq 0$.

In the vector-matrix form, the conditions will be written in the following form: $U_{[3]}(t) \times B_{ikp}^{-1}(t) \times Q_{i[3]} - U_j \geq 0$, where $B_{ikp}^{-1}(t)$ is the matrix inverse to $B_i(t)$, $Q_{i[3]}$ – is the matrix composed of the constraint coefficients.

$$(2) \quad b_s(t) - V_{ar} \geq 0.$$

The following condition will be fulfilled: $B_{ikp}^{-1}(t) \times b_s(t) = 0$.

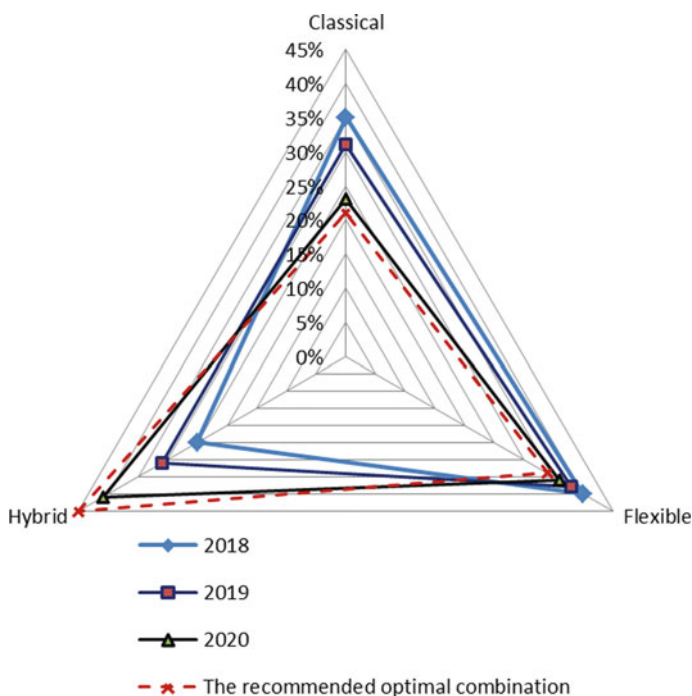
Thus, the use of parametric linear programming methods allows us to solve the problem taking into consideration the dynamics of changes in the coefficients of the objective function and the equations of constraints, which in its turn allows us to introduce forecasting elements.

4 Practical Experience in Project Management

The experiment on the obtained results implementation was carried out on the basis of three universities: Bauman Moscow State Technical University (National Research University), Moscow Pedagogical State University and Togliatti State University (TSU). The tools study used in universities project offices in 2018–2020s allowed us to obtain the following (average) results of using the tools of classical, flexible and hybrid methodologies (Table 1; Fig. 4).

Table 1 The range of assessed indicators (fragment)

	2018	2019	2020
Classical (%)	35	31	23
Flexible (%)	40	38	36
Hybrid (%)	25	31	41

**Fig. 4** Results of the study of project office tools in smart university

According to the project management research in universities, we can conclude that the triangle of optimal tools combination of classical, flexible and hybrid methodologies for a smart university is confirmed by the trends of practical PM tools application in 2020 (23%, 36%, 41%) and it is close to optimal (21%, 34%, 45%). The closest value was achieved in TSU.

5 Conclusion and Next Steps

Conclusions

1. The system of management development in smart university should include a strategic planning group. On its basis the project office is created. The set of projects, the SmU development portfolio, should be formed in accordance with the digital transformation program and the speed of external changes must be taken into account.
2. The expected optimal combination of project management methods should include a ratio of classical methods (21%), hybrid methods (34%) and flexible methods (45%). In this case, the appearance of unidentifiable SmU project management processes is minimal.

Next Steps

1. Studying the information infrastructure of the project office support.
2. Analysis of the applicability of project management information systems for SmU.

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Strategic Analysis of Smart University Resource Potential for Management Objectives



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Dmitrii V. Zmievskii, and Natalya A. Igoshina

Abstract Currently, solving the problem of limited resources requires a constant search for possible reserves. The availability of sufficient resources forms the resource potential. The effectiveness of its use predetermines the success of the Smart University development not only in the current period, but also in the long term. Strategic analysis of the resource potential, taking into account changes in the external and internal environment of the Smart University, is of particular importance for management. The studies have shown that the existing organizational and methodological approaches to the strategic analysis of the resource potential do not take into account the specifics of the Smart University, which confirms the relevance of the scientific article. The article reveals the concept of resource potential in relation to a Smart University. The result of the study is the proposed stages of strategic analysis of the resource potential of Smart-University, as well as the development of a logical model for strategic analysis of the resource potential of Smart-University for management purposes. The results obtained on the strategic analysis of the resource potential for management purposes were tested in the Smart divisions of the university using the example of the department.

Keywords Smart university · Strategic analysis · Resource potential · Resources · Management

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1 Introduction

In today's volatile environment, most organizations experience a series of ups and downs in their development. Due to the influence of various environmental factors, limited resources, universities need to use the existing potential opportunities in the most effective way. It is the availability of the required amount of resources and their effective use that forms the resource potential of an economic entity. In turn, the development of the resource potential of a smart university will allow to increase the results of its functioning, to form the main competitive advantages over other universities.

In our study, a smart university is understood as a university of the future, focused on modern technologies, intellectual and innovative development [1].

Currently, such scientists as Uskov et al. [2, 3], Serdyukova N. A. have made a significant contribution to the conceptual foundations of the intellectual university [4].

The works of Glukhova et al. [5], Berdnikova et al. [6], Tikhomirov and Dneprovskaya [7], Shikhnabieva and Beshenkov [8], Burlea and Burdescu [9] are devoted to the development of a smart university.

Currently, for the purposes of managing a smart university, it becomes relevant to conduct a strategic analysis of its resource potential. The study showed that this issue in the economic and scientific literature has not received sufficient attention, which requires additional development.

Strategic analysis of the resource potential is aimed at researching and assessing the activities of a smart university in the current period and at predicting its development in the future, by identifying opportunities and risks. The results of such an analysis are the basis for the development of strategic goals and are an effective tool for the management of the university's activities. Thus, the main goal of this article is the development of organizational and practical provisions of the strategic analysis of the resource potential of a smart university for management.

2 Formulation of Problem and Its Connection to the Important Scientific and Practical Tasks

Currently, for effective management, increasing the competitiveness of a smart university, it is necessary to have information about not only environmental changes, but also about the available resource potential, the ability to respond to external challenges. Having such information, the management can measure the potential of a smart university with the needs of the market, objectively assess the situation and develop informed management decisions, outlining the right path for strategic development.

Today, we are witnessing the reforming of systems and their transformation, taking into account the influence of the external and internal environment, in which more and

more attention is paid to smart technologies, the intellectualization of knowledge, the development of communication skills in using a high-tech environment, including the need to organize effective feedback [10]. In our opinion, the resource potential of a smart university is a combination of human, material and technical, information, intellectual, innovative and financial resources, as well as reserves and opportunities, the rational use of which will improve performance and ensure effective functioning in the future.

Figure 1 shows the main components of the resource potential of a smart university.

We believe that a strategic analysis of the resource potential of a smart university should be carried out from the perspective of researching the current availability and state of resources, opportunities and risks. Based on the current assessment of the resource potential, it is possible to predict its state in the future, taking into account the influence of external and internal factors, and, if necessary, adjust the university development strategy. Thus, the importance of strategic analysis of the resource potential of a smart university for management purposes should not be underestimated, which requires additional research and development in this direction.

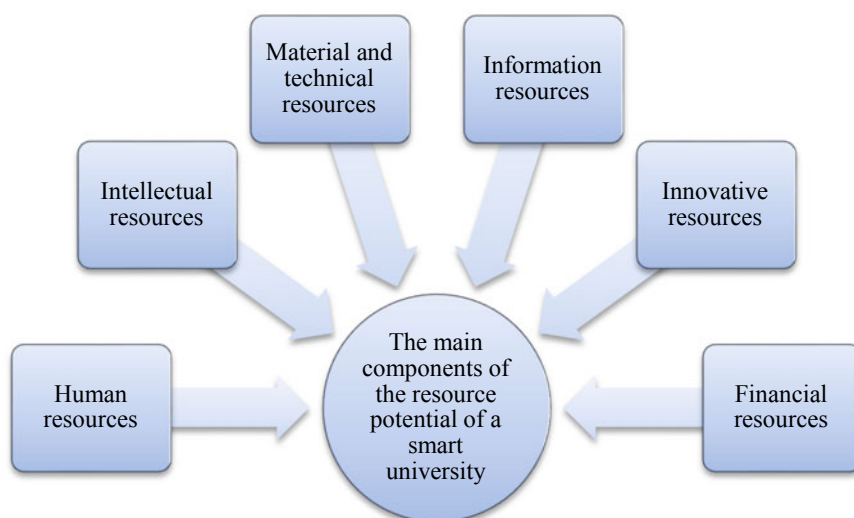


Fig. 1 The main components of the resource potential of a smart university

3 Research Outcomes

3.1 Visualization of a Statistical Study of Key Components that Form the Resource Potential of a Smart University

The development of the resource potential of a smart university is possible through the intellectualization of the labor of its employees, the introduction of smart technologies, the digitalization of basic processes and the use of intelligent systems in professional activities. The most powerful resource component is personnel [11], capable of carrying out scientific developments on the basis of knowledge and experience, thereby enhancing the intellectual and innovative resources of a smart university. However, one cannot fail to note the role of financial resources in the formation of the potential of the university, ensuring all processes of its activities, including the development and maintenance of the information environment.

Based on the statistical data [12] in Fig. 2, we will visually represent the dynamics of the number of personnel engaged in scientific research and development by categories in the Russian Federation.

Figure 2 shows that in 2019 compared to 2015, there was a significant decrease in all categories of personnel engaged in research and development in the Russian Federation. This trend may indicate a possible reduction in the total resource component represented by scientists. Based on the statistical data [12] in Fig. 3, we present the dynamics of the number of researchers by field of science in the Russian Federation.

Figure 3 shows that the largest number of researchers are involved in the technical sciences. However, in 2019 compared to 2015, data analysis showed a decrease in the number of researchers in all areas of science in the Russian Federation. This trend may indicate a possible reduction in the total resource component represented

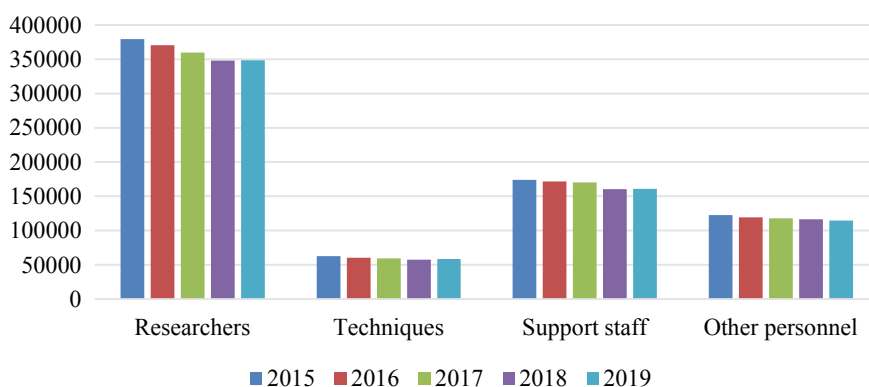


Fig. 2 Dynamics of the number of personnel engaged in research and development by categories in the Russian Federation, (people) [12]

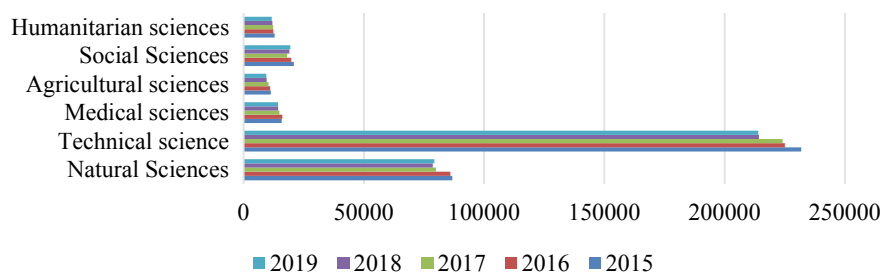


Fig. 3 Dynamics of the number of researchers by fields of science in the Russian Federation, (people) [12]

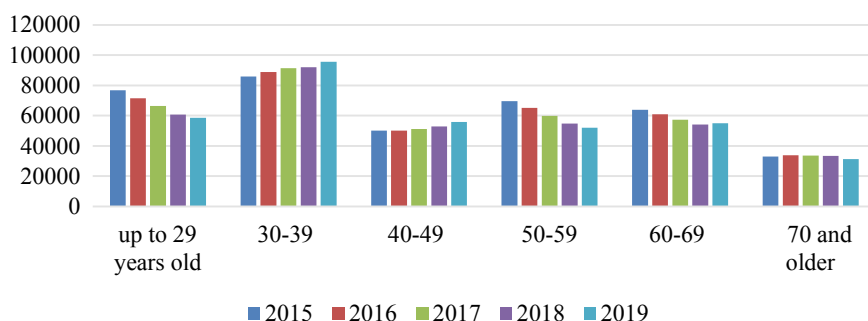


Fig. 4 Dynamics of researchers by age groups in the Russian Federation, (people) [12]

by scientists. To assess the age groups of researchers, we present the data in Fig. 4. The source of information was the data of statistical studies [12].

Figure 4 shows that in 2019 compared to 2015, there was an increase in the total number of researchers in the Russian Federation in the age groups of 30–39 and 40–49. For the rest of the age groups, there is a decrease in the number of researchers in 2019 compared to 2015. Based on statistical data [12] Fig. 5, we reflect the amount of funding for science from the federal budget in the Russian Federation.

Figure 5 shows that funding for applied research decreased in 2019 compared to 2015. However, in 2019, funding for basic research has increased significantly compared to 2015. Such dynamics have a positive effect on the development of the resource potential of smart organizations in terms of fundamental scientific activities.

Thus, the study carried out showed a rather large decrease in the number of personnel engaged in scientific research. Such dynamics can be traced in all areas of science. With regard to funding from the federal budget, there is a positive trend regarding fundamental research. The ongoing changes can significantly affect the resource potential of many smart organizations and affect their main areas of activity—research and educational.

The successful development of smart organizations and their contribution to the overall transformation of the scientific and educational environment will depend on

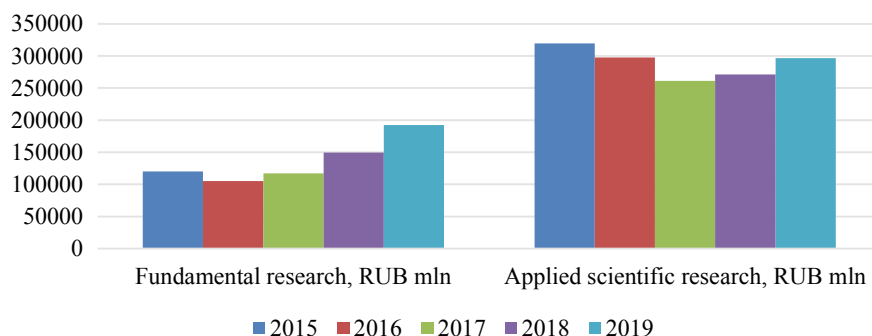


Fig. 5 Financing of science from the federal budget in the Russian Federation [12]

how effectively key resources will be used. All of the above confirms the need for a strategic analysis of the resource potential of a smart university for management purposes.

3.2 Stages of Strategic Analysis of the Resource Potential of a Smart University for Management Purposes

Strategic analysis of the resource potential of a smart university should be aimed not only at studying and establishing the phenomena and processes that are taking place. It should identify favorable and negative trends in the work of a smart organization, contribute to the development of economically sound management decisions for the efficient use of resources and identified opportunities for improving performance.

We believe that the strategic analysis of the resource potential of a smart university for management purposes should be carried out in stages. In this regard, Fig. 6 suggests the following stages of its implementation:

- (1) the 1st stage involves setting goals and objectives for a strategic analysis of resource potential at a specific point in time;
- (2) the 2nd stage is devoted to the selection of priority areas for research. At this stage, the key areas for assessing resource opportunities are determined. This stage can also involve the study of the general resource potential of a smart university;
- (3) the 3rd stage involves collecting the necessary information for strategic analysis. At this stage, preliminary information processing is performed;
- (4) the 4th stage is aimed at developing a balanced scorecard that will allow assessing the resource potential of a smart university;
- (5) the 5th stage is devoted to the choice of methods and methods of research;
- (6) at the 6th stage, analytical calculations are carried out, and the current resource potential of the smart university is assessed;

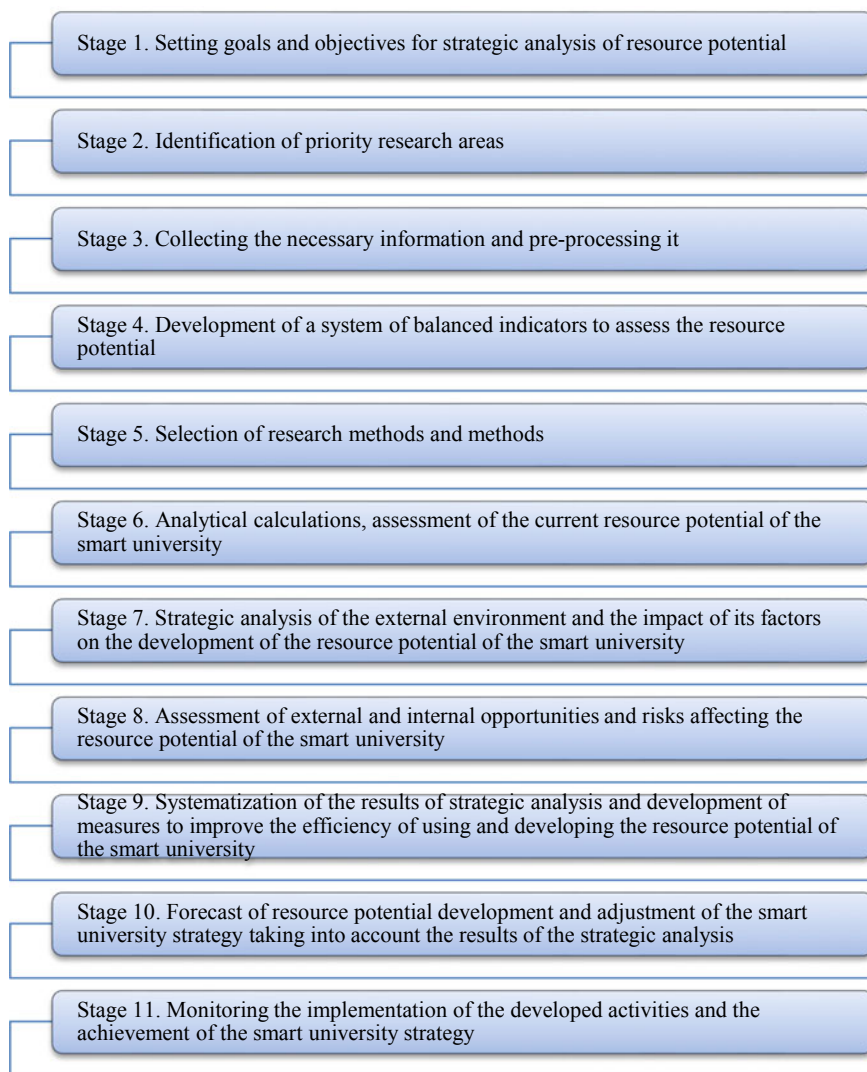


Fig. 6 Recommended stages of strategic analysis of the resource potential of a smart university for management purposes

- (7) the 7th stage is aimed at a strategic analysis of the external environment and the influence of its factors on the development of the resource potential of a smart university;
- (8) the 8th stage involves the assessment of external and internal opportunities, risks affecting the resource potential;
- (9) the 9th stage, the results are systematized and the necessary measures are developed to effectively use the resource potential of the smart university;

- (10) the 10th stage involves forecasting the development of resource potential and adjusting the strategy based on the results obtained;
- (11) the 11th stage is dedicated to monitoring the implementation of the developed activities and the implementation of the smart university strategy.

The stages presented are interrelated. When conducting a strategic analysis of the resource potential, it is necessary to be based on a systematic approach, which assumes that all objects of research are studied as a whole, while identifying the relationships and interdependencies between the factors.

The recommended stages will make it possible to comprehensively assess the existing resource potential of a smart university and analyze the impact of the external environment on its development. The obtained results of strategic analysis will be the basis for developing strategic goals or adjusting an existing strategy, and will also allow the development of measures for reaching a smart organization to a new level of development.

3.3 A Logical Model of Strategic Analysis of the Resource Potential of a Smart University for Management Purposes

To manage a smart university, current and future information is needed. Strategic analysis of resource potential provides such information and contributes to the discovery of internal reserves, identification of opportunities and risks, effective use of the resources of a smart university. The study made it possible to form a logical model of strategic analysis of the resource potential of a smart university for management purposes (Fig. 7).

The proposed model of strategic analysis of the resource potential of a smart university allows us to determine the current state of resources, assess the impact of environmental factors on them, and identify reserves, opportunities and risks. Based on the data obtained, measures are developed for the efficient use of resources, identified reserves and opportunities, and neutralization of risks. The next step of the model is to forecast the resource potential of a smart university, taking into account the implementation of the developed measures. At the end of the logical model, a managerial decision is made to adjust the strategy based on the results obtained and the forecast of resource potential.

At present, the proposed model is being tested in terms of developing a competency map for the purpose of increasing the efficiency of using human resources of the Master's Degree Department of Togliatti State University. As a result of assessing the available human resources and taking into account the development plan of the university, key performance indicators have been developed for employees in the following areas:

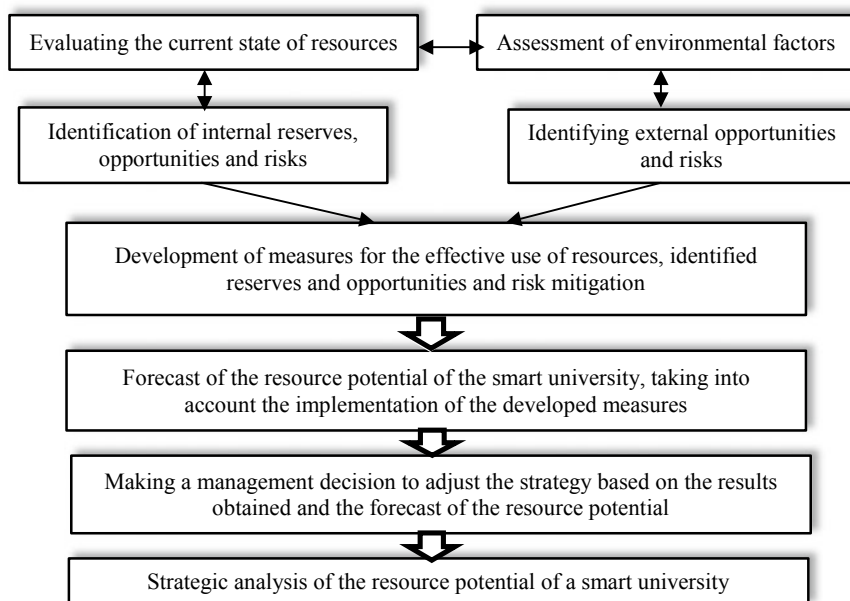


Fig. 7 A logical model of strategic analysis of the resource potential of a smart university for management purposes

- performance of research work;
- publication of scientific works in publications indexed in the SCOPUS database, publications indexed in the Web of Science database of international scientific citation indices;
- training;
- participation in scientific and practical conferences;
- implementation of research work with students;
- the execution of giants.

For each employee, performance plans are established, the implementation of which contributes to the achievement of the strategic development goals of the university. Practice has shown that as a result of using this model, there is an increase in the efficiency of using human resources.

The results of approbation of the proposed model in terms of increasing the efficiency of using other resources and, in general, the resource potential of a smart university will be presented in subsequent scientific works.

4 Conclusions and Next Steps

Conclusions. The study showed that in modern conditions one of the important tasks for management is to maintain activity, competitiveness and efficient use of the resource potential of a smart university. This requires the use of advanced tools that take into account all resources, opportunities and risks to make informed management decisions.

1. The article clarifies the concept of the resource potential of a smart university, taking into account the peculiarities of its activities. The study made it possible to clarify the main components of the resource potential of a smart university.
2. The article presents a statistical study of the dynamics of the number of personnel engaged in research and development by categories, fields of science, age groups in the Russian Federation for 2015–2019. The article presents the results of a statistical analysis of data on the funding of science from the federal budget in relation to fundamental and applied research. The results of the conducted statistical study confirm the need for a strategic analysis of the resource potential of a smart university for management purposes.
3. The article proposes stages and a logical model for conducting a strategic analysis of the resource potential of a smart university for management purposes, taking into account internal capabilities and the influence of external factors. The proposed stages and a logical model for conducting a strategic analysis of the resource potential of a smart university can be applied both at the management level of both a separate structural smart unit and a smart university as a whole.

Next steps. To expand the toolkit for strategic analysis of resource potential, it is necessary to develop a balanced system of indicators that allow assessing the resource potential of a smart university at each stage of its functioning.

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Challenges of Digitalization: Smart Pedagogy for Smart University



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Abstract In the study smart pedagogy is considered as the key element of educational management system at the university which is targeted at designing students' competences and skills including their professional self-determination and social self-organization, self-motivation. The analysis of the key tools of smart-pedagogy and their implementation according to the changing requirements of digitalization age is carried out. Smart education (SmE) is constantly developing today and smart learning is increasingly correlated with the notion of distance education which takes place according to the creed: "at any point, with any educational resource, in a comfortable time and for evolutionary purposes". The study reveals the key notes of smart pedagogy for smart university and describes the examples of its implementation in two universities of the RF: Rosdistant project in Togliatti State University and Kabardino-Balkaria State University has been considered.

Keywords Education · Pedagogy · Sustainability · Evolution · University · Self organization

1 Introduction

Modern universities are implementing their educational activities according to the basics of smart-concept follow its key requirements including Specific, Measurable, Achievable, Relevant and Time bound features. So they have to go through the digital transformation by implementing intelligent systems and tools to support the full cycle of educational activities. The university's digital infrastructure aims to develop and use learning content with new features and better learning options according to the principle of access to educational digital resources "in any place, with any resource, at any time, with any evolutionary purpose". This principle is aimed at reducing

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the systemic complexity of target goals, structures and assessment in smart education. It attracts dynamic methods of learning, monitoring and evaluation of learning outcomes. In other words the methods should be flexible, agile and could be applied to any information technology used by the company or users and support motivation for learning. While implementing targeted curriculums for digital economy, the Russian Federation pays more attention to educational, health and social issues. New models and technologies of modern education are being more widely used and studied. In this work, a systematic analysis of the evolution for smart university in the transition to a digital society and digital economy is revealed.

Smart pedagogy is considered as a control system of educational infrastructure at the university. It is considered as a system of students' competences development, their professional self-determination and social self-organization, self-motivation. Hence, the analysis of key tools of smart pedagogy and the technology of their actualization is carried out. The authors consider both general interpretation of smart education and the special «digital and computer based» description. The system interaction between digital economy and smart universities' models are considered, as well as criteria of transformation in digital economy and education. System contradictions of “content—structure” type of university information environment are analyzed. The dilemma for optimal smart-university's self-adjustment and the assessment for smart university development is offered.

The study is based on the following methodological hypothesis: the implementation of smart technologies into university environment improves the quality of the university educational process both for the social and technical science graduates.

2 Literature Review

According to the modern expert studies and estimations, the pace of digital transformation in Russia has a sufficient evolutionary path, which can be effectively implemented only if there is highly qualified staff. The analysis of numerous studies from KES conferences has shown that most authors adhere to the following 6 distinctive features and indicators which smart university can be characterized by: adaptation, sensing, inferring, anticipation, self-learning and self-optimization [1]. New models and technologies of modern education are to be designed and implemented. According to modern studies the intellectual environment is identified with the physical infrastructure that allows the functioning and development of the surrounding intellectual system [2]. V. L. Uskov, J. P. Bakken, A. Pandey consider smart pedagogy from technological aspects and its actualization in Smart Classroom Systems [3]. Some studies represent smart university with its own key factors and smart interactions [4, 5] that design and develop the competences and personal qualities and metrics [6] of both students and teachers.

Currently a lot of scientists in pedagogy and management make conclusion that if the university does not correspond challenges of digitalization age and doesn't take into account the Generation Z peculiarities it has a lot of weaknesses and treats and

can be pushed out from the educational market requiring the networked educational systems. MOOC (Massive Open Online Courses) classes, task-based MOOC and open distance courses are required for smart university competitiveness [7].

In modern conditions a systematic approach to the analysis and synthesis, composition and decomposition of smart education issues, the creation of advanced smart integrated educational environments [8, 9] is the only way to deal with the modern challenges in education and society. The study is based on countries ranking and statistics [10] in the context of pandemic [11].

3 Methodology: Smart Pedagogy for Educational Management

3.1 Target and Opportunities

In a broad sense, the transition to smart pedagogy is a system process required for the system's highly intelligent level and tools while dealing with the post-industrial (Industry 4.0) challenges.

Statement 1. Transition to smart pedagogy is directly connected with digital transformation of society, education, with impossibility of further development of educational environment and infrastructure without development of smart technologies and methods in society, economy and education. For example, the COVID-19 pandemic has clearly demonstrated the need for a systematic approach to educational infrastructure in the area of medical universities and health care organization.

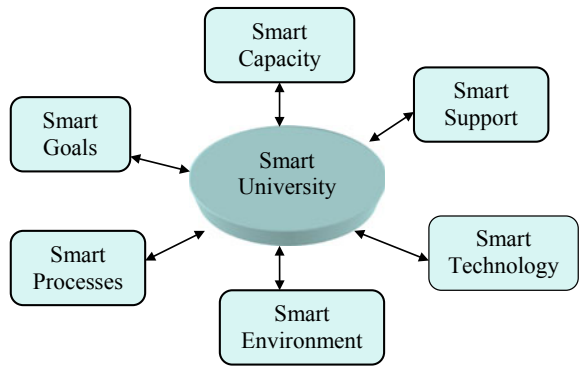
The most common Russian interpretation of the methodological category "SMART" ("Smart", "smart") is similar to foreign interpretation, for example, as a concept that supports Specific, Measurable, Achievable, Relevant and Time bound. This allows us to combine efforts to research and design the foundations of smart pedagogy as a system for managing the scientific and educational process in a post-industrial society in accordance with digital transformation.

Figure 1 represents the structural diagram for the "smart university" system (the author's vision).

The authors consider smart pedagogy and education as the unified and integrated system based on the following levels:

- (1) a set of educational institutions;
- (2) a developed digital infrastructure;
- (3) highly motivated staff at education institutions;
- (4) intellectual approach to education based on intellectual decisions etc.;
- (5) personal development based on motivation approach for all the participants of educational process: learners, educators, business stakeholders.

Fig. 1 The smart university system



This is an educational collaboration of agents, educational factors, supported by both intellectual smart infrastructure and public organizations, partners and employers.

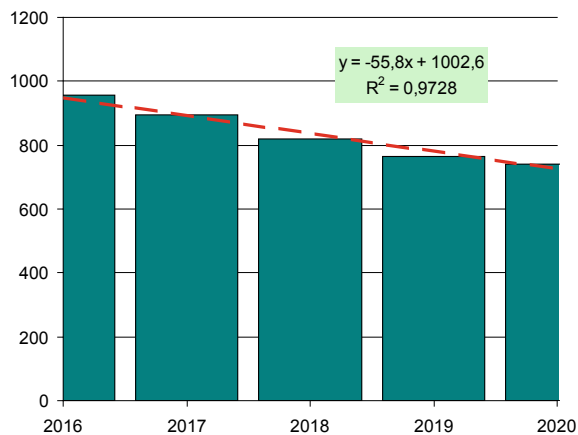
The research problem is the need to assess the readiness of universities to transform traditional pedagogy into smart pedagogy. To solve the identified problem the activity of two universities in the Russian Federation is represented.

3.2 Traditional University Versus Smart University

Digitalization and informatization turned out the necessity for smart systems increasing the requirements for performance and efficiency of higher education.

Figures 2 and 3 represent the tendency for decrease both as the number of universities and students (a fragment of the author’s simulation: according to statistics). The universities’ and students’ trend models are represented.

Fig. 2 The number of universities at the RF (2016–2020 years)



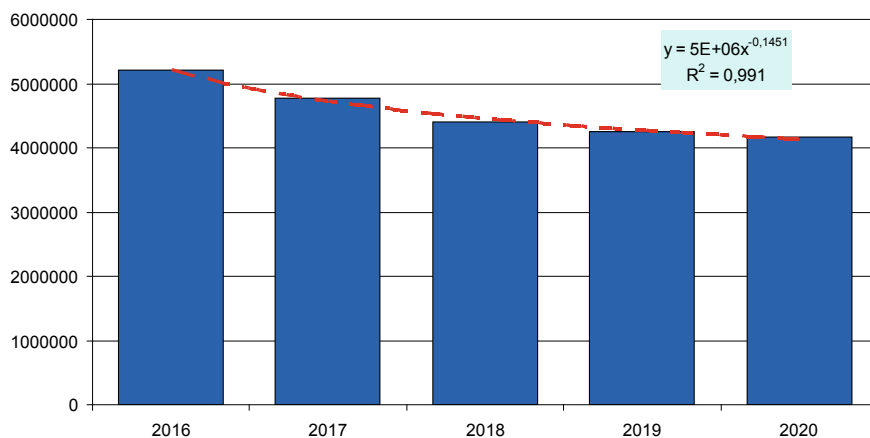


Fig. 3 The number of students (2016–2020 years)

According to the data Table 1 represents the comparison for IT services in the RF and European Union [10].

The pandemic in 2020 forced the indicators increasing both in Russia and EU and proved the necessity and social demand for smart system and smart education [11]. Figure 4 represents comparison of digital economics in the RF and Europe.

Table 2 represents positive changes for IT in an ordinary school located in the RF in Kabardino-Balkarian Republic (KBR) from 2016 to 2020 years.

Smart education implies a proactive approach to innovation and the expansion of competence boundaries, while smart technologies imply a variety and diversity of learning methods.

Table 1 IT service: Russia versus EU

Availability for digital and IT		
Indicator	Russia (%)	EU (%)
Internet connectivity	78	90
Smartphone connectivity	66	71
Mobile Web connectivity	61	73
Businesses with a website	52	81
Citizens receiving public services through the internet	64	69
Citizens making purchases online	39	71
Businesses using CRM systems	14	74
Share of e-commerce in total retail trade volume	5	14

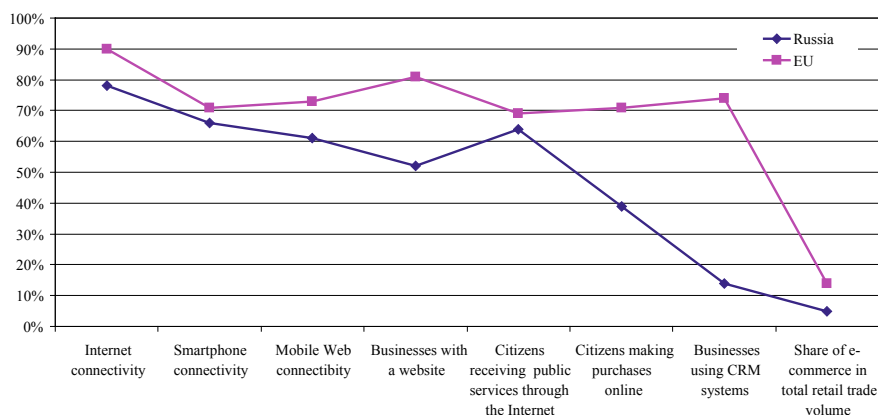


Fig. 4 Availability for digital and IT: Russia versus EU

Table 2 IT in one of the secondary schools in the rural area of the KBR

Figures per 100 employees	2016	2017	2018	2019	2020
Number of computers	35	33	43	37	44
Computers with internet access	27	21	31	28	36
Internet users (teachers)	97	83	79	76	98
Internet users (teachers and pupils)	75	78	79	82	88
Users of web-banking	13	16	18	24	56
Distance learning	18	19	22	32	948

3.3 Smart Pedagogy

Statement 2. Smart pedagogy is to be considered as only an environment updating educational smart products and allowing teachers to design an individual, interactive and adaptive educational trajectories in the conditions of mass learning. Smart education assumes an active appealing and usage of innovations and the expansion of competence boundaries, while smart technologies assume a variety of diverse and versatile methods of learning.

Distance learning in a pandemic is considered to be as the best example here: during the national lockdown more in March 2020 more than 80% of Russian universities activated their distance learning, although in 2019 only 76% of teachers and professor were able to follow all the smart university requirements. Many universities used the MOODLE platform and corporate mail, chat rooms, and “digital” volunteering of advanced students to help “non-professional” teachers master web learning. IT staff literacy is to be embedded in smart university and is key to self-expression and expression of creative ideas as well as practical professional creation. SMART-space of the university in the broad sense includes interacting objects, technologies

and systems of interaction, regardless of space and time conditions. Each university has its own corporate educational space closely related to the socio-economic space of the region.

Statement 3. Smart universities are to be designed for developing both universal core soft skills required in modern professional environment and hard skills by digital tools and means. Thus smart education is known as an adaptive, interactive learning in an open, distributed educational environment targeted to providing the professional or social collaboration for many agents who are motivated to study and help others in their development. For example, the projects of the Joint European University and Rosdistant are designed and developed according to this principle.

Smart Environments

Smart environment is characterised by the following features: hyper-media in real time; SaaS, PaaS and other relationship models; Open Learning Systems (MOOC) and learning-science virtual complexes; adaptive testing systems; intelligent automated workplaces and decision support systems; interactive whiteboards (SMART Boards) and projectors, cameras, monitors; machine learning systems; cloud structures and computing systems; neuro- and fuzzy-system technologies; Big Data and Data Mining; Learning Analytics and Business Intelligence Systems; security assurance systems, etc.

Smart Technologies

Modern e-Learning technologies should be relevant to the principles of smart education. For example, the self-organizational principle of smart learning involves its own adaptive test with maximum efficiency and ease of learning. Adaptive testing is interactive, traceable, analyzable, focused on the complexity indicator for the content to be tested. Situational, simulation techniques based on performance-based testing are needed to develop the cognitive and creative abilities for the tested subjects to a variety of “life” situations.

Some modern studies deal with narrow understanding of the smart infrastructure for the university including the following hardware and software tools:

- (1) servers—file, mail, web-oriented, applications, administration of distance learning, local classes and hyper-media (teleconferences, educational television, etc.);
- (2) switches, scrolls, network communications, video surveillance and provision of scientific and educational laboratories;
- (3) open software solutions such as “Educational Server”, “Applicant”, “Student”, “Testing”, “Electronic Document Management”, “University Branch”, “Big Data”, “Cloud Computing”, etc.
- (4) special training platforms “Moodle”, etc.

Figure 5 represents the scheme for smart pedagogy system. It shows the comparative characteristics of the possibility of implementing smart pedagogy components in universities.

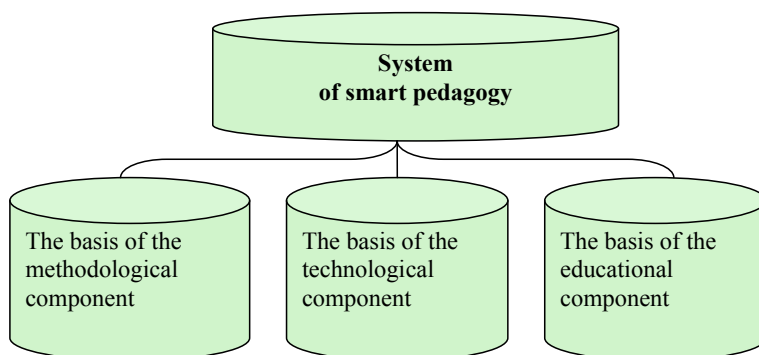


Fig. 5 The smart pedagogy system

The basis of the methodological component includes the following blocks: critical and analytical thinking; system analysis-synthesis; expert approach; creative approach; cognitive approach and others.

The basis of the technological component includes the following requirements: high technologies; algorithmization and programming; virtualization; network organization; security, its control (audit), etc.

The basis of the educational component includes the following elements: remote training, control, monitoring systems; “open” education; self-organization and deliberate self-expression; individual approach, etc.

Benchmarking universities for the availability of a supporting component for smart pedagogy is represented in the Fig. 6.

The evolutionary ability of smart education depends on the above mentioned basic components and is represented in the Fig. 7 (the author’s vision).

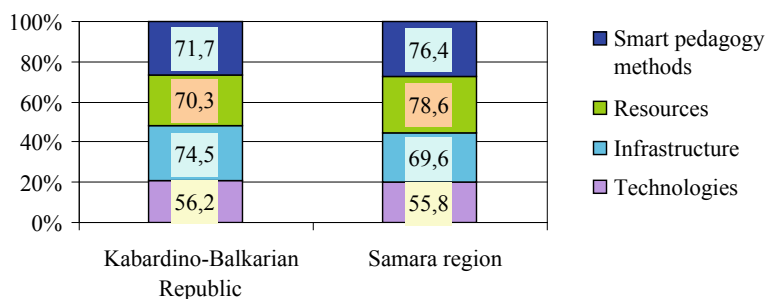


Fig. 6 The comparative analysis (the author’s vision)

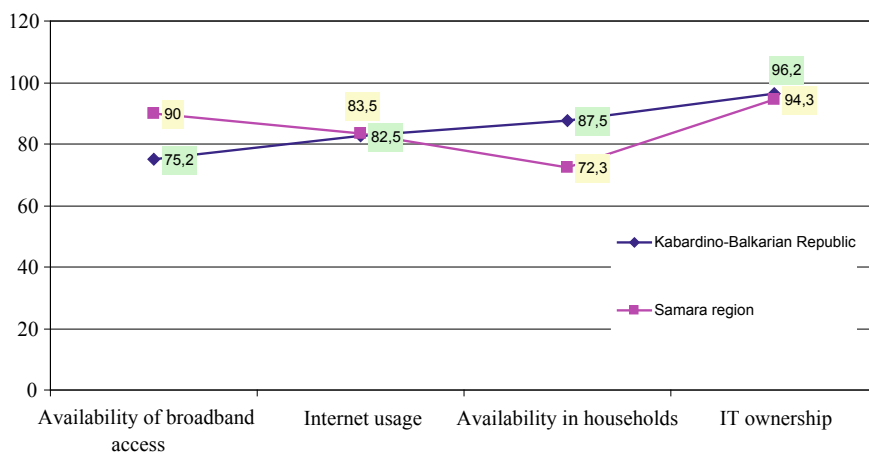


Fig. 7 The evolutionary ability of smart education: Samara region versus KBR

4 Discussion

The originality and novelty of the authors' approaches in the study can be considered as the follows: the resource base of two leading universities from two different regions of the Russian Federation has been analyzed due to the availability of opportunities for rapid implementation of smart pedagogy components and digital transformation in a smart university. The possibility of broadband Internet access, and its availability in households is described and the level of Internet and IT ownership was analyzed. The findings and outcomes reveal the readiness of the analyzed universities for digital transformation and collaboration in scientific and knowledge export activities.

5 Conclusions and Future Steps

Conclusions. Both smart philosophy and the concept of "smart university" are needed for large federal and regional universities. It can be achieved through providing them with flexibility, processability, manageability and the ability to innovative activity. Otherwise the evolution and fostering of smart educational systems is impossible.

The key system conditions for the effective implementation of smart-education is inclusion in the digital infrastructure of modern digital society some important features:

- (1) availability of criteria metrics and tools relevant to the identification of the initial and final level of competencies of trainees;
- (2) availability and possibility of adaptive relevant selection of individual educational trajectory of training;

- (3) availability and relevant adaptive application of learning analytics of the performance for the chosen educational trajectory (e.g. electronic “portfolio” system).

Next Steps. According to the findings, conclusions and the tested components and the enabling capacity of the designated higher education institutions, the following steps in this research project can be taken:

- (1) implementing, testing, validating, and analyzing additional components, particularly analytic learning tools, context-based learning, Personal Inquiry Based Learning (PIBL), and others;
- (2) identifying appropriate software and hardware systems and technologies to effectively support the various smart pedagogy components;
- (3) conducting quality assessments for smart pedagogy components in undergraduate and graduate curricula based on export-based learning programs.

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Smart University: Development of Analytical Management System Based on Big Data



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Abstract In contrast to the existing publications, in this research, the analytical system of smart university management based on big data is considered. Smart components of a smart university generate a very large amount of data continuously and all this data should be taken into account when making management decisions. At the same time, the article does not regard the structure essence of the smart university management system, but rather the analytical management system. The main attention is focused on the smart university information infrastructure. It allows you to use the advantages of intelligent technologies, such as big data. The novelty of the solutions and the practical significance of the research consist in the author's models application for supporting managerial decision-making based on big data and a graphical model and the information infrastructure in the analytical system of smart university management. Mathematical and graphical methods are used for modeling. The experience of managing Russian universities based on big data is also considered.

Keywords Smart university · Management infrastructure · Control system · Big data · Big data analytics · Data lake · API

1 Introduction and Literature Review

1.1 *Smart University Management on the Basis of Data*

Smart university (SmU) is a large, multi-level, complex system that includes many smart components. The classification and detailed description of smart components

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(SmC) are presented in the study by Vladimir Uskov and his colleagues [1, 2]. Considering smart university management system, we should take into account that SmU is a smart system, where smart functions are implemented. These are adaptation, sensing, inferring (logical reasoning), self-learning (self-description, self-analysis), anticipation (awareness), self-organization, and self-optimization [1, 2]. The support and development of these functions is related to big data. Therefore, the issue of building an effective smart university management system based on big data is relevant.

1.2 Control Systems Based on the Cybernetic Approach and Data Processing

Nowadays, the process of accumulating experience, mastering the opportunities and understanding the prospects of new technologies, including big data technologies is going on. Digital data analytics makes it possible to make management decisions in a smart university, based on data. The global Big Data Analytics (BDA) market, 2020, believes that the demand for big data analytics will grow exponentially. Data security is a major concern across all sectors due to the growing deployment of the Internet of Things (IoT) and the increasing number of devices that create huge amounts of data. Worldwide, the BDA market is estimated to grow 4.5 times, generating revenue of \$ 68.09 billion by 2025 from \$ 14.85 billion in 2019, with an average annual growth rate of 28.9% [3]. In addition, in the face of the uncertainty associated with the pandemic, BDA continues to be a top deployment priority for many organizations, as its use will help them remain competitive while accelerating innovation, including in education.

In accordance with the ideas of Industry 4.0, the introduction of a cybernetic approach to management, which is based on making decisions due to the results of objective data analysis (data driven decision), will help to get rid of the “disease” of any management systems such as HiPPOs (Highest-Paid Person’s Opinions) [4, 5]. This rule of decision-making is inherent not only in business, but also in any systems, including educational ones. We mean those, where officials (managers) often make far from optimal decisions. These management decisions are not based on up-to-date data and analytics.

1.3 Research Problem

The analytical system of smart university management is primarily based on data. Smart components of a smart university form a huge amount of data. At the same time, any data is collected and stored. In the ordinary management system of a classical university, data analytics is performed mainly on demand. It does not take into account the huge potential inherent in big data. With the right approach to the

use of big data, they can become the basis of the smart analytics system of smart university management.

That is why the research problem can be described as follows: how to design an optimal smart university management system based on big data to continue to support and promote digital transformation processes and increase smartness.

Our research will focus on the development of the infrastructure and analytical system of smart university management based on big data.

2 Models for Supporting Management Decision-Making Processes in Smart University

2.1 Making Management Decisions in a Multicomponent Smart System

Smart university is a multi-component interconnected structure with a large amount of data at the system input and within the system. So, in our opinion, the process of making a management decision is multi-level [6]. This is also due to the fact that the Smart University management system determines the smart tools set for making managerial decisions based on the consistent implementation set of smart university development projects. Solving the tasks hierarchy within the project portfolio imposes level limits, reducing the uncertainty of the underlying tasks without changing the overall smart university development goal.

Within the framework of these assumptions, it is proposed to build the analytical system of smart university management (smart system) based on either big data or the provisions of Mesarovich's research [7]. Mesarovich proposed a model of multi-layer decision-making system. Layers or levels of complexity of the decision are highlighted to reduce the situation uncertainty. Within a smart university, these levels can be highlighted in accordance with the structure of smart university's project portfolio. So, a set of consistently implemented projects is highlighted. For example, digital transformation. The model of the multi-layer hierarchy of the decision-making system in a smart university is shown in Fig. 1.

Each level is a block D that makes decisions and generates constraints X for the underlying $(i - 1)$ block.

2.2 Main Aspects of Decision-Making in a Multicomponent Smart System Under Conditions of Uncertainty

As an example, let us consider the multi-layer hierarchy of decision-making for managing the smart university business process. In accordance with [7], we can

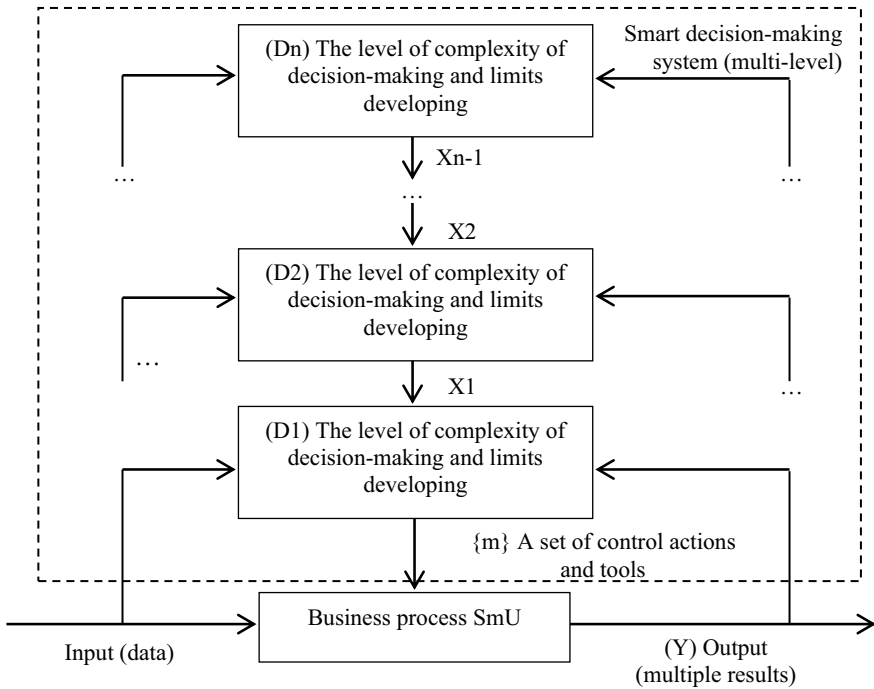


Fig. 1 A model of a multi-layer hierarchy of a decision-making system in a smart university

distinguish three main levels of decision-making under conditions of uncertainty, shown in Fig. 2.

The lower level, the closer one to the managed business process of a smart university, is the level of selection. The main task of this level is to choose the method of action “m”. The decision-making element (block) receives data (information) about the managed process and, using the algorithm obtained at the upper levels, finds the desired method of action, that is, a sequence of control actions on the managed SmU business process. The algorithm can be defined directly as a functional map D , giving a solution for any set of initial data.

If you set the output function of P and the evaluation function G , and the selection of action (m) based on the use of assessment G to P , it means that using set-theoretic representation, the output function can be defined as a mapping $R: M \times U \rightarrow Y$. Where M is a set of alternative actions, and Y is a set of possible outputs (results), U means a lot of uncertainty, reflected the lack of knowledge about the dependencies between the action of «m» and the output Y .

Similarly, the evaluation function G is a mapping $G: M \times Y \rightarrow M$, where V is a set of values that can be associated with the performance characteristics of a smart system (component). If the set U consists of a single element or is empty, so, there is no uncertainty about the output result for a given action «m». The selection can be based on optimization: to find such m' in M that the value $v = G(m', P(m'))$ is

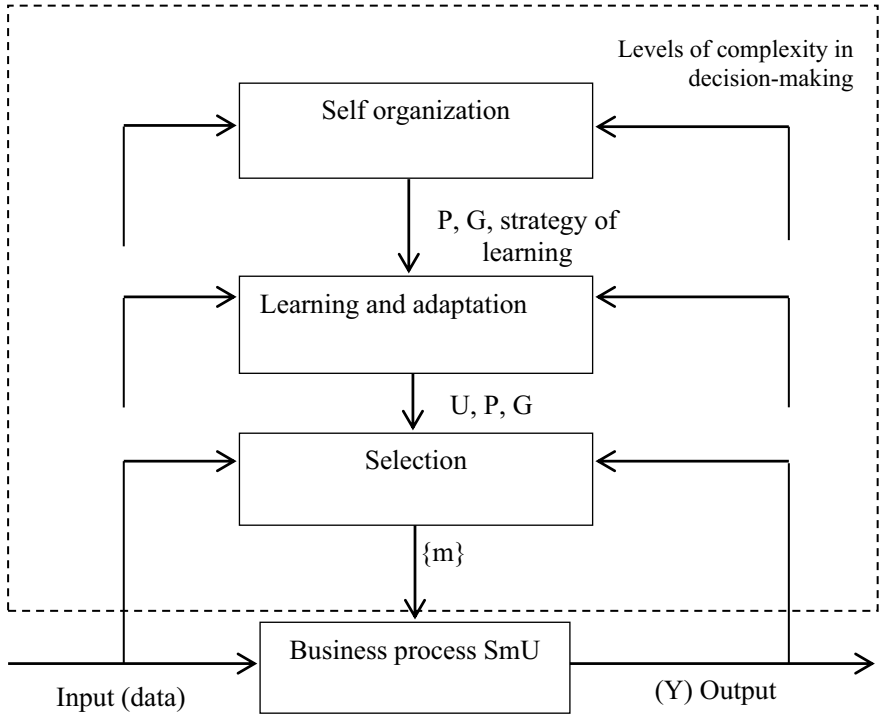


Fig. 2 A model of a three-level decision-making system in a smart university under conditions of uncertainty

less than $v = G(m, P(m))$ for any other action $m \in M$. If U is a richer set, we have to offer some other procedures for choosing the solution method. You may need to enter some other mappings besides P and G . But in the general, in order to determine the selection problem on the first layer, it is necessary to clarify the set of uncertainties and, the required relations P , G , etc. This is done on the upper levels.

The level above the level under consideration is the level of learning or adaptation. The task of this layer is to specify the set of uncertainties U that the selection layer deals with. The uncertainty set U is considered here as a set that includes all possible uncertainties about the behavior of a smart system. It reflects all hypotheses about possible sources and types of such uncertainties. U can be obtained using big data and machine learning. The purpose of the level under consideration is to narrow the set of uncertainties U and thus simplify the model of the selection layer. In the case of a stationary system and environment, the set can be extremely narrowed down to a single element, which corresponds to ideal learning. However, in general, U can include more than just existing ones. But also the uncertainties assumed by the decision-making system, and if necessary, U can be completely changed, expanded, including by changing the previously accepted basic hypothesis. The third, in this case, the upper level is the level of self-organization. At this level, the structure,

functions, and strategies used on the underlying layers are selected. It is being done to get as close as possible to displaying the goal, which is usually set in the form of a verbal description. If the goal is not achieved, the functions P and G at the first level or the learning strategy (technology) at the second level can be changed.

3 The Infrastructure Design of the Analytical Management Subsystem of a Smart University Based on Big Data

The high level of university smartness is characterized by smart tools introduction for working with big data, the use of Smart Learning Analytics, the intellectualization of basic and auxiliary business processes (machine learning, artificial intelligence, neural networks, chatbots, etc.). All interfaces are integrated and intelligent services are created. Digital profiles of staff (teachers, students, employees and management) were formed. Tools for managing digital profiles have been implemented and are working [8]. And all these components should be integrated on the basis of a single infrastructure. This infrastructure should also provide big data and analytics to the smart university's analytical management subsystem.

Smart management information services should provide fast and accurate assessment based on a large volume of data and their analysis.

A graphical infrastructure model of the smart university management system based on big data is presented in Fig. 3.

The component for storing big data should be allocated as a core of the infrastructure. We suggest using a Data Lake [9] as such a component. Smart university Big Data includes a different format for presenting information. It can be Learning Management System (LMS), Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), Application Programming Interfaces (APIs), social network data, sensor data, multimedia files, records from university databases and others. To extract the useful information for smart university management system from all this data, firstly, it is necessary to collect it. A data lake is suitable for this purpose. It is a repository of a large volume of unstructured data collected or generated by a smart university [10]. Unlike enterprise data warehouses or Data Warehouse, a data lake stores unstructured raw data. Such data can be used to make decisions fast and forecast events using Machine Learning algorithms [11, 12].

Smart university's smart services allow you to collect a lot of data based on the digital footprints of students, teachers, management and other university stakeholders. The smart university data lake collects all digital activities related to students, teachers, employees and all business processes and smart components of the university.

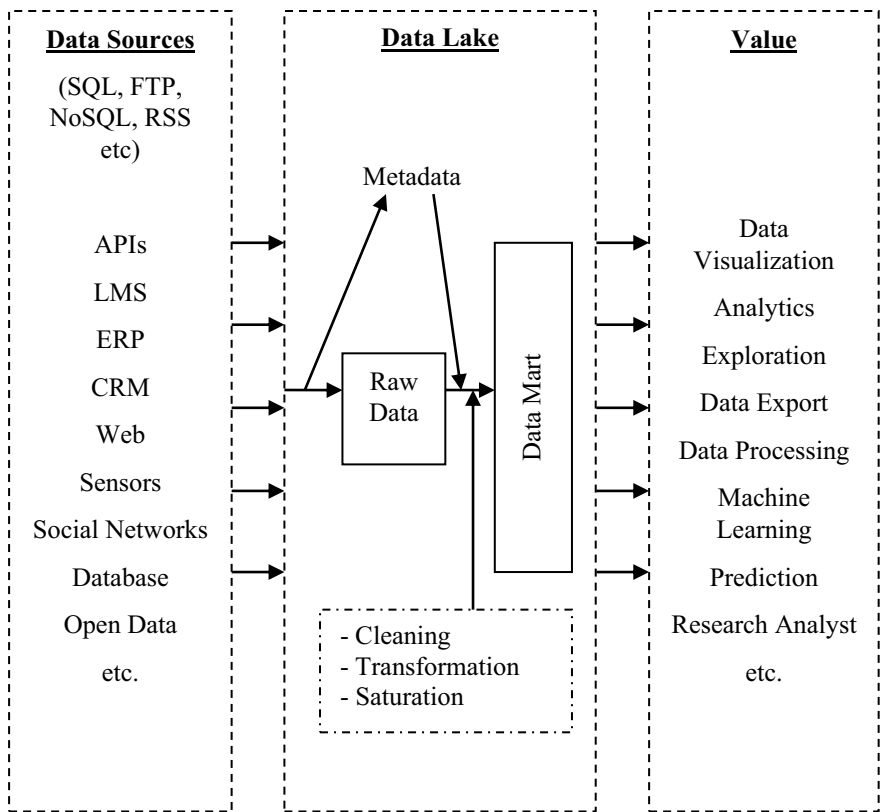


Fig. 3 Graphical model and the infrastructure in the analytical system of smart university management based on big data

4 Practical Experience of Russian Universities in Big Data Management

The practical experience of managing a university based on big data was considered by us on the example of Russian universities. A University consortium of big data researchers has been established in Russia. It is a voluntary association of scientific and educational organizations, their departments, teachers and scientists interested in the development of human and intellectual capital for advanced research implementation and product development in the field of collecting, processing and analyzing large amounts of data (<https://opendata.university/about>). Within the framework of the consortium, leading universities (29 universities) develop software products and analytics tools for working with big data, exchange them and implement them in management systems, increasing smartness. In the future, it is planned to create a unified big data platform for the development of university management systems on the consortium basis.

In order to develop the infrastructure and analytical management system, including within the framework of smart universities, the national standard “Information Technologies. Big data reference architecture. Part 1: Framework and application process”. This standard is a part of series of five standards on the reference architecture of big data. It is a Russian—language adaptation of the international technical report ISO/IEC TR 20547-1:2020. The document describes the structure of the reference architecture of the system for working with big data. It also provides a solution to the problem of displaying possible big data use cases in the reference architecture. The national standard regulations can be applied in universities to describe the architecture of specific systems for working with big data and the implementation of these systems, taking into account the technologies used, as well as the roles/performers and their needs. The standard, along with other parts of the 20547-X series of standards, will promote the effective use of end-to-end digital technology “big data” to solve economic and social problems in the implementation of the national program “Digital Economy”.

Using the consortium universities open data, the main cases of university management in the framework of increasing smartness based on big data (2017–2020) were considered (Fig. 4).

It was done on the base of experience of big data among such universities as Togliatti State University, Bauman Moscow State Technical University, Moscow Pedagogical State University, I. N. Ulyanov Chuvash State University, Tomsk State University and other high educational institutions of consortium. Let’s look in detail: 23 universities out of 29 (80%) use big data technologies for the purposes of managing the educational process (analysis of digital traces, design of an individual educational trajectory, digital doubles development, etc.). It is worth noting that most of the cases of big data analysis are aimed at increasing the smartness of the educational process.

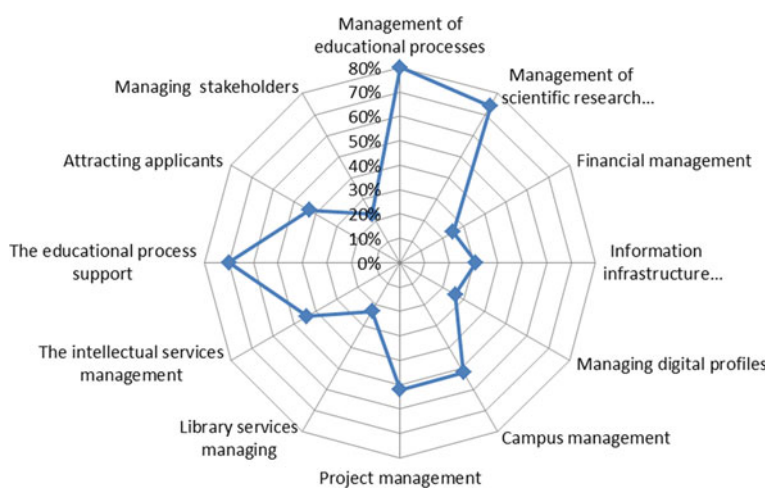


Fig. 4 Results of the study of university management cases based on big data

About half of the universities studied use big data to develop Smart Campus projects. Currently, big data projects are actively developing to attract applicants. In the course of the research, unique cases on library case management were discovered.

5 Conclusion and Next Steps

Conclusions. When implementing Big Data and analytics, there is always an element of risk, since it is complicated to forecast the result from the introduction of more effective control and management tools. But still, within the framework of modern trends, the development of this direction is justified and will provide universities with competitive advantages in the future. Within the framework of the study, the following results were obtained:

1. The models of management decision-making support in the framework of the analytical subsystem of smart university management based on big data are proposed.
2. A graphical model of the infrastructure of smart university management analytical system based on big data has been developed.
3. The experience of managing universities based on big data is considered.

Next Steps. The next planned steps in this research projects are as follows:

- 1 It is possible to speed up the introduction of Big Data technologies and other advanced digital technologies in a smart university by developing soft and hard management skills. So, further it is necessary to study these competencies and the possibilities for their development.
- 2 It is also necessary to explore the designing digital profiles possibilities of the main university stakeholders for more efficient work with big data.

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Smart University Innovation Efficiency Improvement Model



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Abstract Modern conditions are associated with the development of progressive technologies and neural networks, the emergence of smart devices, smart gadgets, and the expansion of information technologies. The innovative activity of any organization determines its competitiveness, the potential for expanding the scope of activities and the growth of efficiency. The rapid development of science and technology is fundamentally changing the needs of society for new products and services that have distinctive characteristics from the existing ones. Consequently, the organization needs to focus on the new consumer and his needs, to offer innovative products and services ahead of competitors. This vector of economic development entails the need to expand the innovative activities of a smart university and transform approaches to its management. At the same time, it becomes necessary to develop models for increasing the efficiency of innovation, which determines the relevance of a scientific article. The article reveals the concept of the specifics of innovative activity in relation to a smart university. The organizational foundations and management principles have been clarified, a model for increasing the efficiency of innovations for a smart university has been proposed. The obtained results were tested in order to increase the innovative activity of the smart-unit of the university on the example of the department.

Keywords Smart university · Innovation activities · Innovations · Innovative capacity · Innovation efficiency · Resources · Innovative development

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1 Introduction and Literature Review

In modern conditions of innovative development of the state, the creation of a new economy of knowledge, high technologies and progressive technology stands out. Institutions of higher education and science make a significant contribution to this development. There is a need to form a new structure of relations between the state, universities and business. Thus, smart universities are becoming market actors that generate innovations. Currently, the higher education system is responsible for the preparation of high-quality labor resources, as well as for the formation of an innovative ecosystem in accordance with the requirements of scientific and technological progress.

A smart university should be understood as the university of the future, focused on modern technologies, intellectual and innovative development [1].

Modern smart universities have reached a radical renewal, formed in a certain innovative environment based on the interaction of the team in the field of fundamental and applied research, the formation of knowledge flows and scientific schools. In these circumstances, the role of innovative development of higher schools increases significantly, which determines the need for their modernization and adaptation to new conditions of functioning. At the same time, a change in the education system is taking place, and modern smart universities are developing new models for increasing the efficiency of innovative activities, taking into account the commercialization of scientific and technical developments.

Currently, a lot of works are devoted to the creation of an innovative environment for higher education. P. Aghion, M. Dewatripont, J. Stein indicate that the transfer of scientific and technological knowledge from universities to business is an unconditional factor in the development of the country [2]. The work of Wu [3], Carayannis [4], I.M.B. is devoted to the study of the role of universities in the formation of innovative products and innovative development of the economy. Freitas et al. [5], Kruss et al. [6], Bhattacharya and Chatterjee [7].

M. Guerrero, J. A. Cunningham, D. Urbano believe that the university is a generator and conductor of knowledge that contributes to economic and social development through the triad of missions—education, research and entrepreneurship [8]. Innovation is born through experimentation between firms, universities, government and the end users of ideas. This fosters business models with value and profitability.

Significant studies of smart-university activities are reflected in the works of scientists such as: N. Serdyukova, V. I. Serdyukov, V. A. Slepov, V. L. Uskov, V. V. Ilyin, J. P. Bakken, R. J. Howlett, L. C. Jain, L. V. Glukhova, S. D. Syrotyuk, A. A. Sherstobitova, S. V. Pavlova [1, 9–12].

The studies have shown that the ongoing changes in the global and national economies, the demands of the business environment require fundamental changes in the models for increasing the efficiency of innovative activities of a smart university. The need to improve the efficiency of innovative activities of a smart university is

increasingly asserting itself, identifying trends associated with determining an operational response to the dynamism of society development, the spread of digitalization conditions [13].

2 Statement of the Problem in General Form and Its Connection with Important Scientific and Practical Tasks

The innovative activity of a smart university is the main source of the formation of the latest developments, innovations, the success of which is largely determined by effective approaches to managing the methods of interaction between the scientific environment and business. The ability to perceive innovations, to determine approaches to increasing the efficiency of the implementation of innovative development contributes to the functioning and development of a smart university in the context of socio-economic changes, digital transformation, and increased competition.

Higher educational institutions, transforming the content and approaches to the implementation of research and teaching activities, choosing the ways of commercializing developments can respond to the demands of the external environment, correspond to global trends.

Currently, a smart university is a dual entity in the economic system. On the one hand, its goal is to preserve the cultural and educational potential of the state. On the other hand, there is the production of personnel for innovative activities in a modern economy. Currently, the smart university is a key economic entity—a commodity producer of innovative products and educational services.

Thus, the problem of the research is to find mechanisms to increase the efficiency of the functioning of a smart university. The expansion of innovative activities of a smart university is focused on the development of personal potential, intellectual abilities of employees, ultimately aimed at the successful use of their capabilities in various areas of the economy, modernization of the higher education system, improving its quality, and growth of innovation.

The effectiveness of innovative activities of a smart university is largely determined by the success of the commercialization of innovative projects and scientific developments. Thus, modern conditions require the search for new approaches to improving the efficiency of innovative activities of a smart university, taking into account the needs of the market environment, society, and global trends.

3 Presentation of the Main Research Material with Full Justification of the Obtained Scientific Results

3.1 *Functional Model of Innovative Activity of a Smart University*

Innovation can be understood as the result of human mental activity, consisting in the form of new or improved objects (products, services, technologies, etc.). In a modern smart university, innovative activity is universal, affecting all areas of its functioning: educational and pedagogical; research; organizational and managerial; financial and economic; international; expert and consulting.

Figure 1 shows the functional components of the innovative environment of a smart university.

In our opinion, the innovative activity of a smart university is a complex of organizational and managerial, educational and pedagogical, research, financial and economic, international expert and consulting components, contributing to the creation of innovative products in the field of management, education, science and practice.

At the same time, the innovative activity of a smart university should be aimed at the possibility of using and commercializing the results of scientific research and development.

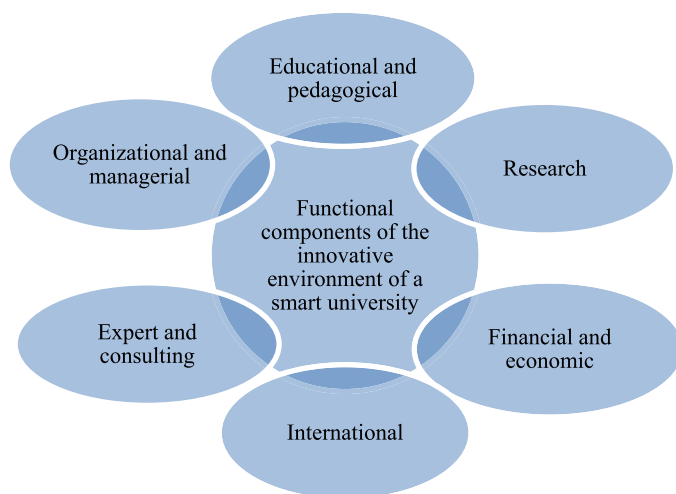


Fig. 1 Functional components of the innovative environment of a smart university

Table 1 Refined principles of smart-university innovation management

Principle	Characteristic
The principle of complexity	involves taking into account all costs and benefits of the innovation process
The principle of planning	is aimed at solving current and future problems of the development of innovative activities
The principle of consistency	suggests that innovation management should be systemic, taking into account changes in external and internal factors
Correctness principle	provides that all indicators and formulas used for evaluating an innovative project must be correct and applicable, have the properties of comparability, transparency
The principle of limited and interchangeable resources	is aimed at the need to select a resource that provides the lowest unit costs
The principle of expanding the need for ongoing innovative projects	involves the search for ways to increase the efficiency of projects based on the assessment of consumer properties and global trends
The principle of taking into account the interests of all participants in the innovation process	suggests the need to take into account the interests of all participants in the development of an innovative project
The principle of efficiency	provides for the excess of the economic effect from the implementation of an innovative project in comparison with the costs of its implementation

3.2 Basic Principles of Managing Innovative Activities of a Smart University

In order to increase the efficiency of innovation, it is necessary to adhere to certain management principles (Table 1).

The application of the refined principles will allow you to effectively manage the innovative activities of a smart university.

3.3 Assessment of the State of Innovation: Substantiation of the Need for the Formation of the Innovative Potential of the University

The study showed that the implementation of innovative activities of Smart-University is associated with the general trend of development of science and innovation. Based on statistical data [14], we visualize research results by the number of

organizations performing research and development by types of organizations in the Russian Federation (Fig. 2).

Figure 2 shows that in 2019, compared to 2015, the number of organizations carrying out research and development by such types as: research organizations, design organizations, design and survey organizations, pilot plants, educational organizations of higher education decreased. However, the growth of industrial organizations with research and development units and other organizations is visible. Such changes were influenced by macroeconomic factors.

Based on statistical data [14], we visualize the research results on the dynamics of the number of organizations performing research and development by sector of activity in the Russian Federation (Fig. 3). Figure 3 shows that in 2019 compared to 2015, the number of organizations carrying out scientific developments in such sectors as public, business and higher education decreased. Significant growth in research and development organizations during this period is in the nonprofit sector.

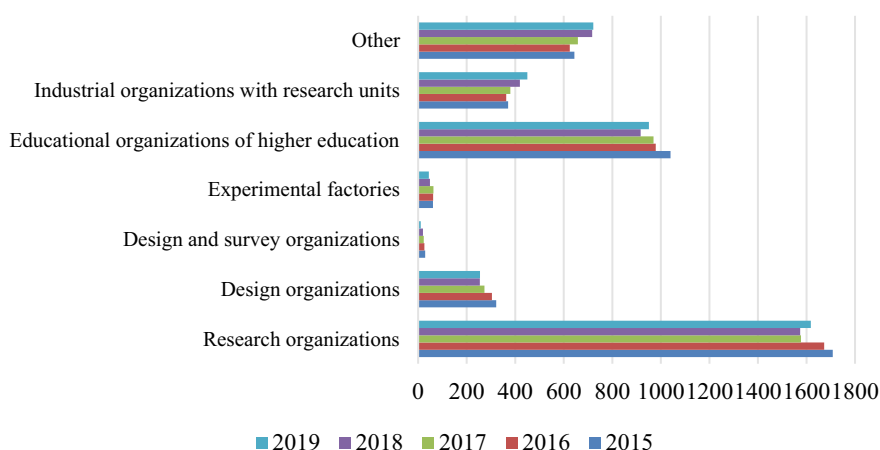


Fig. 2 The number of organizations that carried out research and development, by type of organization in the Russian Federation [14]

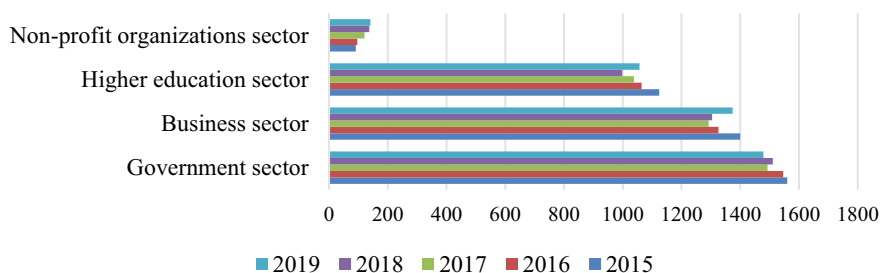


Fig. 3 The number of organizations that performed research and development, by sector of activity in the Russian Federation [14]

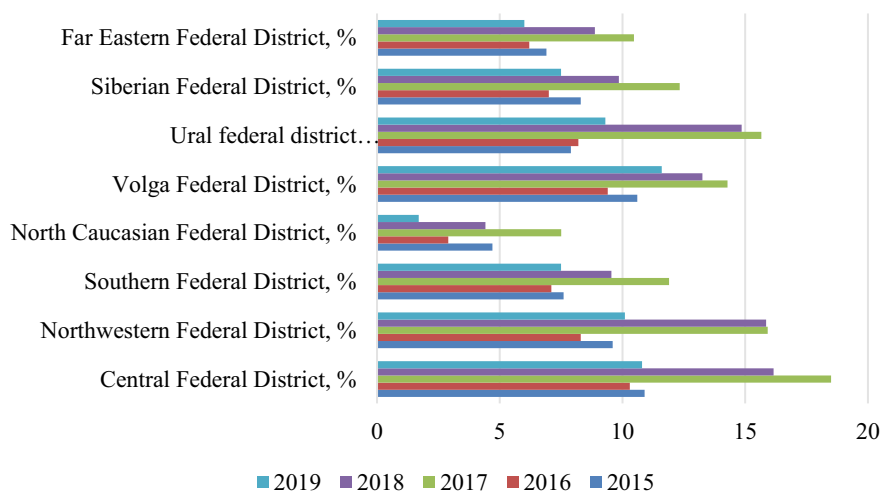


Fig. 4 The level of innovation activity of organizations by constituent entities of the Russian Federation [14]

After examining the statistical information [14], in Fig. 4 we present the dynamics of the level of innovative activity of organizations in the constituent entities of the Russian Federation.

Figure 4 shows that in 2019 the level of innovative activity of organizations decreased approximately to the values of 2015 for all constituent entities of the Russian Federation. The study showed that in the field of innovation activities of organizations, changes and improvements are required in order to develop this direction. In particular, a special role in this is assigned to smart universities, which are key centers of scientific research and innovation.

3.4 *Organizational Model for Improving the Efficiency of Innovative Activities of a Smart University*

The innovative activity of smart university should cover all key vectors of its development. Based on the conducted research, we propose a model for increasing the efficiency of innovative activities of smart university (Fig. 5).

High results of innovative activities of a smart university contribute to improving the image of a higher educational institution, achieving competitiveness and attractiveness for applicants. In this regard, the development of innovative activities is a priority achievement and an important indicator of the development of a smart university and the formation of its brand. The result of innovation should be innovation, which is an innovation that has efficiency.

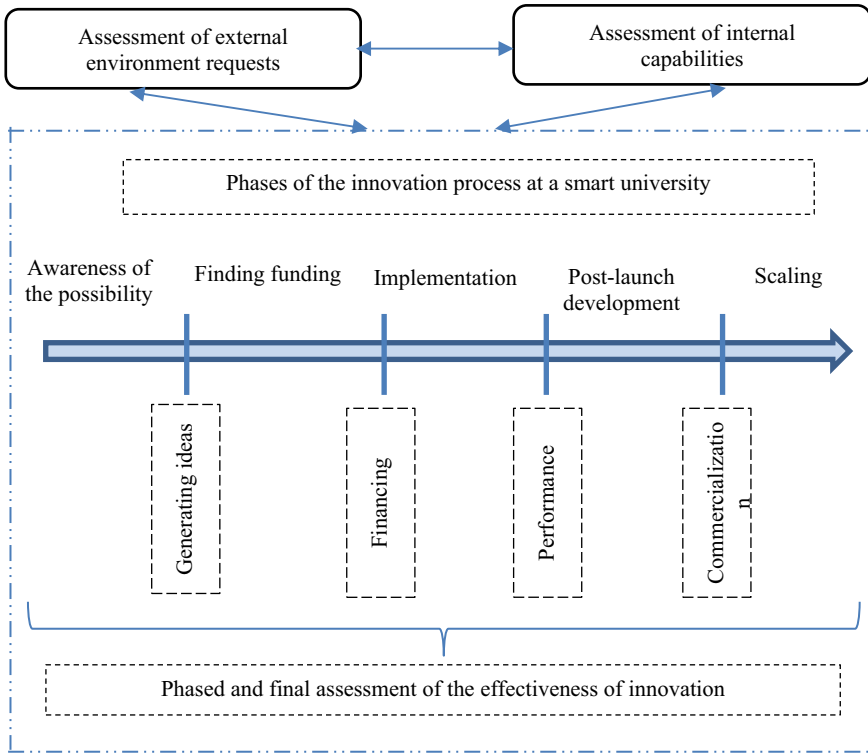


Fig. 5 A model for increasing the efficiency of innovative activities of smart university

To implement innovative activities, a smart university must have the necessary resources and capabilities. The innovative development of a smart university is an imperative of both its current and future activities. The innovative activity of a smart university is closely related to the innovative potential, which complicates the management methodology and requires the improvement of management algorithms. Modern conditions expand the areas of activity of a smart university, pose new management tasks that must meet the needs of modern society and global trends.

The proposed model is focused on the demands of the external environment in innovation and internal possibilities of its development. This model assumes an assessment of innovations at each stage of the innovation process and will improve the efficiency of innovative activities of a smart university.

4 Algorithm of Step-by-Step Implementation of the Model into the Activities of the Smart-Subdivision of the University

The proposed model for increasing the efficiency of innovative activities of a smart university was tested in the activities of Togliatti State University, Department of Master's Degree. The introduction of a model for increasing the efficiency of innovative activities of a smart university assumed the following steps (Fig. 6).

Let's consider the implementation of the model using the example of the department of magistracy of Togliatti State University.

- Step 1 Determination of the need for innovation based on marketing analysis. At this step, the needs of the external environment and internal capabilities were assessed to determine the need for innovation.
- Step 2 Planning innovation activities. During the implementation of this step, stages, phases, deadlines and conditions for the completion of the stage were identified. In the course of the innovation process, at each of its separate phases, the effectiveness of the innovation was assessed. Based on the results obtained, the achievement of critical values was assessed.
- Step 3 Formation of a system for monitoring innovation activity. Based on the monitoring, the expediency of transition to the next phase of the innovation process was determined.
- Step 4 Assessment of the effectiveness of the innovation process and its commercialization. If the assessment carried out indicated the possibility of making a profit and the compliance of the innovation with the request of the external environment, then a decision was made to continue the innovation process and enter the stage of commercialization and scaling.

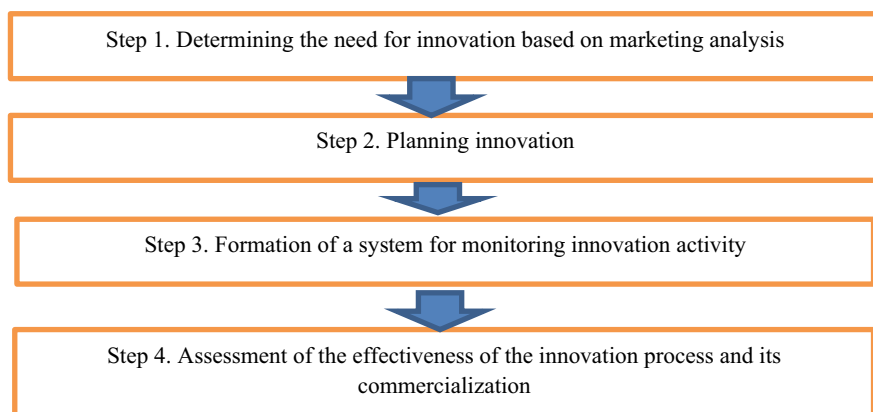


Fig. 6 Steps to introduce a model for improving the efficiency of innovation activities of smart university

As a result of using this model, work was carried out on the innovation process as part of the implementation of innovative projects. Practice has shown that at step 3 “Formation of a monitoring system for innovation activity” out of 5 innovative projects, a decision was made to stop work on 2 projects. At step 4 “Assessment of the effectiveness of the innovation process and its commercialization”, based on the results obtained, a decision was made to refuse to implement another innovative project due to its low efficiency.

Thus, the proposed model makes it possible to increase the efficiency of innovation in a smart university.

5 Conclusions of the Research and Prospects for Further Research in This Direction

Conclusions. The study showed that modern conditions require improving the model for increasing the efficiency of innovative activities of a smart university, taking into account the peculiarities of its functioning.

1. The article clarifies the concept of innovative activity in relation to a smart university. The study made it possible to clarify the principles of managing innovative activities of a smart university, taking into account the peculiarities of the development of a smart environment.
2. The article provides a statistical study of the number of organizations carrying out research and development by type of organization and by sector of activity in the Russian Federation, as well as the level of innovative activity of organizations by constituent entities of the Russian Federation for 2015–2019. The results of the conducted statistical study confirm the need for the development of innovative activities in smart universities.
3. The article proposes a model for increasing the efficiency of innovative activities of a smart university, taking into account the demands of the external environment and internal capabilities. The proposed principles and model for increasing the efficiency of innovative activities can be applied both at the level of a separate structural smart unit and in a smart university as a whole.

Next Steps. To expand the methodological aspects of innovation, it is necessary to develop a system of key indicators that allow evaluating innovations in such areas in a smart university as an educational smart environment; scientific developments; scientific research work; smart university management system.

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Managerial Approach for Foreign Language Learning and Fostering in a Smart University Environment



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Diana Yu. Burenkova, and Inga V. Treshina

Abstract This article reveals both the basic definitions and concepts of managerial theories of motivation implemented into the educational process. Innovative methods for science studies students at smart university are represented. The importance of methodical developments, content and innovative methods used during the English lessons to form a stable positive learning motivation is emphasized. The stages for classroom and extra curriculum educational activities are identified and the main methods used at each of them are described. The well-structured system of feedback enabling the designing of English learning and fostering on the basis of knowledge transfer at Togliatti State University is described.

Keywords Innovative techniques · Motivation · Flipped classroom · Blended learning

1 Introduction

Modern studies reveal that the success and failure of a knowledge transfer for students depend on the motivational component of learning process organized by the teacher. The globalization and digitalization are considered to be the drivers for designing and developing innovative methods for studying and using a foreign language in the professional area. The acquired skills are needed for designing an effective communication and finding common ground in the international business environment. Innovative teaching methods based on managerial and motivation approaches are required for foreign language learning and fostering at smart university. According to the modern standards, smart university is a university enhancing the education,

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research, and work experience of stakeholders by incorporating digital, innovative, and internet-based technologies for the improvement of the society.

In other words smart university requires both smart management and smart pedagogy involving innovative teaching and learning strategies based on motivation approach because motivation is a force that supports students' interest for life-long learning and upgrading their skills [1]. There are a lot of studies representing a similar viewpoint and highlighting the importance of feedback [2], adaptation to the conditions of the educational process implementation [3, 4], and implementation of innovative pedagogical communication means [5, 6].

Having reviewed the above mentioned studies the authors consider that nowadays there are many different theories that can be used for smart universities. However, the possibilities of their application for foreign language teaching and fostering are outlined at an insufficient level. In the conditions of globalization and digital society knowledge of foreign language is considered to be as the key element for career development. Therefore, the authors' experience of motivation theories implementation in teaching a foreign language is represented.

2 Problem Statement: Managerial Theories for Foreign Language Fostering

The authors see the main purpose of this study as expanding the knowledge base of the application of modern management approaches to foreign language teaching in a smart university. According to most theories in pedagogy, psychology, and management, motivation is considered to be the driving force for any activity but most basic motivation theories are limited in their theoretical nature.

Problem Statement and the novelty of this paper is to consider the application of classical motivation theories to the contemporary digital society and possibility of their effective integration into educational process for Generation Z.

The objectives of the study are the expanding of motivation theories usage for the educational process at a smart university and increasing students' interest in a foreign language learning and improving the quality of both hard and soft skills of higher school graduates.

3 Methodology: Managerial Theories for Foreign Language Fostering

There are a lot of studies concerning the necessity for managerial issues of motivation basis for education. The following are the ideas and suggestions introduced by the outstanding researchers. The results of these studies are represented in many

scientific articles but by summarizing all of them the following key motivation theories are considered to be useful for the educational process and foreign language teaching: The Human Motivation and ERG theories by Abraham Maslow and Clayton Paul Alderfer; The Expectancy theory of motivation by Victor Vroom; The Equity theory of motivation by Adams. Nowadays, there are a lot of followers of all the above-mentioned motivation theories. For example, P. Skehan believes that motivation can be «an inner motivation» encouraging, directing, and supporting the behavior patterns that are declared to be the essential condition for students' success in education and career development [7]. R. C. Gardner and W. E. Lambert divide motivation into integrative and instrumental. The integrative motivation is dictated by the desire of a person to become a part of the international professional environment. The instrumental one is related to practical goals or pragmatic attitudes, e.g. a person's desire to master his knowledge and knowledge transfer for getting promotion and career advancement at their workplace, higher salary, and status in society [8]. The researcher Mario Guerrero asserts that motivation is related to commitment, enthusiasm and perseverance in attaining the aim [9].

Zoltán Dörnyei [10] considers the motivation from three points of view: (a) language level engages language and community, cultural and intellectual values; (b) student level consists of the student's personal qualities and cognitive processes; (c) learning situation level focuses on academic programs (AP), teaching methods, content, teachers, course and group. The ideas represented in other modern studies [11] follow the idea of requirements for using the managerial approach and motivation theories while designing the syllabus and knowledge content at smart university.

4 Our Results

4.1 *Application of Motivation Theory for Smart University*

The authors believe that fostering foreign language through motivation approach at smart university is to be applied at all the levels of the educational process. Table 1 represents the authors' vision for implementation of motivation theoretical foundations for foreign language learning and fostering according to the modern studies [12].

4.2 *Applying Motivation Theory to the Smart University in Practice*

Table 2 represents the application of motivation theories for effective planning the teacher's and students' performance during English lessons at Togliatti State University (TSU).

Table 1 Motivation basics for foreign language learning

Motivation theory	Key ideas	Theory implementation for foreign language teaching
The ERG theory	The person needs for self-actualization at the top of his life span The need for career development due to Existence + Relatedness + Growth	According to the European scale of language competences levels for foreign language skills A1-C2 provide the person's capability for achieving all of the targets according to Maslow and Clayton Paul Alderfer Theories
The expectancy theory	Expectancy + Effort" + Remuneration = Motivation Human's performance is based on personality, skills, knowledge, experience and abilities	Students expect that a high level of foreign language skills (B2-C1) achieved while studying at the university is considered to be significant soft skill competence that makes a specialist competitive in both the domestic and international labor markets
The equity theory	Cognitive motivation is stable and increases only if there is a fair reward for the person's performance	The teacher should clearly define and describe the indicators and criteria for assessing the students' performance during the English lessons

The experiment based on the implementation of managerial approach and motivation theories for English studying and fostering has been conducted since 2016 by the professors of Institute for the Humanities and Pedagogy at TSU and some professors of the Russian Presidential Academy of National Economy and Public Administration (RANEPA) since 2020.

4.3 The Innovative Teaching Methods for Foreign Language Learning and Fostering at the Smart University

Nowadays, innovative methods for increasing students' cognitive motivation and fostering foreign language learning is based on the following methods including flipped class, task-based learning, CLIL and blended learning that are considered to be the most effective for learning different subjects, transferring knowledge and developing both hard skills and soft skills in a Smart university. Having analyzed the existing pedagogical methods some of the most efficient tracks for Generation Z were chosen. Table 3 represents the innovative teaching methods for foreign language learning and fostering at the smart university (the authors' vision).

The challenges of modern society promote the usage of blended learning in Smart university due to its following advantages: learning is possible regardless of time

Table 2 Application of motivation theories for English lesson

Stage	Motivation theory	Application at TSU
1. Designing and Simulation	The selection of teaching methods according to ERG theory	The identification of topical themes for discussion and hierarchy of group needs accordingly to the age peculiarities of students and their degree syllabus; Assessment tools (SAT) for the disciplines “Foreign language”, “Professional foreign language” in accordance with the level of language competence of the subgroup and its degree program
2. Choosing training tracks	The selection of methods by learning in accordance with the Vroom’s Theory of expectations and Lock’s Theory of goal setting	The annual lesson planning and the English lesson grade-rating system (GRS) are loaded and represented at the Smart university collaborative portal
3. Lesson and feedback	The selection of methods according to XYZ—McGregor’s Motivation Theory	The using of SAT materials for the English lessons based on CLIL Methodology, Quizlet, Wiser.me, Memrise, Nearpod, Liveworksheets, ISLCollective and Kahoot systems. Individual and team work methods in the organization of the educational process, audiovisual media and game technologies for the development of lexical and grammatical structures
4. Assessment	The selection of methods in accordance with Lock’s theory of goal setting and Adams’ theory of equality	SAT for midterm and final academic assessment according to the TOEIC system. The grade-ranking system, annual students ranking according to the foreign language skills. The discussion of the results of individual and group learning activities in the classroom, as well as the results of midterm and final testing

and place; priority of the learner’s independent activity and flexibility of an educational tracks; integration of online and offline educational and methodical content of multiple use. All of them are targeted to the foreign language learning fostering according to the psychological and managerial aspects of the motivation theory. All the above mentioned tracks support the development of qualities and skills required

Table 3 The methodology

Methodology	Activity and performance
Flipped classroom	The students learn basic concepts related to the new theme before the lesson starts (either working alone or in groups with peers), using digital content prepared by the teacher
Task-based training	After the tasks offered by the teacher are completed, the students work in pairs to create a task on a given topic. The following online services are recommended: Kahoot; Quizizz; LearningApps; QR- codes; Google Form; Plickers
Blending learning: collaborative activities, team work, project work	Individual and face-to-face performance according to the chosen task and targets

in the twenty-first century, such as cooperation, creativity, ability to solve problems, independence, computer, language literacy, troubleshooting.

4.4 The Experiments

Figures 1 and 2 reveal the methodological aids created by the professors of the Institute for Humanities and Pedagogy at Togliatti State University according to the authors' vision and data represented in Tables 1, 2 and 3.

The authors suppose that these are considered to be beneficial and effective for foreign language learning and fostering at smart university. It should be mentioned that all of them are designed according to the managerial approach and motivational theories. For example, tasks represented at Fig. 1 follow the key ideas of the ERG Theory and tasks represented at Fig. 2 correspond to the theory of expectation and XYZ motivation theory and reveal the authors' vision.

5 The Experiment Results

The experiment is conducted in 8 groups of students from different institutes. Each institute has one exposure and one control group. Both control and exposure groups include students of Pre-Intermediate level: Institute of Mechanical Engineering (IME)—139 people, Institute of Mathematics, Physics and Information Technology (IMPIT)—68, Institute of Electrical and Power Engineering (IEPE)—66. Each teacher worked with two exposure groups; different methods for English fostering were designed and used. Quantitatively, the experiment has covered more than 270 students of Engineering degree programs for 3 years (2018–2020). Efficiency is evaluated in terms of attendance increasing and comparing the number of

Instruction for working with the worksheet 1. Read all the tasks attentively. 2. Do the tasks step by step. 3. Check your answers yourself by using QR codes.											
<u>Type 2 Conditionals</u> 11. Match the halves. <table border="1"> <tr> <td>1) If I found a wallet in the street</td> <td>a) they would be offended.</td> </tr> <tr> <td>2) If somebody pointed a gun at me.</td> <td>b) He would lose his temper.</td> </tr> <tr> <td>3) If I did not congratulate them</td> <td>c) if they did not invite us.</td> </tr> <tr> <td>4) If I did not accept any of his proposals.</td> <td>d) I would be scared stiff.</td> </tr> <tr> <td>5) We would be annoyed</td> <td>e) I would go to the police.</td> </tr> </table>	1) If I found a wallet in the street	a) they would be offended.	2) If somebody pointed a gun at me.	b) He would lose his temper.	3) If I did not congratulate them	c) if they did not invite us.	4) If I did not accept any of his proposals.	d) I would be scared stiff.	5) We would be annoyed	e) I would go to the police.	 Please use the QR code to check yourselves.
1) If I found a wallet in the street	a) they would be offended.										
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3) If I did not congratulate them	c) if they did not invite us.										
4) If I did not accept any of his proposals.	d) I would be scared stiff.										
5) We would be annoyed	e) I would go to the police.										
12. How are Type 2 Conditionals formed? Check yourself by following the link: http://bit.ly/2DCdSrW or use the QR code given opposite Ex. 12.											
13. Put the verbs in brackets in the correct form. 1) If Chris _____ (be) here, he _____ (advise) us how to behave. 2) John _____ (lend) you some money if you _____ (ask) him. 3) If mansions _____ (not / cost) so much, we _____ (purchase) one of them. 4) If I _____ in your shoes, I _____ (not / promise) to do it. 5) If I _____ (have) toothache, I _____ (go) to the dentist. 6) If he _____ (take over) the company, he _____ (become) arrogant. 7) If I _____ (see) someone robbing a bank, I _____ (ring) the police. 8) If I _____ (not / work), I _____ travel around the world.	Please use the QR code given below to check yourselves: 										

Fig. 1 The example for QR code and flipped classroom methodology (a fragment)

Part 1

3.4.2 You are going to listen to a song called *If I were a boy* on YouTube.com: please follow the link <https://bit.ly/2LWi3FV> or use the QR-code below:
 Fill in the gaps with the missing words.

IF I WERE A BOY

If I 1 _____ a boy
 Even just for a day
 I 2 _____ of bed in the morning
 And throw on what I wanted and go
 Drink beer with the guys
 And chase after girls
 I'd kick it with who I wanted
 And I 3 _____ for it
 Because they 4 _____ for me
 If I 1 _____ a boy
 I think that I 5 _____
 How it feels to love a girl
 I swear I 6 _____ a better man
 I 7 _____ to her
 Cause I know how it hurts
 When you lose the one you wanted
 Cause he's taken you for granted
 And everything you had got destroyed!

.....

3.4.3 Discuss the following questions with your teacher:

1. What is the topic of the song? What problem is raised?
2. What grammar construction is mostly used in the song? Explain why the character uses this particular grammar construction.
3. What situation has happened to the character? How is she feeling. Underline the words that make you think so.
4. What decision has the character taken? Why?
5. How would the character behave if she was a boy?
6. What occupation would the woman try according to the video?
7. What did the man feel when they changed their places and roles in the video?
8. Why did the woman and the man quarrel in the video?

Fig. 2 The example for QR code and table-based training methodology (a fragment)

students' transition to a higher English skills level. Table 4 represents the results of the experiment during the English study period (2 academic years).

Figure 3 represents the English fostering increase for the experiment period.

Figure 4 represents the English fostering increase for the experiment period by RANEP.

Table 4 Statistics on experimental groups of the students (TSU)

Institute	Number of students in the control group	% of attendance at the beginning (%)	% of attendance in	Student performance (credit 1)	Student performance (credit 2)
			2018 (%)	2019 (%)	2020 (%)
IME	48	73	76	86	90
IMPIT	24	85	85	95	95
IEPE	22	75	80	90	98

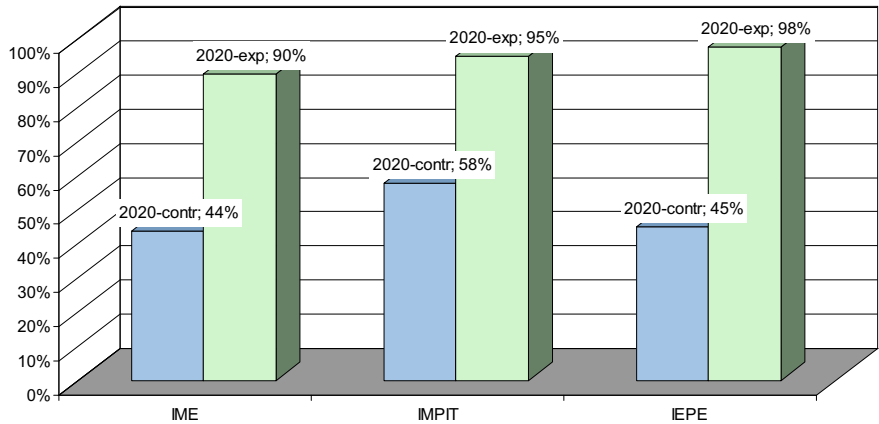


Fig. 3 English fostering increase (TSU)

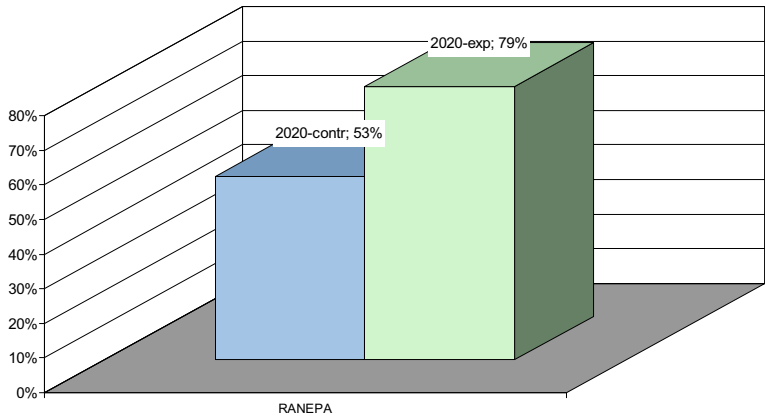


Fig. 4 English fostering increase (RANEPa)

6 Discussion

According to the experimental results of the authors' method based on the theories of motivation it is clear that the objectives of increasing the level of foreign language skills and cognitive activity of "Generation Z" students have been achieved. Thus, managerial approach based on the famous motivation theories is thought to be a fundamental factor in stimulating teaching activity and students' activity at a smart university.

In order to create a positive motivation in the process of learning foreign languages, teachers need to have a clear understanding of the goals and the necessary methods for organizing the educational process for students from «Generation Z». The main purpose of motivation in teaching foreign languages is to stimulate and promote cognitive activities that ensure active participation and work of students during the lessons, contribute to a strong interest in foreign-language culture and the development of foreign-language communication skills. According to the selected stages, the proposed methods have been tested at Togliatti State University. As a result of the implementation of the suggested methods, the growth of students' motivation and performance indicators in the experimental group has been shown.

7 Conclusion and Future Trends

Conclusions. Managerial approach based on the famous motivation theories is thought to be a fundamental factor in stimulating teaching activity and students' activity at a smart university.

1. In order to create a positive motivation in the process of learning foreign languages, teachers need to have a clear understanding of the goals and the necessary methods for organizing the educational process for "Generation Z". The main purpose of motivation in teaching foreign languages is to stimulate and promote cognitive activities that ensure active participation and work of students during the lessons, contribute to a strong interest in foreign-language culture and the development of foreign-language communication skills.
2. According to the selected stages, the proposed methods have been tested at Togliatti State University and the Russian Presidential Academy of National Economy and Public Administration. According to the experimental results of the author's method based on theories of motivation it is obvious that the objectives of increasing the level of foreign language skills and cognitive activity of "Generation Z" students have been achieved.
3. Analysis of the research showed that the modern generation of learners who live in the era of digitalization and use smart technologies and devices has access to global resources. Therefore, new methods and managerial approaches to the formation of motivation for learning activities in a foreign language are required.

4. The article shows the authors' vision of integrative approach, which expands the boundaries of applied use of the basic theoretical models of motivation by enriching them with digital technologies and smart pedagogy techniques (flipped classroom, CLIL, blended learning, etc.). The results of the experiment are shown and confirm the effectiveness of the authors' approaches because it allowed to increase students' motivation by an average of 14.2%

Future trends. The presented authors' methodology is going to be used not only by teachers of a foreign language but also by the staff of different training facilities for attracting new learners of both "Generation Z" and "Generation Alpha" to encourage them to acquire new skills and knowledge.

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Integration of Smart Universities in the Region as a Basis for Development of Educational Information Infrastructure



Yana S. Mitrofanova, Anna V. Tukshumskaya, Valentina I. Burenina, Elena V. Ivanova, and Tatiana N. Popova

Abstract The article offers a unique mechanism for step-by-step integration of smart educational structures into a single integration platform. A special feature of the proposed integration infrastructure in smart universities is the possibility of using various educational institutions as stakeholders in the overall structural composition of interaction. Bringing the integration model into a single educational scheme is carried out taking into consideration the concept of adaptation maximum. A model of the infrastructure platform formation for smart universities integration has been developed. The advantages of the model are the accumulation of a huge digital potential, which can be used by any educational institution to increase smartness. The experience of smart universities integrating is considered on the example of Russian universities.

Keywords Smart university · Smart components · Digital transformation · Integration · Information infrastructure · Pandemic

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1 Introduction and Literature Review

1.1 *The Relevance of Integrated Digital Transformation of Smart Infrastructure*

Nowadays, educational institutions all around the world face a great number of challenges. To cope with all arising ones because of connection with the digitalization of education and the COVID-19 pandemic, schools, colleges and universities have a sharp necessity either to create or implement their own projects for creating smart components [1–3]. They also need to cooperate to exchange experience in building smart educational institutions. One more important task here is creating digital services, cases on augmented and virtual reality, smart models of teaching students and schoolchildren and others [4–6]. The pandemic has intensified the issues of digital education. It has become clear that it is very difficult to implement smart digital learning projects alone, not only for schools, but also for large universities. Therefore, today there are many initiatives to create an integrated digital educational space [7, 8], smart universities are actively uniting [9, 10], and cooperation in this direction is also developing with schools and other educational institutions.

Integration processes all around the world combine different systems based on certain integration schemes. Over the past 10–15 years, unions have been actively created on the basis of scientific and educational centers in various regions. Their aim is to develop technological innovations and grow technological industries and launch startups. Such centers combine the innovative potential of universities and production sites [11]. Regional centers can be combined into a larger innovative hierarchical structure. This experience is also adopted by the education system, which has always followed the integration trend [12].

1.2 *The Problem, Goals and Objectives of the Study*

The problem of the study is to find the mechanisms for building the effective smart universities integrations in the region. A feature of integration is the use of schools and colleges as stakeholders in the overall structural composition of interaction.

The main goal of this study is to obtain a model of smart universities integration interaction. Its focus is concentrated on the spreading the best practices in digital transformation and smart components development, with the possibility of assembling the necessary smart components by educational institutions of various levels (school, college, university, etc.) on request.

The main tasks of the study are:

1. Analysis of opportunities for leading smart universities integration and the educational environment integration.

2. Development of integration union schemes and the model of formation the integration infrastructure in smart university.
3. Modeling the possibilities of assessing the adaptation of the integration model into a single educational scheme.
4. Experimental testing of the results obtained in the region.

2 Modeling the Possibilities of Integration Union in Smart Universities

The purpose of the simulation was to build a smart platform for integration unions. To do this, it was necessary to examine the blocks of influence, blocks of research and blocks of result. Due to the results of the publications analysis [13–15] and according to the survey was to identify the basic blocks of influence [development of education (37%), the rate of the external environment change (83%), especially in the context of a pandemic, the ideas of Industry 4.0 (54%) and integration policy (32%)]. The integration infrastructure management model is based on mathematical and simulation modeling methods. Figure 1 shows a diagram of the study stages and the obtained result.

The integration symbiosis will be further developed by forming a smart integration platform that can be adapted to the conditions of a smart city and a smart region on the way to the smart society development.

3 The Concept of Adaptive Maximum for the Adapting Smart Platform Functioning in Integration Unions

Any integration platform is a developing system. That is why, we considered the concept of adaptive maximum in complex developing systems [16] in the framework of modeling the integration platform. The goal of managing a complex integration system is to keep the system within the adaptive maximum in changing conditions. This condition must be taken into consideration while the developing a management model for the integration infrastructure of smart universities.

The phenomenon of the adaptation maximum consists in the fact that in a system of p variables with a behavior defined up to k intersecting manifolds, the number of arbitrary coefficients in the structure of equivalent equations will be determined by the following relation [16]:

$$S = C_p^{k+1}, p > k,$$

it follows that for multidimensional systems $p > 6$, when restrictions are imposed, the number of arbitrary coefficients will increase at the beginning, reach a maximum,

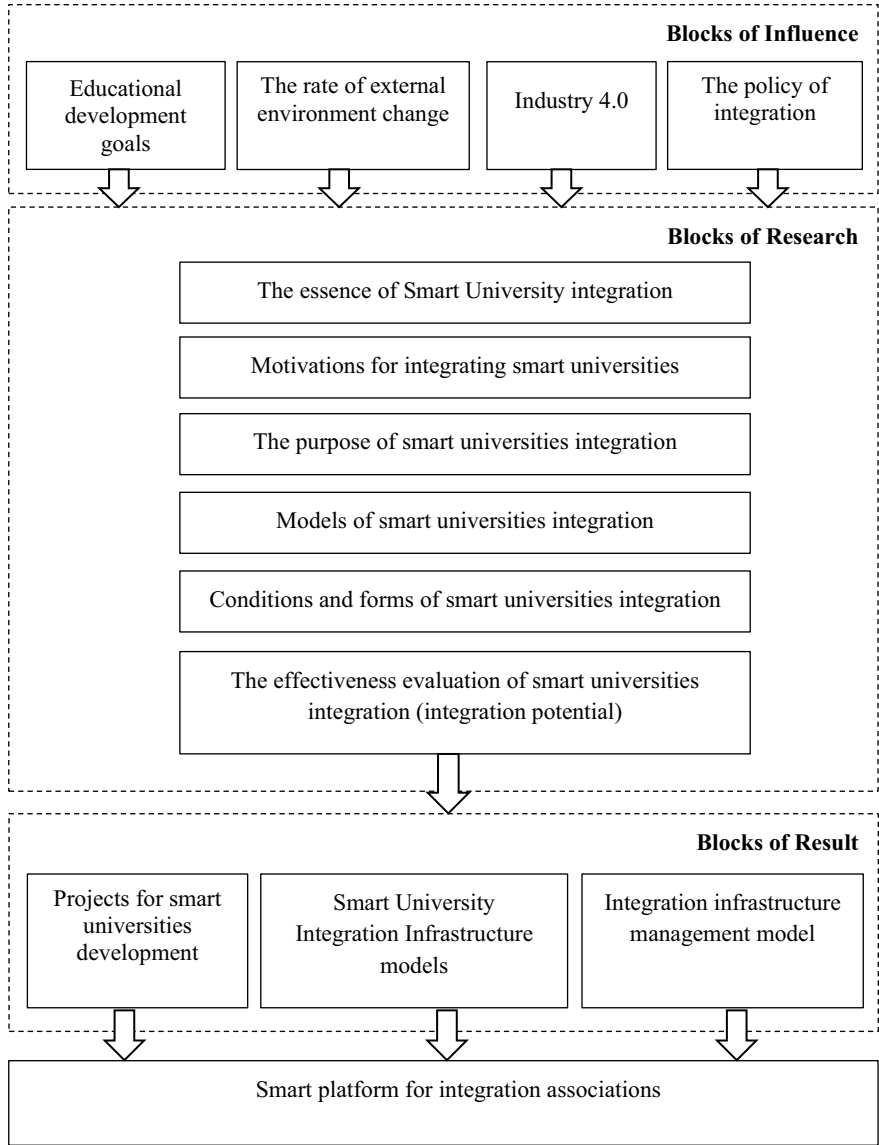


Fig. 1 A model of smart integration platform formation

and start to decrease. This property was called the phenomenon of adaptive maximum [16].

When the system interacts with a changing environment, the presence of arbitrary coefficients in the structure of equivalent equations of the system allows you to

adapt to the flow of changes. The more these coefficients, the higher the adaptive capabilities of the systems.

There are several ways to manage systems in such a way that allows you to keep them in the zone of adaptation maximum in the flow of changes: the imposition of new and removal of old restrictions, integration, etc. In particular, if we have two systems

$$S_1 = C_{p1}^{k1+1}, p1 > k1; S_2 = C_{p2}^{k2+1}, p2 > k2, \quad (1)$$

then, after integrating the systems by imposing restrictions we will have

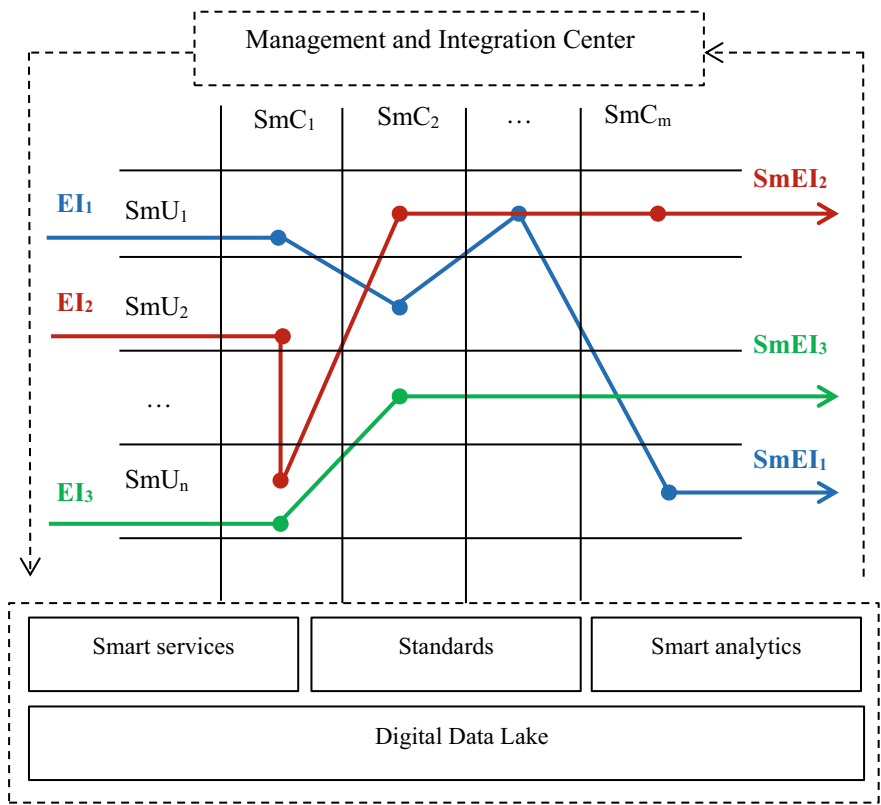
$$S_{col} = C_{p1+p2}^{k1+k2+k}. \quad (2)$$

If $S_{col} > S_1 + S_2$, then the integration gives an increment of adaptive capabilities, and if: $S_{col} < S_1 + S_2$, then integration is impractical.

4 Model of Smart Universities Integration Union

The smart universities integration will allow not only to share best practices, but also to provide access to certain smart components [17] on a commercial or free basis. If we consider smart universities, we can say that they have a huge digital potential, experience in creating and developing a Smart system. This one can be used by any other educational system. Figure 2 shows the author's model of smart universities integration interaction.

The analysis of the presented author's model shows that creating an integrated structure, it is possible to base on the existing experience of developing research centers or scientific and educational centers which accumulate the innovative potential of educational and production systems in the region [18, 19]. The model is designed as a matrix structure. It accumulates smart components from the leading smart universities in the region. A request to increase smartness through the integration platform can come from any educational institution. At the entrance it is a classic educational institution. At the exit it is a smart educational institution with a set of smart components from one or more leading smart universities. The conditions for obtaining a smart component can be different (on a commercial basis, development, connection to a smart service, and other options). The system regulation is carried out in the control center and integration with a smart services use. The integration model will allow us to combine a huge array of data (digital data lake). With the use of smart analytics tools, it will be possible to provide analytical services on the request from educational institutions and not only. The examples of improving smartness by implementing the benefits of an integration infrastructure are given in the next section.



EI – Educational Institution (school, college, university or other educational system)
SmEI – Smart Educational Institution
SmU – Smart University
SmC – Components of SmU

Fig. 2 A model of smart universities integration interaction

5 The Experience of Development the Education Information Infrastructure in the Region on the Integration Unions’ Basis

We will consider the experience of infrastructure integration and creation with the ability to connect to smart components on the example of Russian educational institutions. Togliatti State University (TSU) has entered the stage of digital transformation and has a set of smart components [7, 20]. Togliatti State University has been developing a smart ecosystem for providing higher education “Rosdistant” since 2015.

In 2019, within the framework of the Project Olympus competition, the Rosdistant project became the winner in the category “Project Management in the system of higher education and Science”. The university shares its experience of digital transformation with other Russian universities, providing an opportunity for integration. One of the tools of the smart universities integration association in Russia was the “Charter on Digitalization of the educational space”. In 2019 this document united 29 Russian universities. The document contains a set of rules that encourage integration into a single digital space, including [21]:

- standards compliance (data formats unification generated by universities in the course of their activities, integration protocols unification for developing management and support services for various types of university activities);
- aiming for networking cooperation and spreading the best online courses, practices in the digital services use and proven solutions for building the universities digital architecture.

Universities united by the Charter should provide the necessary prerequisites for successful integration and effective interaction of Russian educational institutions in the interests of developing the region’s educational smart infrastructure [15, 22].

The universities collaboration in the framework of digital transformation is placed on a special marketplace (<https://eduservices.2035.university/#login>). It looks like a platform solution providing formalized rules for the placement and description of digital ecosystem services based on a single markup model.

We will also share the information about experience of providing smart services and smart components of TSU to schools in 2020.

The launch of the Digital Collaboration project enabled access to SmU smart services not only for universities, but also for schools and colleges. The program is valid till 2024 and is aimed to improve all education system levels, which should provide the digital economy with professional staff.

Students and teachers were given the opportunity to develop their digital skills. School teachers got access to the smart technology of the flipped classroom and other smart tools. Teachers’ training on the basis of the integrated infrastructure takes place with the use of simulators and exercises. Within the framework of the Digital Collaboration project (Samara Region, Russia) schools bought 3D printers, special plastic and other equipment for practicing robotics, 3d modelling and prototyping. By 2022, students will be connected to special laboratory workshops and simulators with augmented reality. On the basis of smart equipment, a digital model of teaching will be created in schools in some subjects. Such way will allow spreading the experience not only in the Samara region, but also throughout Russia.

As part of the creation of integration associations based on smart universities and because of all students’ transition to online learning after the outbreak of the COVID-19 pandemic, TSU provided free access to some educational smart components (free.rosdistant.ru). Within the framework of engineering programs, access to virtual laboratories is provided.

In the future, in order for smart universities to guarantee meaningful engineering education, integration is needed within the framework of providing virtual laboratory work. This area is very expensive and lots of schools, colleges and universities can't afford such opportunity to work out this direction. So, it is very important to have access to smart components (virtual laboratories) on a certain integration infrastructure of leading smart universities.

6 Conclusion and Next Steps

The situation with the pandemic made it possible to objectively compare the stability of higher education systems in different countries and test universities for strength, despite their status. Shock was the first reaction of most educational institutions to the need for a quick transition to online. Then the adoption of operational decisions followed. At the same time, the response of the education system as a whole was quite effective. But then it has become obvious that some solutions are unworkable in the long run. In these conditions, the creation of smart university integrations to increase the smartness of educational institutions is a very effective tool taking into consideration the high speed of change.

Conclusions:

1. The developed model of integration interaction of smart universities can serve as an effective tool for transferring the best practices of leading smart universities.
2. The model provides the ability to assemble the necessary smart components by educational institutions of various levels (school, college, university, etc.) on request. Thus, the output is a smart system.

The next steps:

1. The further development of the model of integration interaction of smart universities involves the development of smart platform architecture for the integration system of smart universities.
2. Detailing and development of smart services of the integration platform.

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Controlling as a Tool to Reduce the Risks of Smart University in the Digital Economy



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Abstract Modern conditions are characterized by the uncertainty and volatility of the economic environment. This situation is associated with increased risks, accompanied by uncertainty and uncertainty in achieving the expected results. In this regard, both prompt and preventive measures are needed to reduce the likelihood of the occurrence of risks or localize the negative consequences of their occurrence. The activities of a smart university are subject to the influence of the external environment, which causes certain threats and risks. In addition, one cannot but take into account the influence of internal risks on the results of its functioning. To solve such problems in a smart-organization, it is necessary to strengthen the controlling system, which involves targeted actions to reduce risks. The article analyzes the existing risks of a smart university, clarifies the principles and stages of building controlling in a smart university in a digital economy. The result of the study is the proposed controlling algorithm to reduce risks in a smart university in the context of digital transformation. The obtained results were tested with risk reduction in the Smart-division of the university using the example of the department.

Keywords Smart university · Controlling · Risks · Types of risks · Reducing risks · Risk assessment · Digital economy

1 Introduction

In modern conditions, the activities of each organization are influenced by various factors that can cause threats and risks to its activities. Its successful development will depend on how quickly such factors are identified and how effective the selected risk minimization methods are. The activities of smart universities, described in detail in the work of the authors: Uskov et al. [1] are also subject to the negative impact of external and internal risks. It should be noted that the works of authors such as N.

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Serdyukova, V. I. Serdyukov, V. A. Slepov, V. L. Uskov, V. V. Ilyin, L. V. Glukhova, S. D. Syrotyuk, A. A. Sherstobitova, S. V. Pavlova are devoted to the study of the functioning of the smart university [2–5]. Some issues of strategic development of the smart university, including the influence of the external environment on it, are considered in the work of the authors Berdnikova et al. [6], Tikhomirov and Dneprovskaya [7].

Modern digital transformation processes strongly influence the activities of business units, including smart universities. The risks affecting the functioning of a smart university may be different from the risks affecting other commercial organizations. This is due to the peculiarities of the activities of smart universities, as well as to the conditions of the widespread spread of digital transformation, which requires the improvement of risk management tools.

The unstable conditions of the external environment determine the need to use all management tools to reduce the impact of risks on smart universities. At the same time, a special role is assigned to controlling, which makes it possible to increase the efficiency of management and achieve target indicators when implementing the strategy. In general, the controlling system can be represented as a system of information and analytical support for the management decision-making process in a smart university, including for reducing risks.

The current conditions of the digital economy are transforming the education system, thereby requiring the improvement of diagnostic tools and reducing the risks associated with the activities of a smart university. In such an environment, controlling plays a key role, which, on the one hand, allows control over business processes, and, on the other hand, is an important information source for the application of preventive and operational measures to reduce risks.

Thus, the main goal of this article is to consider controlling as the main tool for reducing the risks of a smart university in the digital economy.

2 Statement of the Problem in General Form and Its Connection with Important Scientific and Practical Tasks

The processes of globalization, digital transformation, mass transition to distance learning technologies, as well as the acceleration of the pace of socio-economic changes have caused a new wave of research in the field of university management and reducing the risks of scientific and educational projects.

During the period of digitalization of the economy, new types of risks appear in the activities of a smart university that affect its educational environment, research, financial and economic, organizational and managerial and other areas of activity.

In Table 1, we will conduct a comparative analysis of the main development risks of Russian universities and Harvard university.

Based on the study, we will single out in Table 2 the risks of developing a smart university within the framework of the main processes of activity.

Table 1 Comparative analysis of the main development risks of Russian Universities and Harvard University

Russian Universities	Harvard University [8]
Planning risks	Academic risks
Risks of changing legal requirements for universities	Compliance risks
Financial risks	Financial risks
Risks in the implementation of educational programs	Operational risks
Risk of reduced research activities	Reputational risks
Risks of reducing the innovative activity of the university	Strategic risks
Strategic risks	

Table 2 Risks of the development of a smart university in the framework of the main processes of activity

Process name at smart university	Risk name
Business planning and marketing	– the risk of changes in applicable legal requirements for higher education institutions – the risk of not achieving the university's quality goals – the risk of failure to meet the student recruitment plan – the risk of failure to meet the main planned indicators of the university – the risk of outflow of students – the risk of reduction in the number of foreign students – the risk of competition from other universities – the risk of non-compliance of the offered educational services with market requirements – the risk of high prices for educational services
Design and development and implementation of educational programs	– the risk of a delay in the development of educational programs – the risk of non-demand for the educational program – the risk of lack of innovative teaching methods – the risk of non-use of modern educational technologies – the risk of insufficient intermediate control of students' knowledge level
Research activities	– the risk of insufficient publication activity of teachers in journals indexed in the Web of Science, Scopus databases – the risk of insufficient citation of university employees in scientometric databases – the risk of insufficient influx of young researchers – the risk of failure to achieve the desired results of scientific activity in the implementation of a scientific project – the risk of non-fulfillment of contractual obligations under government contracts – the risk of low grant activity of employees and students

(continued)

Table 2 (continued)

Process name at smart university	Risk name
Innovative activities	– the risk of reducing the innovative activity of the university – the risk of low demand for innovation in the region and country – the risk of lack of commercialization of scientific research results
Personnel management	– the risk of a shortage of teaching staff of appropriate qualifications – the risk of staff turnover – the risk of non-fulfillment of the plan for advanced training of university staff – the risk of a shortage of teachers with academic degrees and academic titles – the risk of teachers' lack of distance learning skills
Management of the information environment of the university	– the risk of unauthorized access to personal data – the risk of hacking educational portals – the risk of targeted hacker attacks aimed at disrupting the operation of information systems – the risk of lack of digital platforms to support the educational process
University finance and infrastructure management	– the risk of financial and economic activities – the risk of insufficient material and technical base – the risk of non-fulfillment of contractual obligations by counterparties – the risk of lack of funding – the risk of non-use of budget funds in full for the development of the material and technical base

Recently, in addition to theoretical studies, empirical studies of risks in the activities of universities have been widely developed. The growing relevance of risk assessment and management in higher education institutions is associated with a reduction in budget funding, the transfer of universities to self-sufficiency, increased competition, and the conditions of the digital economy. In order for universities to develop effectively, new approaches and tools for identifying and reducing risks are needed, the development of new algorithms for the introduction of a controlling system that contributes to the prompt adoption of management decisions.

3 Presentation of the Main Research Material with Full Justification of the Obtained Scientific Results

3.1 The Principles of Building Controlling in a Smart University in Order to Reduce Risks

Controlling is one of the most effective management tools, including those allowing to reduce various types of risks. By its content, controlling is not limited to only control. The modern concept of system management of a smart university implies the need to ensure its successful development on a long-term basis by:

- compliance of strategic goals with the demands of the external environment, taking into account the requirements of the digital economy;
- coordination of current plans with the strategic direction of development;
- coordination and integration of current plans in different areas of the smart university;
- formation of a system for providing managers with the necessary information about possible risks and threats;
- adaptation of the management system of a smart university in order to increase the responsiveness to changes in the external and internal environment.

Controlling provides an informational and methodological basis for supporting key management functions: planning, accounting, analysis, control, as well as making effective management decisions. The functional components of controlling in the activities of a smart university are:

- focus on high-quality and effective work of a smart university in the current and future periods;
- the formation of an organizational structure, which is aimed at the implementation of strategic goals, taking into account digital transformation;
- informational and methodological support of the management system to reduce risks;
- dividing the key tasks of controlling into cycles in order to ensure interactivity of planning, control of execution and development of corrective decisions.

The study showed that in science and practice, insufficient attention is paid to the development of a controlling system at a smart university in order to reduce risks, taking into account the conditions of the digital economy. For the development of this area, additional research and improvement of algorithms for the implementation of controlling are needed, including preventive and operational measures aimed at reducing risks in the activities of a smart university.

We believe that controlling as a risk mitigation tool is a system that coordinates the relationship between the formation of an information base, risk analysis, risk control, which provides preventive and operational measures in priority areas of risk management in a smart university in a digital economy.

In our opinion, the construction of a risk controlling system in a smart university should be based on the basic principles presented in Fig. 1.

Let's reveal the content of each of the above principles of building risk controlling in a smart university.

The principle of focusing controlling on achieving the strategy of a smart university implies that all actions in the implementation of controlling must correspond to strategic goals.

The principle of multifunctionality of controlling involves ensuring control over risks not only for the smart university as a whole, but also for the priority areas of its activities, in the context of its individual centers of responsibility.

The principle of standardization of control procedures states that the effectiveness of controlling actions will increase if its basic procedures are standardized.

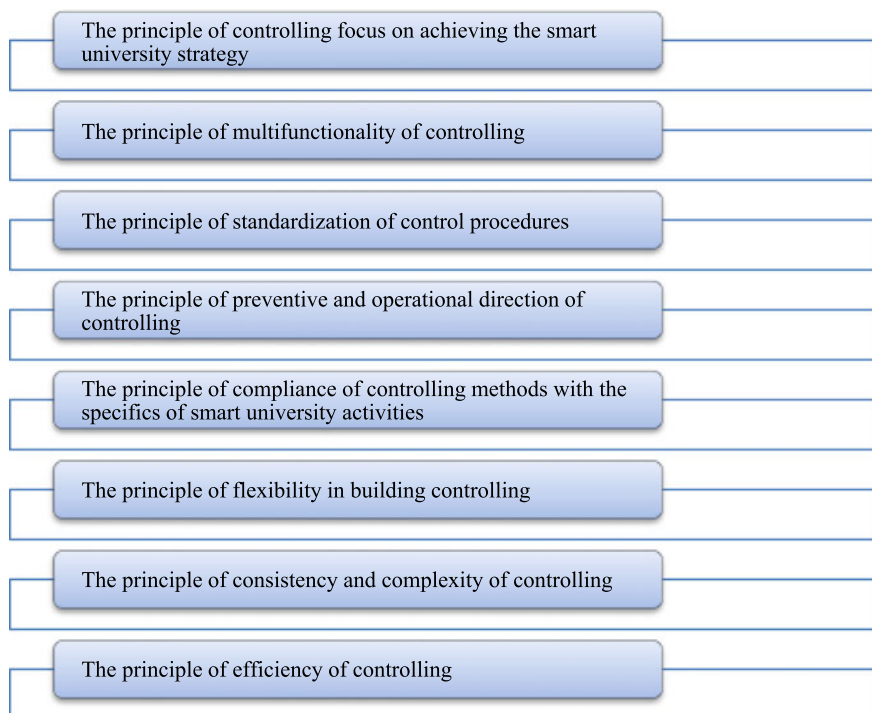


Fig. 1 Principles of building risk controlling in a smart university

The principle of the preventive and operational focus of controlling provides that all actions and measures should be both operational and preventive in nature, working to proactively eliminate risks.

The principle of conformity of controlling methods to the specifics of the activities of a smart university involves the use of the entire arsenal of techniques and methods of control that are adequate for use in the activities of a smart university.

The principle of flexibility in building controlling states that the forms and methods of control should be adaptive to the conditions of the digital economy and changes in the internal environment of a smart university.

The principle of consistency and complexity of controlling assumes that control procedures should cover all key areas of the smart university's activities that are exposed to risks. In addition, control should be based on a system of indicators that allows you to track the dynamics of actual indicators from planned, from indicators of previous years.

The principle of efficiency of controlling indicates that the cost of carrying out control procedures should be less than the effect that is achieved in the process of their implementation.

3.2 Stages of Building Risk Controlling in a Smart University

Based on the study, we recommend the stages of building risk controlling in a smart university, which are presented in Fig. 2.

The first stage involves the establishment of a controlling object from the perspective of the target orientation of a smart university. The second stage is aimed at defining the types of controlling. According to the concept of building a controlling system, it can be divided into types: strategic, operational and preventive.

The third stage involves the development of standards for control procedures to identify risks. The standards should be focused on the specifics of the functioning of a smart university. The fourth stage is aimed at building a system for monitoring indicators to identify risks. This system is the basis for risk controlling. It is a mechanism developed at a smart university for regular monitoring of key performance indicators exposed to risk factors.

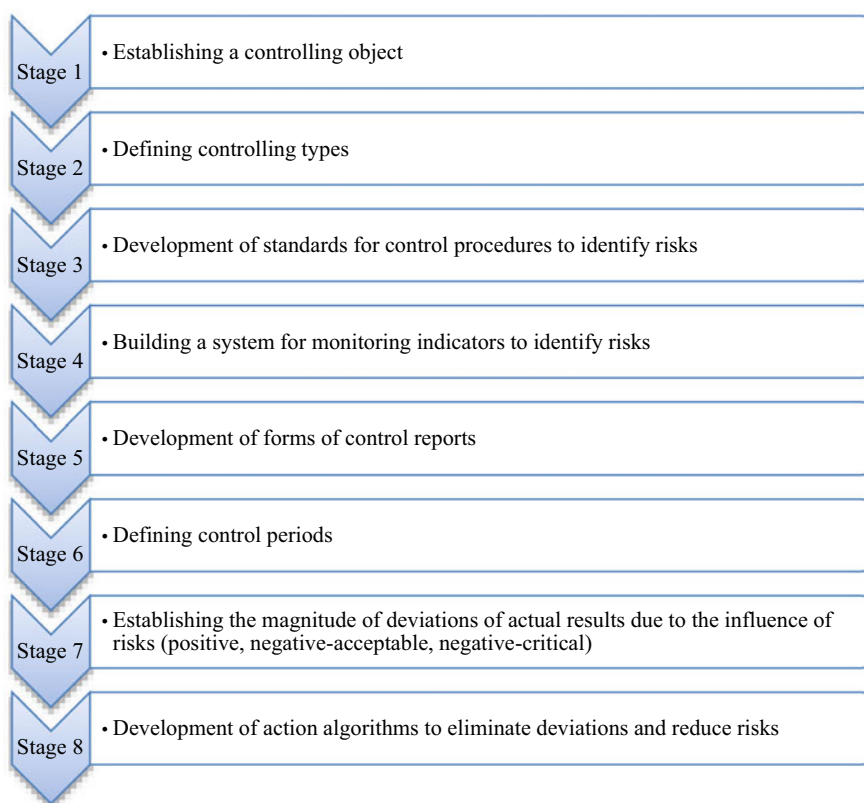


Fig. 2 Recommended stages of building risk controlling in a smart university

The fifth stage involves the development of forms of control reports, which must be formed for the main areas of activity of a smart university that are exposed to risks. The sixth stage involves the definition of control periods for which control reports should be generated. Such periods can be week, month, quarter.

The seventh stage involves establishing the magnitude of deviations in actual results due to the influence of risks. Such influence can be divided into groups: positive, negative-acceptable, negative-critical. It is important to establish the reasons for the resulting deviations. The eighth stage is focused on the development of action algorithms to eliminate deviations and reduce risks. Principled actions can lean towards three algorithms: –do nothing; –eliminate the deviation; –change the system of planned or normative indicators. It should be noted that the introduction of controlling into the activities of a smart university will allow to promptly identify risks with a view to their subsequent reduction or localization.

3.3 Algorithm for Using Controlling to Reduce the Risks of a Smart University in the Digital Economy

In order to develop controlling to reduce risks in the digital economy, it is necessary to improve the mechanism of its application, taking into account the specifics of the work of a smart university. Based on the study, we propose a controlling algorithm to reduce risks in a smart university in a digital economy (Fig. 3).

In a digital economy, the process of identifying and controlling risks should become part of management work, along with the organization and management of educational, scientific, financial and economic activities, etc. In this regard, risk controlling should be considered as a complex of actions for the identification, assessment and neutralization of risks, combined into a system of planning, monitoring, control and correction of actions.

The proposed algorithm contributes to ensuring the flexibility and adaptability of risk controlling in the activities of a smart university through the interaction of each step, forming a management decision-making system. This approach helps to maximize the achievement of risk management and control objectives. This is ensured by the fact that the information obtained at each step makes it possible to adjust the methods of influence on risk, as well as the goals of risk management.

The use of the proposed algorithm will allow a smart university to timely identify risk factors, obtain an assessment of possible damage from risk realization, develop measures to reduce or neutralize risk, and also apply preventive measures that work to proactively eliminate possible risks.

The study showed that the introduction of controlling in the context of the key processes of a smart university will reduce the main internal risks by 65%. Due to the instability of environmental factors, it is rather difficult to predict possible risk macro factors that affect the work of a smart university. However, the introduction of a controlling system will allow regular monitoring of such factors. According

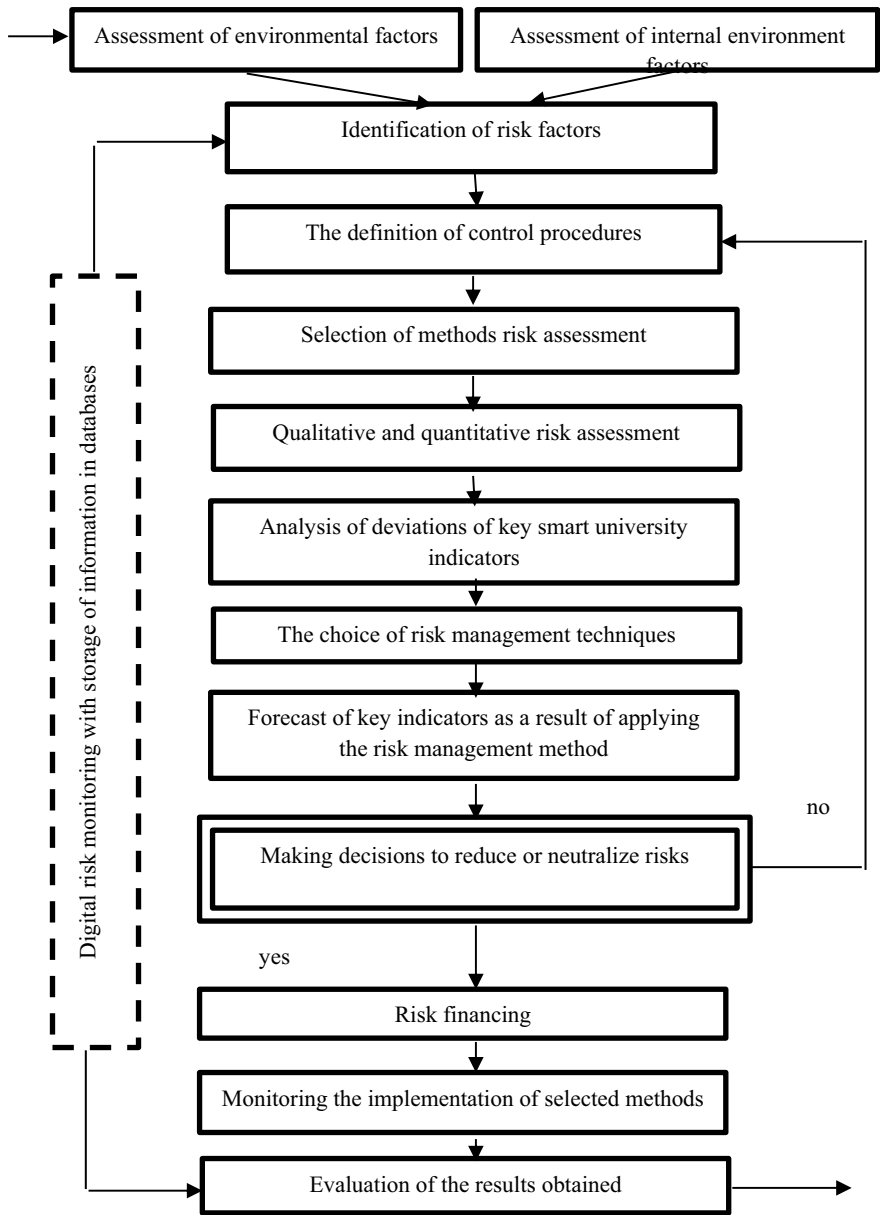


Fig. 3 Controlling algorithm for risk reduction in a smart university in a digital economy

to the study, an effective controlling system will reduce the impact of external risk factors by 35% by applying preventive measures aimed at anticipatory elimination of possible risk.

The proposed algorithm allows you to identify the risk factors of the external environment, controls the risk factors of the internal environment and helps to timely identify unacceptable deviations in order to develop measures to reduce or neutralize risks in the activities of a smart university.

4 Conclusions of the Research and Prospects for Further Research in This Direction

Conclusions. The study showed that the activities of a modern smart university are influenced by various factors that can cause certain risks. In the digital economy, new challenges and threats appear that can cause significant damage to a smart university. This situation necessitates the use of controlling in order to reduce the impact of risks.

1. The article provides a comparative analysis of the main development risks of Russian universities and Harvard university. Based on the study, the risks of developing a smart university in the framework of the main processes of activity are highlighted.
2. On the basis of the study, the main principles of building risk controlling in a smart-university are highlighted. Compliance with these principles will allow timely and full identification of risks and threats.
3. The conducted research made it possible to recommend the stages of building risk controlling in a smart university. These stages take into account the entire cycle of actions aimed at identifying and subsequently reducing risks.
4. The article proposes an algorithm for using controlling to reduce risks in a smart university in a digital economy. The implementation of the proposed algorithm makes it possible to reduce risks in the activities of a smart university.

The proposed principles, stages and an algorithm for using controlling to reduce risks in a smart university can be used both at the level of a separate structural smart unit and in a smart university as a whole.

Next Steps. For the development of the controlling system, it is necessary to develop risk assessment criteria, taking into account the specifics of the activities of a smart university, as well as to improve the methodological tools for managing risks typical for a smart environment.

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Blockchain Methodology for Smart Academic Environment in Russia



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Abstract The cryptocurrency market is only at the beginning of its evolutionary cycle in Russia. Cryptocurrencies are volatile and dependent on many factors. The problem of investing in cryptocurrency exchanges is relevant. The problem of developing and liberalizing the financial system to self-regulatory is important. Society and education need skills of a systems analysis for regulating the cryptocurrency market. In particular analysis of the possibilities of interactions such as P2P (Person-to-Person), P2B (Person-to-Business), G2B (Government-to-Business) are required. The use of blockchain technology in education is also an urgent problem. There are possibilities for liberalizing the educational system especially smart-education. The corresponding system analysis is represented in the study, a model of adaptive cryptocurrency selection and valuation of cryptocurrency reserves are proposed. The task of improving the performance of blockchain transactions by optimizing hash addressing is also considered. The market is seen as an evolving structure and analyzed for its sustainability. It can become the basis for system analytics and research on models of the cryptocurrency market evolution and development for smart education.

Keywords Stability · Crypt currency · Market · Modeling · Blockchain

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1 Introduction

Currently cryptocurrency is considered as a financial decentralized system which is not being controlled but including many pros and cons and the company's or person's financial results mainly depend on the goals or objectives and the chosen way of their application. It is important to know of cryptocurrency transactions and how they affect in the digital society. Therefore the authors consider that the basics of blockchain methodology is considered to be as a must have in the educational process for many tracks. Originally the essence of the blockchain methodology reflects the principle of calculating the value of a cryptographic hash function based on mathematical models, followed by adding a new block to the cryptosystem. According to the above mentioned facts it is necessary to develop relevant skills among university graduates in the tracks of economics, business analysts and IT specialists. To assess the possibility of regulating the cryptocurrency market it is necessary to develop the skills of university graduates in the field of economic profile, business analysts and IT specialists in the system analysis of interaction models P2P, P2B, G2B and many applied problems for the practical application of the blockchain methodology. The paper proposes basic ideas for knowledge base that can be used as the basis for the cryptocurrency market research in the process of training at the higher education.

2 Literature Review and Problem Statement

Sufficient analytical and statistical data [1] about the volatility of financial markets [2], storage of payment accounts, transaction data and their verification on different computers of the distributed peer-to-peer system [3] have already been studied and revealed. However, there are many uncertainties. Even the reference to S. Nakamoto as "Bitcoin's creator" [4] is not objectively confirmed by anyone. Payments, exchanges, donations, trading in cryptocurrency systems [5] are considered more active and profitable in capitalization. Demand, supply, exchange rate issues are considered essential to the popularity of cryptocurrency. Some other studies report that digital coin transactions can lead to conflicts with current national legislation [5, 6]. In January 2019, the Russian State Duma and the Financial Market Committee promised to pass a law on digital assets with 99% probability. The Russian Civil Code has already been amended regarding to the circulation of digital assets and intellectual contracts. The law on crowdfunding came into force.

There are also different points of view among famous financiers and investors: billionaire Warren Buffett, known as the CEO of BerkshireHathaway, believes that Bitcoin has no significant intrinsic value, but Eric Emerson Schmidt, one of the leaders of Google, believes that Bitcoin is a significant cryptographic achievement because blockchain methodology implies security of all transactions in the database, access from each client computer in the payment system [7]. According to [8] by 2025, more than 10% of GDP (\$100 trillion) in the world will be managed in

blockchain systems. The solving and mathematical apparatus for bitcoin is represented [9]. Adaptation of blockchain technology and its potential applications for education is the way to solve complex educational problems [10]. The super-computing and power capacity are considered as key requirements for blockchain technology [11].

It is logical to assume that the size of Russian bitcoins market is incomparably small in comparison with the world. Blockchain methodology assumes safety of all transactions in the database, access from each client computer in the payment system. The authors believe that a modern intelligent university should teach methods of expert heuristic evaluation, statistical and system analysis, synthesis and modeling of opportunities and prospects of secure cryptocurrency 2.0, Blockchain 2.0 evolution.

The research problem is as follows: to transfer key modern trends and indicators, existing knowledge bases and educational tools into the process of preparing meaningful competencies for successful university graduates in the digital economy and digital society.

The main objectives of the study are the following:

1. a brief review of the existing approaches of the blockchain methodology and tools for its implementation;
2. experimental research including analytical review of the needs for digital currency transaction skills among students and their parents, the ranking of responses and their statistical processing, the possibilities of mathematical and algorithmic simulation of individual fragments;
3. representing the example of using the hash function in the learning process.

3 The Methodology

3.1 *System Analytical Component: Review*

Further on all the statements and methods to the Bitcoin system as the most developed and capitalized (over \$126 billion) are to be considered [7]. According to the researchers global investments in Bitcoin—up to \$12.4 billion by 202 are predicted. Thus a lot of modern smart companies are trying to apply blockchain technologies introducing the ability to adapt existing business processes including supply, logistics, payments, staff training, etc. In other words nowadays smart organizations are looking for relevant blockchain models. According to the Constellation Research Data over 67% of U.S. companies have already started to assess the feasibility of projects in the chain of blocks, while 25% have been already implementing them for several years. Among them 68% annually spend more than \$1 million for research.

The latest World Economic Forum reports that more than 40 Central Banks in the world are studying the opportunities for blockchain applications; and in the public sector of 45 countries there are more than 200 large initiative networks connected

with the studied issue. Despite the slowdown in the development of blockchain applications, 45% of blockchain initiatives are funded as innovative research supported by the country's budget. Blockchain is also considered as being a very effective one in the field of accounting, especially when it deals with cross-border transactions and smart contracts due to their possibility of avoiding legal rejections.

3.2 The Tools of Blockchain Methodology

Companies are accustomed to deal with operational issues, and blockchain tools are developing rapidly [7]. In relation to the increasing complexity of business processes (see Table 1, Figs. 1 and 2, which shows data one year after the maximum bitcoin prices were reached in the period from February 2018 to January 2020).

The geographical spread of the popularity for crypto currencies can be estimated on the basis of Exchanges Data. According to the latest CoinMarketCap data [7], there are 316 exchanges operating in the world today trading crypto currencies with different dynamics.

According to financial data nowadays there are more than 1800 crypto currencies on the markets, but many of them are considered to be shit coins because they are not traded and have no liquidity. According to the modern scientific studies Blockchain Methodology allows simultaneous access to a constantly updated digital ledger, which cannot be changed. Although banks were initially skeptical about the blockchain methodology the firms having implemented this technology increased venture capital during the first half of 2019, including the New York Stock Exchange. According to KPMGFintech blockchain venture capital investment almost doubled

Table 1 Comparative analysis of the TOP-10 cryptocurrencies by value (\$) and capitalization (data from [7])

Currency	Value		Capitalization		Increase (%) 01.2019–01.2020
	Start	Current	Start	Current	
Bitcoin	0.06	9183.14	1,000,000	109,936,576,318	+139
Ethereum	0.3	236.87	18,000,000	20,268,570,989	+77
Ripple	0.058	0.240198	45,921,000	17,913,270,592	– 32
BTC	0.008	320.87	–	7,351,538,699	+95
EOS	1	3.88	200,000,000	5,239,561,878	+34
Stellar	0.00008	0.061674	770,000	4,229,635,169	– 48
Litecoin	4.3	66.28	40,000,000	2,905,337,902	+98
Cardano	0.216	0.050241	63,000,000	2,175,079,879	+25
Tether	1.21	1.00	200,000,000	2,051,991,823	– 1
Monero	2.41	71.32	5,000,000	1,692,670,634	+48

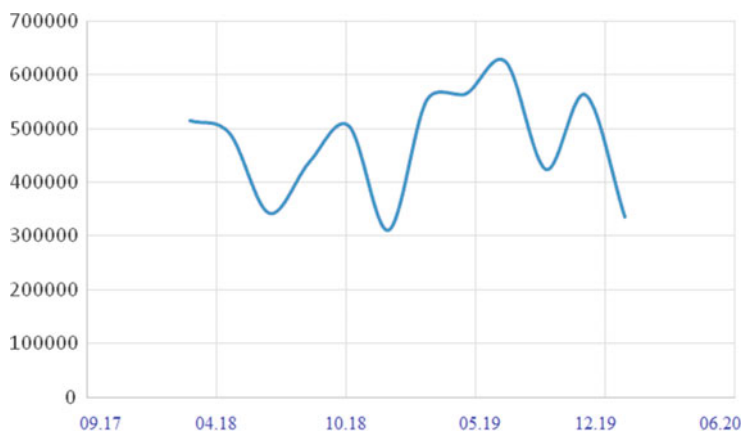


Fig. 1 Number of bitcoin storage location/address [7]

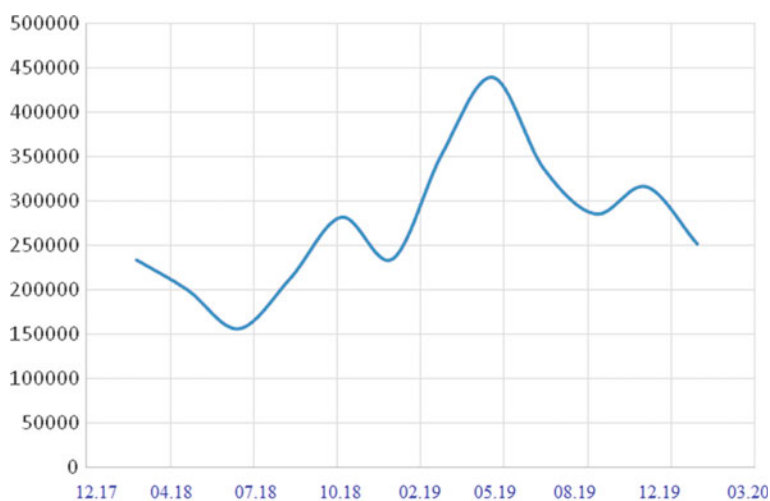


Fig. 2 Daily bitcoin-transactions [7]

and achieved more than \$390 million. Companies are looking for “rendezvous points” or common ground with blockchain technologies and methodology.

3.3 The Key of the Blockchain Methodology Potential

All the studies reveal that the market for crypto currencies is quite volatile, dependent on many factors including just rumors in society or mass media; but it is considered

as being attractive for speculators and users who do not trust currencies and banks, especially in crisis situations. Those willing to test the potential of a blockchain are a significant number. But the state and institutional support of the rate is not expected by players and miners, as in transactions with national currency.

There are a lot of companies that are testing the possibilities of blockchain methodology, but the authors consider that the effective testing should be done according the following statements:

Statement 1: The instable, crypto market trends should be investigated on the base of nontraditional relevant tools, such as quantum informatics, network laws, fuzzy sets, fractal structures, and others.

Statement 2: Purchase and sale of cryptocurrency is a multi-criteria task and the decision is made in conditions of uncertainty.

Advantages of fuzzy logic decision-making are in the smart process of visualization, updating the monitored feedback both for the user and the group, the possibility of analyzing neuro-marketing data including datamining and forecast of consumer preferences, expectations of purchase and sale of cryptographic currency.

4 Our Experiment and Results

4.1 The Statistical Experiment

According to the surveys conducted by the Center of Modern Research and the National Agency of Financial Research of the Russian Federation the popularity of crypto currency among Russians is considered as low, which is confirmed by the following data: those who know well about bitcoin system—4%; those who have heard about it—16%. The results of anonymous survey is based on the Likert type fact-finding questionnaire conducted among students of Kabardino Balkarian State University and Togliatti State University who study at “Applied mathematics and informatics” and “Economics” departments and their parents (Table 2).

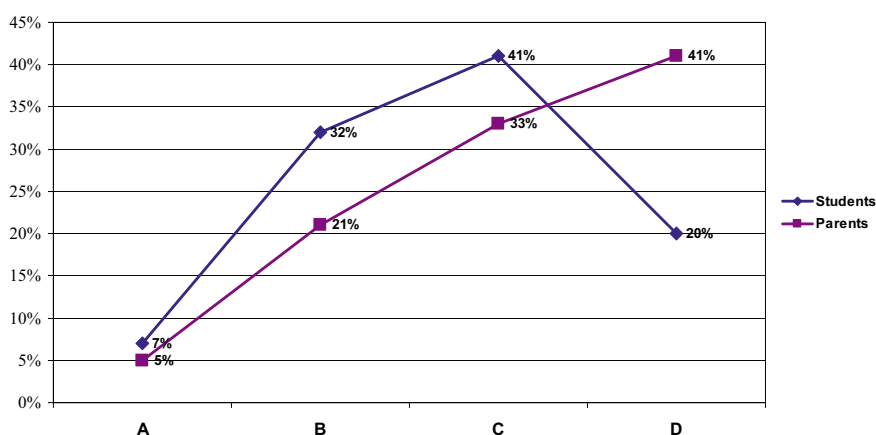
In total, first-year students were interviewed: 52 (Applied Mathematics and Computer Science), 153 (Economics), 119 (Mathematics), 94 (Jurisprudence), 34 (Management). Their parents also participated in the survey—more than 350 people, of which only 40 people were associated with digital technologies. Gender and age differences were not taken into account. For the “purity of the experiment”, we were only interested in the opinions of “ordinary” people, without special education. Therefore, teachers of specialized, financial disciplines were not interviewed.

According to the questionnaire with the suggested alternatives A (“sure”), B (“more sure than sure”), C (“more not sure than sure”), D (“not sure”) showed the following results represented in Table 2 and Fig. 3.

The questionnaire questions concerned cryptocurrency and digital money literacy. These are the patterns for questions used in the survey: «Do you want to get paid for

Table 2 The survey results

	A	B	C	D
Students	7%	32%	41%	20%
Dispersion/variance	0.00388	0.01490	0.01522	0.00786
Deviation	0.01940	0.07448	0.07612	0.03928
Regression	0.03400	−0.02500	−0.04800	0.03900
Excess	−0.93336	−0.94801	−2.15519	−1.25391
Parents	5%	21%	33%	41%
Dispersion/variance	0.00064	0.00102	0.00586	0.01226
Deviation	0.00320	0.00512	0.02928	0.06128
Regression	0.01400	0.00800	−0.01800	−0.00400
Excess	−1.75000	−2.32422	−2.90334	−1.63905

**Fig. 3** The results of statistical processing of the survey. A (“sure”), B (“more sure than sure”), C (“more not sure than sure”), D (“not sure”)

a cryptocurrency wallet?» or «Do you think that paper money is already “outdated” and slows down economic development?». The algorithms of transactions, mining and blockchain are demanding, formalized, self-complicated. You have to look for a balance, holding the “bears”, who threatening to create the basis for new bubbles. Mathematicians come up with various algorithms and hash functions.

4.2 Mathematical Simulation

Nowadays a lot of studies connected with crypto market represent the idea that relevant models describing the current situation at the Crypto Stock Exchange are difficult

for designing. Having analyzed the data represented at Crypto Stock Exchange the authors start simulation with the following logistic model (1):

$$y(x) = \frac{\alpha}{\alpha + e^{-\beta x}}, \quad (1)$$

where x —explanatory variable, α , β —coded features.

Statement 3. The maximum likelihood method can be used for simulation, where p —possible BTC increase but q —possible BTC decrease $y(p = y(x), q = 1 - y(x))$, i -viewing the BTC rate at Stock Exchange (2):

$$(y(x_i))^{p_i} (1 - y(x_i))^{q_i} \quad (2)$$

After taking the logarithm the likelihood function during independent viewing can be represented as follows (3):

$$L(\alpha, \beta | x) = \sum_{i=1}^n (p_i \ln(y(x_i)) + (1 - p_i) \ln(1 - y(x_i))) \quad (3)$$

The set of equations is represented as (4)

$$L_\alpha = 0, L_\beta = 0 \quad (4)$$

And can be solved according to solutions represented at [8], where α , β are to be defined.

Implication. For $x = 1$ we get the growth probability $e^{\alpha+\beta}$, for $x = 0$ we get e^α . The growth is defined by β .

Statement 4. If mining is represented according to Formula (5):

$$\mu_0 = \alpha_0 t + \beta_0 + \gamma_0 \sin v_0 t, \quad (5)$$

$\alpha_0 t + \beta_0$ —unidimensional, but a $\gamma_0 \sin v_0 t$ —cyclic («changes»), then prices for BTC at current time can be defined as the follows (6):

$$p_0(t) = p_0(0) \left(\frac{\alpha_0 t^2}{2} + \beta_0 t + \frac{\gamma_0}{v_0} (1 - \cos v_0 t) \right)^{-1}. \quad (6)$$

Empirical corroboration is to be performed by the following stages:

1. the balance ratio in the domestic and foreign Crypto markets are to be defined [9];
2. the Mining regression ratio (model) is to be set;
3. the balance ratio in differential form is to be formalized;
4. the obtained differential equation is to be solved (based on Cauchy problem).

Adaptation of blockchain technology and its potential applications [10, 11] for education is the way to solve complex educational problems.

4.3 Software Algorithmic Component of the Result

The supercomputing and power capacity are considered as key requirements for blockchain technology. The main principle is to calculate a hash function using crypto algorithms of SHA-256 class (calculating bit string crypto code by hash function). But the most attention is to be paid to the speed in block-transactions, integrity of blocks and data.

Statement 5: Since hashing is achieved by performing simple, basic operations over a chain of characters, you can calculate the sum of the first and last character in the string. This approach is implemented into the software developed for the targeted tasks [11]. The fragment is represented below:

```
public class TAIF1 {
    /* @param args the command line arguments */
    public static void main(String[] args) {
        ArrayList<String> lines = new ArrayList<String>();
        lines.add("str1");
        ....
        private static int HashFunc(String input) {
            int l = input.length();
            if (l == 0) { return 0; }
            if (l == 1) { return ((int) input.charAt(1)) - 64; }
            else if (l == 2) { return ((int) input.charAt(1)) + ((int) input.charAt(l - 1)) - 64; }
            else { return ((int) input.charAt(1)) + ((int) input.charAt(2)) + ((int) input.charAt(l - 1)) - 64; }
        }
    }
}
```

Its key peculiarity is possibility to get rid of probable collisions, conflict of interests, emerging during the computational process or coincidence of hash values at different data blocks. Although there are algorithms of consensus including Proof-of-Work, Proof-of-Stake, Proof-of-Capacity, Proof-of-Elapsed-Time, Proof-of-Activity, Proof-of-Importance the optimal among them are hardly to choose. Anyway at blockchain situation a suitable consensus for safety and speed requirements is chosen. The fragment of designed program representing a hashing algorithm is represented below:


```

private static HashMap<Integer, String> simpleRehash(ArrayList<String> lines) {
    HashMap<Integer, String> table = new HashMap<Integer, String>();
    for (int i = 0; i < lines.size(); i++) {
        int hash = HashFunc(lines.get(i));
        int d = 1;
        //if the cell is busy
        while (table.containsKey(hash)) {
            hash = (hash + d) % Integer.MAX_VALUE;
            d++;
        }
        table.put(hash, lines.get(i));
    }
    return table;
}

```

The hash function is calculated for the data input, then they are added to the table, and the check and assessment are performed: if the hash function value is represented in the table, then there is rehashing. To avoid collisions, some value is added to the hash value (counter D). The loop is to be repeated until the table with unique identifiers is completed.

5 Discussion

A smart contract in the blockchain chain will become a situational imitator of a real contract—from funding a student’s training by the company to his career development in the company. Both parties of the blockchain contract will be able to control their rights and obligations, tracking, cryptographic codes or blockchain blocks. The authors believe this will reduce government spending, attract business to the education area including business education or “credit training”.

As soon as teachers and students can fully master their skills in smart contract many of the training questions will be eliminated.

6 Conclusion and Future Trends

Conclusions

1. The blockchain is the newest technology, the interest to which has grown recently. According to experts and studies, the technology can significantly reduce the costs of business and government. Nowadays it is widely discussed and studied both by the finance specialists and managers from different spheres. Originally created for operations with bitcoins at digital markets it can be used in other areas. Nowadays in the Russian Federation, technology is mostly used by banks, government agencies and innovative enterprises. But there are a lot studies suggesting to use the considered methodology in other directions for smart education and society.

2. The blockchain methodology has the potential to be implemented into smart education and change criteria for educational processes quality both for on-line and off line. The analysis represented in the study including “zero approximation” to the modeling of crypt-currency transactions, educational procedures, course forecasting can become the basis for the development of system analysis and simulation of complex, relevant mathematical and financial-informational models for the development of smart environment in the RF.

Next Steps and Future Trends

1. The blockchain methodology can bring additional benefits to society, its educational system. Cryptocurrencies are activated without special resource investments. The “university” cryptocurrency can even be released.
2. The use of blockchain in training affects educational relationships in the smart system. Any registered student exercises his right to study “at a given place, at a given time, through this channel, excluding falsification of results”.
3. Blockchain, smart contracts will allow educators to get parallel access to the updated digital “magazine” or journal and academic records and students credit books.

Thus opportunity analytics based on blockchain and self-tuning of the trainee’s profile are necessary for liberalization of educational systems.

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Use of Innovation and Emerging Technologies to Address Covid-19-Like Pandemics Challenges in Education Systems



Abdellah Chehri, Tatiana N. Popova, Natalia V. Vinogradova,
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Abstract In this crisis of the COVID19 pandemic, for which the public and private educational system was not prepared, the measures of confinement and social distancing forced us all to experiment with various distance education strategies. The experiences have been diverse, between the absurd and creative. The digital transformation of our societies has been underway for a long time, and it has accelerated in recent years with the advance of many emerging technologies. These emerging technologies to address covid-19-like pandemics challenges in education systems been-or is being-implemented in several countries. This paper introduces the digital transformation of education and the significant trends in technologies applied to education. We track current the education systems' response to COVID-19 across the world and in Canada and Russia, particularly.

Keywords Education systems · Emerging technologies · Pandemics · Online teaching · e-learning

1 Introduction

In early 2020, the COVID-19 (caused by the SARS-CoV-2 virus) pandemic shocked the world, almost bringing it to an unprecedented stop. Until a few months ago, nobody could imagine that the schools and offices in many countries. Until this date, the world is still facing an unprecedented pandemic that will mark a new way of life

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[1]. As far as health is concerned, the novel coronavirus SARS-CoV-2 has affected all age groups, including young children and students [2].

Most governments worldwide have temporarily closed educational institutions to contain the spread of the COVID-19 pandemic. Several other countries have implemented localized closures affecting millions of additional learners.

According to the Nations Educational, Scientific, and Cultural Organization (UNESCO), the world has already exceeded 1.500 million students without attending classes due to the COVID-19 pandemic [3]. These figures are equivalent to 91% of the total number of students enrolled in 188 countries.

In these circumstances, education systems worldwide are trying to offer digital alternatives to continue training remotely. Technology is one of the most critical tools to support remote learning when learners need to remain outside of classrooms. Although online education is not uncharted terrain, the new pandemic situations are necessary to promote its consolidation. By achieving this, not only will the current situation be faced, but it will also become an indispensable element in the education of the future.

Information and communication technologies (ICT) play an increasingly central role in societies' evolution and significantly affect lifestyles. On the educational level, these changes are manifested by the emergence of new characteristics in learners. Academic success, which was measured mainly in cognitive terms, now seems to be more and more determined by young people's techno-cognitive skills.

Several ICT tools and software have been introduced to address these challenges, such as Zoom, Microsoft Teams, Moodle, Blackboard, Kahoot, or Google Classroom. The popularity of these platforms has increased in recent months. Zoom reports that more than 10,000 academic institutions worldwide have opted for its solution, and Google Classroom has already exceeded 50 million downloads. But these tools alone do not solve our problem. Online education implies a set of non-technological challenges to overcome. In the first place, this great offer suggests evaluating which tools are the most appropriate to teach our classes.

This paper aims to highlight the main results of a global survey on the impacts of the COVID-19 pandemic on education systems. This article has compiled the most important challenges that professionals face due to the digital transformation in education and work. Besides, we offer a panorama of future evolution in both areas so that you can prepare today for a post-pandemic world. We give an overview of online and distance training over the past, the current situation (present), and post-Covid-19 (future of online education). Furthermore, we track current education systems' response to COVID-19 across the world and Canada and Russia.

The paper is organized as follows: Sect. 2 presents a review of online education before Covid-19. Section 3 illustrates the current education's challenges in Covid-19's time. Section 4 presents the evolution and current reflections from Canadian and Russian education systems in Covid-19 era. Section 5 concludes the paper.

2 Review of Distance and Online Education Before COVID 19

Distance education is defined as one in which students must not attend a building in-person to develop the teaching-learning process. Indeed, few people know that the first distance learning courses were offered in 1728 by Caleb Phillips [4].

Correspondence training was to remain the norm for distance education for several centuries, to the point that the two concepts would long be virtually equivalent. Later, during the nineteenth century, more institutions appeared offering correspondence courses, mainly in the United States and the United Kingdom. In 1840, Sir Isaac Pitman invented a method of shorthand. To popularize it, he offers correspondence courses. He was soon followed on the continent by two associates from Berlin, Charles Toussaint, and Gustav Langenscheidt, who launched their correspondence school and popularized shorthand. Their business will survive until World War II. In 1858, the University of London was the first to issue diplomas recognizing training obtained by correspondence. In 1873, Anna Ticknor founded the Home Study Encouragement Society: a correspondence school explicitly aimed at a female audience.

It is interesting to know that, in 1938, in Canada, the first International Conference on Education by Correspondence was held [5].

Later, in 1969, in Great Britain, the Open University was established, which began by offering programs, first, via the radio and then by, accompanied by printed material sent to the students' homes.

All institutions that currently operate remotely use information and communication technologies to deliver their courses and programs. During the 70s of the last century, the e-learning modality began to be offered to provide instruction to the student who is separated from the instructor and support substantive and regular interactions between students and teachers synchronously and asynchronously. Technologies used for instruction can include Internet, TV, unidirectional and bidirectional transmission, open or public transmission, closed circuits, cable, microwaves, broadband lines, fiber optics, communication through satellite devices or cable, audio, and video—conferences, DVDs, and CDROMs [6–9].

Since the 1990s, with the development of information and communications technologies (ICT), we have witnessed Educational technology's virtualization (commonly abbreviated as EduTech, or EdTech). The primary forms of which are: synchronous (live), asynchronous (self-supporting modules that can be viewed at the desired time), the hybrid (formula integrating both face-to-face and "remote"), and the co-modal (a procedure which offers the student the possibility of choosing, at any time, to be either face-to-face, synchronous, or asynchronous). As early as 1999, the web offered distance learning courses with BlackBoard, e-College, or even SmartThinking [10–16].

In 2001, the constructivist MOODLE platform made its appearance and offered interactions between online learners. According to the Sloan Consortium, in 2009, 4.9 million people had registered for at least one online course [17].

And then, a newcomer appeared a few years ago: the MOOC. The Massive Online Open Course—online, massive, and open course. Dave Cormier was the first to use the word MOOC to refer to the Connectivism and Connective Knowledge course (also known as CCK08). This course was developed by George Siemens and Steven Downes for the University of Manitoba, Canada.

3 Current Education's Challenges in Covid-19's Time

Education is currently facing a threat of unprecedented magnitude and a crisis of exceptional importance. This situation arises in a global learning crisis: many school students do not acquire the fundamentals of life preparation. According to the World Bank's "learning poverty" indicator, 53% of 10-year-olds in developing countries cannot read and understand the age-appropriate text. The current health crisis that has caused COVID-19 has exposed the educational inequality expressed in the learning gap and the technological gap and socioeconomic inequality by those parents who cannot attend school at home. Figure 1 shows the current education system status worldwide in the Covid-19 era. When studying the comparison of education in different countries, it is clear that some countries have figured it out, and their students are benefiting on a global scale.

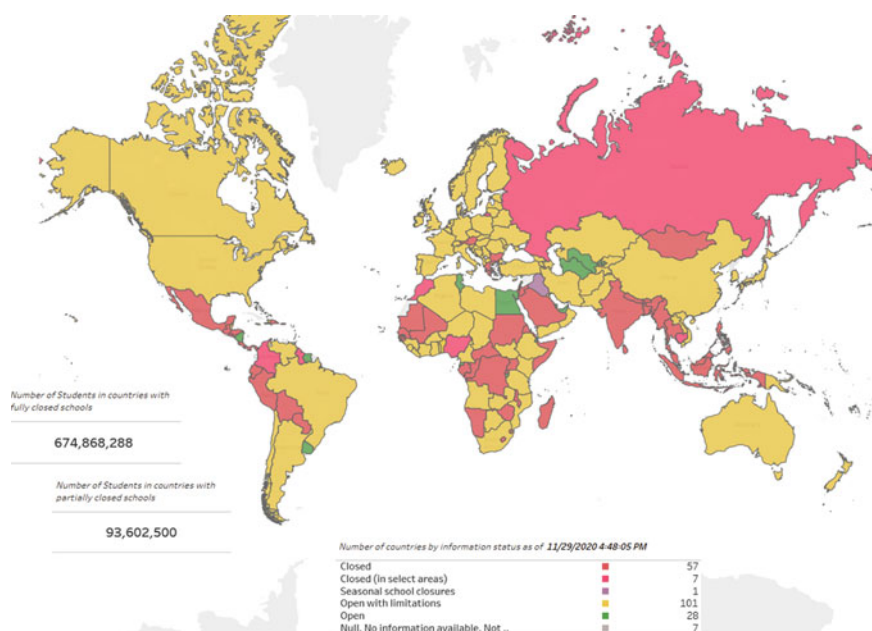


Fig. 1 Education system status worldwide in the Covid-19 era (from <https://www.worldbank.org/en/data/interactive/2020/03/24/world-bank-education-and-covid-19>)

The shift to distance learning has required numerous adjustments because didactics for online learning are different than for in-class learning. Pedagogical advisers have been urgently working to develop guidelines and indicators to support teachers and students through these unpredicted changes. It is essential to note that many students do not currently have the tools needed to take classes online (computer, specialized applications, internet access, etc.) or do not know how to use these technologies for training purposes [18].

However, these situations have brought some benefits. Many advantages can be obtained from remote training, from saving time and money. But of the same magnitude of its benefits are also the challenges it faces, among which are:

- Culture of self-orientation: On the way to consolidating an efficient online education, educational organizations, through the use of Information and Communication Technologies (ICT), must generate a culture of self-orientation, in which each student can be responsible for guiding his path. This self-orientation process will recommend them to enhance their personal, academic, emotional, social, and professional development, taking into account their reality and their immediate environment.
- Project-based methods: The evidence shows that active learning, one in which it is applied, interacts, and reflects individually and collectively, is the best for understanding new skills. The theory is essential to acquire certain concepts, but what will cause learning is practice. Under this dynamic, online education must design its methods based on carrying out projects where students put what they have learned into practice.
- Learning in a team: will always be more enriching than doing it alone. The motivation achieved by being part of a group with a shared mission can make all the difference. The team learning could also be achieved even in times and conditions of social isolation. Similarly, as students would have to do it personally, through the use of technology and digital platforms, students create spaces for collective reflection that help them prepare for collaboration along the way.

Today's goal is to establish online education as an ally of social isolation. Although the courses will be held outside of conventional classrooms, away from desks and colleagues, the academic programs will continue to be the same and make their way among the many advantages of online education [19].

4 Moving Forward in the COVID-19 Era: Reflections from Canadian and Russian Education Systems

4.1 *Russia Case*

Faced with the coronavirus situation, Russia's public health authority-imposed measures to fight the outbreak and protect the population. On 14 March, the Russian

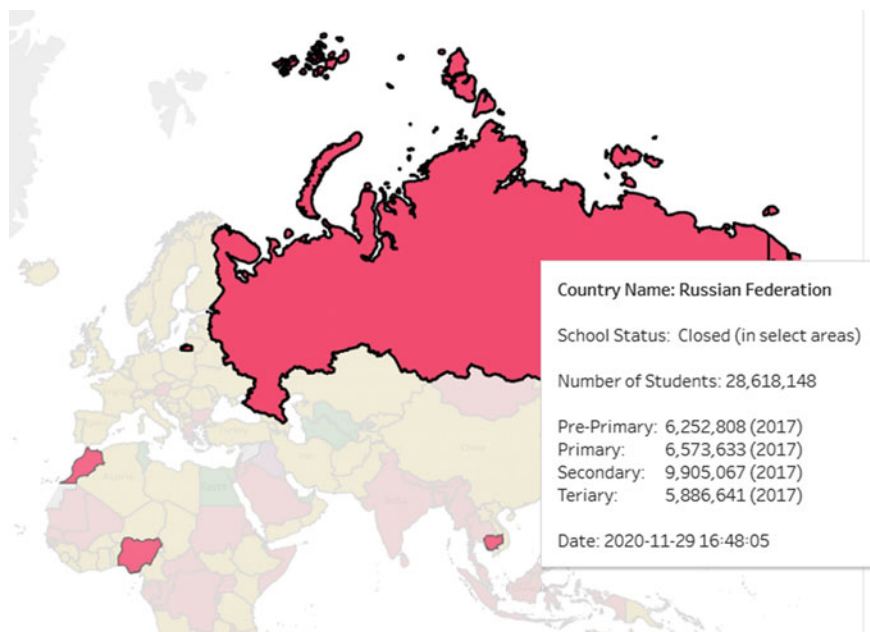


Fig. 2 Education system status in Russian Federation in the Covid-19 era (from <https://www.worldbank.org/en/data/interactive/2020/03/24/world-bank-education-and-covid-19>)

Ministry of Education advised schools across the country to adopt remote learning “as appropriate.” The Moscow region introduced flexible attendance policies at area public schools and kindergartens. However, all regular classes at schools would continue normally, and children who elected to stay home at their parents’ discretion would learn online [20]. Thus, schools will close their doors, events bringing together more than 50 people will be prohibited. These closures are impacting hundreds of millions of students and children (Fig. 2).

4.2 Canada Case

In the period of just a few months, the COVID-19 pandemic caused by a novel coronavirus has radically transformed the lives of millions of Canadians, including higher education students.

In this crisis of the COVID19 pandemic, confinement and social distancing forced all Canadian students to experiment with various education strategies. The experiences have been diverse, between the absurd and creative. Thus, many children only receive endless lists of activities and tasks at home, without any pedagogical mediation, while a few use communication platforms with virtual accompaniment.

On March 16, all schools are closed at the provincial and territorial levels except for British Columbia and Yukon (see Table 1). However, Yukon schools began their spring break on March 16, and on March 18, 2020, the closure was extended until April 15, 2020. On March 17, schools in British Columbia were suspended indefinitely. All educational institutions were closed by late March. Following the summer 2020 break, all schools reopened for the next school year (please refer to Fig. 3).

Table 1 The Canadian provinces measurement to fight the Covid-19 outbreak

Province	Health and safety protocols	In-class and/or online learning
British Columbia	• Mandatory masks for middle and secondary school students in high traffic areas • Minimizing sharing and limiting the use of frequently touched items • Physical distancing (staggered breaks, barriers, individual activities)	• Continuation of online/remote learning opportunities to supplement in-school instruction • In-class instruction for all other students for the maximum time possible within the above limits/targets
Alberta	• Mandatory masks for students and students in Grades 4–12 where physical distancing is not possible, optional masks for K-3 • Reorganize rooms for more physical space • “No sharing” policy	• New restrictions starting November 30, 2020 • Grades 7–12 learning at-home until January 8 (excl. Winter break) • Grades K-6 continue in-person learning until winter break
Ontario	• Masks required for students in grades 4–12 • Encouraged to keep secondary school students in a maximum of two in-person class cohorts • Ontario launches online tracker: “COVID-19 cases in schools and child care centres”. Data is updated at 10:30 am Eastern on weekdays	• Schools will open in September with conventional or delivery, depending upon the designation of the school board • Class cohorts of ~15 • Alternating schedules with >50% in-class instructional days • All school boards are encouraged to adopt timetabling methods that emphasized cohorting of students
Québec	• Students in class with set educator/teacher-student ratios • Teachers will move between classrooms based on the subject being taught, and students will remain in the same classroom • Masks: mandatory for grades 5 and up in common areas and around students from other groups • Maintain one meter between the various subgroups of students (recommended 1 m for 16 and under, 2 for 17 and up) • Minimal school visitors	• Elementary and younger students are expected to attend class in-person • Secondary I, II, III students attend school in person. • Secondary IV, V school boards may opt for alternative delivery if the course schedule cannot be reorganized to maintain stable student groups

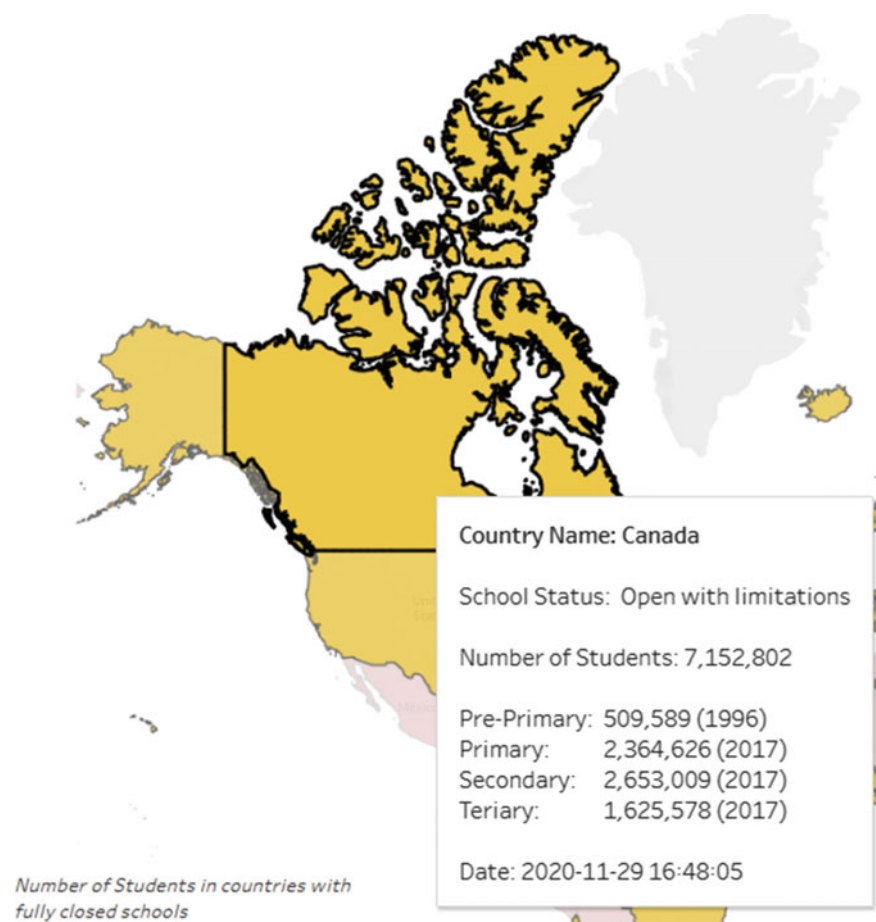


Fig. 3 Current education system status in Canada in the Covid-19 era (from <https://www.worldbank.org/en/data/interactive/2020/03/24/world-bank-education-and-covid-19>)

5 Conclusion

The year 2020 is a year to remember. We are all severely affected by the negative impact of the COVID-19 pandemic, whether rich or poor. Simultaneously, the COVID-19 related challenges united us and the whole world to fight against this pandemic. The role of Science, Technology, and Innovation is more critical than ever.

The coronavirus is instantly changing the way education is delivered, as the school and the home now become the same place after the necessary regulations in place. If we look hard enough at the current situation, we can glimpse some reasons for optimism. Hybrid learning will increase dramatically. Teaching and learning with

asynchronous and synchronous platforms will yield significant benefits when these methods are placed in face-to-face instruction. We will return from COVID-19 with a much more shared understanding that digital tools are complements, not substitutes, for the intimacy and immediacy of face-to-face learning.

Online education will be a strategic priority in each institution. In the future, every educational institution president, professor, teacher, and administrator will understand that online education is not just a potential source of new income. Instead, online education will be recognized as the core of each school's institutional resilience and academic continuity plan.

The role of the educator will be redefined. The students can gain access to knowledge and even learn a technical skill through just a few clicks on their phones, tablets, and computers. This may mean that educators' role should move towards facilitating students' development as contributing members of society.

Besides, we must design new forms of evaluation. These challenges must consider the technological limitations that affect students. Not all of them have the necessary resources, a computer, or an adequate Internet connection.

Finally, considering the young students and children, they require their parents' accompaniment to use the resources provided by their teachers and supervise their use of a computer with Internet access.

This crisis has led us to use online education as an alternative to continue our teaching work. Although the scenario is not ideal and the learning of teachers, students, and parents is taking place daily, we must consider this learning and the experience gained as special tools to complement and strengthen the educational system when we have overcome this crisis.

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Smart Universities and Their Impact on Students with Disabilities

Smart Universities: Assistive Technologies for Students with Visual Impairments



Jeffrey P. Bakken, Prasanthi Putta, and Vladimir L. Uskov

Abstract This paper presents the outcomes of an ongoing research project aimed at systematic identification, analysis and testing of available open source and commercial software systems that could significantly benefit college and university students with visual impairments in highly technological learning environments—smart universities. We analyzed dozens of software systems available for students with visual impairments, including JAWS, VoiceOver, Emacspeak, Orca, NVDA, ChromeVox, Duxbury Braille Translator, Kurzweil 1000, ZoomText, DesktopZoom, Microsoft Narrator, MAGic, ZoomText, Microsoft Magnifier, RoboBraille, Open-Book, Mind Manager, SensusAccess, Index Everest D, Focus 80, FreedomScientific Topaz, VisioBook, TSI Voyager XL, Opti Lite, Optelec ClearView, and other systems. Based on a careful analysis of available open source and commercial products we identified the top software systems that we recommend for implementation in smart universities to benefit students with visual impairments.

Keywords Students with visual impairments · Assistive technology · Software systems · Smart university

1 Introduction: College Students with Visual Disabilities

In accordance with the U.S. Department of Education, National Center for Education Statistics [1], in 2011–2012 9.0% of all undergraduate students in the age range of 15–23 enrolled in postsecondary institutions in the U.S. were students with some

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form of disability, 11.3% of those aged 24–29, and 12.4% of those aged 30 and older. As a result, on average, about 10% of the college student population in all US institutions are identified with a disability. Many of these students need some form of technology to be successful in this environment. In general, students in colleges/universities may experience a variety of different categories of disabilities; they include but are not limited to: (1) learning disabilities, (2) speech or language impairments, (3) health impairments, (4) psychological/neurological impairments, (5) hearing impairments, (6) physical/mobility/motion/orthopedic disabilities; and (7) visual impairments.

Willings [2] describes about 25 different types of common visual impairments. “A visual impairment is the loss of vision that cannot be corrected by refraction (glasses). There are a number of eye disorders that can lead to visual impairments. Visual impairment can also be caused by trauma and brain and nerve disorders. Visual impairments affect people differently. It is important to understand each student’s visual impairment in order to understand the potential impact on the student’s vision and prognosis” [2]. As a result, smart universities need to be equipped with different systems to help students with various types of visual impairments [3–6].

2 Assistive Technologies for Students with Visual Impairments: Literature Review

2.1 Assistive Technologies in Use by Students with Visual Impairments: Examples

Students from Stanford University use Math Player-enabled DAISY player software to read classroom materials in the manner that suits their abilities and preferences best. Math Daisy works with Microsoft Word, Microsoft’s Save as DAISY add-in, and Math Type. As one might guess from its name, Save as DAISY adds a “Save as DAISY” menu item to Word’s File menu. This command saves the document as a DAISY Digital Talking Book ready to be used in an eBook reader. Math Daisy enhances the Word-to-DAISY conversion process, converting the equations in the document to Math ML as required by the DAISY format [7].

Graph Sketching tool of North Carolina State University, was to provide blind screen reader users with a means to create and access graphs as node-link diagrams and share them with sighted people in real-time. Students successfully used GSK in their courses like automata theory, operating systems, software engineering, and artificial intelligence courses to work with graphs, both alone and in conjunction with sighted instructors [8].

Students from University of Pennsylvania uses a combination of screen readers, a refreshable Braille display, and a tactile drawing board, which allow him to access course materials more easily. Students with low vision use CCTV (closed-circuit television), an electronic magnifier—a type of zooming tablet—equipped with a

camera that magnifies text on a piece of paper. It can also, using a camera set up in the classroom before a class starts, enlarge content on a chalkboard or projector screen [9].

Students from Harvard University access course materials in an electronic format such as e-texts and enlarged print materials. Students use screen readers like JAWS, screen reader that enables students who are blind or visually impaired to navigate the Internet and most Windows-based applications by using keystrokes to input data and commands and Kurzweil 1000, the software reads dialogue boxes as well as materials that have been scanned. Students can customize backgrounds and font colors, the appearance of the cursor, and the level of text magnification through MAGic [10].

Most individuals from University of Washington who are blind can use a standard keyboard. Viewing standard screen displays and printed documents is problematic. Specialized voice and Braille output devices can translate text into synthesized voice and Braille output, respectively. Students give input through locator dots on the keyboard for commonly used keys and use screen-reader software, speech output and refreshable Braille displays that allow line-by-line translation of screen text into a Braille display area. Students also use Braille embossers to print tactile Braille [11].

Students from University of Illinois at Urbana-Champaign who are blind or visually impaired use JAWS Screen Reader, Kurzweil 1000, Refreshable Braille Displays to read printed material or to surf the web. Most students with very low vision or who are totally blind use a cane or a dog guide to walk safely around the campus [12, 13].

Students use iPads and iPods with a variety of Bluetooth Braille displays to access electronic documents and they use Optical Character Recognition software to access hard copy prints. Students with low-vision, use CCTV units, Screen reading and magnification software to access information on a computer quickly and effectively [14].

2.2 *Assistive Technologies in Use by Regular Students: Examples*

A Student Group from MIT designed a Tactile which is a portable device that converts any text to Braille in real time. The device works by taking an image of written text when placed above it and then the system converts the text into the appropriate mechanical movements for pins which rise to create the Braille translation. Team Tactile has been working to create a more sophisticated prototype that is the standard Braille size [15].

Student of University of California, Berkeley developed YouDescribe—audio description for Youtube videos. Now people who sign on to YouDescribe can find the video they want, and record their own descriptions of it. Blind people can now

find and listen to this growing collection of YouDescribe-enhanced videos in which recorded voices tell them what they can't see [16].

Engineers at Stanford University have collaborated with members of the blind and visually impaired community to develop a touch-based display that mimics the geometry of 3D objects designed on a computer. This project is about empowering a blind user to be able to design and create independently without relying on sighted mediators because that reduces creativity, agency and availability. The display is reminiscent of a pin art toy in that it forms shapes from a field of tall, rectangular pegs that move up and down. By inputting the specifications of their desired shape in the accompanying 3D modeling program, users can evaluate their creation via the touchable display. And that's the same opportunity that a sighted peer would have, where they too would be able to view various perspectives of their target object [17].

Non-Visual Desktop Access (NVDA), an open source screen reader for Windows. Screen readers help those with vision impairments to access digital content. Teachers in Michigan State University use screen readers to check that web content for lessons is accessible to students with visual impairments [18].

Faculty from University of Pennsylvania makes sure everything on the page is read out loud in a certain order that makes sense. They update metadata in a PDF to include detail descriptions about margin notes, tables, or figures that the screen reader would otherwise skip or read out of order [9].

Purdue University EPICS Team worked on the LEAP and Magnifier projects with the Indiana School for the Blind and Visually Impaired staff and students. The LEAP project creates a better communication line for blind and visually impaired students, enabling their teacher to receive an electronic copy in standard and print what the student type into the Perkin's Braille Writer. It allows the teacher to more easily review student work. A magnification stand and application that will utilize a device like an iPad as a screen to enhance the readability of textbooks and other documents. It is portable, allowing students to take school assignments home for completion. In addition to being mobile, the device is less expensive than others on the market [19].

Education researchers of University of Arizona collaborated to develop Animal-Watch Vi Suite, which includes an iPad application and supplemental materials that enable students with visual impairments to use technology to learn math more easily. AnimalWatch Vi Suite relies on word problems to help students learn math concepts and develop an understanding of endangered species while also training them to interface with emerging technology and accessibility features [20].

Students from University of Missouri invent deep-learning technology to help the visually impaired. DeepLens is a technology to help blind people do their basic, daily life activities. The device is a wearable camera that blind people can attach to a hat or glasses. It works by observing the user's environment and narrating their surroundings to them through an earpiece [21].

3 Research Project Goal and Objectives

During creative literature review of designated and multiple additional available publications no publications were identified with a classification of available open source and commercial software systems using a set of evaluation criteria (i.e. system's functions, technical platform, SWOT analysis, price, etc.) and their ranking in terms of functionality and usability by university/college student.

Project goal. The overall goal of this research project was to identify the best open source and commercially available software systems for university/college students with visual impairments—VI systems. These systems will be recommended for an active use at smart universities.

Project objectives. The objectives of this project included but were not limited to:

1. identification of colleges/universities in the U.S. and what VI systems they provide for college students with various forms of disabilities;
2. analysis of available open source and commercial VI systems and identification of the best systems to implement with college students with various forms of disabilities;
3. the identification of the top three open source and commercial VI systems;
4. creation of recommendations for university/college student assistance centers in terms of VI systems to be implemented to benefit college students with various forms of disabilities.

The obtained up-to-date research outcomes and findings are presented below.

4 Research Project Outcomes

4.1 *VI Systems in Use by Universities/Colleges: Best Examples*

We analyzed available publications about VI systems that are used by student assistance centers at top 10 U.S. universities. A summary of our research outcomes are presented in Table 1. (A note: Due to the limits of the current paper, the references to all VI systems analyzed are omitted in this paper; they are available in the VI bibliography at <http://cs-is1.bradley.edu/uskov/VI/>).

Table 1 A list of VI systems that are used by student assistance centers at top universities (limited to the first 10 universities)

College/University	Assistive technology in use	
MIT—Massachusetts Institute of Technology	• JAWS • VoiceOver • Emacspeak • Orca	• NVDA • ChromeVox • Duxbury Braille Translator
Michigan State University	• JAWS • NVDA • VoiceOver • Kurzweil 1000 • ZoomText	• DesktopZoom [9] • Duxbury Braille Translator • Refreshable Braille Displays
University of Arizona, Tucson/AZ	• JAWS • NVDA • Microsoft Narrator • Mac VoiceOver • MAGic • ZoomText	• Microsoft Magnifier • Mac Vision Tools • Duxbury Braille Translation • RoboBraille
University of Illinois at Urbana-Champaign	• JAWS • SensusAccess • VoiceOver • Dolphin Easy Reader	• ZoomText • OpenBook • CCTV
Purdue University	• JAWS • ZoomText • iZoom USB • Duxbury Braille Translator • Index Everest D • Focus 80	• FreedomScientific Topaz • VisioBook • TSI Voyager XL • Opti Lite • Optelec ClearView
Harvard University	• SensusAccess • Kurzweil 1000	• JAWS • ZoomText
Stanford University	• NVDA	• Mind Manager
University of California, Berkeley	• NVDA • VoiceOver	• SensusAccess
University of Southern California	• NVDA • VoiceOver • JAWS	• ZoomText • NFB Newsline • SensusAccess

4.2 VI Software Systems Analyzed

There are several available VI systems that could be implemented in a highly technological smart classroom at a smart university. The research findings and analysis outcomes of analyzed VI systems are presented in Tables 2 and 3.

Table 2 Open source VI systems analyzed

System's name and developer	Main system's functions
NVDA—Non-Visual Desktop Access by NV Access	• Supports popular applications including web browsers such as Mozilla Firefox and Google Chrome, email clients, internet chat software, music players, and office programs such as microsoft word and excel • Built-in speech synthesizer supporting over 50 languages, plus support for many other 3rd party voices • Reporting of textual formatting where available such as font name and size, style and spelling errors • Automatic announcement of text under the mouse and optional audible indication of the mouse position • Supports many refreshable braille displays, including input of Braille via braille displays that have a braille keyboard • Ability to run entirely from a USB flash drive or other portable media without the need for installation • Translated into more than 50 languages • Support for modern Windows operating systems including both 32 and 64 bit variants • Ability to run on Windows login and other secure screens • Announcing controls and text while interacting with gestures on touch screens
VoiceOver by Apple	• Hear what's happening on your screen • Integrated throughout Mac OS and every built-in app • Improved PDF, web, and messages navigation • Navigate VoiceOver with simple gestures • A virtual controller with customizable commands like headings, links and images • Plug-and-play support for Braille displays • Streamline the things you do every day through Siri, which is integrated with VoiceOver • The Apple Music, Apple Podcasts, Apple Books, and Apple TV apps are compatible with VoiceOver, so you can navigate and play all your content even if you can't see the screen

(continued)

Table 2 (continued)

System's name and developer	Main system's functions
Narrator by Microsoft	• Narrator can now read next, current, and previous sentences. Read by sentence is available whenever you use a keyboard, touch, or braille • Scan mode (navigation and reading mode) lets you navigate apps, email, and webpages using the arrow keys • Supports desktop Windows PCs with touch • Narrator works with braille displays that use a USB or serial port • Narrator provides ways to read text by page, paragraph, line, sentence, word, and character • Supports navigation by the view (links, tables) that you've selected • Supports around 150 languages and their respective voices
Magnifying glass by Workers Collection	• Enhance onscreen and projected detail • Allow modification of the magnified area in real time • Take advantage of multi-monitor systems • Provide you with more control over your business presentations • Save multiple set of viewing preferences as individual Profiles, then switch between those Profiles instantly • Show/hide the desktop magnifying glass simply by shaking the mouse cursor from side to side • Command mode which enables you to quickly change/apply magnifying glass options • The auto-switcher which enables you to associate individual applications, windows, and screen elements with specific user-defined profiles • You can quickly and easily apply a myriad of effects to enhance your viewing experience and solve the sophisticated needs of any text- or graphics-based project

(continued)

Table 2 (continued)

System's name and developer	Main system's functions
Liblouis (GNU Lesser GPL license)	• Supports computer and literary braille • Supports contracted and uncontracted translation for many languages and have support for hyphenation • Supports numerous open-source and proprietary screen readers such as NVDA, Orca, BrailleBack and JAWS • Performs the back-translation, translates braille file to its original format • Reformatting feature helps to format braille file. Example: line length change • Interlining to print original text between the lines of translated braille for proofreading • Input files types accepted by liblouis are xml, html or text • Supports various formats like tables, graphics, mathematics, poetry etc
Braille-Blaster by American Printing House for the Blind, Inc.	• Designed primarily for editing textbooks that meet the specifications published by the Braille Authority of North America • Translate braille accurately in UEB or EBAE • Format braille • Automate line numbered poetry and prose • Split books into volumes • Add transcriber notes • Describe images • Automate braille table of contents, glossaries, preliminary pages and special symbols pages • Automate a variety of table styles • Translate and edit single line math

Table 3 Commercial VI systems analyzed

System's name and developer	Main system's functions
JAWS by Freedom Scientific	• Convenient OCR feature for image files or inaccessible PDF documents • Supports PEARL Camera for direct access to print documents or books • Built-in free DAISY Player and full set of DAISY-formatted basic training books • Works with Microsoft Office, Google Docs, Chrome, Internet Explorer, Firefox, Edge, and much more • Support for MathML content presented in Internet Explorer that is rendered with MathJax • Save time with Skim Reading and Text Analyzer • Includes drivers for all popular Braille displays • Includes voices for over 30 different languages • Fully compatible with ZoomText, Fusion, MAGic, and the OpenBook Scanning and Reading Software
Kurzweil 1000 by Kurzweil Education	• Clear, natural-sounding VoiceText™ voices from Neo Speech and ETI-Eloquence™ voices from Nuance deliver the highest quality text-to-speech (TTS), making your reading a pleasure to hear • Study tools like Add bookmarks, text notes, and annotations; skim, summarize, and create outlines for your work, and have your documents open to the last position read • Two OCR engines, ScanSoft® OCR 18 and ABBYY FineReader 10, provide the most accurate OCR available • Keep track of your schedule with our easy-to-use calendar application. You can add, edit, and delete your calendar entries simply from the desktop taskbar, as well as set audible reminders that will play any sound file on your computer you choose • Based on your criteria, Kurzweil 1000 will search select Internet repositories such as Bookshare.org, NLS Web Braille, AccessWorld, Talking Newspaper Association of the United Kingdom, NFB Publications, and Wikipedia (in multiple languages), and present you with your download choices • Kurzweil 1000 can send your files to a wide variety of hand-held devices so you can read and reference important material when away from your home or office computer • Any file type that you can print through an application installed on your computer can be opened with Kurzweil 1000 (including PDF), which greatly expands your reading choices

(continued)

Table 3 (continued)

System's name and developer	Main system's functions
ZoomText Magnifier/Reader by Freedom Scientific	• xFont technology displays high-definition text that's easy to read at all magnification levels. Smooth, bold and condense settings allow you to fine-tune the thickness and spacing of text for added legibility • ImageReader comes with premium male and female voices for reading captured text. You will have over 70 premium-quality voices for the most commonly spoken languages and dialects from around the world to choose from • ZoomText's powerful screen reading options let you hear what you are doing. As you type text, use the mouse and navigate through your applications, ZoomText narrates and confirms each of your actions • The ZoomText Recorder allows you to turn text from documents, webpages, email, etc. into audio recordings that you can transfer to on your mobile device and listen to at your convenience • ZoomText's Background Reader allows you to listen to documents, webpages, email, etc. while you simultaneously perform other tasks • Innovative color controls improve screen clarity and reduce eyestrain • The ZoomText Camera feature allows you to use any high-definition (HD) webcam to view and magnify printed items and other objects right on your computer screen—including bills, magazines, photographs, medicine labels, craft items and more • ZoomText's logon support provides essential magnification and screen reading features when logging into Windows • With ZoomText's new Smart Invert feature, photos are displayed in their natural colors when ZoomText's Invert Brightness and Invert Colors are active

(continued)

Table 3 (continued)

System's name and developer	Main system's functions
MAGic by Freedom Scientific	• Supports 75 magnification levels from 1× to 60× • View more of the screen content with multiple magnification levels between 1× and 4× • Work longer without fatigue with high-definition text and crisp fonts • Never lose track of the cursor or mouse pointer again with MAGic's high-definition mouse and cursor enhancement options • Eliminate glare and increase contrast with built-in color enhancements • Speech options add human-sounding voices to speak text and echo user actions • Add a MAGic Large Print Keyboard with bold, high-contrast keys for easy access to MAGic's most-used features • Dual-monitor support provides greater productivity • Citrix Remote Access for advanced workplace opportunities
Duxbury Braille Translator by Duxbury Systems Inc.	• Duxbury DBT imports many document formats, translates into Braille, and sends the Braille to your Braille embosser or Braille device • In tune with the latest advances in operating systems and sister applications • Built-in interline printing to have ink-Braille and print together. This makes an easy proofing and teaching tool • DBT imports an extensive array of text and document formats Microsoft Word, HTML, mathematics files, Tactile View graphics files • Duxbury DBT can be used to prepare textbooks to teach foreign languages • Six-key chording software for Braille and print entry, compatible with many keyboards • Bidirectional (print-to-Braille and Braille-to-print) translation for most languages

(continued)

Table 3 (continued)

System's name and developer	Main system's functions
MegaDots by Duxbury Systems Inc.	• Optically scanned documents are automatically recognized and cleaned up—before you even edit your document • MegaDots support your printer's scalable fonts for the most attractive possible large print • Designed specifically to make learning and proofreading Braille easier • Make ink print Braille and text on the same page in three different formats: side by side, word by word, and line by line • Interpret Format to adjust the importation of a document. It shows styles used, running header, print page indicators; and other useful information • MegaDots will automatically squeeze the table to fit on the Braille page • MegaDots support all of the diacritical marks shown in the textbook format codebook. This speeds up another laborious aspect of making Braille

4.3 *Top VI Systems Identified*

We tested, analyzed and evaluated the functionality of almost all the open-source systems and demo versions or trial versions of most of the commercial VI systems. The outcomes of analysis findings as well as the final ratings of those systems are given below in Tables 4, 5 and 6 for open source VI systems and Tables 7, 8 and 9 for commercial VI systems.

4.3.1 Top-Ranked Open-Source VI Systems

See Tables 4, 5 and 6.

4.3.2 Top-Ranked Commercial VI Systems

See Tables 7, 8 and 9.

5 Conclusions. Future Steps

Conclusions. The performed research helped us to identify the current status of VI systems available for college students with visual impairments. The obtained research outcomes and findings enabled us to make the following conclusions:

1. We tested, analyzed and evaluated six open source (Table 2) and six commercial (Table 3) VI systems.
2. We ranked all analyzed open source and commercial VI systems—the research outcomes are presented in Tables 4, 5, 6, 7, 8 and 9.
3. Based on our evaluation the top open source VI system is *Non-Visual Desktop Access* VI system (Table 4), and the top commercial VI system is *JAWS system* (Table 7). Those systems are strongly recommended by our research team for implementation and active use in smart classes of smart universities.
4. More research needs to be completed that directly focuses on the perception of VI systems by actual college students with motion/mobility disabilities.

Next steps. The next steps of this research, design and development project deal with

1. More implementation, analysis, testing and quality assessment of VI systems by actual college students with visual impairments.
2. Implementation, analysis, testing and quality assessment of identified VI systems in everyday teaching of classes in smart classrooms.
3. Organization and implementation of summative and formative evaluations of local and remote college students and learners with and without disabilities with a focus to collect sufficient data on quality of VI systems in smart classrooms.

Table 4 Non-visual desktop access—NVDA—open source VI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• Built-in speech synthesizer supporting over 50 languages, plus support for many other 3rd party voices • Reporting of textual formatting where available such as font name and size, style and spelling errors • Support for many refreshable braille displays, including input of Braille via braille displays that have a braille keyboard • Announcing controls and text while interacting with gestures on touch screens
Strengths and opportunities	• Support for popular applications including web browsers such as Mozilla Firefox and Google Chrome, email clients, internet chat software, music players, and office programs such as Microsoft Word and Excel • Ability to run entirely from a USB flash drive or other portable media without the need for installation • Ability to run on Windows logon and other secure screens
Possible weaknesses	• Lacks in Supporting Unified English Braille (UEB)
Technical platform	• Windows
Price (if any)	• Free
Colleges/Universities that use this system	• MIT—Massachusetts Institute of Technology • Stanford University • University of California, Berkeley • University of Texas, Austin • Michigan State University • University of Arizona, Tucson/AZ • University of Michigan • University of Southern California
Ranking	1

4. Creation of a set of recommendations (technological, structural, financial, curricula, etc.) about which VI systems colleges/universities should get (purchase, if needed) and install to benefit college students with and without visual disabilities at smart university.

Table 5 VoiceOver open source VI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• VoiceOver can switch to the voice for that language automatically when you navigate through different language tabs. And you can add custom commands and workflows to your MacBook Pro with Touch Bar • You can control VoiceOver using many of the same gestures you use with iOS. Touch the trackpad to hear a description of the item under your finger, drag to hear items continuously, and flick to move to the next item • The rotor lists common elements like "headings," "links," and "images," and lets you navigate directly to the element of your choosing
Strengths and opportunities	• VoiceOver is compatible with over 100 refreshable braille displays, you can just plug in or sync your display—even multiple displays—and you're set • Prevent people from seeing what's on your screen by using the screen curtain, which turns the screen black • The braille panel simulates a refreshable braille display, and includes a language translation of the braille • The caption panel shows what VoiceOver is speaking, and can be helpful when sharing your Mac with sighted users
Possible weaknesses and threats	• Unintuitive web navigation features
Technical platform	• Mac OS
Price (if any)	• Free
Colleges/Universities that use this system	• MIT—Massachusetts Institute of Technology • University of California, Berkeley • Michigan State University • University of Arizona, Tucson/AZ • University of Illinois at Urbana-Champaign • University of Michigan • University of Southern California
Ranking	2

Table 6 Windows Narrator: open source VI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• Narrator can read next, current, and previous sentences. Read by sentence is available whenever you use a keyboard, touch, or braille • Scan mode (navigation and reading mode) lets you navigate apps, email, and webpages using the arrow keys • Narrator works with braille displays that use a USB or serial port • Narrator provides ways to read text by page, paragraph, line, sentence, word, and character • Supports navigation by the view (links, tables) that you've selected
Strengths and opportunities	• Audio ducking feature available in Windows Narrator that reduces the volume of other sounds such as YouTube videos when the screen reader is speaking • Narrator uses Microsoft Mobile voices, which I find to be easy to understand and very responsive • Narrator works well with Microsoft's built-in Mail program, as well as Outlook 2016 • Supports around 150 languages and their respective voices
Possible weaknesses	• Users doesn't have much control over what Narrator reads
Technical platform	• Windows
Price (if any)	• Free
Colleges/Universities that use this system	• University of Arizona, Tucson/AZ
Ranking	3

Table 7 JAWS commercial VI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• JAWS provides speech and braille output for the most popular computer applications on your PC • Save time with skim reading and text analyzer • Built-in free DAISY Player and full set of DAISY-formatted basic training books • Convenient OCR feature for image files or inaccessible PDF documents

(continued)

Table 7 (continued)

System's features	System's details
Strengths and opportunities	• Works with Microsoft Office, Google Docs, Chrome, Internet Explorer, Firefox, Edge, and much more • Fast information look-up at your fingertips with Research It • Powerful scripting language to customize the user experience on any application • Distributed worldwide with local sales and support in most countries
Possible weaknesses and threats	• Usage cut to almost 50% with increase in the usage of NVDA and VoiceOver
Technical platform	• Windows
Price (if any)	• Student Annual License—\$90/year • Professional License—\$1200
Colleges/Universities that currently use this system	• MIT—Massachusetts Institute of Technology • Michigan State University • University of Arizona, Tucson/AZ • Harvard University • University of Pennsylvania • University of Illinois at Urbana-Champaign • University of Chicago • Purdue University • Ohio State University • University of Michigan • University of Washington • American University • University of Southern California • NC State University • California State University-Fullerton • University of Wisconsin-Whitewater • University of Connecticut
Ranking	1

Table 8 ZoomText magnifier/reader commercial VI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• ZoomText Magnifier/Reader is a fully integrated magnification and reading program tailored for low-vision users. Magnifier/Reader enlarges and enhances everything on your computer screen, echoes your typing and essential program activity, and automatically reads documents, web pages, email • ZoomText's powerful screen reading options let you hear what you are doing. As you type text, use the mouse and navigate through your applications, ZoomText narrates and confirms each of your actions • xFont technology displays high-definition text that's easy to read at all magnification levels. Smooth, bold and condense settings allow you to fine-tune the thickness and spacing of text for added legibility
Strengths and opportunities	• ZoomText Voices have 70 premium-quality voices for the most commonly spoken languages and dialects from around the world to choose from • Easy to locate and follow the control focus when you navigate through application menus, dialogs and other application controls • ZoomText provides eight zoom window types: Full, Overlay, Lens, Line and four Docked positions. Each of these windows offer a unique way of viewing what is on the screen
Possible weaknesses and threats	• Compatibility Issues with Windows 10 Creators Update
Technical platform	• Windows
Price (if any)	• \$875.00
Colleges/Universities that currently use this system	• University of Texas, Austin • Michigan State University • University of Arizona, Tucson/AZ • Harvard University • University of Pennsylvania • University of Illinois at Urbana-Champaign • Purdue University • Ohio State University • University of Michigan • University of Washington • American University • University of Southern California • NC State University • California State University-Fullerton • University of Wisconsin-Whitewater • University of Connecticut

(continued)

Table 8 (continued)

System's features	System's details
Ranking	2

Table 9 Duxbury Braille translation commercial VI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• Built-in interline printing to have ink-braille and print together. This makes an easy proofing and teaching tool. Great for the braille-impaired too! • Math/Science Code and Computer Braille translation for American, UEB, British, and French Braille • The ability to include tactile graphics files for mixed text-and-graphic documents • A "Quick Find Misspelling" feature for increased speed and ease of use • Embossing to all major braille printers; the first page may be a "banner" for job identification by personnel who don't read braille
Strengths and opportunities	• Accurate presentation of both print or braille in either WYSIWYG (what-you-see-is-what-you-get) or coded (how-you-get-what-you-want) views in the word-processing screen, with easy switching between views • Over 100 formatting and translation codes for a high level of flexibility • Bidirectional (print-to-braille and braille-to-print) translation for most languages • The current translation table menu includes over 130 different languages plus variations, including contracted braille for most jurisdictions where contracted braille is customarily used
Possible weaknesses and threats	• Open-Source software systems like Liblouis, Brailleblaster usage might effects the usage of Duxbury Braille Translation
Technical platform	• Windows • Mac OS
Price (if any)	• \$695.00
Colleges/Universities that currently use this system	• MIT—Massachusetts Institute of Technology • Michigan State University • University of Arizona, Tucson/AZ • Purdue University • American University
Ranking	3

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A Technology for Assisting Literacy Development in Adults with Dyslexia and Illiterate Second Language Learners



Matteo Cristani, Serena Dal Maso, Sabrina Piccinin, Claudio Tomazzoli, Marco Vedovato, and Maria Vender

Abstract Developing good literacy skills is a domain in which two distinct populations of adults display marked difficulties: individuals suffering from developmental dyslexia and second language learners. These weaknesses in literacy can have a negative impact on their quality of life and in particular on their employability and professional carrier. These subjects share, however, advanced *digital skills*, in particular those related to the usage of mobile devices. Leveraging on this skills we developed a platform that allows several different subjects (teachers, project managers, learners) in interacting with each other for assisting the development of the above mentioned literacy proficiency.

Keywords Dyslexia · Language learning · Second language

1 Introduction

The project aims at carrying out accurate investigations and training for strengthening the profile and employment prospects of two populations on the Italian territory for which access to reading and writing is problematic: adults suffering from developmental dyslexia (dyslexia henceforth) and individuals who, due to their migration

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background, have learned Italian as a second language, possibly with a starting condition of first language illiteracy (either full, namely with no reading or writing skills, or partial as in those subjects able to read and write in their L1 language, but unable to do so in L2). Although there are many investigations that have already been carried out in the past on this theme, these needs impact in a novel era, where the widespread diffusion of mobile technologies has brought potential of leverage in the above mentioned populations. Within the project has therefore taken space the idea that it would be useful to support in this way learning and also assistance to learning, control of progress and analyses of the outcomes as driven by the project's concepts. The platform that has been designed, and prototyped in Flask, is illustrated here. The research team, formed by linguists and computer scientists, has devised a model based on the up-to-date methods for the mentioned cases and implemented them in a system able to support various steps of the learning process. The model of continuous progress control is validated in the current literature on acquisition of reading skills in the mentioned populations, and it is devised to support the activities in potential further other groups with similar literacy deficits, but coming from further specifically limited contexts. The solution that we propose does not exist as a means for supporting learning activities in adults with dyslexia or second language illiterate adults. There exist a few solutions for children with dyslexia, and some of these have been adapted to work for adults, whilst for illiterate second language learners this is the first one to be deployed. Moreover, the technology is developed as a prototype exactly as a base for users' feedback by the adults, that are indeed in the experimental perimeter. The technology is developed with a line towards the expression of the ability of users to anticipate the learning processes determined by elements of a specific exercise. In particular, during a training, a teacher or a language specialist could identify limits to the developed exercise and evolve it to satisfy the requests of specific types of users (in particular this holds for second language learners with specific geographic provenance).

The rest of the paper is organised as follows: Sect. 2 discusses the theoretical framework introduced in the project. We then provide a general description of the application in Sect. 3 and specify the functional architecture of the prototype in Sects. 3.1 and 3.2. Section 4 introduces some related work and Sect. 5 takes some conclusions and sketches further work.

2 Theoretical Framework

Adopting a narrow interpretation, literacy refers to the ability to read and write through an automatized process, usually acquired with formal education and schooling. This automatism requires the development and the systematic recourse, during reading, to a lexical route (i.e., global visual representation of the lexical item), in addition to the phonological route (i.e., grapheme-phoneme decoding). The second skill is generally used for unknown or unfamiliar words by skilled readers [1]. This decoding ability is often referred to as “primary literacy”, opposed to “functional

literacy”, which, in a wider sense, has to do with the ability to deal with texts, understanding their meanings, communicative goals and interactions with the contexts. Therefore, successfully reading a text involves the acquisition of a range of complex language structures and the ability to process those structures in real time. Such competences include the awareness of linguistic sounds (phonology) and their relation to graphemes and spelling patterns (orthography), word meaning (semantics), morpho-syntactic structures and textual organization. An automatized processing of all these competences is the necessary condition to develop reading fluency and text comprehension. During the last decades, a large scientific literature has been developed on what should be meant by literacy and how to deal with the problem of assessing literacy skills in education and other social contexts, including assistive learning. In most recent research, a growing emphasis has been put on the application of literacy skills in everyday life. Based on the above mentioned viewpoint the research community has developed a concept of measure on the literacy skill and on their measure by means of continuous scales rather than in terms of dichotomies (literate vs. illiterate). Although dealing with texts in many daily tasks and work routines is extremely common, it remains an obstacle for specific sections of population, both in terms of understanding and production of written texts. This is particularly the case for two specific populations, which will be targeted in our project: (a) Native speakers of Italian with dyslexia and (b) non-native speakers of Italian (i.e., migrants, refugees, asylum seekers) with absent or low literacy skills in their second language. The main characteristics of these two populations, both suffering from reading and spelling impairments which are however related to different conditions, will be now briefly discussed. Dyslexia is a lifelong condition characterized by reading and spelling deficits [2], together with impairments in the phonological [3, 4], morphological [5, 6] and grammatical competence [7, 8]. These difficulties have been found to persist also in adulthood [9], with negative consequences on the individual’s academic and professional careers [10]. Despite representing a rather different type of population, non-literate and low-literate learners of L2 Italian experience similar difficulties in the area of basic literacy skills. Such learners face the double challenge of having to learn another language, while simultaneously trying to grasp the alphabetic principle underlying its writing system. The difficulties encountered by these individuals can locate at rather different levels, because of the highly heterogeneous nature of their profile. These may span from learners still struggling with the understanding of the basic functions of print to those who, being literate in a writing system other than alphabetic, exhibit an inadequate degree of phonemic awareness [11], a skill which is instrumental for cracking the alphabetic code [12, 13]. As social and work-related demands increase continuously, reading and writing skills which would have been considered adequate in the past are often no longer sufficient to gain and maintain employment. Moreover, the current labor market is characterized by both an increase of automatization in the workplace and a higher demand for literacy skills, even for the most manual jobs. Indeed, only few jobs can be performed successfully if employees have problems with reading and writing. The general acknowledgment that a low literacy degree is likely to have negative effects at the level of employability has given rise to a number of programs supporting adult education and literacy in developing

countries (EFA Programs, ‘Education For All’, supported by UNESCO, is one of these). However, Europe is also confronted with a growing challenge for a wider and more equal access to literacy skills and should aim at a more inclusive continuing educational system. So far, the landmarks provided by the Council of Europe and by LESLLA (Literacy Education Second Language Learning for Adults) for both educational policy and research have rarely been integrated into actual intervention guidelines for employability improvement. As a consequence, governmental institutions are not fully resourced to face these new challenges and lack specific tools aiming to support the development of the more solid literacy abilities that are likely to enhance employability. Within the project, some of the experimental activities have been conceived in a way that is strongly supported, in the literacy proficiency, by the usage of the envisioned technology. In particular one important activity shall be the literacy training, which could represent an important opportunity for both individuals with dyslexia and non-native speakers with low literacy skills. Each experimental group is assigned to a specific intervention program, which is developed based on the profiles and on the identified specific learning needs. Literacy skills will be trained by means of specific tools aimed at guiding, supporting and automatizing a fast and accurate recognition and consolidation of syllables, morphemes and single words, and partly therefore by the usage of the platform documented in this work. Both the sub-lexical and the lexical routes for reading [1] are trained, in order to enhance automaticity and reduce reading efforts, while in parallel also reinforcing writing skills. Morphological awareness will also be specifically trained, as both populations have shown sensitivity to the structure of morphologically complex words during visual word recognition tasks [14, 15] and it has been proved to be particularly effective for the rehabilitation of reading and writing disorders [16–18]. Analogously, in the project there will be a specific pursue of literacy training for non-native speakers of Italian.

3 The Application

In the platform we designed and prototyped the core concept is the *project*. A project is a complex activity that involves a project manager, who launched the project itself, and two further groups of users: the *operators* and the *learners*. The users are all managed by the *application managers*. When an instance of the platform is deployed onto a server, an application manager is the one who installed it. She has no specific power different from other application managers who are the further users she can register. When someone involved in a specific learning project asks for deploying it, the application manager registers the new user as a project manager, and therefore gives her the privileges needed to create new projects. The application managers have also the privilege of making a project manager an application manager too. When a project is created with its own description, then the project manager can provide the content of the project itself by acting on the registration of operators, who are in charge of managing the learning activities. In particular, the operator registers

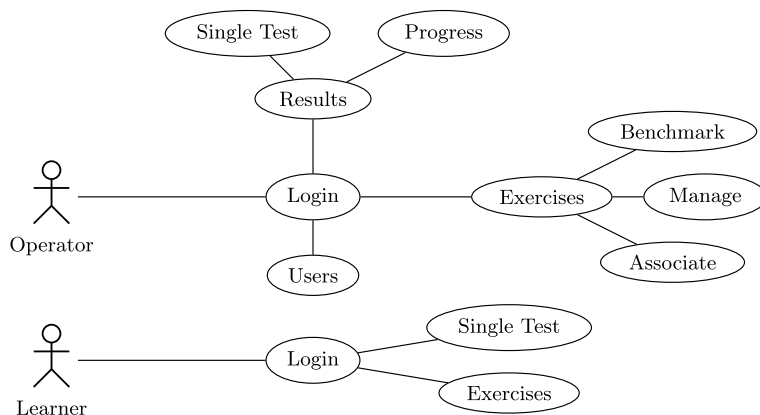


Fig. 1 Use case operator/learner

new *exercises*, that in turn shall be mounted into specific sequences and groups, that constitute the model of deployment for the application, that we describe in details in Sect. 3.1. The exercise constitute the basic brick of the notion of project and the model of learning underlying the application concept. Every exercise is formed by a *task* that is assigned to a learner, a set of *objects* to be displayed on the tool interface, that can be either *texts*, *videos*, *images* or *sounds*. On each of the above objects the operator can add *tags*, that are associated to answers, and that are displayed in form of clickable segments of the rendering. Executing an exercise shall be either one *single click* or a *sequence of clicks*, or a *selection of elements by clicks*, or an input on a further text *input box*. The operator mounts an exercise and the learner executes the exercise by reading the task, and then executing the operations required. To each exercise are associated more than one admissible answers. The operator associates an *evaluation method* for the exercise (either *autonomously* or *twin*, that means that the value of the exercise is assigned by the operator when the learner executes the exercise). To each answer of one exercise, the operator associates a value that is assigned automatically, and can be corrected by the operator when the exercise is in twin modality only. Every execution of an exercise is logged in terms of duration. Every learner registration is under the supervision of a single operator. Once an operator has added a specific learner to a project, the learner is involved in the project itself. On creation, the operator assigns to the learner a *level* of knowledge of the host language (in the specific case, Italian). A learner can perform exercises of a level that is lower than or equal to hers.

3.1 Application Functions

In this section we address the matter of practical functions of the target application, leaving out the managing tasks for the sake of space. We provide a descriptive UML Use Case in Fig. 1 where the use case Manage stands for the basic CRUD (Create, Read, Update, Delete) persistence operations (Fig. 2).

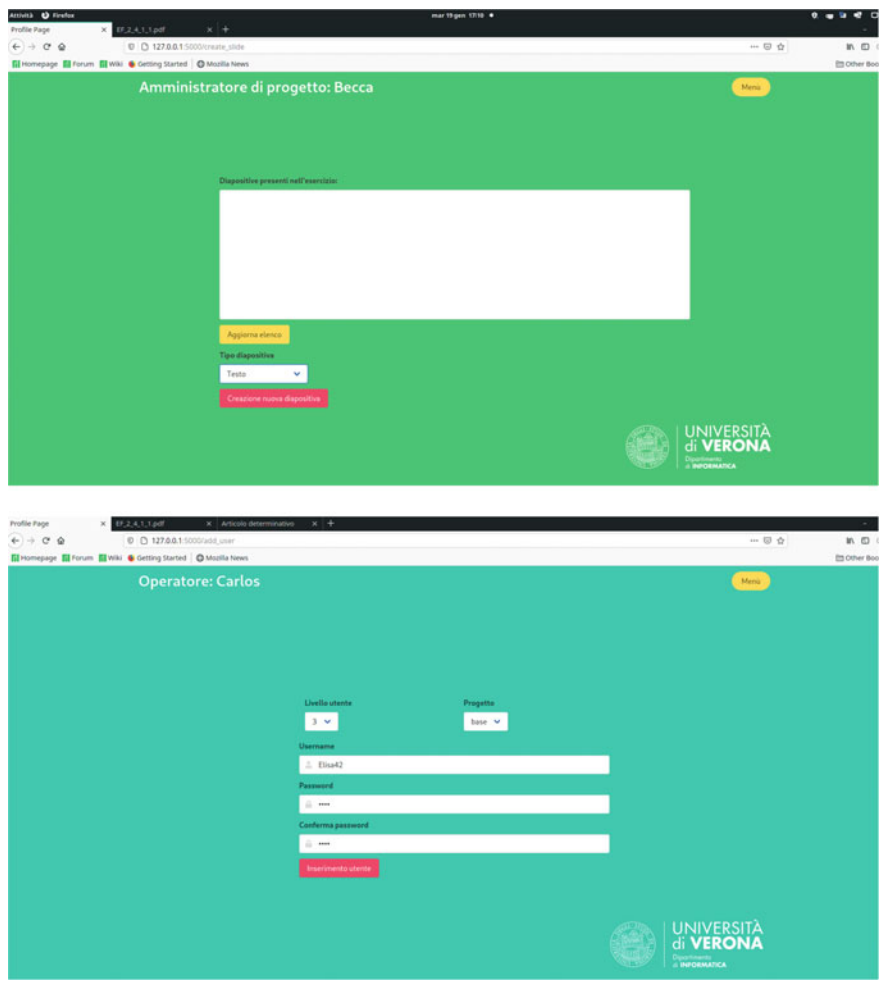


Fig. 2 Two screenshots of the application

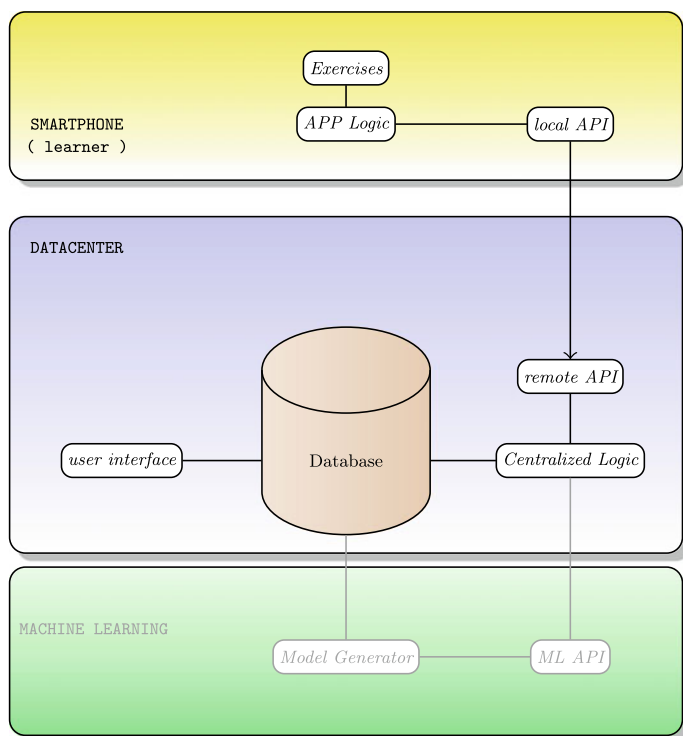


Fig. 3 Logical model of the application

3.2 Application Architecture

The application is a quite common standard *app* whose logical model is described in Fig. 3. Three main layers are described along with their modules: the main interface for *learners* will be responsive. The managing functions are organized in modules which are stored on the cloud. On the server, data are collected and on a background process are passed to a machine learning algorithm used to identify the patterns in common, and use these patterns for the evaluation of progresses.

4 Related Work

Although experimental studies reporting results of reading intervention in adults with dyslexia are still rather sparse, there is evidence indicating that their difficulties in decoding, fluency and comprehension can be successfully compensated for if they are provided with a specific training. The results of a recent systematic review [19] seem to indicate that phonological and morphological interventions, which are typically

beneficial for enhancing reading accuracy, could be integrated with a computerized intervention aiming at improving fluency more specifically. The reading acceleration training, in particular, has provided promising results indicating that imposing time constraints on reading by means of computerized programs can be very effective for automatizing and speeding up the process of decoding, with positive gains extending to reading comprehension as well. Similarly, scientifically validated evidence on effective literacy training for non-literate and low-literate L2 learners is rather scant, as highlighted by a recent review on the topic [20], with most classroom-based studies only providing an impressionistic account of strategies which may be useful with such learners [21]. For languages with a fairly transparent orthography, such as Italian, phonics-based methods, focusing on recognizing and automatizing grapheme-phoneme associations while gradually moving to syllables and words, are generally thought to be among the effective strategies to teach reading (especially if such methods are contextualized, as suggested in [22]). However, a major problem with these methods is represented by the huge amount of instruction hours learners need in order to crack the alphabetic code. Teacher-fronted courses cannot typically allocate many hours to such activities, since this would subtract valuable time to the development of their L2 oral skills, a core need for learners trying to resettle in a new country, and indeed another factor which can boost literacy skills [23]. Even with small groups of students, the time the teacher can dedicate to each learner is inevitably very limited. Moreover, phonics-based activities often turn out to be highly demotivating for adult learners, who already exhibit unusually high dropout rates [23]. Providing students with a tool which promotes autonomous learning would therefore allow teachers to concentrate on interaction rather than focus on time-consuming phonics-based activities, which might be instead more successfully supported by technology, as showed by recent research on technology-assisted learning for LESLLA learners [24]. Furthermore, technology-assisted learning, if properly designed, can guarantee constant and automatic corrective feedback, a practice which has proved to prevent learners from automatizing their errors [25]. Given that such feedback is provided privately, anxiety that learners typically suffer from when corrected in front of their classmates can also be kept at bay, thus creating optimal learning conditions. The application model is inspired by studies by some of the authors on the topic of message deployment in social networks [26, 27], on the topic of multimodal structure of web applications [28, 29] and on the structure of negotiation systems [30, 31]. Technology support to people with dyslexia is a long-term effort since the half of the years 2000 [32]. In 2010, a systematic review of the literature has shown the success (few) and the failures (many) the investigations have reached then [33]. The years between 2010 and 2016 as the most fruitful ones in the study of technological solutions for adults with dyslexia and the learning of language. In particular, some important milestones have been obtained with ‘Dyslexia baca’ [34], an application that has been quite successful and employed in a number of randomised studies thenceforth, and EasyLexia [35], also quite a valid solution essentially obtained in the same period of time. Since then, many studies have been proceeding further. Recently, a new review on the difficulties of these application has been proposed [36] that is concerned with the ability of the adults with dyslexia to work on the web

and mobile devices. It should be also noticed that young children and adults are out of the regular spectrum of technology implementations and only a few methods, such as those cited above have been based upon solutions with this type of feature. Even more limited is the set of studies on illiterate adults' language learning technologies. The very first attempt that is documented in literature back to 2010 [37] and is part of a long-lasting project on the topic. Further on, a more modern approach to the problem is discussed in [38]. Some thought on the principles of design that have to be used in this context are discussed in [39] that have been a reference for the present work.

5 Conclusions and Further Work

In this paper we have discussed the development of an application employed to devise and employ exercises to reinforce the proficiency in language reading and writing for adults with dyslexia and illiterate second language learners. The application is developed as a prototype within a wider intervention for language skill enhancement for employability improvement. The application has shown to be functional for the mentioned purpose from an analytical viewpoint and the prototype is validated in a group of experts, formed by twenty teachers with some training in adults with dyslexia. These have provided a direct feedback on functionalities that reads on a score 8.2 on a scale of 10. One of the major strengths of this research lies in the synergy established between information technology and psycholinguistics and in the integration of different domains of expertise for the development of an original intervention program aimed at enhancing reading skills of people with fragile literacies that can also improve their employability and, more in general, their quality of life. Further steps of this investigation are two: (a) we shall employ and therefore test the application in the real world situations provided as interventions in the project, (b) we shall improve the application in order to achieve higher level quality for further implementations, including the extension to meta-linguistic feedbacks.

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Smart Universities: Assistive Technologies for Students with Hearing Impairments



Jeffrey P. Bakken, Prasanthi Putta, and Vladimir L. Uskov

Abstract Smart universities and smart learning environments can benefit regular students and special students, i.e. students with various types of disabilities including physical, visual, hearing, speech, cognitive and other types of impairments. This paper presents the outcomes of an ongoing research project aimed at systematic identification, analysis, and testing of available open source and commercial software systems that could significantly benefit college students with hearing impairments in highly technological learning environments—smart universities. We analyzed various assistive technologies available for students with hearing impairments, including Dragon, Windows Speech Recognition, Automatic captions in Google Slides, CADET, MovieCaptioner, Dragon Naturally Speaking, Dragon Dictate, Sennheiser MobileConnect, SoundAMP R, Office 365 Dictate, Apple/Mac Voice Recognition, Cochlear implants, C-Print, ViaVoice, Picture Boards, Digitized Speech AAC Devices, Infrared, Inductive Loop Technology, Dragon Dictation App, and other systems. Based on a careful analysis of open source and commercially available products for students with hearing impairments we identified the top software systems that we recommend for implementation and active use in smart universities.

Keywords Students with hearing impairments · Assistive technology · Software systems · Smart university

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1 Introduction: College Students with Disabilities

Smart universities (SmU) and smart classrooms (SmC) have the ability to create multiple opportunities for students to learn material in a multitude of ways. In addition, these environments can give students access to learning content in new and innovative ways that they normally would not have. Although not designed or even conceptualized to benefit students with disabilities, this concept would definitely have an impact on the learning and access to material for students with all different types of disabilities.

In accordance with the U.S. Department of Education, National Center for Education Statistics [1], in 2011–2012 9.0% of all undergraduate students in the age range of 15–23 enrolled in postsecondary institutions in the U.S. were students with some form of disability, 11.3% of those aged 24–29, and 12.4% of those aged 30 and older. As a result, on average, about 10% of the college student population in all US institutions are identified with a disability. Many of these students need some form of technology to be successful in this environment. In general, students in colleges/universities may experience a variety of different categories of disabilities; they include but are not limited to: (1) learning disabilities, (2) speech or language impairments, (3) health impairments, (4) psychological/neurological impairments, (5) hearing impairments, (6) physical/mobility/motion/ orthopedic disabilities, and (7) visual impairments.

Software systems can benefit students with disabilities by providing them equal access in the classroom and different learning environments. These systems can also help them learn more efficiently and effectively and in many cases allow them to interact better with their professor and classmates. Where traditional classrooms do not specifically address software systems and how students with disabilities could be impacted, the implementation of specific advanced software systems in smart universities and smart classrooms would address these learning barriers from the perspective of universal accessibility: providing greater learning opportunities for all students in smart learning environments—including students with disabilities [2–7].

2 Assistive Technologies for Students with Hearing Impairments: Literature Review

2.1 *Assistive Technologies in Use by Students with Hearing Impairments: Examples*

A cochlear implant consists of a tiny receiver placed under the skin behind the ear. The receiver has a probe with electrodes that is implanted into the cochlea, a spiral-shaped portion of the inner ear filled with liquid that transmits vibration to cilia (“hair-cells”) attached to the interior of this coiling structure. A person with a cochlear

implant wears a hearing-aid-like device that features a microphone, a processor, and a transducer. The processor manipulates what the microphone captures and sends a signal to the transducer, usually worn just behind the ear. The transducer changes the signal from electrical to magnetic, that can be received through the skin by the implanted receiver. The receiver then stimulates the probe in the cochlea, causing “hearing” [8].

Professor Bernard Widrow of Stanford University designed a hearing-aid necklace called D-HEAR to aid those with severe to profound hearing loss. The user wears both the necklace and a hearing aid. Sound waves enter from a 60-degree-wide cone-shaped space in front of the user. In a noisy place, the user orients his or her body toward the speaker and surrounding sound is minimized. Microphones in the necklace pick up the sound and transmit it to signal-processing chips that give different weights to input sounds from the various microphones. The microphone array is able to home in on the desired signal and reduce echoes and other undesirable auditory effects while increasing clarity of the dominant signal. The optimized signal is then amplified and sent through a conducting neckloop, which wirelessly transmits a magnetic signal to the telecoil in the user’s hearing aid. Hearing aids commonly feature the telecoil to facilitate use of a telephone by a hearing-impaired person [9].

Kolb, a student at Stanford University is Deaf, wears a hearing aid in each ear, but relies on lip reading and interpreters to understand spoken English. Kolb uses sign language interpreters in all her classes and to help her in her extracurricular activities—she rides for the equestrian team. “The SDRC is really great,” she said, noting how the center is responsible for arranging for the interpretation services she needs [10].

Two University of Washington sophomores Navid Azodi and Thomas Pryor invented “SignAloud,” which is a pair of gloves that can recognize hand gestures that correspond to words and phrases in American Sign Language. Each glove contains sensors that record hand position and movement and send data wirelessly via Bluetooth to a central computer. The computer looks at the gesture data through various sequential statistical regressions, similar to a neural network. If the data match a gesture, then the associated word or phrase is spoken through a speaker. “Our purpose for developing these gloves was to provide an easy-to-use bridge between native speakers of American Sign Language and the rest of the world,” Azodi said [11].

Engineers from Michigan State University developed DeepASL, features a deep learning—or machine learning based on data inspired by the structure and function of the brain—algorithm that automatically translates signs into English. The technology functions through a three-inch sensory device, developed by Leap Motion, that is equipped with cameras to capture the motions of hands and fingers in a continuous manner. Leap Motion converts the motions of one’s hands and fingers into skeleton-like joints. Deep learning algorithm picks up data from the skeleton-like joints and matches it to signs of ASL. “One differentiating feature of DeepASL is that it can translate full sentences without needing users to pause after each sign” [12].

The team, known as MotionSavvy from Rochester Institute of Technology, developed an application. The MotionSavvy case embeds the Leap, and the MotionSavvy software leverages the Leap’s 3D motion recognition, which detects when a person

is using ASL and converts it to text or voice. The software also has voice recognition through the tablet's mic, which allows a hearing person to respond with voice to the person signing. It then converts their voice into text, which the hearing-impaired receiver can understand [13].

2.2 Assistive Technologies in Use by Regular Students: Examples

Four students at the MIT developed a system for captioning online video that was far more efficient than traditional methods, which involve pausing a video frequently to write text and mark time codes. The system used automated speech-recognition software to produce "rough-draft" transcripts, displayed on a simple interface that could easily be edited. Landing a gig to caption videos from five MIT OpenCourseWare (OCW) classes, the students were able to caption 100 h of content in a fraction the time of manual captioning [14].

MIT researchers are developing software and electronics to improve the performance of the implants. Implant recipients wear a small microphone on a hook over the ear that picks up sounds and transmits them along a cable to the processor, a Walkman-size box worn on the body. The processor translates the sounds into multichannel signals and sends them back up the cable to a special connector system behind the ear. The signals are conveyed to the electrodes implanted in the snail-shaped cochlea, where they elicit electric spikes on the auditory nerve fibers that go to the brain [15].

The University of Washington contracts with certified interpreters and CART providers. Sign language interpreters serve as communication facilitators between the student and professors or teaching assistants and other participants in classes and meetings. CART provides an instant translation of spoken English into written English text that is displayed on a laptop monitor. This allows the student to read what is being said during a class session [16].

Instructors from the University of Iowa will be notified in advance before the semester begins by Student Disability Services. Instructors are responsible for offering captioned versions of course materials if they intend to use DVD, VHS, or web-based videos and/or podcasts in their course. Videos are accessible when they are captioned and podcasts are accessible when a written transcript accompanies the audio file. If a video is not captioned or a podcast does not have a transcript, instructors will need to arrange for an accessible version to be produced. Instructors will plan ahead by allowing up to two weeks for transcription and captioning services [17].

Teachers at the University of Southern California report feeling more equipped to help deaf and hard-of-hearing students in the classroom. They receive five days of professional development designed to give them new strategies in teaching phonological awareness, shared reading, and writing. They also benefit from daily coaching and mentoring during the summer program. "Children are more engaged in reading and writing activities. Parents are learning new ways to help their children become better

readers and writers. And teachers are gaining additional skills in helping parents and children during this learning process. Supporting all three groups is critical to child success.” said speech language pathologist Dianne Hammes Ganguly [18].

The students of Oregon State University will approach the faculty before class to provide the FM transmitter and microphone, and will pick them up at the end of class. Some classrooms at Oregon State University are “looped” and in those cases, it is not necessary for the faculty member to wear a microphone or transmitter. The lapel microphone must be placed on a collar or upper lapel area and turned on. Faculty will turn off the microphone when having private conversations. Because the speaker’s voice is transmitted directly to the student using the FM system and other noises are screened out, questions and comments from other students in the class cannot be heard. Faculty should repeat those questions and comments so that the student using the FM system can have access to class discussions [19].

C-Print® is a speech-to-text (captioning) technology and service developed at the National Technical Institute for the Deaf, a college of Rochester Institute of Technology. The basis of C-Print is printed text of spoken English displayed in real time, which is a proven and appropriate means of acquiring information for some individuals who are deaf or hard of hearing. A trained operator, called a C-Print captionist, produces a text display of the spoken information in classroom or other settings. At the same time, one or more students read the display to access the information. A C-Print captionist includes as much information as possible, generally providing a meaning-for-meaning translation of the spoken English content. After class, the text can be provided in paper or electronic format for the student to use as notes [20].

3 Research Project Goal and Objectives

During creative literature review of designated and multiple additional available publications no publications were identified with a classification of available open source and commercial software systems using a set of evaluation criteria (functions, technical platform, SWOT analysis, price, etc.) and their ranking in terms of functionality and usability by university/college student.

Project goal. The overall goal of this research project was to identify the best open source and commercially available software systems for university/college students with hearing impairments—HI systems.

Project objectives. The objectives of this project included but were not limited to:

1. identification of colleges/universities in the U.S. and what HI systems they provide for college students with various forms of disabilities;
2. analysis of available open source and commercial HI systems and identification of the best systems to implement with college students with various forms of disabilities;

3. the identification of the top three open source and commercial HI systems;
4. creation of recommendations for university/college student assistance centers in terms of HI systems to be implemented to benefit college students with various forms of disabilities.

The obtained up-to-date research outcomes and findings are presented below.

4 Research Project Outcomes

4.1 *HI Systems in Use by Universities/Colleges: Best Examples*

We analyzed available publications about HI systems that are used by student assistance centers at top 10 U.S. universities. A summary of our research outcomes are presented in Table 1. (A note: Due to the limits of the current paper, the references to all HI systems analyzed are omitted in this paper; they are available in the HI bibliography at <http://cs-is1.bradley.edu/uskov/HI/> [21]).

Table 1 A list of HI systems that are used by student assistance centers at top universities (limited to the first 10 universities)

College/University	Assistive technology in use	Assistance service providers
MIT—Massachusetts Institute of Technology	• Dragon • Windows Speech Recognition • Mac dictation • Automatic captions in Google Slides • YouTube • CADET • MovieCaptioner	• 3PlayMedia • Rev.com • ACS Captions • Caption First • Rapidtext • Caption Colorado • Casting Words • PostCAP
Stanford University	• Dragon Naturally Speaking • Dragon Dictate • Sennheiser MobileConnect • SoundAMP R	• Rev • Cielo24 • 3Play Media • Deaf Services of Palo Alto (DSPA) • Bay Area Communication Access (BACA) • ONE INTERPRETING • California Relay Service (CRS)
Michigan State University	• Frequency Modulated (FM) systems • LARA's Michigan Online Interpreter System • Dragon Naturally Speaking	• Screen Line, Inc., • Archer Captioning • Karasch & Associates • Sorenson Video Relay Service • Registry of Interpreters for the Deaf (RID) 0 • Purple Communications: IP Relay and VRS

(continued)

Table 1 (continued)

College/University	Assistive technology in use	Assistance service providers
University of Arizona, Tucson/AZ	• FM Systems • Infrared Systems • Office 365 Dictate • Apple/Mac Voice Recognition • Captel Captioned Telephone	• Convo VRS • Sorenson Communications • Z Video Relay Service (ZVRS)
Rochester Institute of Technology	• Cochlear implants • C-Print • ViaVoice • Simply Dictation	• CART • Sorenson • TTY relay services • Harris Communications
University of Illinois at Urbana-Champaign	• Picture Boards • Digitized Speech AAC Devices • Infrared • FM • Inductive Loop Technology • Dragon Dictation App • TTSReader	• VRI Services • TypeWell
Purdue University	• FM Systems	• Purple VRS • On-Site Interpreting • Purple VRI • Sorenson Video Relay Service (SVRS)
Ohio State University	• Captel840i • SComm Ubi-Duo2 TTS™ • UbiDuo2 Wireless • ITY™ • Q90D Digital TTY/VCO by Ameriphone • Uniphone 1140 TTY • Motiva FM Listening System	• Sign language interpreters • Typewell • CART
University of Michigan	• Cochlear Implants • Dragon NaturallySpeaking • Dragon for Mac	• Telecommunication Devices for the Deaf • Closed Captioning • Sign language interpreting
University of Washington	• CaptionSync • FM Systems • Loops • Dragon Naturally Speaking	• Interpreting • Captioning • Amplification services • TTYs
University of Southern California	• FM systems	• Sign Language Interpreter • CART Services • Captioned Media • NoteTake Express

4.2 *HI Software Systems Analyzed*

There are multiple available HI systems that could be implemented in a highly technological smart classroom at a smart university. The research findings and analysis outcomes of analyzed HI systems are presented in Tables 2 and 3.

Table 2 Open source HI systems analyzed

System's name and developer	Main system's functions
Dragon Anywhere—Dictation App by Nuance Communications	• Dragon Anywhere lets you dictate and edit documents by voice on your iOS or Android mobile device quickly and accurately, so you can stay productive anywhere you go • Fast dictation and high recognition accuracy that continually improves as it adapts to your voice. No time or length limits; speak as long as you want to, capturing all of the details needed for complete, accurate documentation • Robust voice formatting and editing options, including the ability to select words and sentences for editing or deletion • Voice navigate through fields of a report template and apply common formatting like underline and bold. Support for auto-texts (frequently used text passages) such as client or work order descriptions • Add custom words for industry-specific terminology for even better dictation accuracy • Simple importing and exporting to and from popular cloud document-sharing tools like Dropbox® and note-taking apps like Evernote®
Ntouch by Sorenson Relay	• The Call Waiting feature lets you answer an incoming call while you are already in a call • The Favorites feature lets you create a list of favorite contacts. You can store your most important contacts in the Favorites list to make them easy to find quickly. Using the Favorites list is especially helpful if you have a large number of contacts. You can add a phone number to your Favorites list at the time you create the contact or afterwards. You can open your Favorites list on the Phonebook screen • The Hide My Caller ID feature lets you make videophone calls that do not send Caller ID information in your outgoing calls. When this feature is enabled, you will not be able to call Sorenson users who have enabled the “Don’t Accept Anonymous Calls” feature (described below) on their endpoints • The Don’t Accept Anonymous Calls feature lets you reject incoming videophone calls that do not have Caller ID information. This feature can help reject solicitation calls or calls from people who are hiding their Caller ID information. Remember, if you turn on the Don’t Accept Anonymous Calls feature, other Sorenson users will not be able to call you if they have enabled the “Hide My Caller ID” feature (described above) on their endpoints • The Yelp Search feature lets you search for phone numbers of local businesses. This feature replaces the Sorenson Directory feature in previous releases • The New VRS Call feature gives you more control when making Sorenson VRS calls. After the interpreter ends the hearing side of a VRS call, you can then dial the phone number for a new call yourself instead of having to sign the number to the interpreter and you stay connected to the same interpreter

(continued)

Table 2 (continued)

System's name and developer	Main system's functions
P3 Mobile by Purple Communications	• P3 Mobile is the on-the-go VRS (Video Relay Service) app that makes it easy for deaf and hard-of-hearing individuals to make and receive VP calls anywhere using a Wi-Fi or a cellular data connection • Multiple Logins—Do you have separate accounts for home and work? No problem! You can now easily switch between up to three accounts logged in on your P3 mobile • Access your iPhone contacts—P3 mobile is now fully integrated with your iPhone contacts so that you can make calls from your contacts list seamlessly • Easily tell your interpreter who you want to call next, no need to switch apps • Contact List Sorting feature helps you Sort your contacts the way you want and Quickly find the name you need to call • PurpleMail Greeting • Quickly customize your PurpleMail greeting from anywhere • Manage your PurpleMail greetings from your phone • Review, record or delete your PurpleMail greeting • With Number Lookup feature, Easily find phone numbers and Quickly conduct map searches
Live Listen by Apple	• With Live Listen, your iPhone, iPad, or iPod touch becomes a remote microphone that sends sound to your Made for iPhone hearing aid • Live Listen can help you hear a conversation in a noisy room or hear someone speaking across the room • Need to pair with your hearing device • If a compatible hearing aid is paired to a user's phone, there are options to turn Live Listen on and off, adjust volume and even set it as their preferred Accessibility Shortcut • Live Listen support in AirPods is key. The inclusion of this feature makes AirPods more capable and more alluring
Microsoft Stream by Microsoft	• Microsoft Stream is an Enterprise Video service where people in your organization can upload, view, and share videos securely • Generate automatic captions and a transcript for your Microsoft Stream videos • Find content that you're looking for in Stream by searching for a video. Video search incorporates deep-search across words spoken in the video transcript. To watch, simply select the video or the timecode • From the playback page of any Microsoft Stream video, you can view a scrolling transcript window so you can easily navigate and follow along with the spoken content of a video

Table 3 Commercial HI systems analyzed

System's name and developer	Main system's functions
Dragon Naturally Speaking by Nuance Communications	• Supports remote use on a computer running Windows Server® 2008 R2, Windows 7 Ultimate, or Windows Server 2012. With Microsoft®'s free Remote Desktop Connection software (formerly called Terminal Services Client), you can use Dragon from a local Windows computer on which Dragon itself is not installed • Includes option to save synchronized audio from dictation done in DragonPad, Word, WordPerfect and OpenOffice Writer (Dragon saves a .dra file along with the transcribed text file); • Includes AutoTranscribe Folder Agent (monitors a specific directory to automatically launch transcription) and Correction Only mode (correctionist can turn on the Correction Only setting within the original dictator's profile) • Dragon 13 Legal is trained using more than 400 million words from legal documents—that delivers optimal out of-the-box recognition accuracy for dictation of legal terms. To further increase accuracy, you can create, import and share custom word lists that are relevant to your clients and areas of specialty • Full Text Control allows you to use voice to perform direct dictation, selection, correction, and cursor movement within text. The Dictation Box is an option in places where Full Text Control is not available
soundAMP R by Ginger Labs	• Sounds are sent to your earbuds in real time. Hear what you'd like to hear. Record what you'd like to record! • Works in many situations, around the table at home, watching TV, in lecture halls, at parties, wherever you'd like to hear, or overhear, the people around you! • We have tuned soundAMP to provide you crystal clear sound at the maximum volume possible. And with its advanced technology, it even reduces volume over the limit • Simple-to-use controls allow you to tune the sound with the equalizer, adjust background sound levels for each situation, and replay the last 30 s • With soundAMP R, record it all—capture lectures, presentations, interviews, or even important information relayed at a doctor's appointment. Bookmark the recording to remember important points! Export recordings to your computer
Cochlear implants by Cochlear Limited	• Cochlear implants are designed to mimic the function of a healthy inner ear or cochlea. They replace the function of damaged sensory hair cells inside the cochlea to help provide clearer sound than what hearing aids can provide • There are two primary components of the Cochlear™ Nucleus® System: the external sound processor and the implant that is surgically placed underneath the skin attached to the electrode array that is inserted in the cochlea • Electrical pulses that represent the energy contained in sound signals are sent from the microphone to the speech processor • The speech processor selects and codes the most useful portions of the sound signals • Code is sent to the transmitter • Transmitter sends code across skin to receiver/stimulator • Receiver/stimulator converts code to electrical signals • Electrical signals are sent to electrode array in the cochlea to stimulate hearing nerve fibers • Signals are recognized as sounds by the brain

(continued)

Table 3 (continued)

System's name and developer	Main system's functions
Williams Sound Personal FM Listening System by Williams AV, LLC	• The PFM PRO RCH personal FM system is perfect for provided hearing assistance in a variety of environments. The PFM PRO RCH model is rechargeable and comes with a recharging base for your receivers. This system is most popular among teachers when dealing with students that need assistance • The user wears the receiver and headphones or earbuds that are provided • Both units are turned on and to the same channel • The system provides a clear channel of communication to prevent distractions or provided amplified assistance of the speaker's voice • The system also comes with a aux cable. This allows the transmitter to be plugged into any audio source to be transmitted to the student or user. This allows video's, assemblies, or other instruction and information to be sent to the receiver
C-Print by National Technical Institute for the Deaf, a college of Rochester Institute of Technology	• C-Print is a speech-to-text (captioning) technology and service developed at the National Technical Institute for the Deaf, a college of Rochester Institute of Technology • C-Print Pro software is specifically designed for providing C-Print speech-to-text services. It allows a captionist to input text using a keyboard abbreviation system • The abbreviation system is based on phonetics, or how words sound. Although spelling-based abbreviations might seem easier to learn, in practice, abbreviations based on how words sound are more instinctive because unlike traditional keyboard typing, a C-Print captionist processes information auditorily. Typing using abbreviations based on how words sound is an extension of the auditory process 1. The software also can accept input from automatic speech recognition applications

4.3 Top HI Systems Identified

We tested and analyzed the functionality of the designated open-source systems and demo versions or trial versions of most of the commercial HI systems. The outcomes of analysis findings as well as the final ratings of those systems are given below in Tables 4, 5, 6 for open source HI systems and Tables 7, 8, 9—for commercial HI systems.

4.3.1 Top-Ranked Open-Source HI Systems

See Tables 4, 5, 6.

4.3.2 Top-Ranked Commercial HI Systems

See Tables 7, 8, 9.

Table 4 ntouch open source HI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• The Favorites feature lets you create a list of favorite contacts. You can store your most important contacts in the Favorites list to make them easy to find quickly. Using the Favorites list is especially helpful if you have a large number of contacts. You can add a phone number to your Favorites list at the time you create the contact or afterwards. You can open your Favorites list on the Phonebook screen • The Hide My Caller ID feature lets you make videophone calls that do not send Caller ID information in your outgoing calls. When this feature is enabled, you will not be able to call Sorenson users who have enabled the "Don't Accept Anonymous Calls" feature (described below) on their endpoints
Strengths and opportunities	• The New VRS Call feature gives you more control when making Sorenson VRS calls. After the interpreter ends the hearing side of a VRS call, you can then dial the phone number for a new call yourself instead of having to sign the number to the interpreter and you stay connected to the same interpreter • The Don't Accept Anonymous Calls feature lets you reject incoming videophone calls that do not have Caller ID information
Possible weaknesses	• Purple VRS effects
Technical platform	• iOS • Android
Price (if any)	• Free
Colleges/Universities that currently use this system	• Michigan State University • University of Arizona, Tucson/AZ • Rochester Institute of Technology • Purdue University • California State University-Northridge
Ranking	1

5 Conclusions. Future Steps

Conclusions. The performed research helped us to identify the status of HI systems available for college students with hearing impairments. The obtained research outcomes and findings enabled us to make the following conclusions:

1. We tested, analyzed and evaluated five (5) open source (Table 4) and five commercial (Table 5) HI systems.
2. We ranked the analyzed open source and commercial HI systems—the research outcomes are presented in Tables 4, 5, 6, 7, 8 and 9.
3. Based on our evaluation, the top open source HI system is *ntouch* system (Table 4), and the top commercial HI system is *Cochlear implants* system (Table 7). Those systems are strongly recommended by our research team for implementation and active use in smart classes of smart universities.
4. More research needs to be completed that directly focuses on the perception of HI systems by actual college students with motion/mobility disabilities.

Next steps. The next steps of this research, design and development project deal with

Table 5 P3 Mobile: open source HI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• Multiple Logins—Do you have separate accounts for home and work? No problem! You can now easily switch between up to three accounts logged in on your P3 mobile • Access your iPhone contacts—P3 mobile is now fully integrated with your iPhone contacts so that you can make calls from your contacts list seamlessly • Easily tell your interpreter who you want to call next, no need to switch apps • Contact List Sorting feature helps you Sort your contacts the way you want and Quickly find the name you need to call
Strengths and opportunities	• With Number Lookup feature, easily find phone numbers and Quickly conduct map searches • Quickly customize your PurpleMail greeting from anywhere • With Number Lookup feature, easily find phone numbers and Quickly conduct map searches
Possible weaknesses and threats	• Unavailability of horizontal screen option
Technical platform	• iOS • Android
Price (if any)	• Free
Colleges/Universities that currently use this system	• Michigan State University • Purdue University
Ranking	2

Table 6 Dragon Anywhere-Dictation App: open source HI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• Dragon Anywhere lets you dictate and edit documents by voice on your iOS or Android mobile device quickly and accurately, so you can stay productive anywhere you go • Fast dictation and high recognition accuracy that continually improves as it adapts to your voice. No time or length limits; speak as long as you want to, capturing all of the details needed for complete, accurate documentation • Robust voice formatting and editing options, including the ability to select words and sentences for editing or deletion
Strengths and opportunities	• Add custom words for industry-specific terminology for even better dictation accuracy • Simple importing and exporting to and from popular cloud document-sharing tools like Dropbox® and note-taking apps like Evernote® • Voice commands for fixing errors and exporting finished text
Possible weaknesses	• Must dictate into this one app
Technical platform	• iOS
Price (if any)	• Free
Colleges/Universities that currently use this system	• MIT—Massachusetts Institute of Technology • Stanford University
Ranking	3

Table 7 Cochlear implants commercial HI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• Cochlear implants are designed to mimic the function of a healthy inner ear or cochlea. They replace the function of damaged sensory hair cells inside the cochlea to help provide clearer sound than what hearing aids can provide • There are two primary components of the Cochlear™ Nucleus® System: the external sound processor and the implant that is surgically placed underneath the skin attached to the electrode array that is inserted in the cochlea • Electrical pulses that represent the energy contained in sound signals are sent from the microphone to the speech processor
Strengths and opportunities	• Hearing ranges from near normal ability to understand speech to no hearing benefit at all • Adults often benefit immediately • Many understand speech without lip-reading • Many can make telephone calls
Possible weaknesses and threats	• General Anesthesia Risks
Technical platform	• NA
Price (if any)	• \$100,000
Colleges/Universities that currently use this system	• Rochester Institute of Technology • University of Michigan • California State University-Northridge
Ranking	1

Table 8 Dragon naturally speaking commercial HI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• Includes option to save synchronized audio from dictation done in DragonPad, Word, WordPerfect and OpenOffice Writer (Dragon saves a .dra file along with the transcribed text file); • Includes AutoTranscribe Folder Agent (monitors a specific directory to automatically launch transcription) and Correction Only mode (correctionist can turn on the Correction Only setting within the original dictator's profile) • Full Text Control allows you to use voice to perform direct dictation, selection, correction, and cursor movement within text. The Dictation Box is an option in places where Full Text Control is not available • Menu Tracking provides the ability to use voice to "click" an application's menus, buttons, dialog boxes, etc
Strengths and opportunities	• Supports remote use on a computer running Windows Server® 2008 R2, Windows 7 Ultimate, or Windows Server 2012. With Microsoft®'s free Remote Desktop Connection software (formerly called Terminal Services Client), you can use Dragon from a local Windows computer on which Dragon itself is not installed • Dragon 13 Legal is trained using more than 400 million words from legal documents—that delivers optimal out-of-the-box recognition accuracy for dictation of legal terms • Natural Language Commands let you state your intent within a specific application instead of following the menu-selection and mouse steps of the Windows interface

(continued)

Table 8 (continued)

System's features	System's details
Possible weaknesses and threats	• Expensive, considering you get some dictation features built into the Windows operating system
Technical platform	• Windows
Price (if any)	• \$150.00
Colleges/Universities that currently use this system	• MIT—Massachusetts Institute of Technology • Stanford University • Michigan State University • University of Michigan • University of Washington
Ranking	2

Table 9 C-print—commercial HI system: brief analysis outcomes

System's features	System's details
Main most system's important features and functions	• C-Print Pro software is specifically designed for providing C-Print speech-to-text services. It allows a captionist to input text using a keyboard abbreviation system • The abbreviation system is based on phonetics, or how words sound. Although spelling-based abbreviations might seem easier to learn, in practice, abbreviations based on how words sound are more instinctive because unlike traditional keyboard typing, a C-Print captionist processes information auditorily. Typing using abbreviations based on how words sound is an extension of the auditory process
Strengths and opportunities	• The software also can accept input from automatic speech recognition applications • An interesting and unique offering is captionists who are also skilled interpreters
Possible weaknesses and threats	• Privacy
Technical platform	• Windows • Mac OS • Android • iOS
Price (if any)	• \$330.00
Colleges/Universities that currently use this system	• Rochester Institute of Technology • California State University-Northridge • University of Wisconsin-Whitewater
Ranking	3

1. More implementation, analysis, testing and quality assessment of HI systems by actual college students with hearing impairments.
2. Implementation, analysis, testing and quality assessment of HI systems in everyday teaching of classes in smart classrooms.
3. Organization and implementation of summative and formative evaluations of local and remote college students and learners with and without disabilities with a focus to collect sufficient data on quality of HI systems.
4. Creation of a set of recommendations (technological, structural, financial, curricula, etc.) on what HI systems universities should get (purchase, if needed)

and install to benefit college students with and without disabilities at smart university.

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